Algebra I

Lecture Notes & Transcription

Course: **Algebra I**
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Notation and Facts Assumed by the Lectures

Background from the First-Year Courses

About This Chapter

This chapter is not part of the lectures. It collects what the lectures take for granted from the first-year courses (Linear Algebra, Analysis, and the Foundations course *"Grundstrukturen"*), together with the notation of these notes. The lectures point here wherever they use one of these facts.

- Results are stated as **facts**, without proof: they are standard first-year material. Bézout's identity and Euclid's lemma get a two-line argument, because the lectures build on them.
- The first lecture of the course (Tuesday, 15 September) was not recorded. According to the later lectures, it introduced groups, rings and fields and discussed factorization in $\mathbb{Z}$; these notes start with the second lecture. The lecturer restates groups and fields in Week 2 and rings in Week 1.

Notation, Sets and Maps

Notation 0.1 (*Number Sets and Inclusions*):

- $a := b$ defines a as b, and $A :\iff B$ defines the statement A as B. Where the lecturer writes *"def"* over the sign on the board, these notes keep it: $\overset{\text{def}}{=}$ and $\overset{\text{def}}{\iff}$.
- $\mathbb{N} := \{0, 1, 2, \dots\}$, also written $\mathbb{N}_0$ to stress that 0 is included, and $\mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$. Subscripts restrict a set: $\mathbb{Z}_{>0} := \{1, 2, 3, \dots\}$, $\mathbb{Z}_{\geq 2}$, $\mathbb{R}_{>0}$.
- For a field K, $K^* := K \setminus \{0\}$. The group of units of a ring R (Week 2, Proposition 2.70) appears in these notes as $\mathcal{U}_R$, $U(R)$ and $\mathrm{Unit}(R)$, as on the board (elsewhere also written $R^\times$), and, from Week 3 on, R^* (Proposition 3.3). For a field the two meanings of K^* agree, since its units are exactly its non-zero elements.
- $A \subset B$ means that every element of A lies in B; equality is allowed, as in the lecturer's usage. A proper inclusion is written $A \subsetneq B$.
- $|X|$ (or $\#X$) is the number of elements of a finite set X; for a finite group it is called the *"order"*.
- $X \times Y := \{(x, y) \mid x \in X,\ y \in Y\}$ is the *"Cartesian product"*. A binary operation on X is a map $X \times X \to X$; the set of pairs used for the field of fractions (Definition 3.12) is a subset of $R \times R$.

- Y^X is the set of all maps $X \to Y$. In particular $R^{\mathbb{N}_0}$ is the set of all sequences $(a_0, a_1, a_2, \dots)$ with entries in R; polynomials and power series are defined as such sequences (Definition 3.34, Definition 3.41).

Definition 0.2 (*Maps*): A map $f\colon X \to Y$ is *“injective”* if $f(x_1) = f(x_2)$ implies $x_1 = x_2$, *“surjective”* if every $y \in Y$ is of the form $f(x)$, and *“bijective”* if it is both. The composition of f with $g\colon Y \to Z$ is $g \circ f\colon X \to Z$, $x \mapsto g(f(x))$; the identity of X is Id_X (also written 1_X); the restriction of f to $U \subset X$ is $f|_U\colon U \to Y$.

Fact 0.3 (*Inverse Maps*): A map $f\colon X \to Y$ is bijective if and only if there is a map $g\colon Y \to X$ with $g \circ f = \mathrm{Id}_X$ and $f \circ g = \mathrm{Id}_Y$. This g is unique; it is the *“inverse map”* f^{-1}.

Fact 0.4 (*Cardinality*): Two sets have the same *“cardinality”* if there is a bijection between them. A set is *“countable”* if it is finite or has the cardinality of $\mathbb{N}$: $\mathbb{Z}$ and $\mathbb{Q}$ are countable, while $\mathbb{R}$ is *“uncountable”* (Cantor’s diagonal argument). For finite sets, $|A \times B| = |A| \cdot |B|$, and a bijection preserves the number of elements; this counts the units of $\mathbb{Z}/n\mathbb{Z}$ in Week 5 (Theorem 5.52). The sets $\mathbb{R}_{>0}$ and $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$ have the same cardinality as $\mathbb{R}$; for $\mathbb{R}_{>0}$, a bijection is exp (Fact 0.38). A bijection says nothing about algebraic structure: the groups $(\mathbb{R}^*, \cdot)$ and $(\mathbb{R}_{>0}, \cdot)$ have the same cardinality but are not isomorphic (Week 2, Proposition 2.15).

Definition 0.5 (*Image and Preimage of a Subset*): Let $f\colon X \to Y$ be a map. For $A \subset X$, the *“image”* of A is $f(A) := \{f(a) \mid a \in A\} \subset Y$; the image of f is $\mathrm{Im}(f) := f(X)$. For $B \subset Y$, the *“preimage”* (or pullback) of B is

$$f^{-1}(B) := \{x \in X \mid f(x) \in B\} \subset X.$$

Important Remark 0.6 (*Two Meanings of f^{-1}*): The preimage $f^{-1}(B)$ is defined for **every** map f, bijective or not; it is a set, possibly empty. Only when f is bijective does f^{-1} also denote the inverse map (Fact 0.3), and then the two readings agree: the preimage of B is the image of B under the inverse map. The kernel of a homomorphism is a preimage, $\ker(f) = f^{-1}(\{0\})$.

Fact 0.7 (*Images and Preimages*): Let $f\colon X \to Y$ be a map, $A, A' \subset X$ and $B, B' \subset Y$.

- **(i)** Preimages respect inclusions, intersections and unions: $B \subset B' \implies f^{-1}(B) \subset f^{-1}(B')$, $f^{-1}(B \cap B') = f^{-1}(B) \cap f^{-1}(B')$, $f^{-1}(B \cup B') = f^{-1}(B) \cup f^{-1}(B')$. Images respect inclusions and unions, but in general not intersections.
- **(ii)** $f(f^{-1}(B)) \subset B$, with equality for every B if and only if f is surjective.
- **(iii)** $A \subset f^{-1}(f(A))$, with equality for every A if and only if f is injective.

The Correspondence Theorem (Proposition 4.52, Week 4) uses (ii) for the surjective quotient map, and replaces injectivity in (iii) by the hypothesis that the ideal contains the kernel.

Equivalence Relations and Partial Orders

Definition 0.8 (*Equivalence Relation*): An *“equivalence relation”* on a set T is a relation $\equiv$ on T that is

- **(i)** **reflexive:** $x \equiv x$ for all $x \in T$;
- **(ii)** **symmetric:** $x \equiv y \implies y \equiv x$;
- **(iii)** **transitive:** $x \equiv y$ and $y \equiv z \implies x \equiv z$.

Definition 0.9 (*Equivalence Classes and Quotient Set*): The *“equivalence class”* of $x \in T$ is $[x] := \{y \in T \mid y \equiv x\}$, and the *“quotient set”* $T/{\equiv}$ (written $T_{\equiv}$ on the lecturer’s board) is the set of all equivalence classes. Two classes are either equal or disjoint, $[x] = [y]$ holds exactly when $x \equiv y$, and the classes cover T: they form a *"partition"* of T.

Maps Defined on Classes: A rule that defines a value for a class through one of its elements (a *"representative"*) defines a map on ${}^{T}\!/_{\equiv}$ only if equivalent representatives give the same value; the rule is then called *"well-defined"*. The lectures carry out this check for the operations of ${}^{\mathbb{Z}}\!/_{n\mathbb{Z}}$ (Definition 1.23), for the extension $g\colon \mathbb{Q} \to F$ (Claim 3.9) and for the field of fractions (Proposition 3.22).

Definition 0.10 (*Partial Orders and Maximal Elements*): A *"partial order"* on a set P is a relation $\leq$ that is reflexive, transitive and *"antisymmetric"* ($x \leq y$ and $y \leq x$ imply $x = y$). Two elements need not be comparable; the standard example is a set of subsets, ordered by inclusion. A *"chain"* is a subset in which any two elements are comparable (a totally ordered subset). An *"upper bound"* of $A \subset P$ is an element $b \in P$ with $a \leq b$ for all $a \in A$. An element $m \in P$ is *"maximal"* if nothing in P lies strictly above it: $m \leq x$ implies $x = m$. A maximal element need not be a largest element, and there can be several. The lectures use these notions for Zorn's lemma (Week 5, Lemma 5.14) and for maximal ideals, which are the maximal elements of the set of proper ideals ordered by inclusion (Week 4, Definition 4.54); $\mathbb{Z}$ has infinitely many of them, the ideals $p\mathbb{Z}$ for primes p.

The Integers

Fact 0.11 (*Well-Ordering Principle*): Every non-empty subset of $\mathbb{N}$ has a least element. (This is equivalent to the principle of induction, Fact 0.12.) The lectures use it to find the generator of an ideal of $\mathbb{Z}$ (Theorem 2.35).

Fact 0.12 (*Induction*): Let $P(n)$ be a statement for each $n \in \mathbb{N}$. If $P(0)$ holds, and $P(n)$ implies $P(n+1)$ for every n, then $P(n)$ holds for all $n \in \mathbb{N}$. The same holds with any starting value $n_0 \in \mathbb{Z}$ in place of 0.

Fact 0.13 (*Binomial Theorem*): For $n \in \mathbb{N}$ and $0 \leq a \leq n$, the *"binomial coefficient"* $\binom{n}{a} := \frac{n!}{a!\,(n-a)!}$ is a positive integer, and $\binom{n}{a} = \binom{n-1}{a-1} + \binom{n-1}{a}$ for $1 \leq a \leq n-1$ (Pascal's rule). For all $x, y \in \mathbb{C}$,

$$(x+y)^n = \sum_{a=0}^{n} \binom{n}{a} x^a y^{n-a},$$

by induction on n using Pascal's rule. The proof uses only commutativity and distributivity, so the formula holds in every commutative ring (Theorem 4.2, Week 4).

Fact 0.14 (*Division with Remainder*): For $y \in \mathbb{Z}$ and $n \in \mathbb{Z}_{>0}$ there are unique integers q, r with

$$y = qn + r \quad \text{and} \quad 0 \leq r < n.$$

Definition 0.15 (*Divisibility, Greatest Common Divisor, Primes*): For $a, b \in \mathbb{Z}$, a *"divides"* b, written $a \mid b$, if $b = ac$ for some $c \in \mathbb{Z}$. For a, b not both 0, $\mathrm{GCD}(a,b)$ is the largest integer dividing both (so $\mathrm{GCD}(a,b) \geq 1$, as 1 divides both); the lecturer also writes (a,b). The integers a, b are *"coprime"* (relatively prime) if $\mathrm{GCD}(a,b) = 1$. A *"prime"* is an integer $p \geq 2$ whose only positive divisors are 1 and p.

Fact 0.16 (*Bézout's Identity*): For $a, b \in \mathbb{Z}$, not both 0, there are $x, y \in \mathbb{Z}$ with

$$ax + by = \mathrm{GCD}(a,b).$$

In particular, a and b are coprime if and only if $ax + by = 1$ for some $x, y \in \mathbb{Z}$.

Why It Holds: The set $I := \{ax + by \mid x, y \in \mathbb{Z}\}$ is an ideal of $\mathbb{Z}$ (Definition 1.29), so $I = d\mathbb{Z}$ with $d > 0$ by Theorem 2.35 (Week 2). Since $a, b \in I$, d divides a and b; and every common divisor of a and b divides every element of I, in particular d. Hence $d = \mathrm{GCD}(a, b)$. Conversely, if $ax + by = 1$, every common divisor of a and b divides 1.

Fact 0.17 (*Euclid's Lemma*): Let p be a prime. If $p \mid ab$, then $p \mid a$ or $p \mid b$.

Why It Holds: If $p \nmid a$, then $\mathrm{GCD}(a, p) = 1$, since the only positive divisors of p are 1 and p. By Bézout, $ax + py = 1$ for some $x, y \in \mathbb{Z}$, so $b = ab\,x + p\,by$, and p divides both terms on the right.

Example 0.18 (*The Euclidean Algorithm*): The coefficients in Bézout's identity can be computed. Since $\mathrm{GCD}(y, n) = \mathrm{GCD}(n, r)$ whenever $y = qn + r$, repeated division with remainder (Fact 0.14) shrinks the pair until the remainder is 0; the last non-zero remainder is the GCD. For $a = 252$, $b = 198$:

$$\begin{aligned} 252 &= 1 \cdot 198 + 54, &\qquad 198 &= 3 \cdot 54 + 36, \\ 54 &= 1 \cdot 36 + 18, &\qquad 36 &= 2 \cdot 18 + 0, \end{aligned}$$

so $\mathrm{GCD}(252, 198) = 18$. Solving the equations backwards for the remainders,

$$18 = 54 - 36 = 54 - (198 - 3 \cdot 54) = 4 \cdot 54 - 198 = 4 \cdot (252 - 198) - 198 = 4 \cdot 252 - 5 \cdot 198.$$

If $\mathrm{GCD}(a, m) = 1$, the same back-substitution gives $ax + my = 1$, so $\bar{x}$ is the inverse of $\bar{a}$ in $\mathbb{Z}/m\mathbb{Z}$ (Theorem 2.71).

Fact 0.19 (*Unique Factorization in $\mathbb{Z}$*): Every integer $n \geq 2$ is a product of primes, and this product is unique up to the order of the factors.

Fact 0.20 (*Least Common Multiple, and GCD and LCM from Prime Factorizations*): For non-zero $a, b \in \mathbb{Z}$, $\mathrm{LCM}(a, b)$ is the smallest positive integer divisible by both. Writing $|a| = \prod_i p_i^{n_i}$ and $|b| = \prod_i p_i^{m_i}$ over the same distinct primes p_i (with exponents $n_i, m_i \geq 0$),

$$\mathrm{GCD}(a, b) = \prod_i p_i^{\min(n_i, m_i)}, \qquad \mathrm{LCM}(a, b) = \prod_i p_i^{\max(n_i, m_i)}, \qquad \mathrm{GCD}(a, b) \cdot \mathrm{LCM}(a, b) = |ab|.$$

Every common divisor of a and b divides $\mathrm{GCD}(a, b)$, and every common multiple is a multiple of $\mathrm{LCM}(a, b)$. Week 6 carries all of this over to unique factorization domains (Theorem 6.35, Proposition 6.39).

Definition 0.21 (*Congruence Modulo n*): Let $n \in \mathbb{Z}_{>0}$. Integers a, b are *"congruent modulo n"*, written $a \equiv b \pmod{n}$, if $n \mid a - b$.

Fact 0.22 (*Residue Classes*): Congruence modulo n is an equivalence relation on $\mathbb{Z}$, and it is compatible with the operations: $a \equiv a'$ and $b \equiv b'$ imply $a + b \equiv a' + b'$ and $ab \equiv a'b' \pmod{n}$. Its classes are the *"residue classes"* $\bar{a} := a + n\mathbb{Z}$; by division with remainder there are exactly n of them, $\bar{0}, \bar{1}, \ldots, \overline{n-1}$. The lecturer recalls the resulting ring $\mathbb{Z}/n\mathbb{Z}$ in Week 1 (Definition 1.23).

Groups, Fields and the Complex Numbers

Definition 0.23 (*Group*): A *"group"* $(G, \cdot, e)$ is a set G with an associative operation $\cdot$ and a neutral element e ($ea = ae = a$ for all a) in which every element has an inverse ($ab = ba = e$ for some b). Without the inverses, such a structure is called a *"monoid"*; the multiplication of a ring is one, although the lectures do not use the word. A *"subgroup"* of G is a subset $H \subset G$ with $e \in H$ that is closed under the operation and under inverses; it is a group with the restricted operation (Week 2, Definition 2.19). The group G is *"abelian"* if $ab = ba$ for all a, b; an abelian group is often written additively as $(G, +, 0)$, with inverses $-a$. Examples: $(\mathbb{Z}, +, 0)$ and $(\mathbb{R}^*, \cdot, 1)$. The lecturer restates the definition in Week 2 (Definition 2.1).

Definition 0.24 (*Group Homomorphism*): A map $f\colon G \to H$ between groups is a *"homomorphism"* if $f(ab) = f(a)f(b)$ for all $a, b \in G$, and an *"isomorphism"* if it is moreover bijective (Week 2, Definition 2.10).

Definition 0.25 (*Order of an Element, Cyclic Groups*): Let G be a group with neutral element e and $a \in G$. The *"order"* of a (in Week 7, $\mathrm{Ord}(a)$, Definition 7.39) is the least $n \geq 1$ with $a^n = e$ (or ∞ if there is none); in additive notation the condition reads $na = 0$. The group G is *"cyclic"*, with *"generator"* a, if every element of G is a power a^k, $k \in \mathbb{Z}$. A finite group G is cyclic with generator a exactly when a has order $|G|$, and then $G = \{e, a, a^2, \dots, a^{|G|-1}\}$. Example: $(\mathbb{Z}/n\mathbb{Z}, +, \bar{0})$ is cyclic with generator $\bar{1}$. An isomorphism maps elements of order n to elements of order n; the lectures use this in Week 2 (Proposition 2.15, Proposition 2.73).

Definition 0.26 (*Field*): A *"field"* $(K, +, \cdot, 0, 1)$ is a set with two operations such that $(K, +, 0)$ is an abelian group, $(K \setminus \{0\}, \cdot, 1)$ is an abelian group, and $a(b + c) = ab + ac$ for all a, b, c. In particular $0 \neq 1$. Examples are $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{Z}/p\mathbb{Z}$ for a prime p (in the lectures: Proposition 2.63 and Theorem 2.71). A *"subfield"* of a field is a subring (Week 2, Definition 2.26) that is itself a field, such as $\mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ or the Gaussian rationals $\mathbb{Q}(i) \subset \mathbb{C}$ (Week 3, Example 3.28). The lecturer restates fields in Week 2 (Definition 2.9).

Fact 0.27 (*Complex Conjugation and Modulus*): For $z = a + bi \in \mathbb{C}$ $(a, b \in \mathbb{R},\ i^2 = -1)$, the *"complex conjugate"* is $\bar{z} := a - bi$ and the *"modulus"* is $|z| := \sqrt{a^2 + b^2}$ (the lecturer calls it the length or the norm of z). For all $z, w \in \mathbb{C}$:

$$\overline{z + w} = \bar{z} + \bar{w}, \qquad \overline{zw} = \bar{z}\,\bar{w}, \qquad z\bar{z} = |z|^2, \qquad |zw| = |z|\,|w|,$$

and $\bar{z} = z$ exactly when $z \in \mathbb{R}$. So conjugation respects sums and products, and under multiplication the lengths of complex numbers multiply (and their angles add).

Linear Algebra

Definition 0.28 (*Linear Map, Kernel and Image*): Let V, W be vector spaces over a field K. A *"subspace"* of V is a subset $U \subset V$ with $0 \in U$ that is closed under addition and under multiplication by scalars ($u, v \in U,\ \lambda \in K \implies u + v \in U,\ \lambda u \in U$); Week 1 compares this with the absorption property of ideals. A map $T\colon V \to W$ is *"linear"* if $T(u + v) = T(u) + T(v)$ and $T(\lambda v) = \lambda T(v)$ for all $u, v \in V$ and $\lambda \in K$. Its *"kernel"* is $\ker(T) := \{v \in V \mid T(v) = 0\}$ and its *"image"* is $\mathrm{Im}(T) := T(V)$. The linear maps $V \to W$ form a vector space $\mathrm{Hom}_K(V, W)$ under pointwise operations; the lectures reuse the notation Hom for the morphisms of any category (Definition 2.44).

Fact 0.29 (*Kernel, Image and Rank–Nullity*): For a linear map $T\colon V \to W$, $\ker(T) \subset V$ and $\mathrm{Im}(T) \subset W$ are subspaces, and T is injective if and only if $\ker(T) = \{0\}$. If V is finite-dimensional,

$$\dim V = \dim \ker(T) + \dim \mathrm{Im}(T).$$

Fact 0.30 (*Span, Linear Independence and Bases*): A family of vectors of V *"spans"* V if every vector of V is a finite linear combination of them, and it is *"linearly independent"* if the only finite linear combination of them that equals 0 is the one with all coefficients 0; a *"basis"* is a linearly independent family that spans V. Every vector space spanned by finitely many vectors has a finite basis, and all its bases have the same number of elements, its *"dimension"*.

That every vector space has a basis is proved with Zorn's lemma, which the course takes up itself (Week 1, Question 1.32; Week 5, Lemma 5.14).

Fact 0.31 (*Matrices*): The $n \times n$ matrices with entries in a field K form a ring $M_n(K)$ under matrix addition and multiplication, with the identity matrix as its 1; for $n \geq 2$ it is not commutative. Its units are the invertible matrices, those with non-zero determinant, and they form the group $\mathrm{GL}_n(K)$. Linear maps $K^n \to K^n$ correspond to matrices and composition to the matrix product, so a linear map between vector spaces of the same finite dimension is an isomorphism exactly when its matrix with respect to bases is invertible. The lectures meet matrix rings only as examples of non-commutative rings (Week 1) and of isomorphisms in $\mathbf{Vect}_K$ (Week 2, Definition 2.50).

Fact 0.32 (*Division with Remainder for Polynomials*): Let K be a field, $f, g \in K[x]$ polynomials with coefficients in K, and $g \neq 0$. There are unique $q, r \in K[x]$ with

$$f = qg + r \quad \text{and} \quad r = 0 \text{ or } \deg(r) < \deg(g).$$

This is long division of polynomials: as long as $\deg(f) \geq \deg(g)$, subtracting $\frac{a}{b} x^{\deg f - \deg g}\, g$, where a and b are the leading coefficients of f and g, lowers the degree of f; dividing by b is where K has to be a field. It is the polynomial analogue of Fact 0.14; the lectures use it for $g = x^2 + 1$ (Week 4, Lemma 4.41).

Definition 0.33 (*Quotient Vector Space*): Let $U \subset V$ be a subspace. The relation $v \equiv w :\iff v - w \in U$ is an equivalence relation on V, with classes $v + U$ (in Linear Algebra often called cosets). The *"quotient space"* ${}^{V}\!/_{U}$ is the set of these classes with

$$(v + U) + (w + U) := (v + w) + U, \qquad \lambda(v + U) := \lambda v + U;$$

these operations are well-defined, ${}^{V}\!/_{U}$ is a vector space, and the projection $\pi \colon V \to {}^{V}\!/_{U}$, $v \mapsto v + U$, is linear and surjective with $\ker(\pi) = U$. If V is finite-dimensional, $\dim {}^{V}\!/_{U} = \dim V - \dim U$.

Fact 0.34 (*First Isomorphism Theorem for Linear Maps*): For a linear map $T \colon V \to W$, the map $\bar{T} \colon {}^{V}\!/_{\ker(T)} \to \mathrm{Im}(T)$, $v + \ker(T) \mapsto T(v)$, is a well-defined isomorphism of vector spaces. With $\dim {}^{V}\!/_{\ker(T)} = \dim V - \dim \ker(T)$ this gives rank–nullity again (Fact 0.29).

The Model for Quotient Rings: The quotient ring ${}^{R}\!/_{I}$ of Week 4 follows this pattern step by step, with the ideal I in the role of the subspace U: congruence modulo I (Definition 4.17), the set of classes (Definition 4.19), well-defined operations (Proposition 4.22, Proposition 4.24), and the First Isomorphism Theorem ${}^{R}\!/_{\ker(f)} \cong \mathrm{Im}(f)$ (Theorem 4.27). The new ingredient is multiplication: a subspace only has to absorb scalars, an ideal has to absorb all of R, and that is exactly what makes the product of classes well-defined.

Definition 0.35 (*Dual Space and Dual Map*): The *"dual space"* of a vector space V over K is $V^* := \mathrm{Hom}_K(V, K)$, the vector space of linear maps $V \to K$. A linear map $T \colon V \to W$ has the *"dual map"* $T^* \colon W^* \to V^*$, $T^*(\varphi) := \varphi \circ T$; it satisfies $(S \circ T)^* = T^* \circ S^*$ and $(\mathrm{Id}_V)^* = \mathrm{Id}_{V^*}$.

Fact 0.36 (*The Double Dual*): For every vector space V, the map $\mathrm{ev} \colon V \to V^{**}$, $v \mapsto (\varphi \mapsto \varphi(v))$, is linear and injective. If V is finite-dimensional, then $\dim V^* = \dim V$ and ev is an isomorphism. (Used in Week 2, Example 2.53.)

Analysis

Fact 0.37 (*The Real Numbers*): $\mathbb{R}$ is a complete ordered field: every non-empty subset that is bounded above has a supremum, and every Cauchy sequence converges. It can be constructed from $\mathbb{Q}$ by Cauchy sequences or by Dedekind cuts (Week 3, Wednesday).

Fact 0.38 (*Exponential Function and Logarithm*): The exponential function $\exp\colon \mathbb{R} \to \mathbb{R}_{>0}$ is a continuous, strictly increasing bijection with $\exp(0) = 1$ and $\exp(x + y) = \exp(x)\exp(y)$; its inverse is the natural logarithm $\ln\colon \mathbb{R}_{>0} \to \mathbb{R}$ (Week 2, Proposition 2.16).

Definition 0.39 (*Open Sets and Closure in $\mathbb{R}$*): A subset $U \subset \mathbb{R}$ is *"open"* if around each of its points it contains a whole interval $(x - \varepsilon, x + \varepsilon)$, $\varepsilon > 0$; examples are open intervals and $\mathbb{R}_{>0}$. A subset is *"closed"* if its complement is open. The *"closure"* $\overline{A}$ of $A \subset \mathbb{R}$ is the smallest closed set containing A: the set of all limits of convergent sequences in A. The lectures use these notions for the support of a function and for restricting functions to an open set (Week 2, Definition 2.39, Definition 2.42).

Fact 0.40 (*Compact Sets and Smooth Functions*): A subset of $\mathbb{R}$ is *"compact"* if and only if it is closed and bounded (Heine–Borel); finite unions of compact sets, and closed subsets of a compact set, are compact. $C^\infty(\mathbb{R})$ denotes the infinitely differentiable functions $\mathbb{R} \to \mathbb{R}$; sums and products of such functions are again infinitely differentiable (Week 2, Definition 2.38).

Fact 0.41 (*Limits and Continuity*): A function $f\colon U \to \mathbb{R}$ on an open set $U \subset \mathbb{R}$ is *"continuous"* at $a \in U$ if $\lim_{x\to a} f(x) = f(a)$, and continuous if this holds at every point of U. Sums, products and compositions of continuous functions are continuous, and every differentiable function is continuous, so the functions in $C^\infty(\mathbb{R})$ are continuous. If $g\colon \mathbb{R} \to \mathbb{R}$ is continuous, the one-sided limit $\lim_{x\to 0^+} g(x)$ exists and equals $g(0)$. So a function on $\mathbb{R}_{>0}$ with no limit at 0^+, such as $x \mapsto \sin(1/x)$, is not the restriction of a continuous function on $\mathbb{R}$ (Week 2, Proposition 2.48).

Fact 0.42 (*Smooth Functions Vanishing on a Half-Line*): The function

$$h(x) := \begin{cases} e^{-1/x} & x > 0, \\ 0 & x \leq 0 \end{cases}$$

lies in $C^\infty(\mathbb{R})$: all its derivatives tend to 0 as $x \to 0^+$, because $e^{-1/x}$ goes to 0 faster than any power of x. So h vanishes exactly on $(-\infty, 0]$; the function $x \mapsto h(-x)$ vanishes exactly on $[0, \infty)$, and $h(x - a)\,h(b - x)$ is a smooth function that is non-zero exactly on (a, b) (a *"bump function"*). Such functions show that the ideals of $C^\infty(\mathbb{R})$ in Week 2 are not zero and differ from each other (Proposition 2.40, Proposition 2.41).

Fact 0.43 (*Geometric Series and Cauchy Product*): For $z \in \mathbb{C}$ with $|z| < 1$,

$$\sum_{k=0}^{\infty} z^k = \frac{1}{1 - z}.$$

If $\sum_n a_n$ and $\sum_n b_n$ converge absolutely, then so does their *"Cauchy product"* $\sum_n c_n$ with $c_n := \sum_{i=0}^{n} a_i b_{n-i}$, and $\sum_n c_n = \bigl(\sum_n a_n\bigr)\bigl(\sum_n b_n\bigr)$. The lectures use the same coefficient formula to multiply polynomials and formal power series, where no convergence is involved (Definition 3.36, Definition 3.42).

Wednesday, 16th September 2026

Gaussian Integers, Ring Axioms, Ring Homomorphisms, and Ideals

Course Content Overview:
Gaussian Integers, Ring Axioms, Homomorphisms & Ideals

In this lecture, we bridge arithmetic and abstract algebra by exploring the following core themes:

- **Gaussian Integers and Primes:** Comparing factorization and units in $\mathbb{Z}$ and $\mathbb{Z}[i]$, and establishing how the primality of an ordinary integer prime in $\mathbb{Z}[i]$ connects directly to Fermat's two-square theorem (an odd prime p is a sum of two squares a^2+b^2 if and only if $p \equiv 1 \pmod 4$).
- **Axioms and Elementary Properties of Rings:** Formalizing the ring axioms $(R,+,\cdot,0,1)$, deriving fundamental arithmetic identities such as $0\cdot a = 0$ and $(-1)\cdot a = -a$, analyzing the degenerate zero ring ($1 = 0$), and defining scalar multiples by integers.
- **Ring Homomorphisms:** Defining structure-preserving maps between rings, explaining why identity preservation ($f(1) = 1$) must be explicitly required while $f(0) = 0$ is automatic, and exploring canonical examples including standard subring inclusions, modular reduction $\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$, and the canonical map $\mathbb{Z} \to R$.
- **Ideals and Field Structure:** Introducing the kernel $\ker(f)$, defining ideals $I \subset R$ via additive closure and multiplicative absorption, proving that every kernel is an ideal, and classifying the ideals of a field (establishing that fields possess only $\{0\}$ and F).
- **Outlook:** Connecting the existence of bases in infinite-dimensional vector spaces to Zorn's lemma and the Axiom of Choice.

Gaussian Integers and Primes

Units and Primes in $\mathbb{Z}$ and $\mathbb{Z}[i]$

Spoken Transcript [00:00:00 - 00:00:53] Prof. Rahul Pandharipande

Okay, so we'll try to start. Maybe you could sit down.

On the practical aspects, there was some progress: the website is more or less correct now, I think, so you can look for the information on the website.

The second issue is that people asked, and now the lectures will be recorded. You can see yourself — or maybe you can only see me, but anyway — the lectures will be recorded, so you can find them later on some website somewhere.

At the beginning, I said that the website is more or less in reasonable form; in particular, the assignment for September 25th is there. Some of the technical parts we're still working on, so we'll get back to you on that, but...

Administrative Opening

The lecturer makes brief administrative announcements regarding the course website, the availability of the first homework assignment due on September 25th, and lecture video recordings.

Spoken Transcript [00:00:53 - 00:02:48] Prof. Rahul Pandharipande

If you spent last night thinking about the Gaussian integers [writes "$\mathbb{Z}[i]$" on the blackboard], then... well, maybe for the people who didn't spend the whole night thinking about the Gaussian integers, I'll say a few things here.

As I said in the previous lecture, we discussed $\mathbb{Z}$ and its factorization properties. And I said you can discuss the same thing for the Gaussian integers. It will be part of the theory of the course, but it's nice to think about the example.

The first thing is: what are... for $\mathbb{Z}$ [writes: $\mathbb{Z}$], there were the units, that's ± 1 [writes: $\{\pm 1\}$]. And those confused our multiplication always, because you could always put in as many -1's and $+1$'s as you want.

So, what's the analogue here? What are the units, what are the invertible elements in here? Well, of course, there's still ± 1, but there's also $\pm i$ [writes: $\{\pm 1, \pm i\}$]. These are also invertible, because $i^2 = -1$, and then you can multiply it by 1 (i.e., $(-i) \cdot i = 1$).

Is that clear that these four objects play in some sense the role of the two there? Is that clear?

So, these are the units. It means if you write down any multiplication, like any unique factorization into primes, before you had to worry about ± 1; now you have to worry about more, okay?

In particular, the notion of prime here was that p is prime [writes: p prime] if the factors are ± 1 and $\pm p$ [writes: $\{\pm 1, \pm p\}$]. That's on the $\mathbb{Z}$ side. Here, if we have some element — maybe I call it q is prime [writes: q prime] — then we can consider factorizations here; but now there's a lot more: there's four choices, there could be any one of these. ± 1 and $\pm i$ are factors, and also $\pm q$ and $\pm iq$ [writes "$\{\pm 1, \pm q, \pm i, \pm iq\}$" in two rows]. These are all factors. That's the analogous statement.

This is all some kind of nonsense. It's just saying that, you know, this is more confusion with these extra i's.

Units and Primes in the Gaussian Integers

Definition 1.1 (*Gaussian Integers*): The ring of *"Gaussian integers"*, denoted by $\mathbb{Z}[i]$, is the subring of the complex numbers $\mathbb{C}$ given by:

$$\mathbb{Z}[i] := \{a + bi \in \mathbb{C} \mid a, b \in \mathbb{Z}\}, \quad \text{where } i^2 = -1.$$

Proposition 1.2 (*Units in $\mathbb{Z}$ and $\mathbb{Z}[i]$*): An element u in a ring is called a *"unit"* if it possesses a multiplicative inverse in that ring.

- **Units in $\mathbb{Z}$:** The group of units in $\mathbb{Z}$ consists of exactly two elements:
$$\mathcal{U}_{\mathbb{Z}} = \{1, -1\} = \{\pm 1\}.$$
- **Units in $\mathbb{Z}[i]$:** The group of units in $\mathbb{Z}[i]$ consists of exactly four elements:
$$\mathcal{U}_{\mathbb{Z}[i]} = \{1, -1, i, -i\} = \{\pm 1, \pm i\}.$$

Definition 1.3 (*Prime Elements and Divisors*): In an integral domain (Definition 2.62, Week 2), a non-zero element that is not a unit is called prime (irreducible) if its only divisors are units or units multiplied by the element itself:

- **Primes in $\mathbb{Z}$:** An integer $p \in \mathbb{Z} \setminus \{0, \pm 1\}$ is prime if its only factors are:
$$\pm 1 \quad \text{and} \quad \pm p.$$
- **Primes in $\mathbb{Z}[i]$:** An element $q \in \mathbb{Z}[i] \setminus \{0, \pm 1, \pm i\}$ is prime if its only factors are the four units and their associated multiples:
$$\pm 1, \quad \pm i, \quad \pm q, \quad \pm iq.$$

Algebraic Intuition: The Concept of Associates: Going beyond the lecture: in a unique factorization domain (such as $\mathbb{Z}$, by Fact 0.19, or $\mathbb{Z}[i]$, as the course shows later), factorizations into irreducible elements are unique up to multiplication by units and order. Two elements $x, y \in R$ are called *"associates"* if $x = u \cdot y$ for some unit $u \in \mathcal{U}_R$.

- **Associates in $\mathbb{Z}$:** The only units are ± 1, meaning every prime p has two associates: p and $-p$.
- **Associates in $\mathbb{Z}[i]$:** Since $i \cdot (-i) = 1$ (i.e., $(-i) \cdot i = 1$), the imaginary units i and $-i$ are also units. Consequently, every Gaussian integer q has four associates:
$$q, \quad -q, \quad iq, \quad -iq.$$

Why ± 1 and $\pm i$ Are Invertible: The elements $\pm 1, \pm i$ are invertible in $\mathbb{Z}[i]$ since $(\pm 1)^2 = 1$ and $i \cdot (-i) = -i^2 = 1$. That there are no others is proved in Week 3 (Proposition 3.2).

Beyond the Lecture: Prime vs. Irreducible Elements

In general commutative algebra, a non-zero non-unit q in an integral domain R whose only divisors are units and associates ($q = xy \implies x \in \mathcal{U}_R$ or $y \in \mathcal{U}_R$) is called *"irreducible"*, whereas q is called *"prime"* if $q \mid xy \implies q \mid x$ or $q \mid y$. Every prime element is irreducible, but in an arbitrary integral domain the converse can fail. In unique factorization domains such as $\mathbb{Z}$ and $\mathbb{Z}[i]$ (studied later in the course), the two notions coincide, which is why the lecture uses *"prime"* for both. In Week 6 the two words are separated: what is defined here is an **irreducible** element (Definition 6.10), while a **prime** element is one whose principal ideal is a prime ideal (Definition 6.24); see Exercise 6.42 and Remark 6.12.

When Does a Rational Prime Remain Prime in $\mathbb{Z}[i]$?

Spoken Transcript [00:02:48 - 00:04:08] Prof. Rahul Pandharipande

So, let's do something that's not nonsense. A math question you can ask, which is a very nice question, is: if I take a prime, an actual prime in the integers [writes: $p \in \mathbb{Z}$], and I view that prime, that p, as an element of the Gaussian integers [writes: and view that p as an element of $\mathbb{Z}[i]$], which it is — because remember, the Gaussian integers are the set of integers plus i times an integer [writes: $\mathbb{Z}[i] = \{n + im \mid n, m \in \mathbb{Z}\}$]... So, the integers naturally sit inside the Gaussian integers [writes: $\mathbb{Z} \subset \mathbb{Z}[i]$] by just an integer going here. Is that clear?

If I take a prime, an honest prime in the integers — so this would be, you know, like $2, 3, 5, 7, 11, \dots$ [writes: $p = 2, 3, 5, 7, 11, \dots$] and so on — take one of them, and I consider the Gaussian integers: does it stay prime? Or maybe say: When does p stay prime? [writes: When does p stay prime?]

Meaning: it's born prime in the integers; when does it stay prime when I consider it as a Gaussian integer? That's a real question. And we had an example in the class last time.

Embedding of $\mathbb{Z}$ and the Prime Divisibility Question

The ring of integers embeds naturally into the Gaussian integers via the inclusion map:

$$\mathbb{Z} \hookrightarrow \mathbb{Z}[i], \quad n \mapsto n + 0i.$$

Question 1.4 (*When Does an Integer Prime Stay Prime in $\mathbb{Z}[i]$?*)**:** Let $p \in \mathbb{Z}$ be an integer prime ($p \in \{2, 3, 5, 7, 11, 13, \dots\}$). When viewed as an element of $\mathbb{Z}[i]$:

When does p remain prime (irreducible) in $\mathbb{Z}[i]$?

Why Primes Can Factor in Larger Rings: An element that has no non-trivial divisors in $\mathbb{Z}$ might acquire non-trivial divisors in $\mathbb{Z}[i]$ because $\mathbb{Z}[i]$ contains new elements with non-zero imaginary parts. For example, 5 cannot be factored in $\mathbb{Z}$, but in $\mathbb{Z}[i]$ it factors as $5 = (2 + i)(2 - i)$, where neither factor is a unit.

To investigate whether p factors non-trivially in $\mathbb{Z}[i]$, suppose there exists a non-unit divisor:

$$(a + bi) \mid p \quad \text{in } \mathbb{Z}[i] \quad (a, b \in \mathbb{Z}). \tag{1.1}$$

Complex Conjugation and the Sum of Two Squares

Spoken Transcript [00:04:08 - 00:05:18] Prof. Rahul Pandharipande

But now we try to think about it mathematically. We have p, and we have to consider whether it's divisible in the Gaussian integers. So, we think, okay, maybe it's divisible. How do I do that? Well, maybe that means $a+bi$ divides p [writes: $(a+bi) \mid p$] in the Gaussian integers. Let's try to think what that means. Now, if $a+bi$ divides p, that means $a+bi$ times something equals p — that's the definition of divisibility. If that happened, I could just complex conjugate the whole thing. So, if this happens — you have to think about this — then $a-bi$ also divides p [writes "$\Downarrow$" and below it "$(a-bi) \mid p$"].

Do you believe that statement? This has to do with complex conjugation, and it depends on what your feeling is for the complex numbers. If you don't understand the statement, you should think about it, because what does this mean? This means p is equal to something something something times $a+bi$ [writes: $p = \ldots(a+bi)$]. That's what divisibility means. And I just take the complex conjugate of everything. But the complex conjugate of p is p, and all the something something something gets complex con— I don't care, but at the end, this one becomes that [points to $a-bi$ and writes: $p = \overline{\ldots}(a-bi)$]. So, that means that that also happens, right?

Complex Conjugation of Divisors

Proposition 1.5 (*Conjugate Divisor Property*): Let $p \in \mathbb{Z}$. If $(a+bi) \mid p$ in $\mathbb{Z}[i]$, then:

$$(a-bi) \mid p \quad \text{in } \mathbb{Z}[i]. \tag{1.2}$$

Proof

By definition of divisibility in $\mathbb{Z}[i]$, $(a+bi) \mid p$ means there exists an element $z \in \mathbb{Z}[i]$ such that:

$$p = z \cdot (a+bi).$$

Applying the complex conjugation automorphism (Fact 0.27) to both sides yields:

$$\overline{p} = \overline{z \cdot (a+bi)} = \overline{z} \cdot \overline{(a+bi)}.$$

Since $p \in \mathbb{Z}$ is real, $\overline{p} = p$. Furthermore, $\overline{a+bi} = a-bi$, and $\overline{z} \in \mathbb{Z}[i]$. Thus,

$$p = \overline{z} \cdot (a-bi),$$

which proves that $(a-bi) \mid p$ in $\mathbb{Z}[i]$. **Q.E.D.**

Spoken Transcript [00:05:18 - 00:06:08] Prof. Rahul Pandharipande

Then you might think — and this is the kind of thing we will make precise in this class — that actually then both of them divide [draws a curved arrow and writes: $(a+bi)(a-bi) \mid p$]. That already requires some statement about unique factorization. And this is now a nice integer: a^2+b^2 divides p [writes "$\Downarrow$" and "$a^2+b^2 \mid p$"]. If it's an integer, then p has to be of the form a^2+b^2 [writes: $p = a^2+b^2$]. So, it has to be a prime that's a sum of two squares.

This argument will show that if you want p to stay prime... so, if p factors, then it's a sum of two squares. And this now connects to this world of trying to express primes as a sum of two squares, and there's facts about this.

Reduction to the Sum of Two Squares

Since $a+bi$ and $a-bi$ both divide p, their product divides p; this step already needs a statement about unique factorization:

$$(a+bi)(a-bi) \mid p.$$

The product is an ordinary integer:

$$(a+bi)(a-bi) = a^2+b^2 \in \mathbb{Z}, \qquad \text{so} \qquad a^2+b^2 \mid p.$$

Since p is prime and $a+bi$ is not a unit, this gives:

$$p = a^2+b^2. \tag{1.3}$$

Connection Between Gaussian Primes and Sums of Squares:

- If an integer prime p factors non-trivially in $\mathbb{Z}[i]$, then p must be a sum of two integer squares: $p = a^2+b^2$.
- By contrapositive, if p cannot be expressed as the sum of two squares, it cannot factor non-trivially in $\mathbb{Z}[i]$, and hence p must stay prime in $\mathbb{Z}[i]$.
- Going beyond the lecture: conversely, $p = a^2+b^2$ gives the factorization $p = (a+bi)(a-bi)$, and neither factor is a unit, since the units $\pm 1, \pm i$ have length 1 while $|a \pm bi| = \sqrt{p} > 1$. Hence, p stays prime in $\mathbb{Z}[i]$ **if and only if** p is not a sum of two squares.

Beyond the Lecture: A Proof with the Norm

The step from $a+bi \mid p$ and $a-bi \mid p$ to $(a+bi)(a-bi) \mid p$ needs more than unique factorization (Fact 0.19, or Euclid's lemma, Fact 0.17): for an irreducible $z = a+bi$, the elements z and $\bar{z}$ must also not be associates. For instance, $3 \mid 3$ and $\bar{3} = 3 \mid 3$, yet $3 \cdot \bar{3} = 9 \nmid 3$. For $p = 2$, the elements $1+i$ and $1-i = -i(1+i)$ are associates, and $(1+i)(1-i) = 2 \mid 2$ holds only because $2 = -i(1+i)^2$.

Textbooks avoid this with the *"field norm"* $N(x+yi) := (x+yi)(x-yi) = x^2+y^2$, which is multiplicative: $N(zw) = N(z)\,N(w)$. The lecturer does not use it. His *"norm"* is the length $|x+yi| = \sqrt{x^2+y^2}$ (Week 3, Proposition 3.2), and N is its square. If $p = z(a+bi)$ with non-units z and $a+bi$, taking norms gives

$$p^2 = N(p) = N(z)\,N(a+bi), \qquad N(z) > 1, \quad N(a+bi) > 1.$$

Since p is prime in $\mathbb{Z}$, the only positive divisors of p^2 are 1, p and p^2, so $N(a+bi) = a^2+b^2 = p$. This reaches (1.3) directly.

Fermat's Theorem on Sums of Two Squares

Spoken Transcript [00:06:08 - 00:08:12] Prof. Rahul Pandharipande

The simple fact is that if p is congruent to $3 \pmod 4$ [writes: If $p \equiv 3 \pmod 4$], this is impossible [writes: this is impossible]. You can never write a prime that is $3 \pmod 4$ as a sum of two squares. That's easy to prove; I think it's part of the exercise, I'm not sure.

But the one that's tricky is: if p is congruent to $1 \pmod 4$ [writes: If $p \equiv 1 \pmod 4$], then you can. You can solve $p = a^2 + b^2$ in the integers [writes: then you can solve $p = a^2 + b^2$]. This is a theorem that's due to Fermat [writes: (Fermat)]. And this is one of the problems on the problem set.

So, the answer to this question finally is that if you have a prime, and you want to know whether it stays prime in the Gaussian integers, the whole thing that matters is whether p is equal to 1 or $3 \pmod 4$. It's kind of an interesting question.

And as I said, we will return to these matters in the class, but it already shows us that the very naive idea of thinking about how things factor leads to interesting mathematics, and in this case it goes in a number-theoretic direction. That's this question about whether a prime can be expressed as a sum of two squares — a really basic question.

We had an example yesterday, I remind you; the answer yesterday had to do with $5 = 2^2 + 1^2$ [writes: $5 = 2^2 + 1^2$]. So, 5 is a prime in the integers, 5 is $1 \pmod 4$, and you can express it as a sum of two squares. The thing that's tricky on the board here is this statement: that if p is equal to $1 \pmod 4$, then you can solve this equation, and that is the exercise on the homework, on the first homework set. There's a way to go through that — a very clever way to go through that, not the way that Fermat used.

Fermat's Two-Square Theorem and Gaussian Primes

Theorem 1.6 (*Fermat's Two-Square Theorem*): Let p be an odd integer prime ($p > 2$).

- **(i)** If $p \equiv 3 \pmod 4$, then p cannot be written as the sum of two squares ($p \neq a^2 + b^2$). Consequently, p remains prime in $\mathbb{Z}[i]$.
- **(ii)** If $p \equiv 1 \pmod 4$, then p can be written as the sum of two squares:

$$p = a^2 + b^2 = (a + bi)(a - bi) \quad (a, b \in \mathbb{Z}).$$

  Consequently, p factors in $\mathbb{Z}[i]$ and does not remain prime.

Example 1.7 (*Decomposition of 5 in $\mathbb{Z}[i]$*): Since $5 \equiv 1 \pmod 4$, we have:

$$5 = 2^2 + 1^2 = (2 + i)(2 - i).$$

Thus, 5 does not stay prime in $\mathbb{Z}[i]$.

Proof of part (i) (beyond the lecture)

The lecture calls part (i) easy and leaves it open. Squares modulo 4 are only 0 or 1 ($0^2 \equiv 0$, $1^2 \equiv 1$, $2^2 \equiv 0$, $3^2 \equiv 1$), so a sum of two squares is 0, 1 or $2 \pmod 4$, and a prime $p \equiv 3 \pmod 4$ is not of the form $a^2 + b^2$. By the lecture's argument (1.3), a prime that factors in $\mathbb{Z}[i]$ is a sum of two squares, so p stays prime in $\mathbb{Z}[i]$; for example, 3 does. **Q.E.D.**

Beyond the Lecture: The Three Kinds of Primes in $\mathbb{Z}[i]$

The representation in part (ii) of Theorem 1.6 is unique up to sign and order.
The even prime is not covered by the theorem. Every integer prime falls into one of three classes in $\mathbb{Z}[i]$:

Ramified Prime: $p = 2$: $2 = 1^2 + 1^2 = (1+i)(1-i)$, and since $1 - i = -i(1+i)$, in fact $2 = -i\,(1+i)^2$.

Split Primes: $p \equiv 1 \pmod 4$: $p = a^2 + b^2 = (a+bi)(a-bi)$ splits into two distinct non-associate Gaussian primes (e.g., $5, 13, 17, 29$).

Inert Primes: $p \equiv 3 \pmod 4$: p cannot be written as $a^2 + b^2$, so p remains prime in $\mathbb{Z}[i]$ (e.g., $3, 7, 11, 19, 23$).

The Axioms and Basic Properties of Rings

Definition of a Ring

Spoken Transcript [00:08:15 - 00:09:45] Prof. Rahul Pandharipande

All right. So, then the main direction of the lecture last time was about rings, which we revisit now.

So, we go back to rings and the foundational material. I had defined a ring: R with multiplication, 0, and 1 [writes: $(R, +, \cdot, 0, 1)$]. This is a ring [writes: is a ring]. And the axioms, if we rewrite them now differently: there's the axioms for $+$, and this axiom for $+$, we can say just that $(R, +, 0)$ is an abelian group [writes: $+ \implies (R, +, 0)$ abelian group]. This is just the same axioms written differently; those are just the group axioms. Is that clear?

Then there's the axiom for multiplication [writes: $\cdot \implies$], and these were — now I don't have a more efficient way to say this — it's associative [writes: associative], and then there's the axiom of the unit [writes "$1 \cdot a = a \cdot 1 = 1$", a slip for $= a$]. And then there's the axioms together [writes: $+, \cdot \implies$], and these are two times the distributive law [writes: $\times 2$ Dist], which I don't write, okay?

This is just the definition which we've seen and you've seen before. And the only thing I've done now is abbreviated the $+$ as saying that if I look at the additive structure, it's an abelian group, because that's what the axioms were.

Axiomatic Definition of a Ring

Definition 1.8 (*Ring*): A *"ring"* is a tuple $(R, +, \cdot, 0, 1)$ consisting of a set R, two binary operations $+$ (addition) and $\cdot$ (multiplication), and two distinguished elements $0, 1 \in R$ satisfying the following axioms:

(1) Additive Abelian Group Axioms: $(R, +, 0)$ is an abelian group:

- **Associativity:** $\forall a, b, c \in R\colon (a+b)+c = a+(b+c)$.
- **Identity:** $\forall a \in R\colon a + 0 = 0 + a = a$.
- **Inverse:** $\forall a \in R\ \exists(-a) \in R\colon a + (-a) = (-a) + a = 0$.
- **Commutativity:** $\forall a, b \in R\colon a + b = b + a$.

(2) Multiplicative Axioms: Multiplication is associative and possesses a unit element 1:

- **Associativity:** $\forall a, b, c \in R\colon (a \cdot b) \cdot c = a \cdot (b \cdot c)$.

- **Identity (Unit):** $\forall a \in R \colon 1 \cdot a = a \cdot 1 = a$.

(3) **Distributivity Axioms:** Multiplication distributes over addition from both the left and the right:

- **Left Distributivity:** $\forall a, b, c \in R \colon a \cdot (b + c) = (a \cdot b) + (a \cdot c)$.
- **Right Distributivity:** $\forall a, b, c \in R \colon (a + b) \cdot c = (a \cdot c) + (b \cdot c)$.

☰ **Structural Synthesis of the Ring Axioms:** Summarizing the additive structure compactly as an abelian group $(R, +, 0)$ packages four standard axioms (associativity, commutativity, neutral element, and inverses) into a single standard algebraic concept. The distributive laws are the only link between the additive structure and the multiplicative structure.

Going beyond the lecture: two words are easy to mix up. **Abelian** refers to the addition ($a + b = b + a$ is part of the axioms), whereas **commutative ring** refers to the multiplication. The definition above does not require $a \cdot b = b \cdot a$; the standing convention of this course adds it (Remark 1.17). Nor does it require $1 \neq 0$: the case $1 = 0$ is allowed and treated in Proposition 1.10.

Elementary Property: $0 \cdot a = 0$

Spoken Transcript [00:09:45 - 00:11:32] Prof. Rahul Pandharipande

So, then there's a bunch of things to say at the beginning: some basic properties [writes: Basic Properties]. And these are all in the books, but in these axioms, there are lots of things that you're familiar with from the integers and arithmetic that are not in the axioms. Like, one question is: what happens if I take $0 \times a$ [writes: $0 \cdot a$]? Because 0 is an object that lives on the additive side, and I'm multiplying it, so it's on the multiplicative side. You can wonder what this is, and you can imagine what the answer is going to be.

To prove it — if we're going to formally prove it — you, of course, have to use the distributive law. The distributive law is the only thing that relates the addition to the multiplication. So, we're going to try to prove this, and to start, we know we have to use the distributive law somewhere. One way to start is we can write this as $(0 + 0) \times a$ [writes: $= (0 + 0) \cdot a$]. We have not used the distributive law yet; this uses the fact that $0 + 0 = 0$, which is the neutral part of the group law. And now we use the distributive law [writes: $= (0 \cdot a) + (0 \cdot a)$]: $(0 \cdot a) + (0 \cdot a)$.

Okay. And now we're happy, because we have this equation, and then we go back to using the additive properties, which is the left cancellation — the additive inverse, okay? So, now we can cancel [cancels $0 \cdot a$ on both sides with chalk], and we happily see that $0 = 0 \times a$ [writes "$0 = 0 \cdot a$" and draws a chalk box around it], which is, of course, the basic law of arithmetic. It's not an axiom, but it follows immediately from the axioms, which is why it's not an axiom. You could actually just put it as an axiom, it wouldn't hurt anybody, but... Okay.

Derivation of Multiplication by Zero

Proposition 1.9 (*Multiplication by Zero*): In any ring R, for every element $a \in R$:

$$0 \cdot a = 0 \quad \text{and} \quad a \cdot 0 = 0. \tag{1.4}$$

Proof

Because 0 is the additive identity of $(R, +, 0)$, we have $0 + 0 = 0$. Substituting this into $0 \cdot a$:

$$\begin{aligned} 0 \cdot a &= (0+0) \cdot a \\ &= (0 \cdot a) + (0 \cdot a) \qquad \text{(by right distributivity).} \end{aligned} \tag{1.5}$$

Because $(R, +, 0)$ is an abelian group, the element $0 \cdot a \in R$ possesses a unique additive inverse $-(0 \cdot a) \in R$. Adding $-(0 \cdot a)$ to both sides of (1.5):

$$\begin{aligned} -(0 \cdot a) + (0 \cdot a) &= -(0 \cdot a) + \big((0 \cdot a) + (0 \cdot a)\big) \\ 0 &= \big(-(0 \cdot a) + (0 \cdot a)\big) + (0 \cdot a) \qquad \text{(by associativity)} \\ 0 &= 0 + (0 \cdot a) \\ 0 &= 0 \cdot a. \end{aligned}$$

By the exact same argument using left distributivity, $a \cdot 0 = a \cdot (0+0) = a \cdot 0 + a \cdot 0$, which upon adding $-(a \cdot 0)$ yields $a \cdot 0 = 0$. **Q.E.D.**

Why Distributivity is Indispensable: The zero element 0 is defined exclusively through addition ($a + 0 = a$). To deduce its behavior under multiplication ($0 \cdot a = 0$), one must invoke the distributive law, as it provides the only axiomatic connection linking addition to multiplication.

The Zero Ring $(1 = 0)$

Spoken Transcript [00:11:32 - 00:12:43] Prof. Rahul Pandharipande

In particular, if $1 = 0$ [writes: If $1 = 0$, then] — because I said for a ring we allow this extremely unpleasant boundary case where 1 is equal to 0, the degenerate case... So, if that happens — if we have a ring where $1 = 0$ — then we have a, for any a, is equal to $a \times 1$ [writes: $a = a \cdot 1$], which is the multiplication axiom. And then, if we use $1 = 0$, this is equal to $a \times 0$ [writes: $= a \cdot 0$] using $1 = 0$, and then this is 0 [writes: $= 0$].

So, it's a rather degenerate case: if you have a ring where $1 = 0$, then all elements are equal to 0. It's not a super exciting ring; there's only one element. If I say this formally: if you have $0 = 1$, this implies your ring as a set consists of just one element [writes: If $0 = 1 \implies R = \{0\}$]. Anyway, it's a degenerate case, it's unpleasant to think about, but we carry it around because it fits well in certain statements of theorems.

The Degenerate Zero Ring

Proposition 1.10 (*Characterization of the Zero Ring*): Let R be a ring. If $1 = 0$ in R, then R consists of a single element:

$$R = \{0\}.$$

Proof

Assume $1 = 0$. Let $a \in R$ be an arbitrary element. Using the unit axiom ($a = a \cdot 1$), the hypothesis $1 = 0$, and Proposition 1.9:

$$a = a \cdot 1 = a \cdot 0 = 0.$$

Because every element $a \in R$ must equal 0, the set R contains only the zero element:

$$R = \{0\}.$$

Q.E.D.

Remark 1.11 (*The Zero Ring*): The ring $R = \{0\}$ is called the *“zero ring”*. It is a degenerate case, but it stays allowed because it fits well in certain statements of theorems. Definitions that must exclude it say so explicitly, as for integral domains (Definition 2.62, Week 2).

Beyond the Lecture: Where the Zero Ring Fits

While fields strictly require $1 \neq 0$, the definition of a ring admits $1 = 0$ so that the category of rings has a terminal object (the zero ring: every ring has exactly one homomorphism to it) and quotient rings R/I remain rings even when $I = R$.

Additive Inverses and the Element $-a$

Spoken Transcript [00:12:43 - 00:15:05] Prof. Rahul Pandharipande

Other things that are kind of convention: if you take -1... 1 is in the ring because it’s part of the data [writes: $1 \in R$]. Because of the additive structure, the group axioms say that there exists an additive inverse. So, by convention, -1 is by definition the additive inverse of 1 [writes: -1 by def the add. inv. of 1], and you can prove that it’s unique. Maybe I leave that as an exercise. Given the ring, there’s 1, and there’s the additive inverse. Additive inverses are unique, you can prove that. So, that’s -1.

And then $-a$, for any element, is defined to be $(-1) \times a$ [writes: $-a \overset{\text{def}}{=} (-1) \cdot a$]; this is not a surprising definition.

And then the little lemma — maybe it doesn’t even get to be called a lemma — is that $-a$ is the additive inverse of a [writes: Lemma: $-a$ is the additive inverse of a.]. That’s very convenient; it would be really unpleasant if that weren’t the case.

The proof of this is: we should start with $0 = (1 + (-1)) \cdot a$ [writes: Proof: $0 = (1 + (-1)) \cdot a$]... that’s 0. And we’ve already proven that that’s that (i.e., $0 \cdot a = 0$, Proposition 1.9), okay? I’ve used two steps here: here on this side, this thing is 0, and we’ve already proved $0 \times a = 0$, so... okay? And then I use the distributive law and distribute this side as a, and on this side as plus $(-1) \cdot a$ [writes: $= a + (-1) \cdot a$], and that exhibits this $(-1) \times a$ as the additive inverse of a.

So, there’s a number of things — I think we spent enough time on this — but the way the axioms work, they work so that it fits with your intuition about how multiplication works.

Are you happy with this? You can also look in the book; there’ll be a couple of pages of playing around with these. It's not deep, but it's good to be familiar with it.

Additive Inverses and Multiplication by -1

Definition 1.12 (*The Element -1 and the Notation $-a$*)**:** Let R be a ring.

- **(i)** We denote by $-1 \in R$ the unique additive inverse (Proposition 2.7, Week 2) of the multiplicative identity $1 \in R$, so that:
$$1 + (-1) = 0.$$
- **(ii)** For any element $a \in R$, we define:
$$-a \overset{\text{def}}{=} (-1) \cdot a. \tag{1.6}$$

Lemma 1.13 (*$-a$ is the Additive Inverse of a*)**:** For any element $a \in R$, the element $-a = (-1) \cdot a$ is the additive inverse of a, that is:
$$a + (-a) = 0.$$

Proof

Using Proposition 1.9 ($0 \cdot a = 0$), Definition 1.12 ($1 + (-1) = 0$), and distributivity:
$$\begin{aligned} 0 &= 0 \cdot a \\ &= (1 + (-1)) \cdot a \\ &= (1 \cdot a) + ((-1) \cdot a) && \text{(by distributivity)} \\ &= a + (-a) && \text{(since } 1 \cdot a = a \text{ and } (-1) \cdot a = -a\text{).} \end{aligned}$$
Since $a + ((-1) \cdot a) = 0$, the element $(-1) \cdot a$ satisfies the defining property of the additive inverse of a. By uniqueness of inverses in $(R, +, 0)$, this confirms that $-a = (-1) \cdot a$ is indeed the unique additive inverse of a. **Q.E.D.**

Two Meanings of $-a$, One Element: Going beyond the lecture: the ring axioms (Definition 1.8) already write $-a$ for the additive inverse of a in the group $(R, +, 0)$, while Definition 1.12 defines $-a$ as $(-1) \cdot a$. There is no circularity: the axioms only assert that an additive inverse exists, and it is unique; the definition of -1 uses just that. Lemma 1.13 then shows that the product $(-1) \cdot a$ **is** this unique inverse, so the two meanings of $-a$ name the same element.

Integer Multiples and the Canonical Map $\mathbb{Z} \to R$

Spoken Transcript [00:15:05 - 00:17:10] Prof. Rahul Pandharipande

Another convention about a ring is: I said that if I have a as an element of the ring [writes: $a \in R$], $-a$ [writes: $-a \in R$] we just write as $-a$, and by definition $-a$ is just $(-1) \times a$, and it's a canonical element of the ring.

There's something else: we can also look at $a + a$ [writes: $a + a$], and this is defined as $2a$ [writes: $= 2a$]. You can take $a + a + a$ [writes: $a + a + a$] n times [draws an underbrace labeled n]; this is what this term na is defined to be [writes: $= na$].

So, in any ring, if I have an element a in a ring [writes: $a \in R$], I can take 17 times that element [writes: $17a \in R$]. That is an element of the ring, and I just mean you add a to itself 17 times. It's well-defined, just using the ring structure. You have no objections?

And I can also do -17 times [writes: $-17a \in R$], because I just, you know, either add $-a$ to itself 17 times, or I add a to itself 17 times and take the minus, and they're the same. All right?

In particular, there's a map from $\mathbb{Z}$ to the ring [writes: $\mathbb{Z} \xrightarrow{\phi} R$]. And this map requires no choices. This map sends 0 to 0 [writes: $0 \mapsto 0$], sends 1 to 1 [writes: $1 \mapsto 1$], sends 2 to $1+1$ [writes: $2 \mapsto 1+1$], sends 3 to $1+1+1$ ...whatever [writes: $3 \mapsto 1+1+1$], okay? And it sends -1 to the additive inverse [writes: $-1 \mapsto -1$].

So, if you give me a ring, I know how to map $\mathbb{Z}$ to that ring. There's nothing going on with this map: I just map 1 to 1, and then everything else I just add up.

This is not just any map. It turns out this is what's called a ring homomorphism. So, we have to define this.

Scalar Multiplication by Integers and the Canonical Map $\mathbb{Z} \to R$

Definition 1.14 (*Multiples of Ring Elements by Integers*): Let R be a ring and $a \in R$. For any integer $n \in \mathbb{Z}$, the element $n \cdot a \in R$ is defined inductively:

$$n \cdot a := \begin{cases} 0_R & \text{if } n = 0, \\ \underbrace{a + a + \cdots + a}_{n \text{ times}} & \text{if } n > 0, \\ \underbrace{(-a) + (-a) + \cdots + (-a)}_{|n| \text{ times}} & \text{if } n < 0. \end{cases}$$

In particular, for any specific integer such as $n = 17$ or $n = -17$:

$$17a \in R \quad \text{and} \quad -17a = -(17a) \in R.$$

Remark 1.15 (*Two Ways to Form $-17 \cdot a$*): The equality $n \cdot (-a) = -(n \cdot a)$ holds for $n > 0$: adding $-a$ to itself n times gives the same element as adding a to itself n times and taking the additive inverse, because $(n \cdot a) + (n \cdot (-a)) = n \cdot (a + (-a)) = 0$ by commutativity and associativity of $+$. This is what the lecturer means by *"they're the same"*.

A Harmless Clash of Notation: Going beyond the lecture: in $n \cdot a$ the factor n is an integer, not an element of R, so this dot is not the ring multiplication. The two readings never conflict: $0 \cdot a = 0_R$ holds for the integer 0 by definition and for the ring element 0 by Proposition 1.9, and in general $n \cdot a = (n \cdot 1_R) \cdot a$ by the distributive law.

Definition 1.16 (*Canonical Map from $\mathbb{Z}$ into a Ring*): For every ring R, there exists a canonical map $\phi \colon \mathbb{Z} \to R$ given by the integer multiples of the multiplicative unit 1_R:

$$\phi(n) := n \cdot 1_R.$$

Explicitly, the map sends:

$$\begin{aligned} 0 &\mapsto 0_R, \\ 1 &\mapsto 1_R, \\ 2 &\mapsto 1_R + 1_R, \\ 3 &\mapsto 1_R + 1_R + 1_R, \\ -1 &\mapsto -1_R, \quad \dots \end{aligned}$$

Universal Initial Property of the Integers $\mathbb{Z}$: This map $\phi\colon \mathbb{Z} \to R$ requires absolutely no choices: it is entirely determined by the ring structure of R. As we will see next (Example 1.21), this map is not merely a set-theoretic function, but a ring homomorphism. Going beyond the lecture: it is even the unique ring homomorphism from $\mathbb{Z}$ to R (any such g has $g(1) = 1_R$, and additivity forces $g(n) = n \cdot 1_R$), establishing that $\mathbb{Z}$ is the *"initial object"* in the category of rings with unit (categories: Definition 2.49, Week 2).

Ring Homomorphisms

Standing Convention: Commutativity and Units

Spoken Transcript [00:17:10 - 00:18:44] Prof. Rahul Pandharipande

Let R_1 and R_2 be two rings [writes: let R_1 and R_2 be two rings].

Now, maybe I'll say a little bit more about these rings. For this class, as I said, if I consider totally abstract rings — like totally arbitrary rings here [points to ring axioms on the upper left board] — I have not assumed the multiplication is commutative. For a lot of things we don't, and sometimes we do, but when I say *"rings"* in this class from now on, I will mean a ring that has commutative multiplication [writes: Ring: commutative mult.] and always has a unit: 1 [writes: unit 1].

If you look in the literature about what the definition of a ring is, of course a ring does not require the multiplication to be commutative, and that's reasonable. If you get off the plane in Tokyo and you meet a mathematician and you say, *"I'm working on a ring,"* that person will not know whether it's commutative or not; you have to tell them. And also, you can define rings without units. So, the fancy words for this would be a **commutative ring with unit**. That's just telling the person that that's the version of ring you're signing up for. You can study anything you want, but in this class, we will be considering for the most part commutative rings with unit. I just wanted to say that.

Convention: Commutative Rings with Unit

Remark 1.17 (*Standing Convention for Rings in this Course*): In the general mathematical literature, rings are not necessarily required to have commutative multiplication, nor are they always required to possess a multiplicative identity 1.

Standing Convention: In this course, unless explicitly stated otherwise, the term *"ring"* will always designate a **commutative ring with unit**: a ring with unit 1 in the sense of Definition 1.8 such that $a \cdot b = b \cdot a$ for all $a, b \in R$. The zero ring ($1 = 0$, Proposition 1.10) is still allowed.

Terminology Across the Literature: Rings vs. Commutative Rings

Mathematical literature reflects varying conventions:

- **Commutative Algebra & Algebraic Geometry (Atiyah–Macdonald, Eisenbud):** A *"ring"* implicitly means a commutative ring with multiplicative identity 1.
- **General Algebra (Bourbaki, Lang, Artin):** A *"ring"* possesses a multiplicative identity 1, but multiplication is not required to be commutative.
- **Functional Analysis & Operator Algebras:** Rings often lack a multiplicative identity 1 (sometimes called *"rngs"* or pseudo-rings).

In this course, Prof. Pandharipande adheres to the commutative algebra convention: unless explicitly stated otherwise, all rings are commutative and unital. The zero ring, where $1 = 0$, is allowed (Proposition 1.10).

Beyond the Lecture: What Changes Without Commutativity

Typical non-commutative rings are matrix rings $M_n(K)$ for $n \geq 2$ (Fact 0.31) and endomorphism rings. Because this course assumes commutativity, the following distinctions never come up in the later lectures:

- **Ideals (Definition 1.29):** left ideals ($r \cdot x \in I$), right ideals ($x \cdot r \in I$) and two-sided ideals ($r \cdot x \cdot s \in I$) coincide in a commutative ring. In general they differ, and kernels of ring homomorphisms are always two-sided ideals.
- **Divisibility (Definition 2.65, Week 2):** in general one distinguishes left divisibility ($b = a \cdot c$) from right divisibility ($b = c' \cdot a$).
- **Units (Definition 2.66, Week 2):** in general a unit must have a two-sided inverse ($uw = wu = 1$). The units still form a group, with $(u_1 u_2)^{-1} = u_2^{-1} u_1^{-1}$, but it can be non-abelian, e.g., $U(M_n(K)) = \mathrm{GL}_n(K)$ for $n \geq 2$.
- **Integral domains (Definition 2.62, Week 2):** the term presupposes commutativity. A non-commutative ring without zero divisors is just called a **domain**, e.g., the real quaternions $\mathbb{H}$.

Definition of a Ring Homomorphism

Spoken Transcript [00:18:44 - 00:20:03] Prof. Rahul Pandharipande

Okay, so, the definition of a homomorphism: a homomorphism f from R_1 to R_2 [writes: A homomorphism $f\colon R_1 \to R_2$] — maybe I put here: a ring homomorphism [writes "Ring" above "homomorphism"] — is defined to satisfy some properties.

The first is that $f(a+b) = f(a) + f(b)$ [writes: (i) $f(a+b) = f(a) + f(b)$]. This is saying that this map respects the linear structure, the additive structure of the two rings. So, this should remind you from linear algebra class: there's the idea of linear homomorphisms of vector spaces, and it has this linear condition — that's this linear condition.

The second condition is that it respects the multiplication, so you can imagine what's going to be written here [writes: (ii) $f(ab) = f(a)f(b)$], okay? And the third condition is that it respects the identity: $f(1) = 1$ [writes: (iii) $f(1) = 1$].

Those are the three conditions. That's the definition of a homomorphism of rings.

Definition of a Ring Homomorphism

Definition 1.18 (*Ring Homomorphism*): Let R_1 and R_2 be rings. A mapping $f\colon R_1 \to R_2$ is called a *"ring homomorphism"* if it satisfies the following three conditions for all $a, b \in R_1$:

(i) Additive Compatibility:

$$f(a+b) = f(a) + f(b). \tag{1.7}$$

(ii) Multiplicative Compatibility:

$$f(a \cdot b) = f(a) \cdot f(b). \tag{1.8}$$

(iii) Identity Preservation:

$$f(1_{R_1}) = 1_{R_2}. \tag{1.9}$$

Comparison to Linear Algebra and Group Theory: Condition **(i)** asserts that f is a group homomorphism between the additive groups $(R_1, +)$ and $(R_2, +)$, mirroring the linearity condition $T(u+v) = T(u) + T(v)$ from linear algebra. Condition **(ii)** ensures compatibility with the multiplication. Condition **(iii)** explicitly guarantees that the multiplicative unit of the domain maps to the multiplicative unit of the codomain.

Preservation of the Additive Identity: $f(0) = 0$

Spoken Transcript [00:20:03 - 00:22:25] Prof. Rahul Pandharipande

There are some puzzling aspects about this definition. You could say, *"Well, why don't I say it respects the zero?"* That's a very reasonable question. So, you could say: what about $f(0)$ [writes: What about $f(0)$?]?

Well, we can write $f(0+0)$ [writes: $f(0+0)$]. I want to use property **(i)**: $f(0+0) = f(0) + f(0)$ [writes: $= f(0) + f(0)$], right? That's just applying condition **(i)** of the homomorphism [writes "Condition (i) $\Longrightarrow$" in front of it]. But now, $0+0$ is 0, we know that, because 0 is the neutral element for addition. So, I can rewrite this as $f(0) = f(0) + f(0)$ [writes: $f(0) = f(0) + f(0)$], okay?

And then I can use left cancellation by taking the inverse. I've said that several times, and I hope you know what I mean by that: if I take such an equation... Let me just do it once formally, because I said it several times. I have one $f(0)$ here and two $f(0)$'s here, and I want to get rid of the $f(0)$ on both sides. So, what I do — and this is what I mean by left cancellation — is that there exists some $b \in R_2$ such that $b + f(0) = 0$ [writes "$\exists b \in R_2$" and "$b + f(0) = 0$"]. That's the group property on R_2.

And then I just add b to both sides of this. So, I get $b + f(0) = b + f(0) + f(0)$ [writes: $b + f(0) = b + f(0) + f(0)$]. I use this cancellation, I use the additive inverse of $f(0)$ on both sides, and then, since that equals 0, this is 0 [writes "0" under $b + f(0)$], that's 0 [writes

"$= 0 + f(0)$" under the right side], and finally I conclude that $f(0) = 0$ [writes: $\boxed{f(0) = 0}$].

So, it is true that $f(0) = 0$ (i.e., $f(0_{R_1}) = 0_{R_2}$), but I don't have to put it here. It's just traditional not to. You could put it here, it wouldn't make any difference.

Automatic Preservation of the Additive Identity

Proposition 1.19 (*Preservation of Zero*): Let $f\colon R_1 \to R_2$ be a ring homomorphism. Then f automatically preserves the additive identity:

$$f(0_{R_1}) = 0_{R_2}. \tag{1.10}$$

Proof

Because 0_{R_1} is the additive identity of R_1, we have $0_{R_1} + 0_{R_1} = 0_{R_1}$. Applying property **(i)** of Definition 1.18:

$$f(0_{R_1}) = f(0_{R_1} + 0_{R_1}) = f(0_{R_1}) + f(0_{R_1}). \tag{1.11}$$

Since $(R_2, +, 0_{R_2})$ is an abelian group, the element $f(0_{R_1}) \in R_2$ possesses an additive inverse $b \in R_2$ such that $b + f(0_{R_1}) = 0_{R_2}$. Adding b to both sides of (1.11):

$$\begin{aligned} b + f(0_{R_1}) &= b + \big(f(0_{R_1}) + f(0_{R_1})\big) \\ 0_{R_2} &= \big(b + f(0_{R_1})\big) + f(0_{R_1}) && \text{(by associativity in } R_2\text{)} \\ 0_{R_2} &= 0_{R_2} + f(0_{R_1}) \\ 0_{R_2} &= f(0_{R_1}). \end{aligned}$$

Thus, $f(0_{R_1}) = 0_{R_2}$. **Q.E.D.**

Why Identity Preservation Must Be Axiomatized

Spoken Transcript [00:22:25 - 00:24:16] Prof. Rahul Pandharipande

Then maybe the question is: *"Well, then why do I put $f(1) = 1$?"* That's a reasonable question. They're treated not symmetrically.

And the answer to that... [erases the cancellation calculation for $f(0)$ on the board]

Yeah, it has to do with the fact that I cannot... I mean, maybe this is what you're hinting at, but I don't have this cancellation property in multiplication. I have this cancellation — the existence of an additive inverse — that is an axiom of the ring, so I can use it. But there is not an existence of a multiplicative inverse, so I cannot just follow these steps. So, I have to assume it by hand.

This is about the definitions, and as I said, the strange thing about mathematics is that when it's taught, the definitions are first, but somehow when it's understood, the definitions are last. So, it's somehow a constant difficulty. [erases more board space and begins writing]

So, the point is: we don't have multiplicative inverses — we don't always have multiplicative inverses — in R_1, and so, we can't use this argument [writes "We don't have mult. inverses in R, so we can't use that argument" and draws an arrow pointing to it] — the one that's in the corner there (i.e., the cancellation argument for $f(0) = 0$, Proposition 1.19) [points at the proof of $f(0) = 0$ in the corner of the board].

So, to fix that, we just say at the beginning: $f(1) = 1$. We don't leave it to chance. You could hope, maybe, that even though we couldn't use this argument, maybe you could find some different argument. You could hope that, but it's not true.

Spoken Transcript [00:24:16 - 00:25:08] Prof. Rahul Pandharipande

For example, I could take a bad example [writes: A bad example]: I could take R and map it to... sorry, I could take $R_1 \to R_2$ [writes: $R_1 \to R_2$] and map it so everything goes to zero. Say they're both healthy rings — in fact, they could both be $\mathbb{Z}$ [writes "$\mathbb{Z}$" above R_1 and "$\mathbb{Z}$" above R_2]. I could map every element to zero.

If you map every element to zero, you certainly are not mapping 1 to 1. But you won't get a contradiction with every element being zero; all the other properties will be satisfied.

So, it's not a good idea to think about this, but the fact of the matter is: we need to just assume it. [crosses out the bad example] Okay.

The Asymmetry Between Addition and Multiplication

Board State Note:

> *"We don't have mult. inverses in R, so we can't use that argument."*

Why $f(1) = 1$ Does Not Follow from $f(ab) = f(a)f(b)$: One might wonder why $f(1) = 1$ is required as an explicit axiom in Definition 1.18, whereas $f(0) = 0$ is a provable theorem.

- **Additive Case:** In the additive group $(R_2, +)$, every element has an additive inverse. Hence, from $f(0) = f(0) + f(0)$, we can subtract $f(0)$ to obtain $f(0) = 0$.
- **Multiplicative Case:** Using multiplicativity, we know that $f(1) = f(1 \cdot 1) = f(1) \cdot f(1)$. However, in a general ring R_2, $f(1)$ does not necessarily possess a multiplicative inverse! Going beyond the lecture: while $x^2 = x$ in a field implies $x \in \{0, 1\}$, in a general ring there can be non-trivial **idempotents** ($e^2 = e$ with $e \neq 0, 1$), or $f(1)$ could simply equal 0.

No cleverer argument can help either: the zero map in Example 1.20 satisfies the first two conditions but not $f(1) = 1$.

Counterexample: The Non-Unital Zero Map

Example 1.20 (*The Trivial Zero Map Fails Condition (iii)*): Let $R_1 = R_2 = \mathbb{Z}$ (or any non-zero rings). Consider the zero map:

$$f\colon R_1 \to R_2, \quad f(a) := 0 \quad \forall a \in R_1.$$

This map satisfies conditions **(i)** and **(ii)**:

- **Addition:** $f(a+b) = 0 = 0 + 0 = f(a) + f(b)$,

- **Multiplication:** $f(a \cdot b) = 0 = 0 \cdot 0 = f(a) \cdot f(b)$.

However, $f(1_{R_1}) = 0 \neq 1_{R_2}$. Without demanding condition **(iii)** ($f(1) = 1$), the zero map would be considered a ring homomorphism between non-zero rings.

III Examples of Ring Homomorphisms: The Canonical Map $\mathbb{Z} \to R$

Spoken Transcript [00:25:08 - 00:27:20] Prof. Rahul Pandharipande

Then there's lots of examples [writes: Examples] of these ring homomorphisms, and it's the main way we compare rings — by homomorphisms.

For example, the thing I said here (i.e., the canonical map of Definition 1.16): $\mathbb{Z}$ goes to R [writes: $\mathbb{Z} \xrightarrow{\phi} R$], this basic map ϕ. This is a ring homomorphism [writes: This is a ring hom.]. Not only does $\mathbb{Z}$ map to R by just seeing where 1 goes, it's actually a homomorphism. And to check it's a homomorphism, you have to do something.

I'm not going to do this for you, you have to do it, because it's some kind of internal tautology about addition... Where does 2 go? We said 2 goes to $1+1$ [writes: $2 \mapsto 1+1$]. Where does 3 go? 3 goes to $1+1+1$ [writes: $3 \mapsto 1+1+1$]. So, then we'd have to check that if I add these two, 5 goes to five 1's, which looks pretty good.

And then you'd also have to check: where does 6 go? 6 is 2×3, and this goes to $(1+1) \times (1+1+1)$ [writes: $2 \cdot 3 \mapsto (1+1)(1+1+1)$]. And now you'd use the distributive law a lot of times. If you use the distributive law a lot of times, you'll get this to be $1+1+1+1+1+1$ [writes: $= (1+1+1+1+1+1)$], which is the definition of where 6 goes. Okay?

So, if you do this, you'll see that there's essentially nothing going on except understanding that the distributive law is making this a homomorphism.

This kind of mathematics that's on this kind of foundational, definitional level, you pretty much have to do for yourself to understand, you know, because it has a certain kind of logical... And I hope this is clear. But there is some content — there's a very small amount of content: it's about how this definition of 2 as $1+1$ interacts with the distributive law. I think most people find this not so controversial, but if you want to understand it, you have to go through the mental effort of checking the things yourself. On the board, I could convince you of anything, probably. Okay.

The Canonical Homomorphism $\mathbb{Z} \to R$

Example 1.21 (*Canonical Homomorphism from the Integers*): For any ring $(R, +, \cdot, 0_R, 1_R)$, the canonical map:

$$\phi \colon \mathbb{Z} \to R, \quad \phi(n) := n \cdot 1_R$$

is a ring homomorphism.

Verification of Homomorphism Axioms:

- **Preservation of Unit:** $\phi(1) = 1 \cdot 1_R = 1_R$.

- **Additive Compatibility (e.g., $2+3=5$):**

$$\begin{aligned}\phi(2)+\phi(3) &= (1_R+1_R)+(1_R+1_R+1_R)\\ &= 1_R+1_R+1_R+1_R+1_R = \phi(5) = \phi(2+3).\end{aligned}$$

- **Multiplicative Compatibility (e.g., $2\cdot 3=6$):** Using repeated applications of the distributive laws:

$$\begin{aligned}\phi(2)\cdot\phi(3) &= (1_R+1_R)(1_R+1_R+1_R)\\ &= 1_R(1_R+1_R+1_R)+1_R(1_R+1_R+1_R)\\ &= (1_R+1_R+1_R)+(1_R+1_R+1_R)\\ &= 1_R+1_R+1_R+1_R+1_R+1_R = \phi(6) = \phi(2\cdot 3).\end{aligned}$$

The General Check (beyond the lecture)

The lecture works out two examples and leaves the general check to the students. The two computations above are instances of two identities, valid for all $m,n\in\mathbb{Z}$:

- **Additivity:** $(m+n)\cdot 1_R = m\cdot 1_R + n\cdot 1_R$. For $m,n\geq 0$ both sides are a sum of $m+n$ copies of 1_R (associativity of $+$); the cases with negative m or n follow by adding inverses on both sides (Definition 1.14, Remark 1.15).
- **Multiplicativity:** $(m\cdot 1_R)(n\cdot 1_R) = (mn)\cdot 1_R$. Expanding the left side with the distributive laws gives $|mn|$ products $1_R\cdot 1_R = 1_R$, taken with the sign of mn, exactly as in the computation for $2\cdot 3=6$.

Together with $\phi(1)=1_R$, this proves that ϕ is a ring homomorphism. **Q.E.D.**

Inclusion Homomorphisms and Subrings

Spoken Transcript [00:27:20 - 00:28:24] Prof. Rahul Pandharipande

That's a very simple example of a homomorphism, but in fact there's lots of... all the ones we've seen so far: if you take $\mathbb{Z}$ inside $\mathbb{Q}$ [writes: $\mathbb{Z}\subset\mathbb{Q}$], if you view this as a map — just the embedding map, if you view an integer as a rational number — then this gives me a homomorphism. And that's a homomorphism inside $\mathbb{R}$ [writes: $\subset\mathbb{R}$], and this is a homomorphism inside $\mathbb{C}$ [writes: $\subset\mathbb{C}$]. So, these inclusions are all homomorphisms. And I can also include this inside the ring of polynomials [writes "$\cap$" under each ring and below them "$\mathbb{Z}[x]\subset\mathbb{Q}[x]\subset\mathbb{R}[x]\subset\mathbb{C}[x]$"] — polynomials in one variable.

These are many of the rings that we will study, but if you start at the bottom, $\mathbb{Z}$, they just include, and all these inclusions are homomorphisms. Because they all respect... you know, whenever I include, I'm not changing the definition of addition or multiplication for the smaller object, and 1 is always going to 1.

So, this is an example. Let's try some different kinds of examples... [draws a vertical line dividing the board]

Inclusion Homomorphisms of Standard Rings

Example 1.22 (*Canonical Inclusions*): The canonical inclusion embeddings of the standard number systems:

$$\mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow \mathbb{R} \hookrightarrow \mathbb{C} \tag{1.12}$$

and their single-variable polynomial rings:

$$\mathbb{Z}[x] \hookrightarrow \mathbb{Q}[x] \hookrightarrow \mathbb{R}[x] \hookrightarrow \mathbb{C}[x] \tag{1.13}$$

are all ring homomorphisms: an inclusion does not change addition or multiplication of the smaller ring, and 1 goes to 1.

The Ring of Residue Classes ${}^{\mathbb{Z}}\!/_{n\mathbb{Z}}$ and the Projection Homomorphism

Spoken Transcript [00:28:24 - 00:30:48] Prof. Rahul Pandharipande

One example is that you've studied this ring ${}^{\mathbb{Z}}\!/_{n\mathbb{Z}}$ [writes: ${}^{\mathbb{Z}}\!/_{n\mathbb{Z}}$]... Well, if we haven't, let me at least say what it is, because this also plays a role in the class. So, this ring here, this set here... [writes: $({}^{\mathbb{Z}}\!/_{n\mathbb{Z}}, +, \cdot, \bar{0}, \bar{1})$]. This is a standard name; probably in other courses it's also given other names. But the words are "$\mathbb{Z}$ mod $n\mathbb{Z}$", and this is the set of residue classes mod$n$ [writes: $=$ {set of residue classes mod n}]. If you want to be explicit, this is equal to the set $\bar{0}, \bar{1}, \bar{2}, \ldots, \overline{n-1}$ [writes: $= \{\bar{0}, \bar{1}, \bar{2}, \ldots, \overline{n-1}\}$], because n is zero. Is this notation reasonable notation? It's kind of semi-standard mathematical notation. There's exactly n elements of this; it's just the residue classes mod$n$.

And you can add residue classes, you can multiply them, there's the zero, and there's the one. So, if I take this whole data together, this exactly is a commutative ring with unit [writes: Ring!]. So, it's a ring for us. It's a ring of a slightly different nature than the other ones because it has finitely many elements. It's an interesting ring to think about. In some sense, one of the simplest and most confusing ones is this one [writes: ${}^{\mathbb{Z}}\!/_{2\mathbb{Z}}$]: this is a ring with two elements [writes: $= \{\bar{0}, \bar{1}\}$], zero and one. That's kind of a fun ring, but it's a full-fledged ring. You know, a ring is a ring no matter how small.

And we get here: $\mathbb{Z}$ goes to ${}^{\mathbb{Z}}\!/_{n\mathbb{Z}}$ [writes: $\mathbb{Z} \to {}^{\mathbb{Z}}\!/_{n\mathbb{Z}}$]; this is also a homomorphism [writes "f" over the arrow]. It's a ring homomorphism where I take $f(a)$ — for any integer a — to $\bar{a}$ [writes: $f(a) = \bar{a}$]. Okay? So, that's another example.

Residue Class Rings and the Reduction Modulo n

Definition 1.23 (*Ring of Residue Classes Modulo n*)**:** For any integer $n \geq 1$, the ring of "*residue classes modulo n*" (denoted ${}^{\mathbb{Z}}\!/_{n\mathbb{Z}}$ or $\mathbb{Z}_n$) is the set:

$${}^{\mathbb{Z}}\!/_{n\mathbb{Z}} := \{\bar{0}, \bar{1}, \bar{2}, \ldots, \overline{n-1}\}, \tag{1.14}$$

equipped with modular addition and multiplication:

$$\bar{a} + \bar{b} := \overline{a+b}, \quad \bar{a} \cdot \bar{b} := \overline{a \cdot b},$$

with additive identity $\bar{0}$ and multiplicative identity $\bar{1}$. Here $\bar{a}$ denotes the residue class of $a \in \mathbb{Z}$, and $\bar{a} = \bar{b}$ if and only if $n \mid (a - b)$ (congruence modulo n, Definition 0.21 and Fact 0.22).

Why the Operations Are Well-Defined (beyond the lecture)

The lecture states this without proof. The formulas use representatives a, b of the classes, so they must not depend on that choice. If $\bar{a} = \bar{a}'$ and $\bar{b} = \bar{b}'$, say $a' = a + kn$ and $b' = b + \ell n$, then

$$a' + b' = (a+b) + (k+\ell)n, \qquad a'b' = ab + (a\ell + kb + k\ell n)n,$$

so $\overline{a'+b'} = \overline{a+b}$ and $\overline{a'b'} = \overline{ab}$. The ring axioms are then inherited from $\mathbb{Z}$, and so is commutativity. For $n = 1$ one gets the zero ring $\{\bar{0}\}$. **Q.E.D.**

Example 1.24 (*The Ring* $\mathbb{Z}/2\mathbb{Z}$)**:** For $n = 2$, $\mathbb{Z}/2\mathbb{Z} = \{\bar{0}, \bar{1}\}$ is a ring with two elements, zero and one.

Addition and Multiplication in $\mathbb{Z}/2\mathbb{Z}$**:** Going beyond the lecture: the two tables are:

$+$	$\bar{0}$	$\bar{1}$
$\bar{0}$	$\bar{0}$	$\bar{1}$
$\bar{1}$	$\bar{1}$	$\bar{0}$

$\cdot$	$\bar{0}$	$\bar{1}$
$\bar{0}$	$\bar{0}$	$\bar{0}$
$\bar{1}$	$\bar{0}$	$\bar{1}$

Proposition 1.25 (*Canonical Projection Homomorphism*)**:** The reduction map modulo n:

$$f \colon \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}, \quad f(a) = \bar{a}$$

is a surjective ring homomorphism.

Proof (beyond the lecture)

The lecture states this without proof. For $a, b \in \mathbb{Z}$, the operations on residue classes give $f(a+b) = \overline{a+b} = \bar{a} + \bar{b} = f(a) + f(b)$ and $f(a \cdot b) = \overline{a \cdot b} = \bar{a} \cdot \bar{b} = f(a) \cdot f(b)$, and $f(1) = \bar{1}$. Every residue class $\bar{k}$ is the image of $k \in \mathbb{Z}$, so f is surjective. **Q.E.D.**

The Kernel of a Ring Homomorphism and Ideals

Definition of the Kernel

Spoken Transcript [00:30:49 - 00:32:06] Prof. Rahul Pandharipande

There are lots of interesting examples; I'm not going to get to them all today. But I want to say that [pushes up the lower board] for homomorphisms, like for vector spaces, whenever you have one, you can consider the kernel and the image. [erases the board that came down: the definition of a ring homomorphism and the proof of $f(0) = 0$] And it turns out, in a maybe surprising way, that for ring theory, the notion of the kernel turns out to be much more important than the notion of the image.

In linear algebra, if you have a homomorphism between two vector spaces — did you use the word *"homomorphism"*? You could also say *"linear transformation"*; both those words are used — if you have a linear transformation between two vector spaces, there's the notion of its kernel (those are the elements that go to zero) and the notion of its image. And they're both subvector spaces. In linear algebra, these things are kind of on similar footing. But in ring theory, when you have a homomorphism, there's again the kernel and the image. So, we will start with the kernel first; probably next time we'll say a little bit about the image.

So, this is an interesting object. Here we have: let...

Spoken Transcript [00:32:06 - 00:32:56] Prof. Rahul Pandharipande

Let f from R_1 to R_2 [writes: Let $f\colon R_1 \to R_2$] be a homomorphism of rings [writes: be a hom. of rings]. Then we define the kernel of f: $\ker(f)$ [writes: $\ker(f) =$] — it's those elements in the first ring, because the kernel lives in the first ring [writes: $\{a \in R_1 \mid$], such that $f(a)$ is equal to zero [writes: $f(a) = 0\}$]. Okay? I hope this definition is clear: it's everybody that goes to zero.

In particular, you could say: what is the kernel of this? [writes: $\ker(f) \subset \mathbb{Z} \xrightarrow{f} \mathbb{Z}/_{n\mathbb{Z}}$] What is the kernel here?

Student Answer

Multiples of n.

Spoken Transcript [00:32:57 - 00:33:17] Prof. Rahul Pandharipande

Yeah. The kernel here is exactly all multiples of n, and this is often denoted $n\mathbb{Z}$ [writes "$n\mathbb{Z}$" under $\ker(f)$]. This is equal to the set of multiples of n [writes: $\subset$ mults of n]. That's the kernel.

Definition of the Kernel of a Ring Homomorphism

Definition 1.26 (*Kernel of a Ring Homomorphism*): Let R_1, R_2 be rings and let $f\colon R_1 \to R_2$ be a ring homomorphism. The *"kernel"* of f, denoted by $\ker(f)$, is the pre-image of the zero element 0_{R_2}:

$$\ker(f) := \{a \in R_1 \mid f(a) = 0_{R_2}\} \subseteq R_1. \tag{1.15}$$

Example 1.27 (*Kernel of the Modular Reduction Homomorphism*): For the homomorphism $f\colon \mathbb{Z} \to \mathbb{Z}/_{n\mathbb{Z}}$ given by $f(a) = \bar{a}$, an integer a maps to $\bar{0}$ if and only if a is a multiple of n. Hence:

$$\ker(f) = n\mathbb{Z} := \{k \cdot n \mid k \in \mathbb{Z}\}. \tag{1.16}$$

The Kernel as a Group Kernel: Going beyond the lecture: since f respects addition and $f(0_{R_1}) = 0_{R_2}$ (Proposition 1.19), $\ker(f)$ is the kernel of the group homomorphism $f\colon (R_1, +, 0) \to (R_2, +, 0)$ in the sense of Week 2 (Definition 2.17). The multiplication is what makes the ring kernel special, see Proposition 1.28.

Insight: Kernels versus Images in Ring Theory

In linear algebra, the kernel and image of a linear map $T\colon V \to W$ are on similar footing: both are vector subspaces. In ring theory, the kernel turns out to be much more important than the image: $\ker(f) \subseteq R_1$ is an **ideal** of the domain, closed under addition and absorbing multiplication from the entire ring ($r \cdot a \in \ker(f)$ for all $r \in R_1$ and $a \in \ker(f)$).

Beyond the Lecture: Image and Quotient

In linear algebra, the two are connected by the rank-nullity theorem $\dim V = \dim \ker(T) + \dim \operatorname{Im}(T)$ (Fact 0.29). In ring theory, the image $\operatorname{Im}(f) \subseteq R_2$ is only a subring of the codomain, while the absorption property of the kernel is what the quotient ring $R_1/\ker(f)$ and the First Isomorphism Theorem ($R_1/\ker(f) \cong \operatorname{Im}(f)$, Week 4, Theorem 4.27) need.

Properties of the Kernel

Spoken Transcript [00:33:17 - 00:34:49] Prof. Rahul Pandharipande

So, the kernel has very specific properties [writes: Properties of the $\ker(f)$]. The first property is that it's closed under addition: if we have a and b in the kernel of f [writes: • $a, b \in \ker(f)$], this implies $a + b$ is in the kernel [writes: $\implies (a+b) \in \ker(f)$]. This is the linear algebra property. All of these things are easy to prove. To check whether $a + b$ is in the kernel, I have to calculate $f(a+b)$ [writes: $f(a+b) =$]. Using the definition of homomorphism, this is $f(a) + f(b)$ [writes: $= f(a) + f(b)$]. Then, using the fact that a and b are both in the kernel, well, this one is zero [writes "0" under $f(a)$] and that one is zero [writes "0" under $f(b)$], so the whole thing is zero [writes: $= 0$] — and so, it's in the kernel, right? I should have said even earlier: we already know that zero is in the kernel [writes: • $0 \in \ker(f)$].

If a is in the kernel [writes: • If $a \in \ker(f)$] and r is any element of the ring [writes: $, r \in R$], then $r \times a$ is in the kernel [writes: $\implies ra \in \ker(f)$].

Spoken Transcript [00:34:49 - 00:36:11] Prof. Rahul Pandharipande

That's a wider property. These are the linear properties — we'll say more about them in a moment — but this property is about the multiplication. It says that if I take an element of the kernel, I can then go pick any element of the ring and multiply it, and it'll be in the kernel. So, it's a more interesting property. You can see that here: if you have an element of the kernel, it's n times something. If I multiply it by any integer, it's still n times something.

This is not hard to prove either. We have to calculate $f(r \times a)$ [writes: $f(ra) =$], and then, of course, we use the multiplicative property [writes: $= f(r)f(a)$]. And now, this is the point: in the multiplicative property, in order to deduce that this product is zero, we only need one of the factors to be zero. For the addition, we needed both, but for the multiplication, we only need to know that one of the factors is zero, because that'll kill the product. And that's already there because a is in the kernel. So, this is zero [writes "0" under $f(a)$], and so, this is zero [writes: $= 0$], okay?

And so, in particular, if a is in the kernel, $-a$ is in the kernel [writes: • $a \in \ker(f) \implies -a \in \ker(f)$]. That's the additive inverse, because for $-a$, r is equal to -1. So, those are all properties enjoyed by the kernel...

Algebraic Properties of the Kernel

Proposition 1.28 (*Axiomatic Properties of the Kernel*)**:** Let $f\colon R_1 \to R_2$ be a ring homomorphism. The kernel $\ker(f) \subseteq R_1$ satisfies the following properties:

(1) Containment of Zero: $0_{R_1} \in \ker(f)$.
(2) Closure under Addition: $\forall a, b \in \ker(f)\colon a + b \in \ker(f)$.
(3) Closure under Additive Inverses: $\forall a \in \ker(f)\colon -a \in \ker(f)$.
(4) Multiplicative Absorption (Ideal Property): $\forall a \in \ker(f), \forall r \in R_1\colon r \cdot a \in \ker(f)$.

Proof

We verify each property directly using the axioms of a ring homomorphism:

- **Property (1):** By Proposition 1.19, $f(0_{R_1}) = 0_{R_2}$, so $0_{R_1} \in \ker(f)$.
- **Property (2):** If $a, b \in \ker(f)$, then $f(a) = 0$ and $f(b) = 0$. By additivity:
$$f(a+b) = f(a) + f(b) = 0 + 0 = 0 \implies a + b \in \ker(f).$$
- **Property (4):** If $a \in \ker(f)$ and $r \in R_1$, then $f(a) = 0_{R_2}$. By multiplicativity and Proposition 1.9:
$$f(r \cdot a) = f(r) \cdot f(a) = f(r) \cdot 0_{R_2} = 0_{R_2} \implies r \cdot a \in \ker(f).$$
- **Property (3):** Setting $r = -1_{R_1}$ in Property (4) and using Lemma 1.13:
$$-a = (-1_{R_1}) \cdot a \in \ker(f).$$

Q.E.D.

Why Multiplicative Absorption Works: Additive vs. Multiplicative Absorption: In the additive closure proof (Property (2)), both $f(a)$ and $f(b)$ had to be zero: knowing only $f(a) = 0$ would leave the sum $f(a) + f(b) = f(b)$ arbitrary. In contrast, for Property (4), multiplication by zero in a ring is absorbing: as soon as one factor in a product is zero, the entire product vanishes ($f(r) \cdot 0_{R_2} = 0_{R_2}$), regardless of whether $f(r)$ is zero or not. Thus, multiplying an element of $\ker(f)$ by **any** element $r \in R_1$ from the ambient ring keeps the result inside $\ker(f)$. This strong absorption property distinguishes ring kernels from mere subrings (Definition 2.26, Week 2).

Ideals in a Ring

Spoken Transcript [00:36:11 - 00:38:07] Prof. Rahul Pandharipande

Those properties can be said in a more concise way, which is a more natural way to say it, and it comes with a definition: the definition of an ideal. [pushes up the board and erases the lower board] This is the definition of an ideal I in R [writes: Def: Ideal $I \subset R$].

The properties are: 0 must be in I [writes: $\bullet\, 0 \in I$]. And the second way to say things is that if I take I, and I take the $+$ from R, and I take the 0, this is an abelian group [writes: $\bullet\, (I, +, 0)$ abelian group]. Remember, whenever I write R, I always mean R with the two operations and this data [writes "$= (R, +, \cdot, 0, 1)$" after $I \subset R$]. So, to be an ideal, you have to have zero, and then when I consider this subset with $+$ and 0, it should be an abelian group. This means it's closed under addition, and it's closed under additive inverses, okay?

And the second thing is that for every a in I and for every r in R, ra remains in I [writes: $\bullet\, \forall a \in I, \forall r \in R \implies ra \in I$]. Okay? That's the ideal condition.

This is an extremely important definition for the class. It tells you various subsets of R are ideals, and they have this linear property, which is kind of like linear algebra, but then they also have this multiplicative property, which is not like linear algebra. And the role of an ideal will play a little bit like the role of a subspace in linear algebra.

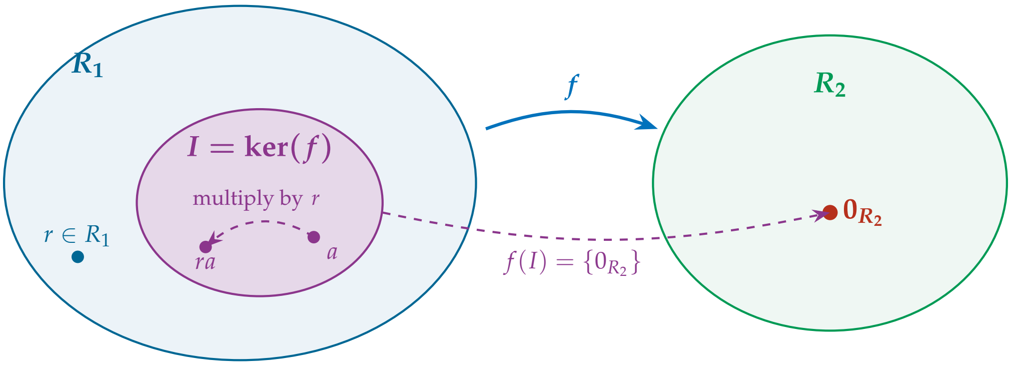
Definition of an Ideal

Definition 1.29 (*Ideal*): Let $R = (R, +, \cdot, 0, 1)$ be a commutative ring with unit. A subset $I \subseteq R$ is called an *"ideal"* of R if it satisfies:

(1) Additive Subgroup: $(I, +, 0)$ is an abelian subgroup (Definition 0.23) of $(R, +, 0)$:
- $0 \in I$,
- $\forall a, b \in I \colon a + b \in I$,
- $\forall a \in I \colon -a \in I$.

(2) Multiplicative Absorption (Ideal Condition):

$$\forall a \in I,\ \forall r \in R \colon r \cdot a \in I. \tag{1.17}$$



Ideals as the Ring-Theoretic Analogue of Normal Subgroups and Subspaces: Going beyond the lecture: in linear algebra, kernels of linear maps are vector subspaces ($T(u + v) = T(u) + T(v)$ and $T(cv) = cT(v)$). A vector subspace $W \subseteq V$ is closed under vector addition ($w_1 + w_2 \in W$) and scalar multiplication by field scalars ($\lambda \cdot w \in W$ for $\lambda \in k$).

In group theory, kernels of group homomorphisms are normal subgroups ($gNg^{-1} \subseteq N$).

In ring theory, kernels of ring homomorphisms are *"ideals"*: they are closed under addition ($a + b \in I$) and absorb multiplication by **any** element from the entire ambient ring R ($r \cdot a \in I$ for all $r \in R$).

Notation: Ideals in the Lecture and in Textbooks

Many textbooks write $I \trianglelefteq R$ (or $I \triangleleft R$) for *"I is an ideal of R"*. The lecture does not use this symbol: on the board, the lecturer writes $I \subset R$ and says in words that I is an ideal (*"Def: Ideal $I \subset R$"*). These notes follow the board. His $\subset$ allows equality, like $\subseteq$; he explains this convention when he defines maximal ideals (Definition 4.54, Week 4).

The Kernel as an Ideal

Spoken Transcript [00:38:07 - 00:39:13] Prof. Rahul Pandharipande

The proposition we've proven with this is:

Proposition [writes: Prop:]: The kernel of a homomorphism $f\colon R_1 \to R_2$ [writes: The ker of a hom $f\colon R_1 \to R_2$] is an ideal [writes: is an ideal] in R_1 [writes: in R_1].

That's what we've proven on the board up there (Proposition 1.28): with the kernel, you always get an ideal. And rings can have... we will do some examples next time, I'm running out of time here, but for $\mathbb{Z}$, if I take inside here $n\mathbb{Z}$ [writes: $\mathbb{Z} \supset n\mathbb{Z} = I$], that's an ideal (Recall: Example 1.27). Rings can have lots of ideals.

What about a field? I'll just say one more thing about this. Is that clear?

We have lots of examples of rings and lots of examples of ideals, but suppose F is a field [writes: F is field], $(F, +, \cdot, 0, 1)$ [writes: $(F, +, \cdot, 0, 1)$]. What are its ideals? [writes: What are the ideals?]

The Kernel is an Ideal

Proposition 1.30 (*Kernel as an Ideal*): Let $f\colon R_1 \to R_2$ be a ring homomorphism. Then $\ker(f)$ is an ideal of R_1:

$$\ker(f) \subset R_1 \text{ is an ideal.}$$

Proof

This is the synthesis of Proposition 1.28:

(1) $(\ker(f), +, 0)$ is an abelian subgroup of $(R_1, +, 0)$ because $0 \in \ker(f)$, $a + b \in \ker(f)$ for all $a, b \in \ker(f)$, and $-a \in \ker(f)$ for all $a \in \ker(f)$.

(2) Multiplicative absorption holds because for all $a \in \ker(f)$ and all $r \in R_1$:

$$f(r \cdot a) = f(r) \cdot f(a) = f(r) \cdot 0 = 0 \implies r \cdot a \in \ker(f).$$

Hence, $\ker(f)$ satisfies all axioms of an ideal. **Q.E.D.**

Ideals in a Field

Spoken Transcript [00:39:13 - 00:40:05] Prof. Rahul Pandharipande

What are its ideals? What are the ideals of a field? Because a field is a ring, so it has the notion of ideals. Every abstract concept of a ring, we can apply to a field, because a field is a particular type of ring: it's a ring where we have multiplicative inverses for non-zero elements. So, what could be an ideal here? What are the ideals?

I'll start with one simple thing: if I just take $I = \{0\}$ [writes: $I = \{0\}$]. Then zero is in I. It is an abelian subgroup just by itself, because $0 + 0 = 0$, and everything... you know, zero is sufficient unto itself [laughter], one might say. And it has this multiplicative property, because if I multiply by anybody, it stays zero. So, zero is always an ideal.

So, what else? Yeah? [points at student]

Student Answer

The whole field?

Spoken Transcript [00:40:07 - 00:40:21] Prof. Rahul Pandharipande

The whole field is always an ideal: ideal equals the whole field [writes: $I = F$]. And, of course, there can be no doubt about that, because if you take the whole thing, it's an abelian group and you're not going to get out of it by multiplying.

Can there be any more? Why not?

Student Answer

Because as soon as you have another element, then you have the inverse...

Spoken Transcript [00:40:26 - 00:40:28] Prof. Rahul Pandharipande

Multiplicative inverse.

Student Answer, continued

...multiplicative inverse, so you have one...

Spoken Transcript [00:40:30 - 00:40:37] Prof. Rahul Pandharipande

One, and then you have everything.

Okay, so, this argument is correct. So, the claim is: for a field, these are the only ideals [writes "only ideals!" next to $I = \{0\}$ and $I = F$].

Spoken Transcript [00:40:37 - 00:42:20] Prof. Rahul Pandharipande

Which is an important thing to always keep in mind. And the reason is the argument he said: suppose that I have an ideal, and suppose this ideal is not this silly zero ideal [writes: $I \neq (0)$]. That means there exists somebody in the ideal that's not zero [writes: $\exists\, a \in I,\ a \neq 0$]. That somebody, because it's a field, has a multiplicative inverse [writes: $\exists\, a^{-1}$ so that]. There exists an a^{-1}, so that $a^{-1} \cdot a$ is also in the ideal [writes: $a^{-1}a \in I$], because you're allowed to take anybody and multiply — this is anybody, and this is somebody in the ideal. So, if we start with a in the ideal, I can multiply by anybody; in particular, I can multiply by its inverse. But then this is one [writes "$1 \in I$" above $a^{-1}a$, joined by "$=$"]!

Once I have one in the ideal, then I can easily get anybody in the ideal, because I can just take any $r \times 1$. Okay?

It's an elementary argument, and it's very good to understand. So, fields have very few ideals — extremely few. But rings can have quite a number of ideals. As I said here, this is the simplest example to see: the ideals in the integers come from multiples of any integer. And ideals in... yeah, we will see some examples of it. But was this argument clear? Sometimes I do these things very quickly, it's just, you know...

The standard thing to say here is that you can view fields as sellouts, because they've lost all their ideals. [laughter]

Classification of Ideals in a Field

Theorem 1.31 (*Ideals in a Field*): Let F be a field. The only ideals of F are the trivial zero ideal $\{0\}$ and the unit ideal F:

$$I \subset F \text{ an ideal} \implies I = \{0\} \quad \text{or} \quad I = F. \tag{1.18}$$

Proof

Let $I \subseteq F$ be an ideal of F. Clearly, $I = \{0\}$ and $I = F$ are ideals.

Now, suppose $I \neq \{0\}$. Then there exists a non-zero element:

$$a \in I \quad \text{with } a \neq 0.$$

Since F is a field, every non-zero element has a multiplicative inverse $a^{-1} \in F$. By the multiplicative absorption property of ideals (Definition 1.29), choosing $r = a^{-1} \in F$:

$$a^{-1} \cdot a \in I \implies 1_F \in I.$$

Once the unit 1_F lies in I, applying absorption once more with any arbitrary element $r \in F$:

$$r = r \cdot 1_F \in I \quad \forall r \in F.$$

Therefore, $F \subseteq I$, which forces $I = F$. **Q.E.D.**

Implications for Field Homomorphisms: Going beyond the lecture: a direct consequence of this theorem is that every ring homomorphism $f\colon F \to R$ from a field F to a non-zero ring R is automatically **injective**:

$$\ker(f) \subset F \text{ an ideal} \implies \ker(f) = \{0\} \quad (\text{since } f(1_F) = 1_R \neq 0_R \implies \ker(f) \neq F).$$

The converse also holds: a non-zero commutative ring whose only ideals are (0) and (1) is a field, that is, R is a field if and only if (0) is a maximal ideal (Week 5, Proposition 5.5); more generally, ${}^{R}\!/_{M}$ is a field exactly when M is maximal (Proposition 5.6).

Because $\ker(f) = \{0\}$, field homomorphisms into non-zero rings are always embeddings ($f(a) = f(b)$ gives $f(a-b) = 0$, so $a = b$). This is why the extension $g\colon \mathbb{Q} \to F$ in Week 3 is automatically injective (Theorem 3.7).

Insight: Fields as *"Sellouts"*

The lecturer's humorous pedagogical quip — that *"fields are sellouts because they have lost all their ideals"* — highlights an essential contrast between field theory and ring theory:

- In a field F, the abundance of units ($\mathcal{U}_F = F \setminus \{0\}$) destroys all non-trivial ideals, as any non-zero element $a \in I$ immediately pulls the identity into the ideal: $1 = a^{-1} \cdot a \in I$ (with $r = a^{-1} \in F$ and $a \in I$), forcing $I = F$.
- Rings can have quite a number of ideals. The simplest example are the ideals $n\mathbb{Z} \subset \mathbb{Z}$, which by Theorem 2.35 are all the ideals of $\mathbb{Z}$.

Outlook: Zorn's Lemma and Vector Space Bases

Existence of Bases in Infinite-Dimensional Vector Spaces

Spoken Transcript [00:42:20 - 00:43:29] Prof. Rahul Pandharipande

For this week, I did some material that was billed to the second week, and I didn't cover Zorn, so I'll say two words about Zorn and then I'll say a little bit more next week.

The first question is about linear algebra, which is this statement which [erases the lower left board and prepares it] probably you've seen in the first year, which is a strange statement, but it's about... So, first of all, I will spend some time also next week on Zorn's lemma, so you can read about that in the book [writes: Zorn's Lemma].

And it's related to the statement:

Let V be a vector space over a field k [writes: Let V be a vector space over a field k].

Does V have a basis [writes: Does V have a basis?]?

Zorn's Lemma and the Existence of Vector Space Bases

Question 1.32 (*Existence of a Basis for Vector Spaces*)**:** Let V be a vector space over a field k. Does V always have a basis?

Remark 1.33 (*Infinite-Dimensional Vector Spaces*)**:** A finite-dimensional vector space has a basis. For an infinite-dimensional vector space, the traditional way to get a basis is an application of *"Zorn's Lemma"*, which is related to the *"axiom of choice"* in logic.

Beyond the Lecture: How Zorn's Lemma Gives a Basis

For finite-dimensional vector spaces, a basis comes from extracting a maximal linearly independent subset, for example via Gaussian elimination or the Steinitz exchange lemma (Fact 0.30). For infinite-dimensional spaces such as $C(\mathbb{R}, \mathbb{R})$ or $\mathbb{R}^{\mathbb{N}}$, the statement that every vector space has a basis (a *"Hamel basis"*) cannot be proven constructively in Zermelo–Fraenkel set theory (ZF): it is equivalent to the axiom of choice (AC). Zorn's lemma says: if P is a non-empty partially ordered set (Definition 0.10) in which every chain (totally ordered subset) has an upper bound in P, then P has a maximal element. For a vector space V over a field k:

(1) Let P be the set of linearly independent subsets of V, ordered by inclusion. It is non-empty, since $\varnothing \in P$.
(2) The union of a chain in P is again linearly independent: a linear relation involves only finitely many vectors, and these all lie in one member of the chain. So every chain has an upper bound in P.
(3) Zorn's lemma gives a maximal element $B \in P$. If some $v \in V$ were not in the span of B, then $B \cup \{v\}$ would be linearly independent and strictly larger than B. Hence B spans V, and B is a basis.

Week 5 states Zorn's lemma formally (Lemma 5.14) and uses the same union-of-a-chain argument (Lemma 5.18) to prove that every non-zero ring has a maximal ideal (Krull's theorem, Theorem 5.19).

Spoken Transcript [00:43:29 - 00:44:01] Prof. Rahul Pandharipande

Did you consider this question in your first year?

Of course, if it's finite-dimensional, of course it has a basis. And this issue is about whether, if you have some massively infinite-dimensional field (i.e., vector space)... has a basis. And the traditional way to do this is an application of Zorn's lemma, which is related to the axiom of choice in logic — not my favorite direction to think about, but we will say a little bit about it. But...

Okay, and I think that's it. All right, so... [puts down the chalk, unclips the microphone, students applaud, and the lecture concludes]

Summary of Key Concepts

This lecture established core bridges connecting concrete arithmetic with axiomatic ring theory:

- **Gaussian Integers $\mathbb{Z}[i]$ and Primes:** The ring $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ possesses four units $\{\pm 1, \pm i\}$. An ordinary prime $p \in \mathbb{Z}$ that factors in $\mathbb{Z}[i]$ is a sum of two squares $p = a^2 + b^2$, and what matters is p modulo 4 (Fermat's two-square theorem): primes $p \equiv 1 \pmod 4$ are sums of two squares, and primes $p \equiv 3 \pmod 4$ remain prime in $\mathbb{Z}[i]$.
- **Rings and Elementary Arithmetic:** A ring $(R, +, \cdot, 0, 1)$ consists of an additive abelian group $(R, +, 0)$ and an associative multiplication with unit 1, linked by the distributive laws. Distributivity yields the universal identities $0 \cdot a = 0$ and $(-1) \cdot a = -a$. When $1 = 0$, the ring collapses to the degenerate zero ring $R = \{0\}$.
- **Canonical Map $\mathbb{Z} \to R$:** Defining integer multiples $n \cdot a$ yields the canonical ring homomorphism $\phi\colon \mathbb{Z} \to R$ with $\phi(n) = n \cdot 1_R$, which requires no choices.
- **Ring Homomorphisms:** A ring homomorphism preserves addition, multiplication, and the multiplicative identity ($f(1) = 1$). While $f(0) = 0$ is automatic via additive group cancellation, $f(1) = 1$ must be explicitly axiomatized because elements in arbitrary rings generally lack multiplicative inverses.
- **Ideals and Kernels:** An ideal $I \subset R$ is an additive subgroup that multiplicatively

absorbs the entire ambient ring ($r \cdot a \in I$ for all $a \in I, r \in R$). The kernel $\ker(f) = f^{-1}(\{0\})$ of every ring homomorphism is an ideal.

- **Ideals in Fields:** Because every non-zero element in a field F is invertible, any non-zero ideal contains 1_F and equals F. Thus, fields have only the ideals $\{0\}$ and F.
- **Outlook on Zorn's Lemma:** Whether every infinite-dimensional vector space has a basis is traditionally settled with Zorn's Lemma, which is related to the axiom of choice.

Tuesday, 22$^{\text{nd}}$ September 2026

Homomorphisms, Substructures, Ideals, and Category Theory

Course Content Overview

This lecture explores the foundational algebraic structures of modern mathematics—groups, rings, and fields—and the structure-preserving mappings between them, concluding with an introduction to category theory:

- **Groups, Rings, and Fields:** Review of the defining axioms; intrinsic uniqueness of group identities and inverses; the axiomatic hierarchy connecting rings and fields to additive abelian groups.
- **Homomorphisms and Isomorphisms:** Group homomorphisms and isomorphisms; characterization via two-sided inverse homomorphisms; comparison of the additive and multiplicative groups of real numbers via the exponential map.
- **Kernels, Images, and Substructures:** Structural properties of kernels and images; formal definition of subgroups ($H \leq G$); ring homomorphisms and subrings ($S \subseteq R$).
- **Ideals and Symmetry Breaking:** The breakdown of structural symmetry in rings, where kernels are ideals ($I \subset R$) and images are subrings; classification of ideals in $\mathbb{Z}$ ($I = n\mathbb{Z}$) via the division algorithm; ideals in rings of smooth functions $C^\infty(\mathbb{R})$.
- **Foundations of Category Theory:** Formal definition of categories (objects, morphisms, composition, unit laws); standard categories ($\mathbf{Grp}$, $\mathbf{Ring}$, $\mathbf{Vect}_\mathbb{C}$); categorical formulation of isomorphisms; covariant functors, including the functor from rings to groups and the double dual functor on vector spaces.

Administrative Remarks and Course Logistics

Spoken Transcript [00:00:00 - 00:01:48] Prof. Rahul Pandharipande

Okay, so, we start, and there has been some progress on the website. There should be instructions there. First of all, you should be able to register for the problem sessions now. The second thing is that there should be instructions on the website about how to upload your homework. And the third thing is about the quiz, and I think there's a link there, too. We'll see if it works on Friday. The rule for the quiz will be that you can take it anytime from 6 in the morning on Friday to midnight. And if this is not enough freedom, we can also discuss it. It's not so...

Another course hack is that if you look at this Metaphor page: in 2018 [writes "Metaphor 2018" on the board] it was the last time I taught this course. That Metaphor page is there. This course is going to follow roughly the same path. In fact, the topics and homework — I mean, there's maybe some slight switching order of lectures — but you can find all the information there. In particular, all the problem sets that Lycka will eventually upload, you can already look at them now if you want to get ahead. And also, there are all the answers to all the problem sets; you can also look at them if you want. But in the modern world, it's easy to find answers to these questions, so, I don't see there's any reason to hide them from you. You can find everything there.

The other thing is that if you want to ask me a question, I'm happy to talk about math with anybody. You have to find me. Normally I have an office, but I was kicked out because of this *"Fenstersanierung"* (i.e., window renovation). And it's going to last forever. So, I have a secret office on this floor. If you find me, you can talk to me. You'd have to find it — it's actually pretty hard to find. But if you do, you can knock on the door. Sometimes I'm busy, sometimes I'm not, but I'm happy to talk to anyone. But you have to find the office yourself.

On the Board

"Metaphor 2018"

Review of Algebraic Structures

Groups and Uniqueness of Identity and Inverses

Spoken Transcript [00:01:48 - 00:03:20] Prof. Rahul Pandharipande

Okay, so, last week we discussed kind of carefully these three objects which are the centerpiece for this course: G is a group. [writes "$(G, \cdot, e)$" on the board] And last week I used a star for the product. No one does that. That was purely for didactic purposes, and probably pointless didactic purposes. The standard for a group is dot, but as I said, you can name this anything you want.

So, this is the standard data of the group. And the group — I won't repeat everything, but it has the three axioms: [draws a brace next to $(G, \cdot, e)$] associativity, [writes "ass."] existence of the neutral element, [writes "exist. neutral"] and existence of inverses. [writes "exist. of inverses"] This is written now in slightly more abstract language than the explicit equations.

And there are many things I didn't say, because they are all in some sense part of the definition. Like, I didn't say explicitly... You could ask yourself: this neutral element, could there be some other element that also has the neutrality property? [writes "Could there be another $\hat{e} \in G$?"] Could there be... These are things you should think about yourself. I can't teach you everything; it's not possible.

Could there be another $\hat{e} \in G$ that has the neutrality property? And the answer is no, [writes "No!"] because then you could look at $e\hat{e}$. [writes "$e\hat{e} = \hat{e}$" and "e" below] If I use the fact that e is a neutral element, this equals $\hat{e}$; and if $\hat{e}$ is a pretend neutral element, then this equals e, okay? Is that clear?

Group Axioms and Uniqueness of the Neutral Element

Definition 2.1 (*Group*): A *"group"* is a triple $(G, \cdot, e)$ consisting of a set G, a binary operation $\cdot\colon G \times G \to G$, and a distinguished element $e \in G$ satisfying the following three axioms:

- **(i) Associativity (ass.):** $\forall a, b, c \in G\colon (a \cdot b) \cdot c = a \cdot (b \cdot c)$.
- **(ii) Existence of Neutral Element (exist. neutral):** $\forall a \in G\colon a \cdot e = e \cdot a = a$.
- **(iii) Existence of Inverses (exist. of inverses):** $\forall a \in G \; \exists b \in G\colon a \cdot b = b \cdot a = e$.

Question 2.2: Could there be another neutral element $\hat{e} \in G$?

Answer 2.3: No!

Proposition 2.4 (*Uniqueness of the Neutral Element*): Let $(G, \cdot, e)$ be a group. The neutral element $e \in G$ is unique.

Proof

Suppose $\hat{e} \in G$ also satisfies the neutrality property:

$$a \cdot \hat{e} = \hat{e} \cdot a = a \quad \forall a \in G.$$

Evaluating the product $e \cdot \hat{e}$:

$$e \cdot \hat{e} = \hat{e} \qquad \text{(since } e \text{ is a neutral element)},$$
$$e \cdot \hat{e} = e \qquad \text{(since } \hat{e} \text{ is a neutral element)}.$$

Equating both expressions yields:

$$\hat{e} = e \cdot \hat{e} = e.$$

Hence, the neutral element in any group is strictly unique. **Q.E.D.**

Spoken Transcript [00:03:20 - 00:05:05] Prof. Rahul Pandharipande

And then other questions you could ask: [writes "If $x \in G$, $\exists y\colon xy = e$"] if x is an element of the group, then this existence of inverses says that there exists a y such that xy equals the neutral element. You could say: Could there be another y, $\hat{y}$, where $x\hat{y} = e$? [writes "Could there be another $\hat{y}\colon x\hat{y} = e$?"] And the answer is no again. [writes "No!"]

And how do you prove this? Well, you look at this equation and you multiply both sides by the first one: $y(x\hat{y}) = y$. And then this thing is equal, by the original property... The neutral element property is on both sides, so, this thing is equal to e, so, this is equal to $\hat{y}$, [writes "$\hat{y} = (yx)\hat{y} = y$" and circles yx] okay? Is that clear?

All of this is buried, in some sense, in the definitions. You could define a group by saying that there exists a neutral element and it's unique, and there exist inverses and they're unique. You don't have to, but you could do that. That's making the definition bigger.

You can also try to make the definition smaller. There's some kind of strange direction where people think that, well, there's three axioms, but we'd like a group to only have two axioms. And some guy in the '60s found some crazy way to write it with just one axiom, but it's just logically equivalent. You can go look that up if you want. I don't remember; it's some complicated axioms, you can imagine.

Uniqueness of Inverse Elements

Question 2.5: Let $x \in G$ and let $y \in G$ satisfy $xy = e$. Could there be another $\hat{y} \in G$ such that $x\hat{y} = e$?

Answer 2.6: No!

Proposition 2.7 (*Uniqueness of Inverses*): Let $(G, \cdot, e)$ be a group, $x \in G$, and $y \in G$ an inverse of x ($xy = yx = e$). If $\hat{y} \in G$ satisfies $x\hat{y} = e$, then $\hat{y} = y$. In particular, the inverse of x is unique.

Proof

Assume there exists $\hat{y} \in G$ such that $x\hat{y} = e$. Since $yx = e$ (circled on the board), associativity gives:

$$\hat{y} = (yx)\hat{y} = y(x\hat{y}) = y \cdot e = y.$$

Thus, for every element $x \in G$, its inverse is uniquely determined; it is denoted x^{-1}.

Q.E.D.

Redundancy in Axiomatic Definitions: Axiom sets are often chosen for pedagogical simplicity rather than strict minimal independence:

- **Uniqueness is Inherent:** Although we only postulated the **existence** of identities and inverses, uniqueness is an algebraic consequence of associativity.
- **Axiom Minimization:** The lecturer mentions that groups can also be written with a single axiom, which is logically equivalent. Going beyond the lecture: such an axiom goes back to Higman and Neumann (1952).

Rings and Fields

Spoken Transcript [00:05:05 - 00:06:54] Prof. Rahul Pandharipande

Okay, so, that's the group, and the other structure we had was this ring (Recall: Definition 1.8), [writes "$(R, +, \cdot, 0, 1)$" on the board] which I won't repeat all over again. But the way to think about this is that if I look at just the additive part, this is an abelian group, [writes: $(R, +, 0)$ abelian group] and then I have some axioms for multiplication, [writes "axioms for $\cdot$"] and then two times the distributive law. [writes "2× Distr"] Those were the axioms for the ring, and we did some examples.

Certainly, one part of this class is really internalizing these axioms. And there's some kind of idea about mathematics which is some kind of asymptotic understanding — which is not just that you understand it, but the really asymptotic understanding that you hope for, which you never really achieve, is that not only do you understand it, but you cannot imagine the state of not understanding it. That's the thing you should really aim for. And you can only do that asymptotically. You can never really achieve that, but...

This is just at the axiomatic level, and then there was the notion of a field. [writes "Field"] The group is by itself an algebraic object; the field is a particular case of a ring, [writes "Field is a particular case of a Ring"] meaning a field is a ring with some additional structure, and the additional structure is that non-zero elements have multiplicative inverses. So, you can think about the whole world of rings: some of them are fields. But rings and groups are separate, although they are related by the fact that the additive part of the ring is an abelian group.

Axiomatic Organization of Rings and Fields

Recap: The ring axioms were introduced in Week 1 (Definition 1.8), together with the standing convention that rings are commutative (Remark 1.17). The lecturer repeats them in the compact form below.

Definition 2.8 (*Ring with Identity*): A *"ring"* is a 5-tuple $(R, +, \cdot, 0, 1)$ such that:

- **(i) Additive Abelian Group:** $(R, +, 0)$ is an abelian group (associative, commutative, with additive identity 0 and additive inverses $-a$).
- **(ii) Multiplication:** Multiplication $\cdot$ is associative and possesses a multiplicative identity $1 \in R$:

$$(a \cdot b) \cdot c = a \cdot (b \cdot c), \quad 1 \cdot a = a \cdot 1 = a \quad \forall a, b, c \in R.$$

- **(iii) Distributive Laws (2× Distr):** Multiplication distributes over addition from both sides:

$$\begin{aligned} a \cdot (b + c) &= (a \cdot b) + (a \cdot c), \\ (a + b) \cdot c &= (a \cdot c) + (b \cdot c). \end{aligned}$$

In this course, the multiplication is moreover commutative (Remark 1.17). The case $1 = 0$ is allowed; it gives the zero ring (Proposition 1.10).

Definition 2.9 (*Field*): A *"field"* is a commutative ring $(R, +, \cdot, 0, 1)$ with $0 \neq 1$ in which every non-zero element possesses a multiplicative inverse:

$$\forall a \in R \setminus \{0\} \; \exists a^{-1} \in R \colon a \cdot a^{-1} = 1.$$

Equivalently, a field is a ring whose non-zero elements form an abelian group under multiplication:

$$(R \setminus \{0\}, \cdot, 1) \text{ is an abelian group.}$$

***Insight:* Asymptotic Understanding in Mathematics**

The lecturer highlights a profound pedagogical concept: true mathematical mastery is not merely recognizing or memorizing formal axioms, but reaching an *"asymptotic understanding"* where a conceptual framework becomes so deeply internalized that one can no longer imagine the mental state of not understanding it.

***Insight:* The Hierarchy of Algebraic Structures**

Algebra organizes mathematical systems by layering structural axioms:

- **Groups $(G,\cdot,e)$:** A single operation with associativity, identity, and inverses.
- **Rings $(R,+,\cdot,0,1)$:** Two operations where $(R,+,0)$ is an abelian group, multiplication is associative with 1 (and, in this course, commutative), and the operations are tied together via the distributive laws.
- **Fields $(F,+,\cdot,0,1)$:** Commutative rings where $(F\setminus\{0\},\cdot,1)$ is also an abelian group.

Group Homomorphisms and Isomorphisms

Definition of a Group Homomorphism

Spoken Transcript [00:07:18 - 00:09:02] Prof. Rahul Pandharipande

Okay, so, those were the structures. And in algebra, after you have structures, there's the notion of the maps that respect the structures, which are the homomorphisms. And you've seen this before, and we discussed it a little bit last week, but I want to put them on equal footing here.

If you have two groups, [writes: $G \to \hat{G}$] then we want to consider maps that respect the group structure. A map f is called a group homomorphism [writes "$f\colon G \to \hat{G}$", "f is a group homomorphism"] if... That's the definition of this term: *"group homomorphism"*. If $f(a\cdot b) = f(a)\cdot f(b)$ [writes: $f(a\cdot b) = f(a)\cdot f(b)$] for all $a,b\in G$. [writes: $\forall a,b \in G$]

Last week, I did something complicated where G was $(G,\star)$ and $\hat{G}$ was $(\hat{G},\hat{\star})$ to be completely clear about the products being different. I'm never going to do that again; no one ever does that. So, here, when you read this, this multiplication is the operation in the first one, and that's the operation in the second one. Is that clear? They're both just called dot.

And in fact, many people just don't write it at all. It would be kind of normal to look in a book and there'd just be nothing, because that's the standard way you deal with algebra. You don't write x times x — that's a bad example — you don't write x times y, you just write xy.

Group Homomorphisms

Definition 2.10 (*Group Homomorphism*): Let $(G,\cdot)$ and $(\hat{G},\cdot)$ be groups. A map $f\colon G\to\hat{G}$ is called a *"group homomorphism"* if it respects the group operations:

$$f(a\cdot b) = f(a)\cdot f(b) \quad \forall a,b\in G. \tag{2.1}$$

In standard algebraic juxtaposition without explicit multiplication symbols:

$$f(ab) = f(a)f(b) \quad \forall a,b\in G.$$

Operational Interpretation of the Definition: In (2.1), the operation on the left-hand side $(a\cdot b)$ is the group multiplication in the domain G, whereas the operation on the right-hand side ($f(a) \cdot f(b)$) is the group multiplication in the codomain $\hat{G}$.

Preservation of the Neutral Element

Spoken Transcript [00:09:02 - 00:10:40] Prof. Rahul Pandharipande

Okay, so, that's the definition of a group homomorphism. It respects the structure. It says you can multiply here and then go over by f, or you can go over separately and multiply. That's what respecting the structure means.

Then you could say: What about the neutral element? Why am I not saying anything about the neutral element? I could. I could add that $f(e)$ is equal to e [writes: $f(e) = \hat{e}$] — or if I want to be clear, $\hat{e}$, the e on that side. I could add that, but I don't have to. Because if I put e here, if I just use this property, I put $f(e \cdot e)$... [writes: $f(e \cdot e) = f(e) \cdot f(e)$] There's only one thing you can do always.

So, I just stick e in here. Do you agree? And then I use the neutral property on this side to get $f(e) = f(e) \cdot f(e)$. [writes: $f(e) = f(e) \cdot f(e)$] And then I can use the cancellation property of the group: $f(e)$ has some inverse. There exists some $y \in \hat{G}$ such that $y \cdot f(e) = \hat{e}$, [writes: $y \in \hat{G}\colon yf(e) = \hat{e}$] right? This is the cancellation property. We've used it many times, and at some point I will stop explaining it, but anyway, here... We multiply by y on both sides, and we get $y \cdot f(e) \ldots$ [writes: $yf(e) = y(f(e)f(e))$]

So, then this is $\hat{e} \ldots$ Then we use the... Sorry, what are we doing here? $\hat{e} \cdot f(e)$, [writes: $\hat{e} = \hat{e}f(e)$] and this is $f(e)$. [writes: $\hat{e} = f(e)$] So, $f(e) = \hat{e}$, okay? So, I don't have to, but you could put it in there.

Automatic Preservation of Identity

Proposition 2.11 (*Identity Preservation*): Let $f\colon G \to \hat{G}$ be a group homomorphism, where $e \in G$ and $\hat{e} \in \hat{G}$ denote the respective neutral elements. Then:

$$f(e) = \hat{e}.$$

Proof

Applying the homomorphism property to $e = e \cdot e \in G$:

$$\begin{aligned} f(e \cdot e) &= f(e) \cdot f(e) \\ \implies f(e) &= f(e) \cdot f(e). \end{aligned} \tag{2.2}$$

Since $f(e) \in \hat{G}$, it possesses an inverse $y \in \hat{G}$ such that $y \cdot f(e) = \hat{e}$. Multiplying both sides of (2.2) on the left by y:

$$\begin{aligned} y \cdot f(e) &= y \cdot \big(f(e) \cdot f(e)\big) \\ \hat{e} &= \big(y \cdot f(e)\big) \cdot f(e) && \text{(associativity in } \hat{G}\text{)} \\ \hat{e} &= \hat{e} \cdot f(e) \\ \hat{e} &= f(e). \end{aligned}$$

Thus, $f(e) = \hat{e}$. **Q.E.D.**

Group Isomorphisms and Characterization

Spoken Transcript [00:10:40 - 00:11:45] Prof. Rahul Pandharipande

So, that's for a group. The group is the algebraic structure, and if you have two groups, you compare them with a homomorphism: it's a map that respects the structure. And you could say that the next thing you see — this is following the pattern in analysis, and in linear algebra also: there you have two vector spaces, and the map is a linear transformation. That's the notion of the map that respects the vector space structure on both sides.

And then there are some standard things to say: we say f is a group isomorphism [writes "f is a group isomorphism"] if f is a group homomorphism [writes "if f is a group hom."] and f is bijective. [writes "and f is bijective"] Okay? This should not surprise you. You've seen these terms in linear algebra, right? Which language did you learn linear algebra in?

Student Answer

English.

Spoken Transcript [00:11:49 - 00:12:43] Prof. Rahul Pandharipande

English, okay. Well, that's no problem then.

And there's another way to say this: f is a group isomorphism [writes "f is group isom."] if and only if [writes: $\iff$] there exists an inverse — there exists a group homomorphism in the other direction. There exists a group homomorphism [writes: $\exists g \colon \hat{G} \to G$ is hom.] such that if I do f first and then g, I get the identity on the first group; and if I do g first and then f, I get the identity on the second group. [writes "such that $g \circ f = 1_G$, $f \circ g = 1_{\hat{G}}$"]

These are equivalent, okay? So, if you have the bottom, it's certainly bijective, because you have the forward and backward maps. And if you have the top, since it's bijective, there exists an inverse function, and you have to show that that's a homomorphism, which I leave to you to show. I hope this is clear.

Group Isomorphisms

Definition 2.12 (*Group Isomorphism*): A group homomorphism $f \colon G \to \hat{G}$ is called a *"group isomorphism"* if f is bijective. Two groups G and $\hat{G}$ are called *"isomorphic"* (written $G \cong \hat{G}$) if there exists a group isomorphism between them.

Proposition 2.13 (*Alternative Characterization of Isomorphism*): A map $f \colon G \to \hat{G}$ is a group isomorphism if and only if there exists a group homomorphism $g \colon \hat{G} \to G$ such that:

$$g \circ f = 1_G \quad \text{and} \quad f \circ g = 1_{\hat{G}}, \tag{2.3}$$

where $1_G = \mathrm{Id}_G$ denotes the identity map of G. (For maps of sets, this is the characterization of bijections in Fact 0.3; the new point is that g is again a homomorphism.)

Proof

Direction ($\Leftarrow$): If homomorphisms f and g satisfy (2.3), then g serves as a set-theoretic two-sided inverse of f. Hence, f is a bijection, and since it is given as a homomorphism, it is an isomorphism.

Direction ($\Rightarrow$): Suppose f is a bijective group homomorphism. As a bijection, there exists a unique set-theoretic inverse map $g = f^{-1}\colon \hat{G} \to G$ satisfying (2.3). That g is a homomorphism is left to the students. **Q.E.D.**

Why the Inverse Is a Homomorphism (beyond the lecture)

The lecture leaves this to the students. Let $\hat{a}, \hat{b} \in \hat{G}$, and let $a = g(\hat{a})$, $b = g(\hat{b}) \in G$. Then $f(a) = \hat{a}$ and $f(b) = \hat{b}$. Since f is a homomorphism:

$$f(ab) = f(a)f(b) = \hat{a}\hat{b}.$$

Applying $g = f^{-1}$ to both sides:

$$g(\hat{a}\hat{b}) = ab = g(\hat{a})g(\hat{b}).$$

Thus, g is a group homomorphism. **Q.E.D.**

Examples: Isomorphisms of the Real Numbers

Spoken Transcript [00:12:43 - 00:14:43] Prof. Rahul Pandharipande

All right, so, maybe an example.

If I take the real numbers, one group is the real numbers with addition and zero: [writes: $(\mathbb{R}, +, 0)$]. This is a rather basic group: real numbers, zero, and you can add them.

Another group, which you maybe don't think about as much, is that you could take the non-zero real numbers, multiplication, and one: [writes: $(\mathbb{R}^*, \cdot, 1)$]. Here, I just remind you — this is pretty standard — when you have a field and you put a star, [writes: note: $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$] this usually, but not always, means you take the field and throw out zero. Is that clear? So, this is the non-zero real numbers with multiplication and one. That is also a group. Are you happy with that? Because you can invert non-zero numbers.

And there's another one here: I can take the non-zero ones, but there's kind of two ways to be non-zero: you can be positive, or you can be negative. So, I could take the positive ones: [writes: $(\mathbb{R}^*_{>0}, \cdot, 1)$]. That's still a group. Is that clear? If I try to take the negative ones, I kind of get in trouble, because if I multiply negative ones, I get positive, and one is not there. So, anyway, there are these three groups.

Which of these are isomorphic, or which ones are not isomorphic? Do you know this? Well, maybe I will write the question: Which are isomorphic? [writes "Which are isomorphic?"] Which means: For which ones can I find an isomorphism?

Three Real Number Groups

Consider the following three groups based on subsets of $\mathbb{R}$:

(1) The additive group of real numbers:

$$(\mathbb{R}, +, 0).$$

(2) The multiplicative group of non-zero real numbers:

$$(\mathbb{R}^*, \cdot, 1), \quad \text{where } \mathbb{R}^* := \mathbb{R} \setminus \{0\}.$$

(3) The multiplicative group of strictly positive real numbers:

$$(\mathbb{R}^*_{>0}, \cdot, 1), \quad \text{where } \mathbb{R}^*_{>0} := \{x \in \mathbb{R} \mid x > 0\}.$$

? **Question 2.14:** Which of these three groups are isomorphic to each other?

Spoken Transcript [00:15:26 - 00:16:21] Prof. Rahul Pandharipande

Let's start with this — this is maybe the more confusing one. If I take the non-zero real numbers, multiplication, and 1, [writes: $(\mathbb{R}^*, \cdot, 1)$] and I take the positive ones, multiplication, and 1, [writes: $(\mathbb{R}^*_{>0}, \cdot, 1)$] could these be isomorphic? [writes "Could these be isom.?"]

It's kind of strange, because if you draw the picture, one of them is kind of both sides, and the other one is just one side. [sketches real lines for $\mathbb{R}^*$ and $\mathbb{R}^*_{>0}$] So, it seems kind of unlikely, although there are still bijections that do this. So, the question is: Is there a bijection which respects... You know, there are set-theoretic bijections, because both sets are just uncountable sets; they have the same cardinality.

What do you think? Yeah?

Student Answer

No, because we run into a problem for -1, because, um, if we had if $f(-1)$...

Spoken Transcript [00:16:31 - 00:16:33] Prof. Rahul Pandharipande

Which way is f going?

Student Answer, continued

So, f goes from, uh, $\mathbb{R}^*$ to $\mathbb{R}^*_{>0}$.

Spoken Transcript [00:16:36 - 00:16:40] Prof. Rahul Pandharipande

Yeah. [writes: $f\colon \mathbb{R}^* \to \mathbb{R}^*_{>0}$]

Student Answer, continued

And then f of minus $f(1)$...

Spoken Transcript [00:16:42 - 00:16:45] Prof. Rahul Pandharipande

Has to be 1, that's the... [writes: $f(1) = 1$]

Student Answer, continued

Yeah, that's 1. But that's also $f(-1)$...

Spoken Transcript [00:16:47 - 00:16:50] Prof. Rahul Pandharipande

Yeah, so, $f(-1)$ equals what? That's a good... [writes: $f(-1) = ?$]

Student Answer, continued

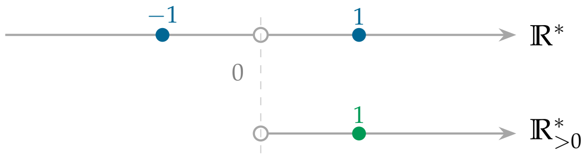
$f(-1) \cdot f(-1)$ will be 1, so the only positive number such that, uh, its times itself is 1 would be 1. So, $f(-1)$ has to be 1. So, $f(1) = f(-1)$, and then it's not bijective.

Spoken Transcript [00:17:10 - 00:18:00] Prof. Rahul Pandharipande

Yeah, this is all correct. The simplest way to say it is that $f(-1)$ has to be a square root of 1, [writes: $f(-1) \cdot f(-1) = f((-1) \cdot (-1)) = f(1) = 1$] by this law. And there's only one square root of 1 in the positive numbers, and that's already taken by 1. So, it violates bijectivity.

Do you understand this statement? This was maybe not said as clearly as possible, but the point is that 1 has two elements in here — there are two elements which, when you multiply them by themselves, you get 1: +1 and −1. And those two elements have to go to two different elements here, because it's a bijection. If there were a bijection, they would have to go to two different elements. But here, there's only one element whose square is 1. So, it's impossible that there is a bijection, okay? This is a solid argument; this is the standard argument.

Non-Isomorphism of $(\mathbb{R}^*, \cdot, 1)$ and $(\mathbb{R}^*_{>0}, \cdot, 1)$



Proposition 2.15 (*Non-Isomorphism of $\mathbb{R}^*$ and $\mathbb{R}^*_{>0}$*)**:** The groups $(\mathbb{R}^*, \cdot, 1)$ and $(\mathbb{R}^*_{>0}, \cdot, 1)$ are not isomorphic:

$$(\mathbb{R}^*, \cdot, 1) \not\cong (\mathbb{R}^*_{>0}, \cdot, 1).$$

Proof

Suppose for contradiction there exists an isomorphism $f \colon \mathbb{R}^* \to \mathbb{R}^*_{>0}$. By Proposition 2.11, any homomorphism must map the identity element to the identity element:

$$f(1) = 1.$$

Now consider the element $-1 \in \mathbb{R}^*$. Since $(-1) \cdot (-1) = 1$, the homomorphism property implies:

$$f(-1)^2 = f(-1) \cdot f(-1) = f\big((-1) \cdot (-1)\big) = f(1) = 1.$$

Thus, $f(-1)$ must be a solution to the equation $y^2 = 1$ in the codomain $\mathbb{R}^*_{>0}$. However, in $\mathbb{R}^*_{>0}$, all elements are strictly positive, and the only positive real number satisfying $y^2 = 1$ is $y = 1$:

$$f(-1) = 1.$$

This implies $f(-1) = f(1) = 1$. Since $-1 \neq 1$, f cannot be injective, contradicting the assumption that f is an isomorphism. **Q.E.D.**

☰ **Algebraic Invariants: Elements of Order 2:** Going beyond the lecture: the proof uses an **algebraic invariant**, a property preserved by every isomorphism. In $(\mathbb{R}^*, \cdot, 1)$, the element -1 has order 2 ($x \neq e$ with $x^2 = e$, Definition 0.25), while $(\mathbb{R}^*_{>0}, \cdot, 1)$ has no element of order 2. An isomorphism maps elements of order 2 to elements of order 2, so the two groups are not isomorphic, although as sets they have the same cardinality (Fact 0.4).

Spoken Transcript [00:18:00 - 00:18:23] Prof. Rahul Pandharipande

Now, what about this one? [writes “$(\mathbb{R}, +, 0)$” and “$(\mathbb{R}^*_{>0}, \cdot, 1)$”] $(\mathbb{R}, +, 0)$, $(\mathbb{R}^*_{>0}, \cdot, 1)$. If I make any mistakes on the board, you have to tell me, all right?

So, what about this? Could these be isomorphic? Yeah?

Student Answer

I think it could. We can take, like, f from the first one, from $\mathbb{R}, +$ to $\mathbb{R}^*_{>0}$ by the function e to the power of x.

Spoken Transcript [00:18:32 - 00:19:27] Prof. Rahul Pandharipande

Yeah, yeah. This is like the most famous function in mathematics, and this is its purpose, actually. This is by exp: [writes “exp” and draws an arrow] $\exp(x)$. So, $f(x) = \exp(x)$. [writes: $f(x) = \exp(x)$] You take a number, and you get a positive number. If you take 0, you get 1.

And the main property of exp is that it intertwines the addition here with the multiplication there. So, it’s actually the most famous function in mathematics, basically. And this is its role: it gives you an isomorphism — a group isomorphism between the additive structure and the multiplicative structure.

So, that’s a kind of weird thing. It’s saying that, from the point of view of the group, the group doesn’t see the difference between the additive structure and the multiplicative structure of the reals.

Isomorphism via the Exponential Map

Proposition 2.16 (*Isomorphism via the Exponential Map*): The additive group of real numbers is isomorphic to the multiplicative group of positive real numbers:

$$(\mathbb{R}, +, 0) \cong (\mathbb{R}^*_{>0}, \cdot, 1).$$

Proof

Define the map $f\colon \mathbb{R} \to \mathbb{R}^*_{>0}$ by the natural exponential function:

$$f(x) := \exp(x) = e^x.$$

We verify that f is a group isomorphism:

(i) **Homomorphism Property:** For any $x, y \in \mathbb{R}$, by the fundamental exponent law (Fact 0.38):

$$f(x+y) = \exp(x+y) = \exp(x) \cdot \exp(y) = f(x) \cdot f(y).$$

Thus, f converts addition in $(\mathbb{R}, +)$ into multiplication in $(\mathbb{R}^*_{>0}, \cdot)$.

(ii) **Identity Preservation:**

$$f(0) = \exp(0) = 1.$$

(iii) **Bijectivity:** The exponential function $\exp\colon \mathbb{R} \to \mathbb{R}^*_{>0}$ is strictly monotonically increasing and continuous, with image $(0, \infty) = \mathbb{R}^*_{>0}$. It possesses a two-sided inverse given by the natural logarithm:

$$g\colon \mathbb{R}^*_{>0} \to \mathbb{R}, \quad g(y) := \ln(y),$$

satisfying $g(f(x)) = \ln(e^x) = x$ and $f(g(y)) = e^{\ln(y)} = y$.

Therefore, $f = \exp$ is a bijective group homomorphism, establishing $(\mathbb{R}, +, 0) \cong (\mathbb{R}^*_{>0}, \cdot, 1)$. **Q.E.D.**

***Insight:* The Fundamental Role of the Exponential Function**

From an algebraic perspective, the exponential function $x \mapsto e^x$ is not merely an analytic power series or a solution to $y' = y$; its primary algebraic purpose is to establish an exact equivalence (*"group isomorphism"*) between the continuous additive world $(\mathbb{R}, +)$ and the continuous positive multiplicative world $(\mathbb{R}^*_{>0}, \cdot)$.

Kernels and Subgroups

The Kernel of a Group Homomorphism

Spoken Transcript [00:19:27 - 00:20:55] Prof. Rahul Pandharipande

Okay, so, maybe I'll say two more things about it.

There's a notion of a subgroup. We did this for the ring (Recall: Definition 1.26, Proposition 1.28); I'm going to repeat it all for the ring in one second. If you have this map, $f\colon G \to \hat{G}$ — and I'm not going to keep saying *"group homomorphism"*; the standard is to say *"homomorphism"*. It is a homomorphism, which is abbreviated hom, [writes "is a hom"] and you're supposed to understand, since I'm talking about groups, that it's a group homomorphism. All right?

If f is a group homomorphism, then in the spirit of linear algebra — which is the algebra you're supposed to be most familiar with — if you have a linear transformation between two vector spaces, the thing you immediately study is the kernel and the image of that linear transformation. We can do that also with groups: we can consider the kernel and the image. [writes "Then we can consider $\ker$, Im"] Let's start with the kernel. What properties does this kernel enjoy?

The kernel of a group homomorphism is defined to be those elements in G such that when I apply f, I get the neutral element on the second side: [writes: $\ker(f) = \{x \in G \mid f(x) = \hat{e}\}$]. In the linear algebra language, that's things that went to zero. Now it's the elements that go to the neutral element. Is that clear?

Definition of the Kernel

Definition 2.17 (*Kernel*): Let $f\colon G \to \hat{G}$ be a group homomorphism, with $\hat{e} \in \hat{G}$ the neutral element of $\hat{G}$. The *"kernel"* of f, denoted $\ker(f)$, is defined as the preimage of the neutral element:

$$\ker(f) := \{x \in G \mid f(x) = \hat{e}\}. \tag{2.4}$$

For a ring homomorphism, viewed as a homomorphism of the additive groups, this is the kernel of Week 1 (Definition 1.26).

Analogy with Linear Algebra: In linear algebra, the kernel of a linear map $T\colon V \to W$ consists of all vectors mapped to the additive identity $\mathbf{0}_W$: $\ker(T) = \{v \in V \mid T(v) = \mathbf{0}_W\}$. In group theory, $\ker(f)$ similarly collects all elements mapped to the neutral element $\hat{e}$ of the target group.

Properties of the Kernel

Spoken Transcript [00:20:55 - 00:22:06] Prof. Rahul Pandharipande

Let's try to look at properties of this. Well, the neutral element itself is in here: [writes: $e \in \ker(f)$]. That's one property, because it's one of the properties of the homomorphism: the homomorphism takes the neutral element here to the neutral element there. So, that's there, right?

The second property: suppose a and b are in the kernel of f. [writes "suppose $a, b \in \ker(f)$, then"] What can we say? We can consider what happens to $f(ab)$. There's only one thing we can do here. These proofs at this level are kind of strong proofs, meaning that there's only one thing you can do at every stage, and you just have to do it.

This is $f(a)f(b)$. [writes: $f(ab) = f(a)f(b)$] And now, the assumption is that they're both in the kernel, so, these are both $\hat{e}$, [writes: $= \hat{e} \cdot \hat{e}$] and then we use the identity property: that's $\hat{e}$. [writes: $= \hat{e}$] What have we learned here? That says that if a and b are in the kernel, we have proven that $a \cdot b$ is in the kernel, [writes: $ab \in \ker(f)$] okay?

Kernel Properties: Identity and Closure Under Products

Property 1 (Identity): The neutral element $e \in G$ always belongs to the kernel:

$$e \in \ker(f) \quad (\text{since } f(e) = \hat{e} \text{ by Proposition 2.11}). \tag{2.5}$$

Property 2 (Closure under Multiplication): If $a, b \in \ker(f)$, then their product ab is also in the kernel:

$$\begin{aligned} f(ab) &= f(a) \cdot f(b) && (\text{homomorphism property}) \\ &= \hat{e} \cdot \hat{e} && (\text{since } a, b \in \ker(f)) \\ &= \hat{e}. \end{aligned} \tag{2.6}$$

Thus, $ab \in \ker(f)$.

Spoken Transcript [00:22:06 - 00:23:03]

Prof. Rahul Pandharipande

So, we're looking at properties of the kernel, and we have discovered that the neutral element is in the kernel, and the kernel is closed under the group law, the product.

We might be curious now whether the inverses are there also. That's the third property of the kernel.

Spoken Transcript [00:23:03 - 00:25:05]

Prof. Rahul Pandharipande

The third property of the kernel is that if x is in the kernel of f, [writes "if $x \in \ker(f)$"] then the inverse... I should make this comment that very often the inverse for the group law is denoted x inverse: [writes "inverse for the group law is denoted x^{-1}"]. That's just the name of it. The axioms say there exists an inverse; this is very often the name that's given to it.

This kind of presupposes that the product you're calling dot. If the product that you're calling is plus, then you usually use the minus sign here. But this is just notation. You could call it anything you want — you could call it Fred or an elephant, anything. We're calling it this, okay?

Then the inverse is also in the kernel: [writes "then $x^{-1} \in \ker(f)$"].

And how are we going to prove this? I'm not going to go through this proof, but you have to start out like this: [writes: $f(x \cdot x^{-1})$] $f(x \cdot x^{-1})$. That's the one thing you know. On the one hand, this is $f(e)$, and that's $\hat{e}$. [writes: $= f(e) = \hat{e}$] On the other hand, this is $f(x) \cdot f(x^{-1})$, [writes: $= f(x) \cdot f(x^{-1})$] and x is assumed to be in the kernel, so, this is equal to... Actually, I'm finishing it now, unfortunately. [writes: $= \hat{e} \cdot f(x^{-1}) = f(x^{-1})$]

So, we get that the neutral element is equal to the neutral element times that, so, that must be the neutral element, okay? This implies that $x^{-1} \in \ker(f)$. [writes: $\implies f(x^{-1}) = \hat{e}$] Is that clear? We have the neutral element, we do all this chain of things, and we get that the neutral element is the neutral element times this. We could go one more if we want to be really explicit: then we have the full chain, the neutral element is equal to that.

So, the kernel enjoys these three properties: it contains the neutral element of G, the source group; it's closed under the group product; and it's closed under inverses.

Consolidated Properties of the Kernel

Proposition 2.18 (*Properties of the Kernel*): Let $f\colon G \to \hat{G}$ be a group homomorphism. The kernel $\ker(f) \subseteq G$ satisfies the following three fundamental structural properties:

(1) Contains the Identity: $e \in \ker(f)$.
(2) Closure under Products: $\forall a, b \in \ker(f)\colon ab \in \ker(f)$.
(3) Closure under Inverses: $\forall x \in \ker(f)\colon x^{-1} \in \ker(f)$.

Proof of Property (3)

Let $x \in \ker(f)$, so that $f(x) = \hat{e}$. Consider the evaluation of $f(x \cdot x^{-1})$:

$$f(x \cdot x^{-1}) = f(e) = \hat{e} \qquad \text{(by Proposition 2.11)}.$$

On the other hand, by the homomorphism property:

$$\begin{aligned} f(x \cdot x^{-1}) &= f(x) \cdot f(x^{-1}) \\ &= \hat{e} \cdot f(x^{-1}) && (\text{since } x \in \ker(f)) \\ &= f(x^{-1}). \end{aligned}$$

Equating both expressions yields:

$$f(x^{-1}) = \hat{e} \implies x^{-1} \in \ker(f).$$

This completes the proof of all three properties. **Q.E.D.**

Definition of a Subgroup

Spoken Transcript [00:25:18 - 00:26:29] Prof. Rahul Pandharipande

This is the definition of a subgroup. We will come back to this in a second.

Definition: [writes "Def:"] If H is a subset of G, [writes: $H \subset G$] then H is a subgroup [writes "then H is a subgroup"] if zero is in H [writes "if $0 \in H$"] — as we've shown — H is closed under the product, [writes "H is closed under the product"] and H is closed under inverses, [writes "and H is closed under inverse"] meaning if an element is in H, then its inverse is also in H, okay?

In particular, this implies that... That's the definition of a subgroup. This implies that if I take H, and I take the group law — which makes sense on H because H is closed under it — and I take the zero, which also makes sense because zero is in H, then this is a group: [writes: $(H, \cdot, 0)$ is group]. Which is a very nice thing: a subgroup is a group. It would be terrible if it weren't, right?

That's the definition of a subgroup.

Definition of a Subgroup

Definition 2.19 (*Subgroup*): Let $(G, \cdot, e)$ be a group. A subset $H \subseteq G$ is called a "*subgroup*" of G (denoted $H \leq G$) if it satisfies the following three conditions:

(1) Contains the Identity: $e \in H$ (often denoted $0 \in H$ in additive notation).

(2) Closure under Multiplication: $\forall a, b \in H\colon a \cdot b \in H$.
(3) Closure under Inverses: $\forall x \in H\colon x^{-1} \in H$.

In particular, the restriction of the group law to H endows $(H, \cdot, e)$ with the structure of a group in its own right.

Inherited Axioms: Associativity holds automatically for elements in H because it holds for all elements in the ambient group G. Therefore, checking identity and closures under products and inverses is necessary and sufficient for H to form a group.

Notation: Subgroups

The lecturer writes $H \subset G$ and says that H is a subgroup. The sign $H \leq G$ in the boxes of these notes is the textbook one.

Kernel and Image as Subgroups

Spoken Transcript [00:26:29 - 00:28:05] Prof. Rahul Pandharipande

And now we've proven a little lemma here — proposition, or however you want to call it: The kernel of f is a subgroup of G: [writes "Prop The $\ker(f)$ is subgroup of G"]. Is that clear?

That's the investigation of the kernel and the notion of a subgroup, but there's also the image. And so, proposition — and I leave this for you, you can prove it: The image of f is a subgroup of $\hat{G}$, [writes "Prop The $\mathrm{Im}(f)$ is subgroup of $\hat{G}$"] okay?

How do you prove this? You just check that the image contains the neutral element. It has to, because the neutral element from the first group goes to the neutral element of the second group, so, the image has to contain the neutral element. And then you have to check that if you have two elements in the image, when you apply the group law for $\hat{G}$, they stay in the image; and the same for inverses. The proofs are all just the same: you just follow your nose. There's only one thing you can do, and you can do it.

The Kernel and Image are Subgroups

Proposition 2.20 (*Kernel as a Subgroup*): Let $f\colon G \to \hat{G}$ be a group homomorphism. Then the kernel of f is a subgroup of the domain:

$$\ker(f) \leq G.$$

Proposition 2.21 (*Image as a Subgroup*): Let $f\colon G \to \hat{G}$ be a group homomorphism. Then the image of f, defined as

$$\mathrm{Im}(f) := f(G) = \{f(x) \in \hat{G} \mid x \in G\},$$

is a subgroup of the codomain:

$$\mathrm{Im}(f) \leq \hat{G}.$$

Why the Image Is a Subgroup (beyond the lecture)

The lecture leaves this to the students. We verify the three subgroup axioms for $\mathrm{Im}(f) \subseteq \hat{G}$:

(1) **Identity:** By Proposition 2.11, $f(e) = \hat{e}$. Since $e \in G$, $\hat{e} \in \mathrm{Im}(f)$.

(2) **Closure under Multiplication:** Let $\hat{a}, \hat{b} \in \mathrm{Im}(f)$. Then there exist $a, b \in G$ such that $f(a) = \hat{a}$ and $f(b) = \hat{b}$. Since G is a group, $ab \in G$, and by the homomorphism property:

$$\hat{a} \cdot \hat{b} = f(a) \cdot f(b) = f(ab) \in \mathrm{Im}(f).$$

(3) **Closure under Inverses:** Let $\hat{x} \in \mathrm{Im}(f)$ with $\hat{x} = f(x)$ for some $x \in G$. Since $x^{-1} \in G$:

$$\hat{x} \cdot f(x^{-1}) = f(x) \cdot f(x^{-1}) = f(x \cdot x^{-1}) = f(e) = \hat{e}.$$

By uniqueness of inverses in $\hat{G}$, $(\hat{x})^{-1} = f(x^{-1}) \in \mathrm{Im}(f)$.

Thus, $\mathrm{Im}(f) \leq \hat{G}$. **Q.E.D.**

Pedagogical Aside: Mathematical Reading and Textbooks

Spoken Transcript [00:28:05 - 00:29:38] Prof. Rahul Pandharipande

This is a nice situation for groups: the understanding of the group is given by these very simple axioms. There are lots of examples — in fact, part of this class is the examples for groups, which will come later. But anyway, there's the notion of a homomorphism, which respects the group structure; and whenever you have one, there's the kernel and the image.

This one I leave to you. You can look in the books, too, but it's not worth looking in the books. Because if you read a book... Actually, I'm very much for reading books; this is maybe the wrong way to order the discussion. You should read the books — I take it back! But the problem with reading such a simple thing in a book is that you have to learn what notation the author is using and all that. It's much simpler just to write the proof of this yourself.

Actually, I'm a big fan of books. That's one way to understand this kind of classical history of mathematics: read a lot of math books. And I know that somehow in the modern world this is less popular, but in my generation we read a lot of math books.

And on the topic of books: there's the book for the course, which was picked a while ago — not by me even — and it has the advantage that it's freely available on the ETH website, so, you can just click it. Then there are some other references I gave, which are, I think, a little bit better books: they're more advanced in perspective, they're written better. But I think they're not freely available on the webpage, so, it wasn't chosen for this class. But it's a good idea to read books; you'll learn something.

Textbook Study and Literature Recommendations

The lecturer reflects on the pedagogical value of reading mathematical literature: while textbooks are invaluable for developing deep perspective, elementary verifications (such as checking subgroup axioms for the image) are best proven independently by the student to avoid the cognitive overhead of unfamiliar author-specific notations.

Ring Homomorphisms, Ideals, and Subrings

Definition of a Ring Homomorphism

Spoken Transcript [00:29:39 - 00:31:07] Prof. Rahul Pandharipande

Okay, so, for the ring, [writes: $(R,+,\cdot,0,1)$] we had also defined what a homomorphism is (Recall: Definition 1.18) — a ring homomorphism. If I have two rings, [writes: $R \to \hat{R}$] then we had defined that f is a... This is from last week: a ring hom, [writes "is a ring hom"] which, as I said, I will just say *"hom"*, and you're supposed to understand they're rings. And the axioms that a ring hom has to satisfy, if you remember: the first one is this one: $f(a+b) = f(a) + f(b)$ [writes: $f(a+b) = f(a) + f(b)$].

The interpretation of this is: the additive part of this is a group, and the additive part of that is a group, and the ring hom should be a group hom for those parts. Is that clear? In words, additively, f is a group hom for the plus part, okay? [writes "additively f is group hom for $+$"]

Of course, we also have that it respects multiplication: $f(ab) = f(a)f(b)$, [writes: $f(ab) = f(a)f(b)$] and then we also have that the identity has to be respected: $f(1) = 1$ [writes: $f(1) = 1$].

Those are the notions of a ring homomorphism from last time. I just write it here.

Definition of a Ring Homomorphism

Recap: Ring homomorphisms were defined in Week 1 (Definition 1.18), where it was also shown that $f(0) = 0$ follows from additivity (Proposition 1.19) while $f(1) = 1$ must be required (Example 1.20). The lecturer repeats the definition and now phrases it with the group notions of this lecture.

Definition 2.22 (*Ring Homomorphism*): Let $(R,+,\cdot,0,1)$ and $(\hat{R},+,\cdot,\hat{0},\hat{1})$ be rings. A map $f\colon R \to \hat{R}$ is called a *"ring homomorphism"* if it satisfies:

- **(i) Additivity:** $f(a+b) = f(a) + f(b)$ for all $a,b \in R$ (i.e., $f\colon (R,+,0) \to (\hat{R},+,\hat{0})$ is a group homomorphism of additive groups).
- **(ii) Multiplicativity:** $f(a \cdot b) = f(a) \cdot f(b)$ for all $a,b \in R$.
- **(iii) Identity Preservation:** $f(1) = \hat{1}$.

Why Must $f(1) = \hat{1}$ Be Explicitly Required?: For group homomorphisms, identity preservation $f(e) = \hat{e}$ is automatic from cancellation in a group (Proposition 2.11). In a ring, multiplicative inverses are generally not available; hence $f(1) \cdot f(1) = f(1)$ does not imply $f(1) = \hat{1}$ (for instance, the zero map $f(x) = \hat{0}$ preserves addition and multiplication, but sends $1 \mapsto \hat{0}$). Therefore, identity preservation $f(1) = \hat{1}$ must be postulated as an explicit axiom for rings with identity.

Ring Isomorphisms and Symmetry Breaking

Spoken Transcript [00:31:07 - 00:32:57] Prof. Rahul Pandharipande

That's the notion of a ring homomorphism. And again, we can think about ring isomorphisms: a ring isom is a bijective ring hom. [writes "Ring isom is a bijective ring hom."] I won't write it, but the same thing is true: another way to think about it is that if there's an isomorphism between two rings, that means that the inverse is a ring homomorphism in the other direction, just as I said with groups.

Then we can do the parallel study: if $f\colon R \to \hat{R}$ is a ring homomorphism, [writes: $f\colon R \to \hat{R}$ is ring hom.] there are two things I can study as before: the kernel of f and the image of f. [writes: $\ker(f), \operatorname{Im}(f)$]

We studied the kernel last week (Recall: Definition 1.26), if you remember. And now the matter is different. For groups, there was the notion of a subgroup, and it happened that both the kernel and the image were subgroups: the former was a subgroup of the source group, and the latter was a subgroup of the target group.

But for rings, the symmetry is broken. So, one has to concentrate a bit here. And by the way, that's also true for vector spaces: for vector spaces, you have a linear map between two vector spaces, the kernel is a subspace of the first one, and the image is a subspace of the second one. From the point of view of the group, that's a very familiar structure. For rings, something different happens, so, it requires some concentration.

Ring Isomorphisms and Symmetry Breaking

Definition 2.23 (*Ring Isomorphism*): A ring homomorphism $f\colon R \to \hat{R}$ is called a *"ring isomorphism"* if f is bijective. Equivalently, f is an isomorphism if and only if there exists a ring homomorphism $g\colon \hat{R} \to R$ such that $g \circ f = \mathrm{Id}_R$ and $f \circ g = \mathrm{Id}_{\hat{R}}$ (as for groups, Proposition 2.13).

Symmetry Breaking: Groups and Vector Spaces vs. Rings: In groups and vector spaces, kernel and image share the exact same algebraic classification:

- **Groups:** $\ker(f) \leq G$ (subgroup) and $\operatorname{Im}(f) \leq \hat{G}$ (subgroup).
- **Vector Spaces:** $\ker(T) \leq V$ (subspace) and $\operatorname{Im}(T) \leq W$ (subspace).
- **Rings (Symmetry Broken!):** The image $\operatorname{Im}(f)$ is a **subring** of $\hat{R}$, but the kernel $\ker(f)$ is **not** a subring (unless $\hat{R} = \{0\}$, since $1 \notin \ker(f)$ because $f(1) = \hat{1} \neq \hat{0}$). Instead, the kernel is an **ideal**!

Ideals in Rings and the Kernel

Spoken Transcript [00:32:57 - 00:35:06] Prof. Rahul Pandharipande

And for the kernel of f, we had proven last week — I think this is on the videotape even, in the second lecture last week — that the kernel of f is an ideal (Recall: Definition 1.29, Proposition 1.30). [writes "for the $\ker(f)$, we had proven last week that $\ker(f)$ is an ideal"] We proved that.

An ideal — and this is an extremely important concept — [writes "Ideal $I \subset R$ satisfies"] an ideal I of a ring satisfies the following. We defined it last time; I'll just repeat it. It satisfies one condition on the additive structure: if I take $(I, +, 0)$ inside $(R, +, 0)$, this is a subgroup: [writes: $(I, +, 0) \subset (R, +, 0)$ subgroup]. That's the additive structure.

We wrote the axioms differently last time, but we interpret it now. For an ideal, there are kind of two pieces: one is how it works additively, and it must be a subgroup. That is not a surprise, because we already said that f on the additive structures is a group homomorphism, and we already know that the kernel from the point of view of the group homomorphism is an additive subgroup, so, it should be an additive subgroup. This is logically the same as what I explained last week.

And the second condition has to do with the multiplicative part. You can see that it's going to be different now, because we have these constraints here. The kernel satisfies this property: if I take any element of the ring and any element of the ideal, then $r \cdot x$ is in the ideal: [writes: $\forall r \in R, x \in I \implies rx \in I$].

These are the two conditions that define an ideal. And the proposition last week was: the kernel is an ideal. All right? That's just review.

Definition of an Ideal

Recap: Ideals were defined in Week 1 (Definition 1.29), and the kernel of a ring homomorphism was shown to be one (Proposition 1.30). The lecturer restates both; with the group notions of this lecture, the additive half of the proof becomes a one-liner.

Definition 2.24 (*Ideal of a Ring*): Let $(R, +, \cdot, 0, 1)$ be a commutative ring. A subset $I \subseteq R$ is called an *"ideal"* if it satisfies:

(1) Additive Subgroup: $(I, +, 0) \leq (R, +, 0)$, meaning:

$$0 \in I, \quad \forall x, y \in I\colon x + y \in I, \quad \text{and} \quad \forall x \in I\colon -x \in I.$$

(2) Absorption of Multiplication:

$$\forall r \in R,\ \forall x \in I\colon r \cdot x \in I. \tag{2.7}$$

Proposition 2.25 (*The Kernel of a Ring Homomorphism is an Ideal*): Let $f\colon R \to \hat{R}$ be a ring homomorphism. Then:

$$\ker(f) := \{x \in R \mid f(x) = \hat{0}\}$$

is an ideal of R.

Proof

Since f is a group homomorphism between the additive groups $(R, +, 0)$ and $(\hat{R}, +, \hat{0})$, by Proposition 2.20, $(\ker(f), +, 0)$ is an additive subgroup.

To verify multiplicative absorption, let $r \in R$ and $x \in \ker(f)$ (so $f(x) = \hat{0}$). By the multiplicativity of f:

$$f(r \cdot x) = f(r) \cdot f(x) = f(r) \cdot \hat{0} = \hat{0}.$$

Hence, $r \cdot x \in \ker(f)$. Thus, $\ker(f)$ is an ideal of R. **Q.E.D.**

Spoken Transcript [00:35:06 - 00:36:04] Prof. Rahul Pandharipande

What about the image? Is the image an ideal? What is the image? The answer is: that's not true.

The ideal is the notion for the kernel. The image is something else: the image is a subring.

Definition of a Subring

Spoken Transcript [00:36:04 - 00:38:17] Prof. Rahul Pandharipande

So, what is a subring?

Let R be a ring: [writes "Let R be a ring $(R,+,\cdot,0,1)$"]. Whenever I say *"let R be a ring"*, that means I have in mind all the data. In this class, when I say *"let R be a ring"*, it means I have in mind the whole data — you can ask me about it, or maybe we share the idea of which ring it is in common — but when you say *"let R be a ring"*, it means the whole thing.

A subring S of R satisfies the following properties — this is the definition: [writes "A subring $S \subset R$ satisfies:"]. And it shouldn't be surprising to you. The first property is that 0 and 1 should be in the subring: [writes "(i) $0,1 \in S$"]. Just like in a group, you should start with the neutral elements.

The second is that S is closed under the two operations: [writes "(ii) S is closed under $+,\cdot$"]. That means whenever I have elements of the subring, I can either add them or multiply them, and they stay in the subring, okay?

And the third is that $(S,+,\cdot,0,1)$ is a ring: [writes "(iii) $(S,+,\cdot,0,1)$ is a ring"]. Now it makes sense to ask this thing to be a ring.

You can say it in different ways, but that's what a subring is. A subring inside here has 0 and 1, it's closed under the operations, and together it's a ring. This means it has additive inverses and things like that, okay?

And so, the proposition here is: [writes "Prop:"] if I have a ring homomorphism $f\colon R \to \hat{R}$, [writes: $f\colon R \to \hat{R}$ is ring hom.] then the image of f in the second one is a subring: [writes: $\mathrm{Im}(f) \subset \hat{R}$ is a subring].

As I said, the kernel is an ideal, and the image is a subring.

Definition of a Subring and Image Property

Definition 2.26 (*Subring*): Let $(R,+,\cdot,0,1)$ be a ring. A subset $S \subset R$ is called a *"subring"* of R if:

- **(i) Neutral Elements:** $0 \in S$ and $1 \in S$.
- **(ii) Closure under Operations:** For all $x,y \in S$, $x+y \in S$ and $x \cdot y \in S$.
- **(iii) Ring Structure:** $(S,+,\cdot,0,1)$ forms a ring with the operations restricted from R (in particular, S contains additive inverses: $\forall x \in S\colon -x \in S$).

Proposition 2.27 (*Image as a Subring*): Let $f\colon R \to \hat{R}$ be a ring homomorphism. Then the image $\mathrm{Im}(f) \subset \hat{R}$ is a subring of $\hat{R}$.

Verification of the Subring Property for the Image

Spoken Transcript [00:38:20 - 00:39:53] Prof. Rahul Pandharipande

Formally, we should prove this. But this is again a strong proof: you get on this road and there are just no turns, you just have to go on it. So, I'm not really going to give you this proof; you should think about it. But you have to check... [writes "Check:"]

Let's just start — I'm definitely not going to finish it. You just have to check that the image satisfies these properties. Does the image have 0? The answer is yes, because on the additive structure it's a homomorphism of groups, and 0 goes to 0. So, the image has 0; that's correct: [writes: $\mathrm{Im}(f) \ni \hat{0}$ (grp hom)].

Does the image have 1? [writes: $\mathrm{Im}(f) \ni \hat{1}$] Well, that's part of the definition of a homomorphism, so, it has 1. [writes: ✓]

That's kind of a start. Then you have to check whether it's closed under plus and dot, and you would just check that by writing out the equations if you have two elements here. I leave this for you. And whether this is a ring: then you have to check whether it has additive inverses, but that's also clear. So, I leave this as an exercise: [writes "left as an exercise"].

Formally, you just have to take this, start checking everything, and use the definitions. That's the only thing you could do.

Verification of the Subring Axioms for the Image

Why the Image Is a Subring (beyond the lecture)

The lecture checks $\hat{0}$ and $\hat{1}$ and leaves the rest to the students. We verify conditions (i)–(iii) of Definition 2.26 for $\mathrm{Im}(f) \subset \hat{R}$:

(i) Neutral Elements:

- Since $f\colon (R,+,0) \to (\hat{R},+,\hat{0})$ is a group homomorphism, $f(0) = \hat{0}$, so $\hat{0} \in \mathrm{Im}(f)$.
- By condition (iii) of Definition 1.18, $f(1) = \hat{1}$, so $\hat{1} \in \mathrm{Im}(f)$.

(ii) Closure under Addition and Multiplication: Let $\hat{x}, \hat{y} \in \mathrm{Im}(f)$ with $\hat{x} = f(x)$ and $\hat{y} = f(y)$ for some $x, y \in R$. Then:

$$\hat{x} + \hat{y} = f(x) + f(y) = f(x+y) \in \mathrm{Im}(f),$$
$$\hat{x} \cdot \hat{y} = f(x) \cdot f(y) = f(x \cdot y) \in \mathrm{Im}(f).$$

(iii) Additive Inverses: For $\hat{x} = f(x) \in \mathrm{Im}(f)$, we have $-\hat{x} = -f(x) = f(-x) \in \mathrm{Im}(f)$; here $f(-x) = -f(x)$ because group homomorphisms preserve inverses (shown in the proof of Proposition 2.21). Associativity, commutativity and distributivity hold identically as they are inherited from $\hat{R}$.

Therefore, $\mathrm{Im}(f)$ is a subring of $\hat{R}$. **Q.E.D.**

***Insight:* Broken Symmetry: Subrings vs. Ideals**

In group theory and linear algebra, substructures are uniform: the kernel and image of a homomorphism are both subgroups (or vector subspaces). In ring theory, this symmetry is broken:

- The **kernel** $\ker(f)$ is an **ideal** (it absorbs multiplication from the entire ring, but does not contain 1 unless $\hat{R} = \{0\}$).
- The **image** $\operatorname{Im}(f)$ is a **subring** (it contains the identity 1, but in general does not absorb multiplication from outside the image).

Examples of Subrings and Ideals

Even Integers in the Ring of Integers

Spoken Transcript [00:39:53 - 00:40:57] Prof. Rahul Pandharipande

It's good to have these ideas extremely clear in your head: you have the group, you have the ring, there's a notion of group hom, there's a notion of ring hom, and there's kernel and image in both cases. That's four different cases to keep in your head. For the group, the kernel and image are subgroups. For the ring, something different happens: the image is a subring, but the kernel is an ideal.

Now we should do some examples. Let's try here: [writes: $2\mathbb{Z} \subset \mathbb{Z}$].

$2\mathbb{Z}$ inside $\mathbb{Z}$. $\mathbb{Z}$ is a ring: [writes "$\mathbb{Z}$ ring"]. I can take the even integers: [writes "$2\mathbb{Z}$ even integers"]. Even integers — that's kind of standard notation for the even integers.

What is this inside that? Is it an ideal? Is it a subring? What's the answer? Yeah?

Student Answer

I think it is an ideal.

Spoken Transcript [00:40:59 - 00:41:32] Prof. Rahul Pandharipande

It's an ideal, so, claim: this is an ideal: [writes "claim: $2\mathbb{Z}$ ideal"]. Why is it an ideal? We have to check that additively it's a subgroup, and that's just saying that if you add two even numbers, you get an even number; zero is an even number; and if you have an even number, the negative of it is an even number. So, you're totally solid on the subgroup.

Then you have to check that if you take any other element of the integers and multiply by an even number, you stay even. That looks pretty good. The answer is yes: [writes "YES"].

What about a subring? Is it a subring? Yeah?

Student Answer

No, because we don't have the 1.

Spoken Transcript [00:41:37 - 00:41:44] Prof. Rahul Pandharipande

Yeah, so, it's not a subring. Ideal: yes; subring: no: [writes "subring no"].

The Even Integers: Ideal vs. Subring

Example 2.28 (*The Even Integers $2\mathbb{Z} \subset \mathbb{Z}$*): Consider the set of even integers $2\mathbb{Z} := \{2k \mid k \in \mathbb{Z}\}$ as a subset of the ring $(\mathbb{Z}, +, \cdot, 0, 1)$.

- **$2\mathbb{Z}$ is an ideal of $\mathbb{Z}$ (YES):**
  - **(a)** Additively, $2\mathbb{Z} \leq \mathbb{Z}$ since $0 \in 2\mathbb{Z}$, the sum of two evens is even ($2k + 2\ell = 2(k+\ell)$), and the negative of an even is even ($-(2k) = 2(-k)$).
  - **(b)** Multiplicatively, $2\mathbb{Z}$ absorbs products from all of $\mathbb{Z}$: for any $r \in \mathbb{Z}$ and $2k \in 2\mathbb{Z}$, we have $r \cdot (2k) = 2(rk) \in 2\mathbb{Z}$.
- **$2\mathbb{Z}$ is not a subring of $\mathbb{Z}$ (NO):** By Definition 2.26, a subring must contain the multiplicative identity of the ambient ring ($1 \in \mathbb{Z}$). Since 1 is odd, $1 \notin 2\mathbb{Z}$.

Intermediate Subrings between $\mathbb{Z}$ and $\mathbb{Q}$

Spoken Transcript [00:41:44 - 00:42:37] Prof. Rahul Pandharipande

Let's try something else: [writes: $\mathbb{Z} \subset \mathbb{Q}$].

This is maybe a trickier question. Certainly $\mathbb{Z}$ is a subring of $\mathbb{Q}$: [writes: $\mathbb{Z} \subset S \subset \mathbb{Q}$]. I hope that's not controversial. You accept that $\mathbb{Z}$ is a subring of $\mathbb{Q}$?

Can you find a subring that's in between $\mathbb{Z}$ and $\mathbb{Q}$? Does there exist a subring in between? [writes "Does there exist a subring in between?"]

That means that these are strictly not equal: [writes: $\mathbb{Z} \subsetneq S \subsetneq \mathbb{Q}$].

How about that? Can you do that? You have three minutes.

Student Proposal

Look at pairs of integers, and the denominator is just powers of two.

Spoken Transcript [00:42:43 - 00:44:07] Prof. Rahul Pandharipande

Okay, that's good. He is saying that we let S be the set of rational numbers of this form: [writes: $S = \{\frac{a}{2^n} \mid a \in \mathbb{Z}, n \geq 0\}$] where a is an integer and n is also an integer. Is this exactly what you're saying?

That's a proposal. One thing that's good about this proposal is that this set S is certainly more than $\mathbb{Z}$ because it has $3/2$ in it, and it's certainly less than $\mathbb{Q}$ because it doesn't have $2/3$ in it. So, that part is solid.

Then the claim is that you have to check that this satisfies all these properties. What are the properties? Having 0 and 1 — basically the answer is yes, it's going to satisfy them, because you need to know how to add fractions, which I don't teach you. I've never taught anyone how to add fractions; I'm not going to teach you! But when you think about the rule for adding fractions, you keep 2^n in the bottom; that's the main thing here. When you add fractions, the denominator doesn't give you things that you didn't put in. And so, the answer is yes, this is true: [writes "YES"].

That's an example, and that's a very important one. Of course, 2 is not important here: you could have put 17 here, and it would be fine: [writes: 17^n]. That's another example. And in fact, being prime is not that important either.

A Subring between $\mathbb{Z}$ and $\mathbb{Q}$

Example 2.29 (*Denominators that Are Powers of 2*): There exists a strict intermediate subring $\mathbb{Z} \subsetneq S \subsetneq \mathbb{Q}$. Consider the set of rational numbers whose denominator is a power of 2:

$$S := \left\{ \frac{a}{2^n} \;\middle|\; a \in \mathbb{Z},\, n \in \mathbb{Z}_{\geq 0} \right\} = \mathbb{Z}\left[\frac{1}{2}\right]. \tag{2.8}$$

- **Proper Inclusions:**
  - For $n = 0$, $\frac{a}{2^0} = a \in S$, so $\mathbb{Z} \subseteq S$. Moreover, $\frac{1}{2} = \frac{1}{2^1} \in S \setminus \mathbb{Z}$, so $\mathbb{Z} \subsetneq S$.
  - Clearly $S \subset \mathbb{Q}$. Moreover, $\frac{1}{3} \in \mathbb{Q} \setminus S$ (since $3 \nmid 2^n$ for any $n \geq 0$), so $S \subsetneq \mathbb{Q}$.
- **Subring Axioms:**
  - **(i)** $0 = \frac{0}{2^0} \in S$ and $1 = \frac{1}{2^0} \in S$.
  - **(ii)** Closed under addition: For $\frac{a}{2^n}, \frac{b}{2^m} \in S$ with $n \leq m$:
  $$\frac{a}{2^n} + \frac{b}{2^m} = \frac{a \cdot 2^{m-n} + b}{2^m} \in S.$$
  - **(iii)** Closed under multiplication:
  $$\frac{a}{2^n} \cdot \frac{b}{2^m} = \frac{a \cdot b}{2^{n+m}} \in S.$$
  - **(iv)** Closed under additive inverses: $-\left(\frac{a}{2^n}\right) = \frac{-a}{2^n} \in S$.

Generalization: Other Denominators: As observed by the lecturer, the prime 2 can be replaced by any integer $k \geq 2$ (e.g., 17):

$$\mathbb{Z}\left[\frac{1}{k}\right] := \left\{ \frac{a}{k^n} \;\middle|\; a \in \mathbb{Z},\, n \geq 0 \right\}$$

forms a valid intermediate subring between $\mathbb{Z}$ and $\mathbb{Q}$. Going beyond the lecture: the numbers with denominator a power of 2 are called **dyadic rationals**, and this construction is the simplest instance of the **localization** of a commutative ring at a multiplicative set, here $\{1, k, k^2, \ldots\}$: the elements of that set are made invertible. The case $k = 3$ returns in Week 3 (Example 3.29).

Polynomial Rings: $\mathbb{Z}[x^2] \subset \mathbb{Z}[x]$

Spoken Transcript [00:44:07 - 00:45:04] Prof. Rahul Pandharipande

Another example — I should have maybe not... Do we get a bell? I don't remember. Do I get to go to the bell, or do I have to stop?

Let's try one more example before the bell, hopefully. If I look at polynomials here, these are polynomials in the variable x — that's a ring: [writes: $\mathbb{Z}[x]$]. Inside it are polynomials in the variable x^2: [writes: $\mathbb{Z}[x^2] \subset \mathbb{Z}[x]$].

Every polynomial in the variable x^2 is a polynomial in x, so, this is a subset. I hope that's clear. Is this an ideal, or a subring, or both, or neither, or what?

You ran out of time. Okay, we take a break; you can think about it.

Polynomial Ring Example Question

Question 2.30: Let $\mathbb{Z}[x]$ denote the ring of polynomials with integer coefficients in the indeterminate x. Consider the subset of polynomials in x^2:

$$\mathbb{Z}[x^2] := \left\{ a_0 + a_1x^2 + a_2x^4 + \cdots + a_mx^{2m} \,\middle|\, m \geq 0,\ a_i \in \mathbb{Z} \right\} \subset \mathbb{Z}[x].$$

Is $\mathbb{Z}[x^2]$ an ideal, a subring, both, or neither in $\mathbb{Z}[x]$?

Mid-Lecture Break

The lecturer announces a short pause for the students to consider whether $\mathbb{Z}[x^2] \subset \mathbb{Z}[x]$ is an ideal or a subring. The lecture resumes at timestamp 00:45:05.

Spoken Transcript [00:45:05 - 00:45:20]

Prof. Rahul Pandharipande

Okay, so, maybe we start. Or is it a minute? I don't know whether we get a bell; I don't remember, but anyway. What's the answer to this question?

Student Answer

I think it's a subring, but not an ideal.

Spoken Transcript [00:45:24 - 00:45:44]

Prof. Rahul Pandharipande

Yeah, so, subring: yes; ideal: no: [writes "Subring YES", "Ideal no"].

For the subring, again, you just have to check that everything's there, and if you look at it, well... Oh, we do get a bell! [bell rings in background]

Why is it not an ideal? Yeah?

Student Answer

Take x^2, and multiply by x.

Spoken Transcript [00:45:48 - 00:46:21]

Prof. Rahul Pandharipande

x^2 is in this, [writes: $x^2 \in \mathbb{Z}[x^2]$] and if it were an ideal, then I should be able to take any r in the ring and multiply it by this x^2. So, if I take any r in the ring and multiply by x^2, it should stay in here if it were an ideal: [writes: $r \in \mathbb{Z}[x] \implies r \cdot x^2 \in \mathbb{Z}[x^2]$].

I could take r to be x, and then this would be odd and wouldn't be in there, okay? You can check for yourself.

These are not supposed to be hard questions; it's just about whether you have absorbed the definitions. Which, as I said, is not an easy task.

Resolution: Subring vs. Ideal for $\mathbb{Z}[x^2]$

Example 2.31 (*Polynomials in x^2: $\mathbb{Z}[x^2] \subset \mathbb{Z}[x]$*)**:**

- **$\mathbb{Z}[x^2]$ is a subring of $\mathbb{Z}[x]$ (YES):**
  - **(a)** Contains $0 = 0 \cdot x^0$ and $1 = 1 \cdot x^0$.
  - **(b)** Closed under addition: The sum of two polynomials with only even powers of x again has only even powers of x.
  - **(c)** Closed under multiplication: The product of two even powers is an even power: $x^{2k} \cdot x^{2\ell} = x^{2(k+\ell)}$.
  - **(d)** Closed under additive inverses: Negating coefficients preserves even powers.
- **$\mathbb{Z}[x^2]$ is not an ideal of $\mathbb{Z}[x]$ (NO):** It fails the absorption property (2.7): for $x^2 \in \mathbb{Z}[x^2]$, an ideal would need $r \in \mathbb{Z}[x] \implies r \cdot x^2 \in \mathbb{Z}[x^2]$. Taking $r = x$:

$$x \cdot x^2 = x^3 \notin \mathbb{Z}[x^2],$$

because x^3 has an odd degree.

III Subrings of the Integers $\mathbb{Z}$

Spoken Transcript [00:46:22 - 00:46:52] Prof. Rahul Pandharipande

Okay, so, let's go to the integers: [writes: $\mathbb{Z}$]. What are the subrings? [writes "What are the subrings?"]

We've seen that the rational numbers have lots of interesting subrings, because we found some: there are the integers, and you could invert 2, invert 3, 17. But what are the subrings of $\mathbb{Z}$? Yeah?

Student Answer

Just $\mathbb{Z}$.

Spoken Transcript [00:46:53 - 00:47:40] Prof. Rahul Pandharipande

Yeah. There are no real interesting subrings: only $\mathbb{Z}$ in $\mathbb{Z}$. [writes "Only $\mathbb{Z} \subset \mathbb{Z}$"] That's the only subring.

The whole ring is always a subring, just by definition. And this is the only one. Why is that? Because the subring would have to contain 1. Once it contains 1, it contains every positive number. But the subring also is a subgroup, so, it has to contain the negative ones, too.

So, there's nothing you can do.

Subrings of $\mathbb{Z}$

Proposition 2.32 (*Uniqueness of the Subring of $\mathbb{Z}$*): The only subring of $(\mathbb{Z}, +, \cdot, 0, 1)$ is $\mathbb{Z}$ itself.

Proof

Let $S \subseteq \mathbb{Z}$ be a subring.

(1) By condition (i) of Definition 2.26, $1 \in S$.

(2) Since S is closed under addition, by induction $n = \underbrace{1 + 1 + \cdots + 1}_{n \text{ times}} \in S$ for all positive integers $n \in \mathbb{N}_{>0}$.

(3) Since $0 \in S$ and S contains additive inverses (condition (iii)), $-n \in S$ for all $n \in \mathbb{N}_{>0}$.

Thus, S contains all integers: $S = \mathbb{Z}$. **Q.E.D.**

Contrast: Ideals vs. Subrings in $\mathbb{Z}$: This result highlights the sharp contrast between subrings and ideals in the ring of integers $\mathbb{Z}$:

- **Subrings of $\mathbb{Z}$:** There is only **one** subring, namely $\mathbb{Z}$ itself (because a subring must contain 1, which generates all of $\mathbb{Z}$ additively).
- **Ideals of $\mathbb{Z}$:** There are **infinitely many** distinct ideals of $\mathbb{Z}$, each of the form $n\mathbb{Z}$ for $n \in \mathbb{N}_{\geq 0}$ (because an ideal does not need to contain 1; see Theorem 2.35 below).

Ideals in the Ring of Integers

Classification of Ideals in $\mathbb{Z}$

Spoken Transcript [00:47:40 - 00:47:46] Prof. Rahul Pandharipande

So, what are the ideals? [writes "What are the ideals of $\mathbb{Z}$?"] What would you guess the answer to this question is? Yeah?

Student Answer

The set of multiples...

Spoken Transcript [00:47:48 - 00:49:02] Prof. Rahul Pandharipande

Yeah, so, you pick some n — [writes: $n \in \mathbb{Z}$] you pick some integer — and you look at the multiples of n: [writes: $n\mathbb{Z} \subset \mathbb{Z}$].

The first comment is: this is certainly an ideal. [points at $n\mathbb{Z} \subset \mathbb{Z}$] That's the standard notation: n times $\mathbb{Z}$ are the multiples. If $n = 2$, it's the even numbers; when $n = 3$, it's the numbers divisible by 3, and so on. So, various comments on this: [writes "Comment 1. Certainly an ideal"]

1. This is certainly an ideal. Just because of the way we checked carefully that 2 — the even ones — were an ideal; it's the same argument, okay? I hope you understand that.

And the second is that it's not sensitive to whether you put $+n$ or $-n$ here, because inside $\mathbb{Z}$ is -1, so, if you put $+7$ or -7, it's the same set. Is that also clear? [writes: • $\pm n$]

So, this doesn't matter. Typically, when you write $n\mathbb{Z}$, [writes: $n \geq 0$] you should take $n \geq 0$. You don't have to, but that's the convention. In the literature, you'll never see minus... It's just a weird thing to do. I guess you could do it, but it's just a weird thing to do; there's no reason.

Ideals Generated by Multiples of Integers

Recap: The set $n\mathbb{Z}$ first appeared in Week 1 as the kernel of the reduction map $\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ (Example 1.27).

Definition 2.33 (*Principal Ideal Generated by an Integer*): For any integer $n \in \mathbb{Z}$, the set of all multiples of n is denoted:

$$n\mathbb{Z} := \{n \cdot k \mid k \in \mathbb{Z}\} \subset \mathbb{Z}. \tag{2.9}$$

Remark 2.34 (*Properties and Conventional Normalization of $n\mathbb{Z}$*):

- **$n\mathbb{Z}$ is an ideal:** For any $n \in \mathbb{Z}$, $n\mathbb{Z}$ is an additive subgroup of $\mathbb{Z}$ and absorbs multiplication: for any $r \in \mathbb{Z}$ and $nk \in n\mathbb{Z}$, $r \cdot (nk) = n(rk) \in n\mathbb{Z}$.
- **Sign Invariance:** Since $-1 \in \mathbb{Z}$ is an invertible unit, $n\mathbb{Z} = (-n)\mathbb{Z}$ for every $n \in \mathbb{Z}$.
- **Non-negative Normalization:** Typically, one writes $n\mathbb{Z}$ with $n \geq 0$.

Historical Context: Ideal Numbers and Hilbert's Epitaph

Spoken Transcript [00:49:02 - 00:49:48] Prof. Rahul Pandharipande

We're going to prove these are all the ideals, but this is an important example because of the word *"ideal"*: Where does it come from? The idea is that it's an idealized number, an idealized integer. For the integers, the ideals are the integers; but when you think about other rings, the ideals are some kind of idea of a number. Maybe this is a slightly old-fashioned point of view, but...

In some sense, the origin of this word *"ideal"* here is *"ideal number"*. This terminology goes back to about 1900, and there were lots of people around, but certainly one person who played a role was the great David Hilbert, who was a mathematician in Göttingen. A piece of math culture: Does anyone know what the epitaph on the grave of David Hilbert is?

Spoken Transcript [00:49:48 - 00:49:52]

Prof. Rahul Pandharipande

Not one person? Yeah?

Student Question

What is an epitaph?

Spoken Transcript [00:49:52 - 00:50:20]

Prof. Rahul Pandharipande

Ah, epitaph! Very good. You have a gravestone, [draws a gravestone shape on the board with name and motto lines] and what's written here: there's the person's name, and then there's like a little motto or slogan. That's the epitaph. What's the word for that in German?

This is a real piece of math culture: What's the epitaph on David Hilbert's grave? These are the kinds of things you learn to be a mathematician, you know; it's part of being in the culture. Yeah, do you know it?

Student Answer

Wir müssen wissen, wir werden wissen.

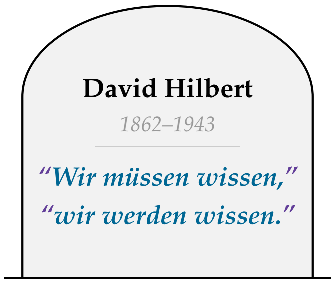
Spoken Transcript [00:50:20 - 00:50:43]

Prof. Rahul Pandharipande

Yeah! *"Wir müssen wissen, wir werden wissen."* Two lines. It's in Göttingen. [erases the gravestone drawing]

I have to teach you some culture, too, you know! Okay, but that's the origin of this name. So, we have to prove something.

Mathematical Culture: David Hilbert's Gravestone



Beyond the Lecture: Where the Word *"Ideal"* Comes From

The lecturer places the term around 1900 and names Hilbert. The history goes further back: Ernst Kummer introduced **ideal numbers** in the 1840s to restore unique prime factorization in cyclotomic integer rings, and Richard Dedekind formalized them as ideals. For the integers $\mathbb{Z}$, every ideal is simply the set of multiples $n\mathbb{Z}$ of an actual integer.

In more general rings where unique factorization fails, ideals serve as the generalized, *"idealized"* numbers that recover unique prime decomposition.

Proof: Every Ideal of $\mathbb{Z}$ is Principal

Classification Theorem for Ideals in $\mathbb{Z}$

Theorem 2.35 (*Ideals of $\mathbb{Z}$ are Principal*): Every ideal $I \subset \mathbb{Z}$ is of the form:

$$I = n\mathbb{Z}$$

for a unique integer $n \geq 0$.

Proof of Theorem 2.35

Spoken Transcript [00:50:43 - 00:52:00] Prof. Rahul Pandharipande

[writes "Proof:"] What do we have to prove? Let I be an ideal in $\mathbb{Z}$: [writes "Let $I \subset \mathbb{Z}$ be an ideal."].

I want to prove it's this. What am I going to do here? Maybe you prove it.

I don't like proofs that much, you know. Actually, about math, what I like is truth: proof is good, but truth is better. And this stuff is true, so, you know... But anyway, we should prove it; it's a math course.

If you want to prove it, you have to think: well, okay, Case 1... [writes "Case (i) $I = (0) \implies 0\mathbb{Z}$"]

Case 1 is that I could be just the zero set. That's fine. And this means it's 0 times $\mathbb{Z}$. It's kind of a degenerate case, but that's possible. And that's an ideal: zero is an ideal.

Case 2 is that I has some non-zero element. [writes "Case (ii) I has some non-zero element"] It has to have zero, because I is an additive subgroup. If it has only zero, you're in this case. If you're not in this case, you have to have some non-zero element. Let's call that element x: [writes: $x \in I$, $x \in \mathbb{Z}$].

Case Analysis: Zero vs. Non-Zero Ideal

Case (i) ($I = (0)$): If I contains only 0, then $I = (0) \implies I = 0\mathbb{Z}$, which corresponds to $n = 0$.

Case (ii) (I has some non-zero element): There exists $x \in I$ with $x \neq 0$.

Spoken Transcript [00:52:00 - 00:53:05] Prof. Rahul Pandharipande

Now, x is an integer. It has a non-zero element, so, x is an integer; then we can multiply by ±1 and get a positive integer: [writes: $\to \pm 1$, get $\pm x \in \mathbb{Z}_{>0}$]. One of these is a positive integer.

So, if it has a non-zero element, then I has a positive integer: [writes "I has positive integer"]. I hope that's clear.

Existence of Positive Elements in I

Since $(I,+,0)$ is an additive subgroup of $\mathbb{Z}$, for any $x \in I$, its additive inverse $-x = (-1)\cdot x$ is also in I. Since $x \neq 0$, exactly one of x or $-x$ is strictly positive:

$$x \in I \setminus \{0\} \implies |x| \in I \cap \mathbb{Z}_{>0}.$$

Therefore, the subset of positive integers in I is non-empty:

$$I \cap \mathbb{Z}_{>0} \neq \emptyset.$$

Spoken Transcript [00:53:05 - 00:53:19] Prof. Rahul Pandharipande

And if it has a positive integer, what am I going to say now? This is where you pretend to be like an LLM: What am I going to say?

Yeah?

Student Answer

It has a minimum.

Spoken Transcript [00:53:19 - 00:53:46] Prof. Rahul Pandharipande

Yeah, it has a least positive integer. That's the next line: [moves to the lower left board and writes: I has a least positive integer]. As I said, for basically every property of the integers, at some point you have to use this principle. Once it has a least positive integer, let's call that one n. So, n is equal to the least positive integer of I: [writes: $n =$ least pos. int. of I].

The Least Positive Element of I

Every non-empty subset of positive integers has a least element (Fact 0.11), so the non-empty set $I \cap \mathbb{Z}_{>0}$ contains a unique minimum element:

$$n := \min(I \cap \mathbb{Z}_{>0}). \tag{2.10}$$

Spoken Transcript [00:53:46 - 00:54:50] Prof. Rahul Pandharipande

Now what we have to show is that I is actually $n\mathbb{Z}$. That's what we want to show, all right? [writes "We have now to show $I = n\mathbb{Z}$"]

Suppose not: [writes "Suppose not"]. Then there's some element $y \in I$ that's not a multiple. Well, maybe I don't want to suppose not — you can do it two different ways. You look at y: y is some element of I: [writes: $y \in I$].

There's only one thing you can do now, which is the division algorithm. The division algorithm says y is equal to some multiple of n plus a remainder term, right? [writes: $y = qn + r$ by division] That's long division.

And the important part of this remainder term is that $0 \leq r < n$: [writes "$0 < r < n$", a slip for $0 \leq r < n$].

Division Algorithm Setup

We show $I = n\mathbb{Z}$:

- **Inclusion $n\mathbb{Z} \subseteq I$:** Since $n \in I$ and I is an ideal, by multiplicative absorption $\forall k \in \mathbb{Z}: kn \in I$, so $n\mathbb{Z} \subseteq I$.
- **Inclusion $I \subseteq n\mathbb{Z}$:** Let $y \in I$ be arbitrary. By the *"division algorithm"* (Fact 0.14), there exist unique quotient and remainder integers $q, r \in \mathbb{Z}$ such that:

$$y = q \cdot n + r \quad \text{with} \quad 0 \leq r < n. \tag{2.11}$$

Spoken Transcript [00:54:50 - 00:56:10] Prof. Rahul Pandharipande

Consider this element $y \in I$. Suppose it's not in $n\mathbb{Z}$: that means that it's not divisible, so, there is a remainder term. But this remainder term is equal to $y - qn$, and so, the remainder term is also in I. Because y is in I, and n is in I, so, qn is in I; and since I is an additive subgroup, this is also in I: [writes: $r = y - qn \in I$].

It's not here, so, we can't have zero, because otherwise it would be divisible, right? So, this r is strictly between zero and n. That's the logic: we say *"suppose not"*, and we use the division algorithm. It can't be zero, because otherwise it would divide, so, long division says r is between zero and n. But then r is a lesser positive integer in I, which is a contradiction: [writes "but r is a less pos. int. in I (contradiction)"].

I don't know what the symbol for contradiction is. Anyway. Okay, that's the proof.

Spoken Transcript [00:56:10 - 00:56:31] Prof. Rahul Pandharipande

We used what you have to use for integers: you use this fact that non-empty subsets of positive integers have a least element. Once you have that least element, you know what you're trying to prove, and then you use the long division algorithm and the remainder. This is what you always do with these kinds of elementary properties of integers.

That proves this claim. [draws a separating line on the board]

Consolidated Proof: Ideals of $\mathbb{Z}$

Complete Derivation of Theorem 2.35

Let $I \subset \mathbb{Z}$ be an ideal.

Case 1: $I = \{0\}$: Then $I = 0\mathbb{Z}$, and $n = 0$ is unique.

Case 2: $I \neq \{0\}$: Choose any non-zero $x \in I$. Since I is an additive subgroup, $-x = (-1) \cdot x \in I$. Thus, $|x| \in I \cap \mathbb{Z}_{>0} \neq \varnothing$.

The non-empty set of positive elements in I has a least element:

$$n := \min(I \cap \mathbb{Z}_{>0}) > 0.$$

We show that $I = n\mathbb{Z}$:

- Since $n \in I$ and I is an ideal, $k \cdot n \in I$ for all $k \in \mathbb{Z}$, whence $n\mathbb{Z} \subseteq I$.

- Conversely, let $y \in I$ be arbitrary. Applying the division algorithm to divide y by n:
$$y = q \cdot n + r \quad \text{with} \quad 0 \leq r < n.$$
Rearranging for the remainder yields:
$$r = y - q \cdot n.$$
Since $y \in I$ and $n \in I$ (which implies $q \cdot n \in I$), closure under subtraction implies $r \in I$.
If $r > 0$, then $r \in I \cap \mathbb{Z}_{>0}$ is a strictly positive element of I satisfying $r < n$, directly contradicting the minimality of $n = \min(I \cap \mathbb{Z}_{>0})$.
Therefore, we must have $r = 0$, which gives $y = q \cdot n \in n\mathbb{Z}$.

Hence, $I = n\mathbb{Z}$. Uniqueness follows because distinct non-negative integers generate distinct sets of positive multiples ($n = \min(n\mathbb{Z} \cap \mathbb{Z}_{>0})$ for $n > 0$).

Q.E.D.

Q.E.D.

Trivial Ideals and Comparison with Fields

Spoken Transcript [00:56:31 - 00:57:05] Prof. Rahul Pandharipande

That's the most basic property of ideals in the integers. For any ring R, we've seen before that we have: zero is an ideal, and the whole ring is an ideal: [writes: $R \to (0)$ is ideal, R is ideal].

We haven't discussed fields at all today, but we discussed last time that if you have a field, these are the only two ideals (Recall: Theorem 1.31). But the integers have a lot of ideals: all these $n\mathbb{Z}$'s are ideals. These are all ideals.

Trivial Ideals in Rings and Fields

Recap: That a field has only these two ideals was shown in Week 1 (Theorem 1.31).

Proposition 2.36 (*Trivial Ideals*): For any ring $(R, +, \cdot, 0, 1)$, the zero ideal $\{0\} = (0)$ and the entire ring $R = (1)$ are always ideals (called *"trivial ideals"*). Here $(a) := aR = \{ra \mid r \in R\}$ denotes the ideal generated by a, defined in Week 4 (Definition 4.32).

Proposition 2.37 (*Ideals in a Field*): If F is a field, its only ideals are (0) and F.

Proof

The lecturer recalls this from the previous lecture, where the argument is given (Theorem 1.31).

Q.E.D.

Beyond the Lecture: PID vs. Field Structure

An integral domain (Definition 2.62) in which every ideal is generated by a single element (i.e., $I = aR$) is called a **principal ideal domain** (**PID**), a name the lecture introduces later. By Theorem 2.35, $\mathbb{Z}$ is a PID, but not a field, because it contains infinitely many non-trivial proper ideals $n\mathbb{Z}$ for $n \geq 2$. In contrast, in a field F, every non-zero element is invertible, instantly forcing any non-zero ideal to absorb 1 and expand to the entire ring.

Ideals in Rings of Smooth Functions

The Ring $C^\infty(\mathbb{R})$ and Compactly Supported Functions

Spoken Transcript [00:57:05 - 00:58:27] Prof. Rahul Pandharipande

Let's try another example. As I said, this course has several purposes, but one of them is to give you some way of looking at objects you know in mathematics and thinking about them in terms of algebra. An object you certainly know in mathematics, if I take the real numbers — I think there's a whole class on real numbers — is that you can take C^∞ functions: [writes: $\mathbb{R} \to C^\infty(\mathbb{R})$].

Are you happy with this? You don't have to take C^∞; you could take 17 times differentiable functions, but we'll just go to infinity. These are infinitely differentiable functions: [writes "$C^\infty(\mathbb{R})$ ∞ diff. functions"].

Are you happy with this terminology? Is it what you use? It's standard. An element here is some function $f\colon \mathbb{R} \to \mathbb{R}$, which I can differentiate as many times as I want, so, it has a nice graph. I can graph it like this: [draws a wave curve]. There it is.

And this beautiful object is a ring, [writes: $(C^\infty(\mathbb{R}), +, \cdot, 0, 1)$] because why? You can add functions, you can multiply functions, zero is just the constant function zero, and one is the constant function one. Is that clear? It's a ring: [writes "Ring"].

Yeah?

Student Question

What is written after $C^\infty(\mathbb{R})$?

Spoken Transcript [00:58:27 - 00:58:45] Prof. Rahul Pandharipande

Here? Which board?

Here? Infinitely differentiable functions.

Yeah, sorry, good. My handwriting's not good: functions. And it's not because I have this wrist injury — it's not because of that! I just have bad handwriting.

Spoken Transcript [00:58:45 - 00:59:33] Prof. Rahul Pandharipande

And I don't even know how I got this wrist injury. It's not like I'm an MMA fighter or anything like that. It just started hurting at one point.

I thought it was going to be a career-ending injury, but it turns out that if I wear this thing, it's fine.

The Ring of Smooth Functions

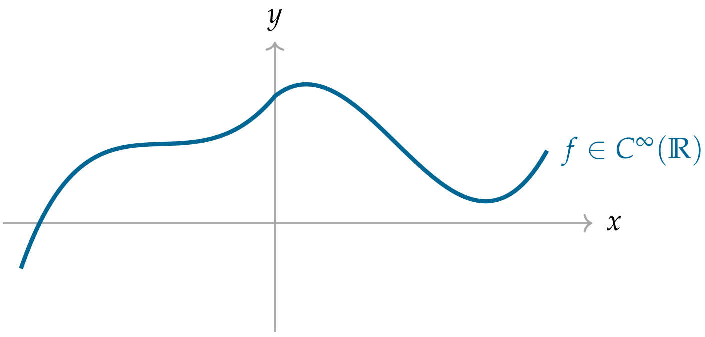
Definition 2.38 (*Ring of Smooth Functions $C^\infty(\mathbb{R})$*)**:** The set of infinitely differentiable real-valued functions on $\mathbb{R}$,

$$C^\infty(\mathbb{R}) := \{f\colon \mathbb{R} \to \mathbb{R} \mid f \text{ is smooth}\},$$

forms a commutative ring $(C^\infty(\mathbb{R}), +, \cdot, 0, 1)$ under pointwise operations:

$$(f+g)(x) := f(x) + g(x),$$
$$(f \cdot g)(x) := f(x) \cdot g(x),$$

with additive identity $0(x) \equiv 0$ and multiplicative identity $1(x) \equiv 1$. Sums and products of smooth functions are smooth (Fact 0.40), so the operations stay inside $C^\infty(\mathbb{R})$.



The Ideal of Compactly Supported Smooth Functions

Spoken Transcript [00:59:33 - 01:01:30] Prof. Rahul Pandharipande

There are a lot of things to say about this ring. This class is not primarily about such rings of differentiable functions — that's some other class — but I can show you something interesting in it.

There's an interesting ideal inside $C^\infty(\mathbb{R})$: [writes: $I \subset C^\infty(\mathbb{R})$] functions with compact support: [writes "functions with compact support"].

First, you have to understand what this is as a set: it's the set of functions with compact support. A general function can do whatever it wants, totally unconstrained. But a function with compact support means that... Are you happy with this analysis terminology: *"compact support"*? It means that its entire life happens somewhere compact. It has to be zero here, and then suddenly it comes to life, and then it goes to zero. It's like these hat functions that look like that, okay? [draws a bump function graph on the board] That's a function with compact support.

Is this an ideal? You have to ask: Who's in here? To be an ideal, it has to be the case that zero is there; and very fortunately, zero has compact support. What does not have compact support is the constant one function; that does not have compact support. So, whatever it is, it's surely not a subring, because it does not have the constant one function.

Then you have to check that if you add compactly supported things, they have compact support; that's true. And the interesting thing is: if I take a function of compact support and multiply it by any function, it still has compact support. Because once this function is zero, it just cannot be brought back to life, no matter how much you multiply. That's an example.

Compactly Supported Smooth Functions as an Ideal

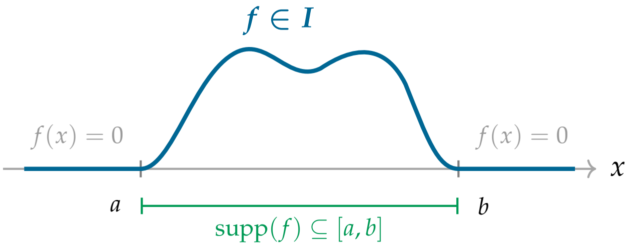
Definition 2.39 (*Compactly Supported Smooth Functions*): The *"support"* of a function $f\colon \mathbb{R} \to \mathbb{R}$ is the topological closure (Definition 0.39) of the set where f does not vanish:

$$\operatorname{supp}(f) := \overline{\{x \in \mathbb{R} \mid f(x) \neq 0\}}.$$

The set of smooth functions with compact support is denoted:

$$I := \{f \in C^\infty(\mathbb{R}) \mid \operatorname{supp}(f) \text{ is compact in } \mathbb{R}\}.$$

By Heine–Borel, $\operatorname{supp}(f)$ is compact if and only if it is bounded (Fact 0.40).



Proposition 2.40 (*I is an Ideal, but not a Subring*): The set $I \subset C^\infty(\mathbb{R})$ of compactly supported smooth functions is an ideal of $C^\infty(\mathbb{R})$, but is not a subring.

Proof

We check the ideal properties:

(1) Additive Subgroup: The zero function has $\operatorname{supp}(0) = \emptyset$, which is compact, so $0 \in I$. For $f, g \in I$, $\operatorname{supp}(f+g) \subseteq \operatorname{supp}(f) \cup \operatorname{supp}(g)$. The union of two compact sets is compact, so $f + g \in I$ and $-f \in I$.

(2) Multiplicative Absorption: Let $g \in C^\infty(\mathbb{R})$ be an **arbitrary** smooth function, and let $f \in I$. Since $(g \cdot f)(x) = 0$ whenever $f(x) = 0$, we have:

$$\operatorname{supp}(g \cdot f) \subseteq \operatorname{supp}(f).$$

As a closed subset of the compact set $\operatorname{supp}(f)$, $\operatorname{supp}(g \cdot f)$ is compact. Thus, $g \cdot f \in I$.

However, the multiplicative identity $1(x) \equiv 1$ has support $\operatorname{supp}(1) = \mathbb{R}$, which is not compact. Hence $1 \notin I$, so I is not a subring. **Q.E.D.**

The lecturer returns to this ideal in Week 5 and shows that it is not a prime ideal (Example 5.28).

Notation: Compactly Supported Functions

Many textbooks write $C_c^\infty(\mathbb{R})$ for this ideal. The lecturer calls it I.

A Strictly Larger Ideal: Functions Vanishing on a Right Half-Line

Spoken Transcript [01:01:30 - 01:03:45] Prof. Rahul Pandharipande

You can ask other questions in analysis about this. I'm not going to talk about that so much, but this is a kind of interesting ideal.

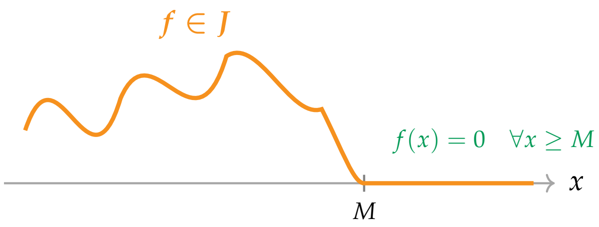
Maybe I'll ask you a question: Is there some ideal J that's bigger? [writes: $I \subsetneq J \neq C^\infty(\mathbb{R})$] Can you find an ideal that's bigger than this, and not the whole ring, of course? Because that's always an ideal that's bigger.

One of the strange things about the way math is taught is that we take all these subjects and divide them. The traditional vision of classical education in mathematics is that there's an algebra direction, an analysis direction, and a geometry direction. But actually, it doesn't really make sense to separate them all: it's all just one way of looking at numbers. I don't know why I said that, but can you think of a way to make this ideal bigger?

We used the fact that it's zero on both sides here, but that's not necessary, right? Can't we just say: functions that eventually go to zero on the right? Those are more, and it satisfies the same thing. So, here you can say J — I'll write it colloquially: functions that become zero for all sufficiently large x: [writes: $J = \{f$ that become 0 for all suff. large $x\}$].

Those are functions that look like... They do whatever they want, and then eventually they become zero: [draws a second graph: function wiggles on the left then becomes identically zero on the right]. These functions are certainly inside there, but there are more of them! As these drawings perfectly prove, there are more of these functions. And these things are also an ideal, because if you multiply by something that's zero here, it's going to stay zero. You might wonder: Are there bigger and bigger ideals? These are kind of interesting questions we're not going to address now, unfortunately.

Construction of a Strictly Larger Proper Ideal



Proposition 2.41 (*A Strictly Larger Proper Ideal*)**:** Let J be the set of smooth functions that vanish identically on some right ray $[M, \infty)$:

$$J := \{f \in C^\infty(\mathbb{R}) \mid \exists M \in \mathbb{R} \text{ such that } f(x) = 0 \text{ for all } x \geq M\}. \tag{2.12}$$

Then J is an ideal of $C^\infty(\mathbb{R})$, and with the ideal I of Definition 2.39:

$$I \subsetneq J \neq C^\infty(\mathbb{R}).$$

Proof

- **J is an ideal:** If $f(x) = 0$ for $x \geq M_1$ and $g(x) = 0$ for $x \geq M_2$, then $(f+g)(x) = 0$ for $x \geq \max(M_1, M_2)$; also $0 \in J$ and $-f = (-1) \cdot f \in J$. Furthermore, for any $h \in C^\infty(\mathbb{R})$, $(h \cdot f)(x) = h(x) \cdot 0 = 0$ for all $x \geq M_1$.
- **Proper inclusions:** If $f \in I$, its support is bounded, so it vanishes for large positive x, whence $I \subseteq J$. A smooth function that is zero for $x \geq 0$ but non-zero on all of $(-\infty, 0)$, such as $x \mapsto e^{1/x}$ for $x < 0$ and 0 for $x \geq 0$ (Fact 0.42), belongs to J but not to I, so $I \subsetneq J$. Finally, the constant function $1 \notin J$, so $J \subsetneq C^\infty(\mathbb{R})$.

Q.E.D.

The Restriction Homomorphism on Function Spaces

Spoken Transcript [01:03:45 - 01:05:45] Prof. Rahul Pandharipande

Maybe I'll say one more thing, just for fun. If I look at C^∞ functions on $\mathbb{R}$, [writes: $C^\infty(\mathbb{R})$] there's a ring homomorphism to C^∞ functions on $\mathbb{R}_{>0}$: [writes: $\to C^\infty(\mathbb{R}_{>0})$].

This is a good example of a homomorphism, and this is called restriction; maybe I'll call it Res: [writes "Res" over the arrow].

If I have a C^∞ function here, it's a function from all of $\mathbb{R}$ to $\mathbb{R}$: [writes: $f\colon \mathbb{R} \to \mathbb{R}$]. And if I have one, I can restrict it: $f|_{\mathbb{R}_{>0}}\colon \mathbb{R}_{>0} \to \mathbb{R}$: [writes: $f_{\mathrm{Res}}\colon \mathbb{R}_{>0} \to \mathbb{R}$].

Is that clear? I can restrict a function. I have a function here; this is a ring. Every element here I can just restrict, and that gives me a map. First, it's a map on sets, and then you have to show it respects zero — of course, the zero function restricts to the zero function; and if I restrict a function that is constant one, it's the function here that is constant one. Also, addition and multiplication behave well under restriction. So, this is a legitimate ring homomorphism: [writes "Ring hom"].

For function spaces, restriction is a very useful way of making ring homomorphisms.

Okay, now your analysis question: Is this surjective? That's properly a question in your analysis class. You can think about that. If you understood all the examples in your analysis class, you should be able to tell me the answer.

Spoken Transcript [01:05:45 - 01:06:13] Prof. Rahul Pandharipande

It's just been a long time since I taught analysis — like 25 years, I think. [begins cleaning the center and left chalkboards] Are you prepared to tell me?

Okay, you can think about it.

The Restriction Homomorphism

Definition 2.42 (*Restriction Homomorphism*): The *"restriction map"*

$$C^\infty(\mathbb{R}) \xrightarrow{\mathrm{Res}} C^\infty(\mathbb{R}_{>0}), \qquad (f\colon \mathbb{R} \to \mathbb{R}) \longmapsto (f_{\mathrm{Res}} := f|_{\mathbb{R}_{>0}}\colon \mathbb{R}_{>0} \to \mathbb{R}),$$

is a ring homomorphism:

(i) $\mathrm{Res}(f+g) = (f+g)|_{\mathbb{R}_{>0}} = f|_{\mathbb{R}_{>0}} + g|_{\mathbb{R}_{>0}} = \mathrm{Res}(f) + \mathrm{Res}(g)$.
(ii) $\mathrm{Res}(f \cdot g) = (f \cdot g)|_{\mathbb{R}_{>0}} = f|_{\mathbb{R}_{>0}} \cdot g|_{\mathbb{R}_{>0}} = \mathrm{Res}(f) \cdot \mathrm{Res}(g)$.
(iii) $\mathrm{Res}(1_{\mathbb{R}}) = 1_{\mathbb{R}_{>0}}$.

Question 2.43: Is the restriction homomorphism $\mathrm{Res}\colon C^\infty(\mathbb{R}) \to C^\infty(\mathbb{R}_{>0})$ surjective? (The lecturer leaves this to the students; the answer follows in Proposition 2.48.)

Introduction to Category Theory

Definition of a Category

Spoken Transcript [01:06:13 - 01:07:09] Prof. Rahul Pandharipande

All right, so, there's a topic that's in some of the reading. There are two foundational topics which I've avoided discussing, but I think I'll discuss now, and it has to do with category theory and Zorn's lemma.

You had this foundations, this Grundstrukturen class. Did you discuss categories there?

You can say no.

Student Answer

No.

Spoken Transcript [01:07:09 - 01:07:57] Prof. Rahul Pandharipande

Okay, so, category. We discuss categories: [writes "Category"].

A category consists of objects and homs [writes: Category consists of Objects and Homs].

Homs are sometimes called morphisms, [writes: $\left\{\begin{array}{l}\text{morphisms}\\ \text{arrows}\end{array}\right.$] so, the language differs. Different authors take a different choice of language: sometimes homs are called morphisms; sometimes they're called arrows. The first thing to know about a category: it has objects and it has homs.

More specifically, given two objects in the category — [writes "Given A, B objects in Category,"] given A and B objects in the category — there is a set of homs, and this set has a name: $\mathrm{Hom}(A, B)$: [writes: $\mathrm{Hom}(A, B)$].

Definition of a Category

Definition 2.44 (*Category*): A *"category"* Cat consists of the following data:

(1) A collection of *"objects"*, denoted $\mathrm{Obj}(\mathrm{Cat})$.
(2) For every pair of objects $A, B \in \mathrm{Obj}(\mathrm{Cat})$, a set of *"morphisms"* (also called *"homs"*

or *"arrows"*), denoted:

$$\mathrm{Hom}(A,B).$$

Terminology Across Literature: In modern mathematical literature:

- **Objects:** The structural spaces (e.g., sets, groups, rings, vector spaces, topological spaces).
- **Homs / Morphisms / Arrows:** The structure-preserving mappings between objects ($f\colon A \to B$).

Notation: Categories

The lecturer writes Cat for any category and $\widehat{\mathrm{Cat}}$ for a second one, and these notes follow him. Textbooks write $\mathcal{C}$ and $\mathcal{D}$, and they keep **Cat** for one particular category, that of all small categories.

The Category of Groups

Spoken Transcript [01:07:57 - 01:09:08] Prof. Rahul Pandharipande

Of course, there are some conditions this has to satisfy, but before I give you the conditions, I will give you the model: for example, the category of groups: [writes "category of Groups"].

The category of groups is a category. This category has objects, [writes "Objects ← are"] and the objects are groups: [writes: $(G,\cdot,e)$].

An object is a group; that means it's a group with an operation and a neutral element. Those are the objects. When you think about the category of groups, somehow you think about this world where all these groups are: they're all over the place, lots of different groups; every possible group is in here. That's in the category.

And what are the arrows going to be? The homs: if I have two particular groups, [writes: $\mathrm{Hom}(G,\hat{G})$] then this is going to be the set of group homomorphisms [writes "= set of group homomorphisms"] from G to $\hat{G}$: [writes "from G to $\hat{G}$"].

The Category of Groups Grp

Example 2.45 (*The Category of Groups*): In the category of groups, standardly denoted **Grp**:

- **Objects:** All groups $(G,\cdot,e)$.
- **Morphisms:** For any two groups $G,\hat{G}$, the set of morphisms from G to $\hat{G}$ is the set of all group homomorphisms:

$$\mathrm{Hom}_{\mathbf{Grp}}(G,\hat{G}) := \{f\colon G\to\hat{G} \mid f \text{ is a group homomorphism}\}. \tag{2.13}$$

Categories of Rings and Vector Spaces

Spoken Transcript [01:09:10 - 01:10:24] Prof. Rahul Pandharipande

Okay, that's the model. Is that clear?

The idea of a category is that we want to think about groups and their relationships abstractly. We can think about the category of groups, we think of all groups, and if I pick two groups in that world of all groups, I get the set of homs, okay?

As I said, there are going to be some conditions. But another example is the category of rings: [writes "Category of Rings"]. That's also fine.

There, the objects are rings, as we've defined in this class, meaning the objects are rings with the full data: [writes: $(R, +, \cdot, 0, 1)$].

And the homs, which will not surprise you, are ring homomorphisms: [writes "are ring homs"].

From last year, you could consider the category of vector spaces over whatever field you want — say, vector spaces over the complex numbers: [writes "Category of Vector Spaces/$\mathbb{C}$"]. In the category of vector spaces over the complex numbers, the objects are vector spaces over the complex numbers, and the homs are linear maps.

Those are all examples of categories.

Categories of Rings and Vector Spaces

Example 2.46 (*The Category of Rings* Ring): In the category of rings, standardly denoted **Ring**:

- **Objects:** Rings $(R, +, \cdot, 0, 1)$, as defined in this class.
- **Morphisms:** Ring homomorphisms $f\colon R \to \hat{R}$ (satisfying additivity, multiplicativity, and $f(1) = \hat{1}$).

Example 2.47 (*The Category of Complex Vector Spaces* $\mathbf{Vect}_{\mathbb{C}}$): In the category of complex vector spaces, denoted $\mathbf{Vect}_{\mathbb{C}}$:

- **Objects:** Vector spaces V over the field $\mathbb{C}$.
- **Morphisms:** $\mathbb{C}$-linear transformations $T\colon V \to W$.

Interlude: Non-Surjectivity of the Restriction Map

Spoken Transcript [01:10:24 - 01:10:54] Prof. Rahul Pandharipande

Then, of course, I have to give you the conditions. To be a category, if I want to tell you what a category is, I have to tell you what the objects are; given two objects, I have to tell you what the homs are; and the resulting structure has to have some properties.

All of these will have that property.

Spoken Transcript [01:10:54 - 01:11:01] Prof. Rahul Pandharipande

[points at the restriction map equation on the right board] Did you figure this out?

Yeah?

Student Answer

I think $\sin(1/x)$...

Spoken Transcript [01:11:01 - 01:11:29] Prof. Rahul Pandharipande

Yeah, so, the answer is no, it's not surjective.

I'll draw the picture. A picture is worth a thousand words!

The picture is: you have an element here that's not there. This is zero, and this is this element — that one: [draws a sketch of $\sin(1/x)$ oscillating wildly as $x \to 0^+$].

You should recognize that from your analysis class, I would think. Maybe not drawn with such enthusiasm!

You know this function, right?

Resolution: The Restriction Map is Not Surjective

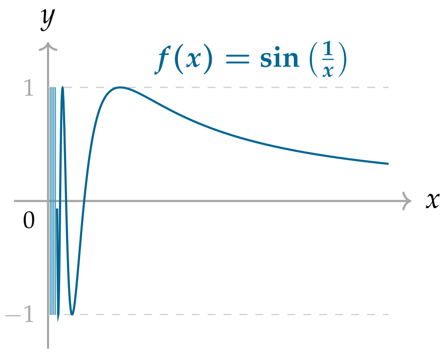
Proposition 2.48 (*Non-Surjectivity of Smooth Restriction*): The restriction homomorphism $\mathrm{Res}\colon C^\infty(\mathbb{R}) \to C^\infty(\mathbb{R}_{>0})$ is not surjective.

Proof

Consider the function $f\colon \mathbb{R}_{>0} \to \mathbb{R}$ defined by:

$$f(x) := \sin\left(\frac{1}{x}\right).$$

By standard calculus rules, f is infinitely differentiable on $(0, \infty)$, so $f \in C^\infty(\mathbb{R}_{>0})$. However, as $x \to 0^+$, $f(x)$ oscillates infinitely often between -1 and 1, so $\lim_{x\to 0^+} f(x)$ does not exist. Consequently, f cannot even be extended continuously to $x = 0$ (Fact 0.41), let alone smoothly to all of $\mathbb{R}$. Thus, $f \notin \mathrm{Im}(\mathrm{Res})$. **Q.E.D.**



Spoken Transcript [01:11:29 - 01:12:07] Prof. Rahul Pandharipande

[erases the graph of $\sin(1/x)$ and thoroughly cleans the lower blackboard to prepare for the category axioms]

Axioms of a Category: Homs, Composition, and Identity

Spoken Transcript [01:12:07 - 01:14:06] Prof. Rahul Pandharipande

Okay, so, the properties that the homs have to satisfy for a category are very simple properties.

The first property, as I said, is that $\mathrm{Hom}(A,B)$ is a set: [writes "• $\mathrm{Hom}(A,B)$ is a set"]. That's almost a ridiculous property, and we're not going to think about it too much, but it's...

The second is that there is composition: [writes "• Composition"].

Composition means that if I have three objects, A, B, and C, [writes: $A \quad B \quad C$] then here I have the set of homs from A to B, [writes: $\mathrm{Hom}(A,B)$] and I also have homs from B to C: [writes: $\times\,\mathrm{Hom}(B,C)$]. This composition means that if I'm giving you the category, I have to tell you the composition rule. Whenever I have this diagram: A, B, and C, there has to be a rule, which is called composition, that tells you how, if you have a hom from A to B and a hom from B to C, this gives me a hom from A to C: [writes: $\to \mathrm{Hom}(A,C)$].

That is part of the data of the category. If I'm telling you a category, it's up to me to tell you how to compose homs. And in all examples, you just compose them, okay?

So, what's the content? For groups, of course, if I have a hom from G to $\hat{G}$ and another one from $\hat{G}$ to G'', you can compose them as functions — you can always compose functions — but then you have to show that the composition is a group hom, all right? Which I leave to you to check.

That's the content of this: to say that this category of groups is a category, there are groups, there are homs, and there's a natural composition of homs; and it's very important that when I have a group hom here and a group hom here, when I compose them, I get a group hom. Okay?

Axioms of a Category: Hom-Sets and Composition

Axiom 1 (Hom-Sets): For any two objects $A,B \in \mathrm{Obj}(\mathrm{Cat})$, the collection $\mathrm{Hom}(A,B)$ is a well-defined set.

Axiom 2 (Composition Law): For any three objects $A,B,C \in \mathrm{Obj}(\mathrm{Cat})$, there is a composition mapping:

$$\begin{aligned} \circ\colon \mathrm{Hom}(A,B)\times\mathrm{Hom}(B,C) &\longrightarrow \mathrm{Hom}(A,C) \\ (f,g) &\longmapsto g\circ f. \end{aligned}$$

Spoken Transcript [01:14:06 - 01:16:30] Prof. Rahul Pandharipande

And there's one more: inside $\mathrm{Hom}(A,A)$ — [writes: $\bullet\,\mathrm{Hom}(A,A) \ni \mathrm{Id}_A$] which is legal to do; I said you pick two objects of the category, and they could be the same object — there is a distinguished element called the identity. That's part of the data, okay?

There's one more condition, which is how the identity relates to composition, which is not going to be a surprise to you.

If we go back to this example of groups: objects are groups, homs are group homs, composition is composition of functions — and we're lucky that composition of functions takes group homs to group homs — and there is an identity: if you take the group and you do nothing, that is definitely a group hom. So, that's here.

And then there's the final property: if you take the identity on A — we take here A, A, and B [writes: $A \quad A \quad B$] — then I can take the identity here, [writes: $\mathrm{Id}_A \in \mathrm{Hom}(A, A)$] and I can take any element f here, [writes: $f \in \mathrm{Hom}(A, B)$] and I can compose it, because composition is given data. I do the identity first and then I do f: $f \circ \mathrm{Id}_A$. [writes: $f \circ \mathrm{Id}_A$] This has to be an element of $\mathrm{Hom}(A, B)$. [writes: $\mathrm{Hom}(A, B)$] And what do you think the axiom is going to say?

What do you think the axiom is going to say here? If I do the identity first, and then I do f, what do you think this should be equal to?

Yeah: f! [writes: $= f$] That's the identity axiom. This is an axiom of the category.

And also the other way: you can also have A, B, and B. [writes: $A \quad B \quad B$] If I put f here and I put Id_B, [writes: $A \xrightarrow{f} B \xrightarrow{\mathrm{Id}_B} B$] then if you do f first and then Id_B, this should just be f, okay? [writes: $\mathrm{Id}_B \circ f = f$] That's actually the full category stuff.

Axioms of a Category: Identity and Unit Laws

Axiom 3 (Identity Morphisms): For every object $A \in \mathrm{Obj}(\mathrm{Cat})$, there exists a distinguished morphism, called the *"identity morphism"*:

$$\mathrm{Id}_A \in \mathrm{Hom}(A, A).$$

Axiom 4 (Identity Laws / Unit Axioms): For every morphism $f \in \mathrm{Hom}(A, B)$:

$$f \circ \mathrm{Id}_A = f \in \mathrm{Hom}(A, B), \tag{2.14}$$

$$\mathrm{Id}_B \circ f = f \in \mathrm{Hom}(A, B). \tag{2.15}$$

Spoken Transcript [01:16:30 - 01:17:26]

Prof. Rahul Pandharipande

The data of this category is supposed to capture what happens when we think about these algebraic objects.

Of course, it can be applied to many other settings. But if you think about groups, what is the abstract structure of the group? We have objects, we have homs, the homs compose, there is an identity hom, and they satisfy these identity properties.

It's the same for rings, and for vector spaces.

And then there's a whole development which most people find kind of dull, but anyway...

Is there a question about what the definition of a category is?

I'll just say it again: a category has objects, it has homs, the homs have to be endowed with a composition rule, there is an identity element, and we have this property. Okay?

Consolidated Definition of a Category

Definition 2.49 (*Formal Category*): A *"category"* Cat consists of:

(1) A collection of objects $\mathrm{Obj}(\mathrm{Cat})$.

(2) A set of morphisms $\mathrm{Hom}(A,B)$ for every pair $A,B \in \mathrm{Obj}(\mathrm{Cat})$.

(3) A composition operation:

$$\circ\colon \mathrm{Hom}(A,B) \times \mathrm{Hom}(B,C) \longrightarrow \mathrm{Hom}(A,C), \quad (f,g) \longmapsto g \circ f,$$

(4) For each object $A \in \mathrm{Obj}(\mathrm{Cat})$, an identity morphism $\mathrm{Id}_A \in \mathrm{Hom}(A,A)$ satisfying the **unit laws**:

$$f \circ \mathrm{Id}_A = f \quad \text{and} \quad \mathrm{Id}_B \circ f = f \quad \forall f \in \mathrm{Hom}(A,B).$$

$$\begin{tikzcd}
 & C & \\
A \arrow[ur, "g \circ f"] \arrow[rr, "f"'] \arrow[loop left, "\mathrm{Id}_A"] & & B \arrow[ul, "g"'] \arrow[loop right, "\mathrm{Id}_B"]
\end{tikzcd}$$

Beyond the Lecture: Associativity of Composition

The board lists Hom-sets, composition, identity morphisms and the two unit laws, and not associativity: $h \circ (g \circ f) = (h \circ g) \circ f$ for all $f \in \mathrm{Hom}(A,B)$, $g \in \mathrm{Hom}(B,C)$, $h \in \mathrm{Hom}(C,D)$. The lecturer adds this axiom the next day (Definition 2.55). It holds in all of the examples above, since composition of maps is associative.

Categorical Definition of Isomorphism

Spoken Transcript [01:17:26 - 01:19:18]

Prof. Rahul Pandharipande

I just assert that all these things — groups, rings, fields, vector spaces... [writes "Groups, Rings, Fields, Vector spaces..."] and many other structures — give you categories.

If you have a category, built into the category is a notion of isomorphic objects.

Suppose we have a category: Cat is a category [writes: Suppose Cat is a category]. I want to define the notion of isomorphism purely in categorical terms [writes: I want to define the notion of isomorphism purely in categorical terms.].

There is a very straightforward way to do this. That's why I say just "Cat *is a category*" — that means you don't know anything about it. All you know is that there are some objects, they have homs, and they satisfy these principles.

Suppose I have a category, and I take two objects. I want to know: Is there a notion of isomorphism in that category? Which is to say: Is there a purely arrow-theoretic notion of isomorphism? Category-theoretic.

Sorry... [adjusts microphone wire] Oh, here we go.

And the answer is yes, and I will explain this.

Spoken Transcript [01:19:18 - 01:20:08] Prof. Rahul Pandharipande

In the reading section, there are a few pages on the definition of these categories — the reading for this week.

Are the readings for this week already on the website? No? Maybe they are. In any case, as I said, if you ever want to know what's on the website, you can go to 2018; it's all there. You don't have to wait.

Course Literature and Reading Assignments

The lecturer reminds students that reading assignments detailing category theory are provided in the syllabus materials, which are also archived on the 2018 course website.

Spoken Transcript [01:20:08 - 01:21:17] Prof. Rahul Pandharipande

I was in the middle of trying to explain the categorical definition of isomorphism.

I have this category Cat, [writes: Cat] and I have two objects: A and B are in the objects of my Cat: [writes: $A, B \in \mathrm{Obj}(\mathrm{Cat})$]. Definition: [writes "Def:"] if I have $f \in \mathrm{Hom}(A,B)$ — that means if I have an arrow here: [writes "$f \in \mathrm{Hom}(A,B)$", draws $A \xrightarrow{f} B$] — then f is an isomorphism — this is a definition, not a theorem — [writes "then f is an isomorphism"] if... [writes "if"]

Now you're supposed to complete this for me: What would you do here?

What do you normally need for an isomorphism? You need a map the other way. If there exists a g, which is an arrow in the other direction, [writes: $\exists g \in \mathrm{Hom}(B,A)$] — that means there exists a g going in the other direction: [draws $B \xrightarrow{g} A$] — such that what?

Yeah?

Student Answer

The two compositions are the identity.

Spoken Transcript [01:21:21 - 01:21:46] Prof. Rahul Pandharipande

Yeah, the two compositions give the identity. There exists a g in the other direction such that [writes "such that"] if I do g first, then f — then you have to concentrate: this is the identity on B: [writes: $f \circ g = \mathrm{Id}_B$] — and if I do f first, then g, this is the identity on A: [writes: $g \circ f = \mathrm{Id}_A$].

That's the definition! That's a purely category-theoretic definition of isomorphism.

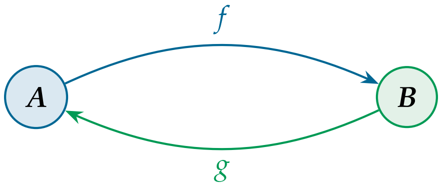
Is that confusing or clear?

Categorical Definition of Isomorphism

Definition 2.50 (*Isomorphism in a Category*): Let Cat be a category, and let $A, B \in \operatorname{Obj}(\mathrm{Cat})$. A morphism $f \in \operatorname{Hom}(A, B)$ (denoted $f\colon A \to B$) is called an *"isomorphism"* if there exists a morphism $g \in \operatorname{Hom}(B, A)$ (denoted $g\colon B \to A$) such that:

$$f \circ g = \mathrm{Id}_B \quad \text{and} \quad g \circ f = \mathrm{Id}_A\,. \tag{2.16}$$

Two objects A and B are called *"isomorphic"* (written $A \cong B$) if there exists an isomorphism between them.



Arrow-Theoretic vs. Element-Theoretic Isomorphism: Going beyond the lecture: in standard concrete algebra, an isomorphism is typically introduced as a **bijective homomorphism**. In abstract category theory, where objects may not have *"elements"* and morphisms are not necessarily *"functions"*, bijectivity cannot even be formulated. Instead, category theory characterizes isomorphisms purely through arrows via the existence of a two-sided inverse morphism g. In concrete categories this recovers the familiar notions:

- In **Grp**, categorical isomorphisms are precisely bijective group homomorphisms (Proposition 2.13).
- In **Ring**, categorical isomorphisms are bijective ring homomorphisms (Definition 2.23).
- In $\mathbf{Vect}_K$, categorical isomorphisms are bijective linear transformations (invertible matrices in finite dimension, Fact 0.31).

Spoken Transcript [01:21:46 - 01:22:27] Prof. Rahul Pandharipande

In particular, once I tell you the categories of groups or rings, I tell you the homs, the identity, and this property, that already tells you what the notion of isomorphism is. I told you anyway what the notion of isomorphism is, but in some sense I didn't have to: it was just this categorical notion of isomorphism.

And there's one more thing that I hope I'll finish before this ends, because this is kind of boring and dry, this category stuff. But it's kind of an organizational way to think about these structures.

There's nothing deep happening here, really nothing. It's just language.

Definition of a Functor

Spoken Transcript [01:22:27 - 01:23:55] Prof. Rahul Pandharipande

There's the notion of a functor: [writes "Functor"].

This is maybe cooler: [writes "Functor"].

A functor is kind of a notion of a map from one category to another. A category is a thing, and you might have two categories, and you want to compare them — just like we have groups, and we might want to compare groups. Functors, in some sense, compare two different categories [writes "Functor Compare categories"] — or it could be the same category: compare categories.

The idea of a functor is: if I have some category here, and I have another category, $\widehat{\text{Cat}}$, [writes: $\text{Cat} \xrightarrow{F} \widehat{\text{Cat}}$] then the functor F is written as a map from one category to the other.

And what's the data of the functor? [writes "Data:"]

If I'm giving you the functor, then it's on me to provide all this data. The data is that for every object A in the first category, [writes "for A object in Cat,"] I have to give you an object in the second category: $F(A) \in \widehat{\text{Cat}}$: [writes: $F(A) \in \widehat{\text{Cat}}$].

That's part of the data of the functor. This is like, you know, if I give you a function, it's my responsibility to tell you where every point goes; that's totally on me if I'm giving you the function. Here, if I'm giving you a functor, I have to tell you, whenever I have an object on one side, how to get the object on the other side. But that's not enough.

Spoken Transcript [01:23:55 - 01:25:22] Prof. Rahul Pandharipande

[cleans a section of the right chalkboard] I have to tell you whenever I have a hom for Cat, the first one: [writes: $\text{Hom}(A, B)$ for Cat]. If I have A and I have an arrow here — [writes: $A \xrightarrow{f} B$] that's on the Cat side — I've already told you what these objects are, [writes: $F(A) \quad F(B)$] but I also have to tell you how this hom goes to a hom between these on the other side: [writes: $F(A) \xrightarrow{F(f)} F(B)$].

That's, again, on me to do that. So, it's a lot of things I have to tell you.

And it can't just be anything: it has to respect all of the structure, which means this rule must take the identity to the identity, and it must respect composition — which I don't write, but it's kind of clear what it is: [writes "must respect identity, composition"].

You can look that up, and I will give you an example. Let me just say it: it must respect identity and composition. You can look in the book, or you can imagine it yourself: there's only one thing you can do here.

That's the notion of a functor.

Definition of a Functor

Definition 2.51 (*Covariant Functor*): Let Cat and $\widehat{\text{Cat}}$ be categories. A *"functor"* (or *"covariant functor"*) $F\colon \text{Cat} \to \widehat{\text{Cat}}$ consists of the following data:

(1) Object Assignment: To every object $A \in \text{Obj}(\text{Cat})$, an object $F(A) \in \text{Obj}(\widehat{\text{Cat}})$.

(2) Morphism Assignment: To every morphism $f \in \mathrm{Hom}(A,B)$ of Cat, a morphism of $\widehat{\mathrm{Cat}}$:

$$F(f) \in \mathrm{Hom}\big(F(A),F(B)\big),$$

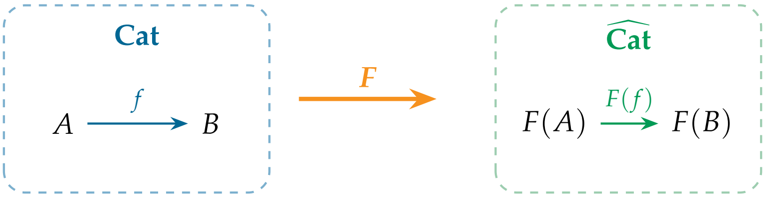
satisfying the following structural compatibility axioms:

(i) Preservation of Identity: For every object $A \in \mathrm{Obj}(\mathrm{Cat})$:

$$F(\mathrm{Id}_A) = \mathrm{Id}_{F(A)}.$$

(ii) Preservation of Composition: For all morphisms $f \in \mathrm{Hom}(A,B)$ and $g \in \mathrm{Hom}(B,C)$:

$$F(g \circ f) = F(g) \circ F(f).$$



Examples of Functors: From Rings to Groups

Spoken Transcript [01:25:22 - 01:27:44] Prof. Rahul Pandharipande

And what's an example of a functor? [erases the middle chalkboard]

There's one that's been staring us in the face the whole time: if I take the category of rings, there's a way to get to the category of groups. This is going to be totally underwhelming, but it's just taking the additive structure of the ring. That's a rule to go from rings to groups, and that is a functor. I'll write it.

You can criticize this whole discussion, this lecture, as being basically nonsensical, and that's basically the correct point of view, I think. But it's kind of useful eventually. Maybe I should say: it's not an incorrect point of view; let's put it like that.

The first example of a functor — this whole direction is not really the point of this class — I take the category of rings: [writes: $\underline{\text{Ring}}$]. Sometimes people take terminology like Ring to denote the category of rings, [underlines $\underline{\text{Ring}}$] and Grp to denote the category of groups: [writes: $\underline{\text{Group}}$]. There's a functor F: [writes: $\underline{\text{Ring}} \xrightarrow{F} \underline{\text{Group}}$]. This functor is defined by: if I take a ring, which is all of this data, $(R,+,\cdot,0,1)$, [writes: $F(R,+,\cdot,0,1)$] I need to make a group out of it. And so, I could just do this: $(R,+,0)$: [writes: $= (R,+,0)$]. Totally underwhelming, but anyway, that's definitely a group.

This functor associates the ring to a group, so, it gives an object here to an object there. And then you have to check that it does all the things that you want. I have to also tell you: if I have a hom on the ring side, how do I get a hom on the group side? Well, the hom on the ring side is just a way to get from elements from one ring to the other; it also gives you an additive homomorphism here.

Again, almost nothing is happening; it's just reinterpretation. But that's an example, kind of a dumb example.

The Functor from Rings to Groups

Example 2.52 (*The Additive Functor F*: $\mathbf{Ring} \to \mathbf{Grp}$): There is a canonical functor:

$$F\colon \mathbf{Ring} \longrightarrow \mathbf{Grp}$$

defined by forgetting the multiplicative structure:

- **On Objects:** For every ring $(R,+,\cdot,0,1) \in \mathrm{Obj}(\mathbf{Ring})$:
$$F(R,+,\cdot,0,1) := (R,+,0) \in \mathrm{Obj}(\mathbf{Grp}).$$
- **On Morphisms:** For any ring homomorphism $f\colon R \to \hat{R}$, $F(f)$ is the same underlying mapping regarded as an additive group homomorphism:
$$F(f) := f\colon (R,+,0) \longrightarrow (\hat{R},+,\hat{0}).$$

Since ring homomorphisms satisfy $f(a+b) = f(a)+f(b)$ and $f(0) = \hat{0}$, $F(f)$ is a group homomorphism. Furthermore, $F(\mathrm{Id}_R) = \mathrm{Id}_{(R,+)}$ and $F(g \circ f) = F(g) \circ F(f)$, verifying all functor axioms.

Forgetful Functors in Algebra: Going beyond the lecture: this construction is the standard prototype of a **forgetful functor**: it strips away (forgets) a layer of algebraic structure (the multiplication $\cdot$ and unit 1), leaving only the underlying additive group $(R,+,0)$.

The Double Dual Functor on Vector Spaces

Spoken Transcript [01:27:44 - 01:30:03] Prof. Rahul Pandharipande

A more interesting one you know — this is actually a much more interesting example that you learned, but probably...

If I take vector spaces... I can do anything I want, but let's just take complex vector spaces, because I like the complex numbers: [writes "VectorSpace/$\mathbb{C}$"]. Is that clear? This is the category of complex vector spaces: the objects are complex vector spaces, and the homs are linear maps. You should be totally happy to think about this category; in some sense, your whole linear algebra class was about this category.

There's a very nice functor here to itself: [writes "VectorSpace/$\mathbb{C}$ $\xrightarrow{F}$ VectorSpace/$\mathbb{C}$"]. There's no reason you can't have a functor from the category to itself.

This functor takes V: [writes: V]. What can you do with V? One thing you can do is take its dual: [writes: V^*]. If you take its dual, it reverses the arrows. Technically, what I've defined is a covariant functor, which means that the composition goes forward. The dual switches the arrows; it's confusing. But something that doesn't switch the arrows is the double dual: [writes: $(V^*)^*$].

That's a really beautiful thing you can do: take a vector space to its double dual. To every vector space, you can associate its double dual. And the claim is that this is a functor: [writes "is a functor"].

It takes an object here and canonically associates an object here, and also the maps. In linear algebra, if you have a map between vector spaces, $T\colon V \to W$, [writes: $V \xrightarrow{f} W$] that gives you a map the other way: $T^*\colon W^* \to V^*$: [writes: $W^* \xrightarrow{f^*} V^*$]. You get a map this way — the dual — and then if you do it again, you get a map in the correct direction: $(V^*)^* \xrightarrow{(T^*)^*} (W^*)^*$: [writes: $(V^*)^* \xrightarrow{f^{**}} (W^*)^*$] V double dual to W double dual.

This is to say that this functor takes arrows to the T double dual arrow: [writes: $F(f) = f^{**}$]. You can think about it.

Student Question

Is the dimension of vector spaces also only finite or can it also be infinite?

Spoken Transcript [01:30:08 - 01:30:50]

Prof. Rahul Pandharipande

Well, I didn't say anything.

You could, if you want, say you want the category of finite-dimensional vector spaces. That's a nice category inside, but the way I said it, it's anything.

For finite-dimensional vector spaces, it happens that the double dual is isomorphic. I'm not saying any of that; I'm just saying there's a motion (notion?) giving a vector space another one, and that's the double dual, and it's functorial in this sense.

You've seen this double dual business, right? Usually, in linear algebra, you spend all your time dualing and double dualing.

Okay, good, that's it. [students applaud]

The Double Dual Functor on Vector Spaces

Example 2.53 (*The Double Dual Functor*): Let $\mathbf{Vect}_{\mathbb{C}}$ denote the category of complex vector spaces and $\mathbb{C}$-linear maps. The double dual construction defines a functor from this category to itself:

$$F\colon \mathbf{Vect}_{\mathbb{C}} \longrightarrow \mathbf{Vect}_{\mathbb{C}}$$

defined by:

- **On Objects:** For every vector space $V \in \operatorname{Obj}(\mathbf{Vect}_{\mathbb{C}})$:
$$V \longmapsto V^{**} := (V^*)^*, \quad \text{where } V^* := \operatorname{Hom}_{\mathbb{C}}(V, \mathbb{C}).$$
- **On Morphisms:** For any linear map $f\colon V \to W$, the dual map $f^*\colon W^* \to V^*$ (Definition 0.35) reverses direction:
$$f^*(\varphi) := \varphi \circ f \quad \forall \varphi \in W^*.$$
Applying the dual operation a second time reverses the arrow direction once more, restoring covariance:
$$f^{**}\colon V^{**} \longrightarrow W^{**}, \quad (f^{**}(\psi))(\varphi) := \psi(f^*(\varphi)) = \psi(\varphi \circ f) \quad \forall \psi \in V^{**}.$$

$$\begin{tikzcd}
V \arrow[r, "f"] & W \\
V^* & W^* \arrow[l, "f^*"'] \\
V^{**} \arrow[r, "f^{**} = (f^*)^*"] & W^{**}
\end{tikzcd}$$

Covariance, Contravariance, and Canonical Isomorphism: Going beyond the lecture:

- **Single Dual is Contravariant:** The mapping $V \mapsto V^*$ and $f \mapsto f^*$ reverses arrow directions ($f\colon V \to W \implies f^*\colon W^* \to V^*$), making the dual functor a *"contravariant functor"*.
- **Functoriality:** $(\mathrm{Id}_V)^{**} = (\mathrm{Id}_{V^*})^* = \mathrm{Id}_{V^{**}}$, and for $U \xrightarrow{g} V \xrightarrow{f} W$ the dual satisfies $(f \circ g)^* = g^* \circ f^*$, so
$$(f \circ g)^{**} = (g^* \circ f^*)^* = f^{**} \circ g^{**}.$$
- **Double Dual is Covariant:** Dualizing twice reverses the arrows twice, so F preserves morphism directions ($f\colon V \to W \implies f^{**}\colon V^{**} \to W^{**}$), forming a valid covariant functor.
- **Finite vs. Infinite Dimensions:** For finite-dimensional vector spaces, the evaluation mapping $\mathrm{ev}\colon V \to V^{**}$, $v \mapsto (\varphi \mapsto \varphi(v))$, is a canonical isomorphism ($V \cong V^{**}$, Fact 0.36). For infinite-dimensional spaces, ev remains an injective linear map, but is generally not surjective. The functorial construction $V \mapsto V^{**}$, however, is universally defined across all vector spaces regardless of dimension.

Summary of Key Concepts

This lecture established the following central algebraic and categorical foundations:

- **Uniqueness of Identity and Inverses:** In any group $(G, \cdot, e)$, associativity inherently guarantees that both the identity element and inverse elements are strictly unique.
- **Homomorphisms and Isomorphisms:** Group homomorphisms preserve products and automatically map identities to identities ($f(e) = \hat{e}$). Two groups are isomorphic if and only if there exists a two-sided inverse homomorphism. Notably, the exponential function establishes an isomorphism between the additive reals $(\mathbb{R}, +, 0)$ and the strictly positive multiplicative reals $(\mathbb{R}^*_{>0}, \cdot, 1)$.
- **Subgroups and Symmetry Breaking in Rings:** For group homomorphisms, both the kernel $\ker(f) \leq G$ and image $\mathrm{Im}(f) \leq \hat{G}$ are subgroups. In ring theory, this symmetry breaks: the image $\mathrm{Im}(f) \subset \hat{R}$ is a subring, whereas the kernel $\ker(f) \subset R$ is an ideal absorbing ring multiplication.
- **Ideals of $\mathbb{Z}$ and $C^\infty(\mathbb{R})$:** Every ideal of $\mathbb{Z}$ is principal ($I = n\mathbb{Z}$ with $n \geq 0$), proven via the least positive element of an ideal and the division algorithm. In the ring of smooth functions $C^\infty(\mathbb{R})$, the compactly supported functions form an ideal I that is strictly contained within larger proper ideals, such as functions vanishing on a right half-line $[M, \infty)$.
- **Categories and Functors:** A category Cat formalizes collections of objects and morphism sets with composition, identity morphisms and unit laws. Isomorphisms are characterized purely arrow-theoretically by two-sided inverses. Functors formalize structure-preserving maps between categories, exemplified by the functor $\mathbf{Ring} \to \mathbf{Grp}$ and the covariant double dual functor $(-)^{**}\colon \mathbf{Vect}_{\mathbb{C}} \to \mathbf{Vect}_{\mathbb{C}}$.

Wednesday, 23rd September 2026

Categories, Integral Domains, and the Group of Units

Course Content Overview

This lecture begins with administrative guidance regarding enrollment, problem set submissions, and exam scope before reviewing the foundational abstractions of category theory and advancing into commutative ring theory. The main topics covered are:

- **Category Theory Review:** Foundational data (objects, hom-sets, composition laws, identity morphisms), category axioms (associativity and identity laws), and categorical formulations of isomorphisms, monomorphisms (injectivity), and epimorphisms (surjectivity).
- **Integral Domains:** Definition of integral domains (with explicit treatment of the non-triviality condition $R \neq \{0\}$), followed by a complete proof that $\mathbb{Z}/m\mathbb{Z}$ is an integral domain if and only if m is prime, along with edge-case considerations for $m \leq 1$.
- **Divisibility and Units in Commutative Rings:** Generalization of divisibility to arbitrary commutative rings, definitions of units and multiplicative inverses, linguistic conventions (*"the units"*, *"the unit"*, and *"a unit"*), and the proof that the units form an abelian group $(\mathcal{U}_R, \cdot, 1)$.
- **Units in Modular Arithmetic:** The units of $\mathbb{Z}/m\mathbb{Z}$ are the classes $\bar{x}$ with $\mathrm{GCD}(x, m) = 1$; explicit computation of $\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z})$, and the construction of an explicit group isomorphism with the additive cyclic group $(\mathbb{Z}/6\mathbb{Z}, +, \bar{0})$ generated by powers of $a = \bar{3} \pmod 7$.

Course Logistics and Announcements

Spoken Transcript [00:00:00 - 00:01:26] Prof. Rahul Pandharipande

Okay, so, we start. I wanted to make this comment: There were several questions about the course. The first one is, if you want to watch the lectures, you have to be enrolled. So, if you're not enrolled, it will stop you. And there's not much I can do about that, or if there is, I don't know what to do. But the idea is you should enroll in the course if you want to watch the lectures.

Other people had some difficulty uploading the problem sets. Maybe you have to be enrolled in the course for that, too. I don't know, but many people were successful; so, if you have difficulty, find someone who was successful, because I don't know how to solve the problem in another way.

A question that's more about the actual course was that every week there are these suggested pages to read in the book. And the question is: How do they relate to the course? That's a good question.

And the answer is that, roughly speaking, I do that so you can find the parallel material on which I'm lecturing in the book. And I don't always follow exactly the order there, I don't follow exactly the notation, I don't follow exactly the text of the definitions. But I do something that's logically equivalent, so it can be helpful. As far as matters like the exam, I will only test you on material that has appeared in the lecture. Okay? So, this is my best statement about this so far. Okay.

Administrative Policies

The lecturer outlines three organizational guidelines:

- **Lecture Access:** Enrolled status is required to stream or access recorded lecture videos.
- **Problem Sets:** Submission portals are restricted to enrolled students; technical submission difficulties should be resolved peer-to-peer or via course administrators.
- **Course Literature vs. Exam Scope:** Weekly reading assignments provide parallel textbook treatment, but examinations are strictly based on definitions, theorems, and proofs presented during the live lectures.

Review of Categories: Data and Axioms

Spoken Transcript [00:01:26 - 00:02:53] Prof. Rahul Pandharipande

So, the last topic yesterday was categories, and I wanted to say a few things.

When I had this discussion about categories yesterday, it was mixed in with all the examples. So, I want to just very quickly give the pure abstract definition and two small points. But this course is not about category theory. It's a topic here because it's natural to introduce when you talk about these algebraic objects; so, anyway, the course is not about category theory.

A category has data, two types of data [writes "Category" and "Cat $\Rightarrow$"]: It has objects, as we said, and it has homs [writes "Obj $\ni$ A" and "Homs $\mathrm{Hom}(A, B)$"]. So, maybe A can be an object; and for homs, if you pick any two objects, you get a hom-space [writes "$\mathrm{Hom}(A, B)$" with a brace], sometimes called morphisms.

And the properties, the abstract properties of this hom-space — I said them, but... — is that it's a set [writes "• Set"]. That's a technical property; that's to avoid some contradictions in foundations, which I'm not going to discuss now. The second is that there's a composition law [writes "• Comp Law"]. And the third is that there are identity elements — existence of identity, so [writes "• $1_A \in \mathrm{Hom}(A, B)$", a slip for $\mathrm{Hom}(A, A)$]...

Okay, that's the data. Is there any question about this data?

Constituents of a Category: Data

Recap: Categories were introduced the day before, mixed in with examples (Definition 2.44, Definition 2.49). The lecturer now repeats the abstract definition on its own.

Definition 2.54 (*Category Data*): A *"category"* Cat consists of the following mathematical data:

- **Objects:** A class of objects, denoted $\mathrm{Obj}(\mathrm{Cat})$. We write $A \in \mathrm{Obj}(\mathrm{Cat})$ (or simply $A \in \mathrm{Cat}$) to denote that A is an object of Cat.
- **Morphism Sets (Homs):** For any pair of objects $A, B \in \mathrm{Obj}(\mathrm{Cat})$, a set of morphisms from A to B, denoted:
  $$\mathrm{Hom}(A,B).$$
  An element $f \in \mathrm{Hom}(A,B)$ is represented diagrammatically as an arrow $f\colon A \to B$.
- **Composition Law:** For any three objects $A, B, C \in \mathrm{Obj}(\mathrm{Cat})$, a binary composition mapping:
  $$\begin{aligned} \circ\colon \mathrm{Hom}(B,C) \times \mathrm{Hom}(A,B) &\to \mathrm{Hom}(A,C), \\ (g,f) &\mapsto g \circ f. \end{aligned}$$
- **Identities:** For each object $A \in \mathrm{Obj}(\mathrm{Cat})$, a designated morphism called the identity morphism on A:
  $$\mathrm{Id}_A \in \mathrm{Hom}(A,A) \quad (\text{or } 1_A \in \mathrm{Hom}(A,A)).$$

Foundational Remark on Hom-Sets: Going beyond the lecture: the requirement that $\mathrm{Hom}(A,B)$ forms a legitimate set (rather than a proper class, a collection too large to be a set, like the collection of all sets) is a standard set-theoretic convention defining a **locally small category**. This prevents Russell-style self-referential paradoxes when indexing collections of arrows.

Spoken Transcript [00:02:53 - 00:03:45]

Prof. Rahul Pandharipande

And then there are the axioms. The axioms... [writes "Axioms"] The first one is that the composition law is associative [writes "– Comp Law Associative"], which means, you know, if you have three things, you can compose them in the two different ways. I hope I don't have to say that more.

And then the second is the axioms about the identity, which is roughly: If I do this, $f \circ \mathrm{Id}_A$, this is f [writes "$f \cdot \mathrm{Id}_A = f$"], and the other way around also [writes "$\mathrm{Id}_B \cdot g = g$"].

So, the identity doesn't change the... We just discussed this last time, so, I wanted to say.

Associativity and Identity Axioms

Definition 2.55 (*Category Axioms*): The data of a category Cat must satisfy the following two fundamental axioms:

(1) Associativity of Composition: For all objects $A, B, C, D \in \mathrm{Obj}(\mathrm{Cat})$ and all morphisms $f \in \mathrm{Hom}(A,B)$, $g \in \mathrm{Hom}(B,C)$, and $h \in \mathrm{Hom}(C,D)$, composition is associative:

$$(h \circ g) \circ f = h \circ (g \circ f). \tag{2.17}$$

(2) Identity Axioms (Neutrality): For all morphisms $f, g \in \mathrm{Hom}(A,B)$, composition with the respective identity morphisms changes nothing (on the board, with a dot for the composition: $f \cdot \mathrm{Id}_A = f$ and $\mathrm{Id}_B \cdot g = g$):

$$f \circ \mathrm{Id}_A = f, \tag{2.18}$$

$$\mathrm{Id}_B \circ g = g. \tag{2.19}$$

$$\begin{tikzcd} & B \arrow[dr, "g"] \arrow[loop above, "\mathrm{Id}_B"] & \\ A \arrow[ur, "f"] \arrow[rr, "g \circ f"'] \arrow[loop left, "\mathrm{Id}_A"] & & C \arrow[loop right, "\mathrm{Id}_C"] \end{tikzcd}$$

Categorical Isomorphisms

Spoken Transcript [00:03:45 - 00:05:37] Prof. Rahul Pandharipande

So, that's the abstract... There's really not that much here. We discussed this last time, and we had lots of examples.

And so, you could say: Well, why do we want to do this? And as I said, many concepts that we talk about when we talk about groups, rings, fields, vector spaces, they're categorical concepts, meaning that they only depend on this structure. So, it simplifies some kind of thought to think about them all together.

I already told you about isomorphism. [writes "• Isomorphism"]

The categorical isomorphism is that if I have this arrow from A to B [writes "$A \xrightarrow{f} B$"] — this picture means $f \in \mathrm{Hom}(A,B)$ [writes "$f \in \mathrm{Hom}(A,B)$"]. In category theory, you can say this, or you can say that; you mean the same thing.

This f is an isomorphism [writes "f is an isom."] — that's a definition — if and only if there exists g the other way [writes "$\iff \exists A \xleftarrow{g} B$"], such that [writes "such that"] the two compositions are identities.

And if you want to say which identity, you have to be careful. So, g starts with B, so you do g first, then f back — this should be the identity in B [writes "$f \circ g = \mathrm{Id}_B$"]. And this is the identity in A [writes "$g \circ f = \mathrm{Id}_A$"].

So, that's the abstract definition of isomorphism. And in all of the categories we discussed, that coincides with the notion of isomorphism in the category that you already know, like, for example, isomorphism of vector spaces.

Categorical Isomorphism

Recap: The categorical isomorphism was defined the day before (Definition 2.50); the lecturer restates it.

Definition 2.56 (*Isomorphism*): Let Cat be a category and let $f \in \mathrm{Hom}(A, B)$ (written $f\colon A \to B$). The morphism f is an *"isomorphism"* if and only if there exists a morphism $g \in \mathrm{Hom}(B, A)$ (written $g\colon B \to A$) such that:

$$f \circ g = \mathrm{Id}_B, \tag{2.20}$$

$$g \circ f = \mathrm{Id}_A\,. \tag{2.21}$$

If such a morphism g exists, it is uniquely determined, called the *"inverse"* of f, and denoted $f^{-1} := g$. Two objects A and B are said to be *"isomorphic"* (written $A \cong B$) if there exists an isomorphism between them.

$$\begin{tikzcd} A \arrow[r, bend left, "f"] & B \arrow[l, bend left, "g = f^{-1}"] \end{tikzcd}$$

$f \circ g = \mathrm{Id}_B$ and $g \circ f = \mathrm{Id}_A$

Consistency across Concrete Categories: Going beyond the lecture: in concrete categories where objects are sets endowed with structure and morphisms are structure-preserving maps (homomorphisms):

- In the category **Set** (objects: all sets; morphisms: all maps between them), isomorphisms are precisely bijective functions (Fact 0.3).
- In **Grp**, **Ring**, and $\mathbf{Vect}_K$, isomorphisms are bijective homomorphisms (for groups: Proposition 2.13).

The power of the categorical definition is that it never inspects elements $x \in A$; it characterizes invertibility purely through diagrammatic composition.

Monomorphisms and Epimorphisms

Spoken Transcript [00:05:37 - 00:07:22] Prof. Rahul Pandharipande

There are two other notions, and that's the last thing I want to say. And this is the notion of monomorphism. [writes "• Monomorphism"]

So, f from A to B is a monomorphism [writes "$A \xrightarrow{f} B$ is a monomorphism"] — that's a long word, that's a definition — if [writes "if"] for all g_1 arrows... this is another object X, and another one... [writes "$\forall X \xrightarrow{g_1} A, X \xrightarrow{g_2} A$"] We take any two, then the composition [writes "then $f \circ g_1 = f \circ g_2$"] implies $g_1 = g_2$ [writes "$\implies g_1 = g_2$"].

I hope I wrote that clearly here, okay?

It's a little bit complicated to say. So, I'm defining: If I have an arrow in the category, it may or may not be a monomorphism. I'm defining a monomorphism.

And the condition of being a monomorphism means that whenever I take an object X and map it two ways to A, the composition with f being equal implies they were equal to start with. Of course, if they're equal to start with, the composition is equal, but it's the other implication. That's the definition of a monomorphism.

You know, you have to think about this and unwind it. But once you have this definition, you can imagine: What is a monomorphism in all of those categories you know? And what do you think it is?

Spoken Transcript [00:07:22 - 00:08:27]

Prof. Rahul Pandharipande

So, I leave this for you to think about, but roughly speaking, a monomorphism is injective. [writes "(injective)" and circles it on the blackboard]

Just imagine they're sets. If this is injective, if you take any two maps, you can compare them after you do f. So, you have to think about it.

But this is the definition. Once you take this definition, you can say: Okay, I can take the category of sets and functions, and what is a monomorphism there? And the answer is injective. That's the *"mono"* in monomorphism.

This is a categorical formulation of being injective.

I don't know if you find this appealing or not, but that's what this discussion is about. Because the nice thing of this formulation is that it applies to all examples of categories. Every single one we discussed, and all the other ones you can think about, there's a notion of monomorphism.

And you can imagine this one has a sibling, or a partner — I don't know their relationship — which is... Whenever you talk about injective, you can also talk about surjective, and the categorical term there is epimorphism.

Monomorphisms

Definition 2.57 (*Monomorphism*)**:** Let Cat be a category. A morphism $f \in \mathrm{Hom}(A, B)$ (or $f\colon A \to B$) is called a *"monomorphism"* if for every object $X \in \mathrm{Obj}(\mathrm{Cat})$ and all pairs of parallel morphisms $g_1, g_2 \in \mathrm{Hom}(X, A)$,

$$f \circ g_1 = f \circ g_2 \implies g_1 = g_2. \tag{2.22}$$

$$\begin{tikzcd}
X \arrow[r, bend left, "g_1"] \arrow[r, bend right, "g_2"'] & A \arrow[r, "f"] & B
\end{tikzcd}$$

$$f \circ g_1 = f \circ g_2 \implies g_1 = g_2 \quad \text{(Categorical Injective)}$$

Why Monomorphisms Characterize Injectivity (beyond the lecture)

The lecture leaves this to the students (*"I leave this for you to think about"*). In the category of sets (**Set**), suppose $f\colon A \to B$ is a monomorphism. To test whether f is injective on points $a_1, a_2 \in A$, let $X = \{*\}$ be a one-element set and define two maps $g_1, g_2\colon \{*\} \to A$ by $g_1(*) := a_1$ and $g_2(*) := a_2$. Then $(f \circ g_1)(*) = f(a_1)$ and $(f \circ g_2)(*) = f(a_2)$. Thus:

$$f(a_1) = f(a_2) \implies f \circ g_1 = f \circ g_2 \implies g_1 = g_2 \implies a_1 = a_2.$$

Hence, left-cancellation with respect to arbitrary test objects X precisely recovers element-wise injectivity without mentioning elements. Conversely, an injective f is a monomorphism: if $f \circ g_1 = f \circ g_2$, then $f(g_1(x)) = f(g_2(x))$ for every $x \in X$, so $g_1(x) = g_2(x)$ by injectivity, that is, $g_1 = g_2$. **Q.E.D.**

Spoken Transcript [00:08:27 - 00:09:20] Prof. Rahul Pandharipande

So, you can define epimorphism [writes: epimorphism]. If you have f from A to B, it is an epimorphism if for all [writes: $A \xrightarrow{f} B$ is an epimorphism if]...

And this is now about this kind of feeling of mathematics: How do you think I'm going to define the opposite notion of injective?

Like here we had injective by X mapping to A. So, what do you think I should do here?

You should guess. Guessing is like one of the most important tools of a mathematician.

Yeah?

Student Answer

Well, you do the same, except we only switch like with the arrows...

Spoken Transcript [00:09:25 - 00:10:37] Prof. Rahul Pandharipande

This is a long answer. You should just say... Here we map X to A, so the opposite of that is mapping from B. It's like the other side.

So, for every $B \xrightarrow{g_1} X$, $B \xrightarrow{g_2} X$ [writes "$\forall B \xrightarrow{g_1} X, B \xrightarrow{g_2} X$"], then...

This is the kind of opposite of that, by just flipping everything on the other side. So, for every one of these, then we can consider f. In this one, it was the g's first, then the f. Now it's the f first, then the g's: $g_1 \circ f = g_2 \circ f$ implies $g_1 = g_2$ [writes "then $g_1 \circ f = g_2 \circ f \implies g_1 = g_2$"].

So, that's the categorical notion of epimorphism, and it's somehow just dual to the monomorphism. If you like this world, it almost makes epimorphism and monomorphism the same definition up to this flipping. And thinking about things categorically is good for such understanding.

If you look at this definition, and you look at it for the examples we know, this is surjectivity.

Epimorphisms

Definition 2.58 (*Epimorphism*)**:** Let Cat be a category. A morphism $f \in \mathrm{Hom}(A, B)$ (or $f\colon A \to B$) is called an *"epimorphism"* if for every object $X \in \mathrm{Obj}(\mathrm{Cat})$ and all pairs of parallel morphisms $g_1, g_2 \in \mathrm{Hom}(B, X)$,

$$g_1 \circ f = g_2 \circ f \implies g_1 = g_2. \tag{2.23}$$

$$\begin{tikzcd}
A \arrow[r, "f"] & B \arrow[r, bend left, "g_1"] \arrow[r, bend right, "g_2"'] & X
\end{tikzcd}$$

$$g_1 \circ f = g_2 \circ f \implies g_1 = g_2 \quad \text{(Categorical Surjective)}$$

Beyond the Lecture: Epimorphisms Need Not Be Surjective

The lecturer says that for the examples we know, epimorphism means surjectivity. This holds in **Set**, **Grp** and $\mathbf{Vect}_K$, but not in **Ring**: the inclusion $\iota\colon \mathbb{Z} \hookrightarrow \mathbb{Q}$ is an epimorphism without being surjective. Indeed, if $g_1, g_2\colon \mathbb{Q} \to R$ are ring homomorphisms with $g_1 \circ \iota = g_2 \circ \iota$, then for $b \neq 0$ the element $g_k(b)$ is invertible with inverse $g_k(\frac{1}{b})$, so $g_1(\frac{a}{b}) = g_1(a)\,g_1(b)^{-1} = g_2(a)\,g_2(b)^{-1} = g_2(\frac{a}{b})$. For monomorphisms the slogan is safe: in **Set**, **Grp**, **Ring** and $\mathbf{Vect}_K$ they are exactly the injective homomorphisms.

Beyond the Lecture: The Opposite Category

The lecturer calls the epimorphism dual to the monomorphism. This can be made precise: if we construct the opposite category $\mathrm{Cat}^{\mathrm{op}}$ by reversing all arrows ($f^{\mathrm{op}}\colon B \to A$ for every $f\colon A \to B$), then

$$f \text{ is a monomorphism in Cat} \iff f^{\mathrm{op}} \text{ is an epimorphism in } \mathrm{Cat}^{\mathrm{op}}.$$

Every statement about monomorphisms yields a dual statement about epimorphisms by reversing all arrows.

Spoken Transcript [00:10:37 - 00:11:03] Prof. Rahul Pandharipande

Okay, so, that's the end of this discussion about categories. If you like it, you can go read more. As I said, this course is not a course in categories. There's a bunch of other things that I would say here, but I'm not going to, because... yeah.

Maybe I will at the end if I have time.

Integral Domains

Zero Divisors and the Definition of Integral Domains

Spoken Transcript [00:11:03 - 00:12:35] Prof. Rahul Pandharipande

So, let us return to rings now. [writes "Rings $(R, +, \cdot, 0, 1)$"]

This was kind of a foundational side, so, we have this ring $(R, +, \cdot, 0, 1)$ (Recall: commutative with unit, Remark 1.17).

And the rings here, there's kind of a different flavor of rings. There's the rings that are like $\mathbb{Z}$ [writes "$\mathbb{Z}$"], where when you have things that are non-zero and you multiply them, they stay non-zero. That's one of the principles of $\mathbb{Z}$.

But it's not true for rings like $\mathbb{Z}/6\mathbb{Z}$, this ring [writes "$\mathbb{Z}/6\mathbb{Z}$"]. I can take two elements here, like the residue class of 2 times the residue class of 3, neither of which is zero. So, this is not zero [writes "$\bar{2} \neq \bar{0}, \bar{3} \neq \bar{0}$ in $\mathbb{Z}/6\mathbb{Z}$"], and that is not zero.

So, I take two elements that are legitimately not zero, but I can multiply them; and when I multiply these residue classes, this is six, and that is zero [writes "$\bar{2} \cdot \bar{3} = \bar{6} = \bar{0}$"].

That's a different type of multiplication. It's a multiplication where when you multiply things, you get zero, which is unexpected from the point of view of the integers. And this is also related to the fact that there's no hope of having a multiplicative inverse if you're mapping things to zero.

So, there are these two differences, and there's a definition for this.

Multiplication in $\mathbb{Z}/6\mathbb{Z}$

Let $(R, +, \cdot, 0, 1)$ be a commutative ring (Remark 1.17).

Example 2.59 (*Multiplication in $\mathbb{Z}/6\mathbb{Z}$*): In the ring of residue classes modulo 6 (Definition 1.23), denoted $\mathbb{Z}/6\mathbb{Z}$ (or $\mathbb{Z}_6$):

$$\bar{2} \neq \bar{0},$$
$$\bar{3} \neq \bar{0}.$$

However, their product in the ring is:

$$\bar{2} \cdot \bar{3} = \overline{2 \cdot 3} = \bar{6} = \bar{0} \in \mathbb{Z}/6\mathbb{Z}.$$

Zero Divisors vs. Invertibility: Going beyond the lecture: non-zero elements whose product is zero are called *"zero divisors"*. If an element $a \in R$ is a zero divisor (with $a \cdot b = 0$ for some $b \neq 0$), it can never possess a multiplicative inverse a^{-1}:

$$b = 1 \cdot b = (a^{-1} \cdot a) \cdot b = a^{-1} \cdot (a \cdot b) = a^{-1} \cdot 0 = 0,$$

which contradicts $b \neq 0$. Hence, rings containing zero divisors fail to exhibit regular arithmetic cancellation.

Spoken Transcript [00:12:35 - 00:14:26] Prof. Rahul Pandharipande

A ring R is an integral domain [writes "Def. A ring R is an integral domain"] — so it's a definition of this term, integral domain. And some authors would just say it's a domain. Also, if you speak about rings, very often mathematicians will say it's a domain.

If... whenever I have a and b, neither of which is zero — so, maybe I'll just be really explicit: whenever I have $a \neq 0$ and $b \neq 0$ [writes "if $\forall a \neq 0, b \neq 0$"] — this implies the product is not equal to zero [writes "$\implies ab \neq 0$"].

So, that's the definition. That's a really basic definition in ring theory, and it separates this type of ring from that type of ring. Is that clear?

Fields, for example — for example, fields are always domains [writes "For example: Fields are always domains."] because you have the multiplicative inverse, if you think about it, okay?

So, see, I did that already there: I called it a domain, not an integral domain, because it takes a lot of syllables to say integral domain. Formally, this is defined as an integral domain; colloquially, almost everyone says domain.

Definition of an Integral Domain (Initial Board State)

Definition 2.60 (*Integral Domain / Domain*): A commutative ring R is called an *"integral domain"* (or simply a *"domain"*) if for all $a, b \in R$:

$$a \neq 0 \quad \text{and} \quad b \neq 0 \implies a \cdot b \neq 0. \tag{2.24}$$

Equivalently, taking the contrapositive:

$$a \cdot b = 0 \implies a = 0 \quad \text{or} \quad b = 0.$$

A few minutes later the lecturer adds the condition $R \neq 0$; the definition used from then on is Definition 2.62.

Example 2.61 (*Fields are Domains*): Every field F is an integral domain: if $a \neq 0$ and $a \cdot b = 0$, then multiplying by the inverse $a^{-1} \in F$ immediately yields:

$$b = 1 \cdot b = (a^{-1}a)b = a^{-1}(ab) = a^{-1} \cdot 0 = 0.$$

Student Discussion: Monomorphisms in Linear Algebra

Spoken Transcript [00:14:26 - 00:14:43] Prof. Rahul Pandharipande

These terms, monomorphism and epimorphism, have you ever heard these terms before? In which context?

Student Answer

Linear algebra.

Spoken Transcript [00:14:45 - 00:14:50] Prof. Rahul Pandharipande

Ah, okay. And they were just used synonymously with injection and surjection?

Student Answer

They were introduced, but then we went back to using injective and surjective.

Spoken Transcript [00:14:54 - 00:14:57] Prof. Rahul Pandharipande

Okay, yeah.

Student Answer

Still they were like injective and surjective homomorphisms.

Spoken Transcript [00:15:02 - 00:15:33] Prof. Rahul Pandharipande

Yes, exactly. Yes, injective and surjective homomorphisms.

That's precisely what this categorical thing says, right? Because in the category of groups, the arrows are homomorphisms; and when a homomorphism is a monomorphism, it's an injective homomorphism.

In my experience with undergraduates, some really like the idea of categories, and they really like to think in that language, and others like the examples. So, it can be... yeah, either way is okay.

Characterization of $\mathbb{Z}/m\mathbb{Z}$ as a Domain

Spoken Transcript [00:15:33 - 00:16:03] Prof. Rahul Pandharipande

All right, so, here we come to the first proposition, which is something my guess is you are already familiar with, but we do it again.

We take this ring, $(\mathbb{Z}/m\mathbb{Z}, +, \cdot, \bar{0}, \bar{1})$ [writes "Prop. $(\mathbb{Z}/m\mathbb{Z}, +, \cdot, \bar{0}, \bar{1}) \quad m \in \mathbb{Z}_{>0}$"], and the claim is that this $\mathbb{Z}/m\mathbb{Z}$ is a domain if and only if... And what am I going to write here?

Student Answer

m is prime.

Spoken Transcript [00:16:05 - 00:17:09] Prof. Rahul Pandharipande

Yeah, m is prime. m is prime. [writes "$\mathbb{Z}/m\mathbb{Z}$ is a domain $\iff$ m is prime"]

Oh, yeah, I should say one thing about domains which is important, which I didn't say, and it's kind of annoying. Part of the domain... I should put that in because it's... [pulls down the middle blackboard to modify the earlier definition]

It's a super annoying thing to talk about, but I talk about any ring here, so my definition has to deal with the zero ring — the one that, you know, is unpleasant to think about. And you could think about: What would this definition be for the zero ring? Because for all a and b, there are no a's and b's that are not equal to zero. So, formally this statement, in the formal deductive logical sense, is true for a zero ring.

So, it is part of the definition: a ring R which is not the zero ring is a domain [adds $\neq 0$ under the R of the definition on the board]. Zero rings are not allowed to be domains, by fiat, in the definition. [pushes the blackboard back up]

Correction: Exclusion of the Zero Ring

The lecturer corrects Definition 2.60 by adding condition (i):

Definition 2.62 (*Integral Domain (Formal Standard Definition)*): A commutative ring R is an *"integral domain"* (or *"domain"*) if:

(i) $R \neq 0$ (that is, $1_R \neq 0_R$: the zero ring is excluded; cf. Proposition 1.10).

(ii) For all $a, b \in R$:

$$a \neq 0 \quad \text{and} \quad b \neq 0 \implies a \cdot b \neq 0.$$

Why the Zero Ring Satisfies the Conditional Vacuously: In the zero ring $R = \{0\}$, the only element is 0. Therefore, there exist no non-zero elements $a, b \in R$. The premise $(a \neq 0 \wedge b \neq 0)$ is vacuously false for all elements, making the logical implication $(a \neq 0 \wedge b \neq 0 \implies ab \neq 0)$ vacuously true! To avoid having the degenerate zero ring classified as a domain, mathematicians add $R \neq 0$ explicitly by fiat.

Proposition: When Residue Class Rings are Domains

Proposition 2.63 (*When Residue Class Rings are Domains*): Let $m \in \mathbb{Z}_{>0}$ be a positive integer. The residue class ring $(\mathbb{Z}/m\mathbb{Z}, +, \cdot, \bar{0}, \bar{1})$ is an integral domain if and only if m is a prime number:

$$\mathbb{Z}/m\mathbb{Z} \text{ is an integral domain} \iff m \text{ is prime.} \tag{2.25}$$

Proof of Proposition 2.63

Spoken Transcript [00:17:09 - 00:19:24] Prof. Rahul Pandharipande

Okay, $\mathbb{Z}/m\mathbb{Z}$ is a domain if and only if m is prime. So, let's try to prove this.

Proof. [writes "Proof"]

This is like one of the issues I said: this issue about the zero ring is a little bit problematic everywhere in the statements. Because for any statement you write, *"let R be a ring"*, you have to think — if you want that statement to be literally true in the mathematical sense from the definitions in this class — what that statement could mean for the zero ring and whether it's true. And, as I said, I don't view it as a deep or really important thing to do, but in terms of bookkeeping and saying things correctly, you have to say it.

Okay, so, let's try to prove this: it is a domain if and only if m is prime.

(1): Suppose m is not prime. [writes "(1) Suppose m is not prime"]

We have m here as some positive integer. If m is not prime, then this means that m admits a factorization into ab [writes "$m = ab$"], where a is strictly greater than 1, strictly less than m [writes "$1 < a < m$"], and b is strictly like this: $1 < b < m$ [writes "$1 < b < m$"]. That's what it means: If it's not prime, I can factor it; and if I can factor it, I break it in two, into smaller pieces.

And then if I look at this equation as a residue class, this equals that, which is equal to zero in $\mathbb{Z}/m\mathbb{Z}$ [writes "$\bar{a} \cdot \bar{b} = \bar{m} = \bar{0} \in \mathbb{Z}/m\mathbb{Z}$"]. And this shows that $\bar{a}$ is not zero [writes "$\bar{a} \neq \bar{0}$"], and this shows that $\bar{b}$ is not zero [writes "$\bar{b} \neq \bar{0}$"].

Okay? So, this says that if m is not prime, there's no way for it to be a domain, because it factors, and those factors precisely give the violation of that property (i.e., condition **(ii)** of Definition 2.62). All right?

Proof of Proposition: Direction 1 (Non-Prime Case)

Direction 1: m Not Prime $\implies \mathbb{Z}/m\mathbb{Z}$ Not a Domain.

Suppose m is not prime. If $m = 1$, then $\mathbb{Z}/1\mathbb{Z} = \{\bar{0}\}$ is the zero ring, which is not a domain by condition (i) of Definition 2.62 (see also Remark 2.64). Otherwise $m > 1$ is composite and admits a proper non-trivial factorization in $\mathbb{Z}$:

$$\begin{aligned} m &= a \cdot b, \qquad (2.26)\\ 1 &< a < m,\\ 1 &< b < m. \end{aligned}$$

Passing to the residue classes in the quotient ring $\mathbb{Z}/m\mathbb{Z}$:

$$\bar{a} \cdot \bar{b} = \overline{a \cdot b} = \bar{m} = \bar{0} \in \mathbb{Z}/m\mathbb{Z}. \qquad (2.27)$$

However, because $1 < a < m$ and $1 < b < m$, neither a nor b is divisible by m. Therefore:

$$\bar{a} \neq \bar{0} \quad \text{and} \quad \bar{b} \neq \bar{0}. \qquad (2.28)$$

The existence of non-zero elements $\bar{a}, \bar{b} \in \mathbb{Z}/m\mathbb{Z}$ satisfying $\bar{a} \cdot \bar{b} = \bar{0}$ demonstrates that $\mathbb{Z}/m\mathbb{Z}$ is not an integral domain.

Spoken Transcript [00:19:24 - 00:21:50]

Prof. Rahul Pandharipande

So, then the other thing is: Suppose it is prime.

Suppose $m = p$ is prime [writes: Suppose $m = p$ is prime]. Then let a and b be integers such that $\bar{a} \cdot \bar{b} = \bar{0}$ in $\mathbb{Z}/p\mathbb{Z}$ [writes: Then let $a, b \in \mathbb{Z}$ such that $\bar{a} \cdot \bar{b} = \bar{0} \in \mathbb{Z}/p\mathbb{Z}$].

All right? What are we going to do here? If I translate this equation on residue classes to integers, this means that $a \cdot b$ is equal to some integer $d \cdot p$ [writes "$ab = dp$"].

I am using this notation and arithmetic in $\mathbb{Z}/m\mathbb{Z}$ freely here, because I assume you've seen this before. Is this true? If it's not, you should tell me. All right. If you haven't seen it, you should study it.

So, if I take these two residue classes and they multiply to zero, that exactly means that the product is divisible by p, and that is that difference. Is that clear?

And then this means that p divides ab, because ab is d times p [writes "$p \mid ab$"]. And then there's the principle of Euclid we discussed in the first week that says: either $p \mid a$ or $p \mid b$ [writes "$\implies$ Euclid either $p \mid a$ or $p \mid b$"]. It wasn't such a long time ago; it was just last week, right?

And if this is true, then that means $\bar{a} = \bar{0}$ [writes "$\implies \bar{a} = \bar{0}$"]; and if this is true, that means $\bar{b} = \bar{0}$ [writes "or $\bar{b} = \bar{0}$"].

So, we've proven that if the product is zero in $\mathbb{Z}/p\mathbb{Z}$, the only way that happens is one of them is zero. And that's logically equivalent to this: If they were both not zero, then the product is not zero. Or the logical equivalent, which is the contrapositive, is that if the product is zero, then one of the factors must be zero, which we've established. So, this proposition is proven.

Proof of Proposition: Direction 2 (Prime Case via Euclid's Lemma)

Direction 2: $m = p$ Prime $\implies \mathbb{Z}/p\mathbb{Z}$ Is a Domain.

Suppose $m = p$ is a prime number. Let $\bar{a}, \bar{b} \in \mathbb{Z}/p\mathbb{Z}$ and assume that their product vanishes:

$$\bar{a} \cdot \bar{b} = \bar{0} \in \mathbb{Z}/p\mathbb{Z}. \tag{2.29}$$

By definition of modular multiplication, this implies $\overline{a \cdot b} = \bar{0}$, meaning that $a \cdot b$ is congruent to $0 \pmod{p}$. Hence, there exists some integer $d \in \mathbb{Z}$ such that:

$$a \cdot b = d \cdot p \implies p \mid (a \cdot b). \tag{2.30}$$

By Euclid's Lemma (which holds whenever p is prime, Fact 0.17):

$$p \mid (a \cdot b) \implies p \mid a \quad \text{or} \quad p \mid b. \tag{2.31}$$

Translating each case back into modular equivalence in $\mathbb{Z}/p\mathbb{Z}$:

- If $p \mid a$, then $\bar{a} = \bar{0} \in \mathbb{Z}/p\mathbb{Z}$.
- If $p \mid b$, then $\bar{b} = \bar{0} \in \mathbb{Z}/p\mathbb{Z}$.

Thus, $\bar{a} \cdot \bar{b} = \bar{0}$ implies $\bar{a} = \bar{0}$ or $\bar{b} = \bar{0}$. Because $p \geq 2$, $\mathbb{Z}/p\mathbb{Z}$ contains at least two distinct elements ($\bar{0} \neq \bar{1}$), so $\mathbb{Z}/p\mathbb{Z}$ is not the zero ring. Therefore, $\mathbb{Z}/p\mathbb{Z}$ is an integral domain.

Q.E.D.

Boundary Cases and Principle of Lecturing

Spoken Transcript [00:21:50 - 00:23:18] Prof. Rahul Pandharipande

You could say again in this proposition: When someone writes this, I here put m to be a positive number, and now it matches carefully. But these boundary cases... There's some aspect of this foundational part where you have to be careful yourself about the boundary cases.

So, if I put $m = 1$, then I get the zero ring; but 1 is not considered a prime number, so it matches. You could say: What happens if I put zero? It's not here. We don't think about the primality of zero, so zero is left out of this. But you could still say: If m is zero, this is $\mathbb{Z}$, and it is a domain. Okay?

All right.

Sometimes being too clear is actually more confusing in lectures. There's this kind of principle of lecturing where you lie about all the little details, and the lectures come out cleaner if you do that. Nah. And it's not clear didactically that that isn't a better way to do it. [erases the middle and right blackboards]

There's a certain kind of mathematician that takes pleasure in these edge cases, and you can tell that I'm not one of them.

Pedagogical Analysis of Boundary Cases

Remark 2.64 (*Boundary Cases: $m = 1$ and $m = 0$*)**:** The proposition specifies $m \in \mathbb{Z}_{>0}$. Let us examine the boundary cases:

- **Case $m = 1$:** The ideal is $1\mathbb{Z} = \mathbb{Z}$. The quotient ring is the zero ring:
  $$\mathbb{Z}/1\mathbb{Z} \cong \{0\},$$
  where $\bar{1} = \bar{0}$. Since the definition of an integral domain explicitly requires $R \neq \{0\}$, $\mathbb{Z}/1\mathbb{Z}$ is **not** an integral domain. This perfectly matches the number-theoretic convention that 1 is not a prime number.
- **Case $m = 0$:** The zero ideal is $0\mathbb{Z} = \{0\}$. The quotient ring is canonically isomorphic to the integers:
  $$\mathbb{Z}/0\mathbb{Z} \cong \mathbb{Z}.$$
  The integers $\mathbb{Z}$ are the prototypical integral domain.

Divisibility and Units in Commutative Rings

Divisibility in General Rings

Spoken Transcript [00:23:18 - 00:24:29] Prof. Rahul Pandharipande

Okay, so, something more fun is we get to talk about units, and this is actually pretty fun.

This has to do with every ring; so, let's say here, we take a ring R [writes "R ring"]. It's our ring.

And this is now related to last week; in fact, it was the point of the discussion last week. There we discussed the ring $\mathbb{Z}$, but now we can take any ring.

So, the definition here is that if we have elements a and b in the ring [writes: Def. if $a, b \in R$], then we say — this is the definition of the term — we say a divides b [writes: we say a divides b]. What this means is there exists some element of the ring c such that $a \cdot c = b$ [writes "$\exists c \in R$" and "$ac = b$"].

Okay? That's just the actual definition of division, and it's the same definition as with the integers. So, this is a notion of division for any ring.

Divisibility Relations

Let R be a ring (commutative, as always in this course).

Definition 2.65 (*Divisibility in a Ring*)**:** Let $a, b \in R$. We say that *"a divides b"*, denoted:

$$a \mid b, \tag{2.32}$$

if there exists an element $c \in R$ such that:

$$a \cdot c = b. \tag{2.33}$$

In this case, we also say that b is a *"multiple"* of a, or that a is a *"divisor"* of b.

Connection with Principal Ideals: Going beyond the lecture: in ideal-theoretic language, $a \mid b$ in a commutative ring R is equivalent to the statement that the principal ideal generated by b is contained in the principal ideal generated by a (principal ideals $(a) = aR$: Definition 4.32, Week 4):

$$a \mid b \iff b \in aR \iff (b) \subset (a).$$

Units and Multiplicative Inverses

Spoken Transcript [00:24:29 - 00:26:22] Prof. Rahul Pandharipande

Now, $u \in R$ is a unit [writes "$u \in R$ is a unit"] — that's the definition of a unit. The definition is: if u divides 1 [writes: $\overset{\text{def}}{\iff}$ $u \mid 1$].

Oh, I should say that also: *"a divides b"*, that's how you say it in English, and the symbol for that is $a \mid b$ [writes "$(a \mid b)$"].

So, here this is now a definition: An element is a unit if u divides 1. And that is equivalent to: There exists a w such that $u \cdot w = 1$ [writes: $\iff$ $\exists w$ such that $uw = 1$]. I never have to worry about the order here, because I said that for the rings for this course, unless I tell you explicitly otherwise, we're always in commutative rings. Okay?

So, this is the definition, this is just the equivalence unwinding division; and if I want to unwind this using English — so, this is the definition of division [writes "Def of div"] — this is just English, which we say: if and only if u has a multiplicative inverse [writes "$\iff$ u has mult. inverse in R"]. Multiplicative inverse in R.

This is a really fundamental definition: a unit in a ring.

Are you happy with this? That's the definition, this is just translation, and that's what you say: u has a multiplicative inverse, w is the multiplicative inverse. Is there any problem with this?

Invertible Elements

Definition 2.66 (*Unit in a Ring*): Let R be a commutative ring with unity 1. An element $u \in R$ is called a *"unit"* if u divides 1:

$$u \text{ is a unit} \overset{\text{def}}{\iff} u \mid 1. \tag{2.34}$$

By the definition of divisibility (Definition 2.65), this is equivalent to:

$$u \mid 1 \overset{\text{def}}{\iff} \exists w \in R \quad \text{such that} \quad u \cdot w = 1. \tag{2.35}$$

In words, u is a unit if and only if u has a multiplicative inverse in R. The element w is uniquely determined and denoted $u^{-1} := w$.

Equivalence of Characterizations on the Blackboard: The blackboard outlines three equivalent perspectives on units:

1. **Divisibility Perspective:** $u \mid 1$ (the element divides the multiplicative identity).
2. **Algebraic Perspective (Unwinding Division):** $\exists w \in R$ such that $u \cdot w = 1$.
3. **Verbal Perspective (English):** u possesses a multiplicative inverse in R.

> Since all rings considered here are commutative, left-inverses and right-inverses coincide.

Linguistic Conventions: The Units, The Unit, and A Unit

Spoken Transcript [00:26:22 - 00:28:16] Prof. Rahul Pandharipande

Actually, people asked me some questions about language, and, you know, English is a language I know, which means that I actually speak it without thinking. It was pointed out to me that the use of *"unit"* is subtle; and I was doing it already in the first week, and if you were sensitive to language, you'd have seen.

So, for $\mathbb{Z}$ here, what are the units for $\mathbb{Z}$? [writes "What are units in $\mathbb{Z} =$"] In $\mathbb{Z}$, and it comes as no surprise, this is $+1$ and -1 [writes "$\{1,-1\}$"]. Those are the only invertible elements in it.

So, the inertia of language: *"The units"* [writes "The units:"] — if I say *"the units"* here, that means all of them: $\{1,-1\}$ [writes "$\{1,-1\}$"]. The units.

If I say *"the unit"* [writes "The unit:"] — this is standard use of mathematical English — if I say *"the unit"*, that's just 1 [writes "1"]. That's the one. If I have a ring and say, *"let's consider the unit"*, that means 1.

You could say *"a unit"* [writes "A unit:"]. If I say, *"let's consider a ring, and let's consider a unit"*, that could be either one [writes "could be either"]. We don't know; it could be either. A unit: this could be either.

So, that's actually standard usage. It's a little confusing, but once you get used to it, you know, it's just language. Is it clear? It was pointed out to me that I was doing that, and anyway, that's right. But if someone comes to you with a ring and says *"consider the unit"*, that person means 1.

Units of the Integers and Linguistic Distinctions

Example 2.67 (*Units of the Ring of Integers $\mathbb{Z}$*): In the ring of integers $\mathbb{Z}$, the only elements possessing multiplicative inverses are 1 and -1:

$$\text{The units of } \mathbb{Z} = \{1,-1\}. \tag{2.36}$$

Remark 2.68 (*Linguistic Distinction in Mathematical English*): In mathematical English, the word *"unit"* carries three distinct grammatical usages depending on the article:

- ***"The units"* (Plural):** Refers to the entire set of all invertible elements in the ring R, denoted $\mathcal{U}_R$ (also written $R^\times$). For $\mathbb{Z}$, the units are $\{1,-1\}$.
- ***"The unit"* (Singular, Definite Article):** Refers specifically to the multiplicative identity element $1_R \in R$.
- ***"A unit"* (Singular, Indefinite Article):** Refers to an arbitrary invertible element $u \in R$ (i.e., any element $u \in \mathcal{U}_R$). For $\mathbb{Z}$, *"a unit"* could be either 1 or -1.

The Group of Units $\mathcal{U}_R$

Spoken Transcript [00:28:16 - 00:29:52] Prof. Rahul Pandharipande

We discussed the units of the Gaussian integers (Recall: $\pm 1, \pm i$, Week 1, Proposition 1.2), and now, since everything is defined, I recommend proving what the units of the Gaussian integers are. We wrote them last time, but let's do something else now. [erases upper middle board and writes "Ex. Prove units of $\mathbb{Z}[i]$ are $\{\pm 1, \pm i\}$"]

This is an exercise: Prove the units of the Gaussian integers are $\pm 1, \pm i$. You need an argument for that, and that argument comes from the norm in complex analysis so that you prove it.

Okay, so, observation [writes "Observation"] — and it's really an important observation: If I take any ring R, and let $\mathcal{U}_R$ be the set of units of R [writes "R ring" and "let $\mathcal{U}_R$ be the set of units of R"].

Then this observation is: If I take the units, I take multiplication — that's multiplication in the ring — and I take the unit 1, this is a group [writes "$(\mathcal{U}_R, \cdot, 1) \leftarrow$ Group"]...

Exercise: Units of the Gaussian Integers

Exercise 2.69 (*Units of the Gaussian Integers*)**:** Prove that the units of the ring of Gaussian integers $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ are precisely:

$$\mathcal{U}_{\mathbb{Z}[i]} = \{\pm 1, \pm i\}. \tag{2.37}$$

The lecturer writes out this proof in Week 3 (Proposition 3.2).

Spoken Transcript [00:29:52 - 00:30:47] Prof. Rahul Pandharipande

What does it say? If I take two units and I multiply them, I get a unit.

And that's clear because, you know, if I have two elements here, $u_1, u_2 \in \mathcal{U}_R$ [writes "$u_1, u_2 \in \mathcal{U}_R$"], if I multiply them [writes "$u_1 u_2$"], I have to show that's a unit. That means I have to show that if I multiply by some more stuff, I get 1.

But u_1 is a unit, so there's already its multiplicative inverse; and u_2 is a unit, so there's already its multiplicative inverse [writes "$u_1 u_2 w_1 w_2 = 1$"]. So, when I multiply these together, I get 1, because I can flip the order around. This says that the units are closed under multiplication.

And there's the neutral element; and then you have to show the existence of an inverse, but a unit is a unit because it has an inverse (Recall: Definition 2.66).

So, that's the proof of this. This is a good thing to think about: this is always a group.

The Group of Units

Proposition 2.70 (*Group of Units*)**:** Let R be a (commutative) ring with unity 1_R, and let $\mathcal{U}_R$ (also denoted $U(R)$ or $R^\times$) be the set of units of R:

$$\mathcal{U}_R := \{u \in R \mid \exists w \in R \colon u \cdot w = 1_R\}. \tag{2.38}$$

Then $(\mathcal{U}_R, \cdot, 1_R)$ is a group (Definition 2.1) under the multiplication of R.

Proof

We verify the group axioms for $(\mathcal{U}_R, \cdot, 1_R)$:

- **Closure under Multiplication:** Let $u_1, u_2 \in \mathcal{U}_R$. By definition, there exist $w_1, w_2 \in R$ such that $u_1 w_1 = 1_R$ and $u_2 w_2 = 1_R$. Since R is commutative:
$$u_1 u_2 w_1 w_2 = (u_1 w_1)(u_2 w_2) = 1_R \cdot 1_R = 1_R.$$
Thus, $u_1 u_2$ has a two-sided multiplicative inverse in R, which shows $u_1 u_2 \in \mathcal{U}_R$.
- **Associativity:** Multiplicative associativity in $\mathcal{U}_R$ is directly inherited from the ring R.
- **Identity Element:** Since $1_R \cdot 1_R = 1_R$, the unity 1_R is its own inverse, so $1_R \in \mathcal{U}_R$. For all $u \in \mathcal{U}_R$, $1_R \cdot u = u$.
- **Existence of Inverses:** For any $u \in \mathcal{U}_R$, there exists $w \in R$ such that $uw = 1_R$. By symmetry of the relation $wu = 1_R$, w is also invertible with inverse u, so $w = u^{-1} \in \mathcal{U}_R$.

Hence, $(\mathcal{U}_R, \cdot, 1_R)$ is a group in the sense of Definition 2.1. If R is commutative, $\mathcal{U}_R$ is an abelian group. **Q.E.D.**

Notation: Units in the Lecture and in Textbooks

Many textbooks write $R^\times$ for the group of units of R. The lecture does not use this symbol: the board has $\mathcal{U}_R$ here, then Unit_R and $\mathrm{Unit}(R)$ from Week 3 on, where the lecturer also calls $(\mathbb{Z}/m\mathbb{Z})^*$ the usual notation for the units of $\mathbb{Z}/m\mathbb{Z}$ and adds that R^* could denote units (Proposition 3.3). These notes follow the board.

Units in Residue Class Rings $\mathbb{Z}/m\mathbb{Z}$

Characterization of Units via Coprimality

Spoken Transcript [00:30:47 - 00:32:27] Prof. Rahul Pandharipande

A very natural question, and one of the first things to think about regarding $\mathbb{Z}/m\mathbb{Z}$... If I take this ring $R = \mathbb{Z}/m\mathbb{Z}$ [writes "$R = \mathbb{Z}/m\mathbb{Z}$"] — and that was implicit there — I will always consider $m > 0$ here when I do this [writes "$m > 0$"], because there's no reason to consider the negative ones, because it's just counting everything twice.

And then there's the confusing thing when $m = 0$, which is usually a special case, so I exclude it. Usually we allow $m = 1$, which is the zero ring, but anyway, this is kind of standard. When I write $\mathbb{Z}/m\mathbb{Z}$, I always mean this; and typically I also mean it's bigger than 1, but anyway.

So, the question is: What are the units of $\mathbb{Z}/m\mathbb{Z}$ [writes: Question: What are the units of $\mathbb{Z}/m\mathbb{Z}$?]?

Suppose we have $\bar{x} \in \mathbb{Z}/m\mathbb{Z}$, and $\bar{x}$ is a unit [writes: Suppose $\bar{x} \in \mathbb{Z}/m\mathbb{Z}$ and $\bar{x}$ is a unit]

— a unit. This means that there exists $\bar{y}$ such that $\bar{x}\bar{y} = \bar{1}$ in $\mathbb{Z}/m\mathbb{Z}$ [writes: $\exists \bar{y}$ such that $\bar{x}\bar{y} = \bar{1} \in \mathbb{Z}/m\mathbb{Z}$]. That's what it means to be a unit; I'm just repeating it, right?

Spoken Transcript [00:32:27 - 00:33:32] Prof. Rahul Pandharipande

And then I can translate: What does this mean in this ring? What does it mean for the integers x, y, and 1?

This is equivalent to: xy is equal to 1 plus some multiple of m [writes "$xy = 1 + dm$"]. Do you agree with that? That's what it means, for some d. Okay?

And this, if you remember from the first week (Recall: Bézout's Identity), means that x and m have GCD equal to 1 [writes "(x, m) have GCD $= 1$"]. The only integer that could divide both x and m would also have to divide 1. So, that's what we say: GCD equal to 1.

And in English, this condition is usually said as: x and m are relatively prime [writes "x, m are relatively prime"]. Those are just words that express GCD $= 1$, okay?

Spoken Transcript [00:33:32 - 00:35:02] Prof. Rahul Pandharipande

We have now discovered what the units of $\mathbb{Z}/m\mathbb{Z}$ are: They're all the elements which are relatively prime. Because if it is a unit, it's the residue class of a relatively prime integer. If you have GCD $= 1$, then you go backwards in this diagram.

If you write this as a result, then it says that the units of $\mathbb{Z}/m\mathbb{Z}$ consist of all elements $\bar{x}$ where $\mathrm{GCD}(x, m) = 1$ [writes: The units of $\mathbb{Z}/m\mathbb{Z}$ consist of $\{\bar{x} \mid \gcd(x, m) = 1\}$].

Did I write GCD here? Maybe GCD. Okay? Is that clear?

And so, the thing we've learned is that this unit group, the units of $\mathbb{Z}/m\mathbb{Z}$, are a group [writes "$\mathcal{U}_{\mathbb{Z}/m\mathbb{Z}}$ are a group"]. In particular, if you take $\mathbb{Z}/m\mathbb{Z}$ and you take the residue classes that are relatively prime to m, they form a group under multiplication.

Units of the Quotient Ring $\mathbb{Z}/m\mathbb{Z}$

Theorem 2.71 (*Units in* $\mathbb{Z}/m\mathbb{Z}$)**:** Let $m \in \mathbb{Z}_{>0}$ be a positive integer. An element $\bar{x} \in \mathbb{Z}/m\mathbb{Z}$ is a unit if and only if x and m are coprime (this does not depend on the representative x, since $\mathrm{GCD}(x + km, m) = \mathrm{GCD}(x, m)$):

$$\mathcal{U}_{\mathbb{Z}/m\mathbb{Z}} = \left\{ \bar{x} \in \mathbb{Z}/m\mathbb{Z} \;\middle|\; \mathrm{GCD}(x, m) = 1 \right\}. \tag{2.39}$$

Proof

We establish the equivalence by unwinding the definition of divisibility:

- **Forward Direction ($\Longrightarrow$):** Suppose $\bar{x} \in \mathcal{U}_{\mathbb{Z}/m\mathbb{Z}}$. Then there exists $\bar{y} \in \mathbb{Z}/m\mathbb{Z}$ such that:
$$\bar{x} \cdot \bar{y} = \bar{1} \in \mathbb{Z}/m\mathbb{Z}.$$
In terms of integer divisibility, this means $xy \equiv 1 \pmod{m}$. Hence, there exists

an integer $d \in \mathbb{Z}$ such that:

$$xy = 1 + d \cdot m \iff xy - d \cdot m = 1.$$

Any common divisor $g \in \mathbb{Z}$ of x and m must divide the linear combination $xy - dm = 1$. Therefore, $g \mid 1$, which forces $\operatorname{GCD}(x, m) = 1$.

- **Reverse Direction ($\Longleftarrow$):** Conversely, suppose $\operatorname{GCD}(x, m) = 1$. By Bézout's Identity (Fact 0.16; computed in Example 0.18), there exist integers $y, k \in \mathbb{Z}$ such that:

  $$x \cdot y + m \cdot k = 1 \implies x \cdot y \equiv 1 \pmod{m}.$$

  Projecting this relation into residue classes in $\mathbb{Z}/_{m\mathbb{Z}}$, we obtain $\bar{x} \cdot \bar{y} = \bar{1}$. Thus, $\bar{x}$ is invertible, so $\bar{x} \in \mathcal{U}_{\mathbb{Z}/_{m\mathbb{Z}}}$.

By Proposition 2.70, $(\mathcal{U}_{\mathbb{Z}/_{m\mathbb{Z}}}, \cdot, \bar{1})$ forms a finite abelian group. **Q.E.D.**

The Units of $\mathbb{Z}/_{7\mathbb{Z}}$ and Cyclic Generator Calculations

Spoken Transcript [00:35:02 - 00:36:08] Prof. Rahul Pandharipande

Let's do an example: $\mathbb{Z}/_{7\mathbb{Z}}$ [writes "$\mathbb{Z}/_{7\mathbb{Z}}$"]. That's actually a prime one, but let's do this example.

The elements of this are $\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}$ [writes "$= \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}\}$"], right? And which of these residue classes come here? They're classes of x where x and 7 are relatively prime.

So, the question is: What are the units? [writes "Unit of $\mathbb{Z}/_{7\mathbb{Z}} =$"]

Of course, it's some of these. Well, which of these residue classes correspond to x and 7 being relatively prime? What would you say? As I said, if you don't know, guess.

Student Answer

All of them.

Spoken Transcript [00:36:11 - 00:36:14] Prof. Rahul Pandharipande

All of them? Well, what about this one? [points at $\bar{0}$]

Student Answer

Only 1, 1 to 6 are the units.

Spoken Transcript [00:36:16 - 00:36:50] Prof. Rahul Pandharipande

Exactly. The answer to this question is these: [writes "$\{\bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}\}$"].

And the reason is because, look, this was the answer: It's the residue classes of those numbers that are relatively prime to 7. If you're $0 \pmod{7}$, then you're not relatively prime to 7. As an example, $0 \pmod{7}$ would be 14, and that's not relatively prime to

7. So, it's these.
   Is that clear?

The Unit Group of ${}^{\mathbb{Z}}\!/\!_{7\mathbb{Z}}$

Example 2.72 (*Units of* ${}^{\mathbb{Z}}\!/\!_{7\mathbb{Z}}$): Consider the quotient ring ${}^{\mathbb{Z}}\!/\!_{7\mathbb{Z}}$. The elements of the ambient ring are:

$$ {}^{\mathbb{Z}}\!/\!_{7\mathbb{Z}} = \{\bar{0},\bar{1},\bar{2},\bar{3},\bar{4},\bar{5},\bar{6}\}. $$

Since 7 is a prime number, every non-zero residue class $\bar{x} \in \{\bar{1},\bar{2},\bar{3},\bar{4},\bar{5},\bar{6}\}$ satisfies $\text{GCD}(x,7) = 1$. The zero class $\bar{0}$ is not coprime to 7 (since $\text{GCD}(0,7) = 7 \neq 1$, or for example $14 \equiv 0 \pmod 7$ where $\text{GCD}(14,7) = 7$). Consequently, the group of units is:

$$ \text{Unit}({}^{\mathbb{Z}}\!/\!_{7\mathbb{Z}}) = \{\bar{1},\bar{2},\bar{3},\bar{4},\bar{5},\bar{6}\}. \tag{2.40} $$

The order of the unit group is $|\text{Unit}({}^{\mathbb{Z}}\!/\!_{7\mathbb{Z}})| = 6$.

Beyond the Lecture: Euler's Totient Function

The order of $\mathcal{U}_{\mathbb{Z}/m\mathbb{Z}}$ is Euler's totient function $\phi(m)$, which the lecturer names in Week 3 (Definition 3.5). For example, $\phi(7) = 7 - 1 = 6$.

Spoken Transcript [00:36:50 - 00:37:28] Prof. Rahul Pandharipande

Okay, and the claim is this is a group under multiplication. Well, anyway, ${}^{\mathbb{Z}}\!/\!_{7\mathbb{Z}}$ is a group under addition; but if you take the relatively prime ones, this is a group under multiplication.

So, we should be able to figure out: What's the group?

What's the neutral element of this group? Yeah, 1 — it's going to be $\bar{1}$.

What is this group? It's a commutative group of order 6. And so, later we'll see that we know all commutative groups of order 6. We'd like to know which one this is. It's isomorphic to some group we know.

And so, it matters which element you pick. I propose we pick 3.

Spoken Transcript [00:37:28 - 00:38:51] Prof. Rahul Pandharipande

Let's say we take this element in this group, $a = \bar{3}$ [writes "$a = \bar{3} \in \text{Unit}({}^{\mathbb{Z}}\!/\!_{7\mathbb{Z}})$"]. That's an element of this unit group.

And let's try to consider its powers. That's the neutral element [writes "$\bar{1}$"], then I can take $a, a^2, a^3, a^4, a^5, a^6$ [writes "$\bar{1}, a, a^2, a^3, a^4, a^5, a^6$"]. You should know the powers of 3. Okay, let's see how many powers of 3 you know.

So, what is this one? I'm just going to do this: This is $\bar{1}$ [writes "$\bar{1}$" under $\bar{1}$].

What is a? Well, that's just $\bar{3}$ [writes "$\bar{3}$" under a].

What is this? That's the square of a. That's the residue class of 3^2, which is 9. So, I should put a 2 here, right? [writes "$\bar{2}$" under a^2] Are you happy with this?

And this is a^3: 3^3 is 27. For 27, the residue modulo 7 is 6, right? [writes “$\bar{6}$” under a^3] a^4 is 3^4, which is 81; and the residue modulo 7 of 81 is 4, I think [writes “$\bar{4}$” under a^4]. 3^5 is 243, and its residue is going to be 5 [writes “$\bar{5}$” under a^5].

Spoken Transcript [00:38:51 - 00:39:45] Prof. Rahul Pandharipande

What do you think this is going to be? [points at a^6]

Do you know what 3^6 is? 3^6 is 729 [writes “$729 = 3^6$”]. That's 3^6. And if I look at its residue class modulo 7 — well, 7 is already here, and 28 is divisible — its residue class is 1 [writes “$\bar{1}$” under a^6]. This is back to 1.

So, this is understanding the group. You should know the powers of 3; they're very useful.

This is what happens when you multiply in this group: You take a; a^2 is the next one; a^3 the next one; a^4 the next one; a^5 this one; and then when you go to the sixth, you go back to the identity.

What group do you know this happens in? Does this remind you of anything? Yeah.

Student Answer

It's a cyclic group.

Spoken Transcript [00:39:48 - 00:39:54] Prof. Rahul Pandharipande

It's $\mathbb{Z}$... The proposition that we've kind of proven here...

Powers of the Generator $a = \bar{3}$ in $\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z})$

To examine the internal structure of the abelian group $\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z})$ of order 6, we choose the candidate generator $a := \bar{3} \in \mathrm{Unit}(\mathbb{Z}/7\mathbb{Z})$ and compute its successive powers:

Power	a^0	a^1	a^2	a^3	a^4	a^5	a^6
Symbolic	$\bar{1}$	a	a^2	a^3	a^4	a^5	a^6
Value in $\mathbb{Z}$	1	3	9	27	81	243	729
Residue modulo 7	$\bar{1}$	$\bar{3}$	$\bar{2}$	$\bar{6}$	$\bar{4}$	$\bar{5}$	$\bar{1}$

Modular Arithmetic Derivations for Powers of 3:

$$\begin{aligned}
a^0 &= \bar{1}, \\
a^1 &= \bar{3}, \\
a^2 &= \overline{3^2} = \bar{9} = \bar{2} && (9 = 7 \cdot 1 + 2), \\
a^3 &= \overline{3^3} = \overline{27} = \bar{6} && (27 = 7 \cdot 3 + 6), \\
a^4 &= \overline{3^4} = \overline{81} = \bar{4} && (81 = 7 \cdot 11 + 4), \\
a^5 &= \overline{3^5} = \overline{243} = \bar{5} && (243 = 7 \cdot 34 + 5), \\
a^6 &= \overline{3^6} = \overline{729} = \bar{1} && (729 = 7 \cdot 104 + 1).
\end{aligned}$$

Because the powers of $a = \bar{3}$ exhaust all 6 elements of $\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z})$ before returning to the identity $\bar{1}$ at exponent 6, the element $\bar{3}$ is a *"generator"*. Consequently, $\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z})$ is a cyclic group of order 6 (Definition 0.25).

Isomorphism with the Additive Cyclic Group $(\mathbb{Z}/6\mathbb{Z}, +, \bar{0})$

Spoken Transcript [00:39:54 - 00:41:19] Prof. Rahul Pandharipande

We've proven... that calculation of powers of 3 — and I picked 3, you know, I picked the 3; if you picked another one, it would have maybe been slightly different — but that calculation proves a group isomorphism, which maybe I write here and we think about. [erases lower left blackboard]

You should check my powers of 3, and then after you take the power of 3, you have to take the residue modulo 7.

This great 20th-century Hungarian mathematician (Paul Erdős) said that if numbers aren't beautiful, I don't know what is. That's kind of the mathematical perspective on numbers.

We have here the units of $\mathbb{Z}/7\mathbb{Z}$ [writes "$(\mathrm{Unit}_{\mathbb{Z}/7\mathbb{Z}}, \cdot, \bar{1})$"], and then there's another group, which is $\mathbb{Z}/6\mathbb{Z}$ with $+$ and $\bar{0}$ [writes "$(\mathbb{Z}/6\mathbb{Z}, +, \bar{0})$"], and I claim these are isomorphic [writes "$\cong$"].

This is the units with multiplication and 1. This group is isomorphic to that group. And what is the isomorphism? It's written on the board.

***Insight:* Paul Erdős on the Beauty of Numbers**

The lecturer references the celebrated quote by the Hungarian mathematician Paul Erdős (1913–1996): *"If numbers are not beautiful, I don't know what is."* The unexpected isomorphism between the unit group $\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z})$ and the simple additive clock arithmetic of $\mathbb{Z}/6\mathbb{Z}$ is a classic embodiment of this aesthetic: multiplication modulo 7 transforms into addition modulo 6 via the powers of $\bar{3}$.

Spoken Transcript [00:41:19 - 00:42:48] Prof. Rahul Pandharipande

The isomorphism... This is a slightly painful way to write the isomorphism, but let's just be painful about it.

Here $\bar{1}$ is going to go to $\bar{0}$ [writes "$\bar{1} \longrightarrow \bar{0}$"]. And bar means something different: It means modulo 7 here, and modulo 6 here, right?

a — that's the 3 here, a is $\bar{3}$. So, the nice way to do it is: a goes to $\bar{1}$ [writes "$a \longrightarrow \bar{1}$"], a^2 goes to $\bar{2}$ [writes "$a^2 \longrightarrow \bar{2}$"] — I guess I could write this in slightly better notation, but anyway, you get the pattern.

Well, there's not so many more: a^3, a^4, a^5 [writes "a^3, a^4, a^5"]. This goes to $\bar{3}, \bar{4}, \bar{5}$ [writes "$\longrightarrow \bar{3}, \bar{4}, \bar{5}$"].

This is the simple way to write it. And the reason this is true is because on this side, when we multiply, we just add up the exponents; on this side, we add up those. And the important thing is the sixth one goes to zero, which is what we have here, okay?

And if you want to say this explicitly here, you have to write these a's in terms of the elements of the units, which I've done there. So, this is totally explicit.

Okay, that's that. [lecture concludes; students applaud]

Group Isomorphism: Unit($\mathbb{Z}/7\mathbb{Z}$) $\cong$ ($\mathbb{Z}/6\mathbb{Z}, +, \bar{0}$)

Proposition 2.73 (*Cyclic Group Isomorphism*)**:** The multiplicative group of units $(\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z}), \cdot, \bar{1})$ is isomorphic to the additive cyclic group $(\mathbb{Z}/6\mathbb{Z}, +, \bar{0})$:

$$(\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z}), \cdot, \bar{1}) \cong (\mathbb{Z}/6\mathbb{Z}, +, \bar{0}). \tag{2.41}$$

Proof

Let $a = \bar{3} \in \mathrm{Unit}(\mathbb{Z}/7\mathbb{Z})$. We define the mapping $\Phi \colon \mathrm{Unit}(\mathbb{Z}/7\mathbb{Z}) \to \mathbb{Z}/6\mathbb{Z}$ by sending powers of the generator a to their integer exponents modulo 6:

$$\begin{aligned} \Phi \colon \mathrm{Unit}(\mathbb{Z}/7\mathbb{Z}) &\to \mathbb{Z}/6\mathbb{Z}, \\ \bar{1} = a^0 &\mapsto \bar{0}, \\ \bar{3} = a^1 &\mapsto \bar{1}, \\ \bar{2} = a^2 &\mapsto \bar{2}, \\ \bar{6} = a^3 &\mapsto \bar{3}, \\ \bar{4} = a^4 &\mapsto \bar{4}, \\ \bar{5} = a^5 &\mapsto \bar{5}. \end{aligned}$$

Homomorphism Property: For any exponents $j, k \in \{0, 1, 2, 3, 4, 5\}$, multiplication in the unit group corresponds to addition of exponents:

$$a^j \cdot a^k = a^{j+k} = a^{(j+k) \bmod 6},$$

since $a^6 = \bar{1}$. Applying the map Φ:

$$\Phi(a^j \cdot a^k) = \overline{j+k} = \bar{j} + \bar{k} = \Phi(a^j) + \Phi(a^k) \in \mathbb{Z}/6\mathbb{Z}.$$

Hence, Φ is a group homomorphism. Because Φ maps the generator a of order 6 to the generator $\bar{1}$ of order 6, Φ is a bijective homomorphism, establishing the isomorphism.

Q.E.D.

$$\begin{tikzcd}
(\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z}),\cdot) & (\mathbb{Z}/6\mathbb{Z},+) \\
a^0=\bar{1} \arrow[r, mapsto, "\Phi"] & \bar{0} \\
a^1=\bar{3} \arrow[r, mapsto] & \bar{1} \\
a^2=\bar{2} \arrow[r, mapsto] & \bar{2} \\
a^3=\bar{6} \arrow[r, mapsto] & \bar{3} \\
a^4=\bar{4} \arrow[r, mapsto] & \bar{4} \\
a^5=\bar{5} \arrow[r, mapsto] & \bar{5}
\end{tikzcd}$$

Disambiguating Residue Class Notations Across Groups: The bar notation $\bar{\cdot}$ operates over two distinct moduli in this isomorphism:

- In the domain $(\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z}),\cdot)$, bars denote residue classes modulo 7 under multiplication: $\bar{x}=x+7\mathbb{Z}$.
- In the codomain $(\mathbb{Z}/6\mathbb{Z},+)$, bars denote residue classes modulo 6 under addition: $\bar{k}=k+6\mathbb{Z}$.

Going beyond the lecture: the isomorphism Φ is the **discrete logarithm** with respect to the base 3 modulo 7.

Summary of Key Concepts

- **Category Data and Axioms:** A category Cat consists of objects $\mathrm{Obj}(\mathrm{Cat})$, morphism sets $\mathrm{Hom}(A,B)$, an associative composition law $\circ$, and identity morphisms $\mathrm{Id}_A \in \mathrm{Hom}(A,A)$ satisfying left and right identity laws.
- **Categorical Morphisms:**
  - **Isomorphism:** A morphism $f\colon A \to B$ admitting a two-sided inverse $g\colon B \to A$ satisfying $f\circ g=\mathrm{Id}_B$ and $g\circ f=\mathrm{Id}_A$.
  - **Monomorphism:** A morphism with $(f\circ g_1 = f\circ g_2 \implies g_1=g_2)$, corresponding to injectivity in standard concrete categories.
  - **Epimorphism:** A morphism with $(g_1\circ f = g_2\circ f \implies g_1=g_2)$, corresponding to surjectivity in **Set**, **Grp** and $\mathbf{Vect}_K$.
- **Integral Domains:** An integral domain is a commutative ring with $R\neq\{0\}$ in which $ab=0$ implies $a=0$ or $b=0$. In modular arithmetic, the residue class ring $\mathbb{Z}/m\mathbb{Z}$ $(m\in\mathbb{Z}_{>0})$ is an integral domain if and only if m is prime.
- **Divisibility and Units:** An element $u\in R$ is a unit if $u\mid 1$ (i.e., $\exists w\in R$ such that $uw=1$). The units of any commutative ring R form an abelian group $(\mathcal{U}_R,\cdot,1)$ under ring multiplication.
- **Modular Units and Cyclic Generators:** The unit group $\mathcal{U}_{\mathbb{Z}/m\mathbb{Z}}$ consists precisely of the residue classes coprime to m: $\mathcal{U}_{\mathbb{Z}/m\mathbb{Z}}=\{\bar{x}\in\mathbb{Z}/m\mathbb{Z}\mid \mathrm{GCD}(x,m)=1\}$. For $p=7$, $\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z})=\{\bar{1},\dots,\bar{6}\}$ is a cyclic group of order 6 generated by

$a = \bar{3}$, isomorphic to the additive cyclic group $(\mathbb{Z}/6\mathbb{Z}, +, \bar{0})$.

Tuesday, 29th September 2026

Groups of Units, Fields of Fractions, and Polynomial Rings

Course Content Overview

This lecture investigates the transition from commutative rings to fields and polynomials, focusing on the invertible elements of rings, the construction that produces the smallest field containing an integral domain, and the algebraic construction of polynomial rings:

- **The Group of Units $\mathrm{Unit}(R)$:** Structure of multiplicatively invertible elements, characterization of units in $\mathbb{Z}$ and the Gaussian integers $\mathbb{Z}[i]$ via the norm, and the group of units of residue class rings $({}^{\mathbb{Z}}\!/\!_{m\mathbb{Z}})^*$.
- **Euler's Totient Function $\phi(m)$:** Definition as the order of $({}^{\mathbb{Z}}\!/\!_{m\mathbb{Z}})^*$, with explicit computations for prime powers $\phi(p) = p-1$ and $\phi(p^2) = p(p-1)$.
- **Universal Property of $\mathbb{Q}$:** Rigorous formulation of $\mathbb{Q}$ as the smallest field containing $\mathbb{Z}$, and the unique extension of ring embeddings into fields.
- **Construction of the Field of Fractions $\mathrm{Frac}(R)$:** Generalization to arbitrary integral domains via equivalence classes of formal pairs $(x,y) \in R \times (R \setminus \{0\})$, verification of well-defined operations, and canonical embedding $i\colon R \hookrightarrow \mathrm{Frac}(R)$.
- **Concrete Quotient Fields:** Field of fractions of $\mathbb{Z}$, the Gaussian rationals $\mathbb{Q}[i] = \mathrm{Frac}(\mathbb{Z}[i])$, subrings of $\mathbb{Q}$ such as $\mathbb{Z}_{1/3}$, and self-isomorphism for existing fields $\mathrm{Frac}(R) \cong R$.
- **Polynomial Rings $R[x]$:** Intuitive formulation via formal linear combinations, rigorous foundation using sequences in $R^{\mathbb{N}_0}$ that are eventually zero, componentwise addition, and Cauchy product multiplication.

Review: The Group of Units of a Ring

Spoken Transcript [00:00:00 - 00:01:40] Prof. Rahul Pandharipande

Okay, so, we start. About the quiz: as I said, I'm very much pro-guessing in mathematics — you're never penalized for getting the wrong answer. Lycka said that there was some implementation problem and some people were penalized for getting the right answer, and she's going to fix that.

If you think the quiz was too easy, too hard, or too medium, tell us. As I said, I've never gone through this quiz thing before.

Last week we discussed rings, [writes: $(R,+,\cdot,0,1)$] and associated to a ring, there is an additive group. [draws arrow to $(R,+,0)$] This is easy, this is part of the definition. But there's another group, which was also discussed in the last lecture (Recall: Proposition 2.70), which is the group of units. [draws arrow and writes "the group of unit"] And I remind you that this group... [writes: Unit_R] Various people use different notation for describing it, but here I just say $\mathrm{Unit}(R)$, so there's no question: it's the group of units of the ring R.

It's those elements such that there exists somebody else where the product is 1 [writes: $= \{x \mid \exists y, xy = 1\}$] — so, it's those elements with a multiplicative inverse. And this is a group; we discussed this last week. So, this group here is $\mathrm{Unit}(R)$, and it has, well, the units by this definition, multiplication, and 1. [writes: $\mathrm{Unit}_R = (\mathrm{Units}, \cdot, 1)$]

Whenever you see a ring, you immediately have these two groups. And this one is often the more complicated one to think about, because it's not part of the definition in some sense.

The Group of Units

Recap: Units and the group of units were introduced in Week 2 as $\mathcal{U}_R$ (Definition 2.66, Proposition 2.70), with $\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z})$ in the example.

Let $(R,+,\cdot,0,1)$ be a ring with unity. Associated with R are two natural canonical groups:

(1) The Additive Group: $(R,+,0)$, which forms an abelian group by the ring axioms.

(2) The Group of Units: The set of multiplicatively invertible elements in R:

$$\mathrm{Unit}_R = \mathrm{Unit}(R) := \{x \in R \mid \exists y \in R \text{ such that } xy = 1\}. \tag{3.1}$$

Under ring multiplication, $(\mathrm{Unit}(R), \cdot, 1)$ forms a group with identity element 1.

Multiplicative Group Structure: The group of units is closed under multiplication: if $x, z \in \mathrm{Unit}(R)$ with inverses $x^{-1}, z^{-1} \in R$, then $(xz)(z^{-1}x^{-1}) = 1$, so $xz \in \mathrm{Unit}(R)$. Associativity is inherited from the ring, and the multiplicative identity 1 is invertible since $1 \cdot 1 = 1$.

Notation: The Lecturer Changes $\mathcal{U}_R$ to Unit_R

The lecturer changes his own notation here. Week 2 wrote $\mathcal{U}_R$ for the group of units (Proposition 2.70); from now on the board has Unit_R, and he says $\mathrm{Unit}(R)$. Both mean the same group.

Units in the Integers and the Gaussian Integers

Spoken Transcript [00:01:40 - 00:02:45] Prof. Rahul Pandharipande

We'd seen this — I just review, but you know this: if you take $\mathbb{Z}$, [writes: $\mathbb{Z} \to$ Units $\{\pm 1\}$] the units are ± 1. And I said this many times, but maybe since I didn't actually ever give a proof: [writes: $\mathbb{Z}[i] \to$ Units $\{\pm 1, \pm i\}$] the units are $\pm 1, \pm i$ (Recall: Exercise 2.69).

If you're taking complex analysis, you should be able to prove this. But let's actually write a proof; I don't think I ever wrote the proof for this.

So, proof. [writes "Proof"] This is proof that these are the only invertible elements in this ring. Is that clear what I'm trying to prove?

Proof: let $a + bi \in \mathbb{Z}[i]$ be a unit. [writes "Let $a + bi \in \mathbb{Z}[i]$ be a unit."]

Spoken Transcript [00:02:45 - 00:03:55] Prof. Rahul Pandharipande

When I write this, of course that means that a and b are integers, okay? [writes: $(a, b \in \mathbb{Z})$] That's what this being in there means.

All right, so, then there exists somebody else, $c + di$, also here, [writes: Then $\exists\, c + di \in \mathbb{Z}[i]$] such that $(a + bi)(c + di) = 1$. [writes: s.t. $(a + bi)(c + di) = 1$] And this is multiplication of complex numbers. The whole idea of the proof is understanding multiplication of complex numbers.

If you've taken a complex numbers class, or you've met complex numbers, however you know: when you multiply complex numbers, there's a length of a complex number [sketches axes and a complex number vector] — the norm of the complex number. So, if you multiply two complex numbers, their angles add and their lengths multiply, right?

So, we just take the norm — the lengths — take the norms. [writes "take the norms:"] The norm here is the square root of $a^2 + b^2$, [writes: $\sqrt{a^2 + b^2}$] that's the length of this one. I hope that's not controversial — that's Euclid or something. This one is the square root of $c^2 + d^2$, [writes: $\cdot\sqrt{c^2 + d^2}$] and their norms multiply, so this has to be just 1. [writes: $= 1$] Okay?

Spoken Transcript [00:03:55 - 00:05:14] Prof. Rahul Pandharipande

This is pretty restrictive: a, b, c, and d are all integers. So, we can square that equation: [writes: $(a^2 + b^2)(c^2 + d^2) = 1$] $(a^2 + b^2)(c^2 + d^2) = 1^2$, which is 1.

Now you have an equation where you have integers all multiplying to 1, and you know the only way this can happen is if this integer itself is ± 1 — because you can't factor 1 in integers without having just $1 \cdot 1$ or $(-1)(-1)$. Is that clear?

So, how could this be 1? Well, a^2 for an integer is either 0, 1, or something bigger. So, the only way this could be ± 1 is if $a = 0$ or ± 1, and $b = 0$ or ± 1 [writes "$a = 0, \pm 1$" and "$b = 0, \pm 1$"] — those are the only cases. If these are both positive integers, it's going to be bigger than 1. So, you only have to check these cases. Is that clear? And if you look at these cases, well, these are the answers. [points at $\{\pm 1, \pm i\}$]

Okay? So, that's a complete proof — that's as complete a proof as you get in this class. I think this proof was 100 percent complete. But that's a nice fact, and if you draw them, they look like this: [draws a coordinate plane with four points at $1, -1, i, -i$] Those are the four units in the Gaussian integers.

Classification of Units in the Gaussian Integers

Recap: The units of $\mathbb{Z}$ were determined in Week 2 (Example 2.67), and those of $\mathbb{Z}[i]$ were stated in Week 1 and set as an exercise (Exercise 2.69). The proof below settles that exercise.

Example 3.1 (*Units of the Integers*): For the ring of integers $\mathbb{Z}$:

$$\mathrm{Unit}(\mathbb{Z}) = \{1, -1\}.$$

Proposition 3.2 (*Units of the Gaussian Integers*): The group of units of the Gaussian integers $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ is:

$$\mathrm{Unit}(\mathbb{Z}[i]) = \{1, -1, i, -i\}.$$

Proof

Let $z = a + bi \in \mathbb{Z}[i]$ be a unit with $a, b \in \mathbb{Z}$. By definition, there exists $w = c + di \in \mathbb{Z}[i]$ ($c, d \in \mathbb{Z}$) such that:

$$(a + bi)(c + di) = 1. \tag{3.2}$$

Taking the norms (the lengths of the complex numbers) on both sides of (3.2):

$$|a + bi| \cdot |c + di| = |1| = 1 \implies \sqrt{a^2 + b^2} \cdot \sqrt{c^2 + d^2} = 1.$$

Squaring this relation yields an equation in non-negative integers:

$$(a^2 + b^2)(c^2 + d^2) = 1. \tag{3.3}$$

Since $a^2 + b^2 \in \mathbb{Z}_{\geq 0}$ and divides 1, we must have:

$$a^2 + b^2 = 1.$$

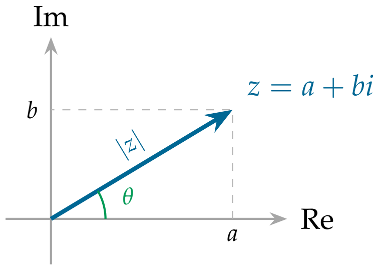
Because $a, b \in \mathbb{Z}$, the only integer solutions to $a^2 + b^2 = 1$ are:

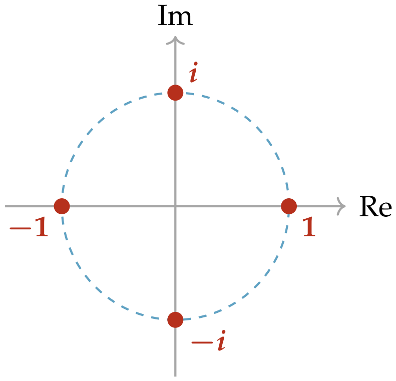
$$a = \pm 1, \quad b = 0 \implies z = \pm 1,$$
$$a = 0, \quad b = \pm 1 \implies z = \pm i.$$

Conversely, $1 \cdot 1 = 1$, $(-1)(-1) = 1$, and $i(-i) = 1$, so all four elements are invertible. Thus:

$$\mathrm{Unit}(\mathbb{Z}[i]) = \{1, -1, i, -i\}.$$

Q.E.D.





Beyond the Lecture: The Norm in Textbooks

The lecturer's *"norm"* is the length $\sqrt{a^2+b^2}$ of a complex number, and he squares the equation only at the end. Textbooks usually work with the square directly: the **field norm** $N(a+bi) = a^2+b^2$ of $\mathbb{Q}[i]$ over $\mathbb{Q}$ is multiplicative, $N(zw) = N(z)N(w)$, and takes non-negative integer values on $\mathbb{Z}[i]$. So $zw = 1$ gives $N(z)N(w) = N(1) = 1$ in one step, hence $N(z) = 1$. The only lattice points on the circle of radius 1 are the four points shown above.

The Group of Units of $\mathbb{Z}/m\mathbb{Z}$

Spoken Transcript [00:05:14 - 00:07:18] Prof. Rahul Pandharipande

We did a much more interesting example last time, which was also with proof, but since various people asked me about it afterwards, I want to do just another example. And that was this ring: $\mathbb{Z}/m\mathbb{Z}$. [writes: $\mathbb{Z}/m\mathbb{Z}$] That's a nice ring, and so it has a group of units: $\mathrm{Unit}(\mathbb{Z}/m\mathbb{Z})$. [writes: $\to$ Units $\mathbb{Z}/m\mathbb{Z}$]

This was a result I proved last time (Recall: Theorem 2.71): the units, this is equal to those classes such that $\mathrm{GCD}(x,m) = 1$. [writes: $= \{\bar{x} \mid (x,m) = 1\}$] And as I said, it's standard here to say m is at least 2, so we're not with the zero ring. [draws a box around $m \geq 2$] That's the units, and these are the residue classes which are relatively prime. And I did an example last time (Recall: $\mathbb{Z}/7\mathbb{Z}$, Example 2.72). This is a very interesting thing to study; it's just not a trivial thing.

In the sense that, one might say that if I take here the additive structure, [writes: $\mathbb{Z}/m\mathbb{Z} \to (\mathbb{Z}/m\mathbb{Z}, +, 0)$] this is very simple to think about. It's just adding up integers mod$m$. In some sense, most people learn about this under one name or the other in school. But this thing — the units — are more complicated, are more interesting. [writes "units are more interesting"]

And so I wanted to do another... there were various people asking me some questions, so, let's try this: $10\mathbb{Z}$, just to say a little bit.

The standard notation for this is that: [writes: $(\mathbb{Z}/m\mathbb{Z})^*$] This is the usual notation for units of $\mathbb{Z}/m\mathbb{Z}$, and also this notation is sometimes used in general. R^* sometimes denotes — could denote — units, depends on the author. So, I've just been writing Unit, so there's no question about what I mean. But in this case, it's so common to use the star here, I also use the star. Is that clear? So, this is it.

Example: Units of $\mathbb{Z}/10\mathbb{Z}$

Spoken Transcript [00:07:18 - 00:08:43] Prof. Rahul Pandharipande

How to think about this? Well, I'll just do an example because, as I said, there were various questions. This is all the residue classes that are relatively prime. So, you can just write them all first and cross them out: $\bar{0},\bar{1},\bar{2},\bar{3},\bar{4},\bar{5},\bar{6},\bar{7},\bar{8},\bar{9}$. [writes: $\bar{0},\bar{1},\bar{2},\ldots,\bar{9}$] Maybe I should have picked something less than 10.

These are all the residue classes, and then we have to cancel all the ones that are not relatively prime.

Which one? Is $\bar{0}$ relatively prime? No, so, we cross that out. [crosses out $\bar{0}$] $\bar{1}$ is relatively prime — $\bar{1}$ is always relatively prime. What about $\bar{2}$? No. [crosses out $\bar{2}$] $\bar{3}$? Yes, so, we skip it. $\bar{4}$? That has a 2 in it and 10 has a 2, so, we cross that out. [crosses out $\bar{4}$] $\bar{5}$ — 10 has a 5 in it, we cross that one out. [crosses out $\bar{5}$] Cross that one out, [crosses out $\bar{6}$ and $\bar{8}$] and what's left? Four, yeah, so, these are the four residue classes left. [circles $\bar{1},\bar{3},\bar{7},\bar{9}$]

The group of units here consists of these residue classes: [writes: $\{\bar{1},\bar{3},\bar{7},\bar{9}\} \subset (\mathbb{Z}/10\mathbb{Z})^*$] $\{\bar{1},\bar{3},\bar{7},\bar{9}\}$. And the claim is: this is a group under multiplication. That's what that claim is.

For example, $\bar{3}$ is in this group, so, what is its inverse? [writes: $\bar{3} \in (\mathbb{Z}/10\mathbb{Z})^*$, what is its inverse?]

Student Answer

7.

Spoken Transcript [00:08:44 - 00:09:45] Prof. Rahul Pandharipande

Seven, because $7 \times 3 = 21$, and so on. So anyway, you can play with this.

This is a more interesting object, and in fact you could ask: what is the order? This is a basic question in elementary number theory: what is the order of this group? [writes "What is the order of $(\mathbb{Z}/m\mathbb{Z})^*$?"]

And by order, this word means for groups the number of elements. [writes "# of elements"] So, when I say order of a group, if it's a finite group, I just mean the number of elements. What is the order of this? Sometimes this order is denoted by just straight lines — the bars: [writes: $|(\mathbb{Z}/m\mathbb{Z})^*|$] You should be familiar with this set-theoretic notation; these two bars usually denote order. Is that clear? That's kind of standard set-theoretic notation.

First of all, let's do 10. What is this? Well, the answer is 4 — you just have to compute these. [writes: $|(\mathbb{Z}/10\mathbb{Z})^*| = 4$]

Units in $\mathbb{Z}/m\mathbb{Z}$ and the Example $\mathbb{Z}/10\mathbb{Z}$

Recap: This characterization was proved in Week 2 (Theorem 2.71); the lecturer restates it with the notation $(\mathbb{Z}/m\mathbb{Z})^*$.

Proposition 3.3 (*Units in the Residue Class Ring*): Let $m \in \mathbb{Z}_{\geq 2}$. The group of units of the residue class ring $\mathbb{Z}/m\mathbb{Z}$ (frequently denoted by $(\mathbb{Z}/m\mathbb{Z})^*$ or R^* for general rings) consists of those residue classes whose representatives are coprime to m (on the board: $(x,m) = 1$):

$$(\mathbb{Z}/m\mathbb{Z})^* = \mathrm{Unit}(\mathbb{Z}/m\mathbb{Z}) = \{\bar{x} \in \mathbb{Z}/m\mathbb{Z} \mid \mathrm{GCD}(x,m) = 1\}. \tag{3.4}$$

Example 3.4 (*Computing the Units of* $\mathbb{Z}/10\mathbb{Z}$): Consider the ring $\mathbb{Z}/10\mathbb{Z} = \{\bar{0},\bar{1},\bar{2},\bar{3},\bar{4},\bar{5},\bar{6},\bar{7},\bar{8},\bar{9}\}$. To find the units, we eliminate all classes that share a common divisor with 10 greater than 1:

- **Class $\bar{0}$:** $\mathrm{GCD}(0,10) = 10 \neq 1$ (eliminate $\bar{0}$),
- **Even classes:** $\mathrm{GCD}(2,10) = 2$, $\mathrm{GCD}(4,10) = 2$, $\mathrm{GCD}(6,10) = 2$, $\mathrm{GCD}(8,10) = 2$ (eliminate $\bar{2},\bar{4},\bar{6},\bar{8}$),
- **Class $\bar{5}$:** $\mathrm{GCD}(5,10) = 5$ (eliminate $\bar{5}$).

The remaining coprime residue classes form the group of units:

$$(\mathbb{Z}/10\mathbb{Z})^* = \{\bar{1},\bar{3},\bar{7},\bar{9}\}.$$

This is an abelian group of order 4 under multiplication modulo 10. For instance, the inverse of $\bar{3}$ is $\bar{7}$, since:

$$\bar{3} \cdot \bar{7} = \overline{21} = \bar{1} \in \mathbb{Z}/10\mathbb{Z}.$$

Notation: The Lecturer Adds $(\mathbb{Z}/m\mathbb{Z})^*$

The lecturer changes his notation again here. For the units of $\mathbb{Z}/m\mathbb{Z}$ he now writes $(\mathbb{Z}/m\mathbb{Z})^*$, which the board calls the usual notation, and adds that R^* could denote units. The condition $\mathrm{GCD}(x,m) = 1$ of Week 2 now appears on the board as $(x,m) = 1$. The next lecture goes back to $\mathrm{Unit}(R)$.

Euler's Totient Function

Spoken Transcript [00:10:36 - 00:12:03] Prof. Rahul Pandharipande

You can ask this question in general, and that's a fun thing to think about. We'll say more about this a bit later, but the definition can be given now: [writes "Order of $(\mathbb{Z}/m\mathbb{Z})^* \overset{\text{def}}{=} \phi(m)$"] the order of $(\mathbb{Z}/m\mathbb{Z})^*$, the group of units, has a name.

Do you know the name of this function? Yes, yeah, it's ϕ... It's Euler's totient function, that's just the definition. That's the definition of this function — it's just the order.

And you could ask: can you compute this? There's different ways to compute it, but it has to do with the factorization, and we will get to that later. So, you could say — this is maybe a question for you now: [writes "Question: Try to compute $\phi(m)$"] Question: Try to compute Euler's totient function.

Euler is like the most famous Swiss person that ever lived. A few years ago it might have been Federer, but now he's retired, so now it's Euler. And it turns out that Euler and Federer are both from Basel — you can check it.

You know, my first trip to Switzerland was when I was a kid in the '80s, and the Swiss still had Euler on the 10 Swiss franc note. I don't know if you've ever seen that; it's a beautiful note. You should go look it up.

Evaluating $\phi(p)$ and $\phi(p^2)$ for a Prime p

Spoken Transcript [00:12:03 - 00:12:15] Prof. Rahul Pandharipande

Okay, so, let's do one example to compute this. Well, let's do two examples. Suppose you have a prime p [writes: $\phi(p) =$] — then what is this?

Student Answer

$p-1$.

Spoken Transcript [00:12:16 - 00:12:20] Prof. Rahul Pandharipande

Yeah, $p-1$. And why is that?

Student Answer

Because any number less than p is coprime to p.

Spoken Transcript [00:12:22 - 00:13:48] Prof. Rahul Pandharipande

In other words, if you write this, you'll write p elements, and you'll cross out 0, and everybody else is relatively prime. So, it's $p-1$. [writes: $p-1$]

And so, what about this one? [writes: $\phi(p^2) = ?$] Do you know how to do this? What you do — I mean, of course once you do this, you can do the general case for powers of a prime — you write all the elements. This is a really primitive way to do it, but let's just do it: $0, 1, 2, \ldots, p^2-1$. [writes: $0, 1, 2, \ldots, p^2-1$] That's what we did there, right?

So, it starts out, and then you cross out all of the... here you're going to cross out 0, so, it's p^2-1, but you have to take away... of course you don't want 0, but you have to take away all the multiples of p, right?

And so, now you have to concentrate really hard: how many multiples of p? Well, there's p here, there's $2p$ here, and what's the last one? The last one you have to cross out is $p(p-1)$ — that's the last one. [writes: $p, 2p, \ldots, p(p-1)$] Is that clear?

So, we have to take away $p-1$, so, the answer is p^2-p. [writes: $(p^2-1)-(p-1) = p^2-p$] The answer to this is p^2-p. Okay, that's the beginning; you can play with it a bit. All right?

Euler's Totient Function and Prime Power Formulas

Definition 3.5 (*Euler's Totient Function*): The *"Euler totient function"* (or Euler's ϕ-function) $\phi\colon \mathbb{Z}_{\geq 1} \to \mathbb{Z}_{\geq 1}$ is defined as the order of the group of units of ${}^{\mathbb{Z}}/_{m\mathbb{Z}}$:

$$\phi(m) \overset{\text{def}}{=} |({}^{\mathbb{Z}}/_{m\mathbb{Z}})^*| = \#\{x \in \{0,1,\dots,m-1\} \mid \mathrm{GCD}(x,m) = 1\}. \tag{3.5}$$

Proposition 3.6 (*Totient Values for Primes and Prime Squares*): Let p be a prime number. Then:

$$\phi(p) = p - 1, \tag{3.6}$$
$$\phi(p^2) = p^2 - p = p(p-1). \tag{3.7}$$

Proof

We evaluate the number of coprime residue classes in each case:

- **For a prime p:** The representatives are $\{0,1,2,\dots,p-1\}$. The only representative sharing a non-trivial factor with p is 0. Thus, the remaining $p-1$ elements are all coprime to p, giving $\phi(p) = p - 1$ (e.g., $\phi(7) = 6$, as in Example 2.72).
- **For p^2:** The representatives are $\{0,1,2,\dots,p^2-1\}$. An element x shares a non-trivial factor with p^2 if and only if $p \mid x$. The multiples of p in this range are:

  $$0,\ p,\ 2p,\ \dots,\ (p-1)p.$$

  There are exactly p such multiples. Subtracting these from the total of p^2 elements yields:

  $$\phi(p^2) = p^2 - p = p(p-1).$$

Q.E.D.

Generalization to Prime Powers: Going beyond the lecture (he returns to this in Week 5): for any prime power p^k with $k \geq 1$, an integer $x \in \{0,1,\dots,p^k-1\}$ shares a factor with p^k if and only if $p \mid x$. The multiples of p are $0, p, 2p, \dots, (p^{k-1}-1)p$, totaling p^{k-1} elements. Hence, $\phi(p^k) = p^k - p^{k-1} = p^k(1-1/p)$. For a general n, the Chinese Remainder Theorem splits ${}^{\mathbb{Z}}/_{n\mathbb{Z}}$ into prime-power factors, which gives $\phi(n) = n\prod_{p|n}(1-1/p)$ (Week 5, Theorem 5.52).

Fields of Fractions

Spoken Transcript [00:13:48 - 00:15:30] Prof. Rahul Pandharipande

The next topic in the course has to do with fields of fractions. [writes "Fields of fractions"] And this is a peculiar subject.

It starts with this idea that $\mathbb{Z}$ is inside $\mathbb{Q}$. [writes: $\mathbb{Z} \subset \mathbb{Q}$] So, $\mathbb{Z}$ is a ring, it's an integral domain [writes arrow from $\mathbb{Z}$ to "int. domain"] and $\mathbb{Q}$ is a field, [writes arrow from $\mathbb{Q}$ to "field"] and $\mathbb{Z}$ sits inside $\mathbb{Q}$. But the relationship is not just of any ring inside any field. In some sense, $\mathbb{Q}$ is the smallest field that contains $\mathbb{Z}$. This understanding will lead to the study of the field of fractions.

Let's try to first make this precise; let's try to make precise the qualitative statement that... [writes "Make precise the statement that "$\mathbb{Q}$ is the smallest field which contains $\mathbb{Z}$""] Here's the statement: *"$\mathbb{Q}$ is the smallest field which contains $\mathbb{Z}$"*. That's a statement that probably you feel is true. It's not a 100 percent mathematical statement, so, let's try to make this precise.

You don't contradict the meaning of this statement, right?

Spoken Transcript [00:15:30 - 00:17:25] Prof. Rahul Pandharipande

[moves the boards and begins erasing the lower board] One thing that when one starts studying these fields of fractions and $\mathbb{Q}$'s, you get this kind of strange feeling that maybe you never checked everything in $\mathbb{Q}$ works. Have you checked everything in $\mathbb{Q}$ works?

It's kind of a strange feeling, because you learn about the rational numbers in school, but it's not like the definitions there are checked to work. You just accept them, which is pretty reasonable — they are correct.

And what do I mean by that? $\mathbb{Q}$ is a much more complicated thing than $\mathbb{Z}$ conceptually, because $\mathbb{Z}$ is really great: there are these objects and they have names, and everybody has one name. That's what's great about $\mathbb{Z}$.

The thing that's confusing about $\mathbb{Q}$ is that they're objects, but they have many names. You know, you can have... [writes: $5/7 = 60/84$] That's kind of confusing: all these numbers have many, many different names. And when I said, *"Have you ever checked it works?"*, it means that whatever operations you do, you have to be able to do them with any name, and it has to give the same answer, so to speak. That's what I mean. And so, the question is: have you ever checked that?

It's a weird question. But kind of the ideal math question is in first grade or second grade, whenever you're adding fractions, you should ask, say: *"But why is it well-defined?"* That's the question you should ask. And if you'd asked that question, then we wouldn't have to give this lecture today, actually.

Multiple Representations of Rational Numbers and Well-Definedness

A fundamental conceptual distinction between $\mathbb{Z}$ and $\mathbb{Q}$ is the uniqueness of representation:

- **Integers ($\mathbb{Z}$):** Each element has a unique canonical name: $0, 1, -1, 2, -2, \ldots$
- **Rationals ($\mathbb{Q}$):** Each number has infinitely many equivalent representations as a quotient of integers:

$$\frac{5}{7} = \frac{60}{84} = \frac{10}{14} = \ldots$$

Consequently, every operation on fractions (addition, multiplication, evaluation) must be proved to be *"well-defined"*, meaning the resulting value is strictly independent of the chosen fractional representative.

The Universal Property of $\mathbb{Q}$

Spoken Transcript [00:17:25 - 00:19:32] Prof. Rahul Pandharipande

I try to make precise the statement that $\mathbb{Q}$ is the smallest field which contains $\mathbb{Z}$. And there's different ways to do it, but one way is to say it like this:

Suppose I have $f\colon \mathbb{Z} \to F$, [writes: Suppose $f\colon \mathbb{Z} \to F$] where F is a field [writes arrow from F to "field"] — this big F is a field, and this little f is an injection of rings, so, an injective homomorphism of rings. [writes "injection of rings / homomorphism of rings"]

So, I start with this: I take any possible field — maybe a field that I might think is smaller than $\mathbb{Q}$, or whatever field, take any field you want — and I start with a ring homomorphism which is injective. That means that I view this as $\mathbb{Z}$ sitting inside F, okay?

Then I claim... [writes "Then I claim $\exists!\, g$"] The typical way you write this is with a diagram: this is f, then I claim there exists a unique g... So, this is $\mathbb{Z}$ sitting inside $\mathbb{Q}$, and here there exists a unique g [draws diagram with $\mathbb{Z} \to \mathbb{Q}$, $\mathbb{Z} \to F$, and dashed arrow $\mathbb{Q} \to F$ labeled g] such that g is an injective homomorphism of fields. [writes "such that g is inj. hom of fields"]

And the diagram commutes, meaning I can take an integer here and go here, or I consider it in $\mathbb{Q}$ and go this way. We would say the diagram commutes: if you have x here, I can get to here going straight across, or I can go inside $\mathbb{Q}$ and come around, so, that would say that $f(x) = g(x)$. [writes "s.t. $\forall x \in \mathbb{Z}$, $f(x) = g(x)$"]

Universal Property of the Rational Field

Qualitative Observation: $\mathbb{Q}$ is the *"smallest"* field containing the integral domain $\mathbb{Z}$.

Theorem 3.7 (*Universal Property of $\mathbb{Q}$*): Let F be a field, and let $f\colon \mathbb{Z} \to F$ be an injective ring homomorphism (embedding of rings). Then there exists a unique field homomorphism $g\colon \mathbb{Q} \to F$ such that the following diagram commutes:

$$\begin{tikzcd}
\mathbb{Q} \arrow[dr, dashed, "\exists!\, g"] & \\
\mathbb{Z} \arrow[u, hook, "\iota"] \arrow[r, "f"'] & F
\end{tikzcd}$$

that is, $g|_{\mathbb{Z}} = f$, meaning $g(x) = f(x)$ for all $x \in \mathbb{Z}$. Furthermore, g is automatically injective.

Injectivity is automatic, because every ring homomorphism from a field to a non-zero ring is injective (see Theorem 1.31 and the steps after it, Week 1).

Universal Property as Categorical Minimality: Going beyond the lecture: in category theory, this universal mapping property identifies $\mathbb{Q}$ as the initial object among the fields containing a copy of $\mathbb{Z}$, that is, the fields of characteristic 0. Any field F containing a copy of $\mathbb{Z}$ must necessarily contain a subfield isomorphic to $\mathbb{Q}$, showing that $\mathbb{Q}$ cannot be compressed or embedded into any strictly smaller field.

Construction and Well-Definedness of the Field Extension

Spoken Transcript [00:19:32 - 00:21:20] Prof. Rahul Pandharipande

I'll give a little proof of this. This is now totally precise mathematics; it makes that statement, which is somehow slightly qualitative, into a mathematical statement.

Do you accept that? When I say $\mathbb{Q}$ is the smallest field which contains $\mathbb{Z}$, it means that if I have any other candidate field F, I can actually squeeze $\mathbb{Q}$ into it. And that means that $\mathbb{Q}$ is smaller than F. Does that make sense?

Okay, but I haven't proven it, so, let's try to prove it. I won't prove the whole thing, because it's of course kind of painful.

So, let us see: I have to prove this, which means I get to start with the data I suppose. That's what *"suppose"* is about — *"suppose"* means I get to start with this data, which is here, all right? And then the whole claim is existence. To prove this, what I have to do is produce g.

Proof, more or less: [writes "Proof I have to construct g"] I have to construct g. So, I write down... [self-correction] a complex number, x over y, where x and y are integers and $y \neq 0$ — sorry, a rational number, that's what a rational number is: x/y, where $x, y \in \mathbb{Z}$ and $y \neq 0$. [writes: $g(x/y) =, x, y \in \mathbb{Z}, y \neq 0$] I hope you're happy with that; that's what a rational number is.

This is my responsibility: I have to define this, or we define. [writes "we define"] So, how am I going to define this?

Student Answer

$f(x)/f(y)$.

Spoken Transcript [00:21:23 - 00:21:51] Prof. Rahul Pandharipande

Yeah, so, formally, of course we'll write it the way you said, but formally we write it like this: [writes: $= f(x)f(y)^{-1}$] $f(x)f(y)^{-1}$.

Because $f(x)$ is in the field F, and $f(y)$ is in the field F, and I want to take its inverse. This is the kind of field language. Why am I allowed to take its inverse? You cannot take inverses of everything.

Student Answer

Because F is a field and we chose, um... $y \neq 0$.

Spoken Transcript [00:21:58 - 00:23:11] Prof. Rahul Pandharipande

Yeah, but you have to check — and this is the whole thing, which is that you have to check. This is legal so long as $f(y) \neq 0$. [writes "legal as long as $f(y) \neq 0$"] And why is $f(y)$ not zero? It's because f is injective, and $y \neq 0$. [writes "and this is because f is injective and $y \neq 0$"] That's actually using this thing here. So, this is legal. Is that clear?

And as was implicit in the answer, we will often write this as that, [writes: $= \frac{f(x)}{f(y)}$] because it's charming to do that. And in a field, if you see this, all that means is that the fellow downstairs can be put upstairs by multiplicative inverse. And you're only legally allowed to write it if you know this is not zero — just like in normal arithmetic, you can't divide by zero.

When we write this, it means various things: it means in particular this one's not zero, and you can put it up here, and you can put it on either side, it doesn't matter because our fields are commutative. Okay?

Candidate Definition of the Field Homomorphism g

Definition 3.8 (*Candidate Map for Field Embedding*)**:** Let $f\colon \mathbb{Z} \to F$ be an injective ring homomorphism into a field F. For a rational number represented by $\frac{x}{y} \in \mathbb{Q}$ with $x, y \in \mathbb{Z}$ and $y \neq 0$, we define:

$$g\left(\frac{x}{y}\right) \overset{\text{def}}{=} f(x) \cdot f(y)^{-1} = \frac{f(x)}{f(y)} \in F. \tag{3.8}$$

Legitimacy of Inversion in the Field F: The inverse $f(y)^{-1}$ exists in F if and only if $f(y) \neq 0_F$:

(1) Non-Zero Denominator: In the rational representation $\frac{x}{y}$, we require $y \neq 0$ in $\mathbb{Z}$.

(2) Injectivity of Ring Embedding: Since $f\colon \mathbb{Z} \to F$ is an injective ring homomorphism, its kernel is trivial: $\ker(f) = \{0\}$.

(3) Non-Vanishing Image: It follows immediately that $y \neq 0 \implies f(y) \neq f(0) = 0_F$ (Proposition 1.19). Since F is a field, every non-zero element has a unique multiplicative inverse $f(y)^{-1} \in F$.

Spoken Transcript [00:23:11 - 00:25:18] Prof. Rahul Pandharipande

We managed to prove... But is this a definition? The answer is, of course, not: all you did was pick a particular name of this, and you gave a definition. You have to check that all the names give the same definition. That's the business about the rational numbers, which no one ever thinks about — and it's better not to think about it, but unfortunately today we have to think about it. [moves to the lower board to erase and write]

What do we have to do? [writes "We are done with the Def of g"] We are done with the definition of g. But we must further check that if $x_1/y_1 = x_2/y_2$, [writes "We must check that if $\frac{x_1}{y_1} = \frac{x_2}{y_2}$"] then $g(x_1/y_1) = g(x_2/y_2)$. [writes "then $g(\frac{x_1}{y_1}) = g(\frac{x_2}{y_2})$"]

You understand we have to check this, because that's the slipperiness of the rational numbers: they don't have a unique name.

Spoken Transcript [00:25:18 - 00:27:08] Prof. Rahul Pandharipande

All right, but you accept we have to check this? Otherwise, if we can't check this, whatever was there was nonsense.

So, we have to check it. And how are we going to check it? Well, so, we have here:

[writes "$g\left(\frac{x_1}{y_1}\right) = \frac{f(x_1)}{f(y_1)}$" and "$g\left(\frac{x_2}{y_2}\right) = \frac{f(x_2)}{f(y_2)}$"]

And we want to figure out whether these things are equal: do we have... [writes "do we have $\frac{f(x_1)}{f(y_1)} \overset{?}{=} \frac{f(x_2)}{f(y_2)}$"] That's the question.

Using the fact that these are invertible, and we can multiply by them, this question is equivalent to what's called clearing the denominators. So, by clearing the denominators, this is equivalent [draws $\iff$ and writes: $f(x_1)f(y_2) = f(x_2)f(y_1)$] to $f(x_1)f(y_2) = f(x_2)f(y_1)$ by just multiplying both sides by $f(y_1)f(y_2)$ [writes: $\cdot f(y_1)f(y_2)$] — that's a legal move in fields. And since these are invertible, this implies that, and this implies the other way, because they're both non-zero. Everything here is invertible.

We're using all the time that when we write this, [writes: $y_1, y_2 \neq 0 \implies f(y_1), f(y_2) \neq 0$] neither y_1 nor y_2 is zero, which implies $f(y_1)$ and $f(y_2)$ are not zero, okay?

Spoken Transcript [00:27:08 - 00:27:50] Prof. Rahul Pandharipande

And now, these things are all integers, and everything's upstairs — there's no division. This is if and only if [draws $\Updownarrow$] you can use the homomorphism property: [writes: $f(x_1y_2) = f(x_2y_1)$] $f(x_1y_2) = f(x_2y_1)$, right? Because f is a homomorphism.

And this is if and only if [draws $\Updownarrow$ and writes: $f(x_1y_2 - x_2y_1) = 0$] $f(x_1y_2 - x_2y_1) = 0$, okay?

Why would this be the case? Yeah?

Student Answer

We could rearrange...

Spoken Transcript [00:27:52 - 00:28:48] Prof. Rahul Pandharipande

Well, you have to say what this means. What does this mean? This goes to the heart of the issue about whether you learned the rational numbers correctly, and the answer is: this means [points to $\frac{x_1}{y_1} = \frac{x_2}{y_2}$ and writes below: $x_1y_2 = x_2y_1$] that $x_1y_2 = x_2y_1$. It means nothing more or less than that; it just actually means that.

So, if we start with this, we have this equation: [writes: $x_1y_2 - x_2y_1 = 0$ in the integers] $x_1y_2 - x_2y_1 = 0$ in the integers. That's what this means.

And very fortunately, that's what we need: we need this to be zero, so, this is true [places a checkmark next to $f(x_1y_2 - x_2y_1) = 0$] — and so, everything is true, and this is well-defined. All right?

That means I have defined a map, but there's a lot of things to check, which of course I'm not going to check. You have to do it, so to speak, on your own time.

Well-Definedness of the Extension Map g

Claim 3.9 (*Well-Definedness of g*)**:** The map $g\colon \mathbb{Q} \to F$ defined in (3.8) is well-defined; that is, if $\frac{x_1}{y_1} = \frac{x_2}{y_2}$ in $\mathbb{Q}$, then $g\left(\frac{x_1}{y_1}\right) = g\left(\frac{x_2}{y_2}\right)$ in F.

Proof

Let $\frac{x_1}{y_1}, \frac{x_2}{y_2} \in \mathbb{Q}$ with $y_1, y_2 \in \mathbb{Z} \setminus \{0\}$ such that $\frac{x_1}{y_1} = \frac{x_2}{y_2}$.

1. Meaning of Equality in $\mathbb{Q}$: By definition of the rational numbers as fractions, equality $\frac{x_1}{y_1} = \frac{x_2}{y_2}$ means precisely that the cross-multiplied relation holds in $\mathbb{Z}$:

$$x_1y_2 = x_2y_1 \iff x_1y_2 - x_2y_1 = 0 \in \mathbb{Z}. \tag{3.9}$$

2. Algebraic Chain in F: We want to establish whether:

$$g\left(\frac{x_1}{y_1}\right) \overset{?}{=} g\left(\frac{x_2}{y_2}\right) \iff \frac{f(x_1)}{f(y_1)} \overset{?}{=} \frac{f(x_2)}{f(y_2)}.$$

Since $y_1, y_2 \neq 0$ and f is injective, $f(y_1) \neq 0$ and $f(y_2) \neq 0$. Multiplying both sides by $f(y_1)f(y_2) \in F \setminus \{0\}$ (clearing denominators) yields an equivalent equality in F:

$$\begin{aligned}
\frac{f(x_1)}{f(y_1)} = \frac{f(x_2)}{f(y_2)} &\iff f(x_1) \cdot f(y_2) = f(x_2) \cdot f(y_1) \\
&\iff f(x_1y_2) = f(x_2y_1) \qquad \text{(since } f \text{ is a ring homomorphism)} \\
&\iff f(x_1y_2) - f(x_2y_1) = 0_F \\
&\iff f(x_1y_2 - x_2y_1) = 0_F. \qquad (3.10)
\end{aligned}$$

3. Conclusion: By (3.9), $x_1y_2 - x_2y_1 = 0 \in \mathbb{Z}$. Since any ring homomorphism maps 0 to 0 (Proposition 1.19):

$$f(x_1y_2 - x_2y_1) = f(0) = 0_F.$$

Thus, condition (3.10) holds, which proves that $g\left(\frac{x_1}{y_1}\right) = g\left(\frac{x_2}{y_2}\right)$. **Q.E.D.**

Core Pedagogical Principle: Non-Unique Names of Fractions: A rational number has infinitely many equivalent names (e.g., $\frac{1}{2} = \frac{2}{4} = \frac{-3}{-6}$). Any function defined on rational numbers using numerator and denominator symbols must be proven invariant under replacement of representatives. As shown above, this reduces entirely to the cross-multiplication identity in the underlying ring $\mathbb{Z}$.

Spoken Transcript [00:28:48 - 00:30:05] Prof. Rahul Pandharipande

I have now defined the map g, finally. Then you have to check that this map g is actually... I mean, all of the properties of g, that it's a homomorphism of fields. You have to check that it respects addition, and you have to check that it respects multiplication.

All of it is slightly painful, because at first blush it's obvious: you just pick representatives for $\mathbb{Q}$ as x_1/y_1, and then you just check it. But then you have to check that all of that stuff respects the different choices, the different names of elements of $\mathbb{Q}$, just like we did here.

And the checks are all the same, so you can do it yourself. That's the end of what I'm going to give as the proof of this; you can check the rest. We're going to have to do many more things like this, so...

Spoken Transcript [00:30:05 - 00:31:50] Prof. Rahul Pandharipande

We have now, successfully, defined g. Then we need to check that g is indeed a homomorphism of fields. [writes "We have defined g. We need to check that g is indeed a hom. of fields."] I leave this to you. We're going to have to do things like this anyway, and it's all just unraveling the definitions.

Meaning you have to check... [writes "You have to check"] You check something like: [writes: $g\left(\frac{x_1}{y_1}\right) + g\left(\frac{x_2}{y_2}\right) = g\left(\frac{x_1}{y_1} + \frac{x_2}{y_2}\right)$] $g(x_1/y_1) + g(x_2/y_2) = g(x_1/y_1 + x_2/y_2)$. You have to go check that; that's what you have to check.

And you can see when you write this, this side... Okay, I'm going to do part of it for you — I don't know why I'm feeling compassionate today here. This one is $f(x_1)/f(y_1) + f(x_2)/f(y_2)$. [writes: $\overset{\text{def}}{=} \frac{f(x_1)}{f(y_1)} + \frac{f(x_2)}{f(y_2)}$] That's the definition here. And this has a different definition.

First of all, how do we do it? We first have to add this fraction, right? Because our definition only applies to names of rational numbers, and this is the sum. So, we first have to do this, but you know how to add fractions: this is equal to [writes: $= g\left(\frac{x_1y_2+x_2y_1}{y_1y_2}\right)$] g of $(x_1y_2 + x_2y_1)/(y_1y_2)$. At least I hope you know how to add fractions — at least I hope I know how to add fractions, right? Okay, so, that's the addition of the fraction.

Spoken Transcript [00:31:50 - 00:32:50] Prof. Rahul Pandharipande

And then this, by definition — use the definition — is $f(x_1y_2 + x_2y_1)/f(y_1y_2)$. [writes: $\overset{\text{def}}{=} \frac{f(x_1y_2+x_2y_1)}{f(y_1y_2)}$] Okay?

So, that's the one side to check, and that's supposed to equal this side, all right? And how are you going to check this? Well, you use the homomorphism property, you move everything upstairs, and then finally you're going to use the thing I erased — this, okay?

So, I leave it to you. There's nothing really I can add to this; you have to do it. But I do say it's painful — painful for me to write it on the board even.

Verification of the Field Homomorphism Axioms

Claim 3.10 (*Preservation of Field Operations*): The map $g\colon \mathbb{Q} \to F$ is a field homomorphism; that is, for all $q_1, q_2 \in \mathbb{Q}$:

$$g(q_1 + q_2) = g(q_1) + g(q_2), \tag{3.11}$$

$$g(q_1 \cdot q_2) = g(q_1) \cdot g(q_2), \tag{3.12}$$

$$g(1_{\mathbb{Q}}) = 1_F. \tag{3.13}$$

Why g Is a Field Homomorphism (beyond the lecture)

The lecture leaves this to the students. Let $q_1 = \frac{x_1}{y_1}$ and $q_2 = \frac{x_2}{y_2}$ with $x_i, y_i \in \mathbb{Z}$ and $y_i \neq 0$.

1. Verification of Additivity (3.11)**:** Adding fractions according to standard rational arithmetic:

$$q_1 + q_2 = \frac{x_1}{y_1} + \frac{x_2}{y_2} = \frac{x_1y_2 + x_2y_1}{y_1y_2}.$$

Applying the definition of g and the ring homomorphism properties of f:

$$\begin{aligned} g(q_1+q_2) &= g\left(\frac{x_1y_2+x_2y_1}{y_1y_2}\right) \\ &\overset{\text{def}}{=} \frac{f(x_1y_2+x_2y_1)}{f(y_1y_2)} \\ &= \frac{f(x_1)f(y_2)+f(x_2)f(y_1)}{f(y_1)f(y_2)}. \end{aligned}$$

On the other hand, evaluating the sum of images in the field F:

$$\begin{aligned} g(q_1)+g(q_2) &\overset{\text{def}}{=} \frac{f(x_1)}{f(y_1)}+\frac{f(x_2)}{f(y_2)} \\ &= f(x_1)f(y_1)^{-1}+f(x_2)f(y_2)^{-1}. \end{aligned}$$

Bringing both terms over the common denominator $f(y_1)f(y_2)$ in the field F:

$$\frac{f(x_1)}{f(y_1)}+\frac{f(x_2)}{f(y_2)} = \frac{f(x_1)f(y_2)+f(x_2)f(y_1)}{f(y_1)f(y_2)}.$$

Both expressions coincide identically, establishing (3.11). By Claim 3.9, this value is independent of the choice of fraction representatives.

2. Verification of Multiplicativity (3.12)**:** Multiplying fractions inside $\mathbb{Q}$:

$$q_1\cdot q_2 = \frac{x_1}{y_1}\cdot\frac{x_2}{y_2} = \frac{x_1x_2}{y_1y_2}.$$

Applying the definition of g and the multiplicativity of f:

$$\begin{aligned} g(q_1\cdot q_2) &= g\left(\frac{x_1x_2}{y_1y_2}\right) \\ &= \frac{f(x_1x_2)}{f(y_1y_2)} \\ &= \frac{f(x_1)f(x_2)}{f(y_1)f(y_2)} \\ &= \frac{f(x_1)}{f(y_1)}\cdot\frac{f(x_2)}{f(y_2)} \\ &= g(q_1)\cdot g(q_2), \end{aligned}$$

establishing (3.12).

3. Preservation of Multiplicative Identity (3.13)**:** Representing $1_{\mathbb{Q}} \in \mathbb{Q}$ as $\frac{1_{\mathbb{Z}}}{1_{\mathbb{Z}}}$:

$$g(1_{\mathbb{Q}}) = g\left(\frac{1_{\mathbb{Z}}}{1_{\mathbb{Z}}}\right) = \frac{f(1_{\mathbb{Z}})}{f(1_{\mathbb{Z}})}.$$

Since $f\colon \mathbb{Z}\to F$ is a unital ring homomorphism, $f(1_{\mathbb{Z}}) = 1_F$. Thus:

$$g(1_{\mathbb{Q}}) = \frac{1_F}{1_F} = 1_F,$$

establishing (3.13). Thus g is a field homomorphism. **Q.E.D.**

The Field of Fractions of an Integral Domain

Spoken Transcript [00:32:50 - 00:35:03] Prof. Rahul Pandharipande

That's the story about $\mathbb{Z}$. Their fates are really closely tied: $\mathbb{Z}$ and $\mathbb{Q}$. [writes: $\mathbb{Z} \subset \mathbb{Q}$] $\mathbb{Q}$ is not just any field that $\mathbb{Z}$ is in — it's the smallest field it's in. So, we could ask: can we construct this relationship for other rings? Because we really like this relationship.

If you study the integers, it's very useful sometimes to be able to divide; so, that's the movement to the field. Let's try this with a ring in general.

Suppose R is an integral domain. [writes "Suppose R is an integral domain"] It turns out that this is extremely important, and we will see in the definition — it may not be clear from the discussion so far — but let's say R is an integral domain.

Can we find the analogue of $\mathbb{Q}$, which is the *"smallest"* field containing R? [writes "Can we find "smallest" field containing R?"] Already in this language, if we find a field containing R, [writes: $R \subset F$] that means R has to be a domain (Recall: Definition 2.62), because a domain means that two elements that are non-zero, when you multiply them, stay non-zero. If you're inside a field, that's always the case, okay? This is one of the reasons why we're talking only about integral domains.

So, that's the question: if you find on the street an integral domain, can you put it in its smallest field? The answer is yes, and that construction is called the field of fractions. And I will show you how to do it.

In some ways, there's lots of conceptual nice things about it, but this pain, you can imagine, we're not going to get away from it — it's just going to be more abstract. One of the things you will see here is that when we define the field of fractions of a ring, it will have this property that the elements are not single objects, but equivalence classes. And you should expect that answer, because that already happens for the rational numbers.

Spoken Transcript [00:35:03 - 00:36:16] Prof. Rahul Pandharipande

By the way, for a rational number, the objection — and I'm surprised no one made it — is that you could say that you could fix the name by taking the numerator and denominator and cancelling all common factors. There's still a sign issue, but we leave that aside.

If you try to do arithmetic with that, it's super painful, because you have to do a lot of cancellation. So, that is the argument against it. The argument that that's kind of a lie is that that's not really how you do arithmetic in the rational numbers. But that is the argument against what I was saying about the rational numbers.

In a sense, if you do that, it's kind of even more painful, because then when you write these expressions, you can't write these expressions — you have to cancel everywhere, and you know...

Generalization from Integers to Integral Domains

Remark 3.11 (*Why Integral Domains?*): If a commutative ring R embeds into a field

F via an injective homomorphism $\iota\colon R \hookrightarrow F$, then R must necessarily have no zero divisors:

$$a, b \in R \setminus \{0\} \implies \iota(a), \iota(b) \neq 0 \implies \iota(ab) = \iota(a)\iota(b) \neq 0 \implies ab \neq 0.$$

Moreover, R is not the zero ring: $\iota(1) = 1 \neq 0 = \iota(0)$ in F, so $1 \neq 0$ in R. Thus, the property of being an *"integral domain"* (Definition 2.62) is a strictly necessary prerequisite for any ring to be embedded into a field of fractions.

The Goal: Field of Fractions

Given an arbitrary integral domain R, our goal is to construct the minimal field containing R, called the *"field of fractions"* (or quotient field) $\mathrm{Frac}(R)$, generalizing the classical relationship:

$$\mathbb{Z} \hookrightarrow \mathbb{Q} = \mathrm{Frac}(\mathbb{Z}).$$

Construction of the Set of Formal Fractions

Spoken Transcript [00:36:16 - 00:38:05] Prof. Rahul Pandharipande

Let's think about this in the abstract. The main thing is just the definition. Here is our ring R: we don't know what it is, it's just any ring, but it's an integral domain. [writes: R int. domain] We want to construct a field containing R. [writes "We want to construct a field containing R"]

The field should have elements that are like x over y, where $x, y \in R$ and $y \neq 0$. [writes: $F \ni \frac{x}{y}$, $x, y \in R, y \neq 0$] This should be the element of that field, because if it contains R, I should be able to take x and multiply it by the inverse of y in the field, so, it should have these elements in it. Is that clear? That's what happened with the rational numbers.

So, we can just define it that way. We start with: let S be the set of pairs. [writes "Let $S = \{(x, y) \mid y \neq 0, x, y \in R\} \subset R \times R$"] This whole thing is going to be a big disappointment if you think something fun is going to happen — I just tell you ahead of time, this is going to be a big disappointment. But some things in life are like that.

So, here: $y \neq 0$, and $x, y \in R$. Okay, that's the set of pairs. For the integers, if we start with the integers, we just get the set of pairs of integers where the second one is not zero, okay? But we can think about this set of pairs for any R. Is that clear? This is inside $R \times R$ — it's just some subset of pairs.

Digression: Equivalence Relations and Partitions

Spoken Transcript [00:38:05 - 00:40:21] Prof. Rahul Pandharipande

All right, and now the thing that will come is that there is an equivalence relation. [writes "There is an equivalence relation on S"] There is an equivalence relation on S, so maybe we call this equivalence relation... [writes "$\sim$", then changes to "$\equiv$"] You can use various symbols for equivalence, that's kind of standard. How about three lines?

Do you know what an equivalence relation is? This was in your Foundations class.

An equivalence relation on a set — maybe a little aside [writes "Aside on T"] — an equivalence relation on a set T tells you when two elements are related, and the rules for the equivalence relation are:

x is always related to itself — this is the reflexive property; [writes: $x \equiv x$, reflexive]

if x is related to y, then y is related to x — this is the symmetric property; [writes: $x \equiv y \implies y \equiv x$, symmetric]

and if you know the definition, what's the third one? Yeah?

Student Answer

Transitivity.

Spoken Transcript [00:40:21 - 00:41:00] Prof. Rahul Pandharipande

The transitive property. It says that if we have x related to y and y related to z, then x is related to z. [writes: $x \equiv y \wedge y \equiv z \implies x \equiv z$, transitive] That's the whole thing of equivalence relations. It comes up many times in mathematics, and it will come up again in this course, so, if you have not seen it, think about it.

And the main thing we use about this equivalence relation is that if I have T with this equivalence relation, then I have another set — sometimes it's called this — which is the set of equivalence classes. [writes: $T_{\equiv} = \{\text{set of equiv. classes}\}$] That's going to be very important for us.

Whenever we have a set with an equivalence relation, that equivalence relation divides the set up into equivalence classes — disjoint equivalence classes, okay? And this thing here is the set of equivalence classes. There's three objects in this kind of abstract situation: there's the set that we start with, there's the equivalence relation, and then there's the set of equivalence classes.

The Set of Pairs and General Equivalence Relations

Definition 3.12 (*The Set of Formal Fractions S*)**:** Let R be an integral domain. On the Cartesian product $R \times R$ (Notation 0.1), we define the set of formal numerator-denominator pairs S by:

$$S := \{(x,y) \in R \times R \mid y \neq 0\} \subset R \times R. \tag{3.14}$$

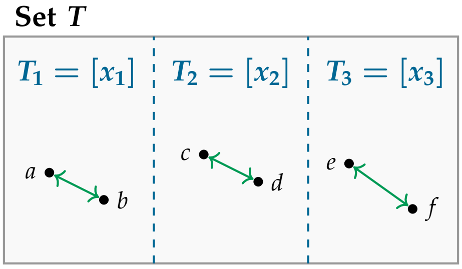
Remark 3.13 (*Review: Equivalence Relations and Partitions*)**:** An *"equivalence relation"* $\equiv$ on a set T is a binary relation satisfying three foundational axioms (Definition 0.8):

(i) **Reflexivity:** $\forall x \in T \colon x \equiv x$.

(ii) **Symmetry:** $\forall x, y \in T \colon x \equiv y \implies y \equiv x$.

(iii) **Transitivity:** $\forall x, y, z \in T \colon (x \equiv y \wedge y \equiv z) \implies x \equiv z$.

The quotient set $T/\equiv$ denotes the set of *"equivalence classes"* (Definition 0.9), which partitions T into mutually disjoint subsets.



Partition of T into disjoint equivalence classes $T/\equiv$

The Equivalence Relation on Formal Fractions

Spoken Transcript [00:41:00 - 00:42:06] Prof. Rahul Pandharipande

We have to define the equivalence relation. We now define an equivalence relation on our set S, which is the set of pairs [writes "We now define $\equiv$ on S"] — and you should think about these pairs as numerator and denominator, but those are just words here.

We need to define when (x_1, y_1) is equivalent to (x_2, y_2). [writes: $(x_1, y_1) \equiv (x_2, y_2)$] We need a definition for this. What's the answer going to be? Yeah?

Student Answer

$x_1 y_2 = x_2 y_1$.

Spoken Transcript [00:42:06 - 00:44:08] Prof. Rahul Pandharipande

Okay, very good. This is the definition: [writes: $\overset{\text{def}}{\iff} \; x_1 y_2 = x_2 y_1$] it's just exactly what you do with integer arithmetic. If you have a fraction, you say they're equal if $x_1 y_2 = x_2 y_1$, okay? That's the definition. There's nothing more or less than that — that's the whole definition.

And it's motivated by the integers. It's very important for this definition that this is only about R. [writes "only about R"] This whole discussion starts with R, so, we can only say things that are about R.

We can make these pairs — that's only about R. We want to define this equivalence relation. At the end, these things are only in R, and the definition is only about some equation in R, okay? We haven't gotten out of R in some sense in this whole discussion. Is that clear?

Now we come to an issue: why is this an equivalence relation? This is the question you should have asked your teacher in first grade or whatever it was. That's like the whole question about the rational numbers in some sense — the foundations. [washes and erases the left part of the bottom chalkboard]

This is the first part where we have to do a little bit of algebra. The first two properties, $x = x$, that's just identically true, okay? That's not a problem. We have to check three properties. [writes "3 properties"]

Definition of the Equivalence Relation on S

Definition 3.14 (*Equivalence Relation on S*): On the set $S = \{(x, y) \in R \times R \mid y \neq 0\}$, we define the relation $\equiv$ by:

$$(x_1, y_1) \equiv (x_2, y_2) \overset{\text{def}}{\iff} x_1 y_2 = x_2 y_1 \quad (\text{in } R). \tag{3.15}$$

Conceptual Motivation: Cross-Multiplication: This relation is the formal algebraic counterpart to equating cross-products of fractions:

$$\frac{x_1}{y_1} = \frac{x_2}{y_2} \iff x_1 y_2 = x_2 y_1.$$

Crucially, formula (3.15) involves only multiplication within the ring R, without invoking any external concept of division or fraction.

Spoken Transcript [00:44:08 - 00:45:05] Prof. Rahul Pandharipande

The first is: if I take (x_1, y_1), this is equivalent to (x_1, y_1). [writes: $(x_1, y_1) \equiv (x_1, y_1)$] And why is this? Because by definition this is equal to $x_1 y_1 = x_1 y_1$, [writes: $\iff x_1 y_1 = x_1 y_1$ ✓] so, we're very happy about that. There was nothing to do here.

You also have to check the symmetric property: that if we have $(x_1, y_1) \equiv (x_2, y_2)$, then the other way around is fine, and it'll just be the same equation. There's nothing to check about that.

The interesting one is the transitive property. [writes "Transitive"] This is the only fun one to check. We start with this one: [writes "$(x_1, y_1) \equiv (x_2, y_2)$" and "$(x_2, y_2) \equiv (x_3, y_3)$"] these are the inputs.

And we must show — this is for us to do — that (x_1, y_1) is congruent (i.e., equivalent) to (x_3, y_3). [writes: $\implies (x_1, y_1) \equiv (x_3, y_3)$]

The thing that's a little funny about this is... let's try to do this. We have to write the first equation. This is what we have: [writes "What we have"] that's what we've assumed, that's in the left column. [the lecture bell rings]

Okay. You'll have to wait until after the break for this excitement to continue.

Verification of Reflexivity and Symmetry on S

Proposition 3.15 (*Reflexivity and Symmetry of $\equiv$*): The relation $\equiv$ defined on $S = \{(x, y) \in R \times R \mid y \neq 0\}$ by:

$$(x_1, y_1) \equiv (x_2, y_2) \overset{\text{def}}{\iff} x_1 y_2 = x_2 y_1 \tag{3.16}$$

is reflexive and symmetric.

Proof

Let $(x_1, y_1), (x_2, y_2) \in S$:

- **Reflexivity:** Trivially $x_1 y_1 = x_1 y_1$, which by (3.16) immediately implies:

$$(x_1, y_1) \equiv (x_1, y_1).$$

- **Symmetry:** Assume $(x_1, y_1) \equiv (x_2, y_2)$. Then $x_1y_2 = x_2y_1$. By symmetry of equality in R:

$$x_2y_1 = x_1y_2 \implies (x_2, y_2) \equiv (x_1, y_1).$$

Q.E.D.

Mid-Lecture Break

The bell rings, and the lecturer pauses the class for a short break. The lecture resumes at timestamp 00:45:05.

Spoken Transcript [00:45:05 - 00:45:33] Prof. Rahul Pandharipande

Okay, so, let's start. I should say that very few people have found my office, and it turns out I'll be in this visitor office for most of the semester, because they're clearing some kind of nuclear waste from my current office. So, my office now is HG G 65.1 [writes "HG G 65.1"] if you want to find me there. And yeah, I'm afraid I'll be there for most of the semester.

Administrative Announcement: Office Relocation

The lecturer writes *"HG G 65.1"* on the far right of the chalkboard and informs students that he will be stationed in this visitor office for the majority of the semester.

Spoken Transcript [00:45:33 - 00:47:00] Prof. Rahul Pandharipande

We come to this point that's kind of fun and interesting. We started with an integral domain, and led by the analogy with the rational numbers, we come to start defining this field as some equivalence class of pairs. And I should have said this — I did say it, but I didn't write it: the way to think about this [writes "Think of as $\frac{x}{y} \leftarrow (x, y) \in S$"] — this is the mathematical text — you think of this pair in S as x/y, okay? Just like with the integers.

As an object, it's this formal object. And then there was this proposal, which was correct, of how to define the equivalence relation. And now we actually need to check something: is it an equivalence relation?

Here we have to prove the transitive property. And so, we start with: what do we have? We have that equivalence, so, if we translate that into equations, it says $x_1y_2 = x_2y_1$ — and you have to be very careful with the subscripts, otherwise you'll confuse yourself: [writes: $x_1y_2 = x_2y_1$] $x_1y_2 = x_2y_1$. And then what we also have here is $x_2y_3 = x_3y_2$. [writes: $x_2y_3 = x_3y_2$]

Spoken Transcript [00:47:20 - 00:48:40] Prof. Rahul Pandharipande

And what we want, [writes "Want"] we want this thing, which is $x_1y_3 = x_3y_1$. [writes: $x_1y_3 = x_3y_1$]

The puzzling thing about this is: we have these quadratic equations and we need to get another quadratic equation. It seems a little bit unfortunate, because you might think you have to do some linear algebra with quadratic equations, and that's not going to work — you're not going to get a new quadratic equation from these.

The correct answer is we're supposed to multiply here. The clever thing is to multiply the top equation by y_3 [writes "mult. by y_3"] — y_3 is what we want. So, we get $x_1y_2y_3 = x_2y_1y_3$. [writes: $x_1y_2y_3 = x_2y_1y_3$]

This is kind of good: we see the x_1y_3 here. Can you see what we have to do next? This is actually algebra. What? [inaudible answer from a student] Yeah, you could do that, but why don't we just take this y_2y_3 here and use this equation? It's the same thing. So, we take the y_2y_3 and use this equation to flip them. This is all about subscripts, and I should just tell you that subscripts have always been difficult for me. Do you accept that? Because I just took this and flipped them using this equation. [writes "$x_3y_1y_2$" under $x_2y_1y_3$]

Spoken Transcript [00:48:40 - 00:50:02] Prof. Rahul Pandharipande

Now what do we do? Well, we have here x_1y_3 — that looks pretty good — and we have y_3y_1, but they're both multiplied by y_2, right?

Now we use the fact that in these pairs, the second one is always born non-zero: to be in this set, the second one has to be non-zero. And this is an integral domain. So, we have this equation — this one equals that one — and then we use the integral domain property. [writes "use int. domain"]

I said this is a place where it's extremely important it's an integral domain. Since $y_2 \neq 0$, [writes "since $y_2 \neq 0$"] we can cancel, [writes "$\implies$ cancel"] okay? Because an integral domain says that's the cancellation property. Is that okay?

If y_2 is not zero, we can cancel it. This is kind of dramatic, but we cancel it here. [crosses out y_2 on both sides] And that's what we want.

We have now proven transitivity. You see there was a little bit of cleverness here; it's easy to confuse yourself.

Proof of Transitivity and the Role of the Integral Domain

Proposition 3.16 (*Transitivity of $\equiv$ on S*): Let R be an integral domain. The relation $\equiv$ on S defined by $(x_1, y_1) \equiv (x_2, y_2) \overset{\text{def}}{\iff} x_1y_2 = x_2y_1$ is transitive. Consequently, $\equiv$ is an equivalence relation on S.

Proof

Let $(x_1, y_1), (x_2, y_2), (x_3, y_3) \in S$, which implies $y_1, y_2, y_3 \in R \setminus \{0\}$.

1. **Assumptions:** We assume the relations:

$$(x_1, y_1) \equiv (x_2, y_2) \implies x_1y_2 = x_2y_1, \tag{3.17}$$

$$(x_2, y_2) \equiv (x_3, y_3) \implies x_2y_3 = x_3y_2. \tag{3.18}$$

2. **Target:** We must prove that $(x_1, y_1) \equiv (x_3, y_3)$, which means showing that:

$$x_1y_3 = x_3y_1. \tag{3.19}$$

3. Derivation via Multiplication and Substitution: Multiplying equation (3.17) on both sides by $y_3 \in R$:

$$(x_1y_2)y_3 = (x_2y_1)y_3 \iff (x_1y_3)y_2 = (x_2y_3)y_1. \tag{3.20}$$

Substituting $x_2y_3 = x_3y_2$ from (3.18) into the right-hand side of (3.20):

$$(x_1y_3)y_2 = (x_3y_2)y_1 = (x_3y_1)y_2. \tag{3.21}$$

Subtracting the right-hand side gives:

$$(x_1y_3 - x_3y_1)y_2 = 0. \tag{3.22}$$

4. Cancellation via the Integral Domain Property: By definition of S, $y_2 \neq 0$. Because R is an **integral domain**, it contains no zero divisors:

$$ab = 0 \text{ with } b \neq 0 \implies a = 0.$$

Applying this cancellation property to (3.22) with $b = y_2 \neq 0$:

$$x_1y_3 - x_3y_1 = 0 \implies x_1y_3 = x_3y_1.$$

Thus, $(x_1, y_1) \equiv (x_3, y_3)$, which completes the proof. **Q.E.D.**

Why Integral Domains Are Essential for Transitivity: Going beyond the lecture: notice where the ring axioms alone would fail: equation (3.22) holds in any commutative ring. However, to deduce $x_1y_3 = x_3y_1$ from $(x_1y_3 - x_3y_1)y_2 = 0$, we must divide out and cancel the non-zero element y_2. If R has zero divisors, the relation $\equiv$ can **fail** to be transitive. For instance, in $\mathbb{Z}/6\mathbb{Z}$ (with $y_2 = \bar{2}$ and $x_1y_3 - x_3y_1 = \bar{3}$): $(\bar{3}, \bar{1}) \equiv (\bar{0}, \bar{2})$ since $\bar{3} \cdot \bar{2} = \bar{0} = \bar{0} \cdot \bar{1}$, and $(\bar{0}, \bar{2}) \equiv (\bar{0}, \bar{1})$ since $\bar{0} \cdot \bar{1} = \bar{0} = \bar{0} \cdot \bar{2}$, but $(\bar{3}, \bar{1}) \not\equiv (\bar{0}, \bar{1})$ since $\bar{3} \cdot \bar{1} = \bar{3} \neq \bar{0} = \bar{0} \cdot \bar{1}$.

Spoken Transcript [00:50:02 - 00:51:06] Prof. Rahul Pandharipande

We have come so far as to define S as a set with an equivalence relation. So, now we can define the points of the field that we're constructing, which will be the equivalence classes. [washes and erases the upper chalkboard] This was almost like the first piece of algebra in this class, I think. I don't remember if we had any algebra before. By that I mean playing with equations. Which in the end — I'm saying a lot of words, but in the end the subject is about equations, you know.

The Field of Fractions and Its Canonical Embedding

Construction of the Field of Fractions

Spoken Transcript [00:51:06 - 00:52:35] Prof. Rahul Pandharipande

Where are we? We started with R as an integral domain. [writes: R int. domain] We constructed S with an equivalence relation, [writes: $S = \dots$ with $\equiv$] and from this we define F. The name of this is the field of fractions of R. [writes "$\implies F = S_{\equiv}$" and "Field of Fractions of R"] Field of fractions of R — that's the name of this object: F is equal to the set of equivalence classes of S. [writes: $F = S_{\equiv}$]

Is it completely clear what this thing is? We defined S, we defined an equivalence relation, and we took great pains to check all of its properties, so, we get the set of equivalence classes and that is F.

The Canonical Embedding of R into F

Spoken Transcript [00:52:35 - 00:53:14] Prof. Rahul Pandharipande

Then we need to do a lot of things. But let's start by defining the map which is the inclusion from R to F, [writes: $R \to F$] because we said this F should be a field that contains R. That means we have to have some way of getting points from R to F. And what's that way going to be?

This ι, the inclusion: we have to put R in F. [writes "We have to put $R \xrightarrow{i} F$"] This is by ι. Normally people don't even write it, but if I have an element of R, we have to put it into F. That means we have to assign it an equivalence class in this set. What are we going to do? Yeah?

Student Answer

The equivalence class of $(r,1)$.

Spoken Transcript [00:53:15 - 00:54:22] Prof. Rahul Pandharipande

Yeah, that's a good idea: $(r,1)$. [writes: $i(r) = (r,1) \in S_{\equiv}$] It has some advantages: $1 \neq 0$, and an integral domain is never the zero ring (Recall: Definition 2.62). So, this is legal: this is an element of S, and then we can take its equivalence class.

I'm not going to write equivalence classes because it becomes too complicated, but you could formally take its equivalence class like this. If you start doing this, you commit to writing things like $[(3,4)]$ for rational numbers, and no one ever does that. We don't want to do it, but this would be kind of more logically correct.

Spoken Transcript [00:54:22 - 00:54:54] Prof. Rahul Pandharipande

When I write this, it's just an equivalence class. I looked at what the book does, and the book says that this is an element of S and that is its equivalence class. That's a very clever solution; I wouldn't say it's totally standard, but anyway. Standard is just to ignore the issue. So, this is the equivalence class here.

Is this reasonable? It better at least be... yeah.

Student Question

The set S, is this all of $R \times R$ or is it just any subset of $R \times R$?

Spoken Transcript [00:55:00 - 00:55:28] Prof. Rahul Pandharipande

No, the set S was very precisely defined on a previous board: S is equal to those elements in $R \times R$ where $y \neq 0$. [writes: $S = \{(x,y) \mid x,y \in R,\ y \neq 0\}$] That's S.

And this is an element of S because $1 \neq 0$; and $1 \neq 0$ because R is an integral domain, and an integral domain is by definition not the zero ring. That's the full logic — that's the full logical certificate if you want it.

Spoken Transcript [00:55:28 - 00:57:05] Prof. Rahul Pandharipande

Still, we'd like more: we'd like this to be an injection. Is ι injective? [writes "Is i injective?"]

This means: when are $(r,1)$ and $(\hat{r},1)$ in the same equivalence class of S? [writes "When are $(r,1)$ and $(\hat{r},1)$ in the same equiv. class of S?"] That's what injectivity is about, right?

How do we check this? The answer is that these are in the same equivalence class if and only if that equivalence class identity holds — that one. [points at $(x_1,y_1) \equiv (x_2,y_2) \iff x_1y_2 = x_2y_1$]

It's if and only if $r \cdot 1 = \hat{r} \cdot 1$, [writes: $\iff r \cdot 1 = \hat{r} \cdot 1$] which is if and only if $r = \hat{r}$, [writes: $\iff r = \hat{r}$] which is great for us because it shows that this is an injection. [places a checkmark]

There's a lot of things to do in this construction — it's like a big house: you have to build all the pieces of it. So, now we not only have defined F, we've defined ι, and we've put R inside F, and this placement is injective. Okay?

Definition of the Field of Fractions and Canonical Embedding

Definition 3.17 (*Field of Fractions*): Let R be an integral domain, and let $S := \{(x,y) \in R \times R \mid y \neq 0\}$. The *"field of fractions"* (or quotient field) of R, denoted by $\mathrm{Frac}(R)$ or F, is defined as the quotient set:

$$F := S_{\equiv} = \{[(x,y)] \mid (x,y) \in S\}. \tag{3.23}$$

The equivalence class $[(x,y)]$ is commonly written as $\frac{x}{y}$.

Definition 3.18 (*Canonical Inclusion Map*): We define the canonical embedding $i\colon R \to F$ by assigning to each $r \in R$ the equivalence class of the pair $(r,1)$:

$$i(r) := [(r,1)] \in F. \tag{3.24}$$

Proposition 3.19 (*Injectivity of the Canonical Inclusion*): The canonical map $i\colon R \to F$ is injective.

Proof

Let $r, \hat{r} \in R$. By definition of equality in the quotient set $F = S_{\equiv}$:

$$\begin{aligned} i(r) = i(\hat{r}) &\iff [(r,1)] = [(\hat{r},1)] \\ &\iff (r,1) \equiv (\hat{r},1) \\ &\iff r \cdot 1 = \hat{r} \cdot 1 \\ &\iff r = \hat{r}. \end{aligned}$$

Therefore, i is injective. **Q.E.D.**

Why $1_R \neq 0_R$ Is Guaranteed: For the pair $(r, 1)$ to belong to S, the denominator must satisfy $1 \neq 0$. By condition (i) of Definition 2.62, an integral domain must satisfy $1_R \neq 0_R$ (i.e., it cannot be the trivial zero ring). Hence, $(r, 1) \in S$ is unconditionally valid for every $r \in R$.

Field Operations on the Field of Fractions

Definition of Zero, One, and Addition on F

Spoken Transcript [00:57:05 - 00:58:38] Prof. Rahul Pandharipande

This is just the beginning of the story. R already is born with all of the data: R, $+$, multiplication, $0, 1$. [writes: $R, +, \cdot, 0, 1$] We need to define all the structures on F and show this is a homomorphism also. [draws arrow to F, gestures, and pushes the upper chalkboard up to reveal the lower board] We have to start talking about all the structures for F as a field. And for many of them we don't have any choice: the zero in F has to correspond to the zero in R for it to be a homomorphism. So, the zero in F has to be $(0, 1)$, [writes "Zero in F has to be $(0, 1)$"] okay? Also, the unit in F has to be $(1, 1)$, [writes "unit in F has to be $(1, 1)$"] because it has to correspond to the unit in R by the rule of getting R to F. We don't really have any choice in this matter.

But we have to define $+$ and multiplication on F. [writes "We have to define $+, \cdot$ on F"] That's something that we have to construct. We have to define it, check it's compatible with R, check it satisfies the field axioms — it's a lot of long and boring work, but the interesting part to start with is we have to define it.

If we have an element here, $(x_1, y_1) \in S$, [writes: $(x_1, y_1) \in S$] and we have another one, $(x_2, y_2) \in S$, [writes: $(x_2, y_2) \in S$] then we have to define this operation. This is up to us: we must define $(x_1, y_1) + (x_2, y_2) = \ldots$ [writes "$(x_1, y_1) + (x_2, y_2) =$" and "we must define"]

That's the problem. How are we going to define this? What do you think? Tell me.

Student Answer

$x_1 y_2 + x_2 y_1$ over $y_1 y_2$.

Spoken Transcript [00:58:39 - 01:00:40] Prof. Rahul Pandharipande

Yeah, so, we just use the standard definition for how you add fractions: this one times that one plus this one times that one over $y_1 y_2$, okay? [writes: $= (x_1 y_2 + x_2 y_1, y_1 y_2)$]

Meaning you pretend that it's this problem, which it kind of is: [writes: $\frac{x_1}{y_1} + \frac{x_2}{y_2} = \frac{x_1 y_2 + x_2 y_1}{y_1 y_2}$] $\frac{x_1}{y_1} + \frac{x_2}{y_2} = \frac{x_1 y_2 + x_2 y_1}{y_1 y_2}$. You know how to do this. I hope I got the subscripts right; you should tell me if I get the subscripts wrong.

This is the definition — that's just it, nothing more, nothing less.

Now, is it a legal definition? There's an easy part: for it to be legal, this better be non-zero, [points at y_1y_2 and writes "non-zero"] because in our set S, the denominator — that's the one in the basement — always has to be non-zero.

We have here: this guy is non-zero, that guy is non-zero, and that implies the product is non-zero, because R is an integral domain. [writes "because R is an int. domain"] We use it again here. Is that clear? Otherwise, if this were zero, this would not be legal.

Well-Definedness of Addition

Spoken Transcript [01:00:40 - 01:02:25] Prof. Rahul Pandharipande

That's the easy part. The hard part is: you have to check that while this is a definition on S, it's well-defined on equivalence classes.

That means if I trade this person for another one in the same equivalence class, and I trade this for another person in the same equivalence class, the result has to be in the same equivalence class. Is that clear? That's the check of rational arithmetic — the check that I claim you should have done in school to understand the rational numbers.

Otherwise, what are you doing? You're doing some random manipulation. So, you have to check this: [writes "You have to check the indep. of choices of equiv. class"] independence of choices of equivalence class.

And we've already done this kind of thing twice already in the lecture. If you just do it and follow your nose, you'll see it all unwinds. You just use the definition: you have to say formally that this result respects our definition of equivalence classes.

In various books they do it; mostly it's left as an exercise. It's really important, so, you should do it. If I do it, it's just a bunch of indices — it doesn't really help, and you have to go through it yourself.

Is it clear what the problem is? I recommend trying this yourself, and if you get stuck, look at the book — the book does it. It'll give you a line of things, and you can imagine it's probably true.

Neutral Elements and Addition on the Field of Fractions

Definition 3.20 (*Neutral Elements in F*)**:** The zero element and multiplicative unit in F are defined by:

$$0_F := i(0_R) = [(0,1)], \tag{3.25}$$
$$1_F := i(1_R) = [(1,1)]. \tag{3.26}$$

Definition 3.21 (*Addition on F*)**:** For any two equivalence classes $[(x_1,y_1)]$, $[(x_2,y_2)] \in F$, addition is defined by:

$$[(x_1,y_1)] + [(x_2,y_2)] \overset{\text{def}}{=} [(x_1y_2 + x_2y_1, y_1y_2)]. \tag{3.27}$$

Proposition 3.22 (*Legitimacy and Well-Definedness of Addition*)**:** Addition on F as given in (3.27) is well-defined.

Why Addition Is Well-Defined (beyond the lecture)

The lecture checks the denominator and leaves the independence of the representatives to the students. We verify two essential requirements:

(1) **Legitimacy of the Denominator:** For $(x_1y_2 + x_2y_1, y_1y_2)$ to lie in S, we must have $y_1y_2 \neq 0$. Since $(x_1, y_1), (x_2, y_2) \in S$, we know $y_1 \neq 0$ and $y_2 \neq 0$. Because R is an **integral domain**, the product of two non-zero elements is non-zero:

$$y_1 \neq 0 \wedge y_2 \neq 0 \implies y_1y_2 \neq 0.$$

Thus, $(x_1y_2 + x_2y_1, y_1y_2) \in S$.

(2) **Independence of Representatives:** Suppose $(x_1, y_1) \equiv (\hat{x}_1, \hat{y}_1)$ and $(x_2, y_2) \equiv (\hat{x}_2, \hat{y}_2)$, meaning:

$$x_1\hat{y}_1 = \hat{x}_1y_1 \quad \text{and} \quad x_2\hat{y}_2 = \hat{x}_2y_2. \tag{3.28}$$

We must show that $(x_1y_2 + x_2y_1, y_1y_2) \equiv (\hat{x}_1\hat{y}_2 + \hat{x}_2\hat{y}_1, \hat{y}_1\hat{y}_2)$, which requires:

$$(x_1y_2 + x_2y_1)(\hat{y}_1\hat{y}_2) \overset{?}{=} (\hat{x}_1\hat{y}_2 + \hat{x}_2\hat{y}_1)(y_1y_2).$$

Expanding the left-hand side and applying (3.28):

$$\begin{aligned}(x_1y_2 + x_2y_1)\hat{y}_1\hat{y}_2 &= (x_1\hat{y}_1)(y_2\hat{y}_2) + (x_2\hat{y}_2)(y_1\hat{y}_1)\\ &= (\hat{x}_1y_1)(y_2\hat{y}_2) + (\hat{x}_2y_2)(y_1\hat{y}_1)\\ &= (\hat{x}_1\hat{y}_2 + \hat{x}_2\hat{y}_1)y_1y_2.\end{aligned}$$

This establishes equivalence, confirming that addition is independent of the choice of representative.

Q.E.D.

III Definition and Well-Definedness of Multiplication on F

Spoken Transcript [01:02:25 - 01:03:40]

Prof. Rahul Pandharipande

Also you have to do it with multiplication. [washes the chalkboard to the left] Multiplication is a bit easier — that was the most painful one.

Also define multiplication, [writes "Also define ·"] which is $(x_1, y_1) \cdot (x_2, y_2)$ [writes: $(x_1, y_1) \cdot (x_2, y_2) \overset{\text{def}}{=}$] — it will surprise no one that this is (x_1x_2, y_1y_2), okay? [writes: (x_1x_2, y_1y_2)] That's the definition.

And again, next is plus check [writes "+ check"] of compatibility with the equivalence classes. [writes "compatibility with ≡"] And this check is easy, so, I'll do it since it's easy.

Spoken Transcript [01:03:40 - 01:05:45]

Prof. Rahul Pandharipande

Suppose we have here $(x_1, y_1) \equiv (\hat{x}_1, \hat{y}_1)$ [writes: $(x_1, y_1) \equiv (\hat{x}_1, \hat{y}_1)$] and $(x_2, y_2) \equiv (\hat{x}_2, \hat{y}_2)$. [writes: $(x_2, y_2) \equiv (\hat{x}_2, \hat{y}_2)$] That's the input, right?

Then we must show [writes "We must show"] that (x_1x_2, y_1y_2) is in the same equivalence class as if we chose the other representatives: $(\hat{x}_1\hat{x}_2, \hat{y}_1\hat{y}_2)$, [writes: $(x_1x_2, y_1y_2) \equiv (\hat{x}_1\hat{x}_2, \hat{y}_1\hat{y}_2)$] okay? That's what we have to do.

What we know is this: that equation that this is in the same equivalence class is

$x_1\hat{y}_1 = \hat{x}_1 y_1$, [writes: $x_1\hat{y}_1 = \hat{x}_1 y_1$] and this equation is $x_2\hat{y}_2 = \hat{x}_2 y_2$ [writes: $x_2\hat{y}_2 = \hat{x}_2 y_2$] — all very confusing.

And now we just multiply them together! These are what we know, and having these two equations, we can multiply them all together, and we get $x_1x_2\hat{y}_1\hat{y}_2 = \hat{x}_1\hat{x}_2 y_1 y_2$. [writes: $x_1x_2\hat{y}_1\hat{y}_2 = \hat{x}_1\hat{x}_2 y_1 y_2$] Now there's subscripts and hats.

And that equation is exactly the certificate that these things are in the same equivalence class, okay? [draws double arrow $\Longleftrightarrow$ and checkmark] This implies that.

So, this shows that if I pick different equivalence classes here, the result doesn't matter. This one I did completely, because it's easy: all you do is multiply. For the other one, you have to do the kind of tricks we were doing before, where you have to clear the denominators, etc. You can do that, so to speak, on your own time.

III Compatibility of the Operations with the Canonical Embedding

Spoken Transcript [01:05:45 - 01:07:25] Prof. Rahul Pandharipande

Then you have to check that these definitions are compatible with R. They're just all defined that way, meaning we have to check [writes "Check $i\colon R \to F$ is a hom of rings"] that this $\iota\colon R \to F$ is a homomorphism of rings.

Again, there's really nothing to this: it means that if you take $r + \hat{r}$ [writes: $r + \hat{r} \xrightarrow{i} (r + \hat{r}, 1)$] and you go by ι, you get $(r + \hat{r}, 1)$. Then you have to check that this thing is equal to $(r, 1) + (\hat{r}, 1)$, [writes: $= (r,1) + (\hat{r},1)$] and that's the same by the definition that we used there. But you have to check it.

You also have to check it for multiplication. So, it's just a bunch of checking. By here, all the cards are played, and you just have to check everything.

Multiplication on F and Ring Homomorphism Properties

Definition 3.23 (*Multiplication on F*): For any $[(x_1, y_1)], [(x_2, y_2)] \in F$, multiplication is defined by:

$$[(x_1, y_1)] \cdot [(x_2, y_2)] \overset{\text{def}}{=} [(x_1x_2, y_1y_2)]. \tag{3.29}$$

Proposition 3.24 (*Well-Definedness of Multiplication*): Multiplication on F is well-defined and independent of the choice of representative.

Proof

The pair (x_1x_2, y_1y_2) lies in S, since $y_1y_2 \neq 0$ in the integral domain R (as for addition, Proposition 3.22).

Let $(x_1, y_1) \equiv (\hat{x}_1, \hat{y}_1)$ and $(x_2, y_2) \equiv (\hat{x}_2, \hat{y}_2)$ in S. By definition of $\equiv$:

$$x_1\hat{y}_1 = \hat{x}_1 y_1 \quad \text{and} \quad x_2\hat{y}_2 = \hat{x}_2 y_2. \tag{3.30}$$

Multiplying these two equations in the commutative ring R:

$$(x_1\hat{y}_1)(x_2\hat{y}_2) = (\hat{x}_1 y_1)(\hat{x}_2 y_2) \implies (x_1x_2)(\hat{y}_1\hat{y}_2) = (\hat{x}_1\hat{x}_2)(y_1y_2).$$

By (3.16), this algebraic identity is precisely the condition for equivalence:

$$(x_1x_2, y_1y_2) \equiv (\hat{x}_1\hat{x}_2, \hat{y}_1\hat{y}_2).$$

Thus, $[(x_1x_2, y_1y_2)] = [(\hat{x}_1\hat{x}_2, \hat{y}_1\hat{y}_2)]$, proving well-definedness. **Q.E.D.**

Proposition 3.25 (*i is an Injective Ring Homomorphism*)**:** The canonical inclusion $i\colon R \hookrightarrow F$ is an injective ring homomorphism:

$$i(r+\hat{r}) = i(r) + i(\hat{r}), \tag{3.31}$$

$$i(r \cdot \hat{r}) = i(r) \cdot i(\hat{r}), \tag{3.32}$$

$$i(1_R) = 1_F. \tag{3.33}$$

Proof

Evaluating the definitions directly:

- **Additivity:**
$$i(r) + i(\hat{r}) = [(r,1)] + [(\hat{r},1)] = [(r\cdot 1 + \hat{r}\cdot 1, 1\cdot 1)] = [(r+\hat{r},1)] = i(r+\hat{r}).$$
- **Multiplicativity:**
$$i(r)\cdot i(\hat{r}) = [(r,1)]\cdot[(\hat{r},1)] = [(r\hat{r}, 1\cdot 1)] = [(r\hat{r},1)] = i(r\hat{r}).$$
- **Identity:**
$$i(1_R) = [(1,1)] = 1_F.$$

Injectivity was already established in Proposition 3.19. **Q.E.D.**

Multiplicative Inverses in F (beyond the lecture)

The lecture leaves the field axioms, inverses included, to the students (*"a lot of long and boring work"*). To confirm that F is indeed a field, let $[(x,y)] \in F$ with $[(x,y)] \neq 0_F = [(0,1)]$. Then $(x,y) \not\equiv (0,1) \implies x\cdot 1 \neq 0 \cdot y \implies x \neq 0$. Thus, $(y,x) \in S$, and:

$$[(x,y)]\cdot[(y,x)] = [(xy,yx)] = [(1,1)] = 1_F.$$

Therefore, every non-zero element in F possesses a multiplicative inverse $[(x,y)]^{-1} = [(y,x)]$. **Q.E.D.**

The Remaining Field Axioms (beyond the lecture)

The lecture leaves these checks to the students as well; they reduce to computations in R on representatives. The key tool is that $(xs,ys) \equiv (x,y)$ for every $s \neq 0$, since $xs\cdot y = x \cdot ys$: a common non-zero factor can be cancelled.

- **Commutativity** of $+$ and $\cdot$ is visible in the formulas of Definition 3.21 and Definition 3.23, since R is commutative.
- **Associativity:** both ways of bracketing a sum of three classes give the same class,
$$\begin{aligned}\big([(x_1,y_1)] + [(x_2,y_2)]\big) + [(x_3,y_3)] &= [(x_1y_2y_3 + x_2y_1y_3 + x_3y_1y_2,\ y_1y_2y_3)] \\ &= [(x_1,y_1)] + \big([(x_2,y_2)] + [(x_3,y_3)]\big),\end{aligned}$$
and both ways of bracketing a product give $[(x_1x_2x_3,\ y_1y_2y_3)]$.
- **Neutral elements and negatives:** $[(x,y)] + [(0,1)] = [(x,y)]$, $[(x,y)]\cdot[(1,1)] = [(x,y)]$, and $[(x,y)] + [(-x,y)] = [(0,y^2)] = [(0,1)]$.
- **Distributivity:** expanding both sides gives
$$[(x_1,y_1)]\cdot\big([(x_2,y_2)] + [(x_3,y_3)]\big) = [(x_1x_2y_3 + x_1x_3y_2,\ y_1y_2y_3)],$$
$$[(x_1,y_1)]\cdot[(x_2,y_2)] + [(x_1,y_1)]\cdot[(x_3,y_3)] = [(x_1x_2y_1y_3 + x_1x_3y_1y_2,\ y_1^2y_2y_3)],$$

and the two right-hand sides are the same class after cancelling the factor $y_1 \neq 0$.

- **$0_F \neq 1_F$:** $(0,1) \not\equiv (1,1)$, since $0 \cdot 1 \neq 1 \cdot 1$ in the integral domain R.

Q.E.D.

Summary of the Field of Fractions Construction

Spoken Transcript [01:07:25 - 01:08:00] Prof. Rahul Pandharipande

That is the construction. When you understand it, you'll realize that actually you knew the whole thing anyway in your understanding of the rational numbers: you were subconsciously doing all of this. Not explicitly, but somehow underneath the surface, you were assuming all of these properties were true — it made the rational number arithmetic work.

There's different ways to try to learn this material, but the way that's foundational is you actually go yourself and check all the things that you're supposed to check. That takes a little bit of time and it's somewhat painful, but you only have to do it once in your life.

You can read the book too. I think that doesn't help so much, because then you just read what the author has written, and it's just a bunch of things with equalities — you can convince yourself of anything if you read what someone else did. The best thing is to go through all of it yourself; then you'll understand it once in your life.

And that's the construction. We're done with this construction.

Spoken Transcript [01:08:00 - 01:09:44] Prof. Rahul Pandharipande

The construction I just summarize here: [writes "Construction"] we start with R is an integral domain, [writes "Start with R int. domain"] and then we construct explicitly — that's as explicit a math construction as you get. You just take R, and you make some sets and equivalence classes and relations; it's totally explicit. Construct this F, which is the field of fractions. [writes "Construct F = field of fractions"]

And R sits inside F by this map: r goes to $(r,1)$. [writes: $R \hookrightarrow F$, $r \mapsto (r,1)$]

When you get used to it, the way an algebraist always thinks about this is: you have elements in R, [writes: $r \in R$] and the elements of F are just r over s. That's the element of F; you just view them as kind of fractions in R. [writes: $\frac{r}{s} \in F$]

This is a perfectly reasonable way to think about it. In our language, when someone writes this, what they really mean is the equivalence class of (r,s) in this set S. That's what it actually means. [writes: $(r,s) \in S$]

And it's only legal if the bottom one is not zero, which you're not allowed to write anyway if the bottom one's not zero. And then you can add fractions, multiply fractions, just as you would normally do, and they exactly correspond to the rules that were prescribed here. So, that's the natural way to think about it.

Once all of this is done, you just start using it like you normally would anyway. But this is the underlying foundation.

Are there any conceptual questions about that?

Summary of the Construction of the Field of Fractions

Theorem 3.26 (*Existence and Structure of the Field of Fractions*): Let R be an integral domain. There exists a field F, called the *"field of fractions"* of R (denoted $\mathrm{Frac}(R)$), and an injective ring homomorphism:

$$i\colon R \hookrightarrow F, \quad r \mapsto [(r,1)],$$

such that:

(1) Field Structure: F is a field with zero $0_F = [(0,1)]$ and multiplicative identity $1_F = [(1,1)]$.

(2) Quotient Representation: Every element of F can be written as a quotient of elements in the image of R:

$$\forall q \in F\ \exists r,s \in R\ (s \neq 0) \quad \text{such that} \quad q = i(r) \cdot i(s)^{-1} = \frac{i(r)}{i(s)}. \tag{3.34}$$

Algebraic View: Working with Fractions

In mathematical practice, once the formal construction of $F = S_{\equiv}$ is established, we identify R with its image $i(R) \subseteq F$ and adopt standard fraction notation:

$$\frac{r}{s} := [(r,s)] \in \mathrm{Frac}(R) \quad (r,s \in R,\ s \neq 0). \tag{3.35}$$

Arithmetic in the Field of Fractions: Under this notation, the field operations on $\mathrm{Frac}(R)$ take the standard form familiar from elementary arithmetic:

$$\text{Equality:} \quad \frac{r_1}{s_1} = \frac{r_2}{s_2} \iff r_1 s_2 = r_2 s_1, \tag{3.36}$$

$$\text{Addition:} \quad \frac{r_1}{s_1} + \frac{r_2}{s_2} = \frac{r_1 s_2 + r_2 s_1}{s_1 s_2}, \tag{3.37}$$

$$\text{Multiplication:} \quad \frac{r_1}{s_1} \cdot \frac{r_2}{s_2} = \frac{r_1 r_2}{s_1 s_2}, \tag{3.38}$$

$$\text{Inverse:} \quad \left(\frac{r}{s}\right)^{-1} = \frac{s}{r} \quad (\text{provided } r \neq 0). \tag{3.39}$$

Examples of Fields of Fractions

Spoken Transcript [01:09:44 - 01:10:46] Prof. Rahul Pandharipande

If I take $\mathbb{Z}$, [writes: $\mathbb{Z}$ → What is field of fractions?] we can ask: what is the field of fractions?

The answer is going to be $\mathbb{Q}$, [writes: $\mathbb{Q}$] which means that if you start with $\mathbb{Z}$ and you do this crazy construction, you get the rational numbers. I hope that's clear by now. All right?

What happens if I start with the ring of Gaussian integers? What is the field of fractions? [writes: $\mathbb{Z}[i]$ → What is the field of fractions?]

That's a strange question, because I've just explained what the field of fractions is:

you take the Gaussian integers, you do this construction, that's the field of fractions. When I say *"what is the field of fractions?"*, what the question is actually asking is: what's another way of thinking about the answer?

Do you have a proposal for what that is? Yeah?

Student Answer

$\mathbb{Q}$ adjoined i.

Spoken Transcript [01:10:46 - 01:11:48] Prof. Rahul Pandharipande

Yeah, so, the answer is $\mathbb{Q}[i]$, [writes: $\mathbb{Z}[i] \subset \mathbb{Q}[i] \rightsquigarrow \{a+bi \mid a,b \in \mathbb{Q}\}$] which is equal to the set $a+bi$ where a and b are in $\mathbb{Q}$.

It's correct, but to check this is correct, you have to do a bunch of things: one is you have to show this is a field — you can invert — and then you have to show that this sits inside, and everything here is a ratio of that. It's all true. But that's the answer here, and requires some checking.

All right, this is correct.

Fields of Fractions of Intermediate Rings and Fields

Spoken Transcript [01:11:48 - 01:12:32] Prof. Rahul Pandharipande

What about this? We saw last week that there are some funny rings that sit inside $\mathbb{Z}$ and $\mathbb{Q}$. [writes: $\mathbb{Z} \subset \mathbb{Z}_{1/3} \subset \mathbb{Q}$]

This ring is the set $a/3^b$, where $a \in \mathbb{Z}$ and $b \geq 0$, right? [writes: $\{a/3^b \mid a \in \mathbb{Z}, b \geq 0\}$]

We discussed this last week (Recall: Example 2.29): the subrings that sit in between. I hope you remember. Do you remember?

This is an integral domain also, [writes "int. domain"] because it sits inside $\mathbb{Q}$. What is the field of fractions of... [writes "What is field of fractions of $\mathbb{Z}_{1/3}$?"] I just gave it a name, $\mathbb{Z}_{1/3}$, because you're allowed to have threes in the bottom. There's two standard names for this: one is $\mathbb{Z}_{1/3}$, and the other's... that's a better name, okay.

So, what is the field of fractions of this integral domain? Yeah?

Student Answer

$\mathbb{Q}$.

Spoken Transcript [01:12:33 - 01:13:22] Prof. Rahul Pandharipande

Also $\mathbb{Q}$. [writes: $\mathbb{Q}$]

Nothing says it can't be $\mathbb{Q}$. It certainly sits inside $\mathbb{Q}$, and everything in $\mathbb{Q}$ is a ratio of numbers here — it's already a ratio of numbers here. So, it's just $\mathbb{Q}$! If you go through this definition, you get $\mathbb{Q}$.

As I said, you think about this field of fractions as the smallest field that contains the integral domain, and you can't really contain something smaller than $\mathbb{Q}$, because it already has $\mathbb{Z}$ in it, and $\mathbb{Z}$ already gives you everything in $\mathbb{Q}$.

So, what is the field of fractions of this? The answer: also $\mathbb{Q}$.

In fact, if you take any ring — any integral domain that sits in between $\mathbb{Z}$ and $\mathbb{Q}$ — the field of fractions is just $\mathbb{Q}$.

Examples: Fractions of the Integers, Gaussian Integers, and Subrings of $\mathbb{Q}$

Example 3.27 (*The Integers*): For the ring of integers $\mathbb{Z}$:

$$\mathrm{Frac}(\mathbb{Z}) = \mathbb{Q}.$$

Example 3.28 (*The Gaussian Integers*): For the ring of Gaussian integers $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$:

$$\mathrm{Frac}(\mathbb{Z}[i]) = \mathbb{Q}[i] := \{a + bi \mid a, b \in \mathbb{Q}\}.$$

The field $\mathbb{Q}[i]$ is called the field of *"Gaussian rationals"*.

Example 3.29 (*Denominators that Are Powers of 3*): Consider the ring $\mathbb{Z}_{1/3}$ (the case $k = 3$ of the generalization after Example 2.29), defined by:

$$\mathbb{Z}\left[\frac{1}{3}\right] := \left\{ \frac{a}{3^b} \;\middle|\; a \in \mathbb{Z},\ b \in \mathbb{Z}_{\geq 0} \right\} \subset \mathbb{Q}.$$

Since $\mathbb{Z} \subset \mathbb{Z}_{1/3} \subset \mathbb{Q}$, any fraction of elements from $\mathbb{Z}_{1/3}$ is already a rational number, and every rational number $\frac{p}{q}$ can be written as a quotient of elements in $\mathbb{Z} \subset \mathbb{Z}_{1/3}$. Thus:

$$\mathrm{Frac}\left(\mathbb{Z}\left[\frac{1}{3}\right]\right) = \mathbb{Q}.$$

Proposition 3.30 (*Intermediate Subrings of $\mathbb{Q}$*): Let R be any subring of $\mathbb{Q}$ with $\mathbb{Z} \subseteq R \subseteq \mathbb{Q}$. Then:

$$\mathrm{Frac}(R) = \mathbb{Q}. \tag{3.40}$$

$$\begin{tikzcd}
\mathbb{Q} \arrow[dr, bend left, "\mathrm{Frac}"] & \\
\mathbb{Z}_{1/3} \arrow[u, hook] \arrow[r, "\mathrm{Frac}"] & \boxed{\mathbf{Frac} = \mathbb{Q}} \\
\mathbb{Z} \arrow[u, hook] \arrow[ur, bend right, "\mathrm{Frac}"'] &
\end{tikzcd}$$

Why All Intermediate Subrings Yield $\mathbb{Q}$: Going beyond the lecture: the universal property was only proved for $\mathbb{Z}$ (Theorem 3.7), so here is a direct argument. Since $R \subseteq \mathbb{Q}$, a formal fraction $\frac{r}{s} \in \mathrm{Frac}(R)$ can be read as an actual rational number, which gives a map

$$\Phi\colon \mathrm{Frac}(R) \to \mathbb{Q}, \qquad [(r,s)] \mapsto rs^{-1}.$$

It is well-defined: $(r,s) \equiv (r',s')$ means $rs' = r's$, hence $rs^{-1} = r's'^{-1}$ in $\mathbb{Q}$. It is a ring homomorphism, because the operations of $\mathrm{Frac}(R)$ (Definition 3.21, Definition 3.23) are the usual fraction rules of $\mathbb{Q}$. As a homomorphism out of a field into a non-zero ring it is injective (Theorem 1.31 and the steps after it). It is surjective, since every rational number is $\frac{p}{q}$ with $p, q \in \mathbb{Z} \subseteq R$, $q \neq 0$. Hence Φ is an isomorphism, $\mathrm{Frac}(R) \cong \mathbb{Q}$.

Beyond the Lecture: Universal Property of $\mathrm{Frac}(R)$ and $\mathrm{Frac}(\mathbb{Z}[i]) \cong \mathbb{Q}[i]$

Textbooks usually write the field $\mathbb{Q}[i]$ of these notes as $\mathbb{Q}(i)$, and the ring $\mathbb{Z}_{1/3}$ as $\mathbb{Z}[\frac{1}{3}]$. The proof of Theorem 3.7 carries over verbatim from $\mathbb{Z} \subset \mathbb{Q}$ to any integral domain R and its canonical embedding $\iota\colon R \hookrightarrow \mathrm{Frac}(R)$, $r \mapsto \frac{r}{1}$: for every injective ring homomorphism $f\colon R \hookrightarrow K$ into a field K, there exists a unique ring homomorphism $g\colon \mathrm{Frac}(R) \to K$ with $g \circ \iota = f$, given by $g\left(\frac{r}{s}\right) = f(r)f(s)^{-1}$ (which is injective because $\mathrm{Frac}(R)$ is a field). In particular, for $\mathbb{Z}[i] \subset \mathbb{Q}[i] := \{a + bi \mid a, b \in \mathbb{Q}\} \subset \mathbb{C}$, every non-zero element $a + bi \in \mathbb{Q}[i]$ has $a^2 + b^2 > 0$ in $\mathbb{Q}$ and inverse $(a + bi)^{-1} = \frac{a}{a^2+b^2} - \frac{b}{a^2+b^2} i \in \mathbb{Q}[i]$, so $\mathbb{Q}[i]$ is a field. By the universal property, the inclusion $\mathbb{Z}[i] \hookrightarrow \mathbb{Q}[i]$ extends to an injective homomorphism $\Phi\colon \mathrm{Frac}(\mathbb{Z}[i]) \hookrightarrow \mathbb{Q}[i]$. Since every $\frac{a_1}{b_1} + \frac{a_2}{b_2} i = \frac{a_1 b_2 + a_2 b_1 i}{b_1 b_2}$ is a ratio of elements in $\mathbb{Z}[i]$, Φ is also surjective, proving $\mathrm{Frac}(\mathbb{Z}[i]) \cong \mathbb{Q}[i]$ (Example 3.28).

Spoken Transcript [01:13:22 - 01:14:49] Prof. Rahul Pandharipande

Now a stranger question: what happens if... well, yeah. [washes the board] You know, on the business of guessing, once I was in a conference, and the conference had some math researchers and some high school teachers. At dinner I was sitting next to some teachers, and I was trying to explain to them my theory of how you should teach children math. And the outcome was that they wanted me disbarred from academics! [laughter]

The theory, by the way, was that on exams you should make the kids just guess the answers: no calculations, just guess everything. That's maybe a little excessive, but maybe a compromise: like, half the exam they should just guess everything. But I think that would be really somehow more in the spirit of mathematics than... [finishes drying the board]

Spoken Transcript [01:14:49 - 01:15:25] Prof. Rahul Pandharipande

A stranger question is: suppose I start with a field. [writes "Suppose R is a field"] Then, of course, it's still an integral domain, [writes "R is int. domain"] because every field is an integral domain (Recall: Example 2.61). Then what is the field of fractions? [writes "What is the field of fractions?"]

What do you think the field of fractions is going to be here? Yeah?

Student Answer

R itself.

Spoken Transcript [01:15:28 - 01:16:00] Prof. Rahul Pandharipande

Yes, R itself. And the answer is R itself. [writes "R itself"]

To prove that: again, this fits in this view that it's the smallest field. If it's already a field, how could there be a smaller field that contains it? It's already a field.

To prove it, you have to go through this construction and show that in the case where R is a field, that construction just reconstructs R. To do that, you have to go through that construction and see it. So, that's also an exercise for you.

Those are really the basic facts.

Proposition 3.31 (*Field of Fractions of a Field*): Let R be a field. Then the canonical inclusion $i\colon R \hookrightarrow \mathrm{Frac}(R)$ is an isomorphism of fields:

$$\mathrm{Frac}(R) \cong R. \tag{3.41}$$

Why i Is an Isomorphism (beyond the lecture)

The lecture leaves this to the students. By Proposition 3.25, $i\colon R \hookrightarrow \mathrm{Frac}(R)$ is an injective field homomorphism. To establish surjectivity, let $[(x,y)] \in \mathrm{Frac}(R)$ with $x,y \in R$ and $y \neq 0$. Since R is a field, y has a multiplicative inverse $y^{-1} \in R$. Then:

$$(x,y) \equiv (xy^{-1},1) \iff x \cdot 1 = (xy^{-1}) \cdot y = x.$$

Consequently:

$$[(x,y)] = [(xy^{-1},1)] = i(xy^{-1}).$$

Every element of $\mathrm{Frac}(R)$ is the image of the element $xy^{-1} \in R$. Hence, i is surjective and constitutes a field isomorphism. **Q.E.D.**

Polynomial Rings

Spoken Transcript [01:16:00 - 01:18:05] Prof. Rahul Pandharipande

The next topic this week is about polynomial rings. These are really fun. [writes "Polynomial rings, Power series rings"] We talk about polynomial rings, and in some sense even more fun than polynomial rings are power series rings. Again, in some sense you've been using all of this already in the first year, but here we take a formal approach.

The idea is: if I take R as a ring [writes "R is a ring"] — and now no assumptions, meaning not an integral domain, so, this just means for us commutative ring with unit: [writes "(comm. ring with unit)"] that's our basic ring. I would say a ring with no assumptions: commutative ring with unit (Recall: Remark 1.17). In particular, everything today so far, R was an integral domain; now I don't care if it's an integral domain, just a commutative ring with unit.

Then I make here a new ring: $R[x]$. [writes: $R[x]$] x is the variable, or sometimes it's called indeterminate [writes "the indeterminate"] — that's kind of a fancy name, but it's just x, right? [writes "or the variable"]

The name of this ring is the polynomial ring with coefficients in R. [writes "(Poly. ring with the coeff's in R)"]

Or if you want to be even more precise: the polynomial ring in one variable with coefficients in R. That's kind of a long sentence, but if you want to be totally precise. If you meet a mathematician and say, *"I'm going to consider the polynomial ring in one variable with coefficients in R"*, as long as you agree what a ring is, it's just absolutely unambiguous: the person will know exactly what you're talking about.

Informal Definition of Polynomials as Formal Expressions

Spoken Transcript [01:18:05 - 01:19:08] Prof. Rahul Pandharipande

What is this? Well, I have to tell you what it is as a set. So, the first thing is: what's the set? [writes "What is the set?"]

Here the definition is: it's the set of polynomials in x with coefficients in R. Polynomials look like this: $a_0 + a_1x + a_2x^2 + \dots$ [writes: $R[x] = \{a_0 + a_1x + a_2x^2 + \dots\}$]

It's these symbols, and one of the properties of a polynomial is that it's not infinite: a polynomial is finite. That's part of the definition always in mathematics of a polynomial. That means we stop somewhere: some n, a_n. So, the elements of this ring are such symbols — just formal combinations of $a_0, \dots, a_n$, where these a_i's are in R, okay? [writes: $+a_nx^n \mid a_i \in R, n \in \mathbb{Z}_{\geq 0}\}$]

And possibly different n's, and this n is some integer greater than or equal to zero.

Examples and Constant Polynomials

Spoken Transcript [01:19:08 - 01:20:48] Prof. Rahul Pandharipande

If I have $\mathbb{Z}[x]$, [writes: $\mathbb{Z}[x] \ni f(x)$] the elements in here are sometimes written $f(x)$, because we like to write polynomials with the variable. It could be, for example, $f(x) = 3 + 2x + 15x^2 + 345x^{200}$. [writes: $= 3 + 2x + 15x^2 + 345x^{200}$] This is a totally legitimate element of it. Other times you'll see people just write this as just f, and just suppress the variable. [writes: or f] Those are two different ways you can write polynomials.

For $R[x]$, it's just the same thing: you just write an element here, $f(x)$ [writes: $R[x] \ni f(x) = a_0 + a_1x + \dots + a_nx^n$] — you pick some elements of R, some finite number of elements of R as the coefficients, all the way to a_n, and then you just put the x's here.

x is a formal variable, so, x is just in some sense a placeholder. Those are the elements of $R[x]$.

And $R[x]$ has a confusing element, just like the integers; there's a confusing polynomial: the constant polynomial. If I put here 17, [writes: $17 \in \mathbb{Z}[x]$] that is also here. That's the polynomial that only has a constant term: it just happens that there's no x. It can happen: there's no x in that. So, it's weird to write this polynomial as a function of x, but it is a function of x — it's a constant function.

In this ring there are also the constant polynomials — the ones with no x.

The Polynomial Ring $R[x]$: Informal Formulation

Recap: Polynomial rings were already used informally: $\mathbb{Z}[x] \subseteq \mathbb{Q}[x] \subseteq \mathbb{R}[x] \subseteq \mathbb{C}[x]$

in Week 1 (Example 1.22) and $\mathbb{Z}[x^2] \subset \mathbb{Z}[x]$ in Week 2 (Example 2.31). Here they are defined formally.

Definition 3.32 (*Polynomial Ring over a Commutative Ring*)**:** Let R be a commutative ring with unit 1_R. The *"polynomial ring in one variable"* (or indeterminate) x with coefficients in R, denoted $R[x]$, is defined as the set of formal expressions:

$$R[x] := \left\{ \sum_{i=0}^{n} a_i x^i = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \;\middle|\; n \in \mathbb{N}_0,\ a_i \in R \right\}. \tag{3.42}$$

Example 3.33 (*Polynomials in $\mathbb{Z}[x]$*)**:** In the ring $\mathbb{Z}[x]$, an element may be denoted $f(x)$ or simply f:

$$f(x) = 3 + 2x + 15x^2 + 345x^{200} \in \mathbb{Z}[x].$$

Constant elements such as $17 \in \mathbb{Z}$ correspond to polynomials of degree 0 with $n = 0$:

$$17 = 17 \cdot x^0 \in \mathbb{Z}[x].$$

The Indeterminate as a Formal Placeholder: Crucially, x is not an element of R, nor does it represent an unknown numerical value in this context. It is an *"indeterminate"* (formal variable) acting as a positional placeholder. The symbols $+$ and x^i in (3.42) are purely typographical markers separating the coefficients $a_i \in R$, rather than evaluations of operations.

Formal Rigorous Definition of $R[x]$ as Sequences

Spoken Transcript [01:20:48 - 01:22:05] Prof. Rahul Pandharipande

Is it clear what this set is? You take R and then you take an x that has nothing to do with R — it's just some kind of Platonic, ideal indeterminate — and then you just consider these formal linear combinations. That's what a polynomial is.

In particular, I don't mean really anything by this plus: it's just a formal linear combination. Some authors don't like to write this, because they think you might think that this plus here means something, which is a legitimate thing to think, actually.

So, if you look at books where they're really formally correct, [pushes upper chalkboard upward] they'll do some weird things just so you don't get confused.

I will explain it for a moment, but I don't do that because I think that it's a little dishonest. When I give these lectures, I try to actually give the lectures in the way I think about the subject: there's some kind of underlying honesty in the lecture.

Spoken Transcript [01:22:05 - 01:23:58] Prof. Rahul Pandharipande

If I were to be dishonest and formally correct, the way you define these polynomials is the following: [washes and dries the board] You say a polynomial — an element of $R[x]$ [writes "An element of $R[x]$ is a tuple of elements of R"] — is a tuple of elements of R: $(a_0, a_1, a_2, \ldots)$, [writes: $(a_0, a_1, a_2, \ldots)$] an infinite tuple such that there exists some big N so that for all $m > N$, the element in this tuple is zero, okay? [writes "and that $\exists N\ \forall m > N,\ a_m = 0$"]

That's a formal definition of a polynomial in one variable, where I just give you the coefficients. The coefficients are $a_0, a_1, a_2, \dots$ — they're all elements of R. [writes: $a_i \in R$]

I just give you the coefficients, and they could be lots and lots of them — any finite number — but eventually this sequence of coefficients is all zero. That's the formal definition of a polynomial.

It's actually the same thing that I'm saying, but it's much more reasonable to think about it with the x's. This way is just so that you don't get confused formally.

That's the set. I hope there's no confusion about the set.

The Indeterminate and the Ring Operations on $R[x]$

Spoken Transcript [01:23:58 - 01:24:24] Prof. Rahul Pandharipande

In this abstract way, what is the variable x? [writes: $x = \dots$]

You have five minutes to figure it out.

What's the variable x? In mine, there's an x. Here there's no x. So, what's x here? Yeah?

Student Answer

$(0, 1, 0, \dots)$

Spoken Transcript [01:24:27 - 01:25:20] Prof. Rahul Pandharipande

Yeah, so, x is this: [writes: $x = (0, 1, 0, \dots)$] zero, one, and then all zeros. That's x if you think about it this way, okay?

That's the set. Once you agree what the set is — and I gave you two ways to think about it: one way, which is a very simple way; this way is more formally correct — I need to give you the addition rule, the multiplication rule, zero, and one. [writes: $R[x], +, \cdot, 0, 1$]

Zero is just the zero polynomial: in my case, just the polynomial zero. In this case, you have to write an infinite number of zeros, which takes a long time. [writes "0" under 0, and $(0, 0, \dots)$]

And one is just the constant one polynomial. Here it's the $(1, 0, 0, \dots)$ polynomial. [writes "1" and "$(1, 0, \dots)$"] Yeah?

Student Question

Is the plus in that definition not the plus from the ring?

Spoken Transcript [01:25:25 - 01:26:05] Prof. Rahul Pandharipande

It is eventually, but I haven't defined the plus for the ring. That's the reason why this definition is a little bit better; this is a good thing to think about.

I'm defining $R[x]$ as a set first. Then I'll define addition. When I finish, it will be, but when it's born, it's just a set. Is that clear? [Student nods] It's a good question. That's the reason why thinking about it this way is slightly confusing. Although I encourage you to think about it this way, because you have a lot of experience with polynomials, and in the end, nothing will go wrong.

Formal Definition of Polynomials as Sequences

Definition 3.34 (*Rigorous Definition of $R[x]$ via Coefficient Tuples*)**:** Let R be a commutative ring with unit. The polynomial ring $R[x]$ is formally defined as the set of sequences $(a_k)_{k\in\mathbb{N}_0}$ of elements in R (elements of $R^{\mathbb{N}_0}$, Notation 0.1) that are eventually zero:

$$R[x] := \left\{(a_0, a_1, a_2, \dots) \in R^{\mathbb{N}_0} \;\middle|\; \exists N \in \mathbb{N}_0 \; \forall m > N \colon a_m = 0\right\}. \tag{3.43}$$

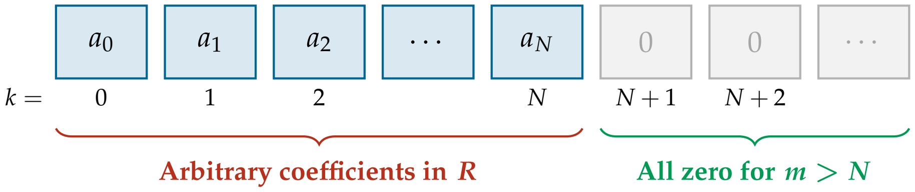
Proposition 3.35 (*Identification of the Indeterminate x and Constants*)**:** Under the sequence model (3.43):

(1) Indeterminate x: The indeterminate x corresponds to the elementary sequence with 1 in position 1:

$$x := (0, 1, 0, 0, \dots). \tag{3.44}$$

(2) Constant Elements: For any $a \in R$, the constant polynomial corresponds to the sequence:

$$a = (a, 0, 0, 0, \dots). \tag{3.45}$$



Addition in $R[x]$

Spoken Transcript [01:26:05 - 01:26:28] Prof. Rahul Pandharipande

You have to agree what the set is, and I gave you two different ways to think about the set. Then you have to define addition. You can define addition in that way as addition of polynomials, which is very good, because I don't have to write it, because you already know how to add polynomials. Is that correct?

If I write it here, then I have to decide how to add polynomials, and that's not hard. How do you add polynomials here? Yeah?

Student Answer

Componentwise addition.

Spoken Transcript [01:26:30 - 01:27:08] Prof. Rahul Pandharipande

Componentwise addition. Which is also how you add polynomials there: componentwise addition.

And it will be the case there that if I take the constant polynomial and I take this polynomial and I add them, [writes "a_0", "x", "$a_0 + x$"] I get the thing I call this, which is also kind of nice. All right?

Addition is easy. Then they have to define multiplication. Multiplication is more of a pain. And if you think about it there, you already know how to multiply polynomials: you have to do this thing where you distribute and whatever — you know how to do it.

How do you define polynomial multiplication here? Let's try to do that.

Multiplication in $R[x]$ and the Cauchy Product

Spoken Transcript [01:27:08 - 01:27:43] Prof. Rahul Pandharipande

If I use this list of coefficients, here's my first polynomial, f, [writes: $f = (a_0, a_1, a_2, \dots)$] and my second polynomial, g. [writes: $g = (b_0, b_1, b_2, \dots)$] Then I have to define this polynomial, which is $f \times g$. [writes: $fg = ()$]

How do I do it? What's this coefficient going to be? What do you think? Yeah?

Student Answer

$a_0 b_0$.

Spoken Transcript [01:27:44 - 01:28:55] Prof. Rahul Pandharipande

Yeah, because it's just the constant... The secret, the way to do this is you write this thing out... This is a definition, but the way to understand this definition is you write this out as a polynomial: [writes: $a_0 + a_1 x + \dots$] that's the easy way to think about it. And so on: write this as a polynomial, [writes: $b_0 + b_1 x + \dots$] and you do polynomial multiplication, and the constant term is that: $a_0 b_0$. [writes: $a_0 b_0$]

What is this term? Well, it's going to be $a_1 b_0 + a_0 b_1$. [writes: $a_1 b_0 + a_0 b_1$] The second term is going to be $a_2 b_0 + a_1 b_1 + a_0 b_2$. [writes: $+a_2 b_0 + a_1 b_1 + a_0 b_2, \dots$]

Formally, this is the definition of polynomial multiplication. Is that clear?

Student Question

This is essentially like doing the Cauchy product?

Spoken Transcript [01:29:05 - 01:29:40] Prof. Rahul Pandharipande

Yeah, you do this thing. [gestures diagonal grouping] That's what you're saying? Yeah, that's what you do. That's how you multiply polynomials.

This is the same rule as you'd get here. Look: if you take the linear term, how do you get an x here? You have to take x times a constant — those are the two ways to get the x. So, you get this rule.

The Ring Structure of $R[x]$

Definition 3.36 (*Ring Operations on $R[x]$*)**:** Let R be a commutative ring with unit 1_R. The polynomial ring $(R[x], +, \cdot, 0_{R[x]}, 1_{R[x]})$ is equipped with the following structures:

(1) Neutral Elements:

$$0_{R[x]} := (0_R, 0_R, 0_R, \dots), \tag{3.46}$$

$$1_{R[x]} := (1_R, 0_R, 0_R, \dots). \tag{3.47}$$

(2) Addition (Componentwise): For $f = (a_k)_{k\geq 0}$ and $g = (b_k)_{k\geq 0}$:

$$(a_0, a_1, a_2, \dots) + (b_0, b_1, b_2, \dots) := (a_0 + b_0,\ a_1 + b_1,\ a_2 + b_2,\ \dots). \tag{3.48}$$

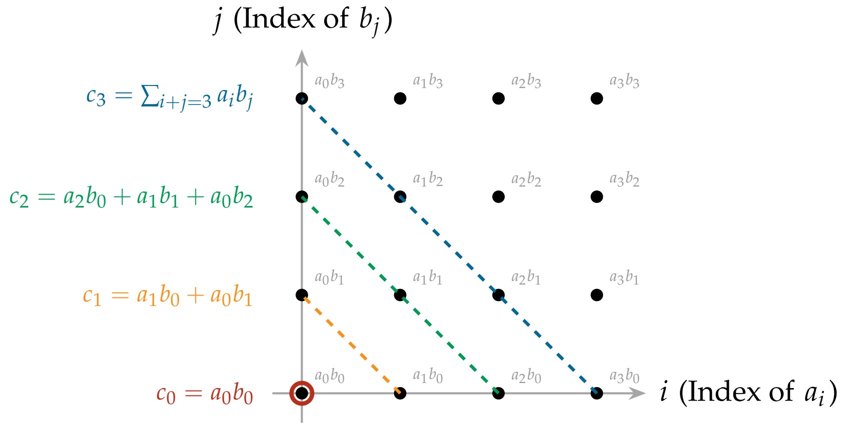
(3) Multiplication (Cauchy Product): The product $f \cdot g = (c_k)_{k\geq 0}$ is defined by:

$$c_k := \sum_{i=0}^{k} a_i b_{k-i} = a_0 b_k + a_1 b_{k-1} + \dots + a_k b_0. \tag{3.49}$$

Explicitly:

$$f \cdot g = \big(a_0 b_0,\ a_0 b_1 + a_1 b_0,\ a_0 b_2 + a_1 b_1 + a_2 b_0,\ \dots\big). \tag{3.50}$$

Both results are again eventually zero: if $a_k = 0$ for $k > N$ and $b_k = 0$ for $k > M$, then $a_k + b_k = 0$ for $k > \max(N, M)$, and $c_k = 0$ for $k > N + M$ (each term $a_i b_{k-i}$ has $i > N$ or $k - i > M$).


j (Index of b_j)
$c_3 = \sum_{i+j=3} a_i b_j$
$c_2 = a_2 b_0 + a_1 b_1 + a_0 b_2$
$c_1 = a_1 b_0 + a_0 b_1$
$c_0 = a_0 b_0$
a_0b_3
a_1b_3
a_2b_3
a_3b_3
a_0b_2
a_1b_2
a_2b_2
a_3b_2
a_0b_1
a_1b_1
a_2b_1
a_3b_1
a_0b_0
a_1b_0
a_2b_0
a_3b_0
i (Index of a_i)

Connection to the Cauchy Product: The formula $c_k = \sum_{i=0}^{k} a_i b_{k-i}$ exactly mirrors:

(1) Distributive Expansion: In formal series notation:

$$\left(\sum_{i=0}^{n} a_i x^i\right)\left(\sum_{j=0}^{m} b_j x^j\right) = \sum_{i=0}^{n}\sum_{j=0}^{m} a_i b_j x^{i+j} = \sum_{k=0}^{n+m}\left(\sum_{i+j=k} a_i b_j\right) x^k.$$

The powers of the indeterminate x naturally collect all terms where the sum of indices equals k.

(2) Cauchy Product of Power Series: In analysis, the product of two absolutely convergent power series $\sum a_n x^n$ and $\sum b_n x^n$ is given by the exact same formula (Fact 0.43).

Conclusion of the Lecture

Spoken Transcript [01:29:40 - 01:30:00] Prof. Rahul Pandharipande

That's the ring; we're going to talk about it more next time. It's a little bit of a subtle matter, and it's good to make sure you completely understand it. Given a ring, there's a polynomial ring. A polynomial is actually just a list of coefficients. It's very helpful to think about the polynomial that way, because that's how you normally think about it, and you won't go wrong thinking about it that way.

Okay, we'll stop. [students applaud]

Outlook: Formal Power Series in the Next Lecture

In the lecture introduction, the professor remarked that *"in some sense even more fun than polynomial rings are power series rings"*. This lecture formalized the polynomial ring $R[x]$ as sequences of coefficients that are eventually zero. Dropping this requirement leads directly to the ring of *"formal power series"* $R[[x]]$, consisting of arbitrary infinite sequences $(a_0, a_1, a_2, \dots)$ equipped with the same Cauchy product multiplication, which is systematically investigated at the start of the next lecture (Definition 3.41).

Summary of Key Concepts

- **The Group of Units $\mathrm{Unit}(R)$ (also denoted $U(R)$, $R^\times$, or R^*):** The set of multiplicatively invertible elements $\mathrm{Unit}(R) = \{x \in R \mid \exists y \in R\colon xy = 1\}$ forms an abelian group under ring multiplication. Key examples include $\mathrm{Unit}(\mathbb{Z}) = \{\pm 1\}$, $\mathrm{Unit}(\mathbb{Z}[i]) = \{\pm 1, \pm i\}$ (characterized via $a^2 + b^2 = 1$), and $(\mathbb{Z}/m\mathbb{Z})^* = \{\bar{x} \mid \mathrm{GCD}(x,m) = 1\}$.
- **Euler's Totient Function $\phi(m)$:** Defined as the order $|(\mathbb{Z}/m\mathbb{Z})^*|$, with explicit values $\phi(p) = p-1$ and $\phi(p^2) = p^2 - p$.
- **Universal Property of $\mathbb{Q}$:** As the minimal field containing the integral domain $\mathbb{Z}$, any injective ring homomorphism $f\colon \mathbb{Z} \to F$ into a field F factors uniquely through an injective field homomorphism $g\colon \mathbb{Q} \to F$ defined by $g(x/y) = f(x)f(y)^{-1}$.
- **Field of Fractions $\mathrm{Frac}(R)$:** For any integral domain R, the field of fractions is constructed as the quotient of formal pairs $S = R \times (R \setminus \{0\})$ by the cross-multiplication equivalence relation $(x_1, y_1) \equiv (x_2, y_2) \overset{\text{def}}{\iff} x_1 y_2 = x_2 y_1$. Transitivity and the non-vanishing of the denominators $y_1 y_2 \neq 0$ crucially rely on the zero-divisor-free property of R.
- **Canonical Inclusion and Examples:** The map $i\colon R \hookrightarrow \mathrm{Frac}(R)$, $r \mapsto [(r,1)]$, is an injective ring embedding. Examples include $\mathrm{Frac}(\mathbb{Z}) = \mathbb{Q}$, $\mathrm{Frac}(\mathbb{Z}[i]) = \mathbb{Q}[i]$, $\mathrm{Frac}(\mathbb{Z}_{1/3}) = \mathbb{Q}$, and $\mathrm{Frac}(R) \cong R$ for any existing field R.
- **Polynomial Rings $R[x]$:** For a commutative ring with unity R, the polynomial ring $R[x]$ is formally defined as the ring of sequences $(a_k)_{k \in \mathbb{N}_0}$ in $R^{\mathbb{N}_0}$ that are eventually zero. It is equipped with componentwise addition and Cauchy product multiplication $c_k = \sum_{i=0}^{k} a_i b_{k-i}$, with indeterminate $x = (0,1,0,0,\dots)$.

Wednesday, 30th September 2026

Field of Fractions, Polynomial Rings, and Formal Power Series

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Context

Continuing from the previous lecture, we review the construction and intuitive arithmetic of the field of fractions $\mathrm{Frac}(R)$ for an integral domain R. We then return to the polynomial ring $R[x]$ over an arbitrary commutative ring with identity, establish its universal mapping property, and determine its units when R is an integral domain. Finally, we introduce the formal power series ring $R[[x]]$, define its ring operations, and characterize its units via formal geometric series.

Field of Fractions and Ring Constructions

Field of Fractions of an Integral Domain

Spoken Transcript [00:00:00 - 00:01:00] Prof. Rahul Pandharipande

Can I start? Okay, we start.

The topic this week was... the first part was R is an integral domain [writes: R int domain], and then it's the construction of the field of fractions [writes "$\implies R \subset F$" and "field of fraction of R"] of R (Recall: Definition 3.17). Sometimes this is denoted by $\mathrm{Frac}(R)$ [writes: $F = \mathrm{Frac}(R)$].

And then the other two topics are: if R is any ring [writes: R any ring], which for us is commutative with unit, then there's the construction of the polynomials [writes: $R[x]$] — we started that, I'll say more — and also of the power series [writes: $R[[x]]$]. This is the universal symbol for polynomial [writes: poly], and this is the universal mathematical way of describing power series, the power series ring [writes: power series].

So, I'll talk about these topics.

Topic Overview: Integral Domains and Ring Constructions

- **Field of Fractions:** If R is an integral domain, there exists an embedding into a field:

$$R \subset F := \mathrm{Frac}(R),$$

where F is called the *"field of fractions"* of R.

- **Polynomial and Power Series Rings:** If R is any commutative ring with identity 1, we construct:
  - $R[x]$: the *"polynomial ring"* in the indeterminate x with coefficients in R,
  - $R[[x]]$: the *"formal power series ring"* in the indeterminate x with coefficients in R.

Spoken Transcript [00:01:00 - 00:01:46] Prof. Rahul Pandharipande

I'll just say one more thing about the field of fractions. I gave you this very formal way to think about the field of fractions, which is you take some set S of pairs [writes: $S \subset R^2$] — I won't review the whole thing now —, a set S of pairs, then you have an equivalence relation [writes: $\equiv$], and you look at the objects in this field of fractions as being equivalence classes in this equivalence relation [writes: $S_{\equiv} \ni f$]. And then you can define everything at this formal level and then check all the properties.

And that is, you know, the underlying mathematical formalism of the field of fractions — meaning that if you're ever thinking about the field of fractions and you're wondering about something, you can always resolve it foundationally by saying, *"I know what the definition is. It's this, this, and this, precisely."*

Formal Construction of the Field of Fractions

Recap: The field of fractions was constructed in the previous lecture: the set of pairs (Definition 3.12), the equivalence relation (Definition 3.14, Proposition 3.16) and the quotient set (Definition 3.17). The lecturer summarizes it again.

Let R be an integral domain. To construct the field of fractions formally:

(1) Set of Pairs: Consider the Cartesian product $S := R \times (R \setminus \{0\}) \subset R^2$.

(2) Equivalence Relation: Define the relation $\equiv$ on S by:

$$(r, s) \equiv (t, w) \overset{\text{def}}{\iff} rw - st = 0.$$

(3) Equivalence Classes: The field of fractions is defined as the quotient set:

$$F = \mathrm{Frac}(R) := S_{\equiv}.$$

Spoken Transcript [00:01:46 - 00:03:25] Prof. Rahul Pandharipande

No one ever thinks about it that way. The way you think about it is: if R is an integral domain [writes: $R \ni r, s$] — these are elements in R, if you have some elements r and s —, then for an element of the field of fractions [writes: $\mathrm{Frac}(R) =$], you just say this is r/s [writes: r/s]. That's it, you know? And when you do this, you make sure s is not zero [writes: $s \neq 0$], because that was the second one there, just like you would with rational numbers.

And if you want to add, multiply, all of this, you just use the rules you did with rational numbers. Like if you want to add $r/s + t/w$ [writes: $\frac{r}{s} + \frac{t}{w}$], you know how to add fractions: you just do it here. It's just the right answer; it's this [writes: $= \frac{rw+st}{sw}$]. And so on. If you multiply, then you multiply fractions; whatever rules you use to manipulate fractions, they're just correct here.

In particular, why is this a field? If I take r/s, if this is not zero [writes: $\frac{r}{s} \neq 0$], that implies that r is not zero [writes: $\implies r \neq 0$]. If r were zero, then that's the zero element of this. And you can go check by the definitions that if r is zero, this is the zero element. So, if this is not zero, r must not be equal to zero. And then it has an inverse, and that's the multiplicative inverse [writes: $(\frac{r}{s})^{-1} = \frac{s}{r}$]. And it's legal because r's not zero, so I can put him downstairs.

Anyway, the way you think about it — I mean, you get to think about it however you want —, the way typically people think about the field of fractions is you just write the fractions. That's why it's called the field of fractions. And you just go forward. And the rules of this formalism are designed so that this actually works. Okay.

Intuitive Arithmetic in the Field of Fractions

Recap: These fraction rules were already listed in the previous lecture (Eqs. (3.36) to (3.39)).

In mathematical practice, elements of $\mathrm{Frac}(R)$ are represented as formal fractions $\frac{r}{s}$ with $r, s \in R$ and $s \neq 0$:

$$\mathrm{Frac}(R) = \left\{ \frac{r}{s} \;\middle|\; r, s \in R,\ s \neq 0 \right\}. \tag{3.51}$$

The field operations are defined analogously to elementary fraction arithmetic:

$$\textbf{Addition:} \quad \frac{r}{s} + \frac{t}{w} = \frac{rw + st}{sw}, \tag{3.52}$$

$$\textbf{Multiplication:} \quad \frac{r}{s} \cdot \frac{t}{w} = \frac{rt}{sw}, \tag{3.53}$$

$$\textbf{Multiplicative Inverse:} \quad \left(\frac{r}{s}\right)^{-1} = \frac{s}{r} \qquad \left(\text{valid whenever } \frac{r}{s} \neq 0, \text{i.e., } r \neq 0\right). \tag{3.54}$$

Why the Integral Domain Assumption Is Essential: For the denominator sw in (3.52) and (3.53) to be non-zero, we must have:

$$s \neq 0 \text{ and } w \neq 0 \implies sw \neq 0.$$

This property holds if and only if R has no zero divisors, which is precisely the defining condition of an **integral domain** (Definition 2.62). Furthermore, the zero element of $\mathrm{Frac}(R)$ is $\frac{0}{1} = \frac{0}{s}$ (Definition 3.20). Thus, $\frac{r}{s} \neq 0$ implies $r \neq 0$, guaranteeing that $\left(\frac{r}{s}\right)^{-1} = \frac{s}{r}$ is a well-defined fraction with non-zero denominator.

Spoken Transcript [00:03:25 - 00:04:45]

Prof. Rahul Pandharipande

This is a basic issue that comes up very often in math education about how to think. The first time this comes up, which you've already passed, is the real numbers. To define the real numbers is complicated: you have to either define them as Cauchy sequences or Dedekind cuts. And that's pretty complicated to define, and then you have to check that it satisfies the properties. That's all fine — that's kind of like some logical underlayer —, but no one thinks about real numbers that way.

And it's just that that's part of learning mathematics: that there's some kind of logical underlayer, and then there's the way you think about it. And usually they have nothing to do with each other. Okay, that's maybe an exaggeration, but... [laughter]

Actually, about the real numbers, I can't help now but comment that Dedekind cuts was discovered by Richard Dedekind, who had moved to Zurich in the second part of the 19th century. And he's an algebraist; he's a very famous mathematician. He's, of course, passed away by now. And he was assigned to teach analysis. He writes it very beautifully in this book that he wrote about the real numbers: he was assigned to teach analysis, and he's an algebraist, and he said, *"Well, I had a difficult time teaching analysis because I don't know what a real number is."* And then he spends the class constructing the real numbers. It's kind of amazing to read the introduction to that. You can find it.

Insight: The Logical Underlayer vs. Working Mathematical Intuition

The lecturer highlights a fundamental pedagogical dichotomy in abstract mathematics: the distinction between the **foundational underlayer** and the **working mathematical intuition**.
Whether constructing $\mathbb{R}$ via Dedekind cuts or Cauchy sequences (Fact 0.37), or constructing $\mathrm{Frac}(R)$ via quotient sets of ordered pairs $S_{\equiv}$ (Definitions 0.9 and 3.14), the foundational machinery serves solely to establish rigorous existence and soundness. Once established, mathematicians operate with the intuitive rules (e.g., standard fraction manipulations $\frac{r}{s}$) without thinking of the underlying equivalence classes.

Beyond the Lecture: Dedekind's Book on the Real Numbers

The book the lecturer refers to is Richard Dedekind's **Stetigkeit und irrationale Zahlen** (*"Continuity and Irrational Numbers"*, 1872). In its preface Dedekind recalls teaching the differential calculus at the Polytechnikum in Zurich (today's ETH) in 1858 and feeling the lack of a truly rigorous foundation of arithmetic; the book introduces what are now called Dedekind cuts.

Polynomial Rings

Definition and Structure of $R[x]$

Spoken Transcript [00:04:45 - 00:06:30] Prof. Rahul Pandharipande

Okay, so we go to polynomials.

I explained yesterday (Recall: Definition 3.32) that the way to think about polynomials is just the way you always think about polynomials. Elements of this ring, I advise you to think about this way: [writes: $R[x] = \{a_0 + a_1x + a_2x^2 + \cdots + a_nx^n \mid a_i \in R\}$]

But in all things, you get your own choice how to think about them. So, they're polynomials that have this formal variable x that's denoted here, and you consider polynomials in x with coefficients in R.

And this polynomial ring, that's the set, and it has addition; and the addition is you add it like polynomials. So, whatever power, you keep the power the same, and you add the coefficients of the power. You know how to add polynomials, so you just do it here.

This is more complicated, multiplying polynomials, but you already know how to multiply polynomials, and you just do it here. You can define it formally, which I did last time (Recall: Definition 3.36), but you just do it.

And it has the zero polynomial and the one polynomial, and this is the ring of polynomials [writes: $(R[x], +, \cdot, 0, 1)$ ring: poly in x with coef in R] — ring of polynomials in the variable x with coefficients in R.

So, we can do this for any ring — I said that. This is important to keep track of: sometimes we make assumptions on rings, sometimes not. Here we needed the assumption (i.e., R an integral domain), that's kind of crucial; here we don't need the assumption.

The Polynomial Ring over a Commutative Ring

Recap: The polynomial ring was introduced in the previous lecture, informally (Definition 3.32) and as sequences of coefficients (Definition 3.34, Definition 3.36). The lecturer recalls it.

Definition 3.37 (*Polynomial Ring*): Let R be an arbitrary commutative ring with identity 1. The *"polynomial ring"* in the variable x with coefficients in R, denoted $R[x]$, is the set of formal sums:

$$R[x] := \left\{a_0 + a_1x + a_2x^2 + \cdots + a_nx^n \;\middle|\; n \in \mathbb{N},\ a_i \in R\right\}. \tag{3.55}$$

Equipped with the standard polynomial addition and Cauchy multiplication:

$$\left(\sum_{i=0}^{n} a_ix^i\right) + \left(\sum_{i=0}^{n} b_ix^i\right) := \sum_{i=0}^{n}(a_i + b_i)x^i, \tag{3.56}$$

$$\left(\sum_{i=0}^{n} a_ix^i\right) \cdot \left(\sum_{j=0}^{m} b_jx^j\right) := \sum_{k=0}^{n+m}\left(\sum_{i+j=k} a_ib_j\right)x^k, \tag{3.57}$$

the structure $(R[x], +, \cdot, 0, 1)$ forms a commutative ring with identity $1 \in R \subset R[x]$ and zero element $0 \in R \subset R[x]$.

Generality of the Polynomial Construction: Unlike the construction of the field of fractions $\mathrm{Frac}(R)$, which strictly requires R to be an integral domain, the polynomial ring $R[x]$ is well-defined for **any** commutative ring with identity R. Zero divisors in R do not obstruct the definition of polynomial addition or multiplication.

The Universal Property of Polynomial Rings

Spoken Transcript [00:06:30 - 00:09:00] Prof. Rahul Pandharipande

Okay, so properties of this polynomial ring. The first is — this is somehow the most basic property of polynomial rings:

Suppose R and S are rings [writes: Suppose R, S are rings] — which I say commutative rings with unit, just for us —, and I have an element $y \in S$ [writes: $y \in S$]. Okay?

Then, given any homomorphism [writes: Then given $\phi \colon R \to S$ homomorphism of rings] — so you have some homomorphism from R to S, a homomorphism of rings —, there exists a unique extension [writes: $\exists!$ extension] — I will say what this means in a moment — from the polynomial ring to S [writes: $\hat{\phi} \colon R[x] \to S$] which satisfies these properties [writes: which satisfies 2 properties], two properties:

Property **(1)** is that if I take a constant polynomial [writes: • $\hat{\phi}(r) = \phi(r)$ (const)]... So, this is just the constant polynomial, which is to say that R sits inside $R[x]$ as the constants [writes: $R \subset R[x]$], okay? If I take a constant polynomial, this should just be the value that I've already specified here, okay?

And the second property is: I have to say what happens to x, because that's not specified here. And I just say x goes to y [writes: • $\hat{\phi}(x) = y$].

Okay, so that's the claim. If I start with these two rings — there's the ring where my maps are starting, and the ring where the target map, where my maps are ending, the homomorphisms are ending — and I pick an element y, I can take any homomorphism from R to S and I can extend it to the polynomial ring by just setting x to y.

The Universal Property of Polynomial Rings

Theorem 3.38 (*Universal Mapping Property of $R[x]$*): Let R and S be commutative rings with identity, let $\phi \colon R \to S$ be a ring homomorphism, and let $y \in S$. Then there exists a unique ring homomorphism $\hat{\phi} \colon R[x] \to S$ extending ϕ such that:

(1) Extension of Constants: For every constant polynomial $r \in R \subset R[x]$:
$$\hat{\phi}(r) = \phi(r). \tag{3.58}$$

(2) Evaluation at y: The indeterminate x is mapped to y:
$$\hat{\phi}(x) = y. \tag{3.59}$$

Evaluation Homomorphism and Factorization: Going beyond the lecture: the map $\hat{\phi}$ is often called the *"evaluation homomorphism"* associated with ϕ and y. In the special case where $S = R$ and $\phi = \mathrm{Id}_R$, $\hat{\phi}$ is simply the evaluation map $\mathrm{ev}_y \colon R[x] \to R$ given by $f(x) \mapsto f(y)$.

The universal property states that $R[x]$ is the *"freest"* ring containing R and an additional generator x: mapping $R[x]$ to any ring S requires only choosing a base ring homomorphism $\phi \colon R \to S$ and choosing where to send x. It is the same kind of statement as the universal property of $\mathbb{Q}$ from the previous lecture (Theorem 3.7): a homomorphism defined on a smaller ring extends uniquely to the bigger one (for $R[x]$, once the image of x has been chosen).

Proof of Theorem 3.38

Spoken Transcript [00:09:00 - 00:09:48] Prof. Rahul Pandharipande

Okay, and the proof of this is just by the formula. So, the proof: [writes: Proof] I have to define $\hat{\phi}$ on any polynomial. So, a polynomial looks like this: [writes: Def $\hat{\phi}(a_0 + a_1x + \cdots + a_nx^n) =$]

And I have to define it. Formally, I would have to first define it, then check that it's a homomorphism — so I can say it extends to a homomorphism: I have to define it, check it's a homomorphism, then show it satisfies these two properties. That's the task at hand, so to speak.

So, how am I going to define this?

Student Answer

Evaluate the coefficients, and then replace each x with y.

Spoken Transcript [00:09:53 - 00:10:45] Prof. Rahul Pandharipande

Yeah. So, this is the definition: I have to somehow take this data and get to S. Any element here — so these a_i's, that's part of the definition of polynomial, they're in R [writes: $a_i \in R$]. And I only have on the board one way to ship elements from R to S: it's here. So, I can't make up another one, that's, you know.

So, all these coefficients I just move by the only way I know how to get to S. So, that's the constant coefficient, then I move this coefficient to S, because this result has to be in S. And then the question is: how do I move x to S? Well, I know one element of S, that's y. And then I do this: $\phi(a_2)y^2$ and so on, to y^n. [writes: $\overset{\text{def}}{=} \phi(a_0) + \phi(a_1) \cdot y + \phi(a_2)y^2 + \cdots + \phi(a_n)y^n$]

Spoken Transcript [00:10:45 - 00:12:20] Prof. Rahul Pandharipande

So, I just define that. The first thing is you have to accept this definition makes sense. And, you know, when you define a map, usually if I define it so that if you give me anything, I tell you what the answer is, that's the definition of the map — I mean, as long as it takes any possibility here, which it does.

In the case when you have equivalence classes, the matter is a little subtle, because, you know, it could be that in equivalence classes things have different names, and you have to be careful. But that's not what's going on here. This is not equivalence classes: the field of fractions was equivalence classes, but the polynomial ring, that is not equivalence classes. I tell you specifically who is a polynomial, and everyone has just one name.

So, that's the definition. Okay. But is that the end? No, I have to check that it satisfies this property, on constant polynomials. But for constant polynomials, well, it does satisfy that property: that would just mean there's none of this, so just by definition it satisfies that property.

So, I have to check constants [writes "• Check consts"], and I have to check x [writes "•

Check x"]. What is $\hat{\phi}(x)$ by this definition? Well, that would be just one x here and everything else is zero or one. Formally, I'd have to put like $\phi(0)$ and $\phi(1)$. I mean, yeah, I'm not going to write that, but you could write that. But $\phi(0) = 0$ and $\phi(1) = 1$, so you don't have to write it. So, this is just y [writes: $\hat{\phi}(x) = y$].

Okay, so I have now checked these things, but the one thing I haven't checked is that it's a homomorphism. Let's think about how you'd check that.

Definition of $\hat{\phi}$ and Verification of Conditions

1. Definition of the Candidate Map: Let $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n \in R[x]$ with $a_i \in R$. We define the map $\hat{\phi}\colon R[x] \to S$ by:

$$\hat{\phi}(a_0 + a_1x + \cdots + a_nx^n) \overset{\text{def}}{=} \phi(a_0) + \phi(a_1) \cdot y + \phi(a_2)y^2 + \cdots + \phi(a_n)y^n. \tag{3.60}$$

Since every polynomial in $R[x]$ has a unique representation as a formal sum of powers of x (unlike equivalence classes in $\operatorname{Frac}(R)$, there are no ambiguous representatives), $\hat{\phi}$ is unconditionally well-defined.

2. Verification of Properties (1) and (2):

- **Constant Polynomials:** If $f(x) = r \in R$, then $a_0 = r$ and $a_i = 0$ for all $i \geq 1$. Hence:
$$\hat{\phi}(r) = \phi(r).$$
- **Image of x:** For $f(x) = x$, we have $a_0 = 0$, $a_1 = 1$, and $a_i = 0$ for all $i \geq 2$. Since $\phi(0) = 0$ and $\phi(1) = 1$:
$$\hat{\phi}(x) = \phi(0) + \phi(1)y = 0 + 1 \cdot y = y.$$

Spoken Transcript [00:12:20 - 00:14:25] Prof. Rahul Pandharipande

To check it's a homomorphism, you have to check some things about zero going to zero, and one goes to one, which it certainly does. So, the main thing to check is that $\hat{\phi}$ of two polynomials — so f and g now are my polynomials: [writes: $\hat{\phi}(f+g) = \hat{\phi}(f) + \hat{\phi}(g)$, $f, g \in R[x]$].

Okay? For this, you need to know... sorry, these are f, right? So, you need to know how to add the polynomials here, and then you need to know how to add elements of S. So, this addition is addition of polynomials [writes: addition of polys], and this addition is addition in S [writes: addition in S], right?

And the way this thing works: what are we going to do here? Let's say that we take this $\hat{\phi}$; if f is given by a's, it's $a_0 + a_1x$, let's say — let's say that's the f, just a linear one —, plus $b_0 + b_1x$ [writes: $\hat{\phi}(\underbrace{a_0 + a_1x}_{f} + \underbrace{b_0 + b_1x}_{g})$].

If we add them first in polynomials, then this is $(a_0 + b_0) + (a_1 + b_1)x$ [writes "$\hat{\phi}((a_0 + b_0) + (a_1 + b_1)x)$" below].

And then, if we go over by the definition, then this is $\phi(a_0 + b_0) + \phi(a_1 + b_1)y$ [writes: $\implies \phi(a_0 + b_0) + \phi(a_1 + b_1)y$], right?

Spoken Transcript [00:14:25 - 00:16:30] Prof. Rahul Pandharipande

And this, I can now use the homomorphism property of R and write as $\phi(a_0) + \phi(a_1)y + \phi(b_0) + \phi(b_1)y$ [writes: $= \phi(a_0) + \phi(a_1)y + \phi(b_0) + \phi(b_1)y$]. I don't know if this is really worth doing all out, but...

And that, at the end, is $\hat{\phi}(a_0 + a_1x) + \hat{\phi}(b_0 + b_1x)$ [writes: $= \hat{\phi}(a_0 + a_1x) + \hat{\phi}(b_0 + b_1x)$].

Okay. So, what did I do here? I just expanded out by the definition, and at a certain point I have to use the homomorphism property of R. Nothing happened here, actually; it was just a lot of waste of chalk. But I just follow... at a certain point I have to use the homomorphism property of R. There. And I did this with linear polynomials, but it's just the same with more coefficients.

And then you have to check something that's more complicated with the indices: that if I take $f \cdot g$, this is equal to $\hat{\phi}(f) \cdot \hat{\phi}(g)$ [writes: $\hat{\phi}(fg) = \hat{\phi}(f) \cdot \hat{\phi}(g)$]. And, again, this follows from the same expansion.

So, I tried to do some checking for you, but in the end I don't know how successful that was. The fact of the matter is: I give you the definition, and then you have to check the compatibility; and when you check the compatibility, all you use is the homomorphism property of this.

That's one way, one abstract property of polynomial rings: that if you can map R to something, then you can pick any place, any landing point for x, and it gives you a homomorphism. Sometimes this is called the universal property of polynomial rings.

Verification of Additivity

3. Verification of Additivity ($\hat{\phi}(f+g) = \hat{\phi}(f) + \hat{\phi}(g)$): Let $f, g \in R[x]$. We verify additivity explicitly for linear polynomials $f(x) = a_0 + a_1x$ and $g(x) = b_0 + b_1x$:

$$\begin{aligned}
\hat{\phi}(f+g) &= \hat{\phi}\big((a_0 + a_1x) + (b_0 + b_1x)\big) \\
&= \hat{\phi}\big((a_0 + b_0) + (a_1 + b_1)x\big) && \text{(addition in } R[x]\text{)} \\
&= \phi(a_0 + b_0) + \phi(a_1 + b_1)y && \text{(by definition of } \hat{\phi}\text{)} \\
&= \big(\phi(a_0) + \phi(b_0)\big) + \big(\phi(a_1) + \phi(b_1)\big)y && (\phi\colon R \to S \text{ is a homomorphism}) \\
&= \big(\phi(a_0) + \phi(a_1)y\big) + \big(\phi(b_0) + \phi(b_1)y\big) && \text{(reordering in ring } S\text{)} \\
&= \hat{\phi}(a_0 + a_1x) + \hat{\phi}(b_0 + b_1x) \\
&= \hat{\phi}(f) + \hat{\phi}(g). && (3.61)
\end{aligned}$$

The same argument works for polynomials of arbitrary degree: writing $f = \sum_{i=0}^{n} a_i x^i$ and $g = \sum_{i=0}^{n} b_i x^i$ (padding the shorter one with zero coefficients),

$$\hat{\phi}(f+g) = \sum_{i=0}^{n} \phi(a_i + b_i)\, y^i = \sum_{i=0}^{n} \phi(a_i)\, y^i + \sum_{i=0}^{n} \phi(b_i)\, y^i = \hat{\phi}(f) + \hat{\phi}(g).$$

Verification of Multiplicativity

4. Verification of Multiplicativity ($\hat{\phi}(fg) = \hat{\phi}(f) \cdot \hat{\phi}(g)$): Let $f(x) = \sum_{i=0}^{n} a_i x^i$ and

$g(x) = \sum_{j=0}^{m} b_j x^j$. Using the Cauchy product in $R[x]$:

$$\begin{aligned}
\hat{\phi}(f \cdot g) &= \hat{\phi}\left(\sum_{k=0}^{n+m}\left(\sum_{i+j=k} a_i b_j\right) x^k\right) \\
&= \sum_{k=0}^{n+m} \phi\left(\sum_{i+j=k} a_i b_j\right) y^k && \text{(by definition of } \hat{\phi}\text{)} \\
&= \sum_{k=0}^{n+m}\left(\sum_{i+j=k} \phi(a_i)\phi(b_j)\right) y^k && \text{(}\phi \text{ is a ring homomorphism)} \\
&= \left(\sum_{i=0}^{n} \phi(a_i) y^i\right) \cdot \left(\sum_{j=0}^{m} \phi(b_j) y^j\right) && \text{(distributivity and Cauchy product in } S\text{)} \\
&= \hat{\phi}(f) \cdot \hat{\phi}(g). && (3.62)
\end{aligned}$$

Furthermore, $\hat{\phi}(1) = \phi(1) = 1_S$. Thus, $\hat{\phi}$ is a ring homomorphism.

Uniqueness (beyond the lecture)

The lecture states uniqueness on the board without proving it. Any ring homomorphism $\psi\colon R[x] \to S$ satisfying conditions **(1)** and **(2)** of Theorem 3.38 must preserve sums and products:

$$\psi\left(\sum_{i=0}^{n} a_i x^i\right) = \sum_{i=0}^{n} \psi(a_i)\psi(x)^i = \sum_{i=0}^{n} \phi(a_i) y^i = \hat{\phi}\left(\sum_{i=0}^{n} a_i x^i\right).$$

Hence $\psi = \hat{\phi}$, proving uniqueness. **Q.E.D.**

Q.E.D.

Units in the Polynomial Ring over an Integral Domain

Spoken Transcript [00:16:30 - 00:18:16] Prof. Rahul Pandharipande

[erases the lower board]

So, let's think about... We know that if you have rings, you can think about their units. So, we can ask: What are the units in a polynomial ring? That's a very reasonable question. And that question actually turns out to have some wrinkles.

Let's start with: Suppose R is an integral domain [writes: Suppose R is an integral domain]. Then the question is: What are the units in the polynomial ring? [writes: What are the units in $R[x]$?] Which is to say: which elements of this polynomial ring have a multiplicative inverse?

So, what do you think the answer to this is? Well, let's say if I have some f in the polynomial ring, it's $a_0 + a_1 x + \cdots + a_n x^n$ [writes: $f = a_0 + a_1 x + \cdots + a_n x^n$]. And if it's a unit, that means that there exists a multiplicative inverse — there exists $g = b_0 + b_1 x + \cdots +$ some different number, $b_m x^m$ [writes: $\exists\, g = b_0 + b_1 x + \cdots + b_m x^m$] —, and I multiply them so that I get 1, the constant polynomial [writes: $fg = 1$].

That would mean that if f is a unit, then there exists a g such that when I multiply them I get the unit, the constant polynomial 1.

How could this happen?

Yeah?

Student Question

If there is an element in R, you can plug it in and then it goes to zero?

Spoken Transcript [00:18:21 - 00:18:25] Prof. Rahul Pandharipande

There exists an element of... Sorry, I didn't... You have to speak louder.

Student Answer

When there exists an inverse of...

Spoken Transcript [00:18:27 - 00:18:31] Prof. Rahul Pandharipande

Well, this x stays. You can't replace x. We're talking about this polynomial ring.

Student Answer

These are the units...

Spoken Transcript [00:18:33 - 00:18:56] Prof. Rahul Pandharipande

Yeah, these are... I mean, but actually the discussion is kind of more simple. Just if I multiply this f with g, I get another polynomial. And for it to be 1, all of its higher terms have to somehow cancel, right? So, the question is: can you arrange it so all the higher terms cancel? That's the real question.

Yeah?

Student Answer

I suppose if f is a constant term?

Spoken Transcript [00:18:59 - 00:20:09] Prof. Rahul Pandharipande

Yeah, so, one way to do it is: the answer is that a unit... if I take the polynomial to be a constant, f equals a constant polynomial [writes: $f = a$], and this happens to be a unit for R [completes the line to $f = a \in \text{Unit}(R)$].

Right? So, that's a good... So, he's saying that just forget it: we could have just the constants, take that to be a constant polynomial, and if I further choose that to be a unit, then I know it has a friend, because it has a unit in R.

This part, I hope, is good: that the units of the original ring sit inside the units of this polynomial ring, and they sit inside as the constant polynomials [writes "Unit(R) ⊂ Unit($R[x]$)" with "Const Polys" below].

Okay. I hope that's clear.

So, the question is: could there be any more units here?

And you should, in the way I presented this, be sensitive to the fact that I didn't take any ring, I took an integral domain. So, you might want to think: why did I do that?

Yeah?

Constant Units in the Polynomial Ring

Let R be a commutative ring with identity, and consider $R[x]$:

- A polynomial $f(x) \in R[x]$ is a *"unit"* (Definition 2.66) if there exists $g(x) \in R[x]$ such that:
$$f(x) \cdot g(x) = 1. \tag{3.63}$$
- Any unit $u \in \text{Unit}(R)$ has an inverse $u^{-1} \in R$. Considered as constant polynomials in $R[x]$:
$$u \cdot u^{-1} = 1 \implies u \in \text{Unit}(R[x]).$$
Thus, the units of R always embed into the units of $R[x]$:
$$\text{Unit}(R) \subset \text{Unit}(R[x]) \quad \text{(as constant polynomials)}. \tag{3.64}$$

Student Answer

There are no more units?

Spoken Transcript [00:20:12 - 00:20:37]

Prof. Rahul Pandharipande

Yes, so, the answer here is that there's no more. It's actually equal. So, the answer is that there are no more units; the units in R equal the units in $R[x]$ [writes "Ans. Unit(R) = Unit($R[x]$)"].

And I just remind you, because otherwise this is going to be wrong: R is an integral domain [writes: R is an int domain].

And how are you going to use this condition?

Student Answer

So, we have the general form of f and g, and if we multiply them, then the term with the highest power...

Spoken Transcript [00:20:45 - 00:22:05]

Prof. Rahul Pandharipande

Yeah, this is the right answer. When you multiply polynomials — and as I said, you must have had some years of experience multiplying polynomials, and you should not forget these years of experience —, if you multiply polynomials, in the middle is a complicated mess of things: you have to add up all the things, it's some complicated mess. But there are two terms which are very simple.

If you want the constant polynomial of the product, that's the product of the constants, right? [writes: $\mathrm{Const}(fg) = a_0 b_0$]

And if you want the highest term in the product, that's the product of the highest terms. [writes "highest term ="]

So, with polynomials, if you're multiplying them, before you do all the multiplication, you should think: is what I'm thinking about resolved by understanding what happens to the constants or the highest term? Because those are the easiest ones, because there's not a big sum there.

As I said, you know polynomials: the constant term of fg is equal to just $a_0 b_0$.

And the highest term, meaning the highest-degree term in x...

Leading and Constant Terms in Polynomial Multiplication

Let R be a commutative ring with identity, and let $f(x), g(x) \in R[x]$ be non-zero polynomials represented in canonical form:

$$f(x) = a_0 + a_1 x + \cdots + a_n x^n, \quad a_n \neq 0 \; (\deg(f) = n), \tag{3.65}$$

$$g(x) = b_0 + b_1 x + \cdots + b_m x^m, \quad b_m \neq 0 \; (\deg(g) = m). \tag{3.66}$$

When multiplying $f(x)$ and $g(x)$, the extremal coefficients take an uncomplicated form:

Constant Term: $$\mathrm{Const}(fg) = a_0 b_0, \tag{3.67}$$

Highest Degree Term: $$\mathrm{highest}(fg) = (a_n x^n) \cdot (b_m x^m) = a_n b_m x^{n+m}. \tag{3.68}$$

Spoken Transcript [00:22:05 - 00:23:22] Prof. Rahul Pandharipande

...the highest term comes from the highest term of f times the highest term of g.

So, if I write it like this, when I take this highest term... Usually when they write polynomials — but you have to be careful about the conventions and the convention of whatever author you're reading —, the typical thing is when you write a polynomial like this, you could say a_n is not zero [writes: $a_n \neq 0$], because if it were zero, then you wouldn't have written it. So, it's kind of standard that when you write a polynomial like this, the highest term you say is not zero, because as I said, if it were zero, you didn't have to write it.

And this one [points at b_m] is not zero [writes: $b_m \neq 0$], because otherwise I wouldn't have to write it.

So, when I take these polynomials, then this is the highest term, and that's the highest term. So, this would be the highest term here: $a_n x^n$. That would be the highest term there: $b_m x^m$. And the highest term would be $a_n \times b_m \times x^{n+m}$ [writes "$(a_n x^n) \cdot (b_m x^m)$", with "highest of f" and "highest of g" below, and "$a_n \cdot b_m\, x^{n+m}$"]: $n+m$, that would be the highest term.

And if we're in an integral domain, this product, if they start out to be not zero, they end up to be not zero. So, this would still be non-zero if R is an integral domain [writes: $\neq 0$ if R int domain].

Units of the Polynomial Ring over an Integral Domain

Theorem 3.39 (*Units in $R[x]$ for an Integral Domain R*): Let R be an integral domain. Then the units of the polynomial ring $R[x]$ are precisely the units of the base ring R, regarded as constant polynomials:

$$\operatorname{Unit}(R[x]) = \operatorname{Unit}(R). \tag{3.69}$$

Proof

The inclusion $\operatorname{Unit}(R) \subset \operatorname{Unit}(R[x])$ holds trivially because any unit $u \in \operatorname{Unit}(R)$ has an inverse $u^{-1} \in R$, which satisfies $u \cdot u^{-1} = 1$ in $R[x]$ as constant polynomials.

Conversely, suppose $f(x) \in \operatorname{Unit}(R[x])$. Then there exists an inverse polynomial $g(x) \in R[x]$ such that:

$$f(x) \cdot g(x) = 1. \tag{3.70}$$

Both f and g are non-zero, since $fg = 1 \neq 0$ (an integral domain has $1 \neq 0$). Write $f(x) = \sum_{i=0}^{n} a_i x^i$ with $a_n \neq 0$ and $g(x) = \sum_{j=0}^{m} b_j x^j$ with $b_m \neq 0$. The leading term of the product is given by:

$$\operatorname{highest}(f \cdot g) = a_n b_m x^{n+m}.$$

Because R is an **integral domain**, the absence of non-trivial zero divisors ensures:

$$a_n \neq 0 \text{ and } b_m \neq 0 \implies a_n b_m \neq 0 \quad \text{in } R.$$

On the other hand, the constant polynomial 1 has degree 0. Thus, comparing degrees on both sides of (3.70) yields:

$$n + m = \deg(fg) = \deg(1) = 0.$$

Since $n, m \in \mathbb{N} = \{0, 1, 2, \dots\}$, this forces:

$$n = 0 \quad \text{and} \quad m = 0.$$

Therefore, $f(x) = a_0$ is a constant polynomial. The constant term relation gives:

$$\operatorname{Const}(fg) = a_0 b_0 = 1 \implies a_0 \in \operatorname{Unit}(R).$$

This proves that every unit in $R[x]$ is a constant polynomial coming from $\operatorname{Unit}(R)$.

Q.E.D.

Degree Formula in Integral Domains: The proof relies on the fundamental degree formula:

$$\deg(f \cdot g) = \deg(f) + \deg(g),$$

which holds for all non-zero polynomials $f, g \in R[x]$ if and only if R has no zero divisors. If R possessed non-trivial zero divisors, the leading coefficient $a_n b_m$ could vanish, causing $\deg(f \cdot g) < \deg(f) + \deg(g)$.

Spoken Transcript [00:23:22 - 00:25:19] Prof. Rahul Pandharipande

This means that if you ever take a polynomial which has a positive-degree part — meaning it's not the constant polynomial, meaning one of these coefficients with x's is not zero —, you will never be able to find an inverse, because when you multiply it by somebody else, you'll always have a term in $n + m$; and if one of them is ≥ 1 (i.e., $n \geq 1$), this will never be a constant term.

Does that make sense? In other words, if your ring is an integral domain, when you multiply two polynomials with x's, the x's will never go away, because you can always look at the highest one, and they will persist.

So, the answer is: if it's an integral domain, you'll have no more units.

Of course, this begs the question: What happens when it's not? [writes "What happen when R is not an int domain."]

Okay, so I leave this to you to think about. That's kind of a fun thing; maybe we'll discuss it later.

You should either prove that this argument somehow can be fixed, or you should find a ring that's not an integral domain where there are more units. Those are the two possible solutions.

But my memory from school — which was really a long time ago — was there were a lot of polynomials in school, weren't there? Huh?

So, you know, you should have some experience with adding, subtracting, multiplying polynomials; and as I said, all that experience is directly applicable here.

The Non-Integral Domain Question

Question 3.40: What happens when R is not an integral domain?

Beyond the Lecture: Non-Constant Units in $R[x]$

When R is not an integral domain, $R[x]$ can have non-constant units. In general, a polynomial $f(x) = a_0 + a_1x + \cdots + a_nx^n \in R[x]$ is a unit if and only if $a_0 \in \mathrm{Unit}(R)$ and all higher coefficients $a_1, \ldots, a_n \in R$ are *"nilpotent"*, that is, $a_i^k = 0$ for some $k \geq 1$. For a concrete counterexample, consider the ring $R = \mathbb{Z}/4\mathbb{Z}$ (which is not an integral domain since $2 \cdot 2 = 0$). In $(\mathbb{Z}/4\mathbb{Z})[x]$, consider the linear polynomial $f(x) = 1 + 2x$. We compute:

$$(1+2x)^2 = 1 + 4x + 4x^2 \equiv 1 \pmod 4.$$

Thus $f(x) = 1 + 2x$ is a unit of degree 1 with $f(x)^{-1} = 1 + 2x$.

Formal Power Series Rings

Definition and Ring Operations in $R[[x]]$

Spoken Transcript [00:25:19 - 00:27:02]

Prof. Rahul Pandharipande

[moves to the lower left chalkboard and writes: Power Series]

Okay, so the next thing is power series. And this is maybe... You may have less experience manipulating power series, I don't know. But it's a beautiful thing.

Here, I have to define for you the power series ring [writes: $R[[x]] \ni$]. And the reason there are two brackets here, somehow, is because there are a lot more x's, you know? [laughter]

An element of the power series ring... And, of course, we now have, as with polynomial rings, two ways to think about it: the formal way, or the way people actually think about it.

Now, the formal way: we had before, a polynomial was given by a sequence of elements of R [writes: $(a_0, a_1, a_2, \dots, a_n)$], and that sequence had a highest term and then had an infinite number of zeros afterwards [continues with $0, 0, 0$]. This was the definition of polynomial, if you remember from yesterday (Recall: Definition 3.34). You could add these, subtract these, multiply them, and it would just be the same thing as the x's, but this is the abstract discussion.

From this point of view, power series are easy: you just don't put this condition of zeros. It's just infinite elements. [erases the zeros and continues the sequence with further coefficients]

So, from this abstract point of view, elements of the power series ring are infinite lists — infinite vectors — of elements of R [writes: Elements of $R[[x]]$ are infinite vectors of elements of R]. Before, we had lists that eventually become all zero, and now we don't; we just lift that condition. You can do anything you want.

Formal Definition of Power Series as Sequences

Definition 3.41 (*Formal Power Series Ring*): Let R be a commutative ring with identity 1.

(1) Sequence Perspective: Formally, a *"formal power series"* with coefficients in R is an infinite sequence of elements of R:

$$(a_0, a_1, a_2, \dots, a_n, \dots) \in R^{\mathbb{N}}. \tag{3.71}$$

In contrast to polynomials in $R[x]$ (sequences $(a_0, a_1, \dots, a_n, 0, 0, 0, \dots)$ with $a_k = 0$ for all sufficiently large k), no vanishing condition is imposed on the terms of a formal power series.

(2) Notation: The set of all formal power series in the indeterminate x with coefficients in R is denoted by:

$$R[[x]]. \tag{3.72}$$

Spoken Transcript [00:27:02 - 00:29:00] Prof. Rahul Pandharipande

Now, the way to think about it as power series: all this means is that before, for polynomials, we had the condition that eventually you stop — that's a polynomial [points to the sequence]. And for power series, you just don't stop: you can go forever [writes "$a_0 + a_1 x + \cdots + a_n x^n + \dots$" after $R[[x]] \ni$].

It's hard to go forever, so typically you would write the following thing: the elements of the power series ring are the sum from $i = 0$ to infinity of $a_i x^i$ [writes: $R[[x]] = \sum_{i=0}^{\infty} a_i x^i$, $a_i \in R$], where the a_i's are in R.

You can see this is exactly the same data as that. The x, as I said, is just the variable; it's not influencing the a's. Like here, it just makes clear that x is just a placeholder, right? And that's all it's doing here also.

Okay, so that's power series. You must have had whole classes on power series — which is good, because I'm not going to teach you about them this way.

So, do you know how to add power series, multiply power series?

The zero is just the zero power series, that's where every coefficient forever is zero. The one power series is just a one and then everybody else is zero [writes "$(R[[x]], +, \cdot, 0, 1)$" and below it "$(0, 0, 0, \dots)$, $(1, 0, 0, 0, \dots)$"].

Do you know how to add power series? I say that you do know how to add power series: in every analysis class, I think you add power series.

Here's the rule, which will surprise no one: [writes: $\sum_{i=0}^{\infty} a_i x^i + \sum_{i=0}^{\infty} b_i x^i \overset{\text{def}}{=} \sum_{i=0}^{\infty} (a_i + b_i) x^i$].

It's a lot of work to write all of these things, so it's... yeah.

Okay, so that's the exciting rule: it's component-wise addition.

Representation as Formal Sums

Every formal power series $(a_n)_{n \in \mathbb{N}} \in R[[x]]$ is represented as a formal infinite sum:

$$a_0 + a_1 x + a_2 x^2 + \cdots = \sum_{i=0}^{\infty} a_i x^i, \quad a_i \in R. \tag{3.73}$$

Algebraic Meaning of the Infinite Sum: In algebra, the expression $\sum_{i=0}^{\infty} a_i x^i$ is purely a formal symbol representing the sequence of coefficients $(a_0, a_1, a_2, \dots)$. There are no questions of convergence, metrics, or limits involved, and the symbol x serves solely as an algebraic placeholder tracking positions.

Ring Structure and Addition in $R[[x]]$

The formal power series ring is equipped with a ring structure $(R[[x]], +, \cdot, 0, 1)$:

- **Additive Identity (0):**

$$0_{R[[x]]} := (0, 0, 0, \dots) = \sum_{i=0}^{\infty} 0 \cdot x^i. \tag{3.74}$$

- **Multiplicative Identity (1):**

$$1_{R[[x]]} := (1, 0, 0, \dots) = 1 + \sum_{i=1}^{\infty} 0 \cdot x^i. \tag{3.75}$$

- **Addition:** Defined componentwise:

$$\sum_{i=0}^{\infty} a_i x^i + \sum_{i=0}^{\infty} b_i x^i \overset{\text{def}}{=} \sum_{i=0}^{\infty} (a_i + b_i) x^i. \tag{3.76}$$

Spoken Transcript [00:29:00 - 00:30:25] Prof. Rahul Pandharipande

More complicated is: how do you multiply power series?

Again, that's a definition, by the way: [writes: $\sum_{i=0}^{\infty} a_i x^i \cdot \sum_{i=0}^{\infty} b_i x^i \overset{\text{def}}{=}$].

Here... I must say that I really dislike indices in mathematics, somehow. Okay.

So, we have to define this. How are we going to define this? As I said, this is how we define it: you define it to be the sum from $i = 0$ to infinity of $c_i x^i$ [writes: $\sum_{i=0}^{\infty} c_i x^i$], and then I have to define for you the c_i's — I have to define what c_i is.

Let's say c_k is... You can write this as a sum from $i = 0$ to k, and I write $a_{k-i}b_i$ [writes: $c_k = \sum_{i=0}^{k} a_{k-i}b_i$].

Okay. All right, I did it.

But you know this: this is how you multiply power series.

Cauchy Multiplication in $R[[x]]$

Definition 3.42 (*Cauchy Product of Power Series*)**:** Let $\sum_{i=0}^{\infty} a_i x^i$ and $\sum_{i=0}^{\infty} b_i x^i$ be formal power series in $R[[x]]$. Their product is defined by:

$$\left(\sum_{i=0}^{\infty} a_i x^i\right) \cdot \left(\sum_{i=0}^{\infty} b_i x^i\right) \overset{\text{def}}{=} \sum_{k=0}^{\infty} c_k x^k, \tag{3.77}$$

where each coefficient $c_k \in R$ is given by the Cauchy product formula:

$$c_k := \sum_{i=0}^{k} a_{k-i} b_i = \sum_{i+j=k} a_i b_j. \tag{3.78}$$

Finiteness of Each Coefficient Calculation: Notice that to determine any individual coefficient c_k of the product, only the finite collection of coefficients $\{a_0, \dots, a_k\}$ and $\{b_0, \dots, b_k\}$ is involved. The sum in (3.78) has exactly $k+1$ terms in R, meaning that no infinite sums in the base ring R are required to define the product (unlike the Cauchy product of analytic power series in $\mathbb{C}$, Fact 0.43, no convergence issues arise).

Spoken Transcript [00:30:25 - 00:31:28] Prof. Rahul Pandharipande

That is a ring. So, is it a ring? Well, then you have to check.

Formally, you have to check all the axioms. The specific axiom that probably you don't see in your head immediately, unless you've thought about it, is... Like, it's kind of obvious this is associative, commutative, everything, because all we're doing is component-wise; really nothing's happening here.

This one may be more interesting if you haven't thought about it: this is somehow a slightly strange rule that tells you how to get the new power series from the old power series. You have to check that this is associative, et cetera.

Commutative is kind of obvious, even though, you know, I've broken the symmetry here; but if you think about it, that's kind of clear that it's commutative. But then you have to check it's associative. And, you know, formally you have to check it, unless somehow you already know that it's true.

And the formal way to check it is you have to multiply three things, and you multiply them in two different ways, and then you go look at the coefficients, and it's an identity, and it will work out.

So, you know, you have to decide how you interact with math; but formally, if you want to say you understand what a power series ring is, you have to check that the multiplication is associative.

Ring Structure of $R[[x]]$

Equipped with addition (3.76) and multiplication (3.77), the tuple $(R[[x]], +, \cdot, 0, 1)$ forms a commutative ring with unit $1_{R[[x]]}$.

Verification of Ring Axioms (beyond the lecture)

The lecture leaves this to the students.

- **Abelian Group under Addition:** Follows immediately from component-wise addition inheriting associativity, commutativity, and inverses from $(R, +)$.
- **Commutativity of Multiplication:** By substituting $j = k - i$:

$$c_k = \sum_{i=0}^{k} a_{k-i} b_i = \sum_{j=0}^{k} b_{k-j} a_j.$$

- **Associativity of Multiplication:** For three series $f = \sum a_i x^i$, $g = \sum b_j x^j$, $h = \sum d_l x^l$ in $R[[x]]$, the n-th coefficient of both $(f \cdot g) \cdot h$ and $f \cdot (g \cdot h)$ evaluates to:

$$\sum_{i+j+l=n} a_i b_j d_l,$$

  which establishes $((f \cdot g) \cdot h)_n = (f \cdot (g \cdot h))_n$ for all $n \in \mathbb{N}$.

Q.E.D.

Spoken Transcript [00:31:28 - 00:32:38] Prof. Rahul Pandharipande

[pushes the power series board up; the board with the units of $R[x]$ comes down, and erases it]

One of the issues about the power series ring is: if you want to communicate to someone else an element of the power series ring, it's a lot harder than the polynomial ring. The polynomial ring, you just write it: it's a finite amount of data, you know, you just write like seven coefficients, that's it.

With the power series ring, a priori, there's like an infinite amount of data there. So, if you want to communicate it, you know, you have to give some rule, or something like that. We'll do some examples eventually, but it's, you know, somehow a more complicated ring.

When you encounter power series in analysis, you know, they could be like the Taylor expansion of trigonometric functions, and they have rules; that's a way to explain it.

Anyway, if you have not encountered these before, you should make yourself comfortable with the idea of polynomials in R, power series in R, how to add, multiply, how to manipulate them.

III Units in the Formal Power Series Ring

Spoken Transcript [00:32:38 - 00:35:07] Prof. Rahul Pandharipande

So, let's try to ask this question: What are the units of this ring? [writes "What are unit $R[[x]]$?"]

So, again, that means we start with a particular power series f: $a_0 + a_1 x + a_2 x^2 + \dots$

[writes: $f = a_0 + a_1x + a_2x^2 + \dots$].

And it goes forever, in principle. Of course, it could also all be zero: a polynomial is a power series, just one that doesn't last too long, because zero is fine.

And g: $b_0 + b_1x + b_2x^2 + \dots$ [writes: $g = b_0 + b_1x + b_2x^2 + \dots$].

Okay, and for this to be a unit — if f is a unit —, then there exists g such that $f \times g = 1$ [writes "If f is a unit, $\exists g \implies f \cdot g = 1$"].

And to be 1 is dramatic! It means when you multiply this, the rule in principle gives you infinitely many terms. If it's 1, it means almost all of them are zero.

If we multiply them, $f \times g$, we get: the first term is a_0b_0 [writes: $fg = a_0b_0 +$]. The second term is $(a_1b_0 + a_0b_1)x$ [writes: $(a_1b_0 + a_0b_1)x +$].

You can tell I've been spending too much time in Europe because I say *"nought"* all the time. [laughter]

Here, a_2... That was a complicated joke. [writes: $(a_2b_0 + a_1b_1 + a_0b_2)x^2 + \dots$]

Okay, so that's how the power series multiplication works, right? How is it possible that this could be 1?

Well, could this be 1? 1, I remind you, is $1 + 0 \times x + 0 \times x^2$, and so on [writes "$1$, $0 \cdot x$, $0 \cdot x^2$" under the terms of fg]. So, the only way it can be 1 is that this is 1 [points to a_0b_0]. So, this would imply that $a_0b_0 = 1$ [writes: $\implies a_0b_0 = 1$]. This means that the constant term of f is a unit [writes: $\implies \mathrm{Const}(f)$ is a unit]. Right?

What about all this other stuff? So, what's your feeling about that?

Setting up the Invertibility Condition in $R[[x]]$

Let $f(x) \in R[[x]]$ be a formal power series:

$$f(x) = a_0 + a_1x + a_2x^2 + \dots = \sum_{i=0}^{\infty} a_i x^i. \tag{3.79}$$

By definition, $f(x)$ is a *"unit"* in $R[[x]]$ if and only if there exists an inverse series $g(x) = \sum_{j=0}^{\infty} b_j x^j \in R[[x]]$ such that:

$$f(x) \cdot g(x) = 1_{R[[x]]} = 1 + 0 \cdot x + 0 \cdot x^2 + \dots \tag{3.80}$$

The Constant Term Invertibility Condition

Multiplying $f(x)$ and $g(x)$ via Cauchy multiplication:

$$\begin{aligned} f(x) \cdot g(x) &= a_0b_0 + (a_1b_0 + a_0b_1)x + (a_2b_0 + a_1b_1 + a_0b_2)x^2 + \dots \\ &\overset{!}{=} 1 + 0 \cdot x + 0 \cdot x^2 + \dots \end{aligned} \tag{3.81}$$

Equating coefficients power by power, the constant term yields:

$$a_0b_0 = 1 \implies a_0 = \mathrm{Const}(f) \in \mathrm{Unit}(R). \tag{3.82}$$

Thus, a necessary condition for $f(x)$ to be invertible in $R[[x]]$ is that its constant term a_0 is a unit in the base ring R.

Spoken Transcript [00:35:07 - 00:36:51] Prof. Rahul Pandharipande

Of course, it's also true that the constants — if we take the units of R — sit inside the units of the power series [writes: $\mathrm{Unit}(R) \subset \mathrm{Unit}(R[[x]])$], by just this unit a going to the constant power series a [writes: $a \mapsto a$]. That's also true, right?

So, the same question as above: Is this an equality?

And now we come to something interesting, because what was the argument before (i.e., the proof of Theorem 3.39)? The argument before was that — at least when R was an integral domain — you go to the two polynomials, take their highest piece, and multiply them together. That was a great argument for polynomials; it absolutely fails for power series, because power series, you know, have no highest piece. So, this is problematic from the point of view of using that argument.

Maybe instead of... Since we don't have so much time, I'll tell you the answer, which I'm always reluctant to do — this is part of my job, actually, but I'm still...

Something that's kind of surprising here is that f is a unit in $R[[x]]$ if and only if the constant term is a unit in R [writes "f is a unit in $R[[x]] \iff \mathrm{Const}(f) \in \mathrm{Unit}$ in R"].

All I have to do to check whether it's a unit is whether the constant term is a unit.

So, that's a weird thing, isn't it?

Characterization of Units in $R[[x]]$

As for polynomials, the units of R sit inside as constant series: $\mathrm{Unit}(R) \subset \mathrm{Unit}(R[[x]])$, $a \mapsto a$. Unlike for $R[x]$, there are many more:

Theorem 3.43 (*Units in the Formal Power Series Ring*)**:** Let R be an arbitrary commutative ring with identity 1. A formal power series $f(x) = \sum_{i=0}^{\infty} a_i x^i \in R[[x]]$ is a unit if and only if its constant term $a_0 = \mathrm{Const}(f)$ is a unit in R:

$$f(x) \in \mathrm{Unit}(R[[x]]) \iff a_0 \in \mathrm{Unit}(R). \tag{3.83}$$

Striking Contrast with Polynomial Rings: Recall from Theorem 3.39 that in $R[x]$ (over an integral domain), a polynomial is a unit if and only if it is a constant unit: no positive-degree terms may appear.

In radical contrast, in $R[[x]]$, a power series can have completely arbitrary non-zero coefficients $a_1, a_2, a_3, \ldots$ for all higher powers of x without restriction; as long as the single scalar a_0 is invertible in R, the entire infinite series $f(x)$ possesses an exact multiplicative inverse in $R[[x]]$!

Spoken Transcript [00:36:51 - 00:37:45] Prof. Rahul Pandharipande

It shouldn't surprise you if you've played with polynomials.

Well, I wanted to give two proofs; we will see. You should always be suspicious when a mathematician says that they will give you two proofs: it usually means both will be sloppy. [laughter]

By the way, since people didn't seem to recognize this Euler note (Recall: the old 10-franc note from the previous lecture), I brought some copies, so you can see if you're interested: the Swiss franc Euler note from the previous century [holds up copies of the former Swiss 10-franc banknote featuring Leonhard Euler]. When I came to Switzerland, I went to a used coin shop and bought a booklet of them.

Beyond the Lecture: The Euler Banknote

The banknote is the 10-franc note of the sixth series of Swiss banknotes, in circulation from 1979 to 2000. Its front shows Leonhard Euler (born 1707 in Basel); its back shows a water turbine, the solar system and the path of light through lenses, three subjects of his work.

Proof of Theorem 3.43

Spoken Transcript [00:37:45 - 00:39:07]

Prof. Rahul Pandharipande

Okay, so I'll try to prove this. I want to prove this statement at the end.

Suppose f has the form $a_0 + a_1x + a_2x^2 + \dots$, where $a_0 \in R$ is a unit [writes: Suppose $f = a_0 + a_1x + a_2x^2 + \dots$, $a_0 \in R$ is a unit]. That means that there exists some $y \in R$ such that $a_0y = 1$ in R [writes: $\exists y \in R$, $a_0y = 1$ in R], right?

So, the first thing I can do is just multiply f by y [writes: $f \cdot y =$].

Is that clear? Whenever I think about this, y is now the constant power series. Are you happy with this?

And who knows what I get here, but I know one thing: it starts with 1, because when I multiply by a constant power series, it just multiplies all these by y, right? So, whatever it is, it starts with 1, and then it's $(a_1y)x$ [writes: $= 1 + (a_1y)x + (a_2y)x^2 + \dots$] — y is an element of the ring, it has nothing to do with x —, $(a_2y)x^2$, and so on.

Okay, so the first step is: I use the fact it's a unit, I multiply by something in R and make this 1.

Step 1: Normalization to Constant Term 1

1. **Forward Implication ($\Rightarrow$):** As established in (3.82), if $f(x) \cdot g(x) = 1$, then $\mathrm{Const}(f) \cdot \mathrm{Const}(g) = a_0b_0 = 1$, forcing $a_0 \in \mathrm{Unit}(R)$.
2. **Reverse Implication ($\Leftarrow$) — Normalization:** Suppose $a_0 \in \mathrm{Unit}(R)$. Then there exists an inverse element $y \in R$ such that:

$$a_0y = 1 \quad \text{in } R. \tag{3.84}$$

Multiplying $f(x)$ by the constant power series $y \in R \subset R[[x]]$:

$$\begin{aligned} f(x) \cdot y &= (a_0 + a_1x + a_2x^2 + \dots) \cdot y \\ &= a_0y + (a_1y)x + (a_2y)x^2 + \dots \\ &= 1 + (a_1y)x + (a_2y)x^2 + \dots \end{aligned} \tag{3.85}$$

Because $y \in \mathrm{Unit}(R) \subset \mathrm{Unit}(R[[x]])$, $f(x)$ is invertible if and only if the normalized series $f(x) \cdot y$ is invertible. Indeed, if $(f(x) \cdot y)^{-1}$ exists, then $f(x)^{-1} = y \cdot (f(x) \cdot y)^{-1}$.

Spoken Transcript [00:39:07 - 00:40:25] Prof. Rahul Pandharipande

Okay? Then it's enough to invert this, because inversion means that there's somebody else I multiply to get 1.

So, then it's sufficient to find $g \in R[[x]]$ [writes: Sufficient to find $g \in R[[x]]$] such that g times this result, $1 + \dots$ [writes: $g \cdot (1 + b_1x + b_2x^2 + \dots) = 1$]. And now I just make these coefficients b's, because I don't want to write a_1y, and I don't know anything about them anyway, so I might as well make it b.

So, it's sufficient to find a g such that whenever I have something of this form — which is what I have there —, this is equal to 1.

Meaning I only have to invert such things. All I've done is use the fact it's a unit to change the first coefficient to 1. And then it's sufficient to invert that; because once I invert that, then I can just multiply it by y, and then, you know...

So, I have to invert this: $(1 + b_1x + b_2x^2 + \dots)^{-1}$ [writes: $(1 + b_1x + b_2x^2 + \dots)^{-1}$].

Okay, well, you never finish, so maybe like this: I have to invert this.

Step 2: Reduction to Inverting $1 + \gamma$

Setting $b_k := a_ky \in R$ for all $k \geq 1$, the normalized series (3.85) reads $1 + b_1x + b_2x^2 + \dots$. It is therefore sufficient to find $g \in R[[x]]$ such that:

$$g \cdot (1 + b_1x + b_2x^2 + \dots) = 1, \tag{3.86}$$

that is, to show that the inverse $(1 + b_1x + b_2x^2 + \dots)^{-1}$ exists in $R[[x]]$. Then $f^{-1} = y \cdot g$.

Spoken Transcript [00:40:25 - 00:41:31] Prof. Rahul Pandharipande

Now, you know how to do this, right? You definitely know how to do this. It's, you know, somehow in your brain; you have to bring it out.

So, you write this whole thing, let's say as γ [writes: $\gamma = b_1x + b_2x^2 + \dots$]: γ is equal to this power series. And this power series has x's in it, okay?

So, you can say: I want to invert this, $1 + \gamma$ [writes: $(1 + \gamma)^{-1}$]. And if I write this as $1/(1+\gamma)$ [writes: $= \frac{1}{1+\gamma}$], surely you know how to invert this: it's equal to $1 - \gamma + \gamma^2 - \gamma^3$, and so on [writes: $= 1 - \gamma + \gamma^2 - \gamma^3 + \dots$].

That's the answer! This actually proves it's invertible. And you have to think: why does it prove...? You know this, this is like some identity from your previous life. I just write it, and I claim this is the formula for the inverse.

Step 3: The Formal Geometric Series

Define the auxiliary power series $\gamma(x) \in R[[x]]$ consisting of all higher-degree terms:

$$\gamma := b_1x + b_2x^2 + b_3x^3 + \dots = \sum_{k=1}^{\infty} b_kx^k. \tag{3.87}$$

The expression to invert is simply $1 + \gamma$. Inspired by the classical geometric series $\frac{1}{1-z} =$

$\sum_{k=0}^{\infty} z^k$ (Fact 0.43), we substitute $z = -\gamma$ to propose the candidate inverse formula:

$$(1+\gamma)^{-1} \overset{\text{formal}}{=} \frac{1}{1+\gamma} = 1 - \gamma + \gamma^2 - \gamma^3 + \gamma^4 - \cdots = \sum_{k=0}^{\infty} (-1)^k \gamma^k. \tag{3.88}$$

Spoken Transcript [00:41:31 - 00:42:50] Prof. Rahul Pandharipande

And the reason is that this thing makes sense. It looks like an infinite sum, but this γ has x's in it. γ is divisible by x, so x divides γ [writes: $x \mid \gamma$]. That means x^2 divides γ^2 [writes: $x^2 \mid \gamma^2$], and x^k divides γ^k [writes: $x^k \mid \gamma^k$].

So, when I expand this, if I want the coefficient of the power series in any particular x, only finitely many of these terms enter.

The whole point of this argument is that this thing is a well-defined element of the power series ring [writes "well defined elem of $R[[x]]$"].

That is what you have to convince yourself of: just that it makes sense to write this. And I gave you the argument, but that's different than you accepting the argument. But it just says that if you do this, every one of these terms is well-defined. And you're not allowed to add infinitely many things in algebra, but since the powers of x get bigger and bigger, you don't have to do it.

So, maybe we'll talk a little bit more about that next time. [claps hands and turns toward desk, students applaud]

Step 4: Well-Definedness via Divisibility by Powers of x

Why the Infinite Sum of Power Series Is Algebraically Sound: In an arbitrary ring, infinite sums are not defined. However, because γ has no constant term (its lowest degree term is $b_1 x$), x divides γ:

$$x \mid \gamma \implies x^2 \mid \gamma^2 \implies \cdots \implies x^k \mid \gamma^k. \tag{3.89}$$

Consequently, the lowest power of x appearing with non-zero coefficient in γ^k is at least x^k:

$$\gamma^k = (b_1 x + b_2 x^2 + \dots)^k = b_1^k x^k + (\text{terms of degree } \geq k+1).$$

Now consider the coefficient of x^N in the formal sum $\sum_{k=0}^{\infty} (-1)^k \gamma^k$:

- For every $k > N$, every term in γ^k has degree at least $k > N$.
- Therefore, terms with $k > N$ contribute strictly 0 to the coefficient of x^N!

Hence, for any fixed power x^N, only the finitely many terms $k \in \{0, 1, \dots, N\}$ contribute to its coefficient:

$$[x^N] \left(\sum_{k=0}^{\infty} (-1)^k \gamma^k \right) = [x^N] \left(\sum_{k=0}^{N} (-1)^k \gamma^k \right) \in R. \tag{3.90}$$

Because every individual coefficient of the resulting power series is computed as a finite sum of elements in R, the series:

$$g(x) := \sum_{k=0}^{\infty} (-1)^k \gamma(x)^k$$

is a rigorous, well-defined element of $R[[x]]$. Multiplying by $(1+\gamma)$ explicitly cancels all terms by telescoping:

$$(1+\gamma)\cdot\sum_{k=0}^{\infty}(-1)^k\gamma^k=\sum_{k=0}^{\infty}(-1)^k\gamma^k+\sum_{k=0}^{\infty}(-1)^k\gamma^{k+1}=1.$$

This manipulation of infinite sums is legitimate because it can be checked coefficient by coefficient: for the coefficient of x^N, only the finitely many terms with $k\le N$ contribute, and for these the identity is the finite telescoping sum $(1+\gamma)\sum_{k=0}^{N}(-1)^k\gamma^k=1+(-1)^N\gamma^{N+1}$, whose last term has no x^N component. Thus, $(1+\gamma)$ is invertible, completing the proof that $f(x)$ is a unit in $R[[x]]$.

Q.E.D.

Beyond the Lecture: The Recursive Coefficient-by-Coefficient Proof

At timestamp 00:36:51, the lecturer mentions a second proof of Theorem 3.43. Instead of substituting γ into the geometric series, one can solve $\left(\sum_{n=0}^{\infty}a_nx^n\right)\left(\sum_{n=0}^{\infty}b_nx^n\right)=1$ directly for the coefficients $b_n\in R$ by induction on n, using the Cauchy product (Definition 3.42):

- **Base case ($n=0$):** $a_0b_0=1$, which has the unique solution $b_0=a_0^{-1}\in R$ since $a_0\in\mathrm{Unit}(R)$.
- **Inductive step ($n\ge 1$):** Having uniquely determined $b_0,\dots,b_{n-1}\in R$, the x^n-coefficient of the product is $\sum_{k=0}^{n}a_kb_{n-k}=a_0b_n+\sum_{k=1}^{n}a_kb_{n-k}=0$, which uniquely determines $b_n=-a_0^{-1}\sum_{k=1}^{n}a_kb_{n-k}\in R$.

Summary of Key Concepts

- **Field of Fractions:** For an integral domain R, $\mathrm{Frac}(R)$ is formally the set of equivalence classes of pairs $(r,s)\in R\times(R\setminus\{0\})$ with $(r,s)\equiv(t,w)\overset{\text{def}}{\iff}rw=st$. In practice one computes with fractions $\frac{r}{s}$ by the usual rules; the integral domain property guarantees that the denominators sw stay non-zero, and every $\frac{r}{s}\neq 0$ has the inverse $\frac{s}{r}$.
- **Polynomial Rings:** $R[x]$ exists for every commutative ring R with unit. Its universal property: for every ring homomorphism $\phi\colon R\to S$ and every $y\in S$, there is a unique ring homomorphism $\hat{\phi}\colon R[x]\to S$ with $\hat{\phi}|_R=\phi$ and $\hat{\phi}(x)=y$, namely $\hat{\phi}(\sum a_ix^i)=\sum\phi(a_i)y^i$.
- **Units of $R[x]$:** If R is an integral domain, then $\mathrm{Unit}(R[x])=\mathrm{Unit}(R)$, because the leading coefficient of a product is the product of the leading coefficients. Whether there can be more units without this assumption is left to the students.
- **Formal Power Series:** $R[[x]]$ consists of all sequences $(a_0,a_1,a_2,\dots)$ of elements of R, written $\sum_{i\ge 0}a_ix^i$, with componentwise addition and the Cauchy product $c_k=\sum_{i=0}^{k}a_{k-i}b_i$; every coefficient of a product is a finite sum.
- **Units of $R[[x]]$:** For every commutative ring R with unit, $f\in R[[x]]$ is a unit if and only if its constant term is a unit in R. After normalizing the constant term to 1, the inverse of $1+\gamma$ is the formal geometric series $\sum_{k\ge 0}(-1)^k\gamma^k$, which is well-defined because $x^k\mid\gamma^k$.

Tuesday, 6th October 2026

Polynomial Rings, Laurent Series, Quotient Rings, and the First Isomorphism Theorem

Course Content Overview

This lecture explores fundamental algebraic ring constructions, bridging polynomial extensions, localized series, and residue structures:

- **Polynomial and Power Series Rings:** Multivariable polynomial rings, the binomial theorem in commutative rings, and the identity $(x+y)^p = x^p + y^p$ over ${}^{\mathbb{Z}}\!/_{p\mathbb{Z}}$.
- **Fields of Fractions and Laurent Series:** The field of rational functions $R(x)$ and formal Laurent series, which over a field K give $K((x)) \cong \operatorname{Frac}(K[[x]])$.
- **Ideals and Quotient Rings:** Review of ideals and kernels, the linear algebra analogy, and the construction of the quotient ring ${}^{R}\!/_{I}$ from scratch (highlighting how ideal absorption ensures well-defined multiplication of classes).
- **The First Isomorphism Theorem for Rings:** Formulation and complete three-step proof of ${}^{R}\!/_{\ker(f)} \cong \operatorname{Im}(f)$.
- **Ideals Generated by Elements and Polynomial Quotients:** Contrasting the quotients ${}^{\mathbb{C}[x]}\!/_{(x^2+1)}$ (which possesses zero divisors) and ${}^{\mathbb{Q}[x]}\!/_{(x^2+1)} \cong \mathbb{Q}[i]$ (which forms the field of Gaussian rationals).

Review: Polynomial Rings and Power Series

Spoken Transcript [00:00:00 - 00:01:55] Prof. Rahul Pandharipande

Okay, so, we start, and I wanted to start with some things from last week. [walks to the board and writes on the blackboard] If R is a ring, we had defined this polynomial ring in one variable, and also the power series ring. [writes: $R \to R[x], R[[x]]$] And just some comments: the polynomial ring is a subring of the power series ring. [writes: $R[x] \subset R[[x]]$] That's important to note, because every polynomial you can view as a finite power series.

So, this is one comment. The second comment is that — and people were asking me after class —: what happens if I have two variables or three variables? [writes: $\to R[x,y]$] There's two ways to approach this. One is you could just define this in the same way as we define this, [writes: $\to$ defined] which is: it's the set of finite sums $a_{i,j}x^iy^j$, where these indices go over some finite set, and the coefficients are in R, [writes: $\{\sum_{0\leq i,j\in\mathbb{N}} a_{i,j}x^iy^j \mid a_{i,j} \in$

$R\}$] and it's just the same definition as before. So, that's one way to do it. I hope that's clear — it's just multiplying with more polynomials.

But there's a more formal way, which is that R is a ring, so then we've already defined this, that's where we add one variable, [writes: $R \Rightarrow R[x]$] and so this is a ring, so we can also define this: this is a ring, and have one more variable. [writes: $\Rightarrow R[x][y]$] And you could take this to be the definition of $R[x,y]$. [writes: $= R[x,y]$] So, it's just: we add the variables one at a time, as this is a polynomial ring over the polynomial ring over R.

So, it's another way. And of course, if you have two different paths, then you have the burden of proving they're both the same, etc., and I leave this burden for you to carry, all right? So, and these are the same, maybe I'll just say that: [draws a double arrow between the two definitions and writes "SAME"] same.

Polynomial Rings in One and Several Variables

Let R be a commutative ring with identity. Recall the construction of the *"polynomial ring"* $R[x]$ (Definition 3.32) and the *"formal power series ring"* $R[[x]]$ (Definition 3.41). Every polynomial can be naturally identified with a formal power series having only finitely many non-zero coefficients, yielding the subring inclusion:

$$R[x] \subseteq R[[x]].$$

Definition 4.1 (*Polynomial Ring in Two Variables*): For two indeterminates x and y, the polynomial ring $R[x,y]$ can be defined in two equivalent ways:

(1) Explicit Definition (Finite Linear Combinations):

$$R[x,y] := \left\{ \sum_{i,j\geq 0}^{\text{finite}} a_{i,j}x^i y^j \;\middle|\; a_{i,j} \in R \right\}.$$

(2) Iterative Definition (Polynomials over Polynomials): Viewing $R[x]$ as the coefficient ring, we form the univariate polynomial ring in y:

$$R \longrightarrow R[x] \longrightarrow (R[x])[y] =: R[x,y].$$

Equivalence of Definitions (beyond the lecture)

The lecture leaves this check to the students. Both constructions yield canonically isomorphic rings:

$$(R[x])[y] \cong R[x,y] \cong (R[y])[x].$$

In the iterative perspective, an element is a polynomial $P(y) = \sum_{j=0}^{m} p_j(x)y^j$, where each coefficient $p_j(x) = \sum_{i=0}^{k_j} a_{i,j}x^i \in R[x]$. Expanding via the distributive law expresses $P(y)$ uniquely as the finite double sum $\sum_{i,j} a_{i,j}x^i y^j$. By induction, this extends to n variables:

$$R[x_1, \ldots, x_n] \cong (R[x_1, \ldots, x_{n-1}])[x_n].$$

Q.E.D.

III The Binomial Formula in Commutative Rings

Spoken Transcript [00:01:55 - 00:03:48] Prof. Rahul Pandharipande

Okay, so, then there's some comments from essentially... Well, something you should know is if I have this polynomial ring in two variables, [writes "$R[x,y]$" on the board] and I take $x+y$, that's in this polynomial ring of two variables. And I can raise it to the n, [writes: $(x+y)^n =$] and this is something you know: this is this binomial expansion, or I don't know how it's called, binomial formula... [writes "binomial expansion"]

This is equal to the sum — so, this is from $a=0$ to n — [writes: $\sum_{a=0}^{n}$] of $x^{n-a}y^a\binom{n}{a}$. [writes: $x^{n-a}y^a\binom{n}{a}$]

Okay? That's the binomial formula. Are you familiar with this binomial formula? So, for example, $(x+y)^5$ [writes: $(x+y)^5 =$] is equal to $1x^5+5x^4y+10x^3y^2+10x^2y^3+5xy^4+y^5$. [writes: $x^5+5x^4y+10x^3y^2+10x^2y^3+5xy^4+y^5$] I guess I should have done a smaller one.

Okay, so, that's an example. And this holds in every ring, because these elements x and y are here, and we've already discussed: these numbers make sense in any ring, they're just how you add up 1 (Recall: Definition 1.16). Is that clear? So, this formula holds in any ring.

And to prove it, it's just the same as by induction we do. If you wanted to prove this formula in any ring, you just write it out as $(x+y)(x+y)$ and use the distributive law, and you get x^2, and you regroup everything, and you get $xy+xy+y^2$, and that gives you the formula; and then you prove it by induction. [writes: $(x+y)^2 = (x+y)(x+y) = x^2+xy+xy+y^2$]

The Binomial Expansion

Let R be any commutative ring with identity 1_R. For integers $k \in \mathbb{Z}$, the integer scalar $k \cdot 1_R$ is canonically defined by repeated addition or negation of 1_R (Definition 1.14). Consequently, the binomial coefficients $\binom{n}{a} \in \mathbb{N}$ act canonically as coefficients in R.

Theorem 4.2 (*Binomial Formula in Commutative Rings*)**:** In the polynomial ring $R[x,y]$, or for any two commuting elements $x,y \in R$ ($xy = yx$), the following identity holds for all $n \in \mathbb{N}$:

$$(x+y)^n = \sum_{a=0}^{n} x^{n-a}y^a\binom{n}{a}. \tag{4.1}$$

Example 4.3 (*Expansion for $n=5$ and $n=2$*)**:** For $n=5$:

$$(x+y)^5 = x^5+5x^4y+10x^3y^2+10x^2y^3+5xy^4+y^5.$$

For $n=2$, distributivity and commutativity ($yx = xy$) give:

$$(x+y)^2 = (x+y)(x+y) = x^2+xy+xy+y^2 = x^2+2xy+y^2.$$

Proof by Induction (beyond the lecture)

The lecture expands $(x+y)^2$ and says that induction does the rest. The general identity follows by induction on n. For $n=0$, both sides equal $1=\binom{0}{0}$. For the induction step:

$$\begin{aligned}(x+y)^{n+1} &= (x+y)\sum_{a=0}^{n}\binom{n}{a}x^{n-a}y^{a}\\ &= \sum_{a=0}^{n}\binom{n}{a}x^{n+1-a}y^{a}+\sum_{a=0}^{n}\binom{n}{a}x^{n-a}y^{a+1}\\ &= x^{n+1}+\sum_{k=1}^{n}\left[\binom{n}{k}+\binom{n}{k-1}\right]x^{n+1-k}y^{k}+y^{n+1}\\ &= \sum_{k=0}^{n+1}\binom{n+1}{k}x^{n+1-k}y^{k},\end{aligned}$$

using Pascal's identity $\binom{n}{k}+\binom{n}{k-1}=\binom{n+1}{k}$ (Fact 0.13). **Q.E.D.**

The Binomial Formula for a Prime Exponent (in textbooks: Frobenius Identity)

Spoken Transcript [00:03:48 - 00:05:44] Prof. Rahul Pandharipande

Okay, so, there's one comment here, which is kind of a fun comment. That holds in any ring, that binomial formula. But what happens if we take this ring $\mathbb{Z}/p\mathbb{Z}$, [writes "$\mathbb{Z}/p\mathbb{Z}$" on the bottom blackboard] where p is a prime? [writes "prime" under it, with an arrow from the p] So, for example, and then in this ring, you could take this ring $(\mathbb{Z}/p\mathbb{Z})[x,y]$, and the fun thing to consider there is $(x+y)^p$. [writes: $(x+y)^p$]

So, that's an example. And so, if I take this formula in $\mathbb{Z}/p\mathbb{Z}$ — $\mathbb{Z}/5\mathbb{Z}$ in this case, where it's the same exponent as here — something kind of nice happens: all these internal coefficients here, they all have a 5 in it, so they all go away. And this is true just in general. In $\mathbb{Z}/p\mathbb{Z}$, if you take $(x+y)^p$, you get this formula that you secretly wanted anyway, which is this: [writes: $=x^p+y^p$] x^p+y^p. It's a simple formula.

And the reason is because for all of the binomial coefficients, $\binom{p}{a}$, p divides $\binom{p}{a}$ [writes: $p \mid \binom{p}{a}$] when a goes from 1 to $p-1$. [writes: when $1\leq a\leq p-1$] All the internal coefficients, p divides that. And why is that? It's because this binomial starts with a $p!$, [writes: $\frac{p!}{a!(p-a)!}$] and there's no p's here to cancel the p in the $p!$, because these are all lesser numbers. Is that clear?

So, you'll always have... if you look at the exponent being prime in this binomial theorem, all of the middle coefficients here will all have that prime in it. So, if I look at it in $\mathbb{Z}/p\mathbb{Z}$, I get this great formula where there's no... So, this is a well-known formula, it will come up. Okay. So, that's an aside about the binomial formula.

The Binomial Formula Modulo p

Lemma 4.4 (*Divisibility of Binomial Coefficients by a Prime*): Let p be a prime number.

For every integer $a \in \{1, 2, \ldots, p-1\}$, the prime p divides the binomial coefficient $\binom{p}{a}$:

$$p \,\Big|\, \binom{p}{a}. \tag{4.2}$$

Proof

Recall the algebraic formula:

$$\binom{p}{a} = \frac{p!}{a!(p-a)!} \implies a!(p-a)!\binom{p}{a} = p! = p \cdot (p-1)!.$$

Thus, p divides the product $a!(p-a)!\binom{p}{a}$. Since $1 \le a \le p-1$, every factor occurring in the factorials $a! = 1 \cdot 2 \ldots a$ and $(p-a)! = 1 \cdot 2 \ldots (p-a)$ is strictly less than p. Because p is prime, Euclid's lemma (Fact 0.17), applied factor by factor, shows that p does not divide $a!(p-a)!$. By Euclid's lemma once more, p must divide the remaining factor $\binom{p}{a}$. **Q.E.D.**

Proposition 4.5 (*The Binomial Formula Modulo p (in textbooks: Frobenius Identity)*): Let p be a prime. In the polynomial ring $(\mathbb{Z}/p\mathbb{Z})[x, y]$, we have:

$$(x+y)^p = x^p + y^p. \tag{4.3}$$

Proof

Expanding $(x+y)^p$ via Theorem 4.2:

$$(x+y)^p = x^p + \sum_{a=1}^{p-1} \binom{p}{a} x^{p-a} y^a + y^p.$$

By Lemma 4.4, $p \mid \binom{p}{a}$ for all $1 \le a \le p-1$, so the coefficient $\binom{p}{a} \cdot 1 = \overline{\binom{p}{a}}$ is $\bar{0}$ in $\mathbb{Z}/p\mathbb{Z}$. Hence each intermediate term vanishes, leaving $(x+y)^p = x^p + y^p$. **Q.E.D.**

Beyond the Lecture: Characteristic p and the Frobenius Endomorphism

Textbooks call this statement the Frobenius identity, informally also the Freshman's Dream. The proof of Proposition 4.5 only uses that $p \cdot 1 = 0$ in the coefficient ring. A non-zero commutative ring R with $p \cdot 1_R = 0$ is said to have *"characteristic"* p (the characteristic is the least $n \ge 1$ with $n \cdot 1_R = 0$); examples are $\mathbb{Z}/p\mathbb{Z}$ and the polynomial rings over it. In every such ring, $(a+b)^p = a^p + b^p$ for all $a, b \in R$, by the same proof. Together with $1^p = 1$ and $(ab)^p = a^p b^p$, this says that $a \mapsto a^p$ is a ring homomorphism $R \to R$, the *"Frobenius endomorphism"* of R.

Integral Domains, Fields of Fractions, and Laurent Series

Polynomial Rings and Power Series Rings as Integral Domains

Spoken Transcript [00:05:44 - 00:06:39] Prof. Rahul Pandharipande

And then there were some definitions: if I take R now to be an integral domain, [writes: R int. domain] we had defined here $R[x]$, that's polynomials in one variable, this is also an integral domain. [writes: $R[x]$ also int. domain] We discussed that (Recall: the highest-term argument in Theorem 3.39). You use the highest term, and if R is an integral domain, when you multiply polynomials, the highest term lives, so it stays an integral domain.

But what I didn't say is this is also an integral domain: $R[[x]]$. [writes: $R[[x]]$ also int. domain] And one needs a different argument here, because power series don't have highest terms. And so, the answer is, well, what do you think the answer is? You can't use the highest term here, so what should you use?

Student Answer

Lowest term.

Spoken Transcript [00:06:41 - 00:07:07] Prof. Rahul Pandharipande

Lowest term, exactly. You could have used the lowest term here, but psychologically with polynomials, you go to the highest term. But it turns out lowest term is a better idea here, because for a power series, a power series does have a lowest term. And when you multiply the two lowest terms of power series, it's the lowest term of the product. So, it's the same argument here, it just uses the lowest term, which is, you know, psychologically a little strange. You could have used the lowest term here, but you didn't.

But, okay, so, this is also an integral domain.

Integral Domain Properties: Highest and Lowest Terms

Theorem 4.6 (*Preservation of the Integral Domain Property*)**:** If R is an integral domain, then:

(1) The polynomial ring $R[x]$ is an integral domain.

(2) The formal power series ring $R[[x]]$ is an integral domain.

Proof

Since $1 \neq 0$ in R, also $1 \neq 0$ in $R[x]$ and in $R[[x]]$; it remains to show that there are no zero divisors. Let f, g be non-zero elements in the respective ring.

(1) **For $R[x]$ (Highest Degree Term):** Let $f = a_n x^n + \cdots + a_0$ and $g = b_m x^m + \cdots + b_0$ with $a_n \neq 0$ and $b_m \neq 0$. As in (3.68), the leading term of the product fg is:

$$a_n b_m x^{n+m}.$$

Since R is an integral domain, $a_n b_m \neq 0$. Thus, $fg \neq 0$ and $\deg(fg) = \deg(f) + \deg(g)$.

(2) For $R[[x]]$ (Lowest Term): A power series does not have a highest term, but every non-zero series has a lowest term. We write $f = x^n(a_n + a_{n+1}x + \dots)$ with $a_n \neq 0$, and $g = x^m(b_m + b_{m+1}x + \dots)$ with $b_m \neq 0$. The lowest non-vanishing term of the Cauchy product (Definition 3.42) is:

$$a_n b_m x^{n+m}.$$

Since R is an integral domain, $a_n b_m \neq 0$. Consequently, $fg \neq 0$.

Q.E.D.

Fields of Fractions: Rational Functions and Laurent Series

Spoken Transcript [00:07:07 - 00:08:35] Prof. Rahul Pandharipande

And the point I wanted to make here is that the main construction last week was the field of fractions (Recall: Theorem 3.26). [writes "field of fractions" over an arrow] We'll have a different construction this week, but that was the field of fractions, and the important thing about a field of fractions: you can only build that on an integral domain. So, if I have an integral domain, I could take the field of fractions of that integral domain. That was a construction.

But I could also take the field of fractions of polynomials, because that's an integral domain, and that is called... well, the symbol for that is R with parentheses, that's brackets: [writes: $R(x)$] $R(x)$. That's the name of the field of fractions of the polynomial ring when R is an integral domain. And it's called the field of rational functions. [writes "field of rational functions"]

So, maybe rational functions with coefficients in R. This is polynomials with coefficients in R, that's the field of rational functions. And you know how to think about this, because we've thought about the field of fractions: an element here in rational functions is a polynomial divided by another polynomial. So, both of these, f and g, are polynomials, and of course g is not allowed to be zero. [writes: $R(x) \ni \frac{f(x)}{g(x)}$, $f, g \in R[x]$, $g \neq 0$] So, that's how to think about an element here, and you know, that's that. Okay.

The Field of Rational Functions

Recall that for any integral domain R, its *"field of fractions"* $\mathrm{Frac}(R)$ consists of equivalence classes of formal quotients $\frac{a}{b}$ with $a, b \in R$ and $b \neq 0$ (Definition 3.17).

Definition 4.7 (*Field of Rational Functions $R(x)$*): Let R be an integral domain, so that $R[x]$ is an integral domain (Theorem 4.6). The field of fractions of the polynomial ring $R[x]$ is denoted by round parentheses $R(x)$ and is called the *"field of rational functions"*:

$$R(x) := \mathrm{Frac}(R[x]) = \left\{ \frac{f(x)}{g(x)} \;\middle|\; f(x), g(x) \in R[x],\ g(x) \neq 0 \right\}. \tag{4.4}$$

Spoken Transcript [00:08:35 - 00:10:15] Prof. Rahul Pandharipande

Now, power series are also an integral domain, so we can again consider the field of fractions, and the symbol for that is two parentheses: $R((x))$. [writes: $R((x))$] All right? So, this is kind of universal math notation. There's the integral domain, you have the polynomial ring, power series, field of fractions, and this one has a special name. Does anyone know what the name of this thing is? It's called the field of Laurent series. [writes "field of Laurent series"]

Actually, I'm having some doubts about how to spell Laurent. Is it with an a? No, I think it's with an e. I'm not 100% sure about that.

So, this is the field of Laurent series, and this will come up in your complex analysis classes for sure, but with a $\mathbb{C}$ here, and usually the argument is called z in complex analysis. So, the field of Laurent series — this is actually something... you could say we've already constructed the field of fractions, so why do I have to talk about it here? You could say this field of Laurent series we think about the same way as $f/g(x)$, where f and g are power series. [writes: $R((x)) = \frac{f(x)}{g(x)}$, $f, g \in R[[x]], g \neq 0$] And this would be absolutely correct. That would be correct, and g is not equal to zero. It's just there's a nicer way to think about it, and that has to do with the last thing I said yesterday... not yesterday, last week (Recall: Theorem 3.43).

The Field of Fractions of Formal Power Series

Definition 4.8 (*Field of Fractions* $R((x))$): Let R be an integral domain. Since $R[[x]]$ is an integral domain (Theorem 4.6), we define its field of fractions, denoted with double round parentheses $R((x))$:

$$R((x)) := \operatorname{Frac}(R[[x]]) = \left\{ \frac{f(x)}{g(x)} \;\middle|\; f(x), g(x) \in R[[x]],\ g(x) \neq 0 \right\}. \tag{4.5}$$

It is called the *"field of Laurent series"*: when $R = K$ is a field, it is isomorphic to the field of formal Laurent series over K (Theorem 4.11).

Formal Laurent Series with Finitely Many Negative Powers

Spoken Transcript [00:10:48 - 00:12:45] Prof. Rahul Pandharipande

Have you seen Laurent series already in the complex analysis class? No? All right, well, we'll see.

So, you won't go wrong if you think about it this way — that's the definition, that's how we constructed it. But there's another way, which is why it's called the field of Laurent series. So, what is a Laurent series? A Laurent series is a power series where you are allowed to have finitely many negative coefficients, that's the definition.

So, I can say this: [writes on the board: Laurent series in x] Laurent series in a variable x, this is a power series with — and this is the crucial thing — [writes "is a power series with"] finitely many negative coefficients... [writes "finitely many negative powers"] You can have as many positive coefficients, positive degree coefficients as you want, it's a power series, but only finitely many negative powers. So, that's the crucial thing for a Laurent series.

So, what is a power series? A power series might be like the power series for e^x, [writes: $1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots \in R[[x]]$] we've seen these also anyway. So, that's a power series; these are elements of the ring, the power series ring.

In Laurent series, you're allowed to have some negative, so you could have some $x^{-12}+3x^{-2}+x^{-1}$, then the constant term... I got too excited about the constant term! $17+x+x^{15}+\ldots$. [writes: $x^{-12}+3x^{-2}+x^{-1}+17+x+x^{15}+\ldots$] So, that is a Laurent series. So, have you seen these objects in mathematics? Okay, well, maybe it's the first time, but you can imagine: you just take power series, you're allowed to have negative terms, but only finitely many. And it's extremely important you have only finitely many.

Formal Definition and Arithmetic of Laurent Series

Definition 4.9 (*Formal Laurent Series*): A *"formal Laurent series"* in the indeterminate x over a ring R is a formal sum of the form:

$$f(x)=\sum_{n=-N}^{\infty} a_n x^n = a_{-N}x^{-N}+a_{-N+1}x^{-N+1}+\cdots+a_0+a_1x+a_2x^2+\ldots \tag{4.6}$$

where $N\in\mathbb{Z}$ and $a_n\in R$ for all $n\geq -N$. That is, $f(x)$ contains at most **finitely many negative powers** of x.

Example 4.10 (*Power Series vs. Laurent Series*):

- **Power Series:** Elements of $R[[x]]$ contain only non-negative powers:
$$1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots\in R[[x]].$$
(For this series, R must contain $\mathbb{Q}$, e.g. $R=\mathbb{R}$, so that the coefficients $\frac{1}{k!}$ lie in R.)
- **Laurent Series:** Elements can have negative exponents up to some lowest power:
$$x^{-12}+3x^{-2}+x^{-1}+17+x+x^{15}+\cdots\in R((x)).$$

Spoken Transcript [00:12:45 - 00:14:25] Prof. Rahul Pandharipande

And you'll find: if you have only finitely many, you can add them, and when you add them, you'll still only have finitely many because two of them only had finitely many. When you multiply them, the lowest term now — you know, if I multiply this with something with lowest term -7 [writes: x^{-7}], its lowest term will be x^{-19} — so, it will still have only finitely many. Is that clear?

And here's the great thing: if you're non-zero here, you can divide. That's why the Laurent series turns out to be a field. And this is something you should think about. We discussed this last time. So, if you have a power series... maybe I just say this, because this is maybe now departing a little bit from the course, but it is the case that if I take this field of Laurent series, [writes "field of Laurent series"] this is isomorphic to $R((x))$, [writes: $\cong R((x))$] the field of fractions of the power series. And what you have to show there is: if you have Laurent series, you can divide. That's the interesting thing.

And that has to do with the last thing I said last week, which is that it's pretty easy to divide power series: all you have to do is find the first non-zero coefficient, then use this rule I explained last week (Recall: Theorem 3.43), and then you can play with this power to take it out. So, I leave this for you.

Spoken Transcript [00:14:25 - 00:15:20] Prof. Rahul Pandharipande

I'll say one thing: why don't we allow power series with infinitely many negative terms? Because if you try to multiply them, then you have to do infinite sums. And we don't do infinite sums in algebra, we only do finite sums. Because if you do infinite sums, you need convergence, and you can try to do that, and that's an interesting subject, but in this class, we're not doing infinite sums. And if you have finitely many negative terms, when you multiply, you see you'll never have to do infinite sums. It's just the same.

So, anyway, that's just this comment here. This field of Laurent series, it turns out, is just a nice way of thinking about this field of fractions for power series. And you can imagine, it's nicer to think about this kind of object than the ratio. And this second way of looking at things is not available for polynomials. There's only one way to think about it there: as ratios of polynomials. Okay.

Isomorphism between Laurent Series and the Field of Fractions

Theorem 4.11 (*Algebraic Characterization of $R((x))$*)**:** When K is a field, the set of formal Laurent series with finitely many negative powers forms a field under formal addition and multiplication, and is naturally isomorphic to the field of fractions of $K[[x]]$:

$$\mathrm{Frac}(K[[x]]) \cong K((x)). \tag{4.7}$$

Why Invertibility Holds and Why Infinitely Many Negative Terms Fail: The lecture sketches both points; in detail:

(1) **Invertibility in $K((x))$:** Any non-zero Laurent series can be factored as $f(x) = x^k u(x)$, where $k \in \mathbb{Z}$ is the minimal index with $a_k \neq 0$, and $u(x) = a_k + a_{k+1}x + \cdots \in K[[x]]$ with $a_k \neq 0$. Since K is a field, the leading coefficient a_k is a unit in K, so $u(x)$ is an invertible unit in the power series ring $K[[x]]$ (Theorem 3.43). Its multiplicative inverse is:

$$f(x)^{-1} = x^{-k}u(x)^{-1} \in K((x)).$$

(2) **Failure of Infinitely Many Negative Terms:** If series with infinitely many negative powers were permitted ($\sum_{n=-\infty}^{\infty} a_n x^n$), the coefficient of x^k in the product of two such series would require evaluating the infinite sum:

$$c_k = \sum_{j=-\infty}^{\infty} a_j b_{k-j}.$$

Without topological convergence, such an infinite sum is undefined in pure algebra. Restricting to finitely many negative powers ensures that for each k, only finitely many non-zero products $a_j b_{k-j}$ occur, keeping all algebraic operations purely finite.

Beyond the Lecture: Why K Must Be a Field

For an integral domain R that is not a field, the Laurent series with coefficients in R do not form a field, so they cannot be all of $R((x)) = \mathrm{Frac}(R[[x]])$. For $R = \mathbb{Z}$, the constant series 2 has no inverse among the Laurent series over $\mathbb{Z}$: comparing constant terms in $2 \cdot g = 1$ would give $2c_0 = 1$ with $c_0 \in \mathbb{Z}$. Yet $\frac{1}{2}$ lies in $\operatorname{Frac}(\mathbb{Z}[[x]])$. In general, the Laurent series over R form a subring of $\operatorname{Frac}(R[[x]])$, via $x^{-N}g \mapsto \frac{g}{x^N}$, and the two coincide precisely when R is a field. For a field K, one can also write $K((x)) = K[[x]][x^{-1}]$.

Ideals and Ring Homomorphisms

Definition and Axioms of an Ideal

Spoken Transcript [00:15:33 - 00:16:29] Prof. Rahul Pandharipande

Okay, so, the main topic this week is about ideals and quotient rings, and the main construction is that of a quotient ring.

If you like polynomial rings, the class will talk about results of polynomial rings, but mostly polynomial rings in one variable. The subject of polynomial rings in several variables is kind of very important in mathematics, and there's a whole class about it more or less, which is... the class is called commutative algebra.

Spoken Transcript [00:16:29 - 00:18:33] Prof. Rahul Pandharipande

Okay, so, now we finish these constructions and we start something new. R is our ring, [writes: R ring] and the first thing: we've already defined I is an ideal. [writes: $I \subset R$ is an ideal] So, we've already defined this (Recall: Definition 1.29), but I'll just review it briefly because now it's crucial for this week.

I is an ideal if it satisfies that, if I take R plus and 0... this is a subgroup of the additive part of the ring. [writes: $\bullet (I,+,0) \subset (R,+,0)$ subgroup] And the second is: if I take anybody in the ring and anybody in the ideal, then their product lies in the ideal. [writes: $\bullet\ \forall r \in R, \forall a \in I$ then $r \cdot a \in I$] This says that the ideal is an additive subgroup under addition, but there's another condition. The ideal has kind of two constraints: one is the addition constraint, and the other is the multiplication constraint. They're just different, because they're talking about two different operations.

And our example we had seen... we did some examples anyway, but a place where we know we always get ideals: if I take a homomorphism... [writes: $f\colon R \to S$] maybe I should put it here... so, f is a homomorphism of rings, [writes: f is a hom of rings] then whenever you have a homomorphism of rings, you get an ideal for free. Sorry. [chalk slips]

And we discussed this also before: then the kernel of f, which is defined to be all the people in the ring such that $f(a) = 0$, is an ideal. [writes: $\implies \ker(f) = \{a \in R \mid f(a) = 0\}$ is an ideal] So, we proved this in one of the early weeks (Recall: Proposition 1.30).

Definition of Ideals and the Kernel of a Homomorphism

Recap: Ideals and the kernel of a ring homomorphism were defined in Week 1 (Definition 1.29, Definition 1.26), where the kernel was also shown to be an ideal (Proposition 1.30); both statements were restated in Week 2 (Definition 2.24, Proposition 2.25). The lecturer reviews them because the quotient ring construction of this lecture rests on them.

Let R be a commutative ring with identity.

Definition 4.12 (*Ideal*): A subset $I \subseteq R$ is called an *"ideal"* of R if it satisfies the following two conditions:

(1) Additive Subgroup Condition: $(I,+,0)$ is a subgroup of the additive group $(R,+,0)$. That is, $0 \in I$, and for all $x,y \in I$, $x-y \in I$.

(2) Absorption / Multiplication Constraint: I absorbs multiplication by arbitrary elements of the ambient ring R:

$$\forall r \in R, \quad \forall a \in I : r \cdot a \in I.$$

Proposition 4.13 (*Kernel of a Ring Homomorphism as an Ideal*): Let $f\colon R \to S$ be a homomorphism of commutative rings. The *"kernel"* of f, defined by:

$$\ker(f) := \{a \in R \mid f(a) = 0_S\}, \tag{4.8}$$

is an ideal of R.

Proof

We verify the two defining conditions of Definition 4.12:

(1) Additive Subgroup: Since $f(0_R) = 0_S$, $0_R \in \ker(f)$. For $x,y \in \ker(f)$, additivity of f yields $f(x-y) = f(x) - f(y) = 0_S - 0_S = 0_S$, so $x - y \in \ker(f)$.

(2) Multiplicative Absorption: For any $r \in R$ and $x \in \ker(f)$, multiplicativity gives:

$$f(r \cdot x) = f(r) \cdot f(x) = f(r) \cdot 0_S = 0_S.$$

Thus, $r \cdot x \in \ker(f)$.

Q.E.D.

The Linear Algebra Analogy: Kernels, Images, and Isomorphisms

Spoken Transcript [00:18:33 - 00:20:12] Prof. Rahul Pandharipande

So, in general, just when you study algebra, even in linear algebra, I remind you in the study of linear algebra... [writes "Reminder" on the middle board] this is a reminder: if you have a vector space V and you have a linear transformation, let's call it ϕ... [writes: $\phi\colon V \to W$] so, if these are two vector spaces — so, V, W are vector spaces over some field, [writes: V, W vector spaces $/\mathbb{C}$] it's not relevant for this example, I don't know, say complex numbers — and this is a linear transformation — I don't know what you called homomorphisms of vector spaces, linear transformation. [writes: ϕ linear transformation]

So, in the abstract study of linear algebra, when you have such a linear transformation, you automatically get two other vector spaces, so to speak for free. So, the two vector spaces are the kernel — this is a subspace of V — and the image, this is a subspace of W. [writes: $\ker(\phi) \subset V$, $\mathrm{Im}(\phi) \subset W$] And there is a theorem that links this data. If you remember, it says that if I take V and I quotient by the kernel... I hope you remember this, it's kind of crucial here... if you take V and you quotient by the kernel, this is isomorphic as vector spaces to the image. [writes: ${}^{V}\!/_{\ker(\phi)} \cong \operatorname{Im}(\phi)$]

All right, so, this is kind of a model. It's some aspect of your... Did you learn it in these terms? I've never taught linear algebra here. Okay.

Linear Algebra Analogy: The First Isomorphism Theorem

For a linear transformation $\phi\colon V \to W$ between vector spaces over a field $\mathbb{F}$:

- **Kernel:** $\ker(\phi) = \{v \in V \mid \phi(v) = 0\} \leq V$ is a vector subspace of V.
- **Image:** $\operatorname{Im}(\phi) = \{\phi(v) \mid v \in V\} \leq W$ is a vector subspace of W.
- **First Isomorphism Theorem for Vector Spaces:**
$$ {}^{V}\!/_{\ker(\phi)} \cong \operatorname{Im}(\phi). $$

In ring theory, we seek an identical structural correspondence for a ring homomorphism $f\colon R \to S$.

Spoken Transcript [00:20:12 - 00:21:10] Prof. Rahul Pandharipande

Yeah, I haven't taught linear algebra for... I don't know, it's possible not in this century. But it's a great class, linear algebra. I used to tell students it's like the most important math class, linear algebra, because humans are actually pretty good at solving linear equations and pretty terrible at solving any other kind of equation.

That's not true, I did teach it. [cleans and wipes the lower board]

Ring Homomorphisms vs. Linear Transformations

Spoken Transcript [00:21:10 - 00:23:08] Prof. Rahul Pandharipande

Okay, so, we would like to think a little bit about this kind of construction for rings. And we see a lot of it to start.

If I have this ring homomorphism... so, these are both rings now, so, I try to make the parallel here: R and S are rings, [writes: $f\colon R \to S$, R, S are rings] and I remind you, they're commutative rings with unit, and f always takes unit... I don't remind you of all the definitions, but it's consistent with our previous... f takes the 1 to 1, that's the definition of a ring (Recall: Definition 1.18). And f is a homomorphism. [writes: f hom] So, we just want to complete that board.

So, we get now in this situation, we get two objects for free. Or I don't know, free might be the wrong word, but anyway, this gives us two objects: one is the kernel inside R, [writes: $\to \ker(f) \subset R$] but this is not... so, there, everything was vector spaces. But this is not a ring now, this is an ideal. [writes "ideal"]

All right, but we also get the image of f inside S, [writes: $\operatorname{Im}(f) \subset S$] and that is a subring. [writes "subring"] All right, so, we saw that was a subring, the image (Recall: Proposition 2.27).

So, that's kind of the parallel there. And then we're going to work on this quotient business, but you see the matter is slightly... you know, in the vector space world, everything is a vector space. But in the ring world, sometimes you get rings, and sometimes you get ideals, and they're slightly different objects, so, we have to come to terms with that.

So, let me just say that the statements there that you know... so, basic statements: [writes "STATEMENT"] One is that f is surjective if and only if this subring is the whole ring. [writes: (i) f is surjective $\iff$ $\mathrm{Im}(f) = S$] That's almost like a tautology, right? That's what surjective means: you get everybody here, and if the image is everybody, it's surjective. So, that's essentially a tautology.

The Morphism Data for Rings: Ideals vs. Subrings

When comparing linear transformations $\phi\colon V \to W$ and ring homomorphisms $f\colon R \to S$, a crucial categorical divergence arises:

Structure	**Linear Algebra ($\phi\colon V \to W$)**	**Ring Theory ($f\colon R \to S$)**
Domain Sub-object	$\ker(\phi) \leq V$ (Vector Subspace)	$\ker(f) \subset R$ (**Ideal**, not a subring with 1_R)
Codomain Sub-object	$\mathrm{Im}(\phi) \leq W$ (Vector Subspace)	$\mathrm{Im}(f) \leq S$ (**Subring**, contains $1_S = f(1_R)$)

Proposition 4.14 (*Surjectivity Criterion*)**:** Let $f\colon R \to S$ be a ring homomorphism. Then f is surjective if and only if:

$$\mathrm{Im}(f) = S. \tag{4.9}$$

Injectivity and the Zero Ideal

Spoken Transcript [00:23:08 - 00:24:34]

Prof. Rahul Pandharipande

The second statement — and that's also true there — the second statement: [writes: (ii)] f is injective if and only if... [writes: f is injective $\iff$] so, what's the statement going to be? It's going to be about the kernel: the kernel of f is the zero ideal. [writes: $\ker(f) = (0)$ zero ideal]

This is the standard terminology for the zero ideal, the standard notation for the zero ideal. The zero ideal is the ideal that has zero and nothing else. If you look at the definition of an ideal, the ideal that has the single element zero... So, this is going to be used a lot, so, it should be completely clear. [moves to the right and writes: Zero ideal, (0), 0, $\{0\}$]

The zero ideal: sometimes this is denoted like this: (0), sometimes it's just denoted 0, but the zero ideal is the ideal that contains one element: zero. So, it's definitely a subgroup, because it's just zero. Zero, when you add it, stays zero; it has inverses; there's nothing wrong with zero. And the second one is: it's closed under multiplication, because if you take zero in the ideal and multiply by anybody, you stay zero. So, that's the zero ideal.

And this last statement says f is injective if and only if the kernel is the zero ideal.

Injectivity and the Zero Ideal

Definition 4.15 (*The Zero Ideal*): In any ring R, the singleton set consisting solely of the additive neutral element:

$$(0) := \{0\} \tag{4.10}$$

is an ideal of R, called the *"zero ideal"*. It is also written simply as 0.

Proposition 4.16 (*Injectivity and the Kernel*): Let $f\colon R \to S$ be a ring homomorphism. Then f is injective if and only if its kernel is trivial:

$$f \text{ is injective} \iff \ker(f) = (0). \tag{4.11}$$

Proof of Proposition 4.16

Spoken Transcript [00:24:34 - 00:25:38] Prof. Rahul Pandharipande

So, let's try to see what the argument for that is; that's almost a tautology. So, proof of **(2)**. [writes: Proof (ii).]

We have to do two things. Suppose f is injective. [writes: Suppose f is injective] Then let a in R be not zero. [writes: let $a \in R, a \neq 0$] Then $f(a)$ is not equal to $f(0)$, because injective means you take different elements, they go to different places — that's what injective means. So, if it's injective, this implies that a... oh, and $f(0) = 0$, so, that means $f(a)$ is not zero, [writes: then $f(a) \neq f(0) = 0$] so, this implies a is not in the kernel of f. [writes: $\implies a \notin \ker(f)$]

So, if we have f is injective, nobody who's not zero is in the kernel of f, so, this implies the kernel of f equals zero. [writes: $\implies \ker(f) = 0$]

So, that's one half of it. I assume it's injective, and I deduce the kernel of f is zero; there can only be one element in it.

Forward Direction: Injectivity Implies Trivial Kernel

Forward Implication ($\implies$)

Suppose f is injective. Let $a \in R$ with $a \neq 0$. Since f is injective and $a \neq 0$:

$$f(a) \neq f(0).$$

Since f is a ring homomorphism, $f(0) = 0_S$ (Proposition 1.19). Thus:

$$f(a) \neq 0_S \implies a \notin \ker(f).$$

Consequently, no non-zero element of R belongs to $\ker(f)$. Since $0 \in \ker(f)$ always holds, we conclude:

$$\ker(f) = (0).$$

Q.E.D.

Spoken Transcript [00:25:38 - 00:26:43] Prof. Rahul Pandharipande

The other way around is: suppose the kernel of f is zero, [writes: Suppose $\ker(f) = 0$] and as I said, it's just standard to use both things, depending on what you want to emphasize.

Suppose the kernel of f is zero; then let $f(a) = f(b)$. [writes: let $f(a) = f(b)$] We're assuming it's zero, and then we pick two elements a, b in the ring that go to the same place.

So, this implies, by the homomorphism property, $f(a) - f(b) = f(a - b)$... [writes: $\implies f(a-b) = f(a) - f(b) = 0$] which equals zero, because I'm assuming they're the same. So, this implies that $a - b$ is in the kernel, [writes: $\implies a - b \in \ker(f)$] but we're assuming the kernel is zero, so, this implies $a - b = 0$, which implies $a = b$. [writes "$\implies a - b = 0 \implies a = b$" and boxes "$a = b$"]

So, that's the second implication.

That's a basic implication, and actually, if you look at this proof, it's the same proof as you had in linear algebra. There's actually no difference.

Reverse Direction: Trivial Kernel Implies Injectivity

Reverse Implication ($\Longleftarrow$)

Suppose $\ker(f) = (0)$. Let $a, b \in R$ such that $f(a) = f(b)$. Using the additive homomorphism property of f:

$$f(a) - f(b) = f(a - b) = 0_S.$$

By definition of the kernel, this implies:

$$a - b \in \ker(f).$$

Since $\ker(f) = (0) = \{0\}$ by hypothesis, it follows that:

$$a - b = 0_R \implies a = b.$$

Thus, f is injective, completing the proof of Proposition 4.16. **Q.E.D.**

Q.E.D.

Spoken Transcript [00:26:43 - 00:28:05] Prof. Rahul Pandharipande

So, we want now... for this linear algebra picture, we've recovered all the pieces except this one, [pushes up the lower blackboard and points at ${}^{V}\!/_{\ker(\phi)} \cong \operatorname{Im}(\phi)$ on the upper board] which says that V mod the kernel is the image, and we now want to recover that. And for that, we need a construction.

And there was a construction already in linear algebra you needed, which was the notion of a quotient vector space, which is, if you remember that, it's going to be very helpful.

The idea of a math education is that, actually, you're not allowed to forget anything you've learned. That's one theory. The other theory is that education is what's left after you've forgotten everything you've learned. You can decide which theory of education you'd like to subscribe to.

The Construction of Quotient Rings

Spoken Transcript [00:28:05 - 00:29:59] Prof. Rahul Pandharipande

Okay, so, the task at hand is to understand this quotient. And now we do this not just for the kernel — we'll get back to the kernel — but for any ideal. So, we let I in R be an ideal. [writes on the lower left board: Let $I \subset R$ be an ideal]

And now, this is the main construction for this week: we will construct the quotient ring. [writes: we will construct the quotient ring] So, that's the name: quotient ring. And the symbol is R mod I. [writes "R/I" and circles it] That's the symbol. So, we have to construct it.

And what's the parallel to linear algebra, of course, is Rmod the kernel; that's the quotient ring. So, we'll get back to that in a moment. We forget about this linear algebra analogy, etc., and we go for this construction. We need to construct this.

So, is that clear? We have I is an ideal, so, our hypothesis is: R is a ring, I is an ideal, and we need to construct the quotient. But we need to construct this object as a ring. That means, as usual, it's a multi-step construction: we have to first construct it as a set, then we have to find zero in it, we have to find one in it, [writes: $(R/I, +, \cdot, 0, 1)$] zero and one, and we have to find an addition and multiplication. So, you have to construct the whole thing. That's just what you have to do; there's no other way around it.

So, Step 1... there's a lot of steps, and as usual, I don't do all of them... [writes: Step 1: What is the set R/I ?]

The Quotient Ring Construction: Roadmap

Let R be a commutative ring with identity 1, and let $I \subset R$ be an ideal. Our objective is to construct a new ring:

$$R/I = (R/I, +, \cdot, 0_{R/I}, 1_{R/I}), \tag{4.12}$$

called the *"quotient ring"* (or residue class ring) of R modulo I.

The Five Stages of Constructing a Ring from Scratch: To endow the quotient R/I with a complete ring structure, we must rigorously execute the following stages:

Step 1: Underlying Set: Define an equivalence relation on R and take R/I to be the set of its equivalence classes.

Step 2: Distinguished Elements: Identify the additive zero $0_{R/I}$ and multiplicative identity $1_{R/I}$.

Step 3: Addition Operation: Define binary addition $+$ on equivalence classes and prove it is **well-defined**.

Step 4: Multiplication Operation: Define binary multiplication $\cdot$ on equivalence classes and prove it is **well-defined** using the ideal absorption property.

Step 5: Ring Axioms Verification: Verify associativity, commutativity, distributivity, and existence of inverses.

Step 1: The Underlying Set and Equivalence Relation Modulo an Ideal

Spoken Transcript [00:29:59 - 00:31:57] Prof. Rahul Pandharipande

Step 1: What is the set R mod I? And here the notation is totally standard. We have I and R, R mod I. Everybody uses just this notation. What is the set? And this should, now if you remember linear algebra, that should help you now.

So, the step for this is: we define an equivalence relation [writes: We define an equivalence relation $\equiv$] on R. Sometimes this equivalence relation could be denoted by this, but you have to carry around a lot of stuff, so, I'm just going to leave that I out. Going to define an equivalence relation on R.

Here's the definition: we say that if I have x and y in R, then x is equivalent to y [writes: Def: $x, y \in R$, $x \equiv y$] — and as I said, if you like, you can put I here because the equivalence relation depends on I — if and only if $x - y$ is in the ideal I. [writes: $\iff x - y \in I$] That's the definition.

And in words — so, as I said, if you want to be really precise, so, this is an aside, [writes: (Aside $\equiv_I$,] if you want to be really precise, you could define it like this, because this thing depends on I — in words, this is usually put: x is equivalent to y mod I. [writes "x is equivalent to y mod I")] This is very standard language: mod$I$. x is equivalent to y mod I. That's what this equivalence relation is. It's just like for the integers.

Anyway, that's the symbol. And so, formally now, you have to check it's an equivalence relation. And to do that, you have to remember the three properties of an equivalence relation (Recall: Remark 3.13). And the three properties are: x is equivalent to x, which is zero being in the ideal; the other property is: if x is equivalent to y, then y is equivalent to x, and that would just put a sign here, but we know that I is a subgroup under addition, so, it's closed under the sign; and the last one is transitive. Maybe I do the last one: the transitive property.

The Equivalence Relation Modulo an Ideal

Definition 4.17 (*Congruence Modulo an Ideal*)**:** Let R be a ring and $I \subset R$ an ideal. For elements $x, y \in R$, we define the relation of *"congruence modulo I"* (denoted $x \equiv_I y$ or simply $x \equiv y$) by:

$$x \equiv y :\iff x - y \in I. \tag{4.13}$$

In words, we say that *"x is equivalent to y modulo I"*.

Connection to Modular Arithmetic in $\mathbb{Z}$: This directly generalizes congruence modulo n in the integers:

$$x \equiv y \pmod{n} \iff x - y \in n\mathbb{Z},$$

where $I = n\mathbb{Z} = (n)$ is the principal ideal generated by n (Definition 2.33).

Spoken Transcript [00:31:57 - 00:33:28] Prof. Rahul Pandharipande

[writes: Transitive:] The transitive property says: if x is equivalent to y and y is equivalent to z, then x is equivalent to z. [writes: If $x \equiv y$ and $y \equiv z$, then $x \equiv z$]

x is equivalent to y means, if I translate this now in the definition, $x - y$ is in I, [writes: $x - y \in I$] and this is $y - z$ is in I. [writes: $y - z \in I$]

What I need to show is: $x - z$ is in I. But these two elements are in I, so, I can add them. [writes: $(x - y) + (y - z) \in I$] These two assumptions, since I is closed under addition, I can add them. Is that clear?

And then happily the y falls out, so, you get $x - z$ is in I, [writes: $x - z \in I$] and we're happy about that, okay? So, that's the proof of the transitivity.

Verification: Equivalence Relation Properties

Proposition 4.18 (*$\equiv_I$ is an Equivalence Relation*): Let $I \subset R$ be an ideal. The relation $\equiv_I$ is an equivalence relation on R.

Proof

We verify the three defining axioms:

(1) Reflexivity ($x \equiv x$): For any $x \in R$, $x - x = 0 \in I$ because $(I, +, 0)$ is a subgroup. Thus, $x \equiv x$.

(2) Symmetry ($x \equiv y \implies y \equiv x$): If $x \equiv y$, then $x - y \in I$. Since $(I, +, 0)$ contains additive inverses:

$$y - x = -(x - y) \in I \implies y \equiv x.$$

(3) Transitivity ($x \equiv y \wedge y \equiv z \implies x \equiv z$): Suppose $x \equiv y$ and $y \equiv z$. By definition:

$$x - y \in I \quad \text{and} \quad y - z \in I.$$

Since I is closed under addition, their sum lies in I:

$$(x - y) + (y - z) \in I \implies x - z \in I \implies x \equiv z.$$

Q.E.D.

Subgroup Property Sufficiency: Notice that verifying that $\equiv_I$ is an equivalence relation relies **exclusively** on the fact that $(I, +, 0)$ is an additive subgroup of $(R, +, 0)$. The multiplicative absorption condition $R \cdot I \subseteq I$ has not yet been utilized!

Spoken Transcript [00:33:28 - 00:34:47] Prof. Rahul Pandharipande

If you think about what we've used: we've used zero is in I, we've used the minus is in I, and we've used I is closed under addition, so, we've kind of used all the group conditions for I. We couldn't get away with less for the ideal.

What we have not used is the multiplicative condition. That's going to come later. Because an ideal has two assumptions: one is the additive structure, it's a subgroup, and we've used it all here. But we have not touched the multiplicative condition; that'll come in a moment.

Okay, so, if it's an equivalence relation, then you can imagine that we're going to take its equivalence classes.

Spoken Transcript [00:34:47 - 00:36:03] Prof. Rahul Pandharipande

Now we can define the set: R mod I as a set is equal to the set of equivalence classes [writes: ${}^{R}\!/_{I}$ = Set of equivalence classes] of this relation, let us say, on R. [writes: of $\equiv_I$ on R]

Okay, so, I've defined this as a set. And now the question of names: how do I name things? And that's a separate discussion. It's the same discussion we had about the rational numbers, the same for the field of fractions, that one of the conceptual issues about the quotient is that people do not have unique names. And this is just, you know, something you have to live with.

So, the question is: how do I name an element? [writes: How do I name an element in ${}^{R}\!/_{I}$?]

Definition 4.19 (*The Quotient Set* ${}^{R}\!/_{I}$)**:** The underlying set of the quotient ring ${}^{R}\!/_{I}$ is defined as the set of all equivalence classes of R with respect to the equivalence relation $\equiv_I$:

$$ {}^{R}\!/_{I} := \{\bar{x} \mid x \in R\}. \tag{4.14}$$

Naming Conventions for Equivalence Classes

Spoken Transcript [00:36:03 - 00:37:33] Prof. Rahul Pandharipande

How do I name an element in R mod I? One way to do it is: you take x in R, [writes: $x \in R$] then x is in R, so, it determines an equivalence class: the equivalence class which x lies in. And that is usually called $\bar{x}$. [writes: $\bar{x} \in {}^{R}\!/_{I}$, the equiv. class that x lies in] This is the equivalence class that x lies in — the equivalence class that x lies in. Okay, that's not beautiful English, but anyway, that's what it is. You take x, you get an equivalence class; that's usually denoted $\bar{x}$. That's what we do with integers also. Is that okay?

Another standard way to denote it... there's kind of three ways to denote this: if x is in R, we can take $\bar{x}$ in R mod I; [writes: $\bar{x} \in {}^{R}\!/_{I}$] if you really want to write things out really explicitly, you can write it like that: [writes: $x + I \in {}^{R}\!/_{I}$] because the equivalence class of x is the set of things where you take... the equivalence class of x is equal to all those things $x + i$ where i is in I. [writes: $\bar{x} = \{x + i \mid i \in I\}$] Is that clear? Because if you're equivalent to x, your difference is in I. So, this is explicitly what the equivalence class is. And since it's x plus something in I, some people will write it like this: [points to $x + I$] okay? It's kind of messy to keep that around, but in some arguments it's nice, so, that's a second way to write it.

The third way to write it, which is by far the most common, is you just call it x. [writes: $x \in {}^{R}\!/_{I}$] That's by far the most common, because you can see that... then if you use this, you have to be very careful. Sometimes, like in this, when x was born, it was an element, and then when you put it here, it becomes an equivalence class, and you have to keep track of the data type of x in your head. But I would say that that's kind of the three ways to do it. They all have their own advantages. Well, this one has limited advantage, but okay.

Three Standard Notations for Residue Classes

An element of ${}^{R}\!/_{I}$ is an equivalence class containing elements of R. Because representations are not unique (different representatives can specify the exact same class), three standard notations are commonly employed:

1. **Overline Notation: $\bar{x} \in {}^{R}\!/_{I}$:** Denoted $\bar{x} \in {}^{R}\!/_{I}$. This is compact and mirrors the notation used for residue classes ${}^{\mathbb{Z}}\!/_{n\mathbb{Z}}$ (Definition 1.23).
2. **Shift Notation: $x + I \subseteq R$:** Explicitly writing the equivalence class as an affine shift of the ideal:
$$\bar{x} = x + I := \{x + i \mid i \in I\} \subseteq R. \tag{4.15}$$
Since $y \equiv x \iff y - x = i \in I \iff y = x + i$, the equivalence class of x is literally the subset $x + I$.
3. **Bare Representative: $x \in {}^{R}\!/_{I}$:** Writing simply $x \in {}^{R}\!/_{I}$, by abuse of notation. While by far the most common, this requires carefully tracking whether x represents an individual ring element in R or an entire equivalence class in ${}^{R}\!/_{I}$ (the *"data type"* in one's head).

Terminology: Cosets

Many textbooks call the set $x + I$ a **coset** of I. The lecture does not use this word: the lecturer says equivalence class or residue class. These notes follow him. In that language, the operations defined below read $(x + I) + (y + I) = (x + y) + I$ and $(x + I) \cdot (y + I) = (x \cdot y) + I$.

Step 2 & 3: Distinguished Elements and the Well-Definedness of Addition

Spoken Transcript [00:37:33 - 00:39:08] Prof. Rahul Pandharipande

All right, so, since we've defined the set, we need to now define addition. So, let's do the zero and one. First, we define zero in R mod I: [writes: $0 \in {}^{R}\!/_{I}$, Def: $0 = \bar{0}$] 0 is equal to $\bar{0}$, okay? Because this is the equivalence class of zero. Zeros are always called zeros.

So, that's the zero here: it's the equivalence class of zero, which is not a surprise. And similarly, the definition of one is: it's the equivalence class of one. [writes: Def: $1 = \bar{1}$] So, now we have at least this R mod I, and we have some distinguished elements: zero, one.

Okay, so, now let us define addition. [writes: Let us define $+$] And it's still called plus. Very formally, you shouldn't call the binary operation in two different rings the same thing, because they're kind of different. But no one does that; everyone calls it plus, and it makes following the definitions... it makes you have to concentrate to follow the definitions.

So, let's define plus. What does that mean? To define plus, I take an element — this is an element of R mod I — and I take another element of R mod I, and I have to define this plus. I have to define it: [writes: $\bar{x}+\bar{y} \underset{\text{DEF}}{=} \overline{x+y}$] okay? This equation here is going to define it. This is the definition. And how am I going to define it? Well, I only know how to add x and y one way, because they were born in R, and it's the only thing I know how to do. So, this is the only thing possibly I could write.

Distinguished Elements and the Definition of Addition

Definition 4.20 (*Zero and Identity in* $^{R}\!/_{I}$)**:** The distinguished neutral elements in the quotient ring $^{R}\!/_{I}$ are defined by the classes of the respective identities in R:

$$0_{R/I} := \bar{0} = 0 + I = I, \tag{4.16}$$

$$1_{R/I} := \bar{1} = 1 + I. \tag{4.17}$$

Definition 4.21 (*Addition on* $^{R}\!/_{I}$)**:** For any two equivalence classes $\bar{x}, \bar{y} \in {}^{R}\!/_{I}$, their sum is defined by adding representatives in R:

$$\bar{x} + \bar{y} \overset{\text{def}}{=} \overline{x+y}. \tag{4.18}$$

Spoken Transcript [00:39:08 - 00:40:48]

Prof. Rahul Pandharipande

So, that's the definition, and as I said, to read it, you have to concentrate on what's being defined in this whole thing. This is being defined; that was already previously known, because it was in R. All right? So, this whole equation is a definition of plus, and it's not a full definition, because you have to check it's well-defined, because these things could have different names. So, that's kind of a pain.

So, let's check it's well-defined. [writes: Let us check $+$ is well defined in $^{R}\!/_{I}$] What could we do? We could have $\bar{x}$ is equal to some $\overline{x'}$, [writes: $\bar{x} = \overline{x'}$] where x' is equal to $x+i$, [writes: $x' = x+i$] where i is in I. [writes: $i \in I$] So, we could have just had a different name for the same equivalence class. And if we had a different name, what we know is that it will just be some shift of x by I, because if they're in the same equivalence class, their difference is an element of I — that's by definition. Okay?

Similarly for y, we could have a different name here: [writes: $\bar{y} = \overline{y'}$, $y' = y+j$, $j \in I$] j, where j is in I.

So, now we have to test that this definition is the same. We already have the formula for the names x and y, so, let's try to see what happened if we use these names. We better get the same answer.

Spoken Transcript [00:40:48 - 00:42:05]

Prof. Rahul Pandharipande

If we use these names, this definition would give us $\overline{x'} + \overline{y'}$; [writes: $\overline{x'} + \overline{y'} = \overline{x'+y'}$] then we use the formulas for $x'+y'$, and then I regroup them — so, this is... well, okay, maybe I don't... [writes: $= \overline{x+i+y+j}$] $x+i$, $y+j$, and then I regroup it: $x+y+i+j$, [writes: $= \overline{x+y+(i+j)}$] but now $i+j$ has to be in the ideal I because I is closed under addition.

And so, this equivalence class is actually the same as that equivalence class, because it's the equivalence class of the same thing shifted by an element of I.

The answer is: we have checked that if I use different names and combined them with addition, we get the same equivalence class on the right side. So, this proves it's well-defined, okay?

And if you remember, I said from linear algebra, you've done all of this before.

Proof that Addition is Well-Defined on ${}^{R}\!/_{I}$

Proposition 4.22 (*Well-Definedness of Addition*): The addition operation defined in Definition 4.21 is well-defined: it is independent of the choice of representatives.

Proof

Suppose $\bar{x} = \bar{x}'$ and $\bar{y} = \bar{y}'$ in ${}^{R}\!/_{I}$. By Definition 4.17, there exist elements $i, j \in I$ such that:

$$x' = x + i, \tag{4.19}$$
$$y' = y + j. \tag{4.20}$$

Evaluating the sum using the primed representatives yields:

$$\begin{aligned} \bar{x}' + \bar{y}' &= \overline{x' + y'} \\ &= \overline{(x+i) + (y+j)} \\ &= \overline{(x+y) + (i+j)}. \end{aligned}$$

Since $(I, +, 0)$ is an additive subgroup, $i, j \in I$ implies $i + j \in I$. Therefore, $(x+y) + (i+j) \equiv (x+y) \pmod{I}$, which implies:

$$\overline{(x+y) + (i+j)} = \overline{x+y} = \bar{x} + \bar{y}.$$

Thus, the sum is independent of representatives. **Q.E.D.**

The Quotient Additive Group (beyond the lecture)

The lecture leaves this to the students. Since addition is well-defined, associativity, commutativity, the identity $\bar{0}$, and inverses $-\bar{x} = \overline{-x}$ all inherit directly from $(R, +, 0)$. Hence, $({}^{R}\!/_{I}, +, \bar{0})$ forms an abelian group. **Q.E.D.**

Spoken Transcript [00:42:05 - 00:43:13] Prof. Rahul Pandharipande

Then formally, we have to show... now I've defined addition and the zero, you have to now formally check this addition is associative, and that zero plays the role of a zero, and there's additive inverses, and I leave all of this to you.

But there is multiplication, and that's worth doing carefully, because so far we have not used the multiplicative part of the hypothesis on the ideal, and it comes out now, and it's kind of crucial. So, I will try to do this carefully, maybe after the break, but let me just say for...

Step 4: Multiplication and the Crucial Role of the Ideal Absorption Property

Spoken Transcript [00:43:13 - 00:45:11] Prof. Rahul Pandharipande

The next step is define multiplication on R mod I. [writes: Define $\cdot$ on ${}^{R}/_{I}$:] That means if we have this x, it's exactly as before, we need to define this: [writes: $\bar{x} \cdot \bar{y} \underset{\text{DEF}}{=} \overline{xy}$] and this is going to be the answer, this is the definition. And how are you going to define it? You don't know... you only know how to do one thing to define things. So, you really have no choice except this. Okay? I could put the dot there too if I want.

This just says that if I want to multiply two equivalence classes, I take one element of one equivalence class, another element of the other equivalence class, I multiply them in the ring, and then take the resulting equivalence class. Is that clear?

And so, then I have to prove this is well-defined. [writes: Prove well defined:]

So, I use the same different choices for x, x'. Suppose I have... if I pick these different names, [writes: $\overline{x'} \cdot \overline{y'} \overset{\text{def}}{=} \overline{x' \cdot y'}$] and I use the definition, then I get... and then I use their formulas: $(x+i)(y+j)$, [writes: $= \overline{(x+i)(y+j)}$] and when I multiply them out, this is the crucial part: if I multiply them out, I get... sorry, there's no primes here, I get xy, and then I get $i \cdot y + j \cdot x$ — it's commutative, so, I can write it either way — plus ij. [writes: $\overline{xy + iy + jx + ij}$]

This is with one name, I get that; with the other name, I get this, and the difference is this. So, the only way this is well-defined is if this whole thing is in I. [underlines $iy + jx + ij$ and writes "is in I"] That's the only way it could be well-defined.

And it is in I, and here finally we use the fact that if you have an element of I and multiply it by anything, you stay in I. So, that's why it's used here. Okay, so...

Definition of Multiplication on ${}^{R}/_{I}$

Definition 4.23 (*Multiplication on ${}^{R}/_{I}$*)**:** For equivalence classes $\bar{x}, \bar{y} \in {}^{R}/_{I}$, their product is defined by multiplying representatives in R:

$$\bar{x} \cdot \bar{y} \overset{\text{def}}{=} \overline{x \cdot y}. \tag{4.21}$$

Proof that Multiplication is Well-Defined on ${}^{R}/_{I}$

Proposition 4.24 (*Well-Definedness of Multiplication*)**:** The multiplication operation defined in Definition 4.23 is well-defined.

Proof

Let $\bar{x}' = \bar{x}$ and $\bar{y}' = \bar{y}$. Then $x' = x + i$ and $y' = y + j$ for some $i, j \in I$. Expanding the product of the alternative representatives:

$$\begin{aligned} \bar{x}' \cdot \bar{y}' &\overset{\text{def}}{=} \overline{x' \cdot y'} \\ &= \overline{(x+i)(y+j)} \\ &= \overline{xy + iy + jx + ij}. \end{aligned}$$

To establish that $\overline{(x+i)(y+j)} = \overline{xy}$, we must verify that the difference belongs to I:

$$(xy + iy + jx + ij) - xy = iy + jx + ij \overset{?}{\in} I.$$

We examine each of the three cross-terms:

- $i \in I$ and $y \in R \implies i \cdot y \in I$ by the **multiplicative absorption property** of ideals ($I \cdot R \subseteq I$).
- $j \in I$ and $x \in R \implies j \cdot x \in I$ by the **multiplicative absorption property** ($R \cdot I \subseteq I$).
- $i, j \in I \implies i \cdot j \in I$ (in fact, $I \cdot I \subseteq I$).

Since $(I, +, 0)$ is closed under addition:

$$iy + jx + ij \in I \implies \overline{xy + iy + jx + ij} = \overline{xy} = \bar{x} \cdot \bar{y}.$$

Therefore, multiplication on ${}^{R}\!/_{I}$ is well-defined. **Q.E.D.**

Why Subrings Fail and Ideals are Necessary: Going beyond the lecture: if I were merely an additive subgroup or a subring (closed under multiplication within itself, but not absorbing arbitrary elements of R), the terms iy and jx would generally fail to lie in I, causing the multiplication on equivalence classes to collapse! **This step is the sole reason why ideals, rather than general subrings, are required to define quotient rings.**

Mid-Lecture Break

The lecturer takes a short pause for the mid-lecture break. The lecture resumes at timestamp 00:45:11 with a pedagogical recap of the essential role played by the ideal axioms.

Recap: The Crucial Role of the Ideal Axioms

Spoken Transcript [00:45:11 - 00:47:25] Prof. Rahul Pandharipande

Okay, so, we start, and the last moment in the first half of the class was about this observation, and it's a really important thing, so, I repeat it now. This is the only time...

If you remember, the definition of an ideal [writes on the right board: Def of Ideal] had two clauses: the first is that [writes: • $(I, +, 0) \subset (R, +, 0)$ Subgroup] this is an additive subgroup, and we've used that repeatedly. This has been used repeatedly in the definition of the equivalence relation, in the definition of addition, if you wanted to prove associativity and you wanted to prove all of those properties that I didn't prove, you'd be using them here.

But it has a second property, which was the multiplicative property, which is that for every element of the ring and for every a in the ideal, the product is in the ideal. [writes: • $\forall r \in R, \forall a \in I \implies ra \in I$] That's just a totally separate property, because this one doesn't have anything to do with multiplication, this one has to do with multiplication, and this is the first time we use it.

And it's used to show that this definition of multiplication — which I tried to say philosophically you didn't have very much choice: if you want to multiply two equivalence classes, you pick an element in one equivalence class, you pick an element in the other, you multiply those elements, and you take the equivalence class. That's pretty much the only choice you have to define it, because you start with just an abstract ring.

And to show this is well-defined, which I did here quickly at the end with different elements in those equivalence classes, you come to this issue: you're taking the equivalence class of this element, it has to be the same as the equivalence class of that element. And the only way it is the same is if the extra terms, when you combine them, are all in I. And to see that: this is a product of two elements of I. So, that's okay, because it's any element of R and any element of I — you only need one element of I. So, this one is in I, and these two have one element of I and one element of R each, because this x and y are arbitrary elements of R. So, these are three times we apply this rule.

Review: The Two Defining Axioms of an Ideal

Recap: This is the definition from the start of this lecture (Definition 4.12), which the lecturer writes on the board again.

Definition 4.25 (*Ideal Axioms Reviewed*): A subset $I \subseteq R$ is an ideal if and only if:

(1) Additive Subgroup Axiom:

$$(I,+,0) \subset (R,+,0) \quad \text{is an additive subgroup.}$$

(2) Multiplicative Absorption Axiom:

$$\forall r \in R, \quad \forall a \in I: r \cdot a \in I.$$

Division of Labor Between the Two Axioms:

- Axiom **(1)** governs all additive aspects: that $\equiv_I$ is an equivalence relation (Proposition 4.18), well-definedness of addition, additive identity, and additive inverses in ${}^{R}\!/_{I}$.
- Axiom **(2)** is reserved solely for multiplication: ensuring that products of classes do not depend on the representative chosen.

Completion of the Quotient Ring Structure

Spoken Transcript [00:47:25 - 00:48:47] Prof. Rahul Pandharipande

Okay, so, that proves the... [walks across the front] And this is one aspect of the construction which is not in your linear algebra class. Although it's not far from what happened in your linear algebra class when you considered the problem of scalar multiplication on the quotient, which I don't repeat. But anyway, even that shouldn't be totally foreign to you.

So, now we've constructed this: we have plus, we have multiplication, we have zero and one. [writes: $({}^{R}\!/_{I},+,\cdot,\bar{0},\bar{1})$] The part that requires the foundational work is the definition as equivalence classes and making sure these are well-defined. The rest just follow the nose, that the identities... I mean, you can imagine: you have to check this times that, so, by definition this is that, and so, that's that, so, that checks one of the identity properties; and there's like 10 other properties you have to check, and they're all checked like this. [writes: $\bar{1} \cdot \bar{x} = \overline{1 \cdot x} = \bar{x}$]

So, that's the construction, and that's the main construction this week. Last week we had the construction of the field of fractions, which you should understand completely, and now there is this construction, which is the construction of the quotient ring, which is your task to understand really completely this week: the construction.

Okay, so, are there any questions about this?

The Complete Quotient Ring Structure

Theorem 4.26 (*The Quotient Ring* $({}^{R}\!/_{I}, +, \cdot, \bar{0}, \bar{1})$)**:** Let R be a commutative ring with identity, and let $I \subset R$ be an ideal. The quintuple:

$$({}^{R}\!/_{I}, +, \cdot, \bar{0}, \bar{1}) \tag{4.22}$$

equipped with the operations on classes:

$$\bar{x} + \bar{y} \overset{\text{def}}{=} \overline{x+y}, \qquad \bar{x} \cdot \bar{y} \overset{\text{def}}{=} \overline{x \cdot y}, \qquad 0_{{}^{R}\!/_{I}} := \bar{0}, \qquad 1_{{}^{R}\!/_{I}} := \bar{1},$$

forms a commutative ring with identity.

The Remaining Ring Axioms (beyond the lecture)

The lecture checks the identity and leaves the rest to the students. Having proven that addition and multiplication are well-defined, the remaining ring axioms follow directly from the ring axioms of R applied to representatives:

- **Multiplicative Identity:** $\bar{1} \cdot \bar{x} = \overline{1 \cdot x} = \bar{x}$ for all $\bar{x} \in {}^{R}\!/_{I}$.
- **Associativity:** $(\bar{x} \cdot \bar{y}) \cdot \bar{z} = \overline{(xy)z} = \overline{x(yz)} = \bar{x} \cdot (\bar{y} \cdot \bar{z})$.
- **Distributivity:** $\bar{x} \cdot (\bar{y} + \bar{z}) = \overline{x(y+z)} = \overline{xy + xz} = (\bar{x} \cdot \bar{y}) + (\bar{x} \cdot \bar{z})$.
- **Commutativity:** $\bar{x} \cdot \bar{y} = \overline{xy} = \overline{yx} = \bar{y} \cdot \bar{x}$.

Q.E.D.

Spoken Transcript [00:48:47 - 00:50:58]

Prof. Rahul Pandharipande

So, people were asking me: what's the relationship of this class to the book? That's a really good question.

Every week I picked out some... there were some pages, and that book is a huge book. And somehow the class covers a very small part of the book, so, it's not your goal to read that book from cover to cover, although it's not a bad thing to do.

So, every week I pick out some topics there, and my lectures are roughly on those topics. Now, my notation is sometimes different, the presentation is sometimes different in the book. The book is useful if you want... I skip some details in lectures; if you want all the details, sometimes they're in the book.

The book also... that selection for the whole semester covers the actual curriculum I'm required to cover more or less for this course, so, it's helpful for me. But I would say: use the book if it's helpful for you. If not, you can use the lectures. What happened when I taught this course at ETH in 2018 was that some students wrote some notes, and they're really great notes. Of course, they sent it to me afterwards, but it's 8 years ago, I couldn't find them. But you know, probably some of your colleagues write some good notes, you can also use that. The lectures are recorded, so, you can also look at that.

Another option is that I gave some really good references, some really classic algebra references. They have the unfortunate quality of not being... their PDF file is not available through the library. But you can look at those books also. But if you look at the other books, the data won't be exactly in the order I'm giving you. If you want a book where it's in the order I'm giving you, then you should follow those page numbers in the actual reference. And that book is freely available on the ETH website.

So, that's a discussion about the book. Use it if it's helpful. For the exam material and things, I will only use material that I lectured in class, not some exotic topic in the book that I didn't lecture.

Okay.

Course References and Exam Scope

The lecturer answers student inquiries regarding the course literature:

- **Textbook Alignment:** Weekly reading assignments correspond to specific sections of the comprehensive course textbook (freely accessible to students via the ETH website). The textbook provides supplementary proofs and alternative presentations.
- **Exam Scope:** The examinations will strictly test the concepts, definitions, and theorems presented during the lectures, rather than extraneous textbook topics.

The First Isomorphism Theorem for Rings

Spoken Transcript [00:50:58 - 00:52:37] Prof. Rahul Pandharipande

All right, so, we go back to this issue now. We've made some progress.

If I have R to S... [writes: $f\colon R \to S$, R, S rings, f is a hom] So, this R and S are rings, f is a homomorphism. Now, this is sometimes called the First Isomorphism Theorem, [writes: First Isom. Thm] and it plays the role of the missing piece that we had in linear algebra that I wanted to develop here, and that says that this ring, R mod the kernel [writes: ${}^{R}\!/\!_{\ker(f)}$] — because when you have here, we have the kernel is an ideal; if we have an ideal, we can take the quotient, because that's what the last 45 minutes were about — this ring is canonically isomorphic to the image. [writes: $\cong \operatorname{Im}(f)$] Okay?

So, that plays the role of the statement in vector spaces. Is that clear? That's the replacement. So, we have to prove this.

So, let us see what we have to do. We start with this f from R to S, [writes: We start $f\colon R \to S$] and we will define a homomorphism [writes: we will define a hom] $\bar{f}$ from R mod the kernel of f to S, [writes: $\bar{f}\colon {}^{R}\!/\!_{\ker(f)} \to S$] just to start.

And how do you think we're going to define this?

Statement of the First Isomorphism Theorem for Rings

Theorem 4.27 (*First Isomorphism Theorem for Rings*): Let $f\colon R \to S$ be a homomorphism of commutative rings. Then the kernel $\ker(f)$ is an ideal of R (Proposition 4.13), the image $\operatorname{Im}(f)$ is a subring of S (Proposition 2.27), and there exists a canonical ring isomorphism:

$$\bar{f}\colon {}^{R}\!/\!_{\ker(f)} \xrightarrow{\cong} \operatorname{Im}(f), \qquad \bar{x} \mapsto f(x). \tag{4.23}$$

In particular, if f is surjective, then ${}^{R}\!/\!_{\ker(f)} \cong S$ (Proposition 4.14).

$$\begin{tikzcd}
R \arrow[r, "f"] \arrow[d, "\pi"'] & S \\
{}^{R}\!/\!_{\ker(f)} \arrow[r, dashed, "{\bar{f}, \cong}"'] & \operatorname{Im}(f) \arrow[u, hook, "\iota"']
\end{tikzcd}$$

Proof of the First Isomorphism Theorem

Step 1: Construction and Well-Definedness of the Induced Map $\bar{f}$

Spoken Transcript [00:53:31 - 00:54:07] Prof. Rahul Pandharipande

Okay, so, we have to define $\bar{f}$. What does it mean? We have to define this, this is the definition. So, define $\bar{f}$: it is a homomorphism that starts from R mod $\ker(f)$. So, what this thing eats is elements of R mod $\ker(f)$. [writes: $\bar{f}(\bar{x})$] We have to define this function, and it eats equivalence classes. [writes: $\bar{x} \in {}^{R}\!/\!_{\ker(f)}$]

What do you think we're going to do here?

Student Answer

Send it to $f(x)$.

Spoken Transcript [00:54:08 - 00:55:09] Prof. Rahul Pandharipande

Yeah. I mean, in this kind of generality, there's very little you can do, because we start... you can imagine R is like one country, S is another, there's only one flight here between them, and you've got to take it. There's no other way. There's just no other way. So, to get in here, you've got to get on f somehow, and that's the only way.

So, that's the definition. [writes: Def: $\bar{f}(\bar{x}) = f(x) \in S$] When you define it like this, as I said, it's the standard pain about this naming business. And you know, I say it slightly facetiously, but this name is an incredibly crucial thing in this. If you study mathematics more and more, you'll see this idea of what you can name, what you can name precisely, how you can name it, turns out to be an incredibly important, subtle point. So, it's good to get used to thinking about that.

Anyway, so, the point: we have to show this is well-defined. [writes: Show $\bar{f}$ is well defined]

Spoken Transcript [00:55:09 - 00:56:41] Prof. Rahul Pandharipande

So, what does that mean? If we take another name with somebody else in the equivalence class, it satisfies this equation; [writes: $x' = x + i$, $i \in \ker(f)$, $\bar{x'} = \bar{x}$] then we must check that $\bar{f}$ of this is equal to $\bar{f}$ of that. [writes: We must check $\bar{f}(\bar{x'}) = \bar{f}(\bar{x})$] We have to check it.

Well, let's see. What is this? $\bar{f}$ of this — here we use the definition — that's f of x', [writes: $\bar{f}(\bar{x'}) \overset{\text{def}}{=} f(x')$] and then we use the definition of x', that's f of $x + i$. [writes: $= f(x+i)$] Okay, now what do we do?

I should have said here very carefully that i is in the kernel. [adds $i \in \ker(f)$] Yeah, so, what do we do here? Well, maybe you should expand this as $f(x) + f(i)$, [writes: $= f(x) + f(i)$] because f itself is a homomorphism. Now i is in the kernel, so, this is equal to $f(x)$, [writes: $= f(x)$] because $f(i)$ is zero. [writes "$f(i) = 0$" underneath] And that's what we wanted, right? And this is equal to $\bar{f}$ of $\bar{x}$. [writes: $= \bar{f}(\bar{x})$] Okay, yeah.

Student Question

Don't you need to first show that $\bar{f}$ is a homomorphism, because we are using the argument that $\bar{f}(\bar{i})$ which is $\bar{f}(\bar{0})$ is zero, but...

Spoken Transcript [00:56:55 - 00:57:33] Prof. Rahul Pandharipande

No, I'm just showing it's well-defined. I only want to show this is that. [points at $\bar{f}(\bar{x'}) = \bar{f}(\bar{x})$]

All this f stuff is fine, because f is a homomorphism. Remember that? There, f is a homomorphism. [points at the top board]

We have to do all the things you're saying, but there's a certain order you do it. So, I define f — well, this is part of the definition of $\bar{f}$. I propose a definition here. The problem with this definition is it may depend on which element I check. And now I check that it doesn't depend. But I haven't proven this is a homomorphism yet. I will have to do that.

Student Question

Where do we know that i is in the kernel?

Spoken Transcript [00:57:37 - 00:57:45] Prof. Rahul Pandharipande

Well, that's here. [points at $x' = x + i$] For another name, that's what it means: it's in the kernel.

Student Answer

Oh yeah, makes sense.

Spoken Transcript [00:57:48 - 00:58:44] Prof. Rahul Pandharipande

Okay, so, maybe if that were confusing, I say it again, because it's kind of important.

I just propose this as a definition, but it has the flaw that I have to check this definition does not depend on which element of the equivalence class I chose. And this is another element of the equivalence class. Every element of the equivalence class... their difference is an element of the kernel. So, that's the equation there. I use this equation to show this thing is well-defined.

And actually I wrote out all the steps. In my experience in teaching, it doesn't help writing all the steps. It just doesn't help. But I did it, maybe it was a mistake. Because there's a lot of steps, and every time you have to check what I'm using at every equality. And you have to do that. I already know, but you have to do it, and if I just write it, it doesn't always help. But...

Definition of $\bar{f}$ and Proof of Well-Definedness

We define the induced map:

$$\bar{f}\colon R/\ker(f) \longrightarrow S, \qquad \bar{f}(\bar{x}) := f(x). \tag{4.24}$$

Claim 4.28 (*$\bar{f}$ is Well-Defined*): The map $\bar{f}$ is independent of the choice of representative in the class $\bar{x}$.

Proof

Suppose $\bar{x}' = \bar{x}$ in $R/\ker(f)$. By definition of congruence modulo $\ker(f)$, there exists an element $i \in \ker(f)$ such that:

$$x' = x + i.$$

Applying the definition of $\bar{f}$ and the additive homomorphism property of f:

$$\begin{aligned}
\bar{f}(\bar{x}') &\overset{\text{def}}{=} f(x') && \text{(by definition of } \bar{f}\text{)} \\
&= f(x+i) && \text{(since } x' = x + i\text{)} \\
&= f(x) + f(i) && \text{(since } f \text{ is a ring homomorphism)} \\
&= f(x) + 0_S && \text{(since } i \in \ker(f) \implies f(i) = 0_S\text{)} \\
&= f(x) \\
&= \bar{f}(\bar{x}).
\end{aligned}$$

Thus, $\bar{f}(\bar{x}') = \bar{f}(\bar{x})$, proving that $\bar{f}$ is well-defined as a set-theoretic function. **Q.E.D.**

Spoken Transcript [00:58:44 - 01:00:17] Prof. Rahul Pandharipande

So, that was a little painful. But we're not done with the pain: this just shows it's well-defined. So, we have $\bar{f}\colon {}^{R}\!/_{\ker(f)} \to S$ is a function now. [writes: $\bar{f}\colon {}^{R}\!/_{I} \to S$ function] And we have to show all the properties to show it's a ring homomorphism.

So, we'd have to show, for example, $\bar{f}(\bar{0})$... well, what is this? We use our definition, and that's equal to $f(0)$, that's what that definition says, and $f(0)$ has to be zero because f itself is a homomorphism. [writes: $\bar{f}(\bar{0}) \overset{\text{def}}{=} f(0) = 0$] So, that's good: it's sending zero to zero. That's maybe what you were worried about. Now we've shown it.

Well, we also have to show $\bar{f}(\bar{x}+\bar{y}) = \bar{f}(\bar{x}) + \bar{f}(\bar{y})$. [writes: $\bar{f}(\bar{x}+\bar{y}) = \bar{f}(\bar{x}) + \bar{f}(\bar{y})$] We also have to show this. And now you just have to expand these sides completely, and if you do that, you'll get a chain like this, more or less similar to that, and you'll find that it's compatible with the definition. Okay, so, expand yourself. [writes "expand yourself"] As I said, you'll get a chain where...

Spoken Transcript [01:01:06 - 01:01:57] Prof. Rahul Pandharipande

That's the additive structure. Then you also have to show that $\bar{f}$ respects the multiplicative structure: [writes: Check $\bar{f}(\bar{x}\cdot\bar{y}) \overset{?}{=} \bar{f}(\bar{x})\cdot\bar{f}(\bar{y})$]

Check this, and if we do that, this is defined to be... first we have the definition of the product, that's this, [writes: $\bar{f}(\overline{xy})$] and then this is the definition of $\bar{f}$, that's $f(xy)$, [writes: $= f(xy)$] and if we expand this side, this is $f(x)\cdot f(y)$. [writes: $= f(x)f(y)$] So, you see, and that's true because f is a homomorphism. [writes: True because f is hom $R \to S$.]

So, true because f was born a homomorphism. Okay.

Spoken Transcript [01:01:57 - 01:02:41] Prof. Rahul Pandharipande

So, you have to concentrate with the bars, but really the proofs of this is just unravelling definitions.

So, that's the first step in the proof. I remind you here: we have this $R \xrightarrow{f} S$, [writes: $f\colon R \to S$] and the kernel of f is my ideal, [writes: $\ker(f) \subset R$] and we've defined this $\bar{f}$ from R mod the kernel of f to S. [writes: $\bar{f}\colon {}^{R}\!/_{\ker(f)} \to S$] Okay? And this is a homomorphism, [writes: is a hom] meaning that we know this is a ring — that was the first part of this class — we've defined $\bar{f}$, and then the claim is that's a ring homomorphism.

Proof that $\bar{f}$ is a Ring Homomorphism

Claim 4.29 (*$\bar{f}$ is a Ring Homomorphism*)**:** The well-defined map $\bar{f}\colon {}^{R}\!/_{\ker(f)} \to S$ preserves zero, identity, addition, and multiplication.

Proof

Using the definitions of the classes in ${}^{R}\!/_{\ker(f)}$ and the ring homomorphism properties of $f\colon R \to S$:

(1) Preservation of Zero:

$$\bar{f}(\bar{0}) \overset{\text{def}}{=} f(0_R) = 0_S.$$

(2) Preservation of Identity:

$$\bar{f}(\bar{1}) = f(1_R) = 1_S.$$

(3) Preservation of Addition:

$$\begin{aligned} \bar{f}(\bar{x}+\bar{y}) &= \bar{f}(\overline{x+y}) && \text{(by definition of addition in } {}^{R}\!/\!_{\ker(f)}\text{)} \\ &= f(x+y) && \text{(by definition of } \bar{f}\text{)} \\ &= f(x)+f(y) && \text{(since } f \text{ is a ring homomorphism)} \\ &= \bar{f}(\bar{x})+\bar{f}(\bar{y}). && \text{(by definition of } \bar{f}\text{)} \end{aligned}$$

(4) Preservation of Multiplication:

$$\begin{aligned} \bar{f}(\bar{x}\cdot\bar{y}) &= \bar{f}(\overline{x\cdot y}) && \text{(by definition of multiplication in } {}^{R}\!/\!_{\ker(f)}\text{)} \\ &= f(x\cdot y) && \text{(by definition of } \bar{f}\text{)} \\ &= f(x)\cdot f(y) && \text{(since } f \text{ is a ring homomorphism)} \\ &= \bar{f}(\bar{x})\cdot\bar{f}(\bar{y}). && \text{(by definition of } \bar{f}\text{)} \end{aligned}$$

Therefore, $\bar{f}$ is a homomorphism of commutative rings. **Q.E.D.**

Step 2: Surjectivity of $\bar{f}$ onto $\operatorname{Im}(f)$

Spoken Transcript [01:02:41 - 01:04:17] Prof. Rahul Pandharipande

All right, so, that's the first step. Next, Step 2: what is the image of $\bar{f}$? [writes: Step 2: What is $\operatorname{Im}(\bar{f})$?]

That means, where do we go here? And the answer is: it's the same as the image of f. So, actually $\bar{f}$ goes from R mod $\ker(f)$ to the image of f. [writes: $\bar{f}\colon {}^{R}\!/\!_{\ker(f)} \to \operatorname{Im}(f)$] This is some subring of S. [writes "$\cap$" and "S" under $\operatorname{Im}(f)$]

Why? Because the definition just says that if I take an equivalence class, I choose an element, I go to $f(x)$. So, everything I get is in the image of f. On the other hand, if I'm in the image of f, that means that I come from some element of R, and I can take its equivalence class. So, I get exactly the image of f.

That's Step 2, and this is a surjection. So, I call $\bar{f}$ the same thing, so, $\bar{f}$ is surjective onto the image of f, okay? [writes: $\bar{f}$ is surjective onto the image of f] Sometimes in math, you write this kind of two arrows for surjectivity; do you follow that kind of stuff or not? Okay, I won't give you more notation now.

Surjectivity of the Induced Homomorphism onto the Image

Claim 4.30 (*Surjectivity of $\bar{f}$ onto $\operatorname{Im}(f)$*)**:** The image of $\bar{f}$ is precisely the subring $\operatorname{Im}(f) \subseteq S$. Consequently, the restricted codomain map:

$$\bar{f}\colon {}^{R}\!/\!_{\ker(f)} \twoheadrightarrow \operatorname{Im}(f) \tag{4.25}$$

is a surjective ring homomorphism.

Proof

We verify mutual inclusion of sets:

- **$\mathrm{Im}(\bar{f}) \subseteq \mathrm{Im}(f)$:** For any class $\bar{x} \in {}^{R}\!/_{\ker(f)}$, $\bar{f}(\bar{x}) = f(x) \in \mathrm{Im}(f)$ by definition of the image of f.
- **$\mathrm{Im}(f) \subseteq \mathrm{Im}(\bar{f})$:** Let $y \in \mathrm{Im}(f)$. By definition of the image, there exists an element $x \in R$ such that $f(x) = y$. Considering the class $\bar{x} \in {}^{R}\!/_{\ker(f)}$, we have:

  $$\bar{f}(\bar{x}) = f(x) = y.$$

  Thus, $y \in \mathrm{Im}(\bar{f})$.

Hence, $\mathrm{Im}(\bar{f}) = \mathrm{Im}(f)$, proving that $\bar{f}$ maps ${}^{R}\!/_{\ker(f)}$ surjectively onto the subring $\mathrm{Im}(f)$. **Q.E.D.**

Step 3: Injectivity and Trivial Kernel of $\bar{f}$

Spoken Transcript [01:04:17 - 01:06:42] Prof. Rahul Pandharipande

So, first we define this $\bar{f}$, and $\bar{f}$ is defined in like the only possible way you can. The second step is we realize it's surjective onto the image of f.

And the third step — and this one requires some real concentration — is: what is the kernel? What is the kernel of $\bar{f}$? [writes: Step 3: What is the $\ker(\bar{f})$?]

Let's try to think about that. That means: suppose I have $\bar{x}$ in R mod I... [writes: Suppose $\bar{x} \in {}^{R}\!/_{I}$ and $\bar{f}(\bar{x}) = 0 \in S$, $\bar{x} \in \ker(\bar{f})$] This is another way of saying $\bar{x}$ is in the kernel of $\bar{f}$. Suppose I have that.

Then what can I conclude? Well, I can use the definition: the definition of this is $f(x)$, right? So, then this implies that actually $f(x)$ is equal to zero in S, right? [writes: Then $f(x) = 0 \in S$] And this implies x is in the kernel of f. [writes: $\implies x \in \ker(f)$]

All right? So, what have we done here? We said: we start with any equivalence class; if it's in the kernel of $\bar{f}$, then that element has to be in the kernel of f. And that means that equivalence class is the zero equivalence class, because it's equivalent to zero. This means that x is equivalent using the kernel relation to zero. [writes: $\implies x \equiv_{\ker(f)} 0$]

And this means that actually, you know, x is equal to zero in R mod I. [writes: $\implies \bar{x} = \bar{0} \in {}^{R}\!/_{I}$]

So, that proves that the kernel of $\bar{f}$ is zero. So, the conclusion here — let me sorry... that's Step 3 — the conclusion is the kernel of $\bar{f}$ is zero.

Injectivity of $\bar{f}$ via Triviality of the Kernel

Claim 4.31 ($\bar{f}$ *is Injective*): The kernel of the homomorphism $\bar{f} \colon {}^{R}\!/_{\ker(f)} \to \mathrm{Im}(f)$ is the zero ideal of the quotient ring:

$$\ker(\bar{f}) = (\bar{0}). \tag{4.26}$$

Consequently, $\bar{f}$ is injective.

Proof

Let $\bar{x} \in \ker(\bar{f}) \subseteq {}^{R}\!/_{\ker(f)}$. By definition of the kernel:

$$\bar{f}(\bar{x}) = 0_S.$$

Using the definition of the map $\bar{f}(\bar{x}) := f(x)$, this means:

$$f(x) = 0_S \implies x \in \ker(f).$$

By definition of classes modulo $\ker(f)$, an element belongs to the ideal if and only if its class coincides with the zero class:

$$x \in \ker(f) \iff x \equiv_{\ker(f)} 0 \iff \bar{x} = \bar{0} \in {}^{R}\!/_{\ker(f)}.$$

Hence, the only element in $\ker(\bar{f})$ is the zero class $\bar{0}$, establishing:

$$\ker(\bar{f}) = (\bar{0}).$$

By the injectivity criterion established earlier in Proposition 4.16, a ring homomorphism is injective if and only if its kernel is trivial. Therefore, $\bar{f}$ is injective. **Q.E.D.**

Conclusion of the Isomorphism: Combining the three steps:

Step 1: Well-Defined Homomorphism: $\bar{f}\colon {}^{R}\!/_{\ker(f)} \to \mathrm{Im}(f)$ is a well-defined ring homomorphism.

Step 2: Surjectivity: $\bar{f}$ is surjective onto $\mathrm{Im}(f)$.

Step 3: Injectivity: $\bar{f}$ is injective, having trivial kernel.

A bijective ring homomorphism is an isomorphism (Definition 2.23). Thus, ${}^{R}\!/_{\ker(f)} \cong \mathrm{Im}(f)$, which completes the proof of the First Isomorphism Theorem (Theorem 4.27).

Q.E.D.

Recapitulation and the Anatomy of the Theorem

Spoken Transcript [01:06:43 - 01:07:37] Prof. Rahul Pandharipande

So, those are the three steps. You have to define $\bar{f}$, you show it's surjective onto the image, and you show its kernel is zero. And the reason why the last one is confusing is: you have to understand who is zero where. The kernel of f is not zero, the kernel of f is the kernel of f. But we quotient by the kernel of f, so, the kernel of $\bar{f}$ is zero.

So, that's an admittedly confusing state of affairs, but you have to somehow come to terms with that.

Spoken Transcript [01:07:47 - 01:09:13] Prof. Rahul Pandharipande

So, I say it in words because it's confusing. We start with f from R to S. This has a kernel, which we call the kernel of f, [writes "$f\colon R \to S$" and "This has kernel $\ker(f)$"] which could be anything — maybe zero, maybe not, could be anything in the world.

Then from this, we make $\bar{f}$ from R mod $\ker(f)$ to S — well, it goes to the image of

f, that subring inside S. [writes: $\bar{f}\colon R/\ker(f) \to \mathrm{Im}(f) \subset S$] So, we study this, and the last step is that the kernel of $\bar{f}$...

This is an ideal in R mod $\ker(f)$, right? [writes: $\ker(\bar{f}) \subset R/\ker(f)$] Because it's an ideal here in this ring. And this ideal is the zero ideal — the zero ideal in this quotient ring. [writes: $\ker(\bar{f}) = (0) \subset R/\ker(f)$]

So, the kernel of f may not be zero, but when I consider the kernel of $\bar{f}$ in the quotient ring, it's zero. And the way to think about that is kind of — and we'll discuss this more tomorrow — is that you can write this kernel of $\bar{f}$ as kind of whatever was in the kernel before, but now we quotient out by f, and then that's just the zero group. [writes: $\ker(\bar{f}) = \ker(f)/\ker(f) = 0$]

That's a nice way to think about it. I'll explain tomorrow how this kind of logic makes sense. But roughly speaking: there was a kernel, but when we quotient out, we quotient out by that kernel, so, after we quotient out, we have no kernel. Okay?

Spoken Transcript [01:09:13 - 01:10:35]

Prof. Rahul Pandharipande

So, these three steps, if you take Steps 1 to 3 together — because first we defined $\bar{f}$, we show it's surjective onto the image of f, and now we show the kernel is zero, so, that means it's also injective. I remind you: the kernel being zero implies that $\bar{f}$ is injective. [writes: $\implies \bar{f}$ is injective] We started the lecture with that, I think (Recall: Proposition 4.16).

So, Steps 1 to 3 show that $\bar{f}$ from R mod $\ker(f)$ to the image of f is well-defined, injective, and surjective, so, $\bar{f}$ is an isomorphism of rings. [writes: STEP 1–3: $\bar{f}\colon R/\ker(f) \to \mathrm{Im}(f)$, $\bar{f}$ is isom.]

So, that's the end of the proof of the First Isomorphism Theorem. Okay.

There was a lot of chalk on the board, but if you really understand this, you'll see there's actually nothing happening in this proof. It's just somehow you have to follow the definitions at every stage. I think it's similar to when you really understand that linear algebra result: the fact that the image is the space mod the kernel, there's almost nothing happening in that proof. You almost... the definitions make it so, and that's what happens here also.

Summary: Anatomy of the First Isomorphism Theorem

The architecture of the First Isomorphism Theorem relies on distinguishing the three rings and their respective kernels:

Mapping	Domain / Codomain	Kernel
f	$R \longrightarrow S$	$\ker(f) \subset R$ (Arbitrary ideal of R)
$\bar{f}$	$R/\ker(f) \xrightarrow{\cong} \mathrm{Im}(f)$	$\ker(\bar{f}) = (\bar{0}) \subset R/\ker(f)$ (The zero ideal in $R/\ker(f)$)

$$f\colon R \longrightarrow S \qquad \text{with kernel } \ker(f) \subset R, \tag{4.27}$$

$$\bar{f}\colon R/\ker(f) \xrightarrow{\sim} \mathrm{Im}(f) \subset S \qquad \text{with kernel } \ker(\bar{f}) = (0) \subset R/\ker(f). \tag{4.28}$$

Heuristically, the kernel of f has been quotiented out by itself (the next lecture makes this precise):

$$\ker(\bar{f}) = {}^{\ker(f)}\!/\!_{\ker(f)} = 0.$$

Ideals Generated by Elements

Spoken Transcript [01:10:37 - 01:12:08] Prof. Rahul Pandharipande

So, let's do an example.

Maybe I have some room here, I can say: one of the things to think about rings and their ideals and quotient rings is that we need to have some way of writing down ideals, otherwise we can't have too much fun with examples.

A standard way of writing an ideal is: let x_1 to x_n be elements of the ring R, [writes on the board: Let $x_1, \ldots, x_n \in R$] then... they could also be infinite, but in this class typically it's finite, but anyway...

Then if you have some list of elements in R, the standard notation is you put parentheses around them, and you define an ideal. This is the definition of this: it is the ideal defined by this equation, [writes: then $(x_1, \ldots, x_n) \subset R$ is the ideal defined by] so, I have to just tell you who is in this ideal: this is equal to those elements that are of the form a_1x_1 ... in language you could say this is R-linear combinations of these elements: a_nx_n, where the a_i's are in R. [writes: $(x_1, \ldots, x_n) = \{a_1x_1 + \cdots + a_nx_n \mid a_i \in R\}$] So, this is a particular set.

Spoken Transcript [01:12:08 - 01:13:39] Prof. Rahul Pandharipande

As I said, you could say in language: they're R-linear combinations of the x_i. [writes: R-linear combs. of the x_i] Other language is that this is the ideal generated by the x_i. That's also language used: ideal generated by the x_i. [writes: ideal generated by the x_i]

Those are two different ways to express the same thing. And it's just this set. So, in order for you to accept that this is an ideal, you have to show that... well, zero's here — of course zero's here if I take all these elements to be zero; you have to show if something is here, then its minus is here — I just flip all the signs of all the a_i's; you have to show if something's here, when I add it to something else, it's still here — and then I just add the coefficients of the linear combination; and finally, I have to show the multiplicative property: if something's here and I multiply it by an r, then it's still there — but view that r as a scalar, and it just multiplies all these coefficients. Okay?

So, you have to check it. Actually, I just checked it for you, but, you know.

So, this is a very nice way: if you take a ring, you can just pick any elements, and you get an ideal. It's the ideal generated by that. Are there questions about this? This is kind of a really fundamental way of generating ideals.

Definition of Ideals Generated by Elements

Let R be a commutative ring with identity 1.

Definition 4.32 (*Ideal Generated by Finitely Many Elements*)**:** Let $x_1, \dots, x_n \in R$. The *"ideal generated by"* $x_1, \dots, x_n$, denoted by $(x_1, \dots, x_n)$, is the set of all R-linear combinations of the generators:

$$(x_1, \dots, x_n) := \left\{ \sum_{i=1}^{n} a_i x_i \;\middle|\; a_1, \dots, a_n \in R \right\}. \tag{4.29}$$

When $n = 1$, the ideal $(x) = \{ax \mid a \in R\} = xR$ is called a *"principal ideal"* (for $R = \mathbb{Z}$, these are the ideals $n\mathbb{Z}$ of Definition 2.33).

Verification of the Ideal Axioms for $(x_1, \dots, x_n)$

Proposition 4.33 (*$(x_1, \dots, x_n)$ is an Ideal*)**:** For any elements $x_1, \dots, x_n \in R$, the set $(x_1, \dots, x_n)$ defined in Definition 4.32 is indeed an ideal of R.

Proof

We check the two defining conditions of an ideal:

(1) Additive Subgroup:

- Setting $a_1 = \cdots = a_n = 0$ yields $0 = \sum_{i=1}^{n} 0 \cdot x_i \in (x_1, \dots, x_n)$.
- For any element $u = \sum_{i=1}^{n} a_i x_i$, its additive inverse is $-u = \sum_{i=1}^{n} (-a_i) x_i \in (x_1, \dots, x_n)$.
- If $u = \sum_{i=1}^{n} a_i x_i$ and $v = \sum_{i=1}^{n} b_i x_i$, their sum is:

$$u + v = \sum_{i=1}^{n} (a_i + b_i) x_i \in (x_1, \dots, x_n).$$

(2) Multiplicative Absorption: For any ring element $r \in R$ and $u = \sum_{i=1}^{n} a_i x_i \in (x_1, \dots, x_n)$:

$$r \cdot u = r \sum_{i=1}^{n} a_i x_i = \sum_{i=1}^{n} (r a_i) x_i.$$

Since R is a ring, each coefficient $r a_i \in R$, so $r \cdot u \in (x_1, \dots, x_n)$.

Q.E.D.

Beyond the Lecture: The Smallest Ideal Containing the x_i

The name *"ideal generated by the x_i"* reflects a minimality property: $(x_1, \dots, x_n)$ is the smallest ideal of R containing $x_1, \dots, x_n$. It contains each $x_i = 1 \cdot x_i$. Conversely, an ideal J containing $x_1, \dots, x_n$ contains every $a_i x_i$ by absorption, hence every sum $\sum_{i=1}^{n} a_i x_i$ by additive closure, so $(x_1, \dots, x_n) \subseteq J$.

Quotients of Univariate Polynomial Rings

Spoken Transcript [01:13:39 - 01:14:41] Prof. Rahul Pandharipande

I have a few minutes, let's do some examples.

So, now you have a very flexible way of generating really a lot of rings. And these rings already — the rings that you have the power to generate — have all sorts of interesting, different behavior. And part of the course is exploring some of that structure, by far not all of it.

Spoken Transcript [01:14:41 - 01:16:10] Prof. Rahul Pandharipande

Okay, so, for example, I can take my ring R to be polynomials in one variable: [writes: $R = \mathbb{C}[x]$] $R = \mathbb{C}[x]$. You should be totally happy with this ring: it's polynomials with complex coefficients, all right?

And I can take an element here, f, to be like x^2+1. [writes: $f = x^2+1 \in R$] That's a polynomial. And then, I'm using the notation there: if you have one element, that's an element; if I put parentheses around it, this is the ideal. [writes "$(f) \subset R$" with an arrow to "ideal"]

Very concretely, using that definition, this is just the ideal of f times a polynomial where the polynomial is in the ring. [writes: $(f) = \{f \cdot p \mid p \in R\}$] That's what it is: it's just one linear combination. So, all this is is polynomials that have an x^2+1 factor. Okay?

What we've done today is that we have a new ring now: $R \bmod$ this ideal. [writes: $R/(f)$] $R \bmod$ this ideal, this is the quotient ring. Are you happy to think about that? We take this polynomial ring, we take this ideal, and we have this quotient ring. I should put here $\mathbb{C}[x]$... [writes: $\mathbb{C}[x]/(f)$] Okay.

Do you accept that all the symbols on this board have been defined?

Spoken Transcript [01:16:10 - 01:16:41] Prof. Rahul Pandharipande

So, I can ask you now any question about this ring. For example: is this an integral domain? [writes: Is this an int. domain?]

What's your opinion about that?

Student Answer

No, because $(x-i)(x+i)$...

Spoken Transcript [01:16:43 - 01:17:31] Prof. Rahul Pandharipande

Yeah. So, the answer is no, [writes "No!" and circles it] and the reason is: I can take $x+i$ times $x-i$, because $x+i$ and $x-i$ are here, so, I can put bars, they're in the quotient ring. [writes: $\overline{x+i} \cdot \overline{x-i}$]

And the claim is that this is equal to, by definition, that, [writes: $= \overline{x^2+1}$] and then this is zero, [writes: $= \bar{0}$] right? This is the beginning of your argument. That's a totally solid start.

But to be an integral domain says that if these are zero... you know, what if one of these are zero? To be an integral... so, the answer is no, then you have to find two non-zero things that multiply to zero. You found two things that multiply to zero. Why are these non-zero?

Student Answer

That would be like their difference to zero... if you multiply x^2+1 by a polynomial, it must have degree at least 2.

Spoken Transcript [01:17:43 - 01:18:56] Prof. Rahul Pandharipande

Okay, that's the correct argument. So, now, this is the beginning to say it's no, then we have to show that $x+i$ is not equal to zero, and $x-i$ is not equal to zero. [writes: But $\overline{x+i} \neq \bar{0}$ and $\overline{x-i} \neq \bar{0}$]

If we have these three statements — that equals zero, the product was zero, and they're both not equal to zero — then it's not an integral domain. You agree with that?

And he gave the correct argument for that: because if $x+i$ were equivalent to zero, [writes: If $x+i \equiv 0$] this means that $x+i$ is in this ideal generated by x^2+1, [writes: $\iff$ $x+i \in (x^2+1)$] but the ideal generated by x^2+1 has x^2+1 times things in it, times polynomials, and you can never lower the degree to a linear polynomial by taking a quadratic polynomial and multiplying it by another polynomial.

So, that's the full argument, the full painful argument that says the answer is no. Okay.

Zero Divisors in the Quotient Ring $\mathbb{C}[x]/(x^2+1)$

Example 4.34 (*Non-Domain Quotient Ring*): Consider the polynomial ring $R = \mathbb{C}[x]$ and the principal ideal generated by $f(x) = x^2+1$:

$$I = (x^2+1) = \{(x^2+1)p(x) \mid p(x) \in \mathbb{C}[x]\}.$$

The quotient ring $\mathbb{C}[x]/(x^2+1)$ is **not an integral domain**.

Proof

In $\mathbb{C}[x]$, the polynomial x^2+1 factors into linear factors:

$$x^2+1 = (x+i)(x-i).$$

In the quotient ring $\mathbb{C}[x]/(x^2+1)$, consider the classes $\overline{x+i}$ and $\overline{x-i}$:

(1) Their Product is Zero:

$$\overline{x+i} \cdot \overline{x-i} = \overline{(x+i)(x-i)} = \overline{x^2+1} = \bar{0} \in \mathbb{C}[x]/(x^2+1).$$

(2) Neither Factor is Zero: Suppose $\overline{x+i} = \bar{0}$. Then $x+i \in (x^2+1)$, which means there exists $p(x) \in \mathbb{C}[x]$ such that:

$$x+i = (x^2+1)p(x).$$

If $p(x) = 0$, the right-hand side is $0 \neq x+i$. If $p(x) \neq 0$, comparing degrees gives:

$$1 = \deg(x+i) = \deg(x^2+1) + \deg(p) = 2 + \deg(p) \geq 2,$$

a contradiction. Thus, $\overline{x+i} \neq \bar{0}$. By identical degree considerations, $\overline{x-i} \neq \bar{0}$.

Since $\overline{x+i}$ and $\overline{x-i}$ are non-zero elements whose product is zero, they are zero divisors. Hence, $\mathbb{C}[x]\big/(x^2+1)$ is not an integral domain.

Q.E.D.

III The Quotient $\mathbb{Q}[x]\big/(x^2+1)$ and the Gaussian Rationals

Spoken Transcript [01:18:56 - 01:20:11] Prof. Rahul Pandharipande

What about this one? [writes: $\mathbb{Q}[x]\big/(x^2+1)$]

This is kind of cutting closer to what this class is about, actually. Is this an integral domain?

His argument you cannot use exactly here, because i is not a rational number, right? You agree you can't use the same argument. But I can ask the same question, though: what do you think the answer to that is?

This is really a question that the last part of the class is about. But I'm going to give you a proof of this. The answer is: this is an integral domain. [writes: is an int. dom.] Actually, more than that: [writes: Field] actually, it's a field!

It doesn't look like it immediately here, but I will try to give you a proof in the next few minutes about this.

There's a lot of ways to prove it, and I said a certain part of the class is about such things called field extensions. And if you take the successor in the spring, that's like entirely about this kind of stuff — at least Galois theory, etc. But it's a very good example to think about.

Spoken Transcript [01:20:11 - 01:20:56] Prof. Rahul Pandharipande

So far, the only thing you're supposed to understand is that the argument proposed for the complex numbers does not work here. But let's try to find an argument that does work, that's related to this lecture.

In some sense, that question is kind of like searching for the complex numbers. Because x^2+1, you know the roots, so, somehow built into that question is this quest for the complex numbers.

The Rational Quotient $\mathbb{Q}[x]\big/(x^2+1)$

Question 4.35: Is the quotient ring $\mathbb{Q}[x]\big/(x^2+1)$ an integral domain?

Answer 4.36: Yes! In fact, it is a field. Notice that the factorization $x^2+1 =$

$(x+i)(x-i)$ used over $\mathbb{C}$ is unavailable in $\mathbb{Q}[x]$, because $i \notin \mathbb{Q}$. We will prove that $\mathbb{Q}[x]\big/(x^2+1)$ is isomorphic to the field of Gaussian rationals $\mathbb{Q}[i]$.

Beyond the Lecture: x^2+1 Has No Factorization over $\mathbb{Q}$

The polynomial x^2+1 is not a product of two non-constant polynomials in $\mathbb{Q}[x]$: such a factorization would consist of two linear factors and give a rational root of x^2+1. In the language of Week 6, it is irreducible in $\mathbb{Q}[x]$ (Definition 6.10).

Evaluation at i

Spoken Transcript [01:20:56 - 01:22:36] Prof. Rahul Pandharipande

We know the complex numbers, so, we write it down. [writes: $\mathbb{C}$] And I take $\mathbb{Q}$... [writes: $\mathbb{Q} \to \mathbb{C}$] $\mathbb{Q}$ embeds in the complex numbers. There is no doubt about that. And now I look at the polynomial ring. [writes "$\cap$" and "$\mathbb{Q}[x]$" under $\mathbb{Q}$]

And we proved that if I can choose any value of x, I get a homomorphism from the polynomial ring (Recall: Theorem 3.38). Do you remember that?

So, I'm going to choose a homomorphism f that takes... for constants, it just commutes, but f on x is i. [draws an arrow f from $\mathbb{Q}[x]$ to $\mathbb{C}$ and writes: $f(x)=i$] I just do that. I get to do it. Because I told you: for the polynomial ring, if I know how the original ring maps to any other ring, I can always extend the map to anything I want for x. So, I get to just do this. It's somehow my right from the previous lectures.

In other words, I define a homomorphism f from polynomials to the complex numbers, [writes: $f\colon \mathbb{Q}[x] \to \mathbb{C}$] where f of a polynomial equals the polynomial evaluated at i. [writes: $f(p(x)) = p(i)$] Okay?

Do you accept this?

And you can, even if you forgot that, you can just check that this is a legal definition: given any polynomial, I just evaluate at i, get a complex number, and it's a ring homomorphism. [writes: Ring hom.]

So, as I said, you can just prove that from scratch in your head here, because I gave you the formula; or you can remember it has to be true because I said you could always do that — I can just take this x going to anywhere. Okay.

Evaluation at i

Let $\iota\colon \mathbb{Q} \hookrightarrow \mathbb{C}$ denote the canonical inclusion of fields. By the universal property of polynomial rings (Theorem 3.38), specifying the image of the indeterminate x uniquely determines a ring homomorphism extending ι.

$$\begin{tikzcd}
\mathbb{Q} \arrow[r, hook, "\iota"] \arrow[d, hook, "\text{inc}"'] & \mathbb{C} \\
\mathbb{Q}[x] \arrow[ur, "f\colon p(x) \mapsto p(i)"']
\end{tikzcd}$$

Definition 4.37 (*Evaluation at i*): We define the homomorphism given by evaluation at the imaginary unit $i \in \mathbb{C}$:

$$f\colon \mathbb{Q}[x] \longrightarrow \mathbb{C}, \qquad f(p(x)) := p(i). \tag{4.30}$$

Explicitly, if $p(x) = \sum_{k=0}^{m} a_k x^k \in \mathbb{Q}[x]$, then $f(p(x)) = \sum_{k=0}^{m} a_k i^k \in \mathbb{C}$.

Homomorphism Property Verification: For any $p(x), q(x) \in \mathbb{Q}[x]$:

$$\begin{aligned}
f(p(x)+q(x)) &= (p+q)(i) = p(i)+q(i) = f(p(x)) + f(q(x)),\\
f(p(x)\cdot q(x)) &= (p\cdot q)(i) = p(i)\cdot q(i) = f(p(x))\cdot f(q(x)),\\
f(1) &= 1.
\end{aligned}$$

The middle equalities, such as $(p\cdot q)(i) = p(i)\cdot q(i)$, are not automatic: they hold because f is the homomorphism $\hat{\iota}$ of Theorem 3.38 for $\phi = \iota$ and $y = i$, whose proof checks them for arbitrary polynomials. Thus, f is a well-defined homomorphism of commutative rings with identity.

The Image of Evaluation at i and the First Isomorphism Theorem

Spoken Transcript [01:22:36 - 01:23:27] Prof. Rahul Pandharipande

What is the image of this? [writes: What is $\text{Im}(f) = ?$]

What is the image, do you know what that is? Yeah?

Student Answer

I mean, if you consider polynomials of degree 1 with the constants, you get the Gaussian integers?

Spoken Transcript [01:23:27 - 01:24:50] Prof. Rahul Pandharipande

Yeah, and you don't get any more than that, because if you take a polynomial, all the quadratic terms give you -1, and all you get is i. So, the answer is: the image of this is what we'd call the Gaussian integers (i.e., Gaussian rationals $\mathbb{Q}[i]$)... [writes: $\text{Im}(f) = \mathbb{Q}[i]$]

If you take a polynomial with rational coefficients and you evaluate it at i, all the even terms just give you rational numbers, and all the odd terms give you i's, and the coefficients are all rational numbers. And you can get anyone you want here.

So, that's kind of interesting. That says that... so, now we'll use our First Isomorphism Theorem! [writes: First Isom. Thm]

That says that $\mathbb{Q}[x]$ mod the kernel of f is isomorphic to the image of f, which is the Gaussian integers (i.e., Gaussian rationals $\mathbb{Q}[i]$), which, by the way, is a field! [writes: $\mathbb{Q}[x]/\ker(f) \cong \mathrm{Im}(f) = \mathbb{Q}[i]$] It's a domain, and it's a field.

So, now it would be nice to know what this kernel is. What is the kernel? [writes: What is the $\ker(f)$?]

Image of f and Application of the First Isomorphism Theorem

Proposition 4.38 (*Image of the Evaluation Map*): The image of the homomorphism $f\colon \mathbb{Q}[x] \to \mathbb{C}$ is the field of Gaussian rationals (Example 3.28):

$$\mathrm{Im}(f) = \mathbb{Q}[i] := \{a + bi \mid a, b \in \mathbb{Q}\} \subset \mathbb{C}. \tag{4.31}$$

Proof

Let $p(x) = a_n x^n + \cdots + a_1 x + a_0 \in \mathbb{Q}[x]$. Since $i^2 = -1$, every even power $i^{2k} = (-1)^k \in \mathbb{Q}$ and every odd power $i^{2k+1} = (-1)^k i$. Collecting the even and odd terms yields:

$$p(i) = \left(\sum_{k \geq 0} a_{2k} (-1)^k \right) + \left(\sum_{k \geq 0} a_{2k+1} (-1)^k \right) i = a + bi,$$

where $a, b \in \mathbb{Q}$. Conversely, for any $a + bi \in \mathbb{Q}[i]$, the linear polynomial $p(x) = bx + a \in \mathbb{Q}[x]$ satisfies $f(p(x)) = a + bi$. Thus, $\mathrm{Im}(f) = \mathbb{Q}[i]$. **Q.E.D.**

Theorem 4.39 (*First Isomorphism Theorem Application*): Applying the First Isomorphism Theorem (Theorem 4.27) to $f\colon \mathbb{Q}[x] \to \mathbb{C}$:

$$\mathbb{Q}[x]/\ker(f) \cong \mathrm{Im}(f) = \mathbb{Q}[i]. \tag{4.32}$$

Since $\mathbb{Q}[i]$ is a subfield of $\mathbb{C}$ (Definition 0.26), the quotient $\mathbb{Q}[x]/\ker(f)$ is an integral domain and a field.

Terminology Note: Gaussian Rationals vs. Gaussian Integers

In the lecture audio, the professor refers to $\mathbb{Q}[i] = \{a + bi \mid a, b \in \mathbb{Q}\}$ as the *"Gaussian integers"*. In standard algebraic terminology, the Gaussian integers are $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ (Definition 1.1), while $\mathbb{Q}[i] = \mathrm{Frac}(\mathbb{Z}[i])$ is the field of *"Gaussian rationals"* (Example 3.28), often written $\mathbb{Q}(i)$. The blackboard notation $\mathbb{Q}[i]$ is preserved verbatim.

Computing the Kernel via Polynomial Long Division

Spoken Transcript [01:24:50 - 01:26:25] Prof. Rahul Pandharipande

Well, one thing that's definitely in the kernel is this polynomial: $x^2 + 1$ is definitely in the kernel. [writes: $x^2 + 1 \in \ker(f)$] Because if I take this polynomial — that's the P — if I take f... I write it here, it's closer to the definition: $f(x^2 + 1)$, by definition this is equal to $i^2 + 1$, which is zero, [writes: $f(x^2 + 1) = i^2 + 1 = 0$] right?

So, whatever the kernel is, it definitely has this polynomial. Okay?

Then we'd like to see whether there's anything else in the kernel. Once this is in the kernel, then the whole ideal generated by this polynomial is also in the kernel. [writes: $(x^2+1) \subset \ker(f)$] Because if I take this polynomial and multiply it by something else, it'll still be in the kernel.

So, the question now is: is there anything else? The question here is: is the ideal generated by this polynomial equal to the kernel of f? [writes: Is $(x^2+1) = \ker(f)$?]

Okay? If the answer is yes, then I've answered that question! Because here it is, it's $\mathbb{Q}[x]$ mod ... and spoiler: it's going to be yes. If it were yes, then I will have answered this question, because I've computed the kernel here, and I've showed it's isomorphic to the ring of Gaussian integers (i.e., Gaussian rationals $\mathbb{Q}[i]$).

Actually, that is a way of writing the ring of Gaussian integers (i.e., Gaussian rationals $\mathbb{Q}[i]$) in slightly hidden form. All I have to do is prove this thing is the kernel.

Containment of the Principal Ideal (x^2+1) in the Kernel

Lemma 4.40 (*Ideal Containment in the Kernel*): The principal ideal (x^2+1) is contained in $\ker(f)$:

$$(x^2+1) \subseteq \ker(f). \tag{4.33}$$

Proof

Evaluating f on the generator x^2+1:

$$f(x^2+1) = i^2+1 = -1+1 = 0 \implies x^2+1 \in \ker(f).$$

Since $\ker(f)$ is an ideal of $\mathbb{Q}[x]$ (Proposition 4.13), it must absorb multiplication by any polynomial $q(x) \in \mathbb{Q}[x]$:

$$\forall q(x) \in \mathbb{Q}[x] \colon q(x)(x^2+1) \in \ker(f).$$

Hence, $(x^2+1) \subseteq \ker(f)$. **Q.E.D.**

Spoken Transcript [01:26:25 - 01:28:30] Prof. Rahul Pandharipande

Let us try to think really what we have to do to prove it's the kernel. You have to actually understand the definitions of the class.

We have f from $\mathbb{Q}[x]$ to $\mathbb{C}$, [writes: $f\colon \mathbb{Q}[x] \to \mathbb{C}$] and f eats polynomials — that's an element here — and it gives back a complex number, which is $p(i)$. [writes: $f(p(x)) = p(i)$]

And the kernel of f: a polynomial is in the kernel of f if and only if $p(i)$ is zero. [writes: $\ker(f) \ni p(x) \iff p(i) = 0$] In real life, you say i is a root of p, [writes "i is root of p"] right? So, the ones in the kernel are those polynomials where i is a root. Okay?

So, we have to see how that's related to this x^2+1.

There's different ways to say this, but we could do the following thing: we try to divide $p(x)$ by x^2+1, [writes: $x^2+1 \mid p(x)$] right? This is some polynomial, and we try to divide it, and everything's happening in this algebra of polynomials $\mathbb{Q}[x]$. [writes: $\mathbb{Q}[x]$]

This is what's called polynomial long division, which you should have learned in school: as you have a polynomial, you can start dividing it. And the answer is that this will be equal to some polynomial $q(x)$ times x^2+1 plus a remainder term. [writes: $p(x) = q(x) \cdot (x^2+1) + r(x)$]

Do you remember polynomial long division? You just keep taking out the highest powers all the way down. I don't know, did you learn this stuff?

If you do polynomial long division, you start dividing, and you'll eventually get some polynomial times this, plus a remainder term. And this remainder term, since this is a quadratic polynomial, this remainder term is a linear polynomial! [writes: linear poly.] So, this remainder term is some $r_1x + r_2$. [writes: $r_1x + r_2$] Of course, these could be zero. All right.

Polynomial Division by $x^2 + 1$

Lemma 4.41 (*Division Algorithm in* $\mathbb{Q}[x]$)**:** For any polynomial $p(x) \in \mathbb{Q}[x]$, there exist unique polynomials $q(x), r(x) \in \mathbb{Q}[x]$ such that:

$$p(x) = q(x)(x^2 + 1) + r(x), \quad \text{with } r = 0 \text{ or } \deg(r) < 2. \tag{4.34}$$

In particular, the remainder $r(x)$ is a linear polynomial with rational coefficients:

$$r(x) = r_1x + r_2, \qquad r_1, r_2 \in \mathbb{Q}. \tag{4.35}$$

This is the case $K = \mathbb{Q}$, $g = x^2 + 1$ of division with remainder for polynomials (Fact 0.32); the lecture uses it without proof.

Spoken Transcript [01:28:30 - 01:29:15] Prof. Rahul Pandharipande

That's polynomial long division, that's something you learn in school, and you should remember it now.

Now, I just stick i in this equation: [writes: $p(i) = q(i)(i^2 + 1) + r(i)$] $p(i)$... I put i's everywhere: $q(i) \cdot (i^2 + 1) + r(i)$.

And now, the assumption is $p(i) = 0$, so, this side is zero. [writes "0" under $p(i)$] This term is also zero, because $i^2 + 1$ is zero. [crosses out $(i^2 + 1)$]

That means that the conclusion is $r(i)$ is zero! [writes: $\implies r(i) = 0$]

How could that be? r is a linear polynomial, the remainder is a linear polynomial; how could it be zero? There's only one way it could be zero. Which is what? Yeah?

Student Answer

Both coefficients must be zero.

Spoken Transcript [01:29:20 - 01:30:14] Prof. Rahul Pandharipande

Yeah, the only way, because i is not... if this were a non-zero polynomial, these are rational numbers, the only way when you stick in i this is zero is if this is the zero polynomial! Because i is not a rational number.

So, this implies that r is the zero polynomial, [writes: $\implies r = 0$] so, actually it's not here, and this p is in the ideal! [crosses out $r(x)$ in the equation]

And that finishes the argument, actually.

So, the conclusion is that $p(i) = 0$ implies $p(x)$ is in this ideal. [writes: $p(i) = 0 \implies$

$p(x) \in (x^2+1)$]

And the final conclusion, using the First Isomorphism Theorem, is this: $\mathbb{Q}[x]$ mod (x^2+1) is isomorphic to this ring of Gaussian integers (i.e., Gaussian rationals $\mathbb{Q}[i]$). [writes: $\mathbb{Q}[x]/(x^2+1) \cong \mathbb{Q}[i]$]

Okay. Made it in time. That's it. [puts down the chalk and concludes the lecture]

Proof that $\ker(f) = (x^2+1)$ and Final Isomorphism

Theorem 4.42 (*Equality of Kernel and Principal Ideal*)**:** The kernel of the homomorphism $f\colon \mathbb{Q}[x] \to \mathbb{C}$ is precisely the principal ideal generated by x^2+1:

$$\ker(f) = (x^2+1). \tag{4.36}$$

Proof

We established in Lemma 4.40 that $(x^2+1) \subseteq \ker(f)$. We now prove the reverse inclusion $\ker(f) \subseteq (x^2+1)$.

Let $p(x) \in \ker(f)$. By definition, $p(i) = 0$. By Lemma 4.41, divide $p(x)$ by x^2+1:

$$p(x) = q(x)(x^2+1) + r(x), \quad \text{where } r(x) = r_1x + r_2 \text{ with } r_1, r_2 \in \mathbb{Q}.$$

Evaluating both sides at $x = i$:

$$p(i) = q(i)(i^2+1) + r(i).$$

Since $p(i) = 0$ and $i^2+1 = -1+1 = 0$, we obtain:

$$0 = q(i)\cdot 0 + r(i) \implies r(i) = 0.$$

Substituting the linear form of $r(x)$:

$$r(i) = r_1 i + r_2 = 0 \implies r_2 + r_1 i = 0.$$

Since $r_1, r_2 \in \mathbb{Q}$ and $i \notin \mathbb{Q}$ (so 1 and i are linearly independent over $\mathbb{Q}$), this forces:

$$r_1 = 0 \quad \text{and} \quad r_2 = 0 \implies r(x) = 0.$$

Therefore, the remainder vanishes identically, and:

$$p(x) = q(x)(x^2+1) \in (x^2+1).$$

Thus, $\ker(f) \subseteq (x^2+1)$, which completes the proof that $\ker(f) = (x^2+1)$.

Q.E.D.

Corollary 4.43 (*Realization of the Gaussian Rationals as a Quotient Ring*)**:** Combining Theorem 4.39 and Theorem 4.42, there is a canonical isomorphism of fields:

$$\mathbb{Q}[x]/(x^2+1) \xrightarrow{\cong} \mathbb{Q}[i], \qquad \overline{p(x)} \mapsto p(i). \tag{4.37}$$

In particular, $\mathbb{Q}[x]/(x^2+1)$ is an integral domain and a field.

Summary of Key Concepts

- **Several Variables and the Binomial Formula:** $R[x,y]$ consists of the finite sums $\sum a_{i,j}x^iy^j$, or equivalently $R[x,y] = (R[x])[y]$. The binomial formula $(x+y)^n =$

$\sum_{a=0}^{n} x^{n-a}y^{a}\binom{n}{a}$ holds in every commutative ring; since $p \mid \binom{p}{a}$ for $0 < a < p$, it becomes $(x+y)^p = x^p + y^p$ over $\mathbb{Z}/p\mathbb{Z}$.

- **Polynomial and Power Series Rings:** The rings $R[x]$ and $R[[x]]$ preserve the integral domain property of R, relying on the highest term and the lowest term, respectively.
- **Laurent Series and Rational Functions:** The field of rational functions $R(x)$ is $\mathrm{Frac}(R[x])$, while over a field K the field of formal Laurent series $K((x))$ (finitely many negative powers) is naturally isomorphic to $\mathrm{Frac}(K[[x]])$.
- **Ideals and Well-Defined Quotients:** An ideal $I \subset R$ requires additive closure and multiplicative absorption ($R \cdot I \subseteq I$). In constructing the quotient ring R/I, the additive subgroup condition guarantees well-defined addition, while the absorption property is strictly required to ensure well-defined multiplication of classes.
- **The First Isomorphism Theorem for Rings:** Every ring homomorphism $f\colon R \to S$ induces a canonical ring isomorphism $\bar{f}\colon R/\ker(f) \xrightarrow{\cong} \mathrm{Im}(f)$ via $\bar{x} \mapsto f(x)$.
- **Generated Ideals and Polynomial Quotients:** Elements $x_1, \ldots, x_n$ generate the ideal $(x_1, \ldots, x_n)$ of their R-linear combinations. Modding out by x^2+1 over $\mathbb{C}$ produces zero divisors via $(x+i)(x-i) \equiv 0$, whereas over $\mathbb{Q}$, evaluation at i, the First Isomorphism Theorem and division with remainder by x^2+1 give $\mathbb{Q}[x]/(x^2+1) \cong \mathbb{Q}[i]$, the field of Gaussian rationals.

Wednesday, 7th October 2026

Quotient Rings, the Correspondence Theorem for Ideals, and Maximal Ideals

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Context and Overview

In this lecture, we explore the algebraic architecture of quotient rings and the structural behavior of ideals under ring homomorphisms:

- **Review of Quotient Rings and the Quotient Map:** We revisit the construction of R/I for an ideal $I \subset R$ and the quotient map $q \colon R \to R/I$ with $\ker(q) = I$. We also recall the First Isomorphism Theorem for rings.
- **Pullbacks vs. Pushforwards of Ideals:** We prove that the preimage (pullback) $f^{-1}(J)$ of an ideal $J \subset S$ under any ring homomorphism $f \colon R \to S$ is always an ideal of R, whereas images of ideals generally fail to be ideals.
- **The Correspondence Theorem for Ideals:** We prove the correspondence of ideals: a bijection between the ideals of the quotient ring R/K and the ideals of R containing K.
- **Maximal Ideals and Fields:** We introduce maximal proper ideals, examine concrete examples in $\mathbb{Z}$, and establish the characterization that a ring R is a field if and only if the zero ideal (0) is maximal.

Review: Ideals and the Quotient Ring Construction

Spoken Transcript [00:00:05 - 00:01:10] Prof. Rahul Pandharipande

Okay, so we start. And I remind you of yesterday — it wasn't a long time ago, so you probably haven't forgotten.

So, if we have an ideal inside a ring... Typically, when I write R, it's a ring; I is an ideal, so it's an ideal [writes "$I \subset R$", with an arrow from I to "ideal" and one from R to "ring"] in a ring.

Then I want to say there are three main things to take away from yesterday. The first [writes: (i) Construction of R/I] is the construction of the quotient R/I (Recall: Theorem 4.26) — and we went through this rather carefully, not fully; there were things to check that are left to you, but this is equal to, as a set, the set of equivalence classes of this relation $\equiv_I$ [writes: $=$ set of equiv. classes of $\equiv_I$]. And then we defined on top of that —

I don't review it here, but — we defined addition, multiplication; there's a zero element and a one element (Recall: Definitions 4.20, 4.21 and 4.23). [writes: $({}^{R}\!/_{I}, +, \cdot, \bar{0}, \bar{1})$]

Review: Construction of the Quotient Ring

Recap: The quotient ring was constructed in the previous lecture: congruence modulo I (Definition 4.17, Proposition 4.18), the quotient set (Definition 4.19), the operations and their well-definedness (Definition 4.21, Proposition 4.22, Definition 4.23, Proposition 4.24), and the resulting ring (Theorem 4.26). The lecturer reviews it as the first of three main points.

Definition 4.44 (*Quotient Ring*): Let R be a ring and $I \subset R$ an ideal. The quotient ring ${}^{R}\!/_{I}$ is defined as the set of equivalence classes under the congruence relation modulo I:

$$x \equiv_I y :\iff x - y \in I.$$

The equivalence class of an element $x \in R$ is the set:

$$\bar{x} := x + I = \{x + a \mid a \in I\}.$$

As a set:

$${}^{R}\!/_{I} := \{\bar{x} \mid x \in R\}.$$

Equipped with the operations:

$$\bar{x} + \bar{y} \overset{\text{def}}{=} \overline{x + y}, \tag{4.38}$$

$$\bar{x} \cdot \bar{y} \overset{\text{def}}{=} \overline{x \cdot y}, \tag{4.39}$$

additive identity $\bar{0} = 0 + I = I$, and multiplicative identity $\bar{1} = 1 + I$, the structure $({}^{R}\!/_{I}, +, \cdot, \bar{0}, \bar{1})$ is a ring.

The Quotient Map

Spoken Transcript [00:01:10 - 00:02:45] Prof. Rahul Pandharipande

So, then something I didn't say as explicitly, but will be the topic today, is that there is a map called the quotient map. We'll discuss this today. I didn't say this yesterday, but: [writes: (ii) $q\colon R \to {}^{R}\!/_{I}$] R to ${}^{R}\!/_{I}$, and this map takes an element here to its equivalence class: [writes: $q(x) = \bar{x}$] $q(x) = \bar{x}$.

Okay? So, it was implicit last time, but now it'll be explicit. So, after we've made this construction, there's a rule that if I take an element here, I can take it to its equivalence class. You accept that, right? It's the most basic way.

And the claim, which is implicit in the construction, but now I make explicit, is that this q — which is called the quotient map, that's why I called it q — is a ring homomorphism. [writes: Claim: q is ring hom.] These are both rings; this is a ring homomorphism. And why is that? That's exactly just by the understanding of the definition: how we define addition here and multiplication here.

So, to check this, we'd have to check something like: what is $q(x + y)$? [writes: $q(x + y) = \overline{x + y}$] This is, by this map, that. And what is $q(x) + q(y)$? That's, by this map, that [writes: $q(x)+q(y)=\bar{x}+\bar{y}$]. And why are they equal? Well, that's how we define addition here. [underlines the two terms] Okay? Is that clear?

And the same is for multiplication. This is in some sense a review of the definition: what is $q(xy)$? By this map, it's the equivalence class of xy. [writes: $q(xy)=\overline{xy}$] What's $q(x)q(y)$? [writes: $q(x)\cdot q(y)=\bar{x}\cdot\bar{y}$] It's that. And why are these equal? Because that's how we define multiplication in the quotient.

Definition 4.45 (*Quotient Map*): Let R be a ring and $I \subset R$ an ideal. The *"quotient map"* is defined by:

$$q\colon R \to {}^{R}\!/_{I}, \quad x \mapsto q(x) := \bar{x} = x + I. \tag{4.40}$$

Claim 4.46 (*Homomorphism Property of the Quotient Map*): The quotient map $q\colon R \to {}^{R}\!/_{I}$ is a ring homomorphism.

Proof

By definition of the algebraic operations in the quotient ring ${}^{R}\!/_{I}$:

$$q(x+y) = \overline{x+y} = \bar{x}+\bar{y} = q(x)+q(y), \tag{4.41}$$

$$q(x\cdot y) = \overline{x\cdot y} = \bar{x}\cdot\bar{y} = q(x)\cdot q(y). \tag{4.42}$$

Furthermore, $q(1) = \bar{1} = 1 + I$. Thus, q preserves addition, multiplication, and the multiplicative identity, so it is a ring homomorphism (Definition 1.18). **Q.E.D.**

Spoken Transcript [00:02:45 - 00:04:13] Prof. Rahul Pandharipande

So, it's another way to think about our construction: it is defined so that the map that takes x to its equivalence class is a ring homomorphism. That's another way to think about it.

What is that? Is that my phone? [microphone emits loud electronic static and whistling noise; lecturer checks pocket and audio pack] What aspect of me is at fault? [checks transmitter] Ah! I think it was just not twisted correctly... I'm not sure it was my fault, but... [fiddles with audio jack]

Anyway, so that's another way of thinking about what we did yesterday: is that we take R, we have the set of equivalence classes, and we want to define a ring structure on it for which this quotient — this map, x goes to its equivalence class — is a ring homomorphism. And we just define all the structure so it is. That's another way of thinking about what happened yesterday.

But anyway, the outcome is that we've constructed it, and this quotient map is a ring homomorphism. And just so we're somehow on the same page here, if it's a ring homomorphism...

Now it's turned it off... [audio receiver clicks off] Okay, it's turned it off. This is not good news.

Technical Difficulties: Microphone Interference

The wireless microphone emits feedback and electronic static. The lecturer pauses the mathematical exposition to troubleshoot the audio transmitter and cable connection.

Spoken Transcript [00:04:21 - 00:05:34] Prof. Rahul Pandharipande

Let's try this one. [untangles replacement microphone cable and clips it on] Okay, now I switch... Can you hear me?

We had this problem before, right? Could it be my phone? It can't be the phone... [checks phone in pocket] Does anyone have an idea? Now it's off. Maybe they were both on, that might have been the problem. All right.

How's that? We will see. So, maybe the solution was turning this one off, maybe.

Technical Difficulties Resolved

The lecturer replaces the faulty microphone transmitter with a secondary unit and confirms the audio is functioning properly before resuming the lecture.

Spoken Transcript [00:05:35 - 00:06:17] Prof. Rahul Pandharipande

Okay, so this is all review from yesterday. So, if we're on the same page, let's say this q, this ring homomorphism, is surjective [writes: q is surjective]. That's clear, because everything here is the equivalence class of somebody. Okay?

So, that's one of the properties of the quotient map: it's always surjective.

What is the kernel of q? [writes: $\ker(q) = ?$] If I have a ring homomorphism, it always has a kernel; the kernel is always an ideal (Recall: Proposition 1.30). So, when you answer this question, you have to give me an ideal in R. And what's the answer to this question?

Yeah?

Student Answer

I.

Spoken Transcript [00:06:17 - 00:07:05] Prof. Rahul Pandharipande

I. It kind of has to be I [writes: I] because, you know, there's no other ideal in the picture. Maybe you could say it could be zero or the full ring, but it's I. And the proof of this — you should really make sure that you understand it, because this is really about whether you understand the construction.

For this, we have to find those elements [writes: Find those $x \in R$ such that] — those x in R — such that $q(x)$, which equals $\bar{x}$, is equal to zero [writes: $q(x) = \bar{x} = \bar{0}$]; and zero is the equivalence class of I. So, this is the same thing as x being in I [writes: $\bar{x} = \bar{0} \iff x \in I$]. So, it's I. It's good to understand that.

Properties of the Quotient Map

Proposition 4.47 (*Surjectivity and Kernel of the Quotient Map*): Let R be a ring, $I \subset R$ an ideal, and $q\colon R \to {}^{R}\!/_{I}$ the quotient map. Then:

- **(i)** q is surjective.
- **(ii)** $\ker(q) = I$.

Proof

We verify each statement directly:

- **(i)** **Surjectivity:** Every element of ${}^{R}\!/_{I}$ is of the form $\bar{x} = x + I$ for some $x \in R$. By definition, $q(x) = \bar{x}$, so q maps R onto ${}^{R}\!/_{I}$.
- **(ii)** **Kernel:** By definition of the kernel of a ring homomorphism (Definition 1.26):
  $$\ker(q) = \{x \in R \mid q(x) = \bar{0}\}.$$
  Since the zero element in ${}^{R}\!/_{I}$ is the class $\bar{0} = 0 + I = I$, we have:
  $$q(x) = \bar{0} \iff \bar{x} = \bar{0} \iff x - 0 \in I \iff x \in I,$$
  where the middle equivalence is the definition of congruence modulo I (Definition 4.17). Thus, $\ker(q) = I$.

Q.E.D.

Review: The First Isomorphism Theorem for Rings

Spoken Transcript [00:07:05 - 00:08:04] Prof. Rahul Pandharipande

All right, so the third property is the one we discussed carefully last time (Recall: Theorem 4.27), which is that if I have a homomorphism [writes: (iii) $R \xrightarrow{f} S$, f is a hom.] — $R \xrightarrow{f} S$, where it's a homomorphism —, then we have this data, which is somehow analogous to the linear algebra situation: that this kernel here is an ideal [writes: $\ker(f) \subset R$ ideal], and the image of f is a subring [writes: $\mathrm{Im}(f) \subset S$ subring], such that the image is canonically — and that's a complicated word in mathematics, *"canonically"*, but anyway, I just wrote it down explicitly — this image is isomorphic to R mod $\ker(f)$ [writes: $\mathrm{Im}(f) \cong {}^{R}\!/_{\ker(f)}$].

So, that's like the three basic properties of the quotient construction, and you should really make sure that you understand them. In some sense, that's the main thing this week.

Recap: The First Isomorphism Theorem was stated and proved in the previous lecture (Theorem 4.27); parts (i) and (ii) are Proposition 4.13 and Proposition 2.27. The lecturer recalls it as the third main point.

Theorem 4.48 (*First Isomorphism Theorem for Rings*): Let $f\colon R \to S$ be a homomorphism of rings. Then:

- **(i)** $\ker(f)$ is an ideal of R.

(ii) $\mathrm{Im}(f)$ is a subring of S.

(iii) There is a canonical ring isomorphism $\mathrm{Im}(f) \cong {}^{R}\!/_{\ker(f)}$, given by:

$$\bar{f}\colon {}^{R}\!/_{\ker(f)} \xrightarrow{\sim} \mathrm{Im}(f), \quad \bar{x} \mapsto f(x). \tag{4.43}$$

Pullbacks of Ideals under Ring Homomorphisms

Spoken Transcript [00:08:04 - 00:10:25] Prof. Rahul Pandharipande

So, we'll explore **(ii)** (i.e., the quotient map $q\colon R \to {}^{R}\!/_{I}$, Definition 4.45) a bit more today. The first statement is: let f from R to S be any homomorphism of rings [writes: Let $f\colon R \to S$ be any hom. of rings], and then let J in S be an ideal [writes: Let $J \subset S$ be an ideal].

All right, so what we have here in the picture is that inside here is J, which is an ideal. Then the claim, so the proposition — and this is kind of an important proposition — is: if I take the inverse image of J [writes: Proposition: $f^{-1}(J) \subset R$ is an ideal]... Are you happy with this set-theoretic terminology? This is just a map: if I take a set here, I can take its inverse image. The inverse image lives here. This is inside R. That's clear — that would not be a proposition, that's just, you know, by definition... It is an ideal.

So, the statement here is: if I have a homomorphism and if I have an ideal here, I can always bring it back here, and it'll be an ideal.

I am not saying the other way around. If I have an ideal here, I can also take the image of that ideal. I'm not saying that's an ideal, and it may not be an ideal. An example, for example: if you take $\mathbb{Z}$ goes to $\mathbb{Q}$ [writes: $\mathbb{Z} \hookrightarrow \mathbb{Q}$], and I take the ideal $2\mathbb{Z}$ [writes: $2\mathbb{Z}$], that's an ideal here. If I view it here — this is an injection, this is still $2\mathbb{Z}$, so to speak [writes: $2\mathbb{Z}$] —, that's not an ideal here. Is that clear?

Because if it were an ideal here, it would have to be closed under multiplication by any element of $\mathbb{Q}$; in particular, it would have to be closed under multiplication by a half. And even numbers are not closed under multiplication by a half.

So, there's kind of a difference: if you have an ideal here, you can bring it back; if you have an ideal here, you're not allowed, without any conditions, to get an ideal there as an image. All right? But anyway, we're not proving the thing that's wrong; we're proving the thing that's correct.

Preimage of an Ideal under a Homomorphism

Proposition 4.49 (*Preimage of an Ideal is an Ideal*): Let $f\colon R \to S$ be a homomorphism of rings, and let $J \subset S$ be an ideal. Then the preimage (or pullback) of J under f:

$$f^{-1}(J) := \{x \in R \mid f(x) \in J\} \tag{4.44}$$

is an ideal of R.

Example 4.50 (*Direct Images of Ideals Need Not Be Ideals*): Consider the canonical inclusion homomorphism:

$$\iota\colon \mathbb{Z} \hookrightarrow \mathbb{Q}, \quad n \mapsto n.$$

In the domain $\mathbb{Z}$, the set of even integers $2\mathbb{Z}$ is an ideal (Example 2.28). However, its direct image $\iota(2\mathbb{Z}) = 2\mathbb{Z}$ is **not** an ideal in $\mathbb{Q}$.

☰ **Why the Direct Image Fails the Ideal Property:** For $2\mathbb{Z}$ to be an ideal of the field $\mathbb{Q}$, it would have to absorb multiplication by all rational scalars:

$$\forall q \in \mathbb{Q},\ \forall x \in 2\mathbb{Z} \colon q \cdot x \in 2\mathbb{Z}.$$

However, taking $q = \frac{1}{2} \in \mathbb{Q}$ and $x = 2 \in 2\mathbb{Z}$, we obtain:

$$q \cdot x = \frac{1}{2} \cdot 2 = 1 \notin 2\mathbb{Z}.$$

Thus, the image of an ideal under a ring homomorphism need not be an ideal. Going beyond the lecture: under a surjective homomorphism it is an ideal, by the argument given below for the quotient map.

Spoken Transcript [00:10:33 - 00:12:45] Prof. Rahul Pandharipande

Okay, so let's try to prove this. Let x and y be in $f^{-1}(J)$, right? [writes: Proof: Let $x, y \in f^{-1}(J)$] This means that $f(x)$ is in J and $f(y)$ is in J [writes "$\Updownarrow$" and below it "$f(x) \in J, f(y) \in J$"]. That's what it means to be in $f^{-1}(J)$.

And to check that $f^{-1}(J)$ is an ideal, I have to show that if I pick any two elements in $f^{-1}(J)$, then certain properties are closed. There's a list of properties, and we'll start them.

So then, for example, if I take $x + y$ and I take f of it, that's equal to $f(x) + f(y)$ — that's using the homomorphism property of f, right? [writes: $f(x+y) = f(x) + f(y)$]

Now, both of these are in J, so that means — and then using the fact that J is an ideal — this is in J [writes: $\in J$]. So, that implies that $x + y$ is in $f^{-1}(J)$ [writes: $\implies x + y \in f^{-1}(J)$].

So, that's, you know, the beginning of checking that this set $f^{-1}(J)$ is an ideal. I've checked already for you that if you pick any two elements in it, their sum is in it, which is a good start.

Spoken Transcript [00:12:45 - 00:14:42] Prof. Rahul Pandharipande

You could also check whether zero is in it, but... let me do something a little fancier. So, I checked that it's closed under addition. Do you accept that?

Let us check also that: let r be any element of R [writes: Let $r \in R$], and let x be an element of $f^{-1}(J)$ [writes: $x \in f^{-1}(J)$]. Now, we have to consider $f(r \cdot x)$ [writes: $f(r \cdot x)$]. Using the homomorphism property of f, this is $f(r) \cdot f(x)$ [writes: $= f(r) \cdot f(x)$]. Now, $f(x)$ is in J, and using that J is an ideal, if I take someone in J and multiply it by any person, it stays in J. So, this stays in J [writes: $\in J$].

Okay? So, this shows that $r \cdot x$ is in $f^{-1}(J)$ [writes: $\implies r \cdot x \in f^{-1}(J)$].

So, we have these two properties of this set: if I take any two elements, it's closed under addition; and the other is, if I take any element and any element of R, it's closed under the product. And the claim is that that's actually all you have to check to be an ideal — only these two properties. The other ones follow from it.

For example, we'd have to check: if x is in $f^{-1}(J)$, we would have to check $-x$ is in

$f^{-1}(J)$ [writes: If $x \in f^{-1}(J) \implies -x \in f^{-1}(J)$]? But $-x$ is $(-1) \cdot x$. So, it's checked by this property. Is that okay? I mean, so we use here that $-x$ is $(-1) \cdot x$ [writes: $-x = (-1) \cdot x$], so we use the second property. That already gets negative.

Once you have x and negative, you can get zero. Zero's in it anyway. Yeah, and then you can add x and $-x$ and get zero. I think formally we should check zero's in it, because otherwise this whole thing could be empty. But anyway. So, let me say here: we also know that $f(0) = 0 \in J$, because zero goes to zero [writes "$0 \in f^{-1}(J)$" above the proof].

Okay? So, anyway, the precise statement is: you need to check zero's in it, these two properties, and then you can check the definition. You can also just check the definition from the beginning.

So, this is a complete argument that if I take J here and I consider $f^{-1}(J)$, this is an ideal if we started with an ideal there.

Proof of Proposition 4.49

Let $f\colon R \to S$ be a ring homomorphism and $J \subset S$ an ideal. To prove that $f^{-1}(J)$ is an ideal of R, we verify the ideal axioms (Definition 1.29):

1. Closure under Addition: Let $x, y \in f^{-1}(J)$. By definition of the preimage (Definition 0.5):

$$f(x) \in J \quad \text{and} \quad f(y) \in J.$$

Evaluating f on the sum $x + y$:

$$f(x+y) = f(x) + f(y).$$

Since J is an ideal, it is closed under addition, so $f(x) + f(y) \in J$. Therefore:

$$f(x+y) \in J \implies x + y \in f^{-1}(J).$$

2. Ideal Absorption (Multiplication by Ring Elements): Let $r \in R$ and $x \in f^{-1}(J)$. Then $f(x) \in J$. Using the multiplicativity of f:

$$f(r \cdot x) = f(r) \cdot f(x).$$

Because J is an ideal in S and $f(r) \in S$, the ideal absorbs multiplication:

$$f(r) \cdot f(x) \in J \implies r \cdot x \in f^{-1}(J).$$

3. Closure under Negatives: For any $x \in f^{-1}(J)$, we have $-x = (-1) \cdot x$ (Lemma 1.13). This is an element of R times an element of $f^{-1}(J)$, so Step 2 with $r = -1$ gives $-x \in f^{-1}(J)$.

4. The Zero Element: Since f is a ring homomorphism, $f(0) = 0$ (Proposition 1.19), and $0 \in J$ because J is an ideal. Thus, $0 \in f^{-1}(J)$; in particular, $f^{-1}(J)$ is not empty.

Hence, $f^{-1}(J)$ is an ideal of R. **Q.E.D.**

The Correspondence Theorem for Ideals

Spoken Transcript [00:14:42 - 00:16:01] Prof. Rahul Pandharipande

Okay, so then we're going to look at this in the case at hand, and then there's something that's kind of pretty that happens there. [erases the right and bottom boards]

So, you know, we have zero in it, it's closed under addition, and closed under minus, and closed under multiplication. I guess I checked them all, the properties there.

So, now we're going to study this pullback of ideals. Often, if we start with J, if you want to say what that $f^{-1}(J)$ is in English, you'd say it's the pullback of J under f — you know, the inverse image or the pullback, something like that.

So, we want to study this for the quotient map, and there we learn something that's kind of fundamental in the behavior of ideals in algebra in general. So, we start with some definitions.

Sets of Ideals and Notation

Spoken Transcript [00:16:01 - 00:18:40] Prof. Rahul Pandharipande

So, we consider now the situation of the quotient map $q\colon R \to R/I$ [writes: $q\colon R \to R/I$]. So, we start with some... I is an ideal in R, we construct the quotient, and we consider the quotient map. And I want to now start pulling back ideals here to here.

Since we have I's in there, let's just — to avoid the notation — call this one K, because we can use I and K [corrects notation on board: changes I to K in $q\colon R \to R/K, K \subset R$]. Sorry about that. So, we start with an ideal K, because I want to keep I and J for my ideals that I play with, rather than fix it.

So, we have an ideal K inside R, and I consider this quotient map. And I want to understand here — and this part of the quotient map is crucial to understand —: I want to understand [writes: I want to understand the pullback of ideals here] what this pullback of ideals looks like here in this situation.

For that, I need some definitions. **Definition 1** [writes: Definition I]: let's look at ideals here. So, let cl of this ring [writes: $\mathrm{cl}(R/K)$] be the set of ideals. So, this equals the set of ideals of this ring [writes: = Set of ideals of R/K]. This is a ring, it has some set of ideals; just let this be the set of ideals of that ring. It could be infinite, could be small, but anyway, it's a set of ideals.

And **Definition 2** [writes: Definition II] is: let $\mathrm{cl}_K(R)$ [writes: $\mathrm{cl}_K(R)$] be the set of ideals of R which contain K [writes: = Set of ideals of R which contain K].

So, these definitions are kind of important. Let me just say it, and I'll write them again in case you don't: $\mathrm{cl}(R/K)$ equals all J in R/K where J is an ideal [writes: $\mathrm{cl}(R/K) = \{J \subset R/K \mid J \text{ is an ideal}\}$]. This is just writing slightly more mathematically what I said there.

And this $\mathrm{cl}_K(R)$ is equal to the set of I, ideals in R, such that K is inside I and I is an ideal [writes: $\mathrm{cl}_K(R) = \{I \subset R \mid K \subset I, I \text{ ideal}\}$]. Okay? So, I just said it again, but the definitions are crucial.

Notation 4.51 (*Ideal Sets for the Quotient Setting*): Let R be a ring and $K \subset R$ an ideal. We consider the quotient map:

$$q\colon R \to {}^{R}\!/_{K}, \quad x \mapsto \bar{x} = x + K.$$

We define two collections of ideals:

(I) Ideals of the Quotient Ring:

$$\mathrm{cl}({}^{R}\!/_{K}) := \{J \subset {}^{R}\!/_{K} \mid J \text{ is an ideal of } {}^{R}\!/_{K}\}. \tag{4.45}$$

(II) Ideals Containing the Kernel:

$$\mathrm{cl}_K(R) := \{I \subset R \mid K \subset I \text{ and } I \text{ is an ideal of } R\}. \tag{4.46}$$

Statement of the Correspondence

Spoken Transcript [00:18:40 - 00:20:07] Prof. Rahul Pandharipande

So, is there any confusion about these definitions? I'm going to consider any ideal here, that's that set; and I consider here not any ideal, just ideals that contain the original K. Is that okay?

So, the proposition, which is kind of fundamental here, is the relationship. And the proposition is that there is a bijection [writes: There is a bijection], and this bijection is between, on the left side, $\mathrm{cl}_K(R)$ [writes: $\mathrm{cl}_K(R)$] — so those are ideals in R which contain K —; they're in bijective correspondence — this is the bijection [draws double arrow $\longleftrightarrow$ with label "bijection"] — with this $\mathrm{cl}({}^{R}\!/_{K})$ [writes: $\mathrm{cl}({}^{R}\!/_{K})$]. That's what we're going to prove.

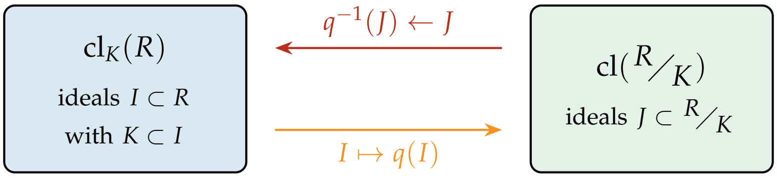
So, we're going to prove this statement.

This is some kind of very nice way of thinking about ideals. We start with R — maybe it's a ring you know, you're comfortable with —, you can make a quotient. And for this quotient ring, you know, it's maybe sometimes more complicated to think about, an interesting object; but if you want to answer the question: *"What do the ideals look like here?"*, this bijection tells you how to think about the ideals in this quotient ring: they're just the same as ideals before you quotient, but the ideals before you quotient have to contain K.

Proposition 4.52 (*Correspondence of Ideals (in textbooks: Correspondence Theorem)*): Let R be a ring and $K \subset R$ an ideal. There is a canonical bijection:

$$\mathrm{cl}_K(R) \longleftrightarrow \mathrm{cl}\left({}^{R}\!/_{K}\right). \tag{4.47}$$

That is, the ideals of the quotient ring ${}^{R}\!/_{K}$ are in exact one-to-one correspondence with the ideals of R that contain K.



Conceptual Meaning of the Bijection: This theorem reveals the internal structure of quotient rings:

- **Ideals in ${}^{R}\!/_{K}$ Are Pre-Existing Ideals in R:** One does not need to search for new, exotic ideals in the quotient ring ${}^{R}\!/_{K}$. Every ideal in ${}^{R}\!/_{K}$ arises directly from an ideal of R that happens to contain K.

Proof of the Correspondence Theorem for Ideals

Well-Definedness of the Correspondence Maps

Spoken Transcript [00:20:07 - 00:22:05] Prof. Rahul Pandharipande

All right, so on this bijection, to prove it, I have to give you maps. If I have J here [writes: $J \in \mathrm{cl}({}^{R}\!/_{K})$], in ${}^{R}\!/_{K}$, I have to take it to get an ideal here, right? [writes: $J \mapsto \dots$] And the map this way is going to be $f^{-1}(J)$ — this is an ideal here [writes: $q^{-1}(J) \in \mathrm{cl}_K(R)$] —, okay?

Let's first understand it. First, we said that if you take any ideal here, when you take f^{-1}, it is an ideal (Recall: Proposition 4.49). I'm saying more now: not only is it an ideal, but it also contains K. And why does it contain K? Because if I take zero here, the inverse image of zero is all of K.

So, this requires some justification that you have to think about. The claim here is that f^{-1} — it's not f here, it's q, sorry: [corrects f^{-1} to q^{-1} on board] q^{-1} for the quotient map —, $q^{-1}(J)$ has to contain this ideal K [writes: Claim: $q^{-1}(J) \supset K$].

And why is that? Well, we've already seen that $q^{-1}(0)$, that's the kernel, and that equals K (Recall: Proposition 4.47) [writes: $q^{-1}(0) = K$]. And this J contains zero [writes: $0 \in J$]. So, $q^{-1}(J)$ has to contain K.

Are you happy with this statement? It's kind of important, because, you know, it's saying that we're trying to prove this bijection between all ideals here and ideals that contain K. And the claim is that if I take any ideal here and I take this q^{-1} to get it to this side, it will always contain K, because this q^{-1} certainly contains all the things that go to zero, because zero is in here, and the things that go to zero are exactly K.

The Leftward Map: Pullback under the Quotient Map

Claim 4.53 (*Well-Definedness of the Pullback Map*): For any ideal $J \in \mathrm{cl}({}^{R}\!/_{K})$, its preimage under q:

$$q^{-1}(J) = \{x \in R \mid q(x) \in J\} \tag{4.48}$$

is an ideal of R that contains K, i.e., $q^{-1}(J) \in \mathrm{cl}_K(R)$.

Proof

By Proposition 4.49, the preimage of any ideal under a ring homomorphism is an ideal of the domain, so $q^{-1}(J)$ is an ideal of R. It remains to show that $q^{-1}(J) \supset K$: Since J is an ideal in ${}^{R}\!/_{K}$, it contains the additive identity $\bar{0} \in J$. By the definition of preimage and the kernel of q (Proposition 4.47):

$$K = \ker(q) = q^{-1}(\bar{0}) \subset q^{-1}(J).$$

Hence, $K \subset q^{-1}(J)$, which proves that $q^{-1}(J) \in \mathrm{cl}_K(R)$. **Q.E.D.**

Spoken Transcript [00:22:05 - 00:23:35]

Prof. Rahul Pandharipande

That's one direction. Now we have to do the other direction. If I have an ideal I here [writes: $I \in \mathrm{cl}_K(R)$] — I have an ideal here, I —, I have to be able to go this way and get an ideal of ${}^{R}\!/_{K}$ [writes: $I \mapsto \dots$].

And the natural thing to do is just to take q of this [writes: $q(I)$].

But I had just told you: in general, if you start with an ideal, you can't just take q of it and get an ideal. But in this case — in the case of a quotient — you can! So, this requires some justification also [writes: requires justification], which we will do in a second.

As I said, if you take an ideal in the domain of a ring homomorphism, you can't always just take the image and have it be an ideal. We did an example where that failed (Recall: $\mathbb{Z} \hookrightarrow \mathbb{Q}$ and $2\mathbb{Z}$, Example 4.50). But for the quotient map, it is true.

So, these are the two motions. I mean, it's a bijection, so you have to say how to go left and how to go right, and here they are explicitly. And so we have to prove some things about them being well-defined and why this is an ideal, etc.; and then I have to prove that the compositions are the same.

Anyway, that's the task at hand.

The Explicit Bijection Maps

To establish the correspondence $\mathrm{cl}_K(R) \longleftrightarrow \mathrm{cl}\left({}^{R}\!/_{K}\right)$, we define two mutually inverse maps:

$$\Phi\colon \mathrm{cl}\left({}^{R}\!/_{K}\right) \to \mathrm{cl}_K(R), \quad J \mapsto q^{-1}(J), \tag{4.49}$$

$$\Psi\colon \mathrm{cl}_K(R) \to \mathrm{cl}\left({}^{R}\!/_{K}\right), \quad I \mapsto q(I) = \{\bar{x} \in {}^{R}\!/_{K} \mid x \in I\}. \tag{4.50}$$

Why the Forward Image $q(I)$ is an Ideal for the Quotient Map: Earlier, Example 4.50 demonstrated that forward images of ideals generally fail to be ideals. However, q is **surjective** (Proposition 4.47). For any ideal $I \subset R$, $q(I)$ is an additive subgroup of ${}^{R}\!/_{K}$ because q is a homomorphism. To check ideal absorption: Let $\bar{r} \in {}^{R}\!/_{K}$ and $\bar{x} \in q(I)$ with $x \in I$. Since q is surjective, $\bar{r} = q(r)$ for some $r \in R$. Then:

$$\bar{r} \cdot \bar{x} = q(r) \cdot q(x) = q(r \cdot x).$$

Since I is an ideal of R, $r \cdot x \in I$. Thus, $q(r \cdot x) \in q(I)$, showing that $q(I)$ absorbs multiplication from all elements of ${}^{R}\!/_{K}$.

Spoken Transcript [00:23:35 - 00:25:06]

Prof. Rahul Pandharipande

To prove the left arrow — this arrow [points at $\longleftarrow$ on the board] — is well-defined [writes: To prove the $\longleftarrow$ is well defined], which we already did (Recall: Claim 4.53), but I'll just write it formally: we must show that K is inside $q^{-1}(J)$ [writes: we must show that $K \subset q^{-1}(J)$].

And the proof of that, the argument for that, is: well, J is an ideal, so 0 is in J [writes: $0 \in J$ so]. So $q^{-1}(0)$ is in $q^{-1}(J)$ [writes: $q^{-1}(0) \subset q^{-1}(J)$], and $q^{-1}(0)$ is exactly K [writes: K] — that's the kernel of the quotient map. So that's the argument there.

Next, to prove this arrow [points at $\longrightarrow$ on the board] is well-defined [writes: To prove $\longrightarrow$ is well defined], we just have to prove that [writes: we have to prove that] $q(I)$ inside this quotient is an ideal [writes: $q(I) \subset {}^{R}\!/_{K}$ is an ideal]. That's all we have to do.

All right, so how are we going to prove it's an ideal? You know, we have to check these properties in general. So we just...

Verification: Well-Definedness of Both Directions

1. Left Arrow ($\longleftarrow$ is Well-Defined): To show $\Phi(J) = q^{-1}(J) \in \mathrm{cl}_K(R)$, we must show that $K \subset q^{-1}(J)$:

$$\bar{0} \in J \implies q^{-1}(\bar{0}) \subset q^{-1}(J).$$

Since $q^{-1}(\bar{0}) = \ker(q) = K$ (Proposition 4.47), we obtain $K \subset q^{-1}(J)$.

2. Right Arrow ($\longrightarrow$ is Well-Defined): To show $\Psi(I) = q(I) \in \mathrm{cl}\left({}^{R}\!/_{K}\right)$, we must show that for any ideal $I \in \mathrm{cl}_K(R)$:

$$q(I) \subset {}^{R}\!/_{K} \quad \text{is an ideal.}$$

Spoken Transcript [00:25:06 - 00:26:54]

Prof. Rahul Pandharipande

So, if I have an element here [writes: $R \to {}^{R}\!/_{K}$] — ${}^{R}\!/_{K}$, and inside here we have $q(I)$ [writes: $q(I)$], and here's the map q [writes: q]...

If we pick two elements here... I want to show it's an ideal, so in particular I have to show it's closed under addition. First of all, let's just get zero out of the way. Zero is definitely in $q(I)$, because zero is in I, and if I apply q to zero, I get zero, right? [writes: $\bar{0} \in q(I)$] Is this clear? Because 0 is in I, so $q(0)$ is in $q(I)$, and $q(0) = 0$ [writes "Because $0 \in I \implies q(0) \in q(I)$", and "$\bar{0}$" above $q(0)$]. So, zero is definitely here.

Then I have to check: if two people are here, that their sum is here, so it's closed under addition. So, let us pick some people — how are we going to say who's here? Let's pick two elements. Well, everything that's in here is in the image of I, so I can pick elements in here such that x and y are in I, so that two general elements here are those that come from I. So, I can write them like this: [writes "$q(I) \ni \bar{x}, \bar{y}$" and "$x, y \in I$"].

So, I pick some elements here; both of them come from I. Since I is an ideal [writes: I ideal], this means $x + y$ is in I [writes: $\implies x + y \in I$], and then that means that that is in $q(I)$ [writes: $\implies \overline{x+y} \in q(I)$].

Okay? So, that shows that $q(I)$ has zero and it's closed under addition.

Proof that $q(I)$ is an Additive Subgroup

We verify that $q(I)$ is an additive subgroup of ${}^{R}\!/_{K}$:

Zero Element: Since I is an ideal of R, we have $0 \in I$. Thus:

$$q(0) \in q(I) \implies \bar{0} \in q(I).$$

Closure under Addition: Let $\bar{x}, \bar{y} \in q(I)$. Since elements of $q(I)$ are images of elements in I, there exist $x, y \in I$ such that:

$$q(x) = \bar{x} \quad \text{and} \quad q(y) = \bar{y}.$$

Because I is an ideal, it is closed under addition:

$$x + y \in I.$$

Applying the homomorphism q:

$$q(x+y) \in q(I) \implies \overline{x+y} = \bar{x} + \bar{y} \in q(I).$$

Thus, $q(I)$ contains $\bar{0}$ and is closed under addition. (Negatives need no separate check: $-\bar{x} = (-\bar{1}) \cdot \bar{x}$ lies in $q(I)$ by the absorption shown next, just as in the proof of Proposition 4.49.)

Spoken Transcript [00:26:54 - 00:28:26] Prof. Rahul Pandharipande

So, finally, we have to do the multiplication part.

Suppose we have to take an element x — that's an arbitrary element here, where x is in I — and we take its equivalence class; and then we have to multiply it by any element of the quotient ring, any single element, nothing to do with I, right? Well, that's some element, but that element is still some element of R.

So, in this multiplication case, we have to show that this set is closed by picking any element in that set — that's here, $\bar{x} \in q(I)$ [writes "$\bar{r} \cdot \bar{x}$" and below it "$\bar{x} \in q(I)$"] — and multiplying it by any element. But for both of those, we can find preimages in R, because the quotient map is surjective. So, whatever element here is, I can write it as $\bar{r}$ of something; and here, since it's in $q(I)$, I can write $\bar{x}$ where $x \in I$.

Okay? And then, since I is an ideal, $r \cdot x$ is in I [writes: $r \in R, x \in I \implies rx \in I$], and that implies that this product is in $q(I)$ [writes: $\implies \bar{r} \cdot \bar{x} \in q(I)$].

So, now I've checked it. I've checked that this is well-defined. So, they're both well-defined. That's a good start. But then I have to show that if you go one way and back, or the other way and back, you get the identity. As usual, there's a bunch of steps.

Ideal Absorption under Multiplication

Let $\bar{r} \in {}^{R}\!/_{K}$ be an arbitrary element of the quotient ring, and let $\bar{x} \in q(I)$ with $x \in I$. Because the quotient map $q\colon R \to {}^{R}\!/_{K}$ is **surjective**, the element $\bar{r}$ has a preimage $r \in R$ such that $q(r) = \bar{r}$.

Since I is an ideal of R, it absorbs multiplication by any ring element:

$$r \in R,\ x \in I \implies r \cdot x \in I.$$

Applying the quotient map q:

$$q(r \cdot x) \in q(I) \implies \overline{r \cdot x} = \bar{r} \cdot \bar{x} \in q(I).$$

Similarly, $\bar{x} \cdot \bar{r} = \overline{x \cdot r} \in q(I)$. Hence, $q(I)$ satisfies all ideal axioms, proving that $q(I) \subset {}^{R}\!/_{K}$ is an ideal.

Bijectivity of the Correspondence

Spoken Transcript [00:28:35 - 00:30:18] Prof. Rahul Pandharipande

And the trick now is to try to prove these steps without erasing the claim.

So, I'm going to erase the definitions of this that you're supposed to remember (Recall: Notation 4.51). [erases Def 1 and Def 2 from the center chalkboard]

To prove it's a bijection, we should prove that if I go here and then back, I get the same thing. That's the first thing; and then I have to do it the other way, you know.

So now... [writes: Now proof of the bijective property] proof of the bijective property. There are two steps. The first step, maybe **Step (A)** [writes: (A)], is: start with an ideal J in the quotient — we start there [writes: Start with $J \subset {}^{R}\!/_{K}$].

And then I have to show that if I take the pullback and then I take q of that, this equals J [writes: $q(q^{-1}(J)) = J$], okay? That's what I have to show. That's saying that if I do the motion left and then the motion right, I have altogether done nothing.

And the claim is: this is clear [writes: This is clear]. This is clear because q is surjective [writes: because q is surjective].

Okay, and I'm going to draw you a picture. So, here's ${}^{R}\!/_{K}$, here's R...

Spoken Transcript [00:30:18 - 00:31:26] Prof. Rahul Pandharipande

And it's surjective. That means that every single point — if you take a point here — is always hit, maybe by several, but it's definitely always hit, right?

So, if I take a point that happens to be in J, then I go over here, that's a point in $q^{-1}(J)$. If I do q, I just go back to J, and in fact I go back to the point I started with. Because every point is hit. So, I take any point in J, it's hit by somebody, so that gives me some point here; and then when I do q, I go back to it.

What would have been a problem is if there was a point here that nobody hit; then when I went back here and forward, I would not get it. But since this is surjective, this is clear just for set-theoretic reasons, not algebra-theoretic reasons. So, this is true. [writes: (TRUE) after $q(q^{-1}J) = J$]

So, we're happy about that. Okay, then we have one more thing to check.

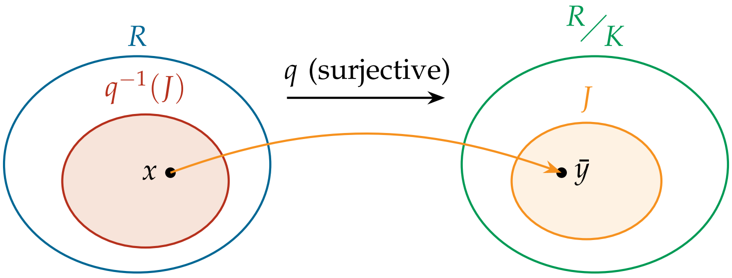
Bijective Property Part (A): $q(q^{-1}(J)) = J$

Let $J \in \mathrm{cl}\left({}^{R}\!/_{K}\right)$ be an ideal of the quotient ring. We show:

$$q(q^{-1}(J)) = J. \tag{4.51}$$

This equality holds purely set-theoretically because the quotient map $q\colon R \to {}^{R}\!/_{K}$ is **surjective**:

- **Inclusion ($\subseteq$):** The relation $q(q^{-1}(J)) \subseteq J$ always holds by the general definition of the preimage (Fact 0.7).
- **Inclusion ($\supseteq$):** Conversely, let $\bar{y} \in J$. By surjectivity of q, there exists $x \in R$ such that $q(x) = \bar{y}$. Thus $x \in q^{-1}(J)$, which implies $\bar{y} = q(x) \in q(q^{-1}(J))$.



Every element $\bar{y} \in J$ has a preimage $x \in q^{-1}(J)$ with $q(x) = \bar{y}$, so $q(q^{-1}(J)) = J$.

Spoken Transcript [00:31:26 - 00:32:06] Prof. Rahul Pandharipande

[erases the upper chalkboard] Okay, I'm going to erase this board.

The usual thing I tell students is that all the stuff I talk about, it's much easier to learn than German. German is a much harder language to learn than this, yeah. And I can tell you that from personal experience, yeah.

Spoken Transcript [00:32:06 - 00:34:06] Prof. Rahul Pandharipande

So, we need now to prove... We started on the right-hand side, so we need to start on this side now. We start with some I that contains K, and this is an ideal [writes: $K \subset I \subset R$] — so it's an ideal that contains that ideal K. And then we go with I to the right-hand side, so we just take $q(I)$; and then we have to go back to q^{-1}, so we take $q(I)$ and then we take the pullback, and we need to show that this is I, okay? [writes: $I \longrightarrow q^{-1}(q(I)) \overset{?}{=} I$]

And now it's not true for set-theoretic reasons. So, with the same diagram [draws two blobs for R and R/K]... Then we have some I here, and here is R and R/K. And when we go forward, we just go with whatever q; so q sends points all over here, right? So it just sends points to some image, that's $q(I)$ [draws arrows from I to $q(I)$]. Okay?

And now you see the problem, because then when we go back, there could be a point here: there's some point x here that goes to $\bar{x}$, but then there could be a y that also goes to that. And we get nervous now, because I want this to be I, but this y makes me nervous, because these two things could have gone here, so this y is definitely in $q^{-1}(q(I))$ [writes: $y \in q^{-1}(q(I))$], okay? And it makes me very nervous.

So, I need to prove: whenever this happens, actually y was already in I. I need to prove that. But it's not true for set-theoretic reasons. That one (i.e., Eq. (4.51)) was true for set-theoretic reasons, but this one is not. Because the q map is not a bijection; you know, it puts things together.

Spoken Transcript [00:34:06 - 00:34:45] Prof. Rahul Pandharipande

All right, so how could this happen? We have this: x is in I [writes: $x \in I$], and so then on this side we have $\bar{x}$ [writes: $\bar{x}$], and we have some y in R [writes: $y \in R$]. And what we have here is that $\bar{y}$ — that's what q does, it takes y to $\bar{y}$ —, $\bar{y}$ might equal $\bar{x}$ [writes: $\bar{y} = \bar{x}$]. That's the situation we have, okay?

And we need to now do something about it. And you see, that gives us a piece of information. What does it mean that $\bar{y} = \bar{x}$? It only means one thing, which you should now repeat to me.

Student Answer

Their difference is in K.

Spoken Transcript [00:34:49 - 00:36:36] Prof. Rahul Pandharipande

Yeah, so what's the equation? [student speaks]

Well, the equation is: this is if and only if there exists some element z in K such that $x - y = z$, right? [writes: $\exists z \in K$: $x - y = z$] That's exactly the equivalence. Or I could write it the other way... Let's write it... okay, anyway, let's write it the other way: $y - x = z$ [corrects to $y - x = z$], because it doesn't matter, it's symmetric. Okay?

So, that's the crucial thing. Since their equivalence classes are the same, it means their difference is in K. So that means that y — if I just now logically conclude this —: $y = x + z$ [writes: $y = x + z$], okay?

But x is in I and z is in K, and so finally we use this funny condition that we're always assuming: K is inside I. So here we have x is in I, z is in K, but K is inside I, so they're both inside I [writes: $x \in I, z \in K \subset I$]. So, this thing is in I [writes: $\implies \in I$]. That means this funny y is actually in I [writes: $y \in I$].

That's the argument. So, what happens: it looks like this can happen, and it can happen set-theoretically, but the algebraic hypothesis says that it actually does not happen. And that's the argument: if it were to happen, they'd have the same equivalence class. If they had the same equivalence class, their difference must be in K, and then I can write y as something in I plus something in K; but since K is in I, that whole thing is still in I. So, that's the argument.

So, now I've proven that bijection. All right.

Bijective Property Part (B): $q^{-1}(q(I)) = I$

Let $I \in \mathrm{cl}_K(R)$ be an ideal containing K (so $K \subset I \subset R$). We show:

$$q^{-1}(q(I)) = I. \tag{4.52}$$

The inclusion $I \subseteq q^{-1}(q(I))$ always holds set-theoretically (Fact 0.7). For the reverse inclusion, let $y \in q^{-1}(q(I))$. By definition of the preimage:

$$q(y) \in q(I) \implies \bar{y} \in q(I).$$

Thus, there exists some $x \in I$ such that:

$$\bar{y} = \bar{x}.$$

By definition of congruence modulo K (Definition 4.17):

$$\bar{y} = \bar{x} \iff y - x \in K \iff \exists z \in K \text{ such that } y - x = z.$$

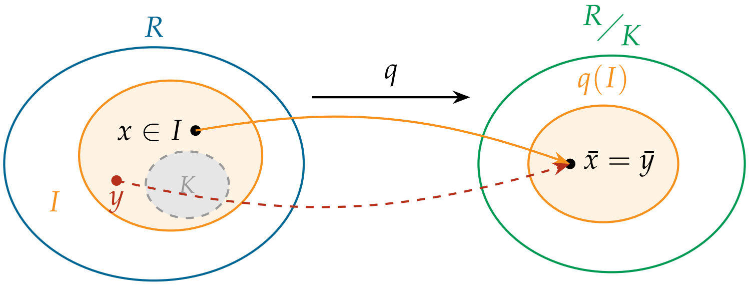
Rearranging yields:

$$y = x + z.$$

Since $x \in I$ and $z \in K$, and by hypothesis $K \subset I$, we have $z \in I$. Because I is an ideal (hence closed under addition):

$$x \in I \quad \text{and} \quad z \in I \implies y = x + z \in I.$$

Therefore, $q^{-1}(q(I)) \subseteq I$, which proves $q^{-1}(q(I)) = I$.



For $y \in q^{-1}(q(I))$: since $\bar{y} = \bar{x}$, we have $y - x = z \in K$.
Since $K \subset I$, $y = x + z \in I$, so y is forced into I.

Together, Steps (A) and (B) establish that Φ and Ψ are mutually inverse bijections, completing the proof of the Correspondence Theorem (Proposition 4.52).

Q.E.D.

Beyond the Lecture: The Correspondence Preserves Inclusions

The bijection of Proposition 4.52 respects inclusions in both directions, which is why it is also called the lattice theorem. For ideals $I_1, I_2 \in \mathrm{cl}_K(R)$, $I_1 \subseteq I_2$ clearly gives $q(I_1) \subseteq q(I_2)$; conversely, $q(I_1) \subseteq q(I_2)$ gives $I_1 = q^{-1}(q(I_1)) \subseteq q^{-1}(q(I_2)) = I_2$ by Step (B) (Eq. (4.52)). In the same way, $J_1 \subseteq J_2$ holds in $\mathrm{cl}\left(R/K\right)$ if and only if $q^{-1}(J_1) \subseteq q^{-1}(J_2)$, using Step (A) (Eq. (4.51)).

Spoken Transcript [00:36:36 - 00:37:56] Prof. Rahul Pandharipande

And this is kind of uncharacteristic: I really proved every single step of it. I don't know whether you like that or not. Normally, in the middle I would start waving my hands, but this time I proved every single step of it. [erases the boards]

But it's very good to understand this, because it shows some non-trivial aspect of this construction. And it really allows you to grasp what the ideals of the quotient are. I wanted to say some more things about fields; I don't have so much time, unfortunately. Anyway, maybe I can start.

So, the next topic, related to the ideas here, is the notion of a maximal ideal. Now we've finished two themes in the class; the third one is the notion of a maximal ideal.

Maximal Ideals

Spoken Transcript [00:37:56 - 00:40:01] Prof. Rahul Pandharipande

Okay, so there are different ways to define a maximal ideal. The most simple one is that [writes: Notion of a maximal ideal $m \subset R$]:

(i) m is an ideal, of course. [writes: (i) m is ideal] This is going to be a definition: maximal ideal. To be a maximal ideal, of course, you have to be an ideal.

(ii) This ideal is not allowed to be the whole thing. So this normally is said that you should be a proper ideal [writes: (ii) $m \neq R$ (proper ideal)]. Normally you'd say that in one: m is a proper ideal, which means it's an ideal that's not the whole thing. Because the whole thing is cheating, it turns out, for a maximal ideal.

(iii) And this is the maximal property: if I have another ideal, another proper ideal [writes: (iii) If $\hat{m} \subset R$, $\hat{m}$ proper ideal], which has the property that m is inside it... [writes: which has the property $m \subset \hat{m}$]

When I use this notation, I hope this is clear: this means that this is a subset of that. It's just silent on whether it's equal or not. Is that how you use it? You know, so this means that this is a subset of that. It could be equal; in fact, it's okay. If you wanted to be specific, you could say like that — that means it's not equal. Anyway, it's important here. So, which has the property that whenever this is inside that, that actually means they're equal [writes: $\implies m = \hat{m}$].

That's the maximality property. It's saying that, well, first of all, it's a proper ideal; and the second one says there's nobody above it that's a proper ideal. Is that clear? So, the reason why you want to rule out R always is because R is the biggest ideal, and no one ever beats R because it's the whole ring. So it kind of has an unfair role in that.

Spoken Transcript [00:40:01 - 00:40:55] Prof. Rahul Pandharipande

So, for a maximal ideal... It's a confusing thing, and, you know, it's a point that's related to the confusion about definitions of what a zero ring is, or whether a field is trivial or not; it's the same kind of confusion there. And we will discuss that in the next lecture. But it's important to understand: it has to be proper, and nobody's bigger.

So, for example, if you take the integers: the multiples of 4 — sometimes we write that as $4\mathbb{Z}$ (Recall: Definition 2.33) —, this is definitely an ideal; is it a maximal ideal? [writes "$\mathbb{Z}$", then "$(4) \subset \mathbb{Z}$" and "$4\mathbb{Z} \subset \mathbb{Z}$"]

Student Answer

No.

Spoken Transcript [00:40:57 - 00:41:18] Prof. Rahul Pandharipande

No, because it sits inside multiples of 2 [writes: $4\mathbb{Z} \subset 2\mathbb{Z}$]. And this is not equal [writes "$\neq$" under the $\subset$, giving $4\mathbb{Z} \subsetneq 2\mathbb{Z}$]. So it would be contradicted by this fact: if there's somebody bigger that's not the full ring, then that's not a maximal ideal.

So, what are the maximal ideals for $\mathbb{Z}$?

Student Answer

Prime numbers.

Spoken Transcript [00:41:21 - 00:43:42] Prof. Rahul Pandharipande

Yeah, so if you have $p\mathbb{Z}$ inside $\mathbb{Z}$, where p is prime [writes: If $p\mathbb{Z} \subset \mathbb{Z}$ where p is prime], then these are the maximal ideals [writes: then these are max ideals]. This is if and only if… yeah, these are the maximal ideals.

In fact, you can kind of draw… Or maybe I'll do it next time; I'll talk about the pictures of this.

If you have a field — so if F is a field [writes: F is a field] —, what are the maximal ideals there?

Well, if you remember, a field only has two ideals (Recall: Theorem 1.31) [writes: has only 2 ideals: $0, F$]: it has the zero ideal and the whole thing. And when we talk about maximal ideals, we don't talk about the whole thing. So, it only has one proper ideal, and that proper ideal is maximal.

So, you can say: field F has 0 as a maximal ideal [writes: Field F has 0 as a max. ideal].

And the converse is maybe the last thing I say today: let R be a ring [writes: Let R be a ring] with 0 in R a maximal ideal [writes: $0 \subset R$ a max. ideal]. So, we take any ring, and suppose you find out, one way or the other, that 0 is a maximal ideal.

Then what am I going to write here? Well, R is a field [writes: $\implies$ R is a field].

And the reason for this is: first of all, you know R is not the zero ring. A field is not the zero ring (i.e., $1 \neq 0$, Definition 2.9). So you know R is not the zero ring because zero is a maximal ideal, and zero in the zero ring is not a maximal ideal because it's the whole thing. So, this hypothesis secretly says that R is not the zero ring. And then I will explain this next time: that has to be a field. [video ends, students applaud]

Definition of Maximal Ideals and Examples

Definition 4.54 (*Maximal Ideal*)**:** Let R be a ring. A subset $m \subset R$ is called a *"maximal ideal"* if:

(i) m is an ideal of R.

(ii) $m \neq R$ (m is a *"proper ideal"*).

(iii) Whenever $\hat{m} \subset R$ is a proper ideal of R with $m \subset \hat{m}$, then:

$$m = \hat{m}.$$

Notation 4.55 (*$\subset$ on the Board*)**:** As the lecturer explains here, $A \subset B$ on the board says only that A is a subset of B and allows $A = B$, like $A \subseteq B$. For a strict inclusion, he writes $A \subsetneq B$.

Examples 4.56 (*Maximal Ideals in $\mathbb{Z}$*)**:** Consider the ring of integers $\mathbb{Z}$:

- **The Ideal $4\mathbb{Z}$ is Not Maximal:** We have the chain of inclusions:
$$4\mathbb{Z} \subsetneq 2\mathbb{Z} \subsetneq \mathbb{Z}.$$
Since $2\mathbb{Z}$ is a proper ideal strictly containing $4\mathbb{Z}$, the ideal $4\mathbb{Z} = (4)$ (Definition 4.32) is not maximal.
- **The Ideals $p\mathbb{Z}$ for Primes p:** The maximal ideals of $\mathbb{Z}$ are the principal ideals $p\mathbb{Z} = (p)$ for prime numbers p.

Proof that the maximal ideals of $\mathbb{Z}$ are the $p\mathbb{Z}$ (beyond the lecture)

The lecture states this without proof. Every ideal of $\mathbb{Z}$ is $I = n\mathbb{Z}$ with $n \geq 0$ (Theorem 2.35), and $p\mathbb{Z} \subseteq n\mathbb{Z}$ means $n \mid p$, so $n = 1$ (then $I = \mathbb{Z}$ is not proper) or $n = p$. The remaining ideals are not maximal: $n\mathbb{Z} \subsetneq a\mathbb{Z} \subsetneq \mathbb{Z}$ for $n = ab$ with $1 < a < n$, and $0\mathbb{Z} \subsetneq 2\mathbb{Z} \subsetneq \mathbb{Z}$, while $1\mathbb{Z} = \mathbb{Z}$ is not proper. **Q.E.D.**

Proposition 4.57 (*Characterization of Fields via the Zero Ideal*): Let R be a ring. Then R is a field if and only if the zero ideal $(0) = \{0\}$ is a maximal ideal in R:
$$R \text{ is a field} \iff (0) \text{ is a maximal ideal of } R. \tag{4.53}$$

Why (0) Being Maximal Characterizes Fields:

- **Forward Direction ($\Longrightarrow$):** If F is a field, its only ideals are (0) and F (Theorem 1.31). Since $(0) \neq F$ (because $1 \neq 0$ in a field, Definition 2.9), (0) is the unique proper ideal, hence maximal: the only proper ideal containing (0) is (0) itself.
- **Backward Direction ($\Longleftarrow$):** If (0) is maximal, then (0) is a proper ideal, so $(0) \neq R \implies R \neq \{0\}$ (ruling out the trivial zero ring). The lecturer stops here and explains in the next lecture why R must then be a field (Proposition 5.5): for a non-zero $x \in R$, the principal ideal (x) (Definition 4.32) strictly contains (0), so $(x) = R$ by maximality; hence $xy = 1$ for some $y \in R$, and every non-zero element is invertible.

Notation: Maximal Ideals

Many textbooks write a maximal ideal as $\mathfrak{m}$. The lecturer writes a plain m on the board, here and in Week 5, and these notes follow him.

Beyond the Lecture: Quotients by Maximal Ideals

Combining Proposition 4.57 with the inclusion-preserving Correspondence Theorem (Proposition 4.52) immediately yields the fundamental characterization of maximal ideals via quotient rings: for any ideal $m \subset R$, the quotient ring ${}^{R}\!/_{m}$ is a field if and only if $(\bar{0}) = q(m)$ is a maximal ideal in ${}^{R}\!/_{m}$, which holds if and only if $m = q^{-1}((\bar{0}))$ is a maximal ideal in R.

Summary of Key Concepts

- **Quotient Rings and the Quotient Map:** For any ideal $I \subset R$, the quotient ${}^{R}\!\diagup_{I}$ is a ring under the operations on classes $\bar{x} + \bar{y} = \overline{x+y}$ and $\bar{x}\bar{y} = \overline{xy}$. The quotient map $q \colon R \to {}^{R}\!\diagup_{I}$ with $x \mapsto \bar{x}$ is a surjective ring homomorphism with $\ker(q) = I$. By the First Isomorphism Theorem, every homomorphism $f \colon R \to S$ induces an isomorphism ${}^{R}\!\diagup_{\ker(f)} \cong \mathrm{Im}(f)$.
- **Pullbacks vs. Pushforwards of Ideals:** Under any ring homomorphism $f \colon R \to S$, the preimage $f^{-1}(J)$ of an ideal $J \subset S$ is always an ideal of R. Conversely, direct images $f(I)$ generally fail to be ideals (e.g., $2\mathbb{Z} \subset \mathbb{Z} \hookrightarrow \mathbb{Q}$).
- **The Correspondence Theorem for Ideals:** For an ideal $K \subset R$, the ideals of ${}^{R}\!\diagup_{K}$ are in canonical bijection with the ideals of R containing K:
$$\mathrm{cl}_K(R) \longleftrightarrow \mathrm{cl}\left({}^{R}\!\diagup_{K}\right), \quad I \mapsto q(I), \quad J \mapsto q^{-1}(J).$$
Surjectivity of q ensures $q(q^{-1}(J)) = J$, while the condition $K \subset I$ ensures $q^{-1}(q(I)) = I$.
- **Maximal Ideals and Fields:** An ideal $m \subsetneq R$ is maximal if no proper ideal strictly contains it. In $\mathbb{Z}$, the maximal ideals are precisely the ideals $p\mathbb{Z}$ for primes p. A commutative ring R is a field if and only if the zero ideal (0) is a maximal ideal.

Week 5

Tuesday, 14th October 2026

Quotient Rings, Maximal and Prime Ideals, and the Chinese Remainder Theorem

Course Content Overview

In this lecture, we explore the structural theory of ideals in commutative rings and their corresponding quotient rings:

- **The Ideal Correspondence and Quotients of Quotients:** We review the bijection between ideals of R containing K and ideals of the quotient ${}^{R}\!/_{K}$, and use it to describe a quotient of a quotient: $\left({}^{R}\!/_{K}\right)\!/_{J} \cong {}^{R}\!/_{q^{-1}(J)}$.
- **Maximal Ideals and Fields:** We define maximal ideals within the poset of proper ideals $(\mathrm{cl}^*(R), \subseteq)$ and prove that ${}^{R}\!/_{m}$ is a field if and only if m is maximal. Furthermore, we show via Zorn's Lemma that every non-zero commutative ring contains at least one maximal ideal, discussing its non-constructive foundational nature.
- **Prime Ideals and Integral Domains:** In parallel, we define prime ideals and establish that ${}^{R}\!/_{P}$ is an integral domain if and only if P is prime, deducing that every maximal ideal is prime. We examine a non-prime ideal: the smooth functions with compact support inside $C^\infty(\mathbb{R})$.
- **The Chinese Remainder Theorem for Rings:** We define ideal arithmetic (sums and products) and coprime ideals ($I + J = (1)$), formulating the general Chinese Remainder Theorem as a ring isomorphism $\phi\colon {}^{R}\!/_{IJ} \xrightarrow{\sim} {}^{R}\!/_{I} \times {}^{R}\!/_{J}$ and completing the proof of injectivity.

Ideals in Quotient Rings

The Correspondence Theorem for Ideals

Spoken Transcript [00:00:01 - 00:01:36] Prof. Rahul Pandharipande

Okay, so, we will start. And I remind you, last week we had this correspondence: If R is a ring and K inside R is an ideal [writes: $K \subset R$ ideal], then we constructed last week this quotient map q to R mod K [writes: $R \xrightarrow{q} {}^{R}\!/_{K}$], so that was one. And then there was this correspondence, which I remind you. For algebraists, this correspondence — it's the correspondence of ideals, and it's really deep in the brain of an algebraist, like almost to the reptilian brain.

This correspondence says: If we take here the ideals of R [writes: $\mathrm{cl}_K(R)$] — and this is the set of ideals inside R, so I is an ideal [writes: $\{I \subset R \mid I \text{ ideal}\}$], and K is inside I [writes: $K \subset I\}$] — on one hand, and on the other hand we take the ideals of R mod K [writes: $\mathrm{cl}({}^{R}\!/_{K})$], and this is just every ideal [writes: $\{J \subset {}^{R}\!/_{K} \mid J \text{ is an ideal}\}$].

And the correspondence we proved last time, which I said, it's just absolutely fundamental. Almost for an algebraist, if you see a ring and this quotient map, this correspondence is so fundamental, one basically identifies these two sets in your brain — just identify they're the same set.

The Correspondence Theorem for Ideals: Setup

Let R be a ring and $K \subset R$ an ideal. Consider the quotient map:

$$q\colon R \to {}^{R}\!/_{K}, \quad x \mapsto \bar{x} = x + K. \tag{5.1}$$

We define the following sets of ideals:

$$\mathrm{cl}_K(R) := \{I \subset R \mid I \text{ is an ideal of } R \text{ with } K \subset I\}, \tag{5.2}$$

$$\mathrm{cl}\left({}^{R}\!/_{K}\right) := \left\{J \subset {}^{R}\!/_{K} \;\middle|\; J \text{ is an ideal of } {}^{R}\!/_{K}\right\}. \tag{5.3}$$

Notation: cl vs. Id

Some textbooks denote the set of ideals of a ring R by $\mathrm{Id}(R)$ or $\mathcal{I}(R)$. The lecture consistently uses the notation $\mathrm{cl}(R)$ (and $\mathrm{cl}_K(R)$ for ideals containing K).

Spoken Transcript [00:01:36 - 00:02:37]

Prof. Rahul Pandharipande

The bijection which we discussed last time is this: There's a bijection here to ideals of R mod K [writes: $\mathrm{cl}_K(R) \longleftrightarrow \mathrm{cl}({}^{R}\!/_{K})$], and it takes a J here, this way goes to $q^{-1}(J)$ [writes: $q^{-1}(J) \leftarrow\!\shortmid J$], and if you take an I here which contains K, this goes to $q(I)$ [writes: $I \mapsto q(I)$]. Sometimes this $q(I)$ is called I mod K [writes: $= {}^{I}\!/_{K}$].

Okay, so, that's the correspondence. We discussed this extremely carefully because I thought it wasn't really explained so carefully in the book.

So, one more step of this is that this is a way to just, if you have ideals here, it's the same thing as certain ideals there. All right? Are there any questions about this? As I said, this is absolutely fundamental; it's just used all the time. And as I said, for the algebraist, these sets are so strongly identified, they're just viewed as like the same set. Yeah.

Recap: The correspondence theorem was proved at the end of the previous lecture (Proposition 4.52); the lecturer restates it as a bijection.

The Correspondence Theorem for Ideals: Bijection

Theorem 5.1 (*Ideal Correspondence Theorem*): Let R be a ring and $K \subset R$ an ideal, with quotient map $q\colon R \to R/K$. The assignment:

$$\begin{aligned} \mathrm{cl}_K(R) &\longleftrightarrow \mathrm{cl}\left(R/K\right) \\ I &\longmapsto q(I) = I/K \\ q^{-1}(J) &\longleftarrow J \end{aligned} \tag{5.4}$$

is an inclusion-preserving bijection between the set of ideals of R containing K and the set of all ideals of the quotient ring R/K.

Properties of the Correspondence Map: Under this bijection, the following structural properties hold for any $I \in \mathrm{cl}_K(R)$ and $J \in \mathrm{cl}(R/K)$:

- **Pre-image of an Ideal:** For any ideal $J \subset R/K$, the full pre-image $q^{-1}(J) = \{x \in R \mid q(x) \in J\}$ is an ideal of R that automatically contains $\ker(q) = K$.
- **Forward Image of an Ideal:** For any ideal $I \subset R$ containing K, the image $q(I) = \{x + K \mid x \in I\}$ is an ideal of R/K, commonly denoted as I/K.
- **Order Preservation:** For ideals $I_1, I_2 \in \mathrm{cl}_K(R)$, one has $I_1 \subseteq I_2 \iff q(I_1) \subseteq q(I_2)$.

Quotients of Quotients (in textbooks: Third Isomorphism Theorem)

Spoken Transcript [00:02:37 - 00:03:48] Prof. Rahul Pandharipande

So, you could ask, for example: What happens if I take a J over here, an ideal J in R mod K [writes: What happens $J \subset R/K$ ideal]? We know that from this, I can construct a quotient: R mod K, that's the ring, and that's an ideal, so this is the quotient [writes: $\to (R/K)/J$].

This is confusing notation sometimes, but this is the ring, this is an ideal in the ring, so I can construct the quotient.

That's on this side of the correspondence: If I have an ideal in this ring, I can construct the quotient. You can ask: Well, what is that on the other side of the correspondence?

To understand that: What is this quotient on the other side of the correspondence? [writes: What is this quotient on the other side of the corr.?] Because for this quotient, I picked an ideal here, now I'm taking the quotient, and you can view that on the R mod K side of the correspondence. So, what is that construction on the R side of the correspondence?

And that has a very nice answer. The answer for this is we look at this quotient map: We have a quotient map R to R mod K [writes: Ans: $R \xrightarrow{q} R/K$].

Quotient of a Quotient Ring: Problem Setup

Given an ideal $J \subset R/K$, the quotient ring $(R/K)/J$ is well-defined.

Question 5.2: What is the structural meaning of the double quotient $\left({}^{R}\!/\!_{K}\right)/_{J}$ in terms of the original ring R? That is, what does this quotient correspond to on the R-side of the ideal correspondence?

Spoken Transcript [00:03:48 - 00:05:06] Prof. Rahul Pandharipande

That's realizing R mod K as a quotient of R, and we can compose that with another quotient map to $(R \bmod K) \bmod J$ [writes: $\xrightarrow{\hat{q}} \left({}^{R}\!/\!_{K}\right)/_{J}$]. This is the quotient map: Since J is an ideal of R mod K, I can consider that quotient map, and I can consider the composition.

Let me consider this composition. For f, I do q first and then I do $\hat{q}$ [writes: $f = \hat{q} \circ q$]. So, this is f going from R all the way to $(R \bmod K) \bmod J$ [writes: $f \colon R \to \left({}^{R}\!/\!_{K}\right)/_{J}$]. And since both of these are surjective — q was surjective, $\hat{q}$ is surjective —, this f is surjective. Is that clear? [writes: f is surjective since both $q, \hat{q}$ are]

Okay. So...

Composition of the Canonical Quotient Maps

We consider the two canonical surjective quotient homomorphisms:

$$q \colon R \longrightarrow {}^{R}\!/\!_{K}, \quad x \longmapsto x + K, \tag{5.5}$$

$$\hat{q} \colon {}^{R}\!/\!_{K} \longrightarrow \left({}^{R}\!/\!_{K}\right)/_{J}, \quad y + K \longmapsto (y + K) + J. \tag{5.6}$$

Their composition is defined by:

$$f := \hat{q} \circ q \colon R \longrightarrow \left({}^{R}\!/\!_{K}\right)/_{J}. \tag{5.7}$$

Since both q and $\hat{q}$ are surjective ring homomorphisms, their composition f is also surjective:

$$f \text{ is surjective since both } q, \hat{q} \text{ are surjective.} \tag{5.8}$$

Spoken Transcript [00:05:06 - 00:06:29] Prof. Rahul Pandharipande

In order to understand this ring homomorphism — we already know it's surjective — we'd like to know what the kernel is. Because if we knew what the kernel is, then we could use our first isomorphism theorem.

So, the question here: What is the kernel? What is the kernel of f? [writes: What is the $\ker(f) = ?$] We'd like to know that. The question is: If I have some person in R, and you know, I go all the way to the right object, when does that person go to zero? Well, that's the question of the kernel.

If I have somebody here [writes "x" under R], when does this person go all the way to zero here? [writes "0" under $\left({}^{R}\!/\!_{K}\right)/_{J}$] Well, that happens... I can take $q(x)$, and then $q(x)$ should be in the kernel [writes: $q(x) \in \ker(\hat{q})$]. So, if x goes all the way to zero here, that means $q(x)$ is in the kernel of this one, right?

What is the kernel of that map? Well, we know the kernel of that map is J [writes: $= J$], because that's the quotient map; the kernel of the quotient map is J. So, the only way x can get to zero here is if $q(x)$ is in J.

Kernel Computation for the Composed Map

To apply the First Isomorphism Theorem to f, we determine its kernel:

$$\ker(f) = \{x \in R \mid f(x) = 0\}.$$

Tracking an element $x \in R$ across the composition:

$$\begin{tikzcd} R \arrow[r, "q"] & {}^{R}\!\big/_{K} \arrow[r, "\hat{q}"] & \left({}^{R}\!\big/_{K}\right)\big/_{J} \\ x \arrow[r, mapsto] & q(x) \arrow[r, mapsto] & \hat{q}(q(x)) = 0 \end{tikzcd} \tag{5.9}$$

By the definition of the quotient map $\hat{q}$, we have:

$$\hat{q}(q(x)) = 0 \iff q(x) \in \ker(\hat{q}) = J.$$

Thus, an element $x \in R$ satisfies $f(x) = 0$ if and only if $q(x) \in J$, which means:

$$x \in q^{-1}(J). \tag{5.10}$$

Spoken Transcript [00:06:29 - 00:07:38] Prof. Rahul Pandharipande

Okay? That tells us what the kernel is in this correspondence. Sorry.

The answer is, the kernel of f is $q^{-1}(J)$, which is that correspondence [writes: $\ker(f) = q^{-1}(J)$]. The kernel is exactly those people in R where, when I move them to R mod K with q, I end up in J, which is exactly $q^{-1}(J)$. Okay?

Then, using that first isomorphism theorem, we can answer this question [writes: First Isom Theorem $\implies$]. The first isomorphism theorem says that if I take the image, which is the whole thing, it's isomorphic to that modulo the kernel.

The first isomorphism theorem says this: R modulo the kernel of f is isomorphic to $(R \bmod K) \bmod J$ [writes: ${}^{R}\!\big/_{\ker(f)} \cong \left({}^{R}\!\big/_{K}\right)\big/_{J}$]. Are you happy that's just recalling what that is?

Now we know what the kernel of f is, so we can restate that as saying $R \bmod q^{-1}(J)$ is isomorphic to $(R \bmod K) \bmod J$ [writes: ${}^{R}\!\big/_{q^{-1}(J)} \cong \left({}^{R}\!\big/_{K}\right)\big/_{J}$]. So, we've proven that.

Application of the First Isomorphism Theorem

From (5.10), we obtain:

$$\ker(f) = q^{-1}(J). \tag{5.11}$$

By the First Isomorphism Theorem for rings applied to the surjective homomorphism $f\colon R \to \left({}^{R}\!\big/_{K}\right)\big/_{J}$:

$${}^{R}\!\big/_{\ker(f)} \cong \operatorname{Im}(f) = \left({}^{R}\!\big/_{K}\right)\big/_{J}. \tag{5.12}$$

Substituting $\ker(f) = q^{-1}(J)$ yields:

$$R/q^{-1}(J) \cong (R/K)/J. \tag{5.13}$$

Spoken Transcript [00:07:38 - 00:09:12] Prof. Rahul Pandharipande

It just used an understanding of quotients and the first isomorphism theorem.

It's usually not written this way [draws a box around $R/q^{-1}(J) \cong (R/K)/J$]. It's usually written on the I side. Equivalently, I can take $q^{-1}(J)$ and call that I. This is just another way of writing it, starting on the I side: This is isomorphic to $(R \bmod K) \bmod q(I)$ [writes: $R/I \cong (R/K)/q(I)$].

This is just writing the same thing, because if we say $q^{-1}(J)$ is equal to some ideal I, this I has to be some ideal that contains K. This is writing it, so to speak, starting with the J side; this is writing it starting with I [boxes $R/I \cong (R/K)/q(I)$]. Okay?

So, you'll see both formulations. And the way you normally see this is you use this notation $q(I) = I \bmod K$, which is a very sensible notation. Then the way you normally see this in books is: $R \bmod I$ is equal to $(R \bmod K)/(I \bmod K)$ [writes: $R/I \cong \dfrac{R/K}{I/K}$]. That's exactly a third way of expressing the same thing, just in new notation. Okay?

And this is very convenient, because you can try to prove it by saying R/K times K/I is equal to R/I [writes: $\frac{R}{K} \cdot \frac{K}{I} = \frac{R}{I}$]. That's a very convenient mnemonic for it. It's not exactly a proof, but you know, the notation works perfectly.

We shouldn't write these things during the lectures, so maybe I cross this out [crosses out $\frac{R}{K} \cdot \frac{K}{I} = \frac{R}{I}$]. But that's why the notation works perfectly.

Theorem 5.3 (*Quotients of Quotients (in textbooks: Third Isomorphism Theorem)*)**:** Let R be a ring, $K \subset R$ an ideal with quotient map $q\colon R \to R/K$, and $J \subset R/K$ an ideal. Then there is a canonical ring isomorphism:

$$R/q^{-1}(J) \cong (R/K)/J. \tag{5.14}$$

Three Equivalent Formulations: Depending on whether one starts from the ideal $J \subset R/K$ or the ideal $I \subset R$ containing K, the isomorphism is expressed as:

(1) Formulation starting with $J \subset R/K$:

$$\boxed{R/q^{-1}(J) \cong (R/K)/J}$$

(2) Formulation starting with $I \subset R$ $(K \subset I)$:

$$\boxed{R/I \cong (R/K)/q(I)}$$

(3) The way it is usually seen in books (writing $q(I) = I/K$):

$$R/I \cong (R/K)\big/(I/K)$$

Beyond the Lecture: The Name

Textbooks call this statement the Third Isomorphism Theorem. The lecturer gives it no name.

On the Board: Informal Fraction Mnemonic

The lecturer writes down an informal arithmetic mnemonic to remember the cancellation of quotient rings:

$$\frac{R}{K} \cdot \frac{K}{I} = \frac{R}{I}$$

While algebraically suggestive (formally mimicking the cancellation of fractions), the lecturer crosses it out immediately, emphasizing that it serves only as a mental mnemonic rather than a valid mathematical deduction.

Spoken Transcript [00:09:12 - 00:09:28] Prof. Rahul Pandharipande

All right, so, that's this correspondence. And as I said, this is really an extremely basic thing that if you study mathematics, it will also go to your reptilian brain eventually, if it's not there already. [takes sponge and begins wiping the middle and right boards]

Maximal Ideals

Definition of Maximal Ideals

Spoken Transcript [00:10:11 - 00:11:04] Prof. Rahul Pandharipande

Okay, so, the other topic that we started last week was that of maximal ideals. We will prove some theorems about this now.

Let R be a ring [writes: Let R be a ring]. We have this notation now $\mathrm{cl}(R)$ — this is the set of all ideals —, but I want this new set $\mathrm{cl}^*(R)$ [writes: $\mathrm{cl}^*(R)$]: This is the set of ideals I in R where I is a proper ideal [writes: $\{I \subset R \mid I \text{ is proper ideal}\}$].

Or equivalently, this just means that I is an ideal not equal to the whole ring, okay? [writes: $I \neq R$] We had this definition last time. So, this is of course a subset of all ideals [writes: $\mathrm{cl}^*(R) \subset \mathrm{cl}(R)$], and more or less, I throw out the ideal which is the whole ring. All right?

Proper Ideals

Let R be a ring. The set of all ideals of R is denoted by $\mathrm{cl}(R)$. We denote the collection of all *"proper ideals"* of R by:

$$\mathrm{cl}^*(R) := \{I \subset R \mid I \text{ is a proper ideal}\} = \{I \in \mathrm{cl}(R) \mid I \neq R\}. \tag{5.15}$$

Clearly, $\mathrm{cl}^*(R) \subset \mathrm{cl}(R)$ is obtained by excluding the unit ideal R itself.

Spoken Transcript [00:11:04 - 00:12:36]

Prof. Rahul Pandharipande

And then we say — and this is a definition —: m is a maximal ideal if it satisfies two properties [writes: $m \subset R$ is a maximal ideal $\overset{\text{def}}{\iff}$]. The first is that m is in this $\mathrm{cl}^*(R)$ [writes: 1. $m \in \mathrm{cl}^*(R)$]. This is just a fancy way of saying that m is not the full ring [writes: $(m \neq R)$], okay?

Maximal ideals are not allowed to be the full ring. This is a source of standard confusion, but it's really important in the definition. Maximal ideals are not the full ring — that's cheating for a maximal ideal.

And the second is it's maximal: That means that for any other ideal in here [writes: 2. $\forall \hat{m} \in \mathrm{cl}^*(R)$], if $\hat{m}$ contains m, this means that $\hat{m} = m$ [writes: if $\hat{m} \supset m \implies \hat{m} = m$]. Okay?

So, if you view this set of ideals as some kind of tournament where one ideal is a better player than the other if it contains that ideal, then this maximal ideal says you're kind of at the top: No one beats you; no one's on top of you. If anyone tries to be on top of you, they have to be equal.

And you can see why we throw out R here, because R would be on top of everybody, and this whole definition would be kind of silly if we involved R. Is that clear? Because R would just be... I mean, there'd only be one maximal ideal, R, and we don't want that.

Recap: Maximal ideals were defined in the previous lecture (Definition 4.54); the lecturer restates the definition here.

Definition 5.4 (*Maximal Ideal*): Let R be a ring and let $m \subset R$ be an ideal. Then m is a *"maximal ideal"* $\overset{\text{def}}{\iff}$ the following two conditions hold:

- **(1) Properness:** $m \in \mathrm{cl}^*(R)$, i.e., $m \neq R$.
- **(2) Maximality:** For every proper ideal $\hat{m} \in \mathrm{cl}^*(R)$,

$$\hat{m} \supseteq m \implies \hat{m} = m.$$

Equivalently, there are no ideals strictly between m and R: if I is an ideal of R with $m \subseteq I \subseteq R$, then $I = m$ or $I = R$.

Insight: The *"Tournament"* Intuition for Maximal Ideals

The lecturer visualizes the partially ordered set of proper ideals $(\mathrm{cl}^*(R), \subseteq)$ as a tournament: an ideal $\hat{m}$ *"beats"* m if $\hat{m} \supset m$. A maximal ideal is an undefeated player at the very top: no proper ideal strictly contains it. The entire ring R is excluded from the competition by definition; otherwise, R would trivialise the concept by containing every ideal.

Fields and the Zero Ideal

Spoken Transcript [00:12:36 - 00:13:28] Prof. Rahul Pandharipande

Okay, so, that's the definition of maximal ideal. We had some examples last time, but now we go...

So, here we have a proposition: R is a field if and only if (0) in R is a maximal ideal [writes: Prop. R is a field $\iff$ $(0) \subset R$ is a max ideal].

This is relating the idea of a field — some rings are fields —, so it's a field if and only if zero is a maximal ideal. Now there are some edge cases here which it's good to think about carefully.

Let's just do this.

Recap: This statement was begun in the previous lecture (Proposition 4.57): one direction was proved there before the bell. The lecturer now proves both directions.

Characterization of Fields via the Zero Ideal

Proposition 5.5 (*Fields via the Zero Ideal*): Let R be a ring. Then R is a field if and only if the zero ideal $(0) \subset R$ is a maximal ideal.

Spoken Transcript [00:13:28 - 00:14:48] Prof. Rahul Pandharipande

Let's try to prove this. This is not hard, but you have to say the words carefully. Proof: We're going to do the first direction, so, assume R is a field [writes: Proof: Assume R is a field]. Okay.

We have — and this is the edge case business — for a field, you're not allowed to be the zero ring, right? So in particular, R is not the zero ring [writes: ($R \neq 0$ ring)].

And then the set of ideals of R consists of two ideals: There's the zero ideal and there's the full ideal [writes: The $\mathrm{cl}(R) = \{(0), R\}$]. And these are different ideals, because R is not the zero ring.

If you take $\mathrm{cl}^*(R)$ here, it consists of just the zero ideal [writes: The $\mathrm{cl}^*(R) = \{(0)\}$]. Okay?

And so, this of course means that (0) is a maximal ideal [writes: $\implies$ $(0) \subset R$ is a max ideal]. Because the set $\mathrm{cl}^*(R)$ only has one element, it's this. And so certainly no one beats it — there's no one there to beat it, anyway.

There's a small meeting of definitions here... I mean, you should think about these edge cases: They're all treated perfectly. When it's a field, it's not the zero ring.

Proof of Proposition: Forward Implication

Proof of Proposition 5.5 ($\Longrightarrow$)

Assume R is a field. By convention, a field is not the zero ring ($R \neq \{0\}$, so $1 \neq 0$). Since every non-zero element $x \in R \setminus \{0\}$ has a multiplicative inverse x^{-1}, any ideal $I \subseteq R$ containing a non-zero element x also contains $1 = x^{-1}x$, which implies $I = R$. Hence, the only ideals of R are the zero ideal and the whole ring:

$$\mathrm{cl}(R) = \{(0), R\}. \tag{5.16}$$

Excluding the improper ideal R, the set of proper ideals contains exactly one element:

$$\mathrm{cl}^*(R) = \{(0)\}. \tag{5.17}$$

Since (0) is the unique element in $\mathrm{cl}^*(R)$, it cannot be strictly contained in any proper ideal. Thus, $(0) \subset R$ is a maximal ideal. **Q.E.D.**

Spoken Transcript [00:14:48 - 00:15:02] Prof. Rahul Pandharipande

Now let's do the other way. There's nothing to these proofs here; it's just somehow organizing the definitions in your head correctly. [takes sponge to clean the rightmost board panel]

Spoken Transcript [00:15:58 - 00:16:17] Prof. Rahul Pandharipande

Okay, so, now the other way: We suppose (0) in R is a maximal ideal [writes: Suppose $(0) \subset R$ is a max ideal]. We want to prove R is a field then. Yeah?

Student Question

I wanted to ask before, for the definition of a maximal ideal, does it always exist? Or can you have two disjoint ideals, or like partially disjoint ideals, where their union would also be an ideal, and therefore you wouldn't have a maximal ideal?

Spoken Transcript [00:16:30 - 00:16:51] Prof. Rahul Pandharipande

Yeah. It's a good question; we're going to prove it. The question is — maybe I say it —: If R is a ring, does there always exist a maximal ideal? Is that your question? That's a good question; we're going to prove it. The answer is going to be yes.

But you might be disappointed. You might nevertheless be disappointed, maybe I'll say.

Beyond the Lecture: Krull's Theorem on the Existence of Maximal Ideals

The assertion that every non-zero unital commutative ring possesses at least one maximal ideal is known as **Krull's Theorem**. Its proof requires Zorn's Lemma (and is in fact equivalent to the Axiom of Choice), which is why the lecturer hints that the student *"might nevertheless be disappointed"* by its non-constructive nature.

Spoken Transcript [00:16:51 - 00:18:05] Prof. Rahul Pandharipande

Suppose (0) is a maximal ideal. I want to prove that R is a field; that's what I have to do here. Do you agree?

And now the edge case is that since (0) is a maximal ideal, this implies that (0) is not R [writes: $(0) \neq R$], because R is not a maximal ideal. So, this implies that R is not the zero ring [writes: $\implies R \neq 0$ ring]. Which is good, because we're going to have to prove that R is a field, and in particular it can't be the zero ring. I'm just trying to point out the edge cases matching here. Okay.

We must prove that R is a field [writes: We must prove that R is a field]. How do I show something is a field if it's already a ring? I have to show the existence of multiplicative inverses for non-zero elements, right?

So, let $x \in R$ be non-zero [writes: Let $x \in R$, $x \neq 0$]. Okay?

Reverse Implication: Setup

Assume $(0) \subset R$ is a maximal ideal.

- **Non-triviality of R:** By definition of a maximal ideal, $(0) \in \mathrm{cl}^*(R)$, meaning $(0) \neq R$. Therefore, R is not the zero ring ($R \neq 0$).
- **Goal:** To show that R is a field, we must establish that every non-zero element $x \in R \setminus \{0\}$ has a multiplicative inverse in R.

Spoken Transcript [00:18:05 - 00:19:16] Prof. Rahul Pandharipande

And I have to produce a multiplicative inverse for this person x, just from the assumptions. So, what can I do here?

Well, I have this ideal (0). Since x is not zero, I can make an ideal that's generated... so I can take the ideal generated by x [writes: $0 \subsetneq (x)$]. We discussed this symbol last time. So, I just take the ideal generated by x [writes: (x) ideal gen by x]. It's given by all linear combinations and multiples of x, right? In this case, just all multiples of x.

This is an ideal, because that is always an ideal: the ideal generated by some elements. It's definitely not the zero ideal, because x is not zero. Okay?

So, what can it be?

Reverse Implication: Principal Ideal Generation

Let $x \in R$ with $x \neq 0$. Consider the principal ideal generated by x:

$$(x) := \{zx \mid z \in R\}. \tag{5.18}$$

Since $x \in (x)$ and $x \neq 0$, the ideal (x) strictly contains (0):

$$(0) \subsetneq (x). \tag{5.19}$$

Spoken Transcript [00:19:16 - 00:20:21] Prof. Rahul Pandharipande

(0) is a maximal ideal, so it's an ideal that's on top of it. There are no proper ideals on top of it. So, it has to be this [writes: $(x) = R$]: That's by the assumption of (0) being maximal [writes: $0 \subset (x) \implies (x) = R$ by max].

That's kind of the crucial step of this argument: (0) is a maximal ideal, this is some ideal, it's not zero, it's bigger than zero, it certainly contains zero, so it has to be the whole thing. Because our assumption is a very strong assumption: (0) is a maximal ideal.

If it's the whole thing, then this has to be the unit ideal. It has to be R. So, in particular, $1 \in (x)$ [writes: $1 \in (x)$].

And as I said, (x) is defined to be all the elements that look like zx where $z \in R$ [writes: $(x) = \{zx \mid z \in R\}$]. That's the ideal generated by x, that's it. And 1 is in it.

Since $1 \in (x)$ [writes: Since $1 \in (x)$], that means there must exist some z such that $zx = 1$ [writes: $\exists z$ such that $zx = 1$]. And that's the multiplicative inverse.

So, z is the multiplicative inverse [writes: z is the multiplicative inverse].

Reverse Implication: Invertibility and Conclusion

Since (0) is a maximal ideal and $(0) \subsetneq (x)$, the ideal (x) cannot be a proper ideal. Therefore:

$$(x) = R. \tag{5.20}$$

Because $1 \in R = (x)$, there exists some $z \in R$ such that:

$$zx = 1. \tag{5.21}$$

Thus, z is the multiplicative inverse of x. Since every non-zero element $x \in R \setminus \{0\}$ has a multiplicative inverse, R is a field.

Quotients by Maximal Ideals

Spoken Transcript [00:20:21 - 00:21:28] Prof. Rahul Pandharipande

So, that's the proof the other way. If I assume that zero is a maximal ideal, I can use this maximality assumption to show that all non-zero elements have a multiplicative inverse.

I have now proven this proposition: R is a field if and only if (0) is a maximal ideal.

The next proposition, which is a kind of strengthening of that one, is: R mod m is a field if and only if $m \subset R$ is a maximal ideal [writes: Prop: ${}^{R}\!/_{m}$ is a field $\iff$ $m \subset R$ is a maximal ideal].

That's a strengthening of this statement: That says R is a field if zero is a maximal ideal; actually I say it for any ideal here: This is a field if and only if that's a maximal ideal. Okay?

In particular, if I take this statement and specialize to $m = (0)$, I get that statement [writes: specialize to $m = 0$], because R mod (0) is R. So, this is the more general one; we prove this next.

Characterization of Maximal Ideals via Field Quotients

Proposition 5.6 (*Maximal Ideals via Field Quotients*): Let R be a ring and $m \subset R$ an ideal. Then the quotient ring ${}^{R}\!/_{m}$ is a field if and only if m is a maximal ideal of R.

Specialization to $m = (0)$: Notice that Proposition 5.6 directly generalizes Proposition 5.5: specializing $m = (0)$ yields:

$$ {}^{R}\!/_{(0)} \cong R \text{ is a field} \iff (0) \subset R \text{ is a maximal ideal.} $$

Spoken Transcript [00:21:28 - 00:21:49] Prof. Rahul Pandharipande

This is somehow one of the reasons that maximal ideals are very important, because they are those ideals such that the quotient is a field. It's just another way of saying what a maximal ideal is.

Spoken Transcript [00:22:36 - 00:23:14] Prof. Rahul Pandharipande

Okay, so, we want to prove that.

The proof is: R mod m is a field if and only if — now we use this proposition that's already on the board — if and only if (0) in R mod m is a maximal ideal [writes "Proof: ${}^{R}\!/_{m}$ is a field $\iff$ $(0) \subset {}^{R}\!/_{m}$ is a max ideal" on the lower board].

We've proven this — already proven, right? [writes: already proven] I just use this proposition here for R mod m. Okay? Yeah.

Student Question

Could you say what you wrote down as specialize?

Spoken Transcript [00:23:17 - 00:23:57] Prof. Rahul Pandharipande

Specialize this thing to the ideal $m = (0)$. If you specialize to (0), then R at the top is R mod (0), which is R, and it says R is a field if and only if (0) is a maximal ideal, which is what we have here. So, that's strictly more general than this.

But we're going to use this — that's why I strategically left it on the board. With my years of professorial experience, you know. [laughter]

So, here R mod m is a field, and we just use the statement for a field: This is a field if and only if (0) is a maximal ideal. All right?

Spoken Transcript [00:23:57 - 00:25:02] Prof. Rahul Pandharipande

Now, the next statement is that (0) is a maximal ideal here if and only if m is a maximal ideal in R [writes: $\iff$ $m \subset R$ is a max ideal].

And this statement is the bijection that I said should be deep in your brain. Because ideals in this quotient correspond to ideals in R which contain m, right? So, if this is maximal, there's nothing over zero, and this means that there's nothing here over m.

Because of that bijection that I started this lecture with, which we proved last time. Okay? So, this has to do with the bijection [writes: $\Longleftrightarrow$ bijection].

The bijection says any ideal here corresponds to ideals here that contain m. And this is saying that if this is a maximal ideal, then there aren't any ideals here above zero, and there's nothing here above m. It's just exactly the bijection.

So, that's the proof of this proposition.

Proof of Proposition: Characterization of Maximal Ideals

Proof of Proposition 5.6

We combine Proposition 5.5 with the Ideal Correspondence Theorem (Theorem 5.1):

$$\begin{aligned} R/m \text{ is a field} &\iff (0) \subset R/m \text{ is a maximal ideal} && \text{(by Proposition 5.5)} \\ &\iff m \subset R \text{ is a maximal ideal} && \text{(by Theorem 5.1).} \end{aligned}$$

Q.E.D.

Why the Bijection Yields the Equivalence: Under the quotient map $q\colon R \to R/m$, the bijection:

$$\mathrm{cl}_m(R) \longleftrightarrow \mathrm{cl}\left(R/m\right)$$

preserves strict inclusions and identifies the zero ideal $(0) \in \mathrm{cl}(R/m)$ with $m = q^{-1}((0)) \in \mathrm{cl}_m(R)$. Therefore:

- (0) is maximal in R/m $\iff$ there are no ideals strictly between (0) and R/m.
- By the correspondence, there are no ideals strictly between m and R $\iff$ m is maximal in R.

Spoken Transcript [00:25:02 - 00:26:03] Prof. Rahul Pandharipande

And that's one of the ways that the idea of a maximal ideal is important. If you have a ring R — this is another way to say it — you have this ring R [writes: R], you have this set of proper ideals $\mathrm{cl}^*(R)$ [writes: $\mathrm{cl}^*(R)$], and the ones in here, the maximal ideals here, are exactly those ideals for which the quotient is a field [writes: max ideals are exactly those ideals for which R/m is a field].

That's actually a very normal way of thinking about it. You could take this as the definition, even.

So, of all these ideals in your ring, some of them have quotients which are fields: Those are exactly the maximal ideals.

Summary Diagram: Maximal Ideals and Field Quotients

On the blackboard, the lecturer summarizes the characterization of maximal ideals within the collection of proper ideals:

$R \qquad \mathrm{cl}^*(R)$

$\downarrow$

max ideals here are exactly those ideals for which ${}^{R}\!/_{m}$ is a field.

Corollary 5.7 (*Alternative Characterization of Maximal Ideals*): Let R be a ring. By Proposition 5.6, the set of maximal ideals of R is precisely the collection of proper ideals whose quotient ring is a field:

$$\{m \subset R \mid m \text{ is a maximal ideal}\} = \left\{m \in \mathrm{cl}^*(R) \;\middle|\; {}^{R}\!/_{m} \text{ is a field}\right\}. \tag{5.22}$$

Zorn's Lemma and the Existence of Maximal Ideals

Partially Ordered Sets (Posets)

Spoken Transcript [00:26:28 - 00:27:00] Prof. Rahul Pandharipande

This existence has to do with chains and posets and Zorn. This is maybe the first time I'll say things clearly about Zorn.

The first thing is: Do you remember what a poset is? Or do you know what a poset is? I will define it anyway, but this comes up in the foundations class, or where have you learned what a poset is? Yeah?

Student Remark

Maybe not with this name, but we definitely know about partially ordered sets.

Spoken Transcript [00:27:04 - 00:28:38] Prof. Rahul Pandharipande

Ah, yeah! Right. Yeah, you could call it a partially ordered set. But the standard mathematical English is poset.

What is a poset? [writes: What is a poset? POSet] This means partially ordered set. You see, one of the things I teach you here is also mathematical English, which happens to be a language I know. As I get older, I feel like it's the only language I know. [laughter]

What is a poset? A poset S is a set, and it has what's called a partial ordering [writes: S is a set and partial ordering].

The partial ordering means that I might be able to compare two elements. So, a partial ordering means if I have x and y in S [writes: $x, y \in S$], I might have this [writes: $x \leq y$], I might have that [writes: $y \leq x$], or they could be not comparable [writes: (or could be incomparable)] — incomparable [writes: incomparable].

And this incomparability is getting to the worry of the question.

It satisfies some properties. I could say this formally: The standard thing for any kind of relation is it's a subset of the product, and the subset of the product says it has... you know. I'll give you an example. But is this okay? I have a set and I have this relation, that's the partial ordering; sometimes I'm allowed to use it, sometimes I'm not. We'll do an example in a second.

Spoken Transcript [00:28:38 - 00:29:58] Prof. Rahul Pandharipande

Maybe the crucial example here is if I take R a ring [writes: R], and I take my set to be the set of all proper ideals [writes: $S = \mathrm{cl}^*(R)$], and I take the partial ordering to be set inclusion, the subset condition [writes "set inclusion" and "subset"]. So, it means if I have ideals I and J that are proper ideals [writes: $I, J \in \mathrm{cl}^*(R)$], then the partial ordering is... the definition is: I is a subset of J [writes: $I \leq J \overset{\text{def}}{\iff} I \subset J$].

That's kind of a standard partial ordering. Because if I have two ideals, one might be a subset of the other, the other might be a subset of the first, or they may have no subset relation at all. That's why it's called a partial ordering: I cannot always compare two elements. That's the partial part of the partial ordering.

This is the abstract poset; this is ours. This is going to be the one that plays a role: It's the set of proper ideals with the subset relation. Okay?

Recap: Partial orders are among the prerequisites (Definition 0.10). The lecturer first describes posets informally and gives the axioms a few minutes later (Definition 5.10).

Remark 5.8 (*Partially Ordered Set / Poset, Informally*)**:** A *"partially ordered set"* (or *"poset"*) is a pair $(S, \leq)$ consisting of a set S and a binary relation $\leq$ on S satisfying the axioms of Definition 5.10. Unlike in a total order, two elements need not be comparable: for $x, y \in S$, any of the following may occur:

- $x \leq y$ (x precedes or is equal to y),
- $y \leq x$ (y precedes or is equal to x),
- x and y are *"incomparable"* (neither $x \leq y$ nor $y \leq x$ holds).

Example 5.9 (*The Canonical Poset of Proper Ideals*)**:** Let R be a ring. The primary poset of interest is the set of all proper ideals of R ordered by set inclusion:

$$S = \mathrm{cl}^*(R) = \{I \in \mathrm{cl}(R) \mid I \neq R\}, \tag{5.23}$$

equipped with the partial order:

$$I \leq J \overset{\text{def}}{\iff} I \subseteq J \quad (\text{written on the board as } I \subset J). \tag{5.24}$$

Two proper ideals $I, J \in \mathrm{cl}^*(R)$ are incomparable if neither $I \subseteq J$ nor $J \subseteq I$ holds.

Notation: $\subset$ vs. $\subseteq$ for Set Inclusion

The lecturer consistently writes $\subset$ on the blackboard for set inclusion, but explicitly clarifies verbally that this symbol allows for equality ($I \subseteq J$).

Spoken Transcript [00:29:58 - 00:31:18] Prof. Rahul Pandharipande

And now, of course, partial orderings are axiomatic. They have three axiomatic properties. So, the axioms of a poset... [writes: Axioms of a poset]

You've never heard the term poset? [laughter]

Okay, the axioms of a poset: **(1)** is that if I take any element, you can compare it to itself, all right? [writes: (i) $x \le x$]

And let's just check that our poset here satisfies it. That says if I have an ideal, it's a subset of itself, which is true, right? [writes: $S = \mathrm{cl}^*(R), I \subset I$]

Do we agree on this symbol? [points to $\subset$] It means subset. It leaves open the possibility of being equal, right?

The second one is — this is sometimes called antisymmetry, I don't know — it says if you have this, if you're so fortunate that you can compare that one and the other way, then it implies that they're equal. [writes: (ii) $x \le y$ and $y \le x \implies x = y$]

That's the second axiom of a poset. How is that here? That says that if I have I and J such that I is inside J and J is inside I, then they're equal, which is true, right? [writes: $I \subset J, J \subset I \implies I = J$] So, we also satisfy that.

Spoken Transcript [00:31:18 - 00:31:22] Prof. Rahul Pandharipande

And the third axiom of a poset: What am I going to write here? Yeah?

Student Answer

Transitivity.

Spoken Transcript [00:31:24 - 00:32:23] Prof. Rahul Pandharipande

Transitivity, which says that if I'm so fortunate that I can compare x and y, and y and z in this order, then this implies that $x \le z$ in the poset. [writes: (iii) $x \le y, y \le z \implies x \le z$]

Those are the full axioms of a poset. A poset is a very simple mathematical structure: It's a set with a partial ordering that satisfies **(1)**, **(2)**, and **(3)**. That's what a poset is.

And for this proof, we're going to use this: It's very important that our set of proper ideals is a poset. It's the proper ideals, and the partial ordering is the subset relation. Certainly I is a subset of itself; in this order we get equality; and this one is also equally simple: If I is inside J and J is inside K, this of course implies that I is inside K. [writes: $I \subset J, J \subset K \implies I \subset K$]

So, we've checked all the poset relations.

Okay, so, we've done nothing really; this is just saying that the set of proper ideals is a poset.

Axioms of a Poset and Verification for Ideals

Definition 5.10 (*Axioms of a Partially Ordered Set*)**:** A binary relation $\le$ on a set S

is a *“partial order”* if it satisfies the following three axioms for all $x, y, z \in S$:

(1) Reflexivity: $x \leq x$.

(2) Antisymmetry: If $x \leq y$ and $y \leq x$, then $x = y$.

(3) Transitivity: If $x \leq y$ and $y \leq z$, then $x \leq z$.

Example 5.11 (*Verification of Poset Axioms for Proper Ideals*): For the set $S = \text{cl}^*(R)$ equipped with the subset relation $\subseteq$:

(1) Reflexivity: For any ideal $I \in \text{cl}^*(R)$, clearly $I \subseteq I$.

(2) Antisymmetry: For any $I, J \in \text{cl}^*(R)$, if $I \subseteq J$ and $J \subseteq I$, then $I = J$ by definition of set equality.

(3) Transitivity: For any $I, J, K \in \text{cl}^*(R)$, if $I \subseteq J$ and $J \subseteq K$, then $I \subseteq K$.

Thus, $(\text{cl}^*(R), \subseteq)$ is a poset.

Maximal Elements in Posets

Spoken Transcript [00:32:38 - 00:33:55] Prof. Rahul Pandharipande

Now, in posets, there’s a notion of a maximal element of a poset. And that, in our example, is going to correspond exactly to the notion of a maximal ideal of a ring.

If S is an abstract poset — we have here $(S, \leq)$ is a poset [writes: $(S, \leq)$ is a Poset] — there’s a definition of a maximal element: $x \in S$ is a maximal element of S [writes: $x \in S$ is a max element of S]. And the definition for that — and again, you can guess this — the definition is: If somebody is on top of x, then this implies actually $x = y$ [writes: $\overset{\text{def}}{\iff}$ if $x \leq y \implies x = y$].

No one is on top of it. That’s what it means. Okay? That’s the definition of a maximal element: In any poset you can define a maximal element; it means that there’s nobody on top of it here.

Spoken Transcript [00:33:55 - 00:34:33] Prof. Rahul Pandharipande

And in our case, the poset here [writes: $\text{cl}^*(R), \subset$], the notion of maximal element of this poset is the same as a maximal ideal [writes: max element of this Poset = max ideal].

It’s just the same definition. If I take this as my poset and I use the abstract definition of maximal element, then that means my element is a maximal ideal. Is that clear? It’s just saying there’s nobody on top of it here. Okay?

Definition 5.12 (*Maximal Element of a Poset*): Let $(S, \leq)$ be a partially ordered set. An element $x \in S$ is a *“maximal element”* of S $\overset{\text{def}}{\iff}$

$$\forall y \in S: \quad x \leq y \implies x = y. \tag{5.25}$$

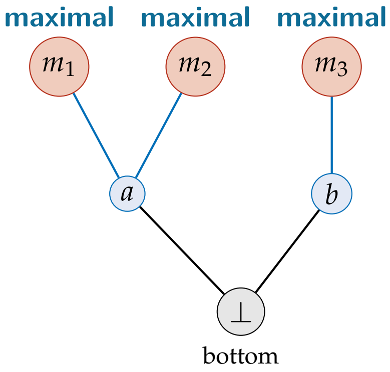
In words: no element in S is strictly greater than x.

Proposition 5.13 (*Equivalence with Maximal Ideals*): Let R be a ring. An ideal $m \subset R$ is a maximal ideal of R in the sense of Definition 5.4 if and only if m is a maximal element of the poset $(\text{cl}^*(R), \subseteq)$.

Maximal Element vs. Maximum / Greatest Element: Going beyond the lecture: it is crucial to distinguish between a **maximal element** and a **greatest element** (or maximum):

- **Greatest Element (Maximum):** An element $g \in S$ such that $y \leq g$ for **all** $y \in S$. A greatest element must be comparable to every element in the poset.
- **Maximal Element:** An element $x \in S$ such that **no** element strictly exceeds it ($x \leq y \implies x = y$). There can be multiple incomparable maximal elements in a poset!

In $(\mathrm{cl}^*(R), \subseteq)$, there is generally no greatest element, but there can be many distinct maximal ideals (e.g., in $\mathbb{Z}$, the ideals $(2), (3), (5), \ldots$ are all maximal elements of $\mathrm{cl}^*(\mathbb{Z})$, and (2) and (3) are incomparable).



Spoken Transcript [00:34:33 - 00:36:33] Prof. Rahul Pandharipande

This existence of a maximal ideal, that was the question. Question of existence in $\mathrm{cl}^*(R)$ of a maximal ideal [writes: Question: existence in $\mathrm{cl}^*(R)$ of a max ideal]: This is really the same as some kind of abstract question of existence of maximal elements of posets [writes: Question of existence of max elts of Posets].

And we haven't done anything yet, because not all posets have maximal elements.

There is a tool, which is essentially in foundations and logic, about when posets have maximal elements, and that's Zorn's lemma [writes: Zorn's Lemma].

Presumably you've all heard of Zorn's lemma? I don't know if you've heard it in this context, but I will explain it, because it's crucial here.

And this is why I would view this whole answer as kind of disappointing, because it's hard to like Zorn's lemma. [laughter]

Actually, his name is Max Zorn, which is kind of ironic, but... [laughter]

Zorn's Lemma and Chains

Spoken Transcript [00:36:33 - 00:38:18] Prof. Rahul Pandharipande

What does Zorn say? Zorn's lemma tells you: If you have a poset that satisfies what's called a chain condition, then it always has a maximal element. Let's first say what that is.

Zorn: Let $(S, \leq)$ be a poset in which every — I have to explain it — every chain is bounded [writes: Zorn: Let $(S, \leq), S \neq \emptyset$ be a Poset in which every chain is bounded]... Somehow my handwriting has deteriorated a bit... every chain is bounded.

I have to explain chain and bounded.

Zorn: We have an abstract poset — I've already explained poset — in which every chain is bounded, then S must contain a maximal element [writes: then S must contain a max element].

And there's one more thing: S not empty, of course, because that would be silly, all right? [writes: $S \neq \emptyset$] So, you have a non-empty poset in which every chain is bounded, then S must contain a maximal element. That's Zorn's lemma.

To explain it...

And by the way, foundationally this is some manifestation of the Axiom of Choice [writes: (Axiom of Choice)]. And if you make the right hypothesis, it's equivalent to the Axiom of Choice.

Lemma 5.14 (*Zorn's Lemma*): Let $(S, \leq)$ be a non-empty partially ordered set ($S \neq \emptyset$). If every chain in S is bounded (i.e., has an upper bound in S), then S contains at least one maximal element.

Beyond the Lecture: Zorn's Lemma and the Axiom of Choice

Within Zermelo–Fraenkel set theory (ZF), Zorn's Lemma is logically equivalent to the Axiom of Choice (AC) and the Well-Ordering Theorem (WOT). Max Zorn formulated this maximum principle in 1935, although equivalent formulations were independently discovered earlier by Kazimierz Kuratowski (1922).

Spoken Transcript [00:38:18 - 00:39:26] Prof. Rahul Pandharipande

Okay, so, I have to explain what *"every chain is bounded"* means.

Let T be a subset of S of our poset [writes: Let $T \subset S$]. Then T is a chain [writes: T is a chain] — that's a definition — if for all elements x and y in T, either $x \leq y$ or $y \leq x$ [writes "if $\forall x, y \in T$" and "either $x \leq y$ or $y \leq x$"].

This means, for a poset, not all elements are comparable, as we've discussed. So, a subset is a chain if every two elements are comparable. Okay?

We say it's a chain because then you kind of view them all being in an order, because you can compare them all and you can sort them and put them in order. And remember, when I say *"either/or"*, the proper mathematical usage of that in English allows for both things to be true; it's not exclusive.

It's not *"entweder ... oder"*, which would be wrong if I wrote that, but in English this is correct: either that or that.

Spoken Transcript [00:39:26 - 00:40:43] Prof. Rahul Pandharipande

This is a chain. What is a chain here in our poset of proper ideals? A chain is a set T of ideals such that if I take any two of them, $I \in T$ and $J \in T$ [writes "$\mathrm{cl}^*(R), T = \{\dots\}$" and "$I, J \in T$"], we either have one inside the other or the other inside one [writes "$I \subset J$" and "$J \subset I$"].

You view this chain as basically... this is not correct, because the index set could be uncountable numbers and all that nonsense from whatever foundations class you took, but the way you think about it is, a chain is like this: [writes: $I_1 \subset I_2 \subset \dots$]

A chain here is a sequence of ideals where they just kind of grow, because what a chain means is that for any two of them, one is inside the other. So, you can think of a chain like this. Of course, as an object it's much more general, because this would be like a countable chain, but I could consider uncountable chains, or the first uncountable ordinal, or whatever you want.

That's the definition of a chain. That's the abstract definition of a chain, and this is what it means to be here.

Definition 5.15 (*Chain*): Let $(S, \leq)$ be a partially ordered set. A subset $T \subseteq S$ is called a "*chain*" if every two elements in T are comparable:

$$\forall x, y \in T: \quad x \leq y \quad \vee \quad y \leq x. \tag{5.26}$$

Example 5.16 (*Chains of Proper Ideals*): In the poset of proper ideals $(\mathrm{cl}^*(R), \subseteq)$, a chain is a collection of proper ideals $T \subseteq \mathrm{cl}^*(R)$ such that for any $I, J \in T$:

$$I \subseteq J \quad \vee \quad J \subseteq I. \tag{5.27}$$

Heuristically, one often visualizes a chain as an ascending nested sequence of ideals:

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots \subseteq I_n \subseteq \dots,$$

although a chain may be arbitrarily infinite and uncountably indexed.

Spoken Transcript [00:40:43 - 00:42:03] Prof. Rahul Pandharipande

Now, for a chain to be bounded: A chain T in S is bounded [writes: a chain $T \subset S$ is bounded] — now, this is the definition of bounded — means there exists a $z \in S$ such that z beats everybody, such that for every x on the chain, $x \leq z$ [writes: $\exists z \in S$ s.t. $\forall x \in T, x \leq z$].

So, z is on top of everybody in the chain. That's what a chain being bounded is. What's important here is this z need only be in S; it doesn't have to be in the chain. Okay?

This is saying that whenever I have this chain, I can find somebody that contains all of them. That's what it means.

Are you happy with this?

For an abstract poset, I have a notion of poset, I have a notion of chain — chain is just a subset where everybody is ordered —, and this chain is bounded if I can reach into S somewhere, find somebody who's on top of everybody in that chain.

Definition 5.17 (*Bounded Chain / Upper Bound*): Let $(S, \leq)$ be a poset and $T \subseteq S$ a chain. The chain T is said to be *"bounded"* (above) in S if there exists an element $z \in S$ such that:

$$\forall x \in T: \quad x \leq z. \tag{5.28}$$

Such an element z is called an *"upper bound"* for T. Crucially, z must belong to the parent poset S, but z is **not** required to belong to the chain T itself.

Proof of the Existence of Maximal Ideals (in textbooks: Krull's Theorem)

Spoken Transcript [00:42:03 - 00:42:52] Prof. Rahul Pandharipande

Now we come to the first question in this discussion which isn't nonsense: If I have this poset of proper ideals, I have a notion of chain: Are chains bounded here? That's actually the crucial thing; this really cuts to the heart of the matter. [writes on board: Are chains in $(\mathrm{cl}^*(R), \subset)$ bounded?]

That means: If I take any chain, can I find another ideal that's on top of all of them? This is actually the only part that isn't formal here.

How would I do that? Yeah?

Student Answer

I think we can take their union.

Spoken Transcript [00:42:54 - 00:44:24] Prof. Rahul Pandharipande

Yeah! This is the only idea in the whole application of Zorn: just one single idea!

You have this T, that's a chain. I want somebody to beat all of them, so, I'll call it K, and K is the ideal which is the union over all the ideals in T of I [writes: $K = \bigcup_{I \in T} I \subset R$]. That's inside R. Okay?

That's a set-theoretic thing: I just take my chain and I take the union of everybody. But then I have to prove — or whoever asserted it has to prove — this K is an element of $\mathrm{cl}^*(R)$ [writes: I have to prove $K \in \mathrm{cl}^*(R)$].

And that's kind of fun!

First, you have to prove this K is an ideal. Now, how do you prove that K is an ideal? Well, it should have zero, but all the ideals have zero. If a and b are in K, we need to show that their sum is in K. That means a is in some I and b is in some I', but either $I \subset I'$ or $I' \subset I$ [writes: $a \in I, b \in I', I \subset I'$ or $I' \subset I$]. So, you take the bigger one, and then their sum is in the bigger one, so it's okay also. And also for closing on the product property.

Finally, to be in here it has to be a proper ideal. Why is this a proper ideal?

Student Question

Does it contain 1?

Spoken Transcript [00:44:26 - 00:45:20] Prof. Rahul Pandharipande

Yeah, because none of these contain 1, so this doesn't contain 1. And if it were not a proper ideal, it would contain 1, it would contain everything. So, it's also a proper ideal.

So, there's a little lemma here which is fun to do. It says that this element K, the union — that's the only part of this that isn't formal —: If I take this poset of proper ideals, in that poset every chain is bounded, and it's by just this construction.

And so, then we can apply Zorn's lemma! In our poset, we satisfy all the conditions here: It's a poset in which every chain is bounded, then it must contain a maximal element, and that's a maximal ideal.

So, that's the official proof that every ring has a maximal ideal. It's an application of Zorn's lemma, and for every mathematician I would view this as deeply disappointing.

Okay, that's it.

Bounding Chains of Ideals and the Existence of Maximal Ideals

Lemma 5.18 (*Chains of Proper Ideals Are Bounded*)**:** Let R be a non-zero ring and let $T \subseteq \mathrm{cl}^*(R)$ be a non-empty chain of proper ideals. Then the union:

$$K := \bigcup_{I \in T} I \tag{5.29}$$

is a proper ideal of R ($K \in \mathrm{cl}^*(R)$) and serves as an upper bound for T under inclusion.

Proof

We verify that $K = \bigcup_{I \in T} I$ satisfies the required conditions:

(1) **K is non-empty:** Since T is non-empty and every ideal $I \in T$ contains 0, we have $0 \in K$.

(2) **Additive Closure:** Let $a, b \in K$. By definition of the union, there exist ideals $I, I' \in T$ such that $a \in I$ and $b \in I'$. Since T is a chain, either $I \subseteq I'$ or $I' \subseteq I$. Assume without loss of generality that $I \subseteq I'$. Then both $a, b \in I'$. Since I' is an ideal, $a + b \in I' \subseteq K$.

(3) **Absorption (Ideal Property):** Let $r \in R$ and $a \in K$. Then $a \in I$ for some $I \in T$. Since I is an ideal, $ra \in I \subseteq K$.

(4) **Properness ($K \neq R$):** Since every ideal $I \in T$ is a proper ideal ($I \in \mathrm{cl}^*(R)$), none of them contains the multiplicative identity 1 (for if $1 \in I$, then $I = R$). Therefore, $1 \notin \bigcup_{I \in T} I = K$. Since $1 \notin K$, it follows that $K \neq R$, so $K \in \mathrm{cl}^*(R)$.

Finally, for every $I \in T$, clearly $I \subseteq K$. Hence, K is an upper bound for T in the poset $(\mathrm{cl}^*(R), \subseteq)$. **Q.E.D.**

Theorem 5.19 (*Existence of Maximal Ideals (in textbooks: Krull's Theorem)*)**:** Every non-zero commutative ring $R \neq 0$ contains at least one maximal ideal.

Proof

Let $R \neq 0$ be a non-zero ring. Consider the poset $S = \mathrm{cl}^*(R)$ of proper ideals ordered by inclusion.

- Since $R \neq 0$, the zero ideal (0) is a proper ideal ($(0) \neq R$), so $S = \mathrm{cl}^*(R)$ is non-empty.
- By Lemma 5.18, every non-empty chain $T \subseteq S$ has an upper bound in S, namely $K = \bigcup_{I \in T} I \in S$; the empty chain is bounded by any element of S.

By Zorn's Lemma (Lemma 5.14), the poset S contains at least one maximal element m. By Proposition 5.13, m is a maximal ideal of R. **Q.E.D.**

Why the Chain Property is Crucial for Unions of Ideals: Going beyond the lecture: in general, the union of two ideals $I_1 \cup I_2$ need not be an ideal. For example, in $\mathbb{Z}$, the union $(2) \cup (3)$ contains 2 and 3, but not $2 + 3 = 5$. For a chain T, the union $\bigcup_{I \in T} I$ is an ideal: any two elements $a, b \in K$ lie in one common ideal of T, so additive closure holds.

Foundational Discussion and Mathematical Philosophy

Board Writing

The lecturer pushes the middle blackboard upward and writes the following foundational summary on the lower board panel:

$$\text{Zorn} \implies \text{Every ring} \neq 0 \text{ has a max ideal} \quad \uparrow \text{(Axiom of Choice)}$$

Spoken Transcript [00:45:22 - 00:46:53] Prof. Rahul Pandharipande

Okay, so, because of this Zorn argument, we have the statement here that every ring has a maximal ideal [writes: Zorn $\implies$ Every ring $\neq 0$ has a max ideal].

And for this, you of course have to take every ring that's not zero, because for the zero ring, $\mathrm{cl}^*(R)$ is empty, and it doesn't satisfy Max Zorn's condition. So, every non-zero ring has a maximal ideal.

The reason I say the proof is deeply disappointing is because this depended on Zorn, which has to do — and you know, formally Zorn is proven by the Axiom of Choice — and so it depends on the foundational organization of mathematics. And it's extremely unfortunate, because maybe you meet a Martian who comes from a different math culture, and the Martians have a different foundation, and it's possible you could have the same definition of rings, where you assert it has a maximal ideal, and the Martian says you can't prove it. That is unfortunate [writes: ↑ (Axiom of Choice)].

There are different ways to think about this, but it is something that I view as unfortunate.

In particular, this argument doesn't give you any hint of how to construct a maximal ideal. It doesn't help you that way. The statement is just that our logical conventions are such that every ring has a maximal ideal. And you know, you could have different logical conventions.

Spoken Transcript [00:46:52 - 00:47:51] Prof. Rahul Pandharipande

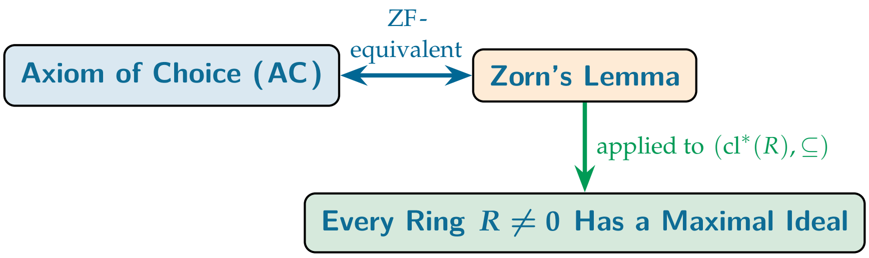
Does this affect me practically? No, because I work on… Some people asked about what my own research is, and Lycka put some links to some public talks, so you can look at it. Part of what I think about is algebra, and the other half is geometry. Actually, I view myself mostly as a geometer.

This issue about Zorn's lemma doesn't affect me personally.

The reason is because I work in certain classes of rings where, rather than the abstract general ring, if you work in some kind of specialized classes of rings, then you can construct maximal ideals and show their existence by some other methods. You don't use Zorn.

Actually, I don't like the idea that the work in math I do is dependent on these logical foundations. I just don't want to do that kind of math. Other people can, but…

Foundational Summary: Zorn and Maximal Ideals


Axiom of Choice (AC)
ZF-equivalent
Zorn's Lemma
applied to $(\mathrm{cl}^*(R), \subseteq)$
Every Ring $R \neq 0$ Has a Maximal Ideal

***Insight:* The Martian Thought Experiment and Non-Constructive Mathematics**

The lecturer highlights a fundamental philosophical dilemma in algebra: the existence of maximal ideals relies on Zorn's Lemma, which is a form of the Axiom of Choice. Consequently, the proof gives no hint of how to construct a maximal ideal. A *"Martian"* from a different mathematical culture, with different foundations, could have the same definition of rings and say that the statement cannot be proved. In the specialized classes of rings the lecturer works with, maximal ideals can be constructed by other methods, without Zorn's Lemma.

Beyond the Lecture: Rings Where No Zorn Is Needed

In coordinate rings of algebraic varieties, for example, maximal ideals correspond to points by Hilbert's Nullstellensatz, so they can be written down without Zorn's Lemma.

Prime Ideals

Definition of Prime Ideals

Spoken Transcript [00:47:51 - 00:49:27] Prof. Rahul Pandharipande

Okay, so, sorry, that was a small discussion there.

That was the end of our introduction to maximal ideals, and there's now a parallel discussion of prime ideals [writes: Parallel discussion of prime ideals]. So, this is a new definition: prime ideals.

Let us say it: An ideal P inside our ring is prime [writes: An ideal $P \subset R$ is prime] — and this is the definition of prime [writes: $\overset{\text{def}}{\iff}$] — and there are two properties:

The first property is that P is in our $\mathrm{cl}^*(R)$ [writes: $* \ P \in \mathrm{cl}^*(R)$]. This means it's an ideal and it's not the whole ring again. It's in the same class. In other words, P is not the whole ring [writes: $P \neq R$]. Maximal ideals can't be the whole ring, and prime ideals can't either, all right?

The second one is the reason why it's called prime. It says that for all elements $a, b \in R$, if $a \notin P$ and $b \notin P$, then this implies their product $ab \notin P$ [writes "$* \ \forall a, b \in R$", "if $a \notin P$", "$b \notin P$" and "$\implies ab \notin P$"]. So, that's the condition that has teeth.

All right, so, that's the definition of a prime ideal, and you have to digest it a little bit. The main thing is, a prime ideal is an ideal, it's not the whole ring, and it has this property: If I take two people outside it and multiply, they stay out.

Definition 5.20 (*Prime Ideal*): Let R be a commutative ring and let $P \subset R$ be an ideal. Then P is a *"prime ideal"* $\overset{\text{def}}{\iff}$ the following two conditions hold:

(1) Properness: $P \in \mathrm{cl}^*(R)$, which means:

$$P \neq R \quad (\text{equivalently, } 1 \notin P).$$

(2) Prime Condition: For all elements $a, b \in R$:

$$a \notin P \text{ and } b \notin P \implies ab \notin P. \tag{5.30}$$

Contrapositive Formulation of the Prime Condition: Going beyond the lecture: the prime condition written on the board:

$$a \notin P \wedge b \notin P \implies ab \notin P$$

is logically equivalent to its contrapositive:

$$ab \in P \implies a \in P \vee b \in P.$$

This directly mirrors Euclid's Lemma for prime numbers: if a prime p divides a product ab, then $p \mid a$ or $p \mid b$.

Prime Ideals in the Integers

Spoken Transcript [00:49:27 - 00:50:54] Prof. Rahul Pandharipande

All right. And so, why is it called a prime ideal?

You can call it anything you want, but the reason it's called a prime ideal... [writes: Reason for the name:]

The reason for the name is that if I take this ideal n... [drops chalk] Oops. You can imagine what the reason is: If I take the ideal generated by n [writes: $(n) \subset \mathbb{Z}$], which we sometimes also call $n\mathbb{Z}$ inside $\mathbb{Z}$ [writes: $n\mathbb{Z}$], then the claim is that... I mean, zero is always special, and we'll discuss that in a second, but if we have some positive integer greater than or equal to 2 [writes: Claim: If $n \in \mathbb{Z}_{\geq 2}$ (or $n \in \mathbb{N}_{\geq 2}$)] — because if $n = 1$, it's not a prime ideal — so, if you have some integer here with $n \in \{2, 3, 4, \dots\}$ [writes: $n = \{2, 3, 4, \dots\}$], then this is a prime ideal if and only if n is a prime number [writes: Then $n\mathbb{Z}$ is prime ideal $\iff$ n is prime number].

There are the boundary cases here: If we take zero, this ideal [writes: $(0) \subset \mathbb{Z}$], is this ideal prime or not?

Student Answer

It is, because $\mathbb{Z}$ is an integral domain.

Spoken Transcript [00:51:03 - 00:51:36] Prof. Rahul Pandharipande

Yeah, the answer is yes: prime [writes: is PRIME].

Because it's not all of $\mathbb{Z}$, it passes this condition, and this condition says if I take two elements and they're both not there, then their product is not there. In this case, it says if I take two elements that are not zero, their product is not zero, which is true here, okay? So, that's one special case.

Is this a prime ideal? [writes: $(1) \subset \mathbb{Z}$]

Student Answer

No, it's the whole space.

Spoken Transcript [00:51:37 - 00:53:20] Prof. Rahul Pandharipande

Yeah, it's not prime [writes: is NOT prime]. And that's because it fails this property: It's the whole thing, it's all of $\mathbb{Z}$.

Anyway, once you're out of these special cases, the notion of prime is it's a prime number, and you should make sure you believe this statement, all right?

I'm not going to give you the proof of it. For example, if you take the ideal (4) inside $\mathbb{Z}$, why is it not prime? Because if I take 2, the number 2 is not in this ideal, because 2 is not a multiple of 4. Actually, maybe I should do 6 then. [erases 4, writes: $(6) \subset \mathbb{Z}$]

$2 \notin (6)$ and $3 \notin (6)$ [writes "$2 \notin (6)$" and "$3 \notin (6)$"], but if I multiply them, $2 \cdot 3 \in (6)$

[writes: $2 \cdot 3 \in (6)$].

This shows that (6) is not prime [writes: NOT Prime!], because to be prime, if you take two people outside and multiply, they have to stay outside. And here we don't: You have two people outside, you multiply, and you got inside. That's bad — you're not prime.

That's an example of this: It's prime if and only if the number is prime; 6 is not prime. This is essentially a proof if you just think about it. I leave this to you: exercise [writes: Exercise].

It's good to understand; that's why it's called a prime ideal. I said that this term ideal comes from the idea of an idealized number: Instead of integers we consider rings, and instead of a number we consider an ideal. This is not perfect, and there are different ways to think about it, but in that way of thinking about the language, this idea of prime corresponds in some sense to a prime number. There are other ways to do it, but that's part of the language.

Prime Ideals in the Ring of Integers $\mathbb{Z}$

Claim 5.21 (*Prime Ideals in $\mathbb{Z}$*)**:** Let $n \in \mathbb{Z}_{\geq 2}$. The principal ideal $(n) = n\mathbb{Z} \subset \mathbb{Z}$ is a prime ideal if and only if n is a prime number.

Boundary Cases for Principal Ideals in $\mathbb{Z}$:

- **The Zero Ideal $(0) \subset \mathbb{Z}$ (PRIME):** Since $\mathbb{Z}$ has no zero divisors ($a \neq 0 \wedge b \neq 0 \implies ab \neq 0$), (0) satisfies (5.30). Moreover, $(0) \neq \mathbb{Z}$, so (0) is indeed a prime ideal.
- **The Unit Ideal $(1) = \mathbb{Z}$ (NOT PRIME):** Since $(1) = \mathbb{Z}$, it is not a proper ideal ($(1) \notin \mathrm{cl}^*(\mathbb{Z})$), violating the properness axiom.

Example 5.22 (*Why $(6) \subset \mathbb{Z}$ is Not Prime*)**:** Consider the principal ideal $(6) = 6\mathbb{Z} \subset \mathbb{Z}$. Taking $a = 2$ and $b = 3$:

$$2 \notin (6) \quad \text{and} \quad 3 \notin (6), \qquad \text{but} \qquad 2 \cdot 3 = 6 \in (6).$$

Because two elements outside the ideal produce a product inside the ideal, (6) is **not** a prime ideal.

Exercise 5.23 (*Characterization of Prime Ideals in $\mathbb{Z}$*)**:** Prove Claim 5.21: Show that for $n \in \mathbb{N}_{\geq 2}$, (n) is a prime ideal in $\mathbb{Z}$ if and only if n is a prime number.

III Quotients by Prime Ideals and Integral Domains

Spoken Transcript [00:53:20 - 00:54:40] Prof. Rahul Pandharipande

Then we want to prove a theorem like the field theorem that says: [writes: Prop: ${}^{R}\!/\!_{P}$ is an int. domain $\iff$ $P \subset R$ is a prime ideal]

R mod P is an integral domain — this is a proposition — R mod P is an integral domain if and only if $P \subset R$ is a prime ideal [writes: $\iff$ $P \subset R$ is a prime ideal].

I want to prove this, and I remind you of the proposition from the first lecture, which I have not had the skill enough to leave on the board: R mod m is a field if and only if

$m \subset R$ is a maximal ideal [writes: Prop: R/m is a field $\iff$ $m \subset R$ is a max ideal].

This was the statement that we already proved just a few minutes ago, and this is the analogous statement for integral domains. Everything is just analogous: Here we want to consider which ideals have quotients that are fields, and the answer is those are the maximal ideals; and here we ask which ideals have the property that the quotient is an integral domain, and those are the prime ideals, okay? So, we have to prove this one. But the parallelism here should be clear.

Characterization of Prime Ideals via Integral Domains

Proposition 5.24 (*Prime Ideals via Integral Domains*)**:** Let R be a commutative ring and $P \subset R$ an ideal. Then the quotient ring R/P is an integral domain if and only if P is a prime ideal of R.

Parallelism: Maximal Ideals vs. Prime Ideals

The two fundamental classes of ideals in commutative ring theory correspond precisely to key algebraic properties of their quotient rings:

$$R/m \text{ is a field} \iff m \subset R \text{ is a maximal ideal,}$$
$$R/P \text{ is an integral domain} \iff P \subset R \text{ is a prime ideal.}$$

Spoken Transcript [00:54:41 - 00:56:19] Prof. Rahul Pandharipande

We prove the proposition about prime ideals [writes: Proof Prop (Prime ideals)].

We have to start somewhere. Let $P \subset R$ be a prime ideal [writes: Let $P \subset R$ be a prime ideal], okay?

Again, we have to think about the ground cases carefully. We must prove that R mod P is an integral domain [writes: $\implies$ we must prove that R/P is an int. domain].

Now, one of the properties in the definition of integral domains is you cannot be the zero ring. That was one of the properties in the definition: Integral domains by definition are not the zero ring [writes: (not zero ring)].

So, the first thing to prove this is an integral domain, we need to know this is not the zero ring. And that's true, because to be prime you can't be the whole ring, okay?

That makes the matching of all of these border cases in the definitions work: To be prime you can't be everything, so that means this thing is not the zero ring. It at least passes that part of the definition. Is that clear?

Okay, but that's not the teeth of the definition of an integral domain.

Spoken Transcript [00:56:19 - 00:57:18] Prof. Rahul Pandharipande

We have to show that given elements $\bar{x}$ and $\bar{y}$ in R mod P [writes: We have to show given $\bar{x}, \bar{y} \in R/P$] — with $x, y \in R$ [writes: (with $x, y \in R$)], these are the equivalence classes of x

and y — such that $\bar{x} \neq \bar{0}$ and $\bar{y} \neq \bar{0}$, then I need to show that $\bar{x}\bar{y} \neq \bar{0}$ [writes: such that $\bar{x} \neq \bar{0}$ and $\bar{y} \neq \bar{0} \implies \bar{x}\bar{y} \neq \bar{0}$]. Is that clear?

Now, I've just used bars everywhere here. Sometimes I do, sometimes I don't, and that's just how it goes. But anyway, in this proof I'm going to use bars everywhere. Yeah?

Student Question

I have a question about the prime ideals... If every field is also a ring, wouldn't that field have a prime ideal that covers the whole thing and therefore has to be the zero ring, therefore break this definition? Or does this proposition exclude...

Spoken Transcript [00:57:40 - 00:57:44] Prof. Rahul Pandharipande

No, it's okay to be a field. A field is also an integral domain.

Student Interjection

What if R is a field?

Spoken Transcript [00:57:46 - 00:57:47] Prof. Rahul Pandharipande

Yeah...

Student Question, continued

Then the prime ideal...

Spoken Transcript [00:57:53 - 00:58:06] Prof. Rahul Pandharipande

Wait. Maybe we discuss this point afterward. If R is a field, there is only one prime ideal, which is zero, because the prime ideals are never allowed to be everything. Does that answer the question? Okay.

Spoken Transcript [00:58:07 - 00:58:38] Prof. Rahul Pandharipande

I need to prove this statement: We assume it's a prime ideal, and I need to prove that R mod P is an integral domain.

And again, there's nothing happening here: This proof has like zero ideas; you just have to unravel the definitions.

We check here these two elements, and assuming that neither of them is zero, we need to know the product is not zero.

Spoken Transcript [00:58:38 - 01:00:16] Prof. Rahul Pandharipande

I warn you in a lot of these proofs that there's nothing happening. It doesn't mean you shouldn't pay attention. It's just that there's a kind of proof and statement in mathematics where nothing is actually happening in it — it's just kind of unraveling the definitions.

It doesn't mean it's not confusing or it doesn't require concentration; it just means that there's no actual interesting logical motion happening in the proof, and this is one of them.

Sometimes as a student it's useful to know whether that's the case or not.

In this kind of math class, there are a lot of definitions. And these definitions have taken a lot of time to be written in a very careful and concise way. Usually, these definitions come after like a century of mathematical exploration before the absolutely perfect and complete definitions are created.

And these definitions contain a lot. That's why all these proofs just unravel themselves: because the definitions are all perfect.

Another way of saying this is, Nietzsche has this great sentence where he says that *"I try to write in ten sentences what other men take a whole book to write."* And you can say this a little bit about mathematics: These definitions in ten sentences take a century to develop.

Nietzsche was also a professor in Basel, if we're keeping track.

Nietzsche's Maxim on Conciseness

While discussing the nature of mathematical definitions, the lecturer quotes Friedrich Nietzsche's aphorism from **Götzen-Dämmerung** (Twilight of the Idols): *"In zehn Sätzen zu sagen, was jeder andere in einem Buche sagt — was jeder andere in einem Buche* ***nicht*** *sagt..."*

Spoken Transcript [01:00:16 - 01:01:22] Prof. Rahul Pandharipande

Okay, so let's try to prove this. That was a long board erase.

$\bar{x} \neq \bar{0}$ means that $x \notin P$ [writes: $\bar{x} \neq \bar{0} \implies x \notin P$]. That's what it means to be non-zero in the quotient, right?

The language here is that when I write things without a bar, they're in R, and with this bar, they're in R mod P.

And $\bar{y} \neq \bar{0}$ implies that $y \notin P$ [writes: $\bar{y} \neq \bar{0} \implies y \notin P$].

Then I have to check that their product is not zero in the quotient ring. Now we use the fact that P is prime [writes: P prime]. This means that $xy \notin P$ [writes: $xy \notin P$] — that's actually using the prime property for the first time.

And this means that $\bar{x}\bar{y} \neq \bar{0}$ [writes: $\implies \bar{x}\bar{y} \neq \bar{0}$]. So, we've shown that R mod P is an integral domain [writes: ${}^{R}\!/_{P}$ is an int. domain].

You will agree that there was no idea in this proof; it was just unraveling.

Spoken Transcript [01:01:22 - 01:02:08] Prof. Rahul Pandharipande

And I leave the other way for you. Exercise: Do the other arrow [writes: Ex $\Downarrow$ (again no ideas)]. You'll find again there's no ideas, just unraveling. The way that prime ideals have been defined just matches perfectly with what it means to be an integral domain.

So, this is an exercise: again, no ideas, which does not mean it's not confusing. It can be very confusing to have no ideas; that's a separate issue. [laughter] But I leave this for you to unravel.

All right?

Proof of Proposition: Quotients by Prime Ideals

Proof of Proposition 5.24

Let R be a commutative ring and $P \subset R$ an ideal.

Forward Direction ($\Longrightarrow$): Assume P is a prime ideal.

- **Non-triviality of the Quotient Ring:** Since P is prime, $P \in \mathrm{cl}^*(R)$, meaning $P \neq R$. Therefore, $1 \notin P$, which implies $\bar{1} \neq \bar{0}$ in ${}^{R}\!/_{P}$. Thus, ${}^{R}\!/_{P}$ is not the zero ring.
- **Absence of Zero Divisors:** Let $\bar{x}, \bar{y} \in {}^{R}\!/_{P}$ with representatives $x, y \in R$. Suppose $\bar{x} \neq \bar{0}$ and $\bar{y} \neq \bar{0}$. By definition of equivalence classes in the quotient ring:
$$\bar{x} \neq \bar{0} \implies x \notin P,$$
$$\bar{y} \neq \bar{0} \implies y \notin P.$$
Since P is a prime ideal, the prime property (5.30) directly implies:
$$xy \notin P.$$
Translating this back to the quotient ring:
$$xy \notin P \implies \bar{x}\bar{y} = \overline{xy} \neq \bar{0}.$$
Hence, ${}^{R}\!/_{P}$ has no zero divisors.

Combining non-triviality and zero-divisor freeness, ${}^{R}\!/_{P}$ is an integral domain.

Q.E.D.

Exercise 5.25 (*Reverse Direction of Proposition 5.24*): Prove the converse implication ($\Longleftarrow$): Assume ${}^{R}\!/_{P}$ is an integral domain.

(1) Since ${}^{R}\!/_{P}$ is not the zero ring, deduce that $P \subsetneq R$ (so $P \in \mathrm{cl}^*(R)$).

(2) Let $a, b \in R$ with $ab \in P$. Show that $\bar{a}\bar{b} = \bar{0}$ in ${}^{R}\!/_{P}$, and use the integral domain property to conclude that $a \in P$ or $b \in P$.

Corollaries: Maximal Ideals Are Prime

Spoken Transcript [01:02:08 - 01:03:02] Prof. Rahul Pandharipande

Some consequences of this — and this is maybe related to the question before —: From this, you can draw some conclusions.

For example: $m \subset R$ is maximal implies that m is prime [writes: $m \subset R$ is max $\implies$ $m \subset R$ is prime].

Because if m is maximal, then R mod m is a field, every field is an integral domain, and that means that m must be prime, all right?

This is just a linguistic consequence of these two.

In particular, that's one consequence: Every ring has a prime ideal [writes "→ Every Ring has a prime ideal. (Zorn)", a slip for every ring $\neq 0$].

This is also Zorn, because every ring has a maximal ideal, and a maximal ideal is prime, so in particular every ring has a prime ideal.

All right.

Maximal Ideals are Prime

Corollary 5.26 (*Maximal Ideals are Prime*)**:** Let R be a commutative ring and $m \subset R$ an ideal. If m is a maximal ideal, then m is a prime ideal:

$$m \text{ maximal} \implies m \text{ prime}. \tag{5.31}$$

Proof

If $m \subset R$ is a maximal ideal, then by Proposition 5.6, the quotient ring R/m is a field. Since every field is an integral domain, R/m is an integral domain. By Proposition 5.24, this implies that m is a prime ideal. **Q.E.D.**

Corollary 5.27 (*Existence of Prime Ideals*)**:** Every non-zero commutative ring $R \neq 0$ contains at least one prime ideal.

Proof

By Theorem 5.19, every non-zero commutative ring $R \neq 0$ possesses at least one maximal ideal m. By Corollary 5.26, m is also a prime ideal.

Q.E.D.

Spoken Transcript [01:03:02 - 01:03:44] Prof. Rahul Pandharipande

We'll do some more things with these later, but that's like the basic aspects of maximal and prime ideals.

They correspond just exactly: A summary of this is that every ideal is associated to a quotient ring, and the ideal is maximal if and only if the quotient ring is a field, and an ideal is prime if and only if the quotient ring is an integral domain.

Those notions are perfectly lined up: maximal and field, and integral domain and prime.

A Non-Prime Ideal: Smooth Functions with Compact Support

Spoken Transcript [01:03:44 - 01:04:41] Prof. Rahul Pandharipande

Maybe we do one example.

It was an example on the quiz. By the way, people do very well on the quizzes, so I asked various people whether I should make the quizzes harder, and the uniform answer was no, I should not make the quizzes harder. [laughter]

Which is fine with me. As I said, I took almost a decade off from teaching because I had some desk job. But now that I've come back, the world has changed a bit, and you know, I'm not used to these kinds of things.

Actually, it's fun to lecture. I hadn't lectured for a while, and it's kind of fun. Yeah.

Spoken Transcript [01:04:41 - 01:05:24] Prof. Rahul Pandharipande

Okay, so, now we do an example that was related to the quiz.

The example is, I take C^∞ functions on $\mathbb{R}$ [writes: $C^\infty(\mathbb{R})$], and I take the ideal of compactly supported functions [writes: $I \subset C^\infty(\mathbb{R})$, compactly supported functions].

That was a problem on the quiz. Question now: Is I a prime ideal? [writes: Is $I \subset C^\infty(\mathbb{R})$ a prime ideal?] This wasn't on the quiz, because... well, is I a prime ideal? I think the quiz question was whether it was a maximal ideal, right? Anyway, we can now ask: Is this a prime ideal?

Spoken Transcript [01:05:25 - 01:07:17] Prof. Rahul Pandharipande

And the answer is no. [writes: Ans: No.]

To show something is not a prime ideal, it's pretty simple: I have to find two things that are not in the ideal, that when I multiply, become in that ideal. That's what I have to do, right?

I will give you two things: Let a be the function that looks... So, here is $\mathbb{R}$, and this is 0, 1... Let a be a smooth function that is non-zero, and then becomes somehow zero here [draws graph of a]. That's a.

And you can imagine what b is going to be; maybe I draw it with another color.

b is the function that comes here and goes like that [draws graph of b]. That's b, all right?

a and b definitely don't have compact support individually. Is that clear? Because the support of a goes all the way that way, and the support of b goes all the way this way.

If I multiply $a \cdot b$ [writes "ab" below the axis], that function... It's the green function, so that's $a \cdot b$, green: It's zero always here, and then it has some life, and then it goes to zero again. So, that's compact support.

This is an example where $a \notin I$ [writes: $a \notin I$], $b \notin I$ [writes: $b \notin I$], but $ab \in I$ [writes: $ab \in I$]. That means that I is not prime [writes: $\implies$ I is not prime].

Okay? That's kind of a fun example.

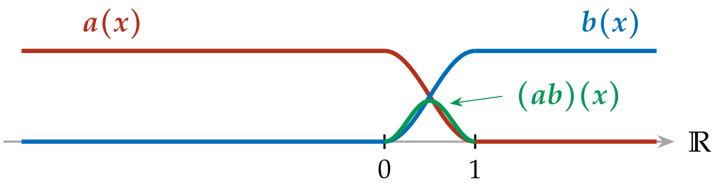
Counterexample: Compactly Supported Smooth Functions

Example 5.28 (*Compactly Supported Functions Do Not Form a Prime Ideal*): Let $R = C^\infty(\mathbb{R})$ be the ring of smooth functions under pointwise operations. Consider the ideal of functions with compact support (Week 2, Definition 2.39, Proposition 2.40):

$$I := \{f \in C^\infty(\mathbb{R}) \mid \operatorname{supp}(f) \text{ is compact}\}. \tag{5.32}$$

Question 5.29: Is I a prime ideal of $C^\infty(\mathbb{R})$?

Answer 5.30: **No**, I is not a prime ideal.



Verification of Counterexample

We construct two smooth functions $a, b \in C^\infty(\mathbb{R})$:

$$a(x) = \begin{cases} > 0 & \text{for } x < 1, \\ 0 & \text{for } x \geq 1, \end{cases}$$

$$b(x) = \begin{cases} 0 & \text{for } x \leq 0, \\ > 0 & \text{for } x > 0. \end{cases}$$

Then:

- $\operatorname{supp}(a) = (-\infty, 1]$ is unbounded, so $a \notin I$.
- $\operatorname{supp}(b) = [0, \infty)$ is unbounded, so $b \notin I$.
- However, their pointwise product satisfies $(a \cdot b)(x) = 0$ for all $x \notin [0,1]$. Thus, $\operatorname{supp}(a \cdot b) \subseteq [0,1]$ is compact, which means:
$$a \cdot b \in I.$$

Since $a \notin I$ and $b \notin I$, yet $ab \in I$, the ideal I is not a prime ideal. **Q.E.D.**

Notation: Compactly Supported Functions

Many textbooks write $C_c^\infty(\mathbb{R})$ for this ideal (Definition 2.39). The lecturer calls it I.

The Chinese Remainder Theorem

Elementary Formulation for the Integers

Spoken Transcript [01:07:17 - 01:08:58] Prof. Rahul Pandharipande

All right, so, the next topic, which I start today, is the famous Chinese Remainder Theorem.

Do you know this theorem? This is kind of standard: Chinese Remainder Theorem [writes: Chinese Remainder Theorem]. We're going to do some fancier version of it, but the plain version of it is kind of where we start, because it's common in schools — I guess particularly in Chinese schools, I don't know.

It's called the Chinese Remainder Theorem because the first shadow of it was found in some manuscript from the fifth century AD in China, I think.

The very basic one, the version you learn in school, is: Let a and b be numbers that are relatively prime. So, a and b are some integers that are relatively prime [writes: Let $(a,b)=1$, $a,b\in\mathbb{Z}$].

Then choose $x,y\in\mathbb{Z}$ to be some other integers [writes: choose $x,y\in\mathbb{Z}$].

The Chinese Remainder Theorem is about the solution of a certain system. We want to solve for a third integer z that satisfies the equations: $z\equiv x \pmod{a}$ and $z\equiv y \pmod{b}$ [writes "We want for $z\in\mathbb{Z}, z\equiv x \bmod a$" and "$z\equiv y \bmod b$"].

That's what we have to solve. Is this familiar to you?

And the answer, the Chinese Remainder Theorem: You can always do it [writes: Ans: You can do it.].

The Chinese Remainder Theorem over $\mathbb{Z}$ (Classical Form)

Theorem 5.31 (*Classical Chinese Remainder Theorem for Coprime Integers*)**:** Let $a,b\in\mathbb{Z}$ be coprime integers, that is, $\mathrm{GCD}(a,b)=1$. Then for any integers $x,y\in\mathbb{Z}$, there exists an integer $z\in\mathbb{Z}$ satisfying the simultaneous system of congruences:

$$\begin{aligned} z &\equiv x \pmod{a}, \\ z &\equiv y \pmod{b}. \end{aligned} \tag{5.33}$$

Spoken Transcript [01:09:07 - 01:10:32] Prof. Rahul Pandharipande

We would like to pursue this in the context of general rings, and let us try to do it in that context. Maybe do an example first if you haven't seen it — it's not necessary that you've seen it.

If you haven't seen it, we'll do a small example, and then I'll do the general case. We take here $\bmod 5$, and say we want $z\equiv 2 \pmod{5}$ and $z\equiv 1 \pmod{7}$ [writes "$z\equiv 2 \bmod 5$" and "$z\equiv 1 \bmod 7$"].

How do you do this? I look at the numbers $\bmod 5$: There's 2. What's 2 mod 5? Well, I can add 7... that doesn't seem to work... 12... that's more failure... 17... 22... Ah, 22 is pretty good! [circles 22]

So, 22: $22 \equiv 2 \pmod 5$ and $1 \pmod 7$, see? And the Chinese Remainder Theorem says you can always do this; you'll always succeed.

Numerical Example: Simultaneous Congruences in $\mathbb{Z}$

Example 5.32 (*Solving a System of Two Congruences*): We seek an integer $z \in \mathbb{Z}$ satisfying:

$$\begin{aligned} z &\equiv 2 \pmod 5, \\ z &\equiv 1 \pmod 7. \end{aligned} \tag{5.34}$$

Since $\mathrm{GCD}(5,7) = 1$, the Chinese Remainder Theorem guarantees a solution.

Testing successive integers congruent to $2 \pmod 5$:

$$z \in \{2, 7, 12, 17, 22, \dots\}$$

Evaluating these candidates modulo 7:

$$\begin{aligned} 2 &\equiv 2 \pmod 7 && \text{(fails)} \\ 7 &\equiv 0 \pmod 7 && \text{(fails)} \\ 12 &\equiv 5 \pmod 7 && \text{(fails)} \\ 17 &\equiv 3 \pmod 7 && \text{(fails)} \\ 22 &\equiv 1 \pmod 7 && \text{(success!)} \end{aligned}$$

Thus, $z = 22$ is a solution.

Beyond the Lecture: Uniqueness of the Solution

The solution is unique modulo ab: this follows from the injectivity of ϕ in the next lecture (Theorem 5.46). In this example, the solutions are exactly the integers $z \equiv 22 \pmod{35}$.

Operations on Ideals: Sum, Product, and Coprimality

Spoken Transcript [01:10:32 - 01:12:09] Prof. Rahul Pandharipande

Okay. So, we want to pursue this in the full context of rings. Let R be a ring [writes: Let R be a ring], and I want to consider two ideals, I and J [writes: $I, J \subset R$ ideals].

In the context here, the ideals would be the ideal generated by a and the ideal generated by b. Those are ideals.

The whole idea of the ideal is some kind of idealized number. So, before we even get to the Chinese Remainder Theorem, there are some things to talk about.

This is totally general: If I have a ring with two ideals, there's an ideal which is the sum — I'm going to define it. I take two ideals, and I can consider their sum. By definition [writes: $I + J \overset{\text{def}}{=} \{i + j \mid i \in I, j \in J\} \subset R$], this is equal to the set of elements $i + j$, where i is in the first ideal and j is in the second ideal, all right?

Is it clear what this definition is? Then you have to check that this is an ideal. In particular, zero should be here, which it is, because $0 \in I$ and $0 \in J$. Then you have to check that if two things are here, when you add them, they're also in here, and you have to check it's also closed under scalar multiplication. I leave all those checks for you: They're all simple; there's nothing going on in those checks. But this is a very important definition.

Are there any questions about this definition? If I take two ideals, I have the ideal sum, $I + J$.

Definition 5.33 (*Sum of Ideals*): Let R be a ring and let $I, J \subset R$ be ideals. The *"sum"* of I and J is defined as:

$$I + J \overset{\text{def}}{=} \{i + j \mid i \in I, j \in J\} \subset R. \tag{5.35}$$

Verification of the Ideal Axioms for $I + J$ (beyond the lecture)

The lecture checks $0 \in I + J$ and leaves the rest to the students (*"I leave all those checks for you"*). To confirm that $I + J$ is an ideal of R:

(1) Non-emptiness and Zero: Since $0 \in I$ and $0 \in J$, we have $0 = 0 + 0 \in I + J$.

(2) Additive Closure: Let $x, y \in I + J$. Then $x = i_1 + j_1$ and $y = i_2 + j_2$ for some $i_1, i_2 \in I$ and $j_1, j_2 \in J$. Using the associativity and commutativity of addition in R:

$$x + y = (i_1 + i_2) + (j_1 + j_2) \in I + J \quad \text{since } i_1 + i_2 \in I \text{ and } j_1 + j_2 \in J.$$

Similarly, $-x = (-i_1) + (-j_1) \in I + J$.

(3) Absorption under Ring Multiplication: For any $r \in R$ and $x = i + j \in I + J$:

$$rx = r(i + j) = ri + rj \in I + J \quad \text{since } ri \in I \text{ and } rj \in J.$$

Thus, $I + J$ is indeed an ideal of R. In fact, $I + J$ is the smallest ideal containing both I and J. **Q.E.D.**

Spoken Transcript [01:12:09 - 01:14:12] Prof. Rahul Pandharipande

And you can imagine what I'm going to write next: There's also IJ, the ideal product [writes: $I \cdot J$]. The definition here is slightly more complicated: [writes: $\overset{\text{def}}{=} \{\sum_{r=1}^{k} i_r j_r \mid i_r \in I, j_r \in J\} \subset R$]

For the ideal product, I take all finite linear combinations of the products, where one element is in one ideal and the other is in the other. Okay? It's slightly different than this one. Here, it's literally the sum of two elements, one in one and the other in the other.

You could say: Why don't I just take this to be the set of elements ij where $i \in I$ and $j \in J$? That's a good question. Are you going to ask this question? You have no curiosity about this fact? Okay, then I... [laughter]

The reason is that that is problematic: If I defined it to just be elements ij, then I'd have to show that if I take two of them [writes: $i_1 j_1 + i_2 j_2$], this is equal to $i_3 j_3$ [writes: $= i_3 j_3$], because the ideal should be closed under addition.

So, if my definition was just products, then I'd have to say that if I take a product plus another product, since an ideal should be closed under addition, it should be a product.

And that doesn't seem 100% likely, right?

We solve the problem by saying we take all sums of products, because this thing here is a sum of products. This is a subtle point, but it's important to understand.

That's the definition. Take any ring and take two ideals: You can take their ideal sum and the ideal product, all right? There's a slight difference in the definition based on this small issue I said.

Definition 5.34 (*Product of Ideals*): Let R be a ring and let $I, J \subset R$ be ideals. The *"product"* of I and J, denoted by IJ (or $I \cdot J$), is defined as the set of all finite sums of products of elements from I and J:

$$I \cdot J \overset{\text{def}}{=} \left\{ \sum_{r=1}^{k} i_r j_r \;\middle|\; k \in \mathbb{N},\ i_r \in I,\ j_r \in J \right\} \subset R. \tag{5.36}$$

Why the Product Ideal Requires Finite Sums: A student might naturally ask: **Why not simply define the product ideal as the naive set of single products $\{ij \mid i \in I, j \in J\}$?**

- **Failure of Additive Closure:** The set of simple products $\{ij \mid i \in I, j \in J\}$ is in general **not** closed under addition!
- **The Sum of Products:** If $i_1, i_2 \in I$ and $j_1, j_2 \in J$, the sum:
  $$i_1 j_1 + i_2 j_2$$
  cannot generally be factored into a single product $i_3 j_3$ with $i_3 \in I$ and $j_3 \in J$.
- **Ideal Generation:** To ensure that IJ forms a legitimate ideal that is closed under addition, one must take the additive group generated by all products $\{ij\}$, which consists precisely of all finite sums $\sum_{r=1}^{k} i_r j_r$.

Spoken Transcript [01:14:12 - 01:14:51] Prof. Rahul Pandharipande

For example, what is the ideal sum here when I consider the ring $\mathbb{Z}$? [writes: $\mathbb{Z}$, $(3) + (5) =$] What is that?

Student Answer

It's the entire $\mathbb{Z}$.

Spoken Transcript [01:14:52 - 01:15:31] Prof. Rahul Pandharipande

It's the entire $\mathbb{Z}$, yeah. This is a weird calculus, all right? It's (1) [writes: (1)], because you can write anything in $\mathbb{Z}$ because they're relatively prime. You have to concentrate when you do this calculus.

What about this: [writes: $(3) \cdot (5) =$]

I just want an answer: What number is it? [laughter]

Student Answer

One?

Spoken Transcript [01:15:33 - 01:16:00] Prof. Rahul Pandharipande

One? Why? The proposal is (1) [writes: (1)]. Well, why would you say that?

Remember the definition: To be in here, you have to be a multiple of 3 times a multiple of 5, and then a sum of things that are multiples of 3 times multiples of 5. So, you can't be 1, because...

Student Answer

Is it 15?

Spoken Transcript [01:16:02 - 01:16:29] Prof. Rahul Pandharipande

It's (15), yeah. This one worked, actually. [erases (1), writes: (15)]

You just have to look at this definition: To be in here, you have to be a sum of things that are all multiples of 3 times multiples of 5, so, every summand is a multiple of 15, so the whole thing must be a multiple of 15; and of course every multiple of 15 is in there.

Although the definition is somehow more delicate, here it behaves better. Anyway, you have to be careful. That was my exercise in showing you have to be careful.

Examples of Ideal Sums and Products in $\mathbb{Z}$

Example 5.35 (*Ideal Sum vs. Ideal Product for (3) and (5) in $\mathbb{Z}$*)**:** Consider the principal ideals $(3) = 3\mathbb{Z}$ and $(5) = 5\mathbb{Z}$ in the ring $R = \mathbb{Z}$:

(1) Ideal Sum:

$$(3) + (5) = (1) = \mathbb{Z}. \tag{5.37}$$

Since $\mathrm{GCD}(3,5) = 1$, Bézout's identity provides $2 \cdot 3 + (-1) \cdot 5 = 1 \in (3) + (5)$. Because $1 \in (3) + (5)$, the sum is the entire unit ideal $\mathbb{Z}$.

(2) Ideal Product:

$$(3) \cdot (5) = (15). \tag{5.38}$$

Every element in $(3) \cdot (5)$ is a finite sum of terms of the form $(3a)(5b) = 15ab$, which is a multiple of 15. Conversely, any multiple $15k = 3 \cdot (5k)$ is a product of an element from (3) and an element from (5).

Spoken Transcript [01:16:29 - 01:17:39] Prof. Rahul Pandharipande

If we go back to the general development: The definition is that I and J in R are called coprime [writes: Def: $I, J \subset R$ are coprime].

The definition of this is that $I + J = (1)$ [writes: $\overset{\text{def}}{\iff} I + J = (1)$].

When I write the unit ideal here, another way of writing it is just R, okay? Those are two equivalent things. That's the definition: It's coprime.

This definition makes sense here: If I specialize this to $\mathbb{Z}$, the notion of coprime is the standard notion of coprime.

All right, so, now we can state the Chinese Remainder Theorem in these definitions.

Definition 5.36 (*Coprime Ideals*): Let R be a ring. Two ideals $I, J \subset R$ are called *"coprime"* if their sum is the unit ideal:

$$I, J \text{ are coprime} \overset{\text{def}}{\iff} I + J = (1) = R. \tag{5.39}$$

Equivalently, I and J are coprime if and only if there exist elements $i \in I$ and $j \in J$ such that:

$$i + j = 1. \tag{5.40}$$

Compatibility with Elementary Coprimality in $\mathbb{Z}$: Going beyond the lecture: in the ring of integers $\mathbb{Z}$, every ideal is principal: $I = (a)$ and $J = (b)$. The sum of two principal ideals is generated by their greatest common divisor:

$$(a) + (b) = (\mathrm{GCD}(a, b)).$$

Thus, $(a) + (b) = (1) \iff \mathrm{GCD}(a, b) = 1$, showing that Definition 5.36 generalizes the elementary notion of coprime numbers to arbitrary rings.

III The Chinese Remainder Theorem for General Rings

Spoken Transcript [01:18:15 - 01:19:50] Prof. Rahul Pandharipande

This is actually a pretty nice theorem in the sense that it's not hard to prove, but there are some ideas. It's not like that vacuous stuff that falls straight from definitions.

For the Chinese Remainder Theorem: We have $I, J \subset R$ [writes: CRT: $I, J \subset R$].

These are the assumptions: $I + J = (1)$ [writes: $I + J = (1)$].

Then — and I have to explain this — there's a natural ring homomorphism from R mod IJ [writes: $\phi\colon R/IJ \to$]. Given these ideals, there's an ideal which is the product, and I can take its quotient; I'm allowed to do that: This is a ring. Do you accept that?

It turns out there's a homomorphism to $(R \bmod I) \times (R \bmod J)$ [writes: $R/I \times R/J$]. I'll explain all this. And ϕ is an isomorphism of rings [writes: ϕ is an isomorphism of rings].

That's the conclusion of the Chinese Remainder Theorem. You take any two ideals in any ring such that they're coprime, then there is this map, and the conclusion of the Chinese Remainder Theorem is that this is an isomorphism of rings.

What is this map? First of all, we know what these three rings are; they've been defined [points at $R/IJ, R/I, R/J$]. These are just our quotients: We have the ideal I, we have the ideal J, and we have the ideal IJ. So, these are well-defined rings. You accept that?

Theorem 5.37 (*Chinese Remainder Theorem for Rings*)**:** Let R be a commutative ring and let $I, J \subset R$ be coprime ideals ($I + J = (1)$). Then there exists a canonical ring isomorphism:

$$\phi\colon R/_{IJ} \xrightarrow{\sim} R/_{I} \times R/_{J}. \tag{5.41}$$

Spoken Transcript [01:19:50 - 01:21:59] Prof. Rahul Pandharipande

If I have any ring, I can consider the product. This is a new ring [writes: $R/_I \times R/_J = \{(\bar{a}, \bar{b}) \mid \bar{a} \in R/_I, \bar{b} \in R/_J\}$]. This ring consists of elements $(\bar{a}, \bar{b})$ where $\bar{a} \in R \bmod I$ and $\bar{b} \in R \bmod J$. And the way you add and multiply them is just componentwise, all right?

We've seen some things like this, but I haven't formally defined it. This definition can be made for any rings: If S_1 and S_2 are rings [writes: (S_1, S_2 are rings $\implies$ $S_1 \times S_2$ is ring) (cpt. wise)], then $S_1 \times S_2$ is a ring in the natural way, where you do componentwise addition, componentwise multiplication, the zero element is of course $(0, 0)$, the identity element is $(1, 1)$, etc., okay?

That's the explanation of what this is: I take the two quotients, and I take the product. And this one you know, so I have to tell you what the map is. This is kind of fun.

I have to tell you, if I start with the equivalence class of some element in $R \bmod IJ$ [writes: $\phi(\bar{z} \in R/_{IJ}) \to$], how to get a product. And it's going to be perfect: It's just $(\bar{z}, \bar{z})$ [writes: $(\bar{z}, \bar{z})$].

Actually, you could write this — and this is maybe a slightly malicious way to write it — but it's just this: $\phi(\bar{z}) = (\bar{z}, \bar{z})$ [writes: $\phi(\bar{z}) = (\bar{z}, \bar{z})$]. We'll discuss this in a second. Many people would just write it without the bars: $\phi(z) = (z, z)$ [writes: $\phi(z) = (z, z)$]. All right?

Let's try to understand what this means. Does it make sense? Because I take an equivalence class here, and I send it to the equivalence class... this is an equivalence class in $R \bmod I$, and this is an equivalence class in $R \bmod J$ [writes "$R/_I$" under first $\bar{z}$, and "$R/_J$" under second $\bar{z}$]. Okay?

Direct Product of Rings and Definition of the Map ϕ

Definition 5.38 (*Direct Product of Rings*)**:** Let S_1 and S_2 be rings. The *"direct product"* $S_1 \times S_2$ is the Cartesian product equipped with componentwise ring operations:

$$(a_1, a_2) + (b_1, b_2) := (a_1 + b_1, a_2 + b_2),$$
$$(a_1, a_2) \cdot (b_1, b_2) := (a_1 \cdot b_1, a_2 \cdot b_2).$$

The zero element is $0_{S_1 \times S_2} := (0_{S_1}, 0_{S_2})$ and the identity element is $1_{S_1 \times S_2} := (1_{S_1}, 1_{S_2})$.

Definition 5.39 (*The Canonical CRT Homomorphism ϕ*)**:** Let R be a ring and $I, J \subset R$ ideals. The canonical assignment:

$$\begin{aligned} \phi\colon R/_{IJ} &\longrightarrow R/_{I} \times R/_{J}, \\ z + IJ &\longmapsto (z + I,\ z + J), \end{aligned} \tag{5.42}$$

is denoted concisely on the blackboard as $\phi(\bar{z}) \overset{\text{def}}{=} (\bar{z}, \bar{z})$ (or informally $\phi(z) = (z, z)$).

Spoken Transcript [01:21:59 - 01:24:09] Prof. Rahul Pandharipande

Let me say a few words about that. You have to prove it's well-defined, etc. I can say, whatever was on that scroll in the fifth century AD, it wasn't this statement. [laughter]

To really understand and get started, we have to be on the same page about this ϕ. As I said, I take some element $z \in R$, and I get a $\bar{z}$ here — this is a class in R mod IJ, right? [writes "$\phi(\bar{z}) \overset{\text{def}}{=} (\bar{z}, \bar{z})$", with "$z \in R$", "$\bar{z} \in R/_{IJ}$", and "$R/_{I}, R/_{J}$" under the tuple]

And I define this to be the class $(\bar{z}, \bar{z})$, where the bars mean different things! That's the confusing thing about this, which I warned you comes: This $\bar{z}$ means this bar is the equivalence class in R mod I, and this bar is the equivalence class in R mod J, okay?

Why is this well-defined? Because this guy is only defined up to shifting by some $\gamma \in IJ$, right? [writes: $\bar{z} = \overline{z+\gamma}, \gamma \in IJ$]

If I shifted it by γ, this should be the same thing: If I write that, I would get $(\overline{z+\gamma}, \overline{z+\gamma})$ [writes: $\phi(\overline{z+\gamma}) = (\overline{z+\gamma}, \overline{z+\gamma})$].

It had better be the case that these shifts respect the different equivalence classes here. And this is okay [writes: OK because $\gamma \in IJ$ and $IJ \subset I$ and $IJ \subset J$], because $\gamma \in IJ$, and $IJ \subset I$ and $IJ \subset J$.

Lemma 5.40 (*Well-Definedness of ϕ*)**:** The mapping $\phi\colon R/_{IJ} \to R/_{I} \times R/_{J}$ given by $\phi(z + IJ) = (z + I, z + J)$ is well-defined.

Proof

Let $z, z' \in R$ such that $z + IJ = z' + IJ$ in $R/_{IJ}$. Then $z' = z + \gamma$ for some $\gamma \in IJ$. To show that $\phi(z' + IJ) = \phi(z + IJ)$, we must verify:

$$(z + \gamma + I,\ z + \gamma + J) = (z + I,\ z + J).$$

This holds if and only if $\gamma \in I$ and $\gamma \in J$. Since $IJ \subseteq I$ and $IJ \subseteq J$ (as established in Lemma 5.41), we have $\gamma \in I$ and $\gamma \in J$. Consequently:

$$z + \gamma + I = z + I \quad \text{and} \quad z + \gamma + J = z + J.$$

Thus, the value of $\phi(z + IJ)$ is independent of the chosen representative z. **Q.E.D.**

Spoken Transcript [01:24:09 - 01:24:55] Prof. Rahul Pandharipande

Why is IJ in I? Remember, IJ is equal to sums of the form $i_1 j_1 + i_2 j_2 + \cdots + i_k j_k$ [writes: $IJ = \{\sum_{r=1}^{k} i_r j_r \mid i_r \in I, j_r \in J\}$]. Since this is in I and that's something else, each one of these terms is both in I and in J.

A very important property of IJ — I just put it in a box here —: This is always inside I and it's always inside J [draws box with $IJ \subset I$ and $IJ \subset J$]. Okay?

The ambiguity here is that I could change the name with a shift of somebody in IJ, and that would change this; but this shift is still in I, and this shift is still in J, so the answer is the same, yeah.

Student Question

Do we need to assume that multiplication is commutative for this or... because if we look at it like this, I would argue that j or the ij's are in J, but then is IJ the same as JI?

Spoken Transcript [01:25:13 - 01:26:33] Prof. Rahul Pandharipande

Yes! We're in commutative rings! An interesting question is: How do you define this for non-commutative rings? What you're getting to is that if you want to do this theory for non-commutative rings — which we're not doing in this class — you have to be, at the very beginning, very careful about what you mean by an ideal, and whether it's left multiplication or right multiplication.

His concern is that if I view this element of J, the i is sitting on the left, which is what we normally do. But if I view this as an element of I, the j is sitting on the right. That's your concern. But we're commutative, so you can always move the chairs.

If you're doing non-commutative ring theory, then you have to be super organized about what you mean by an ideal, etc. But we're not doing it.

Did I convince you this map is well-defined? The matter here is a little bit of confusion, and that confusion goes to this issue that I said: In quotients, there's a deep philosophical confusion that it's hard to name anybody; they all have lots of different names. It'd be like if humans each had a countable number of names, which would make the world pretty interesting, I think.

Once you're on top of this naming ambiguity, you see that this map makes sense: It works. And then you have to check it's a homomorphism, which I leave to you.

Lemma 5.41 (*Inclusion in Component Ideals*): For any two ideals I, J in a commutative ring R:

$$IJ \subseteq I \quad \text{and} \quad IJ \subseteq J \qquad (\text{hence } IJ \subseteq I \cap J). \tag{5.43}$$

Proof

Let $\gamma \in IJ$. By Definition 5.34, $\gamma = \sum_{r=1}^{k} i_r j_r$ with $i_r \in I$ and $j_r \in J$.

- Since I is an ideal, each term $i_r j_r \in I$ by ideal absorption. Because I is closed under finite sums, $\gamma \in I$.
- Since J is an ideal and R is commutative, $i_r j_r = j_r i_r \in J$ by ideal absorption. Since J is closed under finite sums, $\gamma \in J$.

Therefore, $IJ \subseteq I$ and $IJ \subseteq J$, which implies $IJ \subseteq I \cap J$. **Q.E.D.**

Spoken Transcript [01:26:33 - 01:27:32] Prof. Rahul Pandharipande

You must check ϕ is a homomorphism of rings [writes: You must check ϕ is a homomorphism of rings].

There's nothing to this. The only issue is this naming, because you'd have to check something like $\phi(\bar{z}_1\bar{z}_2) = \phi(\bar{z}_1)\phi(\bar{z}_2)$ [writes: $\phi(\bar{z}_1 + \bar{z}_2) = \phi(\bar{z}_1) + \phi(\bar{z}_2)$]. When you do this, you'll see that all that mattered was that this was legal; the rest will work out.

You can see that $0 \mapsto (0,0)$ and $1 \mapsto (1,1)$, so, this is a homomorphism of rings.

Finally, that's the explanation of this: The Chinese Remainder Theorem says that this is an isomorphism of rings [writes: CRT: ϕ is isomorphism of rings].

All right, so, we don't have much time, but I should be able to prove at least half of this. Let's try.

Lemma 5.42 (*ϕ is a Ring Homomorphism*): The map $\phi\colon R/IJ \to R/I \times R/J$ is a ring homomorphism.

Checking the Homomorphism Property (beyond the lecture)

The lecture leaves this to the students. Let $\bar{z}_1 = z_1 + IJ$ and $\bar{z}_2 = z_2 + IJ$ in R/IJ.

(1) Additive Homomorphism:

$$\begin{aligned}\phi(\bar{z}_1 + \bar{z}_2) &= \phi((z_1+z_2)+IJ)\\ &= (z_1+z_2+I,\ z_1+z_2+J)\\ &= (z_1+I,\ z_1+J) + (z_2+I,\ z_2+J)\\ &= \phi(\bar{z}_1)+\phi(\bar{z}_2).\end{aligned}$$

(2) Multiplicative Homomorphism:

$$\begin{aligned}\phi(\bar{z}_1 \cdot \bar{z}_2) &= \phi(z_1z_2+IJ)\\ &= (z_1z_2+I,\ z_1z_2+J)\\ &= (z_1+I,\ z_1+J)\cdot(z_2+I,\ z_2+J)\\ &= \phi(\bar{z}_1)\cdot\phi(\bar{z}_2).\end{aligned}$$

(3) Preservation of Identity:

$$\phi(1+IJ) = (1+I,\ 1+J) = 1_{R/I \times R/J}.$$

Thus, ϕ is a ring homomorphism. **Q.E.D.**

Proof of Injectivity

Spoken Transcript [01:27:55 - 01:29:47] Prof. Rahul Pandharipande

I will in the end explain how this is related to the Chinese Remainder Theorem as I said at the beginning. The answer is, it's directly related.

We want to prove it's an isomorphism, so, this basically always means you have to prove two things: that it's injective and surjective.

Let us prove ϕ is injective [writes: Let us prove ϕ is inj.].

Our hypothesis, our assumption, is that $I + J = (1)$ [writes: hypothesis $I + J = (1)$]. That's our assumption.

This means that there exists an $i \in I$ and a $j \in J$ such that $i + j = 1$ [writes "$\implies \exists i \in I, j \in J$" and "$i + j = 1$"]. That's our basic assumption of these ideals, all right?

And we need to now show… I'll do it quickly and we'll discuss it again next time:

Suppose we have $\phi(\bar{z}) = (\bar{0}, \bar{0})$ [writes: Suppose $\phi(\bar{z}) = (\bar{0}, \bar{0})$].

That's in the kernel. That means that our $\bar{z} \in R \bmod IJ$ is in the kernel: $\bar{z} \in \ker(\phi)$ [writes: $\bar{z} \in {}^{R}\!/_{IJ}, \bar{z} \in \ker(\phi)$]. That's what it means to be in the kernel, right?

Spoken Transcript [01:29:47 - 01:31:12] Prof. Rahul Pandharipande

How can it be in the kernel? That means that if I take $\bar{z} \in R \bmod I$, this has to be zero: $\bar{0} = \bar{z} \in R \bmod I \implies z \in I$ [writes: $\bar{0} = \bar{z} \in {}^{R}\!/_{I} \implies z \in I$]. This first zero says that $z \in I$, okay?

Then zero has to be equal to $\bar{z} \in R \bmod J$: $\bar{0} = \bar{z} \in R \bmod J \implies z \in J$ [writes: $\bar{0} = \bar{z} \in {}^{R}\!/_{J} \implies z \in J$].

And then I want to show that $z \in IJ$. I'll do this next time also: I do a great thing, which is the whole point of this — I take that equation $i + j = 1$ and multiply it by z [writes: $z \in IJ, i + j = 1, iz + jz = z$]: $iz + jz = z$.

This is the main trick in the whole Chinese Remainder Theorem: I just multiply it by z. And this is z, and both of these are in IJ [underlines iz and jz, writes "$\in IJ$" under each].

This means that $z \in IJ$ [writes: $\implies z \in IJ$], and that means that $\bar{z} = \bar{0} \in R \bmod IJ$ [writes: $\implies \bar{z} = \bar{0} \in {}^{R}\!/_{IJ}$].

So, that's the end of it. I'll explain it again next time.

But that's the cool idea in this. We'll discuss it next time. Okay? [students applaud]

Proof of Injectivity of the Canonical Map ϕ

Proof of Injectivity in Theorem 5.37

To prove that the ring homomorphism ϕ is injective, it suffices to show that its kernel is trivial:

$$\ker(\phi) = \{\bar{0}\}.$$

Let $\bar{z} = z + IJ \in \ker(\phi)$ with representative $z \in R$. By definition of ϕ:

$$\phi(\bar{z}) = (z + I,\ z + J) = (\bar{0}, \bar{0}) = (0 + I,\ 0 + J).$$

This yields two conditions on z:

$$z + I = 0 + I \implies z \in I, \qquad \text{and} \qquad z + J = 0 + J \implies z \in J. \tag{5.44}$$

Therefore, $z \in I \cap J$.

We now use the hypothesis that I and J are coprime:

$$I + J = (1) \implies \exists i \in I,\ j \in J \quad \text{such that} \quad i + j = 1.$$

Multiplying this identity through by z:

$$z = z \cdot 1 = z(i + j) = zi + zj = iz + jz. \tag{5.45}$$

We examine the two terms on the right-hand side of (5.45):

- For iz: We know $i \in I$, and from (5.44) we have $z \in J$. Thus, $iz \in IJ$.
- For jz: We know $j \in J$, and from (5.44) we have $z \in I$. By commutativity, $jz = zj \in IJ$.

Since both $iz \in IJ$ and $jz \in IJ$, their sum belongs to the ideal IJ:

$$z = iz + jz \in IJ.$$

Consequently:

$$\bar{z} = z + IJ = 0 + IJ = \bar{0} \quad \text{in } R/IJ.$$

This proves $\ker(\phi) = \{\bar{0}\}$, and hence ϕ is injective. **Q.E.D.**

The Essential Mechanism of the Injectivity Proof: Going beyond the lecture: the proof shows that for any two coprime ideals I and J in a commutative ring:

$$I \cap J = IJ.$$

While the inclusion $IJ \subseteq I \cap J$ holds unconditionally (Lemma 5.41), the reverse inclusion $I \cap J \subseteq IJ$ relies crucially on coprimality:

(1) **Coprimality Decomposition:** $1 = i + j$ with $i \in I, j \in J$.
(2) **Splitting Any Common Element:** Any $z \in I \cap J$ splits as $z = iz + jz$.
(3) **Trapping in the Product:** In iz, the factor i belongs to I and z belongs to J, forcing $iz \in IJ$. In jz, $z \in I$ and $j \in J$, forcing $jz \in IJ$.

This identity $I \cap J = IJ$ ensures that $\ker(\phi) = (I \cap J)/IJ = \{\bar{0}\}$, establishing injectivity.

Summary of Key Concepts

- **Ideal Correspondence Theorem:** For an ideal $K \subset R$, the projection $q\colon R \to R/K$ induces an inclusion-preserving bijection between ideals of R containing K and all ideals of the quotient R/K.
- **Quotients of Quotients:** For an ideal $K \subset R$ and an ideal $J \subset R/K$, there is a canonical ring isomorphism $R/q^{-1}(J) \cong \left(R/K\right)/J$.
- **Maximal Ideals and Fields:** An ideal $m \subset R$ is maximal if and only if R/m is a field. In particular, a commutative ring R is a field if and only if $(0) \subset R$ is a maximal ideal.
- **Maximal Ideals via Zorn's Lemma:** Every non-zero commutative ring contains at least one maximal ideal. The proof uses Zorn's Lemma on the poset of proper ideals $(\mathrm{cl}^*(R), \subseteq)$, where every chain is bounded above by the union of its members.
- **Prime Ideals and Integral Domains:** An ideal $P \subset R$ is prime if and only if R/P is an integral domain. Consequently, every maximal ideal is prime, and every non-zero commutative ring possesses at least one prime ideal.
- **Chinese Remainder Theorem:** For coprime ideals $I + J = (1)$, the canonical homomorphism $\phi\colon R/IJ \to R/I \times R/J$ defined by $z + IJ \mapsto (z + I, z + J)$ is an isomorphism. Injectivity follows from the decomposition $1 = i + j$, multiplied by z.

Wednesday, 15th October 2026

The Chinese Remainder Theorem and Polynomial Rings

Course Content Overview

In this lecture, we complete the general ideal-theoretic formulation of the Chinese Remainder Theorem for commutative rings and begin the theory of polynomial rings over fields:

- **The Canonical Product Homomorphism:** Definition of the canonical map $\phi\colon {}^{R}\!/_{IJ} \to {}^{R}\!/_{I} \times {}^{R}\!/_{J}$ and verification of well-definedness via the ideal inclusions $IJ \subseteq I, J$.
- **Failure of Isomorphism for Non-Coprime Ideals:** A concrete counterexample in $\mathbb{Z}$ with $I = (4)$ and $J = (6)$, illustrating that ϕ fails to be either injective or surjective when ideals are not coprime.
- **The Chinese Remainder Theorem:** Proof that coprimality ($I + J = (1)$) implies that ϕ is an isomorphism, using the identity $i + j = 1$.
- **Applications to $\mathbb{Z}$ and Euler's Totient Function:** Iterated factorization into prime powers, the multiplicative structure of unit groups, and the derivation of Euler's product formula for $\phi^{\text{tot}}(n)$ alongside the prime 691 and the Bernoulli number B_{12}.
- **Polynomial Rings over Fields:** Rigorous definition of the degree function on $k[x]$, the convention of leaving $\deg(0)$ undefined, and polynomial long division with remainder.
- **Principal Ideal Domains (PIDs):** Proof via well-ordering of degrees that every ideal in $k[x]$ is principal, the formal definition of a PID, and non-examples such as $\mathbb{Z}[x]$ and $\mathbb{C}[x, y]$.

The Chinese Remainder Theorem for Commutative Rings

Ideals and the Canonical Homomorphism

Spoken Transcript [00:00:00 - 00:02:30] Prof. Rahul Pandharipande

Okay, so, we start. And yesterday I was in the middle of this discussion of the Chinese Remainder Theorem. [writes: Chinese RT:]

And I will complete it now, but we start from a slightly different perspective. Let's just start with I and J [writes: I, J]... Isn't that loud? Doesn't it seem loud? No? I think that's somehow louder than usual.

Okay, so, we start with I and J inside a ring R [writes: $\subset R$], so, these are ideals [writes: ideals]. And I had discussed... ah, thanks. [student brings chalk] Thanks.

I had discussed last time that there's a notion of adding ideals [writes: $I + J$] and multiplying ideals [writes: IJ]. So, these are both ideals of R [writes: ideals of R].

And without... just assuming just this thing: we take I and J are ideals of R, we can construct this quotient R mod IJ [writes: ${}^{R}\!/_{IJ}$], and we could construct R mod $I \times R$ mod J [writes: ${}^{R}\!/_{I} \times {}^{R}\!/_{J}$]. And moreover, we can construct this homomorphism ϕ [writes: $\phi\colon {}^{R}\!/_{IJ} \to {}^{R}\!/_{I} \times {}^{R}\!/_{J}$], and the rule for ϕ is $\phi(\bar{z}) = (\bar{z}, \bar{z})$ [writes: $\phi(\bar{z}) = (\bar{z}, \bar{z})$]. So, it's kind of almost a silly, tautological formula.

And the interesting thing is you have to remember that this $\bar{z}$ is... the bar here is $\bmod IJ$ [writes "$\bmod IJ$" under the first $\bar{z}$], and the bar here is $\bmod I$ [writes "$\bmod I$" under the second $\bar{z}$], and the bar here means $\bmod J$ [writes "$\bmod J$" under the third $\bar{z}$]. So, the three bars have three different meanings, which you have to be able to read from the place they appear in this formula.

And this is always a homomorphism: ϕ is a homomorphism in general [writes: ϕ is a hom.]. A ring homomorphism in general. This part of the theory just holds in arbitrary generality. Just because the only thing we had to check here was that this definition works, and it works because this is well-defined [writes: Well defined] — I mentioned this, but I'll say it explicitly —, well-defined because IJ is inside I [writes: $IJ \subset I$] and IJ is inside J [writes: $IJ \subset J$]. That makes this association well-defined. All right?

Recap: Sums and products of ideals (Definition 5.33, Definition 5.34), the canonical map ϕ (Definition 5.39) with its well-definedness (Lemma 5.40), and the statement of the Chinese Remainder Theorem (Theorem 5.37) are from the previous lecture. The lecturer recalls them and proves the theorem today.

The Canonical Product Homomorphism

Let R be a commutative ring with identity, and let $I, J \subseteq R$ be ideals.

- **Sum of Ideals:** $I + J \overset{\text{def}}{=} \{i + j \mid i \in I,\, j \in J\}$ is an ideal of R.
- **Product of Ideals:** $IJ \overset{\text{def}}{=} \left\{\sum_{r=1}^{k} i_r j_r \;\middle|\; k \in \mathbb{N},\, i_r \in I,\, j_r \in J\right\}$ is an ideal of R with $IJ \subseteq I$ and $IJ \subseteq J$, so $IJ \subseteq I \cap J$.

Definition 5.43 (*Canonical Product Homomorphism*)**:** The canonical map between quotient rings is defined by:

$$\begin{aligned} \phi\colon {}^{R}\!/_{IJ} &\longrightarrow {}^{R}\!/_{I} \times {}^{R}\!/_{J}, \\ z + IJ &\longmapsto (z + I,\, z + J), \end{aligned} \tag{5.46}$$

which on the blackboard is written compactly with class bars as:

$$\phi(\underbrace{\bar{z}}_{\text{mod } IJ}) = (\underbrace{\bar{z}}_{\text{mod } I}, \underbrace{\bar{z}}_{\text{mod } J}).$$

Proposition 5.44 (*Well-Definedness and Homomorphism Property*)**:** The mapping ϕ

is a well-defined homomorphism of rings with identity.

Proof

Let $z, z' \in R$ with $\bar{z} = \bar{z}'$ in R/IJ, meaning $z - z' \in IJ$. Since $IJ \subseteq I$ and $IJ \subseteq J$, we immediately have:

$$z - z' \in I \implies z + I = z' + I \quad \text{and} \quad z - z' \in J \implies z + J = z' + J.$$

Hence $\phi(z + IJ) = \phi(z' + IJ)$, proving well-definedness. That ϕ preserves addition, multiplication, and the identity $(1 + I, 1 + J)$ follows directly from the componentwise definition of ring operations in the direct product. **Q.E.D.**

Semantic Meaning of the Class Bars: Although the symbol $\bar{z}$ appears on both sides of $\phi(\bar{z}) = (\bar{z}, \bar{z})$, the three occurrences denote equivalence classes in three distinct quotient rings:

(1) **Domain Class:** The argument $\bar{z}$ represents the class $z + IJ \in R/IJ$.

(2) **First Component:** The first component $\bar{z}$ represents the class $z + I \in R/I$.

(3) **Second Component:** The second component $\bar{z}$ represents the class $z + J \in R/J$.

A Counterexample to Isomorphism in $\mathbb{Z}$

Spoken Transcript [00:02:30 - 00:04:48] Prof. Rahul Pandharipande

And so, let's do an example before we go to the Chinese Remainder Theorem. Example is, let's say, we have the ideal (4) [writes: $(4) = I$], that's equal to I, and the ideal (6) that's equal to J [writes: $(6) = J$]. And these are ideals inside, of course, our favorite ring, the integers [writes: $\subset \mathbb{Z}$]. Well, maybe it's your favorite ring, I don't know. It's not my favorite ring, but... okay.

So, then what we claim here is that we have $\mathbb{Z}$... and now this is the product ideal, and we discussed last time that the product ideal here is just (24); it's the product [writes: $\mathbb{Z}/(24) = \mathbb{Z}/(4)(6)$]. That's equal to the product ideal $(4)(6)$ [writes: $(4)(6)$], okay? Do you agree?

And this thing maps, by that map on the board, to $\mathbb{Z}$ mod 4 — so what's this for? — and $\mathbb{Z}$ mod 6 [writes: $\xrightarrow{\phi} \mathbb{Z}/(4) \times \mathbb{Z}/(6)$]. Sorry. So, sometimes we write this (4) as $4\mathbb{Z}$, it just means the same thing; it depends on your mood. Here I write it in this... these are different ways of writing an ideal, right? Today we're writing it like that.

Okay, so, this is the map. So, first of all, is ϕ surjective? [writes: Is ϕ surjective?] And I'm just going to answer this for you: the answer is going to be no [writes: No.].

And why? For example, I could consider the point here that's given, so, consider $(0, 1)$ [writes: consider $(0, 1)$]. All right, that's a point, 0 and 1. Could that come from anything? Well, it would have to come from something divisible by 4, so, it would have to be even. But then it's 1 mod 6, so, it would have to be odd. It's very hard for it to be both even and odd; it's not possible, in fact. So, this is not in the image [writes: $(0, 1) \notin \operatorname{Im}(\phi)$]. Are you happy with that argument?

So, it's not surjective. Is it injective? [writes: Is ϕ inj.?] And the answer is going to be also no. But can you tell me an element here that goes to zero? An element that's not zero that goes to zero? Yeah?

Student Answer

12.

Spoken Transcript [00:04:51 - 00:05:25] Prof. Rahul Pandharipande

Well, zero is zero, so 12 is... So, if you take here $\phi(\bar{12})$ [writes: $\phi(\bar{12})$], this is equal to $(\bar{0}, \bar{0})$ [writes: $= (\bar{0}, \bar{0})$]. So, that shows that $\bar{12}$ goes to zero. And it's not zero because $24 \nmid 12$.

So, it's neither injective nor surjective. That's the typical thing. If you pick some ring with these ideals, you can construct this map, it looks pretty good, it's a homomorphism, but in general it's neither injective, nor is it surjective. And so, what the Chinese Remainder Theorem says...

Failure of Isomorphism for Non-Coprime Ideals

Example 5.45 (*The Homomorphism for $I = (4)$ and $J = (6)$ in $\mathbb{Z}$*)**:** In the ring of integers $R = \mathbb{Z}$, consider the principal ideals $I = (4) = 4\mathbb{Z}$ and $J = (6) = 6\mathbb{Z}$. Their product ideal is:

$$IJ = (4)(6) = (24) = 24\mathbb{Z}.$$

The canonical homomorphism is:

$$\phi \colon \mathbb{Z}/24\mathbb{Z} \longrightarrow \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}, \quad z + 24\mathbb{Z} \longmapsto (z + 4\mathbb{Z},\, z + 6\mathbb{Z}).$$

- **Failure of Surjectivity:** Consider the element $(0 + 4\mathbb{Z}, 1 + 6\mathbb{Z}) \in \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$. If there existed a pre-image $z + 24\mathbb{Z}$, it would have to satisfy the system of congruences:

  $$\begin{aligned} z &\equiv 0 \pmod 4 \implies z \text{ is even}, \\ z &\equiv 1 \pmod 6 \implies z \text{ is odd}, \end{aligned}$$

  which is impossible. Hence:

  $$(0, 1) \notin \operatorname{Im}(\phi),$$

  so ϕ is not surjective.
- **Failure of Injectivity:** The element $12 + 24\mathbb{Z} \in \mathbb{Z}/24\mathbb{Z}$ is non-zero since $24 \nmid 12$. However:

  $$\phi(12 + 24\mathbb{Z}) = (12 + 4\mathbb{Z},\, 12 + 6\mathbb{Z}) = (0 + 4\mathbb{Z},\, 0 + 6\mathbb{Z}) = (\bar{0}, \bar{0}).$$

  Thus, $12 + 24\mathbb{Z} \in \ker(\phi) \setminus \{0\}$, so ϕ is not injective.

Algebraic Reason for the Failure: Going beyond the lecture: notice that $|\mathbb{Z}/24\mathbb{Z}| = 24$, whereas $|\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}| = 4 \times 6 = 24$. Even though the two finite rings have the exact same cardinality, ϕ fails to be bijective because:

(a) The sum of the ideals is $(4) + (6) = \mathrm{GCD}(4, 6)\mathbb{Z} = 2\mathbb{Z} \neq \mathbb{Z}$, so 4 and 6 are not coprime.

(b) The intersection $I \cap J = \mathrm{LCM}(4,6)\mathbb{Z} = 12\mathbb{Z}$ strictly contains the product ideal $IJ = 24\mathbb{Z}$, and the kernel of ϕ consists of the classes of the elements of $I \cap J$, so it is not zero.

Statement and Proof of the Chinese Remainder Theorem

Spoken Transcript [00:05:25 - 00:06:11] Prof. Rahul Pandharipande

Sorry. Maybe I don't want that shadow here. [adjusts the blackboard position]

Okay, the Chinese Remainder Theorem says: if we have the additional assumption that I and J are what we called coprime, that their sum is equal to the unit ideal: $I + J = (1)$ [writes: CRT: If $I + J = (1)$]. If this is true, if that holds, then ϕ is an isomorphism [writes: then ϕ is an isom.].

So, in particular, it means it's both injective and surjective. And we have to prove both; we get neither of them for free, because, you know, they failed here. So, okay, the proof. [writes: Proof]

Let us start with ϕ is injective [writes: ϕ is injective]. So, we have to prove that.

Theorem 5.46 (*Chinese Remainder Theorem for Two Ideals*): Let R be a commutative ring with identity, and let $I, J \subseteq R$ be ideals such that they are coprime, meaning:

$$I + J = (1) = R. \tag{5.47}$$

Then the canonical ring homomorphism:

$$\phi\colon R/IJ \xrightarrow{\sim} R/I \times R/J, \quad \bar{z} \longmapsto (\bar{z}, \bar{z}) \tag{5.48}$$

is an isomorphism of rings.

Proof of Theorem 5.46

Injectivity of ϕ

Spoken Transcript [00:06:11 - 00:07:50] Prof. Rahul Pandharipande

So, suppose we have $\bar{z}$ in R mod IJ [writes: Suppose $\bar{z} \in R/IJ$] such that $\phi(\bar{z}) = (\bar{0}, \bar{0})$ [writes "such that $\phi(\bar{z}, \bar{z}) = (\bar{0}, \bar{0})$", a slip for $\phi(\bar{z})$]. So, we just have to study the kernel for injectivity, okay?

Then what do we know? Since that means that $\bar{z} = 0 \bmod I$ and $z = 0 \bmod J$ [writes: Since $\bar{z} = 0 \bmod I$ and $z = 0 \bmod J$]. That's what those two zeros mean, okay? You agree?

And so, this means that z is in I and z is in J [writes "$z \in I$" and "$z \in J$"]. That's how that's how you're zero $\bmod I$ and J.

Okay, but we need to show that z is in IJ. We have to somehow use the fact that z is in I and z is in J to show it's in IJ, which is a hard thing to do in general, not true; but we have to use this equation. So, but we know — we haven't used it yet, but we know — that there exists some $i \in I$ and there exists a $j \in J$ such that $i + j = 1$ [writes: But we know $\exists i \in I, j \in J$ such that $i + j = 1$]. That's the kind of magic of this assumption. That's exactly what this assumption says: there's an i in I, a little i in I, and a little j in J such that they sum to 1.

Spoken Transcript [00:07:50 - 00:09:25]

Prof. Rahul Pandharipande

And the whole idea of this injectivity is just one idea. Up to here there have been no ideas. And you just multiply this by z. So, the idea is: multiply this by z. And $iz + jz$ — it's a very simple idea — is equal to z [writes: $iz + jz = z$]. There.

So, that's undeniable: I multiply by z, I get this. And then I look at this equation: this is z... This one, this i is in I, but z we know is already in J [draws arrows showing $i \in I$ and $z \in J$]. This j is in J, but z we know is in I [draws arrows showing $j \in J$ and $z \in I$]. So, both of these things on the right are in IJ, in different ways. We use the fact that z is both in I and J in this analysis.

And if I look at it, that means this term is in IJ, that term is in IJ, and since IJ is an ideal, the conclusion is $z \in IJ$ [writes: $z \in IJ$]. That's the conclusion, which is the one we wanted, because if z is in IJ, then it was zero up here. And this implies, so, $\bar{z} = 0$ in R mod IJ [writes: So $\bar{z} = 0 \in {}^{R}\!/_{IJ}$].

So, that shows it's injective. And as I said, this whole proof only has one idea. Here, just following your nose, writing the data, and the only idea is: you take this, multiply by z, that's it.

Proof of Injectivity

Verification of Injectivity

To prove that ϕ is injective, we study its kernel. Suppose $\bar{z} = z + IJ \in {}^{R}\!/_{IJ}$ satisfies:

$$\phi(\bar{z}) = (\bar{0}, \bar{0}) \in {}^{R}\!/_{I} \times {}^{R}\!/_{J}.$$

By the definition of the component classes, this means:

$$z \equiv 0 \pmod{I} \implies z \in I, \qquad \text{and} \qquad z \equiv 0 \pmod{J} \implies z \in J.$$

Thus, $z \in I \cap J$. To show that $\bar{z} = 0 \in {}^{R}\!/_{IJ}$, we must prove that $z \in IJ$.

By assumption, $I + J = (1)$, so there exist elements $i \in I$ and $j \in J$ such that:

$$i + j = 1. \tag{5.49}$$

Multiplying both sides of (5.49) by z yields:

$$z = (i + j)z = iz + jz. \tag{5.50}$$

We now analyze each of the two summands in (5.50):

First Summand: $iz \in IJ$: In the product iz, we have $i \in I$ and $z \in J$, so $iz \in IJ$.

Second Summand: $jz \in IJ$: In the product jz, we have $j \in J$ and $z \in I$, so $jz \in JI = IJ$ (by commutativity of R).

Since IJ is an ideal, it is closed under addition, which implies:

$$z = \underbrace{iz}_{\in IJ} + \underbrace{jz}_{\in IJ} \in IJ.$$

Therefore, $z \in IJ$, which implies $\bar{z} = 0$ in ${}^{R}\!/\!_{IJ}$. This shows that $\ker(\phi) = \{0\}$, so ϕ is injective. **Q.E.D.**

Surjectivity of ϕ

Spoken Transcript [00:09:25 - 00:11:15] Prof. Rahul Pandharipande

Okay, but as I said, we don't get anything for free. Now we have to prove surjectivity, which looks a little more complicated.

So, now we have to prove that this map is surjective [writes: ϕ is surjective].

Surjective means we start: so, choose any $\bar{x} \in R$ mod I and $\bar{y} \in R$ mod J [writes: Choose any $\bar{x} \in {}^{R}\!/\!_{I}, \bar{y} \in {}^{R}\!/\!_{J}$], right? Then we have to find... we have to construct a $\bar{z} \in R$ mod IJ such that [writes: We have to construct a $\bar{z} \in {}^{R}\!/\!_{IJ}$ such that] z is congruent to x mod I and z is congruent to y mod J [writes "$z \equiv x \bmod I$" and "$z \equiv y \bmod J$"]. If I do that, that's surjectivity. That's the Chinese Remainder problem that we talked about at the beginning. So, you pick where that you want, and you have to find the z that satisfies that.

Okay, so, how are we going to do that? Well, what I know is $i + j = 1$ [writes: $i + j = 1$]. That's all I know. So, now I need to do something. And what do you suggest I do here?

Well, before, the only thing we had was z. That was our putative element of the kernel. And we took this equation, we multiplied by z. Now, what we have in our hands are x and y. So, we have this equation, so it's more or less reasonable we multiply this by x and y. What else could we do? Let's multiply it by x.

Spoken Transcript [00:11:15 - 00:12:57] Prof. Rahul Pandharipande

$ix + jx = x$ [writes: $ix + jx = x$]. That's an interesting equation.

If I look at this term here, jx [circles jx], this is congruent to x mod I [writes: $jx \equiv x \bmod I$], because this equation says that if I look mod I, this is congruent... it's to x, okay? So, just this equation shows that jx is congruent to x mod I. Is that clear?

And jx, that's congruent to 0 mod J [writes: $jx \equiv 0 \bmod J$], because it has a j in it. It's in the ideal.

All right, so, that was some progress. You can imagine now we should multiply it by y [writes "$i + j = 1$"]: $iy + jy = y$ [writes: $iy + jy = y$]. And now we want to look at this one [circles iy]: iy is congruent to 0 mod I [writes: $iy \equiv 0 \bmod I$], because it has an i in it. But iy is congruent to y mod J [writes: $iy \equiv y \bmod J$], because of this equation. If I look at this mod J, that goes away, and iy is congruent...

So, that's perfect. We've used this... there was only one trick in this whole thing. We used it once for injectivity, and we're going to use it twice for surjectivity. So, what should z be here? Then I claim that we're now prepared to construct z. What should I write z here? Yeah?

Student Answer

$jx + iy$.

Spoken Transcript [00:12:59 - 00:13:45] Prof. Rahul Pandharipande

Yeah, did you get it right? $jx + iy$, yes: $jx + iy$ [writes: $\implies z = jx + iy$]. Because this thing, you just add them up now. It's x mod I and y mod J, that's it.

So, that's the end of surjectivity. So, here for this one, $\phi(\bar{z})$ is exactly equal to $(\bar{x}, \bar{y})$ [writes: $\phi(\bar{z}) = (\bar{x}, \bar{y})$]. And that's the end of the proof of the Chinese Remainder Theorem. And the proof... so, it's not a proof with no ideas. In some sense, it's one idea used three times.

But if you, I mean, if you understand this proof, you'll never forget it for the rest of your life, I think, somehow, maybe, I don't know.

Proof of Surjectivity and Full Restatement

Verification of Surjectivity

Let $\bar{x} = x + I \in {}^{R}\!/_{I}$ and $\bar{y} = y + J \in {}^{R}\!/_{J}$ be arbitrary elements. We must construct an element $z \in R$ satisfying the simultaneous congruences:

$$z \equiv x \pmod{I} \qquad \text{and} \qquad z \equiv y \pmod{J}. \tag{5.51}$$

We use the elements $i \in I$ and $j \in J$ with $i + j = 1$:

- **Action on x:** Multiplying $i + j = 1$ by x gives $ix + jx = x$. Analyzing the term jx:

$$jx = x - ix \equiv x \pmod{I} \qquad (\text{since } i \in I \implies ix \in I), \tag{5.52}$$

$$jx \equiv 0 \pmod{J} \qquad (\text{since } j \in J \implies jx \in J). \tag{5.53}$$

- **Action on y:** Multiplying $i + j = 1$ by y gives $iy + jy = y$. Analyzing the term iy:

$$iy \equiv 0 \pmod{I} \qquad (\text{since } i \in I \implies iy \in I), \tag{5.54}$$

$$iy = y - jy \equiv y \pmod{J} \qquad (\text{since } j \in J \implies jy \in J). \tag{5.55}$$

Now define:

$$z := jx + iy. \tag{5.56}$$

Evaluating z modulo each ideal:

$$z \equiv jx + iy \equiv x + 0 \equiv x \pmod{I},$$
$$z \equiv jx + iy \equiv 0 + y \equiv y \pmod{J}.$$

Thus, $\phi(z + IJ) = (z + I, z + J) = (x + I, y + J) = (\bar{x}, \bar{y})$. This proves that ϕ is surjective. Having established both injectivity and surjectivity, ϕ is an isomorphism of rings.

Q.E.D.

Q.E.D.

Beyond the Lecture: The Equation $i+j=1$ as a Partition of Unity

The equation $i+j=1$ with $i \in I$ and $j \in J$ works like an algebraic *"partition of unity"*: $j \equiv 1 \pmod{I}$ and $j \equiv 0 \pmod{J}$, while $i \equiv 0 \pmod{I}$ and $i \equiv 1 \pmod{J}$. The explicit solution $z = jx + iy$ combines these two elements with the target values x and y. This mirrors orthogonal projection bases in linear algebra and Lagrange interpolation polynomials in numerical analysis.

The Chinese Remainder Theorem in $\mathbb{Z}$

Spoken Transcript [00:13:45 - 00:15:00] Prof. Rahul Pandharipande

And you can... yeah. This is how it works, the Chinese Remainder Theorem. You just take that $i+j=1$ and use it. [erases the lower board]

Okay, so, let's go back to the integers, because you can use this now many, many times, and you get some fun results about the integers, which, you know, are very closely related to topics in school, but maybe usually not made explicit there.

So, this says that if I just write this theorem, the Chinese Remainder Theorem for $R = \mathbb{Z}$ [writes: CRT in $R = \mathbb{Z}$], then it says that if a and b are relatively prime [writes: $(a,b) = 1$] — this is in the language of the Chinese Remainder Theorem, this says that the ideal of a plus the ideal of b is equal to (1) [writes "$\iff$ $(a)+(b) = 1$", a slip for $= (1)$] —, then $\mathbb{Z}$ mod ab is isomorphic to $\mathbb{Z}$ mod $a \times \mathbb{Z}$ mod b [writes: then $\mathbb{Z}/(ab) \cong \mathbb{Z}/(a) \times \mathbb{Z}/(b)$], okay?

Corollary 5.47 (*Chinese Remainder Theorem for Coprime Integers*): Let $R = \mathbb{Z}$, and let $a, b \in \mathbb{Z}$ be coprime integers, i.e., $\mathrm{GCD}(a,b) = 1$. By Bézout's identity, this is equivalent to coprime principal ideals:

$$(a) + (b) = (\mathrm{GCD}(a,b)) = (1) = \mathbb{Z}.$$

Since the product of principal ideals in $\mathbb{Z}$ is $(a)(b) = (ab)$, Theorem 5.46 yields a ring isomorphism:

$$\mathbb{Z}/ab\mathbb{Z} \xrightarrow{\sim} \mathbb{Z}/a\mathbb{Z} \times \mathbb{Z}/b\mathbb{Z}, \quad z \bmod ab \longmapsto (z \bmod a,\ z \bmod b). \tag{5.57}$$

Spoken Transcript [00:15:00 - 00:17:15] Prof. Rahul Pandharipande

So, you think, okay, but this is only two, what if I factor in more? But you can do them one at a time. So, for example, if I take some number like $3 \times 5 \times 7$, that has... it has 105, it has three factors [writes: $3 \cdot 5 \cdot 7 = 105$].

So, how should I do with a, b, and c? Well, I can always group it as $a = 3$, $b = 35$ [writes: $a = 3, b = 35$]. And then the first step is that $\mathbb{Z}$ mod 105 is isomorphic to $\mathbb{Z}$ mod $3 \times \mathbb{Z}$ mod 35 [writes: $\mathbb{Z}/(105) \cong \mathbb{Z}/(3) \times \mathbb{Z}/(35)$], right?

And then I can again now do it to the 35 and split it further. So, this further isomorphic, I just apply it now to the 35 [writes: $\cong \mathbb{Z}/(3) \times \mathbb{Z}/(7) \times \mathbb{Z}/(5)$]. We could have stated this at the very beginning, but there's no reason to. So, it says I can just write them all out as long as I'm at every time splitting coprime factors out. Is that clear?

And so, then the fanciest way to write it is: suppose I have m that's an integer... let's call it n. There's a prime factorization: $p_1^{k_1}\ldots$, where these are the distinct primes that come here, $p_r^{k_r}$ [writes: $n = p_1^{k_1}p_2^{k_2}\cdots p_r^{k_r}$]. So, this is the prime factorization of n. It has some distinct primes to various powers. There's a unique way to write that. That's the unique factorization of primes.

And then if I just apply this, I'm not allowed to separate the prime factors, because if I separate p and, you know, p^2 and p, they're not relatively prime. But I can separate different prime factors. So, what this shows, this Chinese Remainder Theorem shows that this is isomorphic to the product... [writes: $\mathbb{Z}/(n) \cong \mathbb{Z}/(p_1^{k_1}) \times \cdots \times \mathbb{Z}/(p_r^{k_r})$]. I just separate out the primes. The primes all come apart.

Okay? So, I don't know whether this was... I mean, whatever happens in school is usually not in this language.

Iterated Factoring and Prime-Power Decomposition

Example 5.48 (*Iterated Decomposition of $\mathbb{Z}/(105)$*): Consider $n = 105 = 3\cdot 5\cdot 7$. Grouping the factors as $a = 3$ and $b = 35$, with $\mathrm{GCD}(3,35) = 1$:

$$\mathbb{Z}/105\mathbb{Z} \cong \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/35\mathbb{Z}.$$

Since $\mathrm{GCD}(5,7) = 1$, we decompose $\mathbb{Z}/35\mathbb{Z} \cong \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/7\mathbb{Z}$, which yields:

$$\mathbb{Z}/105\mathbb{Z} \cong \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/7\mathbb{Z}.$$

Corollary 5.49 (*Prime-Power Factorization of $\mathbb{Z}/n\mathbb{Z}$*): Let $n \in \mathbb{N}_{\geq 2}$ have the prime factorization into distinct primes $p_1 < p_2 < \cdots < p_r$:

$$n = p_1^{k_1}p_2^{k_2}\cdots p_r^{k_r}, \quad k_i \geq 1. \tag{5.58}$$

Since $\mathrm{GCD}(p_i^{k_i}, p_j^{k_j}) = 1$ for all $i \neq j$, induction on r gives the canonical ring isomorphism:

$$\mathbb{Z}/n\mathbb{Z} \xrightarrow{\sim} \mathbb{Z}/p_1^{k_1}\mathbb{Z} \times \mathbb{Z}/p_2^{k_2}\mathbb{Z} \times \cdots \times \mathbb{Z}/p_r^{k_r}\mathbb{Z} = \prod_{i=1}^{r} \mathbb{Z}/p_i^{k_i}\mathbb{Z}. \tag{5.59}$$

Why Prime Powers Cannot Be Split Further: Going beyond the lecture: while distinct prime powers $p_i^{k_i}$ and $p_j^{k_j}$ are coprime and can be separated, factors involving the same prime cannot be split via the Chinese Remainder Theorem. For example:

$$\mathbb{Z}/p^2\mathbb{Z} \not\cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z},$$

because $\mathrm{GCD}(p,p) = p \neq 1$. The ring $\mathbb{Z}/p^2\mathbb{Z}$ contains non-zero nilpotent elements ($p + p^2\mathbb{Z} \neq 0$, but $(p + p^2\mathbb{Z})^2 = 0$), whereas $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$ is a direct product of fields and has no non-zero nilpotents.

Units in ${}^{\mathbb{Z}}\!/\!_{n\mathbb{Z}}$ and Euler's Totient Function

Spoken Transcript [00:17:15 - 00:19:16] Prof. Rahul Pandharipande

Now we can consider the units. So, we have this idea of the units. We consider the units of this ring [writes: $\mathrm{Unit}({}^{\mathbb{Z}}\!/\!_{(n)})$] $\mathbb{Z}$ mod $n\mathbb{Z}$... Let me just write it, because I don't want to confuse the notation: the units of this ring $\mathbb{Z}$ mod $n\mathbb{Z}$, $\mathbb{Z}$ mod n.

The units. Those are the invertible elements here. But remember, 1 here is just $(1, 1)$, everybody is just 1, the unit here is the unit everywhere. So, how can you be invertible on this side? Well, you have to be invertible in every factor. So, this says the units here are also just the product here: it's the product of the units [writes: $= \mathrm{Unit}({}^{\mathbb{Z}}\!/\!_{(p_1^{k_1})}) \times \cdots \times \mathrm{Unit}({}^{\mathbb{Z}}\!/\!_{(p_r^{k_r})})$]. [writes "$\mathrm{Unit}({}^{\mathbb{Z}}\!/\!_{(p_r^{k_r})})$" on the far right] This was bad board management.

$p_r^{k_r}$, okay? So, if you're interested in the units of this ring, this is extremely useful, because it just tells you the units are just the product. You have to pick a unit here, a unit here, a unit here, that's all it is. And moreover, then it also tells you how many there are!

Because I hinted at this last... whenever we discussed this a few weeks ago. But this is related to this Euler totient function. That Euler totient function is the number of units. And this says if you want to know the number of units of $\mathbb{Z}$ mod $n\mathbb{Z}$, it's just the product of the number of units of all of these prime ones, prime power ones. But it's easy to calculate the number of units in the prime power, which I just remind you, I think... maybe I left it as an exercise. If so, I'll leave it as an exercise again.

Group of Units of the Direct Product

Proposition 5.50 (*Multiplicative Group of Units of a Product Ring*): Let $R_1, \ldots, R_r$ be commutative rings with identity. An element $(u_1, \ldots, u_r) \in \prod_{i=1}^r R_i$ is a unit if and only if each component $u_i \in \mathrm{Unit}(R_i)$ is a unit. Therefore, taking unit groups commutes with finite direct products:

$$\mathrm{Unit}\left(\prod_{i=1}^{r} R_i\right) \cong \prod_{i=1}^{r} \mathrm{Unit}(R_i). \tag{5.60}$$

Applying this to the prime-power decomposition (5.59) of ${}^{\mathbb{Z}}\!/\!_{n\mathbb{Z}}$:

$$\mathrm{Unit}\left({}^{\mathbb{Z}}\!/\!_{n\mathbb{Z}}\right) \cong \mathrm{Unit}\left({}^{\mathbb{Z}}\!/\!_{p_1^{k_1}\mathbb{Z}}\right) \times \mathrm{Unit}\left({}^{\mathbb{Z}}\!/\!_{p_2^{k_2}\mathbb{Z}}\right) \times \cdots \times \mathrm{Unit}\left({}^{\mathbb{Z}}\!/\!_{p_r^{k_r}\mathbb{Z}}\right). \tag{5.61}$$

Spoken Transcript [00:19:16 - 00:20:45] Prof. Rahul Pandharipande

If I take ϕ of a power of a prime [erases part of the lower board and writes: $\phi^{\mathrm{tot}}(p^k) =$] — and let me just follow the notation here: p, but any prime, no index, with a k —, so, what is this?

Well, this is all the numbers in $\mathbb{Z}$ mod $p^k\mathbb{Z}$ that are relatively... all the numbers that are relatively prime to p^k that are less than p^k. And how are you going to be a common factor? You have to have a p in it. And so, how many people have p's in it? Well, exactly $1/p$ of these has p in it. So, the answer is $p^k - p^{k-1}$ [writes: $= p^k - p^{k-1}$], or you can write this as $p^{k-1}(p-1)$ [writes: $= p^{k-1}(p-1)$]. Okay?

So, that gives a solution, a kind of formula to this Euler totient. It says that if I want to know what this totient function is, that's the order of the units [writes: $\phi^{\text{tot}}(n) =$], this is equal to the product from $i = 1$ to r, in that language, of $p_i^{k_i-1}(p_i - 1)$ [writes: $= \prod_{i=1}^{r} p_i^{k_i-1}(p_i-1)$]. That's a pretty good formula for it, probably the fastest way to calculate it.

Uh, but it's a confusing formula, like you might think... okay, so, this is the part that I view as perhaps related to things that may have been in school.

Recap: Euler's totient function was introduced in Week 3 (Definition 3.5), with its values $\phi(p)$ and $\phi(p^2)$ (Proposition 3.6); the units of $\mathbb{Z}/m\mathbb{Z}$ were determined in Week 2 (Theorem 2.71). The Chinese Remainder Theorem now gives the general formula.

Derivation of Euler's Product Formula

Definition 5.51 (*Euler's Totient Function*)**:** The *"Euler totient function"* $\phi^{\text{tot}}(n)$ counts the number of invertible elements in the ring $\mathbb{Z}/n\mathbb{Z}$:

$$\phi^{\text{tot}}(n) \overset{\text{def}}{=} \left|\text{Unit}\left(\mathbb{Z}/n\mathbb{Z}\right)\right| = \left|\{m \in \{1, \dots, n\} \mid \text{GCD}(m,n) = 1\}\right|. \tag{5.62}$$

Theorem 5.52 (*Euler's Product Formula*)**:** For a prime power p^k ($k \geq 1$), the elements in $\{1, 2, \dots, p^k\}$ that are **not** coprime to p^k are precisely the multiples of p:

$$\{p,\, 2p,\, 3p,\, \dots,\, p^{k-1}p = p^k\}.$$

There are exactly $\frac{p^k}{p} = p^{k-1}$ such multiples. Subtracting these from the total of p^k elements gives:

$$\phi^{\text{tot}}(p^k) = p^k - p^{k-1} = p^{k-1}(p-1). \tag{5.63}$$

Since the order of a direct product of finite groups is the product of their orders, taking cardinalities in (5.61) yields:

$$\phi^{\text{tot}}(n) = \prod_{i=1}^{r} \phi^{\text{tot}}\left(p_i^{k_i}\right) = \prod_{i=1}^{r} p_i^{k_i-1}(p_i - 1). \tag{5.64}$$

Beyond the Lecture: The Formula with Fractions

Textbooks often write these formulas as $\phi(p^k) = p^k\left(1 - \frac{1}{p}\right)$ and $\phi(n) = n\prod_{i=1}^{r}\left(1 - \frac{1}{p_i}\right)$.

Spoken Transcript [00:20:45 - 00:22:20]

Prof. Rahul Pandharipande

It's a formula that almost looks like if you know what this totient thing, since all the primes are here, it seems like you could reconstruct the actual number, because all its prime factorization is here. But it's not true! It's because of these $(p_i - 1)$'s. They confuse the issue. Like if you take a prime number... well, I can tell you a personal thing: my favorite prime number.

If I take 691 [writes: $\phi(691) =$] — which I assume you know is a prime number; this is an extremely famous prime number in mathematics —, but this is a prime number, and if I take that, it's just the $p-1$, so, this is 690 [writes: $= 690$], and that has a lot of factors in it. It has, you know, some 3 and 23, 5 and 2 [writes: $= 3 \cdot 23 \cdot 5 \cdot 2$]. So, you somehow lose... If you start with some nice prime number, this totient function gives you lots of primes and they get mixed up here. So, you can't recover, even though it almost looks like it, you can't recover the n from the totient value, because this $p-1$ kind of factors everything out.

Have you ever heard of this prime number? It's like one of the most famous prime numbers in mathematics. It comes from Bernoulli 12 [writes: $B_{12} =$] which is just a great Bernoulli number. It's $-691/2730$ [writes: $-\frac{691}{2730}$]. But have you learned about the Bernoulli numbers? They're also a local product, made in Basel. Because the Bernoullis were in Basel, that's why.

Okay, so, I think this is more or less what I wanted to say about the Chinese Remainder Theorem.

The Prime 691 and Bernoulli Numbers

Example 5.53 (*Euler's Totient on the Prime 691*): For the prime $p = 691$, the value of Euler's totient function is:

$$\phi(691) = 691 - 1 = 690 = 2 \cdot 3 \cdot 5 \cdot 23. \tag{5.65}$$

The prime 691 is famous in mathematics as the numerator of the 12th Bernoulli number B_{12}:

$$B_{12} = -\frac{691}{2730}. \tag{5.66}$$

Irreversibility of Euler's Totient Function: Going beyond the lecture: even though the prime factorization of n determines $\phi^{\text{tot}}(n)$ via formula (5.64), one cannot generally reconstruct n from $\phi^{\text{tot}}(n)$. The presence of the terms $(p_i - 1)$ mixes new prime factors (such as 23 appearing in $\phi(691) = 690$) that have no relation to the original prime $p = 691$. In modern cryptography (e.g., the RSA cryptosystem), the computational difficulty of finding $\phi^{\text{tot}}(n) = (p-1)(q-1)$ without knowing the prime factors p and q of a large integer $n = pq$ provides the trapdoor mechanism for secure encryption.

Beyond the Lecture: Kummer's Regularity Criterion and B_{12}

The prime 691 is the prototypical example of an *"irregular prime"*. In 1850, Ernst Kummer proved Fermat's Last Theorem for all regular prime exponents. Kummer showed that a prime p is regular if and only if p does not divide the numerator of any of the even Bernoulli numbers $B_2, B_4, \dots, B_{p-3}$. Since 691 divides the numerator of B_{12}, it is irregular, and it plays a prominent role in modern modular forms and arithmetic geometry (e.g., Ramanujan's τ-function congruences $\tau(n) \equiv \sigma_{11}(n) \pmod{691}$).

Polynomial Rings over Fields

Definitions and the Degree Function

Spoken Transcript [00:22:20 - 00:23:46] Prof. Rahul Pandharipande

And so, the last topic of the week, which is going to go on to next week, has to do with polynomials [writes: Polynomials].

And we've talked about these things in various ways, but I'll try to say it now clearly from the beginning. So, we start with k is a field [writes: k is a field].

And we've discussed before there is a ring which is polynomials in one variable [writes: $k[x]$ polys in a single variable] with coefficients in k [writes: with coeffs in k].

Okay. And now here is something that one has to just state a little bit carefully: it's the degree of a polynomial. We've said it kind of informally, but now I'm going to say it carefully, and there's only one point that's... All right, so, we would like to define the degree of a polynomial [writes: $\deg(f(x))$]. This seems kind of silly to even talk about, but we're going to say. So, we have some element of this ring [writes: $f(x) \in k[x]$], and we want to find the degree of the polynomial.

Spoken Transcript [00:23:46 - 00:25:16] Prof. Rahul Pandharipande

And so, the degree of the polynomial is that you write the polynomial, you find some coefficient, the highest coefficient which is not zero [writes "$c_d x^d$" above], so, this is the highest coefficient [writes: highest coeff.] which is not zero [writes: which is not zero].

And that makes sense, because a polynomial is just some list of these coefficients. You find the highest one that's not zero, you look at the d index, and that's the degree. So, this is the degree of this polynomial. Another way to write it informally: the degree of f is equal to this d [writes: $\deg(f) = d$], where d is the coefficient of the highest not zero. Okay, is that clear?

And the only subtle point of this is: what is the degree of the zero polynomial? [writes: $\deg(0)$] That is a super unpleasant thing to think about, because it has no coefficient that's not zero.

Spoken Transcript [00:25:16 - 00:26:46] Prof. Rahul Pandharipande

So, there's some kind of conventions about it, but the convention that's correct is that this is undefined [writes: undefined].

You can make up your own conventions, but the convention that works the best is it's undefined. So, that was my whole point of having the degree, right? It's kind of an annoying thing, but you see that this recipe makes no sense, because you're supposed to write it out, find the highest coefficient which is not zero, and say that's the degree. But if it's zero, all the coefficients are zero, so, you're kind of, you know...

Anyway, so, it's an important thing and it sounds silly, and in a sense it's silly, but when you talk about degrees, you're not allowed to talk about the degree of the zero polynomial.

The degree of a constant is okay. The degree of a constant, like the degree of the polynomial 7, as long as that 7 is not zero, that has degree zero. But the degree of zero, you don't talk about.

Spoken Transcript [00:26:46 - 00:27:36] Prof. Rahul Pandharipande

I hope that wasn't too polemic, but that's just the standard mathematical definition, and since I'm supposed to be teaching you mathematics, I tell you the definition.

So, that should have been also when you learned about polynomials in school, you should have asked your teacher: what is the degree of the zero polynomial? You should have asked that question. [erases the middle chalkboard to the left]

I think that's the only thing I'm going to say about polynomials that most people don't learn in school. The rest I'm going to talk about today is stuff you've learned in school, but this particular point is that...

Recap: Polynomial rings were defined in Week 3 (Definition 3.32, Definition 3.37), division with remainder in $\mathbb{Q}[x]$ was used in Week 4 (Lemma 4.41), and division with remainder for polynomials is among the prerequisites (Fact 0.32). The lecturer now treats $k[x]$ over a field systematically.

Polynomial Ring and the Degree of a Polynomial

Definition 5.54 (*Polynomial Ring over a Field*)**:** Let k be a field. The *"polynomial ring in one variable"* over k, denoted by $k[x]$, is the set of formal expressions:

$$f(x) = \sum_{i=0}^{n} c_i x^i = c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n, \quad c_i \in k, \tag{5.67}$$

equipped with standard polynomial addition and Cauchy product multiplication.

Definition 5.55 (*Degree of a Polynomial*)**:** Let $f(x) \in k[x]$.

(1) Non-Zero Polynomials: If $f(x) \neq 0$, write $f(x) = \sum_{i=0}^{n} c_i x^i$. The *"degree"* of f, denoted by $\deg(f)$ or $\deg(f(x))$, is the highest index $d \in \mathbb{N}_0$ such that the coefficient $c_d \neq 0$:

$$\deg(f) = d \quad :\Longleftrightarrow \quad c_d \neq 0 \quad \text{and} \quad c_i = 0 \quad \forall i > d. \tag{5.68}$$

The non-zero element $c_d \in k \setminus \{0\}$ is called the *"leading coefficient"*.

(2) Non-Zero Constants: For any non-zero constant $c \in k \setminus \{0\}$, we have $\deg(c) = 0$.

(3) The Zero Polynomial: For the zero polynomial $f(x) = 0$, the degree is explicitly:

$$\deg(0) \longleftarrow \text{Undefined.} \tag{5.69}$$

Remark 5.56 (*Conventions for the Degree of the Zero Polynomial*)**:** There are several conventions here. The one the lecturer calls correct, because it works best, leaves the degree of the zero polynomial **undefined**.

Why the Leading Coefficient Recipe Fails for Zero: The algorithmic recipe for the degree requires identifying the *"highest non-zero coefficient c_d"*. For the zero polynomial $0 \in k[x]$, all coefficients in $0 = \sum 0 \cdot x^i$ are identically zero, so no such index d exists.

Beyond the Lecture: The Convention $\deg(0) = -\infty$

Some textbooks set $\deg(0) := -\infty$, so that the degree formula $\deg(f \cdot g) = \deg(f) + \deg(g)$ remains formally valid when one factor is zero (with $-\infty + d = -\infty$).

III Polynomial Long Division

Spoken Transcript [00:27:36 - 00:29:16] Prof. Rahul Pandharipande

Okay, so, very important for us, and I'm not going to review it because you definitely learned it in school. At least I think you did. I think I went to school, but it's been a long time since I remember, but...

So, is polynomial long division [writes: Poly long div].

It says the following. So, suppose... and the twist here is it's now for any field, but the fact of the matter is whatever you learned in school also applies here. So, suppose we have two polynomials f and g in this polynomial ring [writes: $f(x), g(x) \in k[x]$]. So, coefficients in a field, that's important.

And we're going to do division, so when division, you always have to assume somebody's not zero. This one's not zero [writes: $g \neq 0$]. You can't divide by zero.

Then polynomial long division says that there exist unique polynomials [writes: $\exists!$] $q(x)$ and $r(x)$ [writes: $q(x), r(x)$] that satisfy these two properties. The first property is that $f(x)$ — q is for quotient, it's kind of the division — times g, and r is the remainder [writes: $f(x) = q(x)g(x) + r(x)$]. And then there's the constraint on r. So, the constraint is either r is zero [writes: either $r(x) = 0$]... and it's the either here because then I'm going to talk about the degree of r. I cannot talk about the degree of r formally if it's zero. So, either r is zero or r is not zero [writes: or $r(x) \neq 0$] and the degree of r is strictly less than the degree of g [writes: and $\deg(r) < \deg(g)$]. Okay, that's the one we're dividing. And you see, I assumed g is not zero, so, I'm allowed to talk about the degree of g.

Polynomial Long Division (Division with Remainder)

Theorem 5.57 (*Polynomial Long Division in $k[x]$*)**:** Let k be a field, and let $f(x), g(x) \in k[x]$ with $g(x) \neq 0$. Then there exist uniquely determined polynomials $q(x), r(x) \in k[x]$ such that:

$$f(x) = q(x)g(x) + r(x), \tag{5.70}$$

where the remainder $r(x)$ satisfies the dichotomy:

$$r(x) = 0 \quad \text{or} \quad \big(r(x) \neq 0 \text{ and } \deg(r) < \deg(g)\big). \tag{5.71}$$

The polynomial $q(x)$ is called the *"quotient"*, and $r(x)$ is called the *"remainder"*.

Handling the Remainder Condition with Undefined Degree: Because $\deg(0)$ is undefined, one cannot simply state the degree condition as $\deg(r) < \deg(g)$ without excluding the case $r(x) = 0$. The explicit case separation *"either $r(x) = 0$ or $\deg(r) < \deg(g)$"* preserves strict mathematical consistency with the definition of degree.

Spoken Transcript [00:29:16 - 00:30:46] Prof. Rahul Pandharipande

So, that's polynomial long division, and as I said, you know this. I'm not going to talk about how to prove it. But maybe we talk about some things about the uniqueness. So, polynomial long division is maybe you have some like $3x+1$, that's your g, and this is... I can't believe I'm writing this thing, x^4 like this [writes: $3x+1\overline{)\,x^4+\dots}$], and then you divide it, okay? And when you divide it, you get somebody, that's this, and what's left over is that. And if you think about it, they'll satisfy these properties. That's as much as I can possibly say about this.

Okay, so, what about... Did you write it like this?

Okay, good. As I said, I think I went to school, but it's been a long time.

So, maybe something that you didn't emphasize in school is the uniqueness, so, maybe we talk a little bit about that.

So, this says that if I have... [writes "$f(x) = q(x)g(x) + r(x)$" and below "$\hat{q}(x)g(x) + \hat{r}(x)$"]

Spoken Transcript [00:30:46 - 00:32:21] Prof. Rahul Pandharipande

I think if you remember every single thing you learned in school, you could almost be a professor.

So, suppose if I have two of these things here. So, if I have two ways to write it, I have to show that they're the same as long as they satisfy these conditions. So, of course, what do I do? I subtract them [writes: $0 = (q-\hat{q})g + (r-\hat{r})$]. I get zero is equal to q minus q hat — I don't always write the x's, that's just, you know, for emphasis it's a polynomial —: $(q-\hat{q})g + (r-\hat{r})$. Okay.

So, now then the cases are: so, g is not zero, that was the very beginning. So, suppose this is not zero [points to $q-\hat{q}$]. If this is not zero, then this product will have degree at least g [writes "$\deg \geq \deg(g)$" under $(q-\hat{q})g$]. And this thing has... again there's some cases about somebody's being zero or not, but this thing can never have degree g or higher, because they're both either zero or born lower. So, that means that the first one has to be zero. So, the first part of this logic is that $q-\hat{q}$ has to be zero [writes: $q-\hat{q}=0$], which of course means $q=\hat{q}$. Now, once this is zero, then this part dies, then you have this equation, and then again there's some cases, but if they're both not zero, it means they're equal; if one of them is zero, the other one is zero. So, that's the uniqueness. Okay?

All right. [erases the long division example on the blackboard]

Uniqueness of Polynomial Long Division

Proof of Uniqueness for Theorem 5.57

Suppose there exist two representations for $f(x)$ upon division by $g(x) \neq 0$:

$$f(x) = q(x)g(x) + r(x), \tag{5.72}$$

$$f(x) = \hat{q}(x)g(x) + \hat{r}(x), \tag{5.73}$$

where both remainder polynomials $r(x)$ and $\hat{r}(x)$ satisfy condition (5.71). Subtracting (5.73) from (5.72) gives:

$$0 = \big(q(x) - \hat{q}(x)\big)g(x) + \big(r(x) - \hat{r}(x)\big) \implies \big(q(x) - \hat{q}(x)\big)g(x) = \hat{r}(x) - r(x). \quad (5.74)$$

We show that $q(x) - \hat{q}(x) = 0$:

- **Case $q \neq \hat{q}$:** Assume for contradiction that $q(x) - \hat{q}(x) \neq 0$. Since k is a field and $g(x) \neq 0$, the degree of a product equals the sum of the degrees:
$$\deg\Big(\big(q(x) - \hat{q}(x)\big)g(x)\Big) = \deg(q - \hat{q}) + \deg(g) \geq \deg(g).$$
- **Degree Contradiction:** On the other hand, each of $r(x)$ and $\hat{r}(x)$ is either zero or has degree strictly less than $\deg(g)$. The difference $\hat{r}(x) - r(x)$ is therefore either zero or satisfies:
$$\deg(\hat{r} - r) \leq \max\big(\deg(\hat{r}), \deg(r)\big) < \deg(g).$$

This forces $\deg(g) \leq \deg\big((q - \hat{q})g\big) = \deg(\hat{r} - r) < \deg(g)$, an impossible contradiction. Therefore:

$$q(x) - \hat{q}(x) = 0 \implies q(x) = \hat{q}(x).$$

Substituting this back into (5.74) yields:

$$0 \cdot g(x) = \hat{r}(x) - r(x) \implies \hat{r}(x) - r(x) = 0 \implies r(x) = \hat{r}(x).$$

Hence, the quotient $q(x)$ and remainder $r(x)$ are uniquely determined. **Q.E.D.**

Proof of Existence for Theorem 5.57 (beyond the lecture)

The lecture does not prove existence (*"I'm not going to talk about how to prove it"*) and starts one long division example. The existence of $q(x)$ and $r(x)$ is proved by induction on the degree of $f(x)$. If $\deg(f) < \deg(g)$ or $f = 0$, we set $q(x) = 0$ and $r(x) = f(x)$. If $\deg(f) = n \geq m = \deg(g)$, write $f(x) = a_n x^n + \ldots$ and $g(x) = b_m x^m + \ldots$. Since k is a field, $b_m \in k \setminus \{0\}$ is invertible. The polynomial $f_1(x) := f(x) - a_n b_m^{-1} x^{n-m} g(x)$ has degree strictly less than n (or is zero). By the induction hypothesis, $f_1(x) = q_1(x)g(x) + r(x)$ with $\deg(r) < \deg(g)$ or $r = 0$, which yields $f(x) = \big(a_n b_m^{-1} x^{n-m} + q_1(x)\big)g(x) + r(x)$. **Q.E.D.**

III Every Ideal in $k[x]$ is Principal

Spoken Transcript [00:32:21 - 00:34:09] Prof. Rahul Pandharipande

But this is an extremely important way to think about these polynomials. I mean, the long division is just an important thing. It sounds silly, but it's important.

It seems like if you went to high school in Basel, you should learn about the Bernoulli numbers, though. If I were in charge of the curriculum... [erases the middle chalkboard]

Okay, so, let us say now some things maybe you haven't thought about. So, let I be an ideal [writes: Let $I \subset k[x]$ be an ideal], okay? So, any ideal. Then I claim, so, the proposition — and this is totally fundamental about polynomial rings — is that there exists a polynomial in I [writes: Prop. $\exists f \in I$ such] such that the whole ideal is just generated by that single thing [writes: that $I = (f)$]. It's just like the integers. Integers also have this property: if I take any ideal, it's always just the ideal generated by one person.

So, this is called a principal ideal [writes: Principal ideal]. A principal ideal is an ideal generated by one element, and this says that every ideal is principal, just like for the integers. There'll be a definition in a moment. So, let's prove this.

Proposition 5.58 (*Principal Ideals in $k[x]$*): Let k be a field, and let $I \subseteq k[x]$ be an ideal. Then there exists a polynomial $f \in I$ such that:

$$I = (f) = f \cdot k[x] = \{q(x)f(x) \mid q(x) \in k[x]\}. \tag{5.75}$$

That is, every ideal in the polynomial ring $k[x]$ is a *"principal ideal"*.

Proof of Proposition 5.58

Spoken Transcript [00:34:09 - 00:35:48] Prof. Rahul Pandharipande

Proof. [writes: Proof] So, again, we take I [writes: $I \subset k[x]$]. We know nothing about I except it's an ideal.

So, the first case, which is annoying, is that I could be the zero ideal [writes: $I = (0)$]. And then we're okay, we're happy [writes: $\implies$ happy, $0 \in I$], since zero is in I, because I is... So, this case is fine.

So, the second case is there exist elements g in I such that g is not zero [writes: $\exists g \in I, g \neq 0$]. So, either I is just the zero ideal, or there exist some non-zero elements. And now those non-zero elements, we can take their degrees.

So, then, now here's the whole choice, here's the whole proof, actually, in this line: pick f in I, f non-zero, such that the degree of f is minimal [writes: Pick $f \in I, f \neq 0$ s.t. $\deg(f)$ is minimal].

Because the set of degrees are some set of integers. As long as it's not empty, well, we can pick a minimal one, and that's what we do. Okay, that's actually the whole proof, this line. You look in I, you pick an element such that its degree is minimal, meaning it's minimal among all polynomials in I. Maybe I write that: minimal among all elements not zero of I [writes: (among all elts $\neq 0$ of I)]. All right? So, do you agree I can do that? That has to do with the fact that you can pick lowest integers in a set of non-negative integers. But that's the whole proof here.

Case 1 and the Choice of the Minimal Degree Generator

Case 1: $I = \{0\}$: If I contains only the zero polynomial, then $I = (0) = 0 \cdot k[x]$, which is principal.

Case 2: $I \neq \{0\}$: If I contains non-zero elements, pick $f \in I \setminus \{0\}$ whose degree is minimal among all non-zero elements of I; this is possible because every non-empty set of non-negative integers has a least element (Fact 0.11):

$$\deg(f) = \min\{\deg(g) \mid g \in I,\ g \neq 0\}. \tag{5.76}$$

Spoken Transcript [00:35:48 - 00:37:19] Prof. Rahul Pandharipande

So, now we have to prove it. So, then let g in I be any element [writes: Let $g \in I$ be any element]. I'm going to use now polynomial long division. Except I'm going to divide g by f, so, it's $q(x) \cdot f + r$ [writes: $g = q(x) \cdot f + r(x)$], which is the opposite I did there, but nevertheless, okay?

So, I take this g, here's my f that's the lowest, that's the minimal one. I take any g and I use this polynomial long division, all right? So, now remember the cases are... So, the first thing I notice about this polynomial long division is that f is in I [writes "$\in I$" under f], so, this whole product is in I [writes "$\in I$" under $q(x)f$], and g is in I [writes "$\in I$" under g], so, the conclusion is $r(x)$ is in I [writes: $\implies r(x) \in I$], okay?

For this polynomial long division, I spend my time, eventually get this remainder; the remainder is in I, it has to be.

Division of an Arbitrary Element by the Minimal Polynomial

Let $g \in I$ be an arbitrary element. We apply Polynomial Long Division (Theorem 5.57) to divide $g(x)$ by $f(x) \neq 0$:

$$g(x) = q(x)f(x) + r(x), \quad \text{where } r(x) = 0 \text{ or } \deg(r) < \deg(f). \tag{5.77}$$

Since $f \in I$ and I is an ideal, the multiple $q(x)f(x) \in I$. Furthermore, $g \in I$. Rearranging for the remainder yields:

$$r(x) = \underbrace{g(x)}_{\in I} - \underbrace{q(x)f(x)}_{\in I} \in I. \tag{5.78}$$

Thus, the remainder $r(x)$ is itself an element of the ideal I.

Spoken Transcript [00:37:19 - 00:38:47] Prof. Rahul Pandharipande

Okay, so, what are the possibilities for the remainder? The remainder could be zero [writes: $r(x) = 0$]. That's good for us, because then again we're happy again, because then that means that g is in [writes: $\implies g \in (f)$]... because if this is zero, that means g is in the ideal generated by f. Okay?

What's the other possibility? This $r(x)$ is not zero [writes: $r(x) \neq 0$] and the degree of r is strictly less than the degree of f [writes: $\implies \deg(r) < \deg(f)$]. But that's not possible, because that r would then be an element of I with degree strictly less than f, which doesn't exist, because f is minimal with respect to degree.

So, this is not possible [writes: not possible] because of the choice of f [writes: bec of choice of f].

So, this thing can't happen. We can cross it out with purple [crosses out with purple chalk]. So, it has to be this, and that's the proof.

Full Restatement: Proof that Every Ideal in $k[x]$ is Principal

Complete Proof of Proposition 5.58

Let $I \subseteq k[x]$ be an ideal.

Step 1: Zero Ideal: If $I = \{0\}$, then $I = (0)$ is principal.

Step 2: Minimal Degree Choice: If $I \neq \{0\}$, choose $f \in I \setminus \{0\}$ of minimal degree among all non-zero polynomials in I.

Step 3: First Inclusion ($(f) \subseteq I$): Clearly, the principal ideal $(f) = f \cdot k[x] \subseteq I$, because $f \in I$ and I is an ideal.

Step 4: Reverse Inclusion ($I \subseteq (f)$): Conversely, take any $g \in I$. By Theorem 5.57, there exist $q(x), r(x) \in k[x]$ such that:

$$g = qf + r, \quad \text{with } r = 0 \text{ or } \deg(r) < \deg(f).$$

Then $r = g - qf \in I$. If $r \neq 0$, then $r \in I \setminus \{0\}$ would satisfy $\deg(r) < \deg(f)$, directly contradicting the minimality of $\deg(f)$.

Step 5: Conclusion: Therefore, $r = 0$, which implies:

$$g = qf \in (f).$$

Since $g \in I$ was arbitrary, $I \subseteq (f)$.

Combining inclusions gives $I = (f)$, proving that I is principal. **Q.E.D.**

Parallel with the Integers: This proof mirrors the classical proof that every ideal in $\mathbb{Z}$ is principal ($I = (n)$):

- **Ring of Integers $\mathbb{Z}$:** One selects the non-zero element $n \in I$ with the smallest absolute value (or smallest positive integer) and uses integer division with remainder $a = qn + r$ ($0 \leq r < |n|$).
- **Polynomial Ring $k[x]$:** One selects the non-zero polynomial $f \in I$ with the smallest degree and uses polynomial division with remainder $g = qf + r$ ($r = 0$ or $\deg(r) < \deg(f)$).

Q.E.D.

Principal Ideal Domains (PIDs)

Spoken Transcript [00:38:47 - 00:40:01] Prof. Rahul Pandharipande

So, the outcome of this is that one can think — of course, this breaks down at a certain point, but it's useful to think about these polynomial rings a little bit like the way you think about the integers. Integers have... I mean, this kind of polynomial in one variable with a field, that they have this property that all the ideals are just generated by one element, just like in the integers. And there's a name for such rings, and they'll play a role later in the class, but maybe I say it.

So, in words, the definition is: R is... so the ring R is a principal ideal domain, which is called PID, but not *"pid"*. Yeah.

Student Question

Why is q in the ideal?

Spoken Transcript [00:40:04 - 00:40:09] Prof. Rahul Pandharipande

q? I never said q's in the ideal. So, I can't answer that question.

Student Clarification

Ah, das ist ein Elementzeichen... [realizes the chalk inscription was $f \in I$, not $q \in I$]

Spoken Transcript [00:40:09 - 00:40:29] Prof. Rahul Pandharipande

f is in the ideal, and this is for the product.

Student Acknowledgment

Oh, okay. Thank you.

Spoken Transcript [00:40:29 - 00:41:37] Prof. Rahul Pandharipande

Okay, so, R is a principal ideal domain in words if every ideal $I \subset R$ is principal [writes: R is a PID if every ideal $I \subset R$ is principal].

So, that's in words. It means exactly: for every ideal in R, there exists some element of the ideal such that the ideal is generated by that element [writes: [$\forall$ ideal $I \subset R$, $\exists f \in I$ s.t. $I = (f)$]], okay? This is just repeating it.

So, that's a special kind of domain. I should say that when I say R is a principal ideal domain, it should be a domain too. I didn't write it, I guess, because it's already there. R is a principal ideal domain if every ideal of R is principal, and R is a domain [adds: and R is a domain].

Okay? But it's already in the... right. So, to be a principal ideal domain, you have to be a domain, of course.

So, what we've seen in the class so far: $\mathbb{Z}$, I mean, many of them, but let's say $\mathbb{Z}$ is a PID [writes: $\mathbb{Z} \leftarrow$ PID].

What about $\mathbb{Z}$ mod $6\mathbb{Z}$? [writes: ${}^{\mathbb{Z}}\!/_{6\mathbb{Z}}$] Well, no, because it's not a domain [writes: not a domain].

A field is a principal ideal domain. It's kind of a dumb principal ideal domain, because there's not many ideals, but it is [writes: A field is a PID, because $(0), (1)$]. Because there are only two ideals in a field: one ideal is this one, and one ideal is that one. So, it's also a principal ideal domain — not, as I said, not the most interesting way to look at a field. That's an interesting principal ideal domain, and then today we see that if k is a field, then this is a principal ideal domain [writes "$k[x] \leftarrow$ PID" and "field" with an arrow pointing to k].

Definition 5.59 (*Principal Ideal Domain (PID)*)**:** An integral domain R is called a *"principal ideal domain"* (abbreviated *"PID"*) if every ideal $I \subseteq R$ is principal, meaning:

$$\forall \text{ ideal } I \subseteq R,\ \exists f \in I \quad \text{such that} \quad I = (f). \tag{5.79}$$

Examples 5.60 (*Examples and Non-Examples of PIDs*)**:**

- **(1) The Integers $\mathbb{Z}$ (PID):** Every ideal in $\mathbb{Z}$ has the form $n\mathbb{Z} = (n)$. Since $\mathbb{Z}$ is an integral domain, $\mathbb{Z}$ is a PID.
- **(2) Quotient Ring $\mathbb{Z}/6\mathbb{Z}$ (NOT a PID):** $\mathbb{Z}/6\mathbb{Z}$ is **not** an integral domain because it contains zero divisors $(2 \cdot 3 \equiv 0 \pmod 6)$.
- **(3) Any Field k (PID):** If k is a field, its only ideals are (0) and $(1) = k$. Both are principal, and k is an integral domain, so every field is a PID.
- **(4) Polynomial Ring $k[x]$ over a Field (PID):** By Proposition 5.58, every ideal in $k[x]$ is principal. Since $k[x]$ has no zero divisors when k is a field, $k[x]$ is a PID.

Beyond the Lecture: The Ideals of $\mathbb{Z}/6\mathbb{Z}$

Every ideal of $\mathbb{Z}/6\mathbb{Z}$ is principal. The ring fails to be a PID only because it is not an integral domain.

Rings that are Not Principal Ideal Domains

Spoken Transcript [00:41:37 - 00:41:50] Prof. Rahul Pandharipande

Something I should have said kind of clearly here is that that division algorithm that you learn in school uses the fact that you have a field, because, you know, you might have to divide by the leading coefficients. So, that was used in the way that the division algorithm works.

So, some things that are not... rings that are not principal ideal domains [writes: Rings that are not PIDs].

Well, one which is kind of close is that if I take polynomials over $\mathbb{Z}$ in one variable [writes: $\mathbb{Z}[x]$], because you can't use the division algorithm the same way, because you can't divide. So, can you give me an ideal here that's probably not principal?

I don't expect you to give me a proof of it, but what do you think would be probably not principal here?

Student Suggestion

$x^2 + 1$?

Spoken Transcript [00:41:52 - 00:42:03] Prof. Rahul Pandharipande

Yeah, I think that might be principal. But, you know, if you're just using the x part with leading terms 1... yeah, okay, you can think about that ideal, I think that one is principal. The standard example is you use something from $\mathbb{Z}$, like the 2, and something from x on this side [writes: $\supset (2,x)$]. This is like the canonical one.

Yeah, yours is principal, I'd say. I think.

All right, so, this is the standard one that's not principal. And the reason, as I said, you take something from the $\mathbb{Z}$ side and something from the two side. And so, this should be an exercise for you: prove it's not principal [writes: Prove not principal].

And another way you could do it, if you have a field with polynomials in two variables over a field, then you can take the ideal (x,y) [writes "$\mathbb{C}[x,y] \supset (x,y)$" and "Prove not princ."]. And that's also clearly not going to be principal, because there's no way you can find one polynomial that generates both x and y. Okay. That's it. [students applaud]

Non-Principal Ideals in Multi-Variable and Integer Polynomial Rings

Examples 5.61 (*Rings that are Not Principal Ideal Domains*):

(1) Polynomials over the Integers $\mathbb{Z}[x]$ (NOT a PID): The ideal generated by 2 and x:

$$I := (2,x) = \{2p(x) + xq(x) \mid p(x), q(x) \in \mathbb{Z}[x]\} \tag{5.80}$$

cannot be generated by a single polynomial, so $\mathbb{Z}[x]$ is not a PID.

(2) Polynomials in Two Variables $\mathbb{C}[x,y]$ (NOT a PID): In the polynomial ring in two variables over $\mathbb{C}$ (or any field k), the ideal:

$$J := (x,y) = \{xp(x,y) + yq(x,y) \mid p,q \in \mathbb{C}[x,y]\} \tag{5.81}$$

requires at least two generators. No single polynomial in $\mathbb{C}[x,y]$ can divide both x and y simultaneously while generating all elements in J; a proof is given at the start of Week 6 (Answer 6.3).

Exercise 5.62 (*Proving that $(2,x)$ is Not Principal in $\mathbb{Z}[x]$*): Prove that the ideal $(2,x) \subset \mathbb{Z}[x]$ is not principal.

Proof Hint for Exercise 5.62 (beyond the lecture)

The lecture leaves this as an exercise. Suppose $(2,x) = (f(x))$ for some $f(x) \in \mathbb{Z}[x]$. Since $2 \in (f(x))$, $f(x)$ must divide 2 in $\mathbb{Z}[x]$, which forces $f(x) = \pm 1$ or $f(x) = \pm 2$. If $f(x) = \pm 1$, then $(f(x)) = \mathbb{Z}[x]$, but every element $2p(x) + xq(x)$ of $(2,x)$ has an even constant term, so $1 \notin (2,x)$. If $f(x) = \pm 2$, then $x \in (2)$, which would mean $x = 2q(x)$ for some $q(x) \in \mathbb{Z}[x]$, which is impossible since the leading coefficient of x is 1, not divisible by 2. Thus, no single generator exists. **Q.E.D.**

Summary of Key Concepts

- **Canonical Product Homomorphism:** For any ideals $I, J \subseteq R$, the natural map $\phi\colon R/IJ \to R/I \times R/J$ given by $\bar{z} \mapsto (\bar{z}, \bar{z})$ is a well-defined ring homomorphism. Without coprimality, ϕ generally fails to be injective or surjective (e.g., $I = (4)$ and $J = (6)$ in $\mathbb{Z}$).
- **Chinese Remainder Theorem:** When $I + J = (1) = R$, there exist $i \in I$ and $j \in J$ such that $i + j = 1$. This equation ensures that ϕ is an isomorphism, yielding

explicit solutions $z = jx + iy$ to simultaneous systems of congruences.

- **Arithmetic in $\mathbb{Z}$ and Euler's Totient Function:** Iterated application yields ${}^{\mathbb{Z}}\!/_{n\mathbb{Z}} \cong \prod_{i=1}^{r} {}^{\mathbb{Z}}\!/_{p_i^{k_i}\mathbb{Z}}$. Taking unit groups gives $\phi^{\text{tot}}(n) = \prod_{i=1}^{r} p_i^{k_i-1}(p_i - 1)$. The terms $p_i - 1$ mix the primes, so n cannot be recovered from this value (for example $\phi^{\text{tot}}(691) = 690$).
- **Polynomial Rings $k[x]$ over Fields:** For a field k, polynomials have long division with unique quotient and remainder ($r = 0$ or $\deg(r) < \deg(g)$). The degree of the zero polynomial $\deg(0)$ is left strictly undefined.
- **Principal Ideal Domains (PIDs):** Using the well-ordering of degrees, every ideal $I \subseteq k[x]$ is generated by any non-zero element of minimal degree, proving that $k[x]$ is a PID. Multi-variable polynomial rings like $\mathbb{C}[x, y]$ and polynomial rings over non-fields like $\mathbb{Z}[x]$ fail to be PIDs due to non-principal ideals such as (x, y) and $(2, x)$.

Tuesday, 20th October 2026

Unique Factorization Domains and Factorization in PIDs

Course Content Overview

This lecture formally develops the theory of unique factorization domains (UFDs) and establishes their fundamental relationship with principal ideal domains (PIDs):

- **Review and Motivation:** We examine why multivariate polynomial rings fail to be principal ideal domains (e.g., $(x, y) \subset k[x, y]$ is non-principal, and (x^2, xy, y^2) requires three generators), motivating the Hilbert Basis Theorem and Emmy Noether's concept of a Noetherian ring.
- **Foundations of Factorization:** The foundational notions of units $\mathcal{U}_R$ and irreducible elements are established for arbitrary integral domains, highlighting the critical distinction between irreducible elements and prime elements in abstract ring theory.
- **Unique Factorization Domains:** We define UFDs via the existence and uniqueness of irreducible factorizations (up to permutation and unit associates). We examine standard examples ($\mathbb{Z}$, fields k, $k[x]$, and multivariate polynomial rings $k[x_1, \ldots, x_n]$ via Gauss's Theorem B) and investigate an explicit non-example: the subring $R = \mathbb{Q}[x^2, x^3] \subset \mathbb{Q}[x]$ of polynomials lacking a linear term, whose picture, the lecturer says, is a cusp.
- **Proof of Theorem A (PID $\implies$ UFD):** The core theorem is established across three sequential steps:

  Step 1: Uniqueness Property: If every irreducible element in a domain generates a prime ideal, factorizations into irreducibles are unique up to permutation and unit associates.

  Step 2: Irreducible Implies Prime in a PID: In any PID, every irreducible element generates a prime ideal, establishing the uniqueness property for all PIDs.

  Step 3: Existence Property: By analyzing a recursive binary factorization tree, we show that any non-terminating factorization branch yields a strictly ascending chain of principal ideals $(x_0) \subsetneq (x_1) \subsetneq (x_2) \subsetneq \ldots$, which is ruled out by the PID property on the union ideal $J = \bigcup_{n=0}^{\infty} (x_n) = (w)$.
- **GCD, LCM, and Power Series:** We express greatest common divisors and least common multiples in arbitrary UFDs via prime factorizations, analyze associate classes and unit ambiguities, prove the identity $\mathrm{GCD}(x, y) \cdot \mathrm{LCM}(x, y) \sim x \cdot y$, and conclude with the exercise that formal power series rings $k[[x]]$ are PIDs and UFDs.

Review: Polynomial Rings and Ideals in Several Variables

Spoken Transcript [00:00:00 - 00:01:19] Prof. Rahul Pandharipande

Okay, so, we start. And first, some comments from last week.

Last week we had k is a field, [writes: k field] and we proved, using the division algorithm (Recall: Theorem 5.57) — so, this was from last week — that if I take the polynomial ring, this is a PID, [writes: $\to k[x]$ is a PID] a principal ideal domain. And this means that [draws curved arrow] every ideal [writes: any ideal is of the form $(f) = I$] is of the form... an ideal... every ideal is of this form, where f is just one element, is generated by one element.

And so, there were several questions which were interesting, so, I'll say a little bit about it.

So, what about [writes: What about] if I take $k[x,y]$, [writes: $k[x,y]$] so, polynomials in two variables? Is this a PID? [writes: is this a PID]

And the answer is no. [writes: Answer: No]

You can just take the ideal generated by x and y, [writes: $I = (x,y)$] and then you have to prove yourself that this ideal does not have a single generator, which is elementary, it's believable.

Recap: Principal ideal domains were defined in the previous lecture (Definition 5.59), where it was proved that every ideal of $k[x]$ is principal (Proposition 5.58), using polynomial long division (Theorem 5.57). That lecture also named $(x,y) \subset \mathbb{C}[x,y]$ as a non-principal ideal and left the proof as an exercise (Examples 5.61). The lecturer recalls both.

Review of Univariate Polynomial Rings and the Failure of the PID Property in Two Variables

Proposition 6.1 (*$k[x]$ is a Principal Ideal Domain*)**:** Let k be a field. Then the polynomial ring in one variable $k[x]$ is a *"principal ideal domain"* (PID). That is, for every ideal $I \subseteq k[x]$, there exists a single polynomial $f \in k[x]$ such that:

$$I = (f) = f \cdot k[x]. \tag{6.1}$$

Why the Division Algorithm Yields a PID: If $I = (0)$, then $I = (0)$ is principal. If $I \neq (0)$, choose a non-zero polynomial $f \in I$ of minimal degree. For any $g \in I$, polynomial division with remainder yields $g = q \cdot f + r$ with $q, r \in k[x]$ and either $r = 0$ or $\deg(r) < \deg(f)$. Since $r = g - q \cdot f \in I$, the minimality of $\deg(f)$ forces $r = 0$. Hence, $g \in (f)$, proving $I = (f)$.

Question 6.2 (*Is $k[x,y]$ a PID?*)**:** What about $k[x,y]$: is this a PID?

Answer 6.3 (*to Question 6.2*)**:** **No**. The ideal generated by the two variables:

$$I := (x,y) = \{a(x,y)x + b(x,y)y \mid a, b \in k[x,y]\}$$

is not principal.

Proof that (x,y) is not Principal in $k[x,y]$ (beyond the lecture)

The lecture leaves this to the students. Suppose $(x,y) = (f)$ for some $f \in k[x,y]$. Every element of (x,y) has constant term 0, so $1 \notin (x,y)$; hence f is not a non-zero constant, and $f \neq 0$ since $x \in (f)$. So f has total degree ≥ 1. From $x \in (f)$ we get $x = fg$; total degrees add in $k[x,y]$ (the top-degree parts of f and g multiply to a non-zero polynomial, as $k[x,y]$ is a domain by Theorem 4.6), so $\deg f = 1$ and $g = c \in k \setminus \{0\}$, that is, $f = c^{-1}x$. Then $y \in (f) = (x)$, so $y = xh$ for some $h \in k[x,y]$; setting $x = 0$ gives $y = 0$, a contradiction. Hence (x,y) cannot be generated by a single polynomial; this settles the exercise of Examples 5.61. **Q.E.D.**

Spoken Transcript [00:01:19 - 00:02:03]

Prof. Rahul Pandharipande

Then the second question, related, well: are all ideals [writes: Are all ideals] of $k[x,y]$ [writes: of $k[x,y]$] generated by at most two elements? [writes: generated by at most 2 elements?] These are good questions. [inserts "at most" with a caret] So, at most two elements?

That's a reasonable thing to guess from this answer. And the answer is no. [writes: No:] And you can take the ideal [writes: $I = (x^2, xy, y^2)$] x^2, xy, y^2. This one needs at least three elements, which I leave you to prove also.

Ideals Requiring More Than Two Generators

Question 6.4 (*Are All Ideals of $k[x,y]$ Generated by at Most Two Elements?*)**:** Are all ideals of $k[x,y]$ generated by at most 2 elements?

Answer 6.5 (*to Question 6.4*)**: No.** The ideal:

$$I := (x^2, xy, y^2) \subset k[x,y] \tag{6.2}$$

requires at least 3 generators.

Why (x^2, xy, y^2) Needs at Least Three Generators (beyond the lecture)

The lecture leaves this to the students (*"which I leave you to prove also"*). Consider the maximal ideal $m = (x,y) \subset k[x,y]$ and the ideal $mI \subseteq I$ (products, Definition 5.34). The quotient I/mI is a vector space over $k[x,y]/m \cong k$. Every element of I has all its terms of degree ≥ 2, and every element of mI has all its terms of degree ≥ 3; so a k-linear combination of x^2, xy, y^2 lies in mI only if it is 0, and the images of x^2, xy, y^2 are linearly independent: $\dim_k(I/mI) \geq 3$. If $I = (g_1, \ldots, g_s)$, then every element of I is $\sum r_i g_i$, and modulo mI only the constant terms of the r_i matter; so the images of $g_1, \ldots, g_s$ span I/mI, and $s \geq 3$. **Q.E.D.**

Spoken Transcript [00:02:03 - 00:03:30]

Prof. Rahul Pandharipande

So, then, the question is: what could be true? [writes: What could be true] This is the third question, [chuckles] maybe a more general question: what could be true? And there's a great answer here, which really goes somehow to the beginning of commutative algebra and the work of David Hilbert, a hundred years ago, [chuckles] a hundred twenty years ago.

The thing that is true is a famous theorem which is called the Hilbert Basis Theorem. [draws curved arrow and writes: Hilbert Basis Thm] This is not for this class. [chuckles]

And Hilbert proves that if [writes: If] I is an ideal in a polynomial ring over a field in finitely many variables — [writes: $I \subset k[x_1, \ldots, x_n]$] so, this is finitely many variables, that's why I write n, it's an ideal in this ring —, then [writes: then] I has finitely many generators; I is generated by finitely many elements. [writes: I is generated by finitely many elements]

This means that I [writes: $I = (f_1, \ldots, f_m)$] is the ideal generated by some number of elements, $f_1 \ldots f_m$, some finite number. This number is not that, it's just some finite number. So, it says that at least ideals are finitely generated. That's the best you can do for these.

The Hilbert Basis Theorem

Theorem 6.6 (*Hilbert Basis Theorem*): Let k be a field, and let $k[x_1, \ldots, x_n]$ be the polynomial ring in n variables over k. If $I \subseteq k[x_1, \ldots, x_n]$ is an ideal, then I is finitely generated:

$$I = (f_1, \ldots, f_m) \quad \text{for some finite collection } f_1, \ldots, f_m \in k[x_1, \ldots, x_n]. \tag{6.3}$$

Finite Generation vs. Uniform Bounds: Going beyond the lecture: although $k[x_1, \ldots, x_n]$ is not a PID for $n \geq 2$, and there is no universal upper bound on the minimal number of generators an ideal may require, every ideal is guaranteed to admit a finite generating set.

Spoken Transcript [00:03:30 - 00:04:50] Prof. Rahul Pandharipande

And the next step in the story is... This theorem, it's kind of an interesting theorem, it's not even so hard to prove. In fact, I could prove it in this lecture for you, but I'm not going to do that, because it's not for this course. But this theorem kind of changes how to think about rings in many variables. And it leads to a definition by Emmy Noether: [draws curved arrow and writes: Noetherian Ring] Noetherian ring.

And the definition [writes: Def: R is a Noetherian Ring] is that R is a Noetherian ring — this is also on the order of a hundred years ago — Noetherian ring. There are various ways to define it: if and only if every ideal [writes: Every $I \subset R$ is finitely gen.] is finitely generated, in the sense of David Hilbert.

And so, this is an example: if you look at the flow of mathematics, there's kind of a standard slogan, which maybe you've heard, or if not, you should think about it. The slogan is that every great theorem becomes a definition. This is a perfect example of that.

And these ideas are the topic, kind of the central topic of the Commutative Algebra course. We're not going in this direction.

Noetherian Rings

Definition 6.7 (*Noetherian Ring*): A ring R is called a *"Noetherian ring"* if every ideal $I \subseteq R$ is finitely generated:

$$\forall I \subseteq R, \quad \exists f_1, \dots, f_m \in R \quad \text{such that} \quad I = (f_1, \dots, f_m). \tag{6.4}$$

Equivalent Formulation via the Ascending Chain Condition: Going beyond the lecture: equivalently, a ring R is Noetherian if it satisfies the *"Ascending Chain Condition"* (ACC) on ideals: every chain of ideals

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \dots$$

eventually stabilizes, i.e., there exists $N \geq 1$ such that $I_k = I_N$ for all $k \geq N$. (If every ideal is finitely generated, the union of the chain is an ideal, generated by finitely many elements that all lie in some I_N; conversely, a non-finitely-generated ideal yields a strictly ascending chain by adding one element at a time.) Step 3 of the proof of Theorem A below uses exactly this argument for chains of principal ideals. In its general form, the Hilbert Basis Theorem says: if a commutative ring R is Noetherian, then so is $R[x]$, and by induction $R[x_1, \dots, x_n]$; for $R = k$ a field (whose only ideals are (0) and (1), Theorem 1.31) this gives Theorem 6.6.

***Insight:* "Every Great Theorem Becomes a Definition"**

The lecturer gives a standard slogan: *"every great theorem becomes a definition."* This is a perfect example of it: David Hilbert proved that every ideal in $k[x_1, \dots, x_n]$ is finitely generated (the Basis Theorem), and this property became Emmy Noether's definition of a *"Noetherian ring"*, the central topic of the Commutative Algebra course.

Unique Factorization Domains

Spoken Transcript [00:04:50 - 00:06:05] Prof. Rahul Pandharipande

Okay, so, what is the topic of this week? The topic of this week are unique factorization domains. [writes: Unique Factorization Domains]

Also, you don't have to wait to next year to take this course, you can just read about it also.

Factorization domains. [writes: Factorization Domains] So, that's the topic this week. And the first step, of course, is I have to define it.

First, though, it's going to take a little bit of time to define, so, you have to be patient about that.

Integral Domains, Units, and Irreducible Elements

Spoken Transcript [00:06:05 - 00:07:00] Prof. Rahul Pandharipande

So, first of all, we are going to consider domains. [writes: R is a domain] R is a domain.

The whole week is about domains. I will try to say it every time, but of course if I say unique factorization domain, principal ideal domain, there's the domain in the name, and it means that the ring is a domain.

And we want to consider these domains where we have this factorization, and the domain that you should have in your head as kind of motivation, inspiration for this is of course the integers. The integers are a domain, and they have unique factorization (Recall: Fact 0.19), which I don't repeat now. But we want to now consider those ideas in the context of arbitrary domains.

Standing Convention: Integral Domains

Remark 6.8 (*Standing Convention: Domains*): Throughout this chapter, R will always denote an *"integral domain"* (or simply a *"domain"*): a commutative ring with identity $1 \neq 0$ containing no zero divisors, that is:

$$\forall a, b \in R: \quad a \cdot b = 0 \implies a = 0 \text{ or } b = 0. \tag{6.5}$$

The integers $\mathbb{Z}$ serve as the primary archetype and motivation for factorization theory. Integral domains were defined in Week 2 (Definition 2.62); in a domain, a non-zero factor can be cancelled ($ac = bc,\ c \neq 0 \implies a = b$), which the proofs below use repeatedly.

Spoken Transcript [00:07:00 - 00:08:45] Prof. Rahul Pandharipande

About the domain, there's some special elements: zero, [writes: $0 \in R$ zero] and then there's the set of units. [writes: Units $\in R$]

So, these two classes of elements, the zeros and units, play a special role. Like in the integers, when you talk about factorization, you're not talking about zero. Zero is not in the picture.

Units complicate the factorization. The units in the integers are plus/minus one, and they confuse the matter of the unique factorization because you could always take 15 [writes: $15 = 3 \cdot 5$] is equal to 3×5, or 15 equals minus three [writes: $= (-3) \cdot (-5)$] times minus five, and there's this unfortunate confusion about the units. That's because this is the effect of this unit minus one. [writes: -1]

So, we have to be careful about the units when we state unique factorization.

Recap: Units and the group of units were introduced in Week 2 (Definition 2.66, Proposition 2.70), and the units of $\mathbb{Z}$ were found to be ± 1 (Example 2.67).

Zero and Units in an Integral Domain

In any domain R, two classes of elements are distinguished and excluded from the building blocks of factorization:

The Zero Element: $0 \in R$: Factorization of zero is degenerate and excluded.

The Group of Units: $\mathcal{U}_R \subset R$: The set of multiplicatively invertible elements (elsewhere also written $R^\times$):

$$\mathcal{U}_R = \mathrm{Units}(R) := \{u \in R \mid \exists v \in R \text{ such that } u \cdot v = 1\}. \tag{6.6}$$

Example 6.9 (*Units in the Integers*): In $\mathbb{Z}$, the only units are $\mathrm{Units}(\mathbb{Z}) = \{1, -1\}$. They introduce trivial variations in factorizations:

$$15 = 3 \cdot 5 = (-3) \cdot (-5) = (-1) \cdot 3 \cdot (-5).$$

Any rigorous definition of unique factorization must allow for ambiguities arising from units.

Spoken Transcript [00:08:45 - 00:09:50]

Prof. Rahul Pandharipande

All right. So, the first definition — so, I have not defined UFD, I will eventually define it. The first definition here [writes: Def:] is: let x [writes: let $x \in R$, $x \neq 0$, x not unit] be an element of R where x is not zero and not a unit. So, it's going to be one of the elements we will consider factorization. We don't consider factorization of zero or the units.

Then the definition: x is irreducible [writes: x is irreducible] — so, this is the term that's being defined — if and only if, [writes: $\iff$] whenever [writes: whenever] we have a factorization $x = yz$, [writes: $x = yz$] then [writes: then] — whenever $x = yz$, then — either y [writes: either y is a unit] is a unit or z [writes: or z is a unit.] is a unit.

Okay. So, this is the first definition to try to understand.

Definition of Irreducible Elements

Definition 6.10 (*Irreducible Element*): Let R be an integral domain, and let $x \in R$ with $x \neq 0$ and $x \notin \mathcal{U}_R$ (so x is non-zero and not a unit). We say that x is *"irreducible"* if and only if whenever x factors as:

$$x = y \cdot z \quad (y, z \in R), \tag{6.7}$$

then either y is a unit or z is a unit ($y \in \mathcal{U}_R$ or $z \in \mathcal{U}_R$).

Meaning of Irreducibility: An element is irreducible if it cannot be decomposed into a product of two *"strictly smaller"* (non-unit) elements. Any factorization $x = yz$ is trivial in the sense that one factor is an invertible unit and the other is an associate of x.

Spoken Transcript [00:09:50 - 00:11:05]

Prof. Rahul Pandharipande

So, if I have a ring, which I said for this week is always a domain, if I have a domain, I can take an element of it — and as I said, we don't talk about zero or the units — I take this element, it can be either irreducible or reducible. Reducible is not irreducible.

So, irreducible has this property that every time I try to factor it, I don't really make much progress, because I can always factor it where one element is a unit, right? So, it kind of says that I can't reduce it at all, that's why it's called irreducible.

And you should think, of course: why am I calling it irreducible? Why don't I call it prime? Because in the integers... so, this is an aside. [writes: [Aside:]

And this is actually genuine confusion. The terminology is used very precisely here, but it is confusing because of these two notions.

So, the aside is: when R is the integers, [writes: $R = \mathbb{Z}$] when is x irreducible? [writes: when is x irreducible?] And the answer for that is going to be: this is if and only if [writes: $\iff$] x is prime, [writes: x is prime]] the conventional high school notion of prime. Although here it also uses minus prime, you know.

Is that clear? The only numbers in the integers that have this property are the prime numbers, or minus prime numbers is okay.

Irreducibility in the Integers

Remark 6.11 (*Aside: Irreducible vs. Prime in $\mathbb{Z}$*)**:** When $R = \mathbb{Z}$, a non-zero, non-unit integer $x \in \mathbb{Z} \setminus \{0, 1, -1\}$ is irreducible if and only if x is a prime number or the negation of a prime number:

$$x \text{ is irreducible in } \mathbb{Z} \iff x = \pm p \quad (p \text{ is a prime number}). \tag{6.8}$$

Why Distinguish Irreducible from Prime?: In high school arithmetic over $\mathbb{Z}$, the words *"prime"* and *"irreducible"* are used interchangeably. In abstract ring theory, however, they represent two fundamentally different concepts:

Irreducible: $x = yz \implies y \in \mathcal{U}_R \vee z \in \mathcal{U}_R$**:** A multiplicative property stating that x cannot be broken into non-trivial factors.

Prime: $p \mid yz \implies p \mid y \vee p \mid z$**:** A divisibility property stating that if p divides a product, it must divide at least one of the factors (equivalent to the principal ideal (p) being a prime ideal).

Going beyond the lecture: in any integral domain, every prime element is irreducible (Exercise 6.42 in the next lecture). The converse fails in general domains: in $\mathbb{Q}[x^2, x^3]$ below, x^2 is irreducible but not prime (Example 6.23), and in $\mathbb{Z}[\sqrt{-5}]$ the irreducible 2 divides $6 = (1 + \sqrt{-5})(1 - \sqrt{-5})$ but neither factor. It holds in every UFD (Exercise 6.42), and in particular in every PID (Step 2 below, Lemma 6.26).

Remark 6.12 (*Week 1 Used "Prime" for "Irreducible"*)**:** In Week 1, the elements of $\mathbb{Z}[i]$ whose only divisors are units and associates were called *"prime (irreducible)"* (Definition 1.3). In the language of this week, that definition describes **irreducible** elements (Definition 6.10); *"prime"* now means that the principal ideal is a prime ideal (Definition 6.24). For $\mathbb{Z}$ the two notions agree (Euclid's lemma, Fact 0.17); for $\mathbb{Z}[i]$ they agree as well, since $\mathbb{Z}[i]$ has division with remainder, hence is a PID, hence a UFD by Theorem A (Theorem 6.15) — a fact these notes do not prove.

Definition of a Unique Factorization Domain

Spoken Transcript [00:11:05 - 00:12:10] Prof. Rahul Pandharipande

So, the question you should be asking from a terminological point of view is: why don't I call this prime? And that's the kind of confusion that we're going to address. And the reason is that there's already a notion of prime: prime ideals (Recall: Definition 5.20).

All right. So, is there a question about this definition? It's kind of the crucial definition for this week, so, you should ask.

And as I said, part of this week is going to be a certain confusion about this word irreducible, and your idea of what prime is. But anyway, this is already an aspect of that confusion: for in the integers, x is irreducible if and only if it's prime. Okay?

So, now, I can tell you the definition [writes: Def of a UFD] of a UFD. That's a unique factorization domain.

So, R is a UFD [writes: R is a UFD] if — and I don't repeat that it's a domain, because there's the domain in the name — if, for every [writes: if $\forall x \in R$, $x \neq 0$, x not unit] x in R, where x is not equal to zero and not a unit — so, we don't talk about those — two factorization properties hold. [writes: two factorization properties hold:]

Spoken Transcript [00:12:10 - 00:13:40] Prof. Rahul Pandharipande

So, I'll number them, but I need the other board. [pushes the middle blackboard up and pulls down the second board] Hold on. [begins wiping the lower right blackboard with a wet sponge] There's a great picture of Emmy Noether on a boat in Zurich that I found when I taught the Commutative Algebra class. She was here in the '30s for the ICM (i.e., the International Congress of Mathematicians, Zurich 1932), and she gave a lecture in this building.

I don't know where that photo is, but probably if you asked Lycka, she can put it on the web page if you're curious. [finishes wiping the board with the sponge]

***Insight:* Emmy Noether at the 1932 Zurich ICM**

The lecturer mentions a great picture of Emmy Noether on a boat in Zurich: she was here in the 1930s for the International Congress of Mathematicians (ICM), and she gave a lecture in this building.

Beyond the Lecture: Noether's Address

At the ICM in Zurich in September 1932, Noether gave a plenary address, *"Hyperkomplexe Systeme in ihren Beziehungen zur kommutativen Algebra und zur Zahlentheorie"*.

Spoken Transcript [00:13:40 - 00:15:05] Prof. Rahul Pandharipande

Okay, so, the two properties of factorization. So, we're in the middle of the definition of a UFD, I'm still in the middle of it, so, I have to give you the two properties of the factorization.

The first is existence. [writes: $(\exists)$] The existence property says that x [writes: $x = p_1 p_2 \cdots p_n$] is equal to $p_1 p_2 \cdots p_n$, where [writes: where the p_i are irreducibles] the p_i are irreducibles.

So, it's a UFD if for every element that's not zero or a unit, then the first is the existence, that I can factor that person into irreducibles, okay? That's existence.

And the second is the uniqueness. [writes: $(!)$] So, those are the symbols: this is existence, uniqueness. And uniqueness says that if I have two different factorizations of any element... [writes: if $p_1 \cdots p_n = q_1 \cdots q_m$]

So, these are two different factorizations of an element in irreducibles, where all p_i and q_j [writes: where all p_i and q_j are irreducibles] are irreducibles, then [writes: then] the uniqueness statement says that the first thing is that $n = m$, [writes: $n = m$] that means that the number on both sides has to be the same.

And once the numbers are the same... formally you would say that there is a permutation [writes: and there is a permutation of q_j such that] of the q_j such that... after the permutation — I don't know how you write permutations, like σ from the set 1 to n, [writes: $\sigma\colon \{1, \ldots, n\} \to \{1, \ldots, m\}$] 1 to m. So, it's a permutation, I'm just switching the indices around.

There exists a permutation such that p_i [writes: $p_i = u_i q_{\sigma(i)}$] equals $q_{\sigma(i)}$ up to some unit, u_i. So, this is a unit. [labels u_i "unit" with an arrow] This one's irreducible, [labels $q_{\sigma(i)}$ "irr" with an arrow] and that one's irreducible. [labels p_i "irr" with an arrow]

Definition of a Unique Factorization Domain

Definition 6.13 (*Unique Factorization Domain (UFD)*)**:** Let R be an integral domain. We say that R is a "*unique factorization domain*" (UFD) if for every $x \in R$ with $x \neq 0$ and $x \notin \mathcal{U}_R$, two factorization properties hold:

$(\exists)$ **Existence of Factorization:** There exist irreducible elements $p_1, \ldots, p_n \in R$ such that:

$$x = p_1 \cdot p_2 \cdots p_n. \tag{6.9}$$

(!) **Uniqueness of Factorization:** If x has two factorizations into irreducible elements:

$$p_1 \cdot p_2 \cdots p_n = q_1 \cdot q_2 \cdots q_m, \tag{6.10}$$

where all p_i and q_j are irreducible in R, then:

- $n = m$ (the number of irreducible factors is the same), and
- there exists a permutation $\sigma\colon \{1, \ldots, n\} \to \{1, \ldots, n\}$ and units $u_1, \ldots, u_n \in \mathcal{U}_R$ such that:

$$p_i = u_i \cdot q_{\sigma(i)} \quad \text{for all } i \in \{1, \ldots, n\}. \tag{6.11}$$

Uniqueness up to Permutation and Associates: In words: two factorizations of an element into irreducibles can differ only by:

(1) reordering the factors (the permutation σ), and

(2) multiplying individual factors by units ($p_i \sim q_{\sigma(i)}$).

Spoken Transcript [00:15:05 - 00:16:20] Prof. Rahul Pandharipande

Okay. So, colloquially, we'd say that the uniqueness says that if I have two different factorizations of some number into irreducibles, these two factorizations are unique, meaning that I can reorder these and then I can correct by units and then they're equal.

All right? So, this is supposed to explain, as I said: if I take 15 in the integers, [writes: $15 = 3 \cdot 5$] it's 3×5, and it's also minus five [writes: $= (-5) \cdot (-3)$] times minus three. These are two different factorizations — so, I could write 3×5 [writes: $3 \cdot 5 = (-5) \cdot (-3)$] is equal to minus... this looks almost silly to do it, but anyway. And I want a uniqueness statement.

The statement is that I have to permute these to get them in the right order. So, I'd first permute them, [draws curved arrow and writes: $= (-3) \cdot (-5)$] I put minus three, minus five, and then these are the same up to units: minus three is minus one times three, and minus five is minus one times five.

Okay? So, that's the statement here.

So, if these two properties hold, then R is a UFD. So, that's the end of the definition. Long definition: [writes: End of definition] end of definition.

Illustration of Uniqueness up to Associates and Permutation

Example 6.14 (*Matching Factorizations of 15 in $\mathbb{Z}$*): Consider the two factorizations of $15 \in \mathbb{Z}$ into irreducibles:

$$p_1 \cdot p_2 = 3 \cdot 5 \quad \text{and} \quad q_1 \cdot q_2 = (-5) \cdot (-3).$$

To show that these match according to Definition 6.13:

(1) **Equal length:** $n = 2$ and $m = 2$, so $n = m$.

(2) **Permutation:** Swap the order of factors in the second factorization via the permutation $\sigma = (1\,2)$:

$$q_{\sigma(1)} = q_2 = -3, \qquad q_{\sigma(2)} = q_1 = -5.$$

(3) **Unit correction:** Choose units $u_1 = -1$ and $u_2 = -1 \in \text{Units}(\mathbb{Z})$:

$$p_1 = 3 = (-1) \cdot (-3) = u_1 \cdot q_{\sigma(1)},$$
$$p_2 = 5 = (-1) \cdot (-5) = u_2 \cdot q_{\sigma(2)}.$$

Thus, the factorizations are identical up to reordering and unit associates.

Main Theorems on Unique Factorization Domains

Spoken Transcript [00:16:20 - 00:17:25] Prof. Rahul Pandharipande

Okay, so, that's what a UFD is. It has this property that if you take any element that's not zero or a unit, I can find a factorization into irreducibles, and any two factorizations into irreducibles are the same up to these two moves: you might have to spin around one side, and then you might have to correct by units.

All right? So, that's the definition.

All right, so, the theorems this week are the two main theorems. Today, Theorem A, [writes: Thm A:] and that's going to be: if R is a PID, [writes: If R is a PID] then R is a UFD. [writes: then R is a UFD.] Or colloquially you could just say that [writes: (PID $\implies$ UFD)] PID implies UFD. Okay, that's going to be the theorem today, hopefully.

Theorem B, for tomorrow, [writes: Thm B:] is that if R is a UFD... [erases "PID" and writes: UFD] Sorry. Excuse me. Is a UFD, then the polynomial ring is a UFD. [writes: then the $R[x]$ is a UFD.] Then the polynomial ring is a UFD.

So, this says that PID gives us a UFD automatically, and if you have a UFD, you can always add variables to it. Okay?

The Two Main Theorems

Theorem 6.15 (*Theorem A: PIDs are UFDs*)**:** If R is a principal ideal domain (PID), then R is a unique factorization domain (UFD):

$$R \text{ is a PID} \implies R \text{ is a UFD.} \tag{6.12}$$

Theorem 6.16 (*Theorem B: Polynomial Extensions of UFDs*)**:** If R is a unique factorization domain (UFD), then the polynomial ring $R[x]$ is also a unique factorization domain (UFD):

$$R \text{ is a UFD} \implies R[x] \text{ is a UFD.} \tag{6.13}$$

Core Takeaways of Theorems A and B:

- **Theorem A** establishes that principal ideal domains (such as $\mathbb{Z}$ or univariate polynomial rings $k[x]$ over a field) automatically possess unique factorization.
- **Theorem B** (Gauss's Theorem) guarantees that the UFD property is preserved under adjoining polynomial variables. While $k[x, y]$ is not a PID, Theorem B implies that it is nonetheless a UFD. Theorem A is proved in this lecture (Steps 1–3 below), Theorem B in the next one (Theorem 6.41).

Examples and Non-Examples of UFDs

Spoken Transcript [00:17:25 - 00:18:35] Prof. Rahul Pandharipande

So, once we know these two theorems, [pushes board up to reveal lower panel] it gives us a lot of examples, the standard examples in the first algebra course. And this is why we study it, because a lot of the basic examples of rings are UFDs. The basic examples of domains that arise naturally are UFDs — by no means all of them.

But a lot of the basic ones are, and you can imagine when you're working inside a ring, if it's a UFD, you'd want to know that, because it gives you a lot of insight into how the multiplication works in that ring. [wipes the board clean on the left]

So, I hope that I have time today to prove the first one, probably.

Maybe it's good in this context... Oh yeah, so, maybe, first I state some consequences so, you're motivated to pay attention to the proof.

So, first of all: examples. [writes: Ex:] This is the best example: $\mathbb{Z}$ is a UFD. [writes: $\mathbb{Z}$ is a UFD] And the proof of this is what you learned in high school about unique factorization. It just probably wasn't written this formally, but that is the content of the unique factorization you learn in high school.

The Integers as a UFD

Example 6.17 (*$\mathbb{Z}$ is a UFD*): The ring of integers $\mathbb{Z}$ is a unique factorization domain: this is the unique factorization of school (Fact 0.19), now with signs, since the units are ± 1.

Beyond the Lecture: The Integers via Theorem A

This statement is also called the Fundamental Theorem of Arithmetic. It also follows from the theory of this week: every ideal of $\mathbb{Z}$ is principal (Theorem 2.35), so $\mathbb{Z}$ is a PID, and therefore a UFD by Theorem A (Theorem 6.15).

Spoken Transcript [00:18:35 - 00:20:20]

Prof. Rahul Pandharipande

I mean, usually when you learn this in school, no one cares about the signs, because you only talk about positive numbers, and you factor into positive primes, so, the business about the units doesn't play a role. And the second thing is that, of course, there's this reordering, but, you know, you can always with numbers just write them in the increasing order, so, you don't have to worry about that either. So, generally speaking, these matters are silenced in the school treatment.

Anyway, this is formally a UFD.

There's a dumber example: [writes: k is a field] k is a field implies k is a UFD. [writes: $\implies$ k is a UFD]

And why do I say this is a little silly? Because the UFD condition is about factoring elements that are not zero and not units, but in a field everything is zero or a unit, so, the whole condition is empty there. [writes: (silly)] You know, so, there's nothing to check. So, this is kind of silly, but it's formally true.

Those are two basic examples. Then once you have these two, you can start building examples, like the first one: we know that $k[x]$ is a PID. [writes: we know $k[x]$ is a PID] That's what we know from last week (Recall: Proposition 5.58). So, this implies by Theorem A [writes: $\xrightarrow{\text{Thm A}}$ $k[x]$ UFD] that this is a UFD.

Actually, also by Theorem B, [writes: $\xrightarrow[\text{Thm B}]{}$ $k[x]$ UFD] you can also use Theorem B for this, because k is also a UFD, so, either one. And this is kind of a useful thing to know if you haven't thought about it: it says if you have a polynomial ring in one variable, you can think about factorization just like you think about factorization in the integers. There's irreducibles, that have unique factorization into irreducibles. So, that's a useful way of thinking about polynomials in one variable if you haven't somehow incorporated that in your view of polynomials in one variable.

Fields and Univariate Polynomial Rings as UFDs

Example 6.18 (*Fields are Vacuously UFDs*): If k is a field, then k is a UFD. Every non-zero element of k is a unit ($k \setminus \{0\} = \mathcal{U}_k$), so there are no non-zero, non-unit elements to factor. The conditions of Definition 6.13 hold vacuously.

Example 6.19 (*Univariate Polynomial Rings $k[x]$*)**:** Let k be a field. The polynomial ring $k[x]$ is a UFD:

- **Via Theorem A (Theorem 6.15):** $k[x]$ is a PID (Proposition 5.58), and every PID is a UFD.
- **Via Theorem B (Theorem 6.16):** Since k is a UFD, the polynomial ring $k[x]$ is a UFD.

Consequently, every non-constant polynomial in $k[x]$ factors uniquely into a product of irreducible polynomials, up to reordering and scalar multiplication by units $c \in \mathcal{U}_k$.

Spoken Transcript [00:20:20 - 00:22:15] Prof. Rahul Pandharipande

Okay, but if I start using B, I can get many more. So, for example, [writes: for example $B \implies$] Theorem B also implies that $k[x_1, \ldots, x_n]$ [writes: $k[x_1, \ldots, x_n]$ is a UFD] is a UFD, because I just keep using B one time after the other. Why is this? I take k, that's a UFD; that implies the first one, and that's still a UFD, so, then that gives me the next one, [writes: $k \implies k[x] \implies k[x, y] \ldots$] and so on. So, I just keep applying B.

And so, that's kind of fancier, probably you haven't thought about this too much, but if I take a polynomial ring in several variables, it is a UFD, which means your intuition about factorization to a certain extent is correct there — depends of course what you put in that intuition —, but a lot of that intuition about unique factorization, if it just uses these axioms, is correct. I mean, if I have a polynomial in many variables, I can factor it into these irreducibles, there's a unique way of doing it, and it's a very useful way of thinking about these polynomials.

Multivariate Polynomial Rings are UFDs

Example 6.20 (*Multivariate Polynomial Rings $k[x_1, \ldots, x_n]$*)**:** Let k be a field. By iterated application of Theorem B (Theorem 6.16):

$$k \text{ UFD} \implies k[x_1] \text{ UFD} \implies k[x_1, x_2] \text{ UFD} \implies \cdots \implies k[x_1, \ldots, x_n] \text{ UFD}.$$

Therefore, despite failing to be PIDs for $n \geq 2$, multivariate polynomial rings $k[x_1, \ldots, x_n]$ are always unique factorization domains.

PIDs vs. UFDs Hierarchy: Going beyond the lecture: this yields the strict containment of classes of integral domains:

$$\{\text{Euclidean domains}\} \subsetneq \{\text{PIDs}\} \subsetneq \{\text{UFDs}\} \subsetneq \{\text{integral domains}\}.$$

The ring $k[x, y]$ is a prime example of a domain that is a UFD but not a PID (Answer 6.3); so is $\mathbb{Z}[x]$ (a UFD by Theorem B, not a PID by Examples 5.61). The first inclusion is strict as well, but the standard example, $\mathbb{Z}[\frac{1+\sqrt{-19}}{2}]$, is beyond this course; Euclidean domains are not defined in these notes.

Spoken Transcript [00:22:15 - 00:23:45] Prof. Rahul Pandharipande

Okay, so, it might be useful here... pedagogically, anyway, it might be useful to give an example of a ring that doesn't have this unique factorization domain property. So, let me do that before I do the proof. Whether this is a good idea pedagogically, we will see.

So, what is a domain [writes: What is a domain that does not have] that does not have this unique factorization domain property? [writes: this UFD property?]

A simple example is: I look in rational numbers [writes: $\mathbb{Q}[x]$] in one variable. Of course, this one does have the UFD property, because it's an example here. But I'm going to take a cunning subring of this, [writes: $R \subset \mathbb{Q}[x]$] right?

This subring... this is a set of all polynomials. So, this subring is defined to be those polynomials — this is the definition, [writes: Def:] and it's going to be a strange definition, or maybe at least one that you haven't thought of: those polynomials with no linear term. [writes: those polys with no linear term.] So, have you ever thought of this ring?

A Non-UFD Candidate: Polynomials with No Linear Term

Definition 6.21 (*Subring of Polynomials with No Linear Term*)**:** We define the subset $R \subset \mathbb{Q}[x]$ as the set of all polynomials with rational coefficients having no linear term (i.e., the coefficient of x is zero):

$$R := \left\{ a_0 + a_2x^2 + a_3x^3 + \cdots + a_dx^d \;\middle|\; a_i \in \mathbb{Q} \right\}. \tag{6.14}$$

In other words, $R = \mathbb{Q}[x^2, x^3] \subset \mathbb{Q}[x]$.

Spoken Transcript [00:23:45 - 00:25:05] Prof. Rahul Pandharipande

So, who's in this ring, R? Well, you can be any polynomial, something like $13x^3 + 12x^2$ [writes: $R \ni 13x^3 + 12x^2$] plus something, plus 17, [writes: $+17$] but you're not allowed to have a linear term, that's what it is. So, just this one has to not be there, that's it. Okay? Is that clear what this definition is?

So, that's a strange definition, and you can't just make up any definition you want, but we should check [writes: Check: R is a ring] R is a ring. Well, the addition is all fine, because addition is kind of linear algebra, and if it has a zero here, when you add them, they still have zeros here, it's not a problem. And of course 1 is okay, and 0's okay. The thing you should be concerned about is multiplication.

And so, the question is: if a polynomial does not have a linear term, and you multiply it by another polynomial that doesn't have a linear term, can a linear term come? And if you think about it for a moment, it can't come, because where would it come from? That's proof.

All right? Check this is a ring, all right? Because the linear term can't come from nowhere, it certainly can't come from any of these terms. [chuckles]

So, it's a weird ring.

Verification that R is a Subring of $\mathbb{Q}[x]$

Verification of Subring Properties: We verify that R is a subring of $\mathbb{Q}[x]$:

- **Additive Subgroup:** $0 \in R$. If $f, g \in R$, their linear coefficients are both zero, so the linear coefficient of $f - g$ is $0 - 0 = 0$, whence $f - g \in R$.
- **Identity:** The constant polynomial $1 = 1 + 0 \cdot x \in R$.
- **Multiplicative Closure:** Let $f(x) = a_0 + a_2x^2 + \dots$ and $g(x) = b_0 + b_2x^2 + \dots$. In the product $f(x) \cdot g(x)$, the coefficient of x^1 is:

$$a_0 \cdot 0 + 0 \cdot b_0 = 0.$$

  Since the linear term vanishes, $f(x) \cdot g(x) \in R$.

Since $R \subset \mathbb{Q}[x]$ is a subring of an integral domain containing 1, R is an integral domain.

Spoken Transcript [00:25:07 - 00:26:18]

Prof. Rahul Pandharipande

So, who's in this ring? Well, x^2 is in the ring, [writes: $R \ni x^2$] it has no linear term. x^3 is in the ring, [writes: , x^3] it has no linear term. And also, of course, all the numbers are in there, like 17, [writes: , 17] and anything like that.

Okay. So, then, we can think about this ring.

x^2: [writes: x^2] it's definitely not a unit, so, this is not zero, not a unit. [writes: $\neq 0$, unit]

So, now: is x^2 irreducible? [writes: is x^2 irr.?]

And if you think about it, the answer is going to be yes, because how could I have two things and multiply to get x^2?

When you try, you're going to see you're going to want to write some linear terms. And so, it turns out — I'll leave this for you to check: is x^2 irreducible? Yes. [writes: Yes]

Is x^3 irreducible? [writes: is x^3 irr.?] Also, yes. [writes: Yes]

If I have two things to get a three, one of them has got to be a one, roughly speaking.

So, I leave these for you to check; this is an example. In this ring, both these elements are irreducible. [writes "irr" with arrows to x^2 and x^3]

Irreducible Elements in R

The subring R contains all constants $\mathbb{Q}$, as well as monomials of degree ≥ 2:

$$R = \mathbb{Q}[x^2, x^3] \ni x^2, x^3, 17, 13x^3 + 12x^2 + 17.$$

Observation 6.22 (*Irreducibility of x^2 and x^3 in R*)**:** In the domain $R = \mathbb{Q}[x^2, x^3]$:

(1) The units of R are precisely the non-zero rational constants:

$$\mathcal{U}_R = \mathcal{U}_{\mathbb{Q}} = \mathbb{Q} \setminus \{0\}.$$

(2) The elements x^2 and x^3 are non-zero, non-units, and *"irreducible"* in R.

Why x^2 and x^3 Are Irreducible (beyond the lecture)

The lecture leaves this to the students.

- If $x^2 = f(x) \cdot g(x)$ with $f, g \in R$, then by degree considerations in $\mathbb{Q}[x]$ we have $\deg(f) + \deg(g) = 2$. In $\mathbb{Q}[x]$, the only factorizations into non-constant polynomials have $\deg(f) = 1$ and $\deg(g) = 1$. But R contains no polynomials of degree 1. Thus, either $\deg(f) = 0$ or $\deg(g) = 0$, meaning $f \in \mathcal{U}_R$ or $g \in \mathcal{U}_R$. Hence, x^2 is irreducible in R.

- If $x^3 = f(x) \cdot g(x)$ with $f, g \in R$, then $\deg(f) + \deg(g) = 3$. The only partitions of 3 into non-negative integers are $3 + 0$ and $2 + 1$. Since R has no elements of degree 1, any factorization must have one factor of degree 0, which is a unit in $\mathcal{U}_R$. Hence, x^3 is irreducible in R.

Q.E.D.

Spoken Transcript [00:26:18 - 00:27:06]

Prof. Rahul Pandharipande

So, what am I going to do? I can take x^6. [writes: $x^6 \in R$] That is in R, and it's not zero, not a unit. [writes: not 0, not a unit]

And I can write x^6 as $x^2x^2x^2$, [writes: $x^6 = x^2 \cdot x^2 \cdot x^2$] and I can also write it as x^3x^3. [writes: $= x^3 \cdot x^3$]

So, these are two different ways I'm writing an element in terms of irreducibles.

And they're just really different. They're not different up to reordering and units, they're just really different.

Okay? So, this is an example of a ring which is not a UFD.

Failure of Unique Factorization in R

Example 6.23 ($R = \mathbb{Q}[x^2, x^3]$ *is not a UFD*): The element $x^6 \in R$ is non-zero and not a unit, and admits two completely distinct factorizations into irreducible elements of R:

$$x^6 = \underbrace{x^2 \cdot x^2 \cdot x^2}_{\text{3 irreducible factors}} = \underbrace{x^3 \cdot x^3}_{\text{2 irreducible factors}} . \tag{6.15}$$

The first factorization contains $n = 3$ irreducible factors, whereas the second contains $m = 2$ irreducible factors. Since $3 \neq 2$, no permutation and unit modification can transform one factorization into the other. Thus, R is not a unique factorization domain.

The Irreducible x^2 Is Not Prime: Going beyond the lecture: the same equation shows that the irreducible x^2 is not prime in R: x^2 divides $x^3 \cdot x^3 = x^2 \cdot x^4$ in R, but $x^2 \nmid x^3$, since $x^3 = x^2 \cdot h$ forces $h = x \notin R$. This is exactly the property that Step 1 of the proof of Theorem A (Claim 6.25) needs and that fails here.

Spoken Transcript [00:27:06 - 00:28:25]

Prof. Rahul Pandharipande

I mean, to really prove this, you have to check all the assertions. They're all pretty elementary assertions. Anyway, I leave all of these things to check. But that's the example where it's not.

And you could say: where does this come from?

Yeah, this is also kind of an aside. But let me just say that rings have pictures. It's not about this class, a different class, that's algebraic geometry, which is a subject I really love. Rings have pictures. And the picture of this ring is that: [sketches a plane curve with a sharp cusp at the origin]

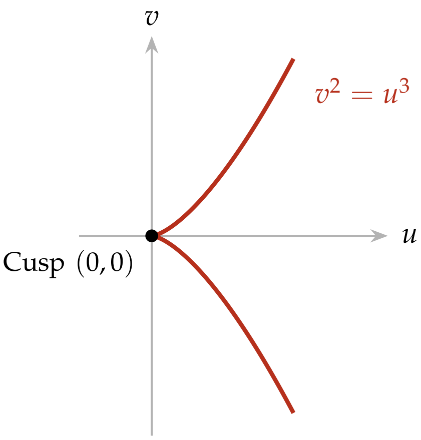
It's a cusp.

And somehow it's that singularity that gives you this problem.

All right, so, if you want to know how rings have pictures, that's a different subject. Not this class. [begins wiping the blackboard with a wet sponge]

But from the point of view of the geometry, this example is not exotic. It's basically an obvious example. Because you consider the picture of the polynomial ring is just a straight line.

Geometric Perspective: The Cusp Singularity



Geometric Intuition from Algebraic Geometry: Going beyond the lecture: The ring $R = \mathbb{Q}[x^2, x^3]$ can be identified via $u = x^2$ and $v = x^3$ with the coordinate ring of the affine algebraic curve:

$$\mathbb{Q}[u,v]/(v^2 - u^3).$$

(The map $\mathbb{Q}[u,v] \to R$, $u \mapsto x^2$, $v \mapsto x^3$, is surjective with kernel $(v^2 - u^3)$; by the first isomorphism theorem, Theorem 4.27, this identifies the two rings.) In algebraic geometry, this curve is the *"cuspidal cubic"* (Neil's parabola), which has a sharp singular point (cusp) at the origin $(0,0)$. The failure of unique factorization in R is the algebraic shadow of this singularity. One way to see the connection: a UFD contains every element of its field of fractions that is a root of a monic polynomial with coefficients in it. Here $t = x^3/x^2 \in \operatorname{Frac}(R)$ is a root of $T^2 - x^2 \in R[T]$, but $t = x \notin R$. The *"missing"* element x is what would resolve the cusp: adding it gives back $\mathbb{Q}[x]$, the ring of the straight line the lecturer mentions.

Proof of Theorem A: PIDs are UFDs

Spoken Transcript [00:28:25 - 00:30:11] Prof. Rahul Pandharipande

All right, so, we now try to prove Theorem A. [continues wiping the boards with a sponge]

To prove Theorem A, to remind you, we start with a PID and we want to show that it has the unique factorization property. And there are several steps there, and the logic is a little bit intricate, so, you have to pay attention. [finishes wiping the middle lower board]

Theorem A: [writes: Thm A] PID [writes: PID] implies [writes: $\implies$] UFD. [writes: UFD]

Okay, so, there's three steps. We start with Step 1; it'll take some time.

Step 1. [writes: STEP I] This concerns the issue that I was saying is somehow a confusing issue: Step 1 concerns this issue that if I have R as a domain, [writes: R is domain] and I take an element in R — and always this element is not zero and not a unit —, [writes: $x \in R$, $x \neq 0$, unit] then I have two ways to talk about this element, two names I can give.

There's two things I can say: x is irreducible... [writes: x irr] We've defined that x is irreducible. Sometimes I'll write irr for irreducible.

x is irreducible, and I remind you, since it was a few minutes ago: this means if $x = yz$, [writes: $(x = yz)$] then at least one [writes: at least one of y, z is a unit)] of y, z is a unit.

Spoken Transcript [00:30:11 - 00:32:16] Prof. Rahul Pandharipande

Then there's another thing, and that's I can take the ideal generated by x, [writes: (x)] right? And this ideal could be prime. [writes: prime]

Those are two different things I can do with x. I can ask whether it's irreducible, and I can ask whether its ideal is prime. And this I said, there's a certain amount of... you have to really concentrate on the notation here.

And I remind you what this means is: if I take ab in this ideal, [writes "$(ab \in (p))$", with p a slip for x] then either a is in the ideal [writes: then either $a \in (p)$] or b is in the ideal. [writes: or $b \in (p)$)]

So, these are two formally different notions. Is that clear?

I take a domain, I have an element, I've defined what irreducible is, and last week or some week, we talked about what prime ideals were (Recall: Definition 5.20).

Okay? And this unique factorization property has something to do with the relationship from our element being irreducible with the ideal being prime.

It's one of the reasons we don't use prime here, [points at irreducible] because we've already used prime — I mean the word prime — for this. [points at (x) prime]

Okay?

Recap: Prime ideals were defined in Week 5 (Definition 5.20): a proper ideal P with $ab \in P \implies a \in P$ or $b \in P$; equivalently, ${}^{R}\!/_{P}$ is an integral domain (Proposition 5.24).

Irreducible Elements vs. Prime Elements and Prime Ideals

Definition 6.24 (*Prime Elements and Principal Prime Ideals*)**:** Let R be an integral domain, and let $x \in R$ with $x \neq 0$ and $x \notin \mathcal{U}_R$. We distinguish two fundamental mathematical concepts:

(1) **x is Irreducible (irr):** Whenever x factors as a product:

$$x = y \cdot z \implies y \in \mathcal{U}_R \text{ or } z \in \mathcal{U}_R. \tag{6.16}$$

(2) **(x) is a Prime Ideal (x is Prime):** The principal ideal $(x) = xR$ is a prime ideal, meaning that whenever a product of elements lies in (x):

$$a \cdot b \in (x) \implies a \in (x) \text{ or } b \in (x). \tag{6.17}$$

In terms of divisibility: $x \mid (a \cdot b) \implies x \mid a$ or $x \mid b$. Since x is not a unit, $(x) \neq R$, so the properness required in Definition 5.20 holds automatically. An x with this property is called a *"prime element"*.

Terminology Distinction: In abstract ring theory, calling an element x *"prime"* strictly refers to the ideal-theoretic condition that (x) is a prime ideal in R. Calling x *"irreducible"* refers to the non-factorability condition. Establishing that irreducible elements generate prime ideals will be the key to proving uniqueness of factorizations.

Step 1: Irreducible Implies Prime Yields Uniqueness

Spoken Transcript [00:32:16 - 00:33:39] Prof. Rahul Pandharipande

Okay. So, **Step 1** [writes: STEP 1] says: suppose — Step 1 in the proof, there's three steps in the proof — Step 1 is: suppose [writes: Suppose] R satisfies the following property: [writes: R satisfies the following property]

For all x not equal to zero or a unit, [writes: $\forall x \neq 0$, unit] x is irreducible [writes: x irr] implies [erases "x is prime" and draws arrow: $\implies$ (x) is prime] (x) is prime.

So, that's the property. [draws a box around: $\forall x \neq 0$, unit x irr $\implies$ (x) is prime]

For an integral domain, we have these two different notions for an element x: an element x could be irreducible, I've defined it, or the ideal generated by x could be prime.

So, you're happy with this terminology: if I write these parentheses, it means the ideal. Are you happy with that?

This is the ideal [writes: ideal gen by x in R] generated by x in R. I've been using it.

And so, Step 1 is: I just assume I have a domain that has this property: every irreducible one, the ideal is prime. Okay?

And then the claim [writes: Claim:] is that R satisfies [writes: R satisfies] the uniqueness property [writes: the (!) of UFD] of the UFD definition.

To be a UFD you have to have two properties: existence and uniqueness. And in Step 1, I'm going to assume this condition, and then claim that if R satisfies this condition, it satisfies the uniqueness property. Okay?

As I said, there's three steps here, so, you have to be patient.

Spoken Transcript [00:33:39 - 00:36:03] Prof. Rahul Pandharipande

Actually, the proof isn't hard, it's just that there's a certain amount of terminology, and this issue with factorization is you have your own terminology in your head from school about factorization, and it's best maybe to silence that. It's very important, you should turn it on later, but in the middle of this proof, there's all of these notions for an arbitrary ring, and one has to keep the terminology straight.

Roughly speaking, the three steps each have their own idea. Maybe the idea in the third step is the coolest, so, you should pay attention to that one at least. [finishes wiping the middle chalkboard]

All right, so, this is what I want to do now: I assume that every irreducible ideal is prime, then I want to show it satisfies the uniqueness property, okay?

So, now, I'm trying to prove the claim: proof of claim. [writes: Proof of claim]

I assume all of that, and now... so, this is by induction.

First, suppose p_1 is equal to $q_1 \cdots q_m$. [writes: $p_1 = q_1 \cdots q_m$]

I remind you that with this uniqueness property, I'm not dealing with existence. The uniqueness property, maybe I should just remind you what that is, because it's been a while. The uniqueness property says that if $p_1 \cdots p_n = q_1 \cdots q_m$, [writes on middle board: (!) If $p_1 \cdots p_n = q_1 \cdots q_m$] all elements here are irreducible, [writes: all irr] then $n = m$, [writes: then $n = m$] and the factorizations [writes: and the factorizations] are the same [writes: are the same] up to reordering [writes: up to reordering] and units. [writes: and units.]

All right? Just to remind you, because that's what we're going to prove here.

Formulation of Step 1 and the Uniqueness Claim

Claim 6.25 (*Step 1: Irreducible Implies Prime Yields Uniqueness*): Let R be an integral domain satisfying the property that every irreducible element is prime:

$$\forall x \in R \setminus (\{0\} \cup \mathcal{U}_R), \quad x \text{ is irreducible} \implies (x) \text{ is a prime ideal.} \tag{6.18}$$

Then R satisfies the uniqueness of factorization property of a UFD (Definition 6.13, condition (!)).

Proof of Claim 6.25

Spoken Transcript [00:36:03 - 00:37:55] Prof. Rahul Pandharipande

Okay, so, the proof is going to be by induction on that n, the number of p's. As such, we have to understand the $n = 1$ case first.

Suppose we have p_1 is equal to some list of q's. [points at $p_1 = q_1 \cdots q_m$]

The assumption is p_1 is irreducible. [writes: p_1 is irr.] That's the assumption.

Because that's the assumption when we have to check the uniqueness: everybody's irreducible. Then by that box, [points to boxed condition] that's the assumption on the ring, this means that this ideal [writes "$\implies$ $(p_1) \subset R$ is prime", with a boxed R over the arrow] is prime.

Of course we have to use that, and we're going to use it in the first step. If this is prime, that means that $q_1 \cdots q_m$ are inside this prime ideal, [writes: $q_1 \cdots q_m \in (p_1)$] because this product is p_1, and p_1 is inside the prime ideal. Okay?

And so, then, I have to use the property of a prime ideal.

This means that there exists some q_j [writes: This means $\exists q_j$] in that list such that [writes: such that] q_j is in the prime ideal. [writes: $q_j \in (p_1)$]

Because that's the property of prime ideals. Normally we say it for two, but you keep splitting it up. Are you happy with that?

That if you have a product in the prime, then at least one person has to be in the prime.

The definition of prime is with two: if you have a product in the prime, then at least one of them has to be in the prime. But if I have a list, I can just keep separating them off one at a time using if two. So, one of them has to be in this prime.

Spoken Transcript [00:37:55 - 00:39:17] Prof. Rahul Pandharipande

What does that mean? Specifically, it means q_j is equal to... let's not bias it: xp_1, [writes: $q_j = xp_1$] for some [writes: for some] x in the ring, or if we've already used x... Anyway, x has also been used a lot, let's do y, just so there's no bias here. [erases x and writes "y" and "for some $y \in R$"]

All right?

Now, this is hypothesis that this is irreducible, that's by hypothesis. [points at q_j]

Because that's what we start with. And the property of being irreducible means that if I factor it, then one of these has to be a unit, right?

So, one of y, p_1, [writes: So one of y, p_1] at least one, [inserts: (at least one)] has to be a unit. [writes: has to be a unit]

Okay. So, which one could be a unit? It can't be p_1, because p_1's irreducible, and to be irreducible, you can't be a unit.

So, there's only one possibility: y has to be a unit. So, we have here: y is a unit. [writes: So y is a unit]

All right, so, this is kind of useful.

Base Case of the Induction ($n = 1$)

We proceed by induction on n, the number of irreducible factors on the left-hand side.

Base Case ($n = 1$): Suppose that:

$$p_1 = q_1 \cdot q_2 \cdots q_m, \tag{6.19}$$

where p_1 and all q_j are irreducible elements in R.

Step 1: Primality of the Ideal (p_1): Since p_1 is irreducible, the hypothesis (6.18) implies that the principal ideal $(p_1) = Rp_1$ is prime.

Step 2: Membership in (p_1): Clearly $p_1 \in (p_1)$, so:

$$q_1 \cdot q_2 \cdots q_m \in (p_1).$$

By splitting off one factor at a time in the defining property of prime ideals, if a finite product lies in a prime ideal, at least one factor must belong to that ideal:

$$\exists j \in \{1, \dots, m\} \quad \text{such that} \quad q_j \in (p_1). \tag{6.20}$$

Step 3: Divisibility and Unit Associate: The condition $q_j \in (p_1)$ means that:

$$q_j = y \cdot p_1 \quad \text{for some } y \in R. \tag{6.21}$$

Since q_j is irreducible, any factorization of q_j into two elements must contain at least one unit. But p_1 is irreducible and therefore cannot be a unit ($p_1 \notin \mathcal{U}_R$). Hence, we must have:

$$y \in \mathcal{U}_R \quad (y \text{ is a unit}). \tag{6.22}$$

Spoken Transcript [00:39:17 - 00:41:44] Prof. Rahul Pandharipande

So, now, we go back to the original equation: [writes on right board: $p_1 =$]

p_1 is equal to... I put the q_j out first, and then I take the product of all of them, [writes: $q_j(q_1 \dots$] q_1... and the standard mathematical way of writing this is $\widehat{q}_j$. [writes: $\widehat{q}_j \dots q_m)$]

I just put q_j in the front, and this means I take it out.

Do you use this notation? It's kind of standard notation. The hat means that it's omitted here.

Anyway, I just put the q_j in the front, and then I write the one thing I know about q_j: [writes: $p_1 = yp_1(q_1 \dots \widehat{q}_j \dots q_m)$]

Now, since it's a domain and p_1 is not zero, because it's irreducible, I can cancel.

We've discussed this before: for domains, there's cancellation (Recall: Definition 2.62).

Since R is a domain, I can just cancel this: [cancels p_1 on both sides and writes: $1 = y(q_1 \dots \widehat{q}_j \dots q_m)$]

Okay, this is using the fact that R is a domain. [writes: R is domain]

That's an interesting equation. It shows that... we take y; y is a unit, we already knew y was a unit.

All of these things are also units then, because I can multiply them to get 1. So, everybody here must be a unit. [underlines $(q_1 \dots \widehat{q}_j \dots q_m)$ and writes: every element is a unit]

But in this list, they're all irreducible! Irreducibles can't be units.

So, that means that this list actually doesn't exist. There's nobody there.

And this means actually then y is 1.

So, this list doesn't exist, and y is equal to 1. [writes: $1 = y \implies y = 1$]

And then we've precisely proved the ground case, because we took this arbitrary possibility, and we picked out a q_j, and we actually said this p_1 is equal to that q_j and there's nothing else. [circles p_1 and writes q_j above it]

So, that proved the uniqueness in this case, the ground case. Okay? That was a modest start.

Conclusion of the Base Case

We return to the equality $p_1 = q_1 \cdots q_m$ and isolate the chosen factor q_j:

$$p_1 = q_j \cdot \prod_{k \neq j} q_k = q_j \cdot (q_1 \cdots \widehat{q_j} \cdots q_m), \tag{6.23}$$

where $\widehat{q_j}$ denotes the omission of q_j. Substituting $q_j = y \cdot p_1$ from (6.21) gives:

$$p_1 = y \cdot p_1 \cdot (q_1 \cdots \widehat{q_j} \cdots q_m). \tag{6.24}$$

Since R is an integral domain and $p_1 \neq 0$, we can cancel p_1 from both sides:

$$1 = y \cdot (q_1 \cdots \widehat{q_j} \cdots q_m). \tag{6.25}$$

This implies that every factor in the remaining product $q_1 \cdots \widehat{q_j} \cdots q_m$ divides 1, and is therefore a unit in $\mathcal{U}_R$. However, every q_k is irreducible by assumption, and by Definition 6.10 an irreducible element cannot be a unit.

Consequently, the product $q_1 \cdots \widehat{q_j} \cdots q_m$ must be the empty product (which equals 1). Hence:

$$m = 1 \quad \text{and} \quad q_j = q_1. \tag{6.26}$$

Equation (6.25) then reduces to $1 = y \cdot 1$, so $y = 1$. Substituting back into (6.21):

$$p_1 = q_1. \tag{6.27}$$

Thus, $n = m = 1$, and the factorizations are identical, establishing the base case.

Spoken Transcript [00:41:44 - 00:43:15] Prof. Rahul Pandharipande

So, now, let's prove the induction case. I should be able to do that before the break. [wipes the middle-right board]

The induction case is the same. I just did this ground case very slowly, so, let's do the induction case.

It is actually just literally the same. [continues wiping the board with sponge]

So, I will do it more quickly.

Now we have $p_1 \cdots p_n$, [writes: $\underline{n > 1}$: $p_1 \cdots p_n$] where n is greater than one, equals $q_1 \cdots q_m$. [writes: $= q_1 \cdots q_m$]

We still concentrate on this p_1.

p_1 is irreducible, [writes: p_1 is irr.] so, by the assumption on R, this implies that this ideal is prime, [writes " $\implies$ $(p_1) \subset R$ is prime", with a boxed R over the arrow] just as before.

These q's still lie in that ideal. So, we still have $q_1 \cdots q_m$ is in this ideal, [writes: $q_1 \cdots q_m \in (p_1)$] because this ideal contains everything that's a multiple of p_1, so, the left-hand side is a multiple of p_1, so, the right-hand side is in it.

Spoken Transcript [00:43:15 - 00:45:19] Prof. Rahul Pandharipande

So, then, by the same logic, there exists some q_j in the list [writes: $\exists q_j$ such that $q_j \in (p_1)$] such that q_j is in the ideal (p_1).

And then we have the same logic here: once q_j is irreducible in the ideal (p_1), this means that q_j... I'll use the same: yp_1. [writes: $q_j = yp_1$] It's just the same. And this is a unit, [writes: y is a unit] by the logic above.

And so, then, I can go back and play the cancelling game, and this will finish it. So, we go up here: I write $p_1 \cdots p_n$, [writes: $p_1 p_2 \cdots p_n$] and I write q_j in the front, and then here's everybody else: [writes: $= q_j(q_1 \cdots \widehat{q_j} \cdots q_m)$]

Then I use this equation, and I'm going to erase it now, which is not outstanding board technique, but since we did it so carefully up there, I'm going to erase it and substitute this: [erases: q_j and replaces it with: yp_1, writing: $p_1 p_2 \cdots p_n = yp_1(q_1 \cdots \widehat{q_j} \cdots q_m)$]

And then I cancel. [cancels p_1 on both sides]

So, then, I have $p_2 \cdots p_n$ is equal to y, and then this is the list of everybody besides q_j. So, it has q_1 all the way to q_{j-1}, and then it has q_{j+1}, and then all the way to q_m. [writes: $p_2 \cdots p_n = y(q_1 \cdots \widehat{q_j} \cdots q_m) = (yq_1) \cdots q_{j-1} q_{j+1} \cdots q_m$]

Now we have basically the same equation with one fewer factor on this side: there are only $n-1$ terms on this side. So, we can apply the induction step.

Because we found the partner of p_1, that's the point. It was q_j. [draws arrow pairing p_1 with q_j]

So, then, apply induction, and then we're done. [writes: Apply induction □]

You might wonder what happens to this unit, but you can just put the unit in q_1. Just change q_1 to be q_1 times the unit; that's another irreducible.

Okay, so, that's the end of Step 1, and I proved that if R has this property that irreducibles are prime, then R satisfies the uniqueness property of the UFD. Okay?

Inductive Step ($n > 1$)

Inductive Step: Let $n > 1$, and assume the uniqueness property holds for any product of fewer than n irreducible elements. Suppose:

$$p_1 \cdot p_2 \cdots p_n = q_1 \cdot q_2 \cdots q_m, \tag{6.28}$$

where all p_i and q_k are irreducible in R.

Step 1: Primality of (p_1) and Finding the Partner: As in the base case, p_1 is irreducible, so (p_1) is prime by hypothesis (6.18). Since the product $q_1 \cdots q_m = p_1 \cdots p_n \in (p_1)$, there exists some index $j \in \{1, \ldots, m\}$ such that:

$$q_j \in (p_1) \implies q_j = y \cdot p_1 \quad \text{for some } y \in R. \tag{6.29}$$

Because q_j is irreducible and $p_1 \notin \mathcal{U}_R$, the factor y must be a unit: $y \in \mathcal{U}_R$. Thus, p_1 and q_j are associate irreducibles.

Step 2: Cancelling p_1 via Domain Property: Commuting q_j to the front of the right-hand side product and substituting $q_j = y \cdot p_1$:

$$p_1 \cdot (p_2 \cdots p_n) = y \cdot p_1 \cdot (q_1 \cdots \widehat{q_j} \cdots q_m). \tag{6.30}$$

Since R is an integral domain and $p_1 \neq 0$, we cancel p_1:

$$p_2 \cdots p_n = y \cdot (q_1 \cdots \widehat{q_j} \cdots q_m). \tag{6.31}$$

Step 3: Absorbing the Unit and Applying Induction Hypothesis: The product $q_1 \cdots \widehat{q_j} \cdots q_m$ is not empty: otherwise $p_2 \cdots p_n = y$ would be a unit, so p_2 would divide 1 and be a unit itself, which an irreducible element is not. So $m \geq 2$, and we absorb the unit $y \in \mathcal{U}_R$ into one of the remaining factors q_k, $k \neq j$, replacing it by the associate irreducible $\tilde{q}_k := y \cdot q_k$ (an associate of an irreducible element is irreducible). Renumbering so that $k = 1$:

$$p_2 \cdots p_n = \tilde{q}_1 \cdots \widehat{q_j} \cdots q_m. \tag{6.32}$$

The left-hand side is now a product of $n-1$ irreducible factors. By the induction hypothesis:

- The number of factors must match: $n-1=m-1 \implies n=m$.
- The remaining factors $\{p_2,\ldots,p_n\}$ match the factors $\{q_k\}_{k\neq j}$ up to a permutation and multiplication by units.

Together with $p_1=y^{-1}q_j$, this proves that the original factorizations in (6.28) are identical up to reordering and unit associates.

This completes the induction and establishes Claim 6.25.

Q.E.D.

Step 2: In a PID, Every Irreducible Element is Prime

Spoken Transcript [00:45:19 - 00:47:07] Prof. Rahul Pandharipande

[pauses and taps the microphone] Can you hear me? Yeah?

Okay, so, we proved Step 1.

And Step 1 has nothing to do with PIDs. The goal is to prove Theorem A. Maybe I keep that on the board so, we keep our eye on the target. [writes on the right board: Goal: Thm A: PID $\implies$ UFD]

Our goal is to prove this; we proved Step 1, and unfortunately Step 1 has nothing to do with PIDs: it's just an abstract statement that says if R satisfies this irreducible implies prime, then R satisfies the uniqueness part of the UFD. So, there are many things not satisfactory about this: first, it has nothing to do with PIDs; second, the conclusion was only half of the unique factorization.

So, we address that in Step 2.

Step 2 [writes: STEP II] is: if R is a PID, [writes: If R is a PID, then] then — now I switch to p here — p irreducible [writes: p irr] implies [writes: $\implies$] (p) prime. [writes: (p) prime]

Okay? This is going to be important, because then that allows us to use this for PIDs. So, after I prove Step 2 and I use Step 1, I will have proved half of the UFD property for PIDs: I will have proved the uniqueness part.

And in Step 3, we'll have to prove the other part, the existence part.

Okay? So, this is a crucial step in the proof, so, let us see.

Formulation of Step 2: Irreducible Elements in a PID Generate Prime Ideals

Lemma 6.26 (*Step 2: Irreducible Implies Prime in a PID*): Let R be a principal ideal domain (PID). If $p\in R$ is an irreducible element, then the principal ideal (p) is a prime ideal:

$$p \text{ is irreducible in } R \implies (p) \text{ is prime in } R. \tag{6.33}$$

Overview of the Architecture of the Proof of Theorem A: The proof of Theorem A (R is a PID $\implies$ R is a UFD) is structured into three distinct steps:

Step 1: General Domains: If in an integral domain every irreducible element generates a prime ideal, then factorizations into irreducibles are unique (Uniqueness Property).

Step 2: PIDs (Uniqueness): In any PID, every irreducible element generates a prime ideal. Combining Step 2 with Step 1 proves the Uniqueness of factorizations in every PID.

Step 3: PIDs (Existence): In a PID, every ascending chain of principal ideals stabilizes (the argument of Definition 6.7); from this one proves that every non-zero, non-unit element factors into a product of irreducibles (Existence Property).

Proof of Lemma 6.26

Spoken Transcript [00:47:07 - 00:48:42]

Prof. Rahul Pandharipande

So, let p in R [writes: Let $p \in R$] be an irreducible element. [writes: be an irr. element]

And then let a, b in R [writes: Let $a, b \in R$] satisfy a times b is in this ideal, all right? [writes: satisfy $ab \in (p)$]

To prove it's prime, we have to show that either a or b is in the ideal themselves, right?

And then further suppose — this is a standard move — further suppose [writes: further suppose] a is not in this ideal. [writes: $a \notin (p)$]

Then we must prove [writes: then we must prove] b is in the ideal. [writes: $b \in (p)$]

Okay? So, that's the logic; it's totally standard logic. We have some assumption: this is irreducible; we want to prove that this ideal is prime. So, we write down any two elements a, b that are in the ideal, and we assume the first one's not in the ideal, then we prove the second one is, right? Because that's logically fine. To prove it's prime, we need to prove one of these two is in: so, if the first one's in, then we go home; if it's not in it, then we use that, okay?

Proof Setup for Step 2

Let R be a PID, and let $p \in R$ be an irreducible element. To prove that (p) is a prime ideal, let $a, b \in R$ satisfy:

$$a \cdot b \in (p). \tag{6.34}$$

We must show that $a \in (p)$ or $b \in (p)$.

By standard logic, we may assume without loss of generality that:

$$a \notin (p). \tag{6.35}$$

Our objective is then to prove that:

$$b \in (p). \tag{6.36}$$

Spoken Transcript [00:48:42 - 00:50:42]

Prof. Rahul Pandharipande

All right, so, how are we going to do this? It's nice to write the assumptions here: [draws a bracket and lists assumptions]

p is irreducible, that's one of the assumptions. a is not in (p), that's another assumption. And the other assumption is ab is in (p). These are the three assumptions, all right? And we need to somehow make it to this faraway place where b is in (p), all right? [writes: $\rightsquigarrow b \in (p)$]

So, we have to traverse this.

And there's one other assumption that's extremely important, at the beginning: that R is a PID. [writes: R is a PID] All right.

We have to use the PID property somehow. We have p and a: this is an ideal. We take the ideal generated by a and p. [writes: $I = (a, p) \subset R$] We're allowed to do that.

This is an ideal in R, okay?

And since R is a PID, [writes: Since R is a PID] this means that there is somebody in R [writes: $\exists z \in R$] such that this ideal is this principal ideal generated by z. [writes: such that $(a, p) = (z)$]

Because that's what it means to be a principal ideal: every single ideal is principal; in particular this ideal is principal, and so, it's generated by this element, okay?

And what does this mean? This means a is equal to some element k times z, and p is equal to some element ℓ times z. [writes: This means: $a = kz$, $p = \ell z$ $(k, \ell \in R)$]

Spoken Transcript [00:50:42 - 00:50:50] Prof. Rahul Pandharipande

All right. So, what do we do with those? Can someone tell me what I'm supposed to do here?

Yeah?

Student Answer

Because p is irreducible, we can conclude that one of the elements is a unit.

The Ideal (a, p) and Principal Generator z

We compile the operative hypotheses:

- **(1)** $p \in R$ is irreducible.
- **(2)** $a \notin (p)$.
- **(3)** $a \cdot b \in (p)$.
- **(4)** R is a principal ideal domain (PID).

Consider the ideal generated by a and p:

$$I := (a, p) = \{r \cdot a + s \cdot p \mid r, s \in R\} \subseteq R. \tag{6.37}$$

Since R is a PID, the ideal I is principal. Thus, there exists an element $z \in R$ such that:

$$(a, p) = (z). \tag{6.38}$$

Since $a \in (a, p) = (z)$ and $p \in (a, p) = (z)$, z divides both a and p. That is, there exist $k, \ell \in R$ such that:

$$a = k \cdot z, \tag{6.39}$$

$$p = \ell \cdot z. \tag{6.40}$$

Spoken Transcript [00:50:50 - 00:52:42] Prof. Rahul Pandharipande

Exactly! So, we have to use that. The next step here, there's only one step you can do, and you have to use the fact that p's irreducible. So, this was the correct answer: use that p is irreducible. [pushes board up and writes on lower board: Use that p is irr.]

And we look at the second one, and it's very natural to use it, because we've deduced here some factoring of p. If you know p's irreducible, and you have a factoring of p, you should definitely use the irreducibility property.

So, p... what was the letter? ℓ, right? ℓz. [writes: $p = \ell z$] But this is irreducible. This means one of these, [writes: one of these is a unit in R] one of these is a unit in R.

And of course we're very curious which one it is. Could it be ℓ? [writes: Could it be ℓ?]

Well, if ℓ is a unit, [writes: If ℓ is a unit] then I can use this equation here to write... then it has an inverse. Then we have ℓ^{-1} times ℓz, that's just... [writes: $\ell\ell^{-1} = 1$] If it's a unit, there's some element where I can multiply to get 1.

And then I can use the second equation: $a = kz$, and I can write that as $z = \ell^{-1}p$, right? [writes: $a = kz = k\ell^{-1}p$]

And what's the problem with this? Well, then a is in the ideal generated by p! [writes: $\implies a \in (p)$]

Because I've just written that a is equal to something times p. [underlines $a = k\ell^{-1}p$] Okay?

Spoken Transcript [00:52:42 - 00:54:02] Prof. Rahul Pandharipande

Is it okay for me to write this ℓ^{-1}? Sometimes people get scared by that. ℓ^{-1} is just some element, is some element such that [writes: is some element such that $\ell^{-1}\ell = 1$] $\ell^{-1}\ell = 1$, because it's a unit. If it's a unit, that means there's somebody; it's traditional to call it ℓ^{-1}, but sometimes people get confused by it. I could call it something else, you know, like W or Fred or anything, doesn't really matter.

Okay, so, this is not possible! This is a contradiction! [writes: Contradiction!]

Because one of our assumptions is a is not in (p). [points at assumption: $a \notin (p)$]

So, as I said, when we're here, we can apply the definition of irreducibility to deduce that one of these is a unit, and as I said, we're very curious which one. And the first path was what we consider ℓ: that is not possible. I showed here by contradiction it could not be ℓ; ℓ couldn't be the unit.

Eliminating the Possibility that ℓ is a Unit

From equation (6.40), we have a factorization of p in R:

$$p = \ell \cdot z.$$

By assumption, p is irreducible. According to Definition 6.10, this forces at least one of the factors ℓ or z to be an invertible unit in $\mathcal{U}_R$.

Case 1: Suppose ℓ is a unit ($\ell \in \mathcal{U}_R$).

If $\ell \in \mathcal{U}_R$, there exists an inverse $\ell^{-1} \in R$ such that $\ell \cdot \ell^{-1} = 1$. Multiplying (6.40) by

ℓ^{-1} gives:

$$z = \ell^{-1} \cdot p. \tag{6.41}$$

Substituting this expression for z into equation (6.39) for a:

$$a = k \cdot z = k \cdot (\ell^{-1} \cdot p) = (k \cdot \ell^{-1}) \cdot p \in (p). \tag{6.42}$$

This contradicts our standing hypothesis (6.35) that $a \notin (p)$.

Therefore, ℓ cannot be a unit ($\ell \notin \mathcal{U}_R$).

Spoken Transcript [00:54:02 - 00:55:24] Prof. Rahul Pandharipande

So, there's only one other possibility: that z is a unit. So, we conclude that z must be a unit. [writes: z is a unit.] z is a unit.

This means that this ideal, which is generated by a and p, is equal to the unit ideal. [writes: This means $I = (a, p) = (1)$]

Because once I have an element z which is a unit, I can multiply it by its inverse and get the unit ideal, so, it's the whole ideal.

This means there exist some elements in the ring such that $na + mp = 1$. [writes: So $\exists n, m \in R \implies na + mp = 1$]

Because this ideal is just the combinations of R times this element and R times that element; that's what that ideal is. Okay?

Okay, so, what do you think we should do now?

Yeah?

Student Answer

We can multiply by b.

Spoken Transcript [00:55:30 - 00:56:32] Prof. Rahul Pandharipande

Yeah! This is kind of a standard trick; we've done it many times in this class (Recall: Euclid's lemma, Fact 0.17, and the Chinese Remainder Theorem, Theorem 5.46). So, you see this, and now we multiply by b. [writes: Mult. by b]

We get $nab + mbp = b$. [writes: $nab + mbp = b$] And that's really great for us, because then this already is in the ideal generated by p, because there's a p, [circles mbp and writes: $\in (p)$]

and this one is in the ideal generated by p, [underlines nab and writes: $\in (p)$]

because that's an assumption! This assumption: $ab \in (p)$. [points at assumption: $ab \in (p)$]

So, we conclude that b is in the ideal generated by p. So, we're very happy about that; that's the end of Step 2. Okay? [writes: $\implies b \in (p)$ $\square$]

Case 2: z is a Unit

Since $p = \ell \cdot z$ is irreducible and $\ell \notin \mathcal{U}_R$, the only remaining possibility is:

$$z \in \mathcal{U}_R \quad (z \text{ is a unit}). \tag{6.43}$$

Since z is an invertible unit, the principal ideal (z) contains $z \cdot z^{-1} = 1$, which implies:

$$I = (a, p) = (z) = (1) = R. \tag{6.44}$$

Since $1 \in (a, p)$, there exist elements $n, m \in R$ such that:

$$n \cdot a + m \cdot p = 1. \tag{6.45}$$

Multiplying both sides of (6.45) by b:

$$n \cdot (a \cdot b) + m \cdot b \cdot p = b. \tag{6.46}$$

We analyze the terms on the left-hand side:

- $a \cdot b \in (p)$ by hypothesis (6.34), so $n \cdot (a \cdot b) \in (p)$.
- $m \cdot b \cdot p = (m \cdot b) \cdot p \in (p)$ is an explicit multiple of p.

Since ideals are closed under addition, the sum lies in (p):

$$b = \underbrace{n \cdot (a \cdot b)}_{\in (p)} + \underbrace{(m \cdot b) \cdot p}_{\in (p)} \in (p). \tag{6.47}$$

Hence, $b \in (p)$. Since p is not a unit, $(p) \neq R$; so (p) is a prime ideal (Definition 5.20), completing the proof of Lemma 6.26.

Q.E.D.

Beyond the Lecture: Bézout, Euclid and Maximal Ideals

The equation $n \cdot a + m \cdot p = 1$ is a *"Bézout identity"*, and Step 2 is the argument of Euclid's lemma in $\mathbb{Z}$ (Fact 0.17), with the PID property in place of Bézout's identity. The proof shows even more: $(a, p) = R$ for every $a \notin (p)$, so (p) is a maximal ideal (Definition 4.54).

Spoken Transcript [00:56:32 - 00:57:42] Prof. Rahul Pandharipande

All right, so, that's Step 1 and Step 2. So, now, we're getting close. [wipes the blackboard on the left with a wet sponge]

Maybe I have the summary of Step 1 and Step 2, and we have one more step. But this argument about Step 2 is kind of like the fun algebra in this class, where we use the things several ways and several times, and of course it works out perfectly.

I always feel a little guilty teaching these classes because everything works out so perfectly.

You could ask: why does everything work out perfectly? That's an interesting question.

And the answer is: because of the definitions.

And if you follow that line, you realize that actually the definitions are an incredibly important part of this class.

In the actual development of mathematics, the definitions come at the end of the development — that's kind of standard — not at the beginning. So, the whole way it's presented is kind of dishonest. Anyway, that's an aside.

Insight: The Organic Evolution of Mathematical Definitions

The lecturer asks why everything in this class works out so perfectly, and answers: because of the definitions. In the actual development of mathematics, the definitions come at the end, not at the beginning, so the way it is presented is, in his words, kind of dishonest.

Step 3: Existence of Factorizations in a PID

Spoken Transcript [00:57:42 - 00:59:02] Prof. Rahul Pandharipande

Step 3. So, what do we have here with Step 1 plus 2? [writes: Step I + II $\implies$ If R is a PID, then we have the (!) of UFD.]

Right? Do you agree with that? That was the logic in Step 1 and 2; it gives us that.

So, the last, Step 3, is that if R is a PID, then we have the existence part of the definition of UFD. [writes: STEP III: If R is a PID, then we have the $(\exists)$ part of UFD.]

And when I finish Step 3, then we've proven the goal. So, then, Steps 1 + 2 + 3 imply this Theorem A: PID implies UFD. [writes: Steps I, II, + III $\implies$ (Thm A) PID $\implies$ UFD]

Okay, so, this is the last thing I have to prove.

Cumulative Progress Towards Theorem A

Proposition 6.27 (_Cumulative Progress Towards Theorem A_): Combining Step 1 and Step 2:

Step 1: Claim 6.25: $(\forall x \text{ irr}\colon (x) \text{ prime}) \implies R$ satisfies Uniqueness $(!)$.

Step 2: Lemma 6.26: R is a PID $\implies (\forall p \text{ irr}\colon (p) \text{ prime})$.

Therefore, every principal ideal domain R satisfies the **Uniqueness Property (!)** of a unique factorization domain.

It remains to establish Step 3:

$$\text{STEP 3}\colon \quad R \text{ is a PID} \implies R \text{ satisfies the Existence Property } (\exists) \text{ of a UFD.} \tag{6.48}$$

Formal Goal of Step 3

Lemma 6.28 (_Step 3: Existence of Factorization into Irreducibles_): Let R be a principal ideal domain (PID). Every non-zero, non-unit element $x \in R \setminus (\{0\} \cup \mathcal{U}_R)$ can be factored into a finite product of irreducible elements:

$$x = p_1 \cdot p_2 \cdots p_n \quad (n \geq 1,\ p_i \in R \text{ irreducible}). \tag{6.49}$$

Proof of Lemma 6.28

Spoken Transcript [00:59:02 - 01:00:37] Prof. Rahul Pandharipande

Specifically, explicitly what I have to prove is that we have R is a PID, and $x \neq 0$, not a unit, is an element. [writes: R is a PID, $x \neq 0$, not a unit, $x \in R$]

So, we must find or construct or show the existence of... we must find a factorization: [writes: We must find a fact. $x = p_1 \cdots p_n$ where the p_i are all irr.]

Okay, that's our goal. Is that kind of clear?

And, you know, this is like the existence part. For integers, you could say: if I have an integer, how do I know I can factor it into primes? And the answer is: well, it could be prime, and then I stop. If it's not prime, it means that I can factor it somehow. And, you know, I break it, and then I keep breaking those parts, and eventually I stop. That's kind of a proof, because the parts get smaller and smaller.

That's the proof for the integers, which you probably have thought about before.

Spoken Transcript [01:00:37 - 01:02:42] Prof. Rahul Pandharipande

And we'd like to use that intuition for a ring. So, let's try to do it.

Somehow the idea is: either x is born irreducible, and then we're done, [writes: Idea: either x is irr. $\implies$ done]

or x has a factorization where neither y nor z are units. [writes: or x has a fact. $x = yz$ where neither y nor z are units.]

Right? Because if every factorization of x had one of them being units, then it would be irreducible. If it's not irreducible, then there's some legitimate, proper factorization where neither are units, okay?

And then we just apply this idea to each factor, and we just keep going down. So, we have to show that this process somehow finishes, terminates. I'll draw a picture of it.

And for integers, it does terminate, because the numbers get smaller, and there's only a limit to how small the integers can get.

But for a ring, the matter is a little bit more confusing, because we don't have this kind of ordering of elements in the ring; that's not part of the data. This is any ring.

So, let's try to... As I said, for me, this is the most interesting part of the proof. It's the most fun to think about. [cleans the blackboard with a sponge]

The Factorization Strategy and Termination Question

The Natural Algorithm: Given $x \in R \setminus (\{0\} \cup \mathcal{U}_R)$:

Case A: x is Irreducible: Factorization complete ($n = 1$).

Case B: x is Reducible: There exists a non-trivial factorization $x = y \cdot z$ with $y, z \notin \mathcal{U}_R$.

We recursively repeat this split on each non-unit factor.

Why General Rings Differ from the Integers: In $\mathbb{Z}$, a proper factorization $x = y \cdot z$ yields $|y| < |x|$ and $|z| < |x|$. The positive integers are well-ordered, so this splitting process must terminate after finitely many steps. In an arbitrary PID, however, there

> is no such ordering of the elements; termination must instead be deduced from the algebraic structure of the ideals.

Spoken Transcript [01:02:42 - 01:04:42] Prof. Rahul Pandharipande

Okay, so, let's try to write this in some kind of algorithmic flowchart.

We start with x. That's the person that we want to find a factorization for. And let's just call this person x_0, because we're going to have some subtrees. [writes: $x = x_0$]

So, x_0 is here, $x = x_0$; that's the beginning.

And so, either x_0 is already irreducible, or we can factor it into some y and z. [draws tree branching down from x_0 to y and z]

And whenever I go down, I always mean this is a non-trivial factorization, a legitimate factorization. In this diagram, whenever I factor, it means I factor where neither of them are units, okay?

So, either this is irreducible, or I can factor. Then I look at these guys. If this is irreducible, maybe I make a block here, and I end it. [draws a square around y]

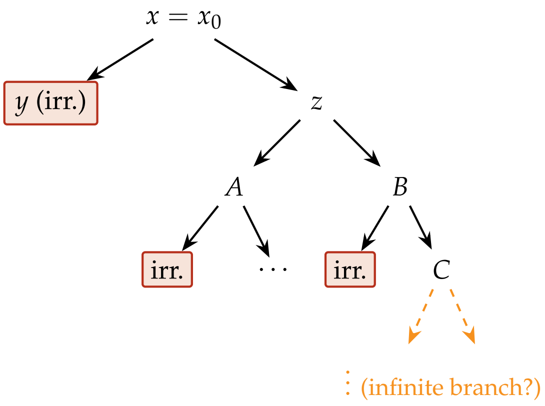
If not... so, let's say y is irreducible, then that's going to be one of my end states. And then I factor this, maybe that's reducible, so, then that will be some... I started kind of late in the alphabet, which was bad; let's go back to the beginning. [erases and labels branches with A and B]

Okay? Every time, if it's not irreducible, I can factor. So, I keep going down, and then maybe go down some more, and some more, and then maybe I find some irreducible people, and then I have some more. And that's pretty good; I have all of these irreducibles. If I found them all irreducible, then I'd be done, right? Then I'd be finished.

But it could be that there's some other C here, and I keep going down... [draws tree continuing downwards with branches and boxes]

You see, the problem is that there's an infinite structure here. It's possible... So, what's the problem? The only problem is this:

The Factorization Tree Diagram



Mechanics of the Factorization Tree: In this binary tree:

- Every downward split represents a non-trivial factorization into two non-units ($u = v \cdot w$ with $v, w \notin \mathcal{U}_R$).
- Whenever a factor is irreducible, the branch terminates in a leaf (boxed in red).
- If every branch terminates, the leaves form a finite factorization $x = p_1 \cdots p_n$.
- The only obstacle to the existence of a factorization is the occurrence of an infinite branch that never terminates.

Spoken Transcript [01:04:42 - 01:06:02] Prof. Rahul Pandharipande

I have some x_0, and I factor it, and I get some x_1, some y. [draws path: x_0 to x_1 and y]

And then I factor this: I get some x_2, z, and maybe these are all... The only problem is if I keep factoring, and I get this infinite chain that goes forever, and I never finish. So, do you accept that?

We will arrive either at a finite factorization, or an infinite chain in this diagram. [writes: We will arrive either at a finite fact. $x = p_1 \cdots p_n$ or an infinite chain]

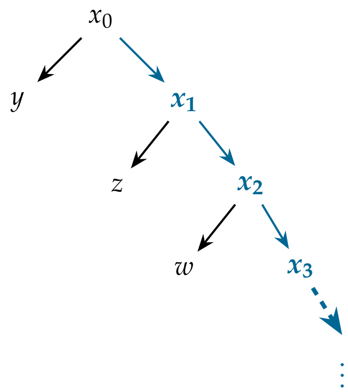
And the infinite chain will have x_0, and then (x_0) will be inside (x_1), because the equation here is $x_0 = x_1 y$, so, it'll be inside that ideal. [writes: $(x_0) \subseteq (x_1)$]

And then (x_1) will be inside the ideal (x_2) by the same logic, and it'll be inside the ideal (x_3) by the same logic: this infinite chain of ideals, okay? [writes: $(x_0) \subseteq (x_1) \subseteq (x_2) \subseteq (x_3) \subseteq \ldots$]

We're going to say more about this. But the first thing you have to accept for this proof is that we try to factor it, and we think about what could happen, and what happens is two outcomes: either we finish, or this process is infinite.

And if this process is infinite, it means we have this chain, because we just take a path which realizes the infinity there.

The Infinite Branch and the Associated Chain of Ideals



The infinite path can be chosen explicitly. Suppose $x_0 = x$ has no factorization into irreducibles. Then x_0 is not irreducible, so $x_0 = x_1 \cdot y$ with x_1, y non-units (and non-zero, as $x_0 \neq 0$). If both x_1 and y had factorizations into irreducibles, their concatenation would factor x_0; so at least one of them has none, and we call that one x_1. Repeating the argument with x_1 in place of x_0, and so on, gives an infinite path of non-trivial factorizations in which no x_k has a factorization:

$$x_0 = x_1 \cdot y, \quad x_1 = x_2 \cdot z, \quad x_2 = x_3 \cdot w, \quad \dots \qquad (y, z, w, \dots \notin \mathcal{U}_R). \tag{6.50}$$

Since each element is a multiple of its successor, this produces an ascending chain of principal ideals:

$$(x_0) \subseteq (x_1) \subseteq (x_2) \subseteq (x_3) \subseteq \dots \tag{6.51}$$

Spoken Transcript [01:06:02 - 01:07:12] Prof. Rahul Pandharipande

Okay, so, now, there's something extremely important, which is... of course, here we're done, so, we don't worry about it. Our concern here, we have to address this case. [circles the infinite chain and writes: Address this case]

And there's something that's extremely important about these ideals, and that is that they actually strictly increase. [modifies $\subseteq$ to $\subsetneq$ on the board: $(x_0) \subsetneq (x_1) \subsetneq (x_2) \subsetneq \dots$]

Let's look at this x_0, x_1 and y, this factorization. [writes: $x_0 = x_1 y$]

The equation is $x_0 = x_1 y$, and this equation shows that x_0 is in the ideal of x_1. That's what I'm saying: x_0 is inside here; that's just from this equation.

So, now, we can ask: [writes: Can $x_1 \in (x_0)$?]

Can x_1 be in the ideal of x_0? This thing just means this is a subset of that. Now I want to wonder whether this is an equality or not. So, this is a question.

Strictness of the Ideal Inclusions

Question 6.29 (*Can* (x_1) *Equal* (x_0)*?*): In a proper factorization $x_0 = x_1 \cdot y$ where $y \notin \mathcal{U}_R$, can x_1 belong to (x_0)? That is, can the inclusion $(x_0) \subseteq (x_1)$ fail to be strict?

Spoken Transcript [01:07:11 - 01:09:39] Prof. Rahul Pandharipande

Suppose yes. Suppose $x_1 = x_0 z$. [writes: Suppose $x_1 = x_0 z$]

That will mean x_1 is in this ideal, all right?

Then what am I going to do? I'll go back to that equation: $x_0 = x_1 y$. [writes: $x_0 = x_1 y$] I can stick this in; I get $x_0 = x_0 z y$, [writes: $x_0 = x_0 z y$] okay? You agree?

And now, I did this in the first half, and someone asked: this is a domain; all of these things are neither zero nor units, so, I can cancel this. [writes: $1 = zy$]

This is the cancellation property for domains. Are you happy with that? The proof of this is: I subtract $x_0 - x_0 z y = 0$, then I factor out the x_0: $x_0(1 - zy) = 0$. This is not zero, so, that must be zero. [writes: $x_0 - x_0 z y = 0 \implies x_0(1 - zy) = 0 \implies 1 - zy = 0 \implies 1 = zy$]

So, this is zero. What do we deduce from that? This is an interesting deduction: that means both z and y are units! [writes: $\implies z, y$ units]

But y here was assumed not to be a unit! So, this cannot happen.

So, the answer is: can x_1 be in (x_0)? The answer is: NO. [writes: NO!]

Because if it were the case, we would deduce y is a unit, and it's not allowed. In this diagram, every time we factor, the factoring is a legitimate factoring at every stage: there are no units produced.

So, that says that not only are these ideals contained, they're never equal. It's the same argument at every stage: they're never equal.

Okay. So, this is actually the crucial... this is like the heart of this proof: first we say that either we succeed or there's an infinite chain, and then the really technical point is that this chain is not only increasing, it is strictly increasing. [pushes the middle board up]

That strictly increasing is what was argued here. [points at the algebraic contradiction on the right]

So, then...

Proof that the Ascending Chain is Strictly Increasing

Answer 6.30 (*to Question 6.29*): No. The inclusion is strictly proper: $(x_0) \subsetneq (x_1)$.

Since $x_0 = x_1 \cdot y$, we have $x_0 \in (x_1)$, which guarantees $(x_0) \subseteq (x_1)$.

Suppose for contradiction that $x_1 \in (x_0)$. Then there exists some element $z \in R$ such that:

$$x_1 = x_0 \cdot z. \tag{6.52}$$

Substituting (6.52) into the factorization $x_0 = x_1 \cdot y$:

$$x_0 = (x_0 \cdot z) \cdot y = x_0 \cdot (z \cdot y). \tag{6.53}$$

Subtracting $x_0 \cdot (z \cdot y)$ from both sides yields:

$$x_0 - x_0 \cdot (z \cdot y) = 0 \implies x_0 \cdot (1 - z \cdot y) = 0. \tag{6.54}$$

Because R is an integral domain and $x_0 \neq 0$, the zero-divisor property forces:

$$1 - z \cdot y = 0 \implies z \cdot y = 1. \tag{6.55}$$

This implies that y is a unit in R ($y \in \mathcal{U}_R$).

However, this directly contradicts our foundational assumption that $x_0 = x_1 \cdot y$ is a proper non-trivial factorization with $y \notin \mathcal{U}_R$.

Hence, $x_1 \notin (x_0)$. By identical reasoning applied at each successive stage $x_k = x_{k+1} \cdot y_k$, every inclusion in the chain must be strictly proper:

$$(x_0) \subsetneq (x_1) \subsetneq (x_2) \subsetneq (x_3) \subsetneq \dots \tag{6.56}$$

Spoken Transcript [01:09:39 - 01:12:09]

Prof. Rahul Pandharipande

[cleans the lower blackboard with a wet sponge]

Then we apply an idea that came up already in the discussion of Max Zorn (Recall: Lemma 5.14, Lemma 5.18)... which is that I have this ascending chain of ideals. And the question is: can this happen in a PID?

And we're going to see that, no, it can't happen; that ascending chain of ideals can never happen in a PID, and that will rule out the second branch and say that the first branch is the only possibility.

So, to find a contradiction here... [writes: To find a contradiction: Let $J = \bigcup_{n \geq 0}(x_n)$]

We have this strictly increasing set of ideals, we just take their union, right?

J inside R is an ideal. [writes: $J \subset R$ is an ideal.]

You have to check that. We discussed this with Zorn: zero is in all of them.

If I have two elements here... this is an aside: [writes: aside] if I have a and b in J, I need to show that $a+b$ is in J.

That means a is in one of these — like a might be in (x_{342}), [writes: $a \in (x_{342})$] and b is in another one, like (x_{759}). [writes: $b \in (x_{759})$] They have to be somewhere.

And then these things are in order, so, one of these is inside the other. You can just take, in this case, the bigger one, so, they're both in the bigger one, and you can add them there because that's an ideal, okay?

So, J is an ideal; all the properties are like this. This is maybe the most interesting one to prove. So, it's an ideal.

The Ascending Union of Ideals

To derive a contradiction from the existence of an infinite strictly ascending chain:

$$(x_0) \subsetneq (x_1) \subsetneq (x_2) \subsetneq (x_3) \subsetneq \dots$$

we define the set J as the union of all ideals in the chain:

$$J := \bigcup_{n=0}^{\infty} (x_n) \subset R. \tag{6.57}$$

Claim 6.31 (*J is an Ideal in R*): The union $J = \bigcup_{n=0}^{\infty}(x_n)$ is an ideal of R.

This is the argument of Lemma 5.18 (Week 5), where the union of a chain of proper ideals was shown to be a proper ideal on the way to Theorem 5.19.

Proof

We verify the three defining properties of an ideal:

(1) **Non-emptiness:** $0 \in (x_0) \subseteq J$, so $J \neq \emptyset$.

(2) **Additive Closure:** Let $a, b \in J$. By definition of the union, there exist indices $j, k \in \mathbb{N}$ such that $a \in (x_j)$ and $b \in (x_k)$. Without loss of generality, assume $j \leq k$. Since the chain is ordered by inclusion:

$$(x_j) \subseteq (x_k) \implies a, b \in (x_k).$$

Since (x_k) is an ideal, $a + b \in (x_k) \subseteq J$.

(3) **Absorption under Multiplication:** Let $r \in R$ and $a \in J$. There exists some $n \in \mathbb{N}$ such that $a \in (x_n)$. Because (x_n) is an ideal:

$$r \cdot a \in (x_n) \subseteq J.$$

Thus, J is an ideal of R. **Q.E.D.**

Spoken Transcript [01:12:09 - 01:14:44] Prof. Rahul Pandharipande

Okay, so, what do we know about ideals? They're principal. That's what we assume, right?

For Step 3, we assume R is a principal ideal domain, and we want to show that it has this existence property.

But we know R is a PID, so, $J = (w)$, $w \in R$. [writes: But we know R is a PID, so $J = (w)$, $w \in R$]

This means that this ideal is generated by one person. I don't know, let's call it w, okay?

That means this whole ideal, the union of everybody, is generated by one person. What this means is that in this chain, if I were taller I could just write it there, but... [writes: $(x_0) \subsetneq (x_1) \subsetneq \ldots (x_{691}) \ldots$]

In this chain, w sits somewhere. w is in this ideal, and this ideal is just the union of all of these (x_i), so, it must sit somewhere.

We can find somewhere, like I don't know, okay, let's have it in (x_{691}), why not there? [writes "(x_{691})" with "w" underneath]

w is there somewhere. But then it generates everything! In particular, this shows that this is equal to the next one: [writes: $= (x_{692}) = \cdots = J$]

Because w generates the whole ideal, so, it means it in particular generates everything here, because that's also in J. So, all of them become equal after that.

We first find where w is, and then everything has to become equal after that, because it contains w, and the whole ideal is just the ideal generated by w. So, they're just all equal. In fact, this is equal to J: J, J, J, they're all equal after that. [writes: $= J = J = J$]

And that contradicts the strictly increasing property there. That's the contradiction!

This contradicts the strictly increasing property! [writes: This contradicts the strictly increasing property! □]

Okay, and that's the end. This is a more delicate argument, so, probably you have to think about this argument. But it's an interesting argument in the sense that, as I said, with integers we don't really face this problem, because it's kind of obvious: when you factor integers you're going to stop. But in an arbitrary principal ideal domain, we have to face it.

And it's handled by using the PID property in a clever way here. Specifically, it exactly rules out... you just cannot have this infinite chain.

Completion of Step 3: Termination of Factorization in a PID

Since R is a principal ideal domain (PID), every ideal in R is principal. In particular, the union ideal J must be generated by a single element $w \in R$:

$$J = (w) = Rw \quad \text{for some } w \in R. \tag{6.58}$$

Since $w \in J = \bigcup_{n=0}^{\infty}(x_n)$, the generator w must belong to some specific ideal in the chain:

$$\exists N \in \mathbb{N} \quad \text{such that} \quad w \in (x_N). \tag{6.59}$$

Since $w \in (x_N)$, the principal ideal generated by w is contained in (x_N):

$$(w) \subseteq (x_N).$$

Combining this with the chain inclusions for any $k \geq N$:

$$J = (w) \subseteq (x_N) \subseteq (x_k) \subseteq J. \tag{6.60}$$

This forces equality across all subsequent ideals in the chain:

$$(x_N) = (x_{N+1}) = (x_{N+2}) = \cdots = J. \tag{6.61}$$

However, equation (6.61) directly contradicts (6.56), which established that every step in the chain is strictly proper ($(x_N) \subsetneq (x_{N+1})$).

Conclusion: No infinite branch can exist in the factorization tree. Hence the assumption that x has no factorization was false: every non-zero, non-unit element $x \in R$ factors into a finite product of irreducible elements:

$$x = p_1 \cdot p_2 \cdots p_n \quad (p_i \text{ irreducible}).$$

This completes Step 3 and establishes Lemma 6.28.

Q.E.D.

Conclusion of Theorem A

✓ **Theorem A proved:** Steps 1–3 together prove Theorem 6.15: every principal ideal domain is a unique factorization domain.

Summary of the Three-Step Proof:

Step 1: Claim 6.25: In any domain where irreducible elements are prime, factorizations into irreducibles are unique up to permutation and associates (**Uniqueness**).

Step 2: Lemma 6.26: In a PID, every irreducible element generates a prime ideal. By Step 1, factorizations in a PID are unique.

Step 3: Lemma 6.28: In a PID, there is no strictly ascending infinite chain of principal ideals (every ascending chain stabilizes), ruling out infinite factorization branches and proving that every non-zero non-unit admits a factorization into irreducibles (**Existence**).

Spoken Transcript [01:14:44 - 01:15:54] Prof. Rahul Pandharipande

[wipes the blackboards with a wet sponge]

All right, so, that finishes the proof of Theorem A, and as I said, the logic there is slightly intricate, maybe more intricate than the results we've proven so far in this class.

Greatest Common Divisors and Least Common Multiples in UFDs

Definition of GCD and LCM in Unique Factorization Domains

Spoken Transcript [01:15:54 - 01:18:14] Prof. Rahul Pandharipande

So, this is Theorem A, and that's PID implies UFD. [writes: Thm A PID $\Longrightarrow$ UFD]

And there's a lot of room to be confused about this result. So, let's just say a couple of things.

If I take something like the rational numbers, if I use this, $\mathbb{Q}[x]$ is a PID, so, this implies that $\mathbb{Q}[x]$ is a UFD. [writes: $\mathbb{Q}[x]$ is a PID $\implies$ $\mathbb{Q}[x]$ is a UFD]

And these rings behave similarly to the integers in the way you expect for this UFD property: which is the fact that they have GCDs and LCMs. Most of those properties hold for any UFD, but there's the possibility of saying things that are incorrect, which I'm trying to explain here. So, let's see, how should I do this?

First of all, let R be a UFD. [writes: R be a UFD]

Then the first thing is these two ideas, which hold for integers, also hold for UFDs: the greatest common divisor [writes: • GCD] and least common multiple. [writes: • LCM] Say you have two elements, x and y. [writes: (x,y)]

Let us try to define these. You must have learned about greatest common divisors and least common multiples for integers (Recall: Definition 0.15, Fact 0.20), yes?

Okay, so, the claim is that if R is a UFD, those principles, those definitions, hold without any change.

Definition: [writes: Def: $z = \mathrm{GCD}(x,y)$ if $z \mid x$]

Recap: Divisibility in a ring was defined in Week 2 (Definition 2.65), and associates — elements that differ by a unit factor — appeared in Week 1 for $\mathbb{Z}$ and $\mathbb{Z}[i]$ (in the discussion after Definition 1.3). The lecturer restates divisibility and then names associates in general (Definition 6.34).

Definition of Divisibility and Greatest Common Divisor

Notation 6.32 (*Divisibility*)**:** Let R be an integral domain, and let $z, x \in R$. We say that *"z divides x"* (written $z \mid x$) if there exists an element $q \in R$ such that:

$$x = q \cdot z. \tag{6.62}$$

Equivalently, $z \mid x \iff (x) \subseteq (z)$.

Definition 6.33 (*Greatest Common Divisor (GCD)*)**:** Let R be a unique factorization domain (UFD), and let $x, y \in R \setminus \{0\}$. An element $z \in R$ is called a *"greatest common divisor"* of x and y (written $z = \mathrm{GCD}(x,y)$) if the following two conditions hold:

- **(i) Common Divisor:** $z \mid x$ and $z \mid y$.
- **(ii) Greatest Divisor:** If $w \in R$ satisfies $w \mid x$ and $w \mid y$, then $w \mid z$.

Spoken Transcript [01:18:14 - 01:21:04] Prof. Rahul Pandharipande

We say z equals the GCD of x and y if z divides x — and when I write this, z divides x, all I mean is that x equals something times z, [writes: $(x = qz)$] I don't know, q, that's what this symbol means.

So, the GCD: z divides x, z divides y, [writes: $z \mid y$] and if w divides x and w divides y, then w divides z. [writes: and if $w \mid x$ and $w \mid y \implies w \mid z$]

That's it: z is the greatest common divisor, it's the biggest person that divides them both. Okay? That's how it was also defined for integers.

And what is the relation to unique factorization?

Well, you can write x as... [writes: $x =$] Now I group everything, so, the standard way to write this is $p_1^{n_1} p_2^{n_2} \cdots p_k^{n_k}$, and then maybe times some unit, [writes: $x = u \cdot p_1^{n_1} \ldots p_k^{n_k}$ (u unit)]

and y is equal to some different unit times $p_1^{m_1} p_2^{m_2} \cdots p_k^{m_k}$. [writes: $y = \hat{u} \cdot p_1^{m_1} \ldots p_k^{m_k}$ ($\hat{u}$ unit)]

This is also what you do with integers: you take x, and the irreducibles you group, so that all the irreducibles that differ just by a unit, you put them all together. Meaning you don't write -3 and 3, you would write that as minus — that's the unit part — and 3^2. [writes: $(-3)(3) = -1 \cdot 3^2$]

So, you put all of the unit stuff here, and then you put all the primes, the distinct classes of primes that do not differ by units.

This is what we do with integers, and what we do also in a UFD.

When you write this for a UFD, there's some kind of background, and the background is a little unpleasant to say, but maybe I should say it: here's the whole UFD: [draws a large closed region representing R]

And if I pick some irreducible, [draws point p] that's an irreducible, it has a lot of people that differ by units, because it depends on how big the unit group is. [draws a small contour surrounding p and other elements] You just take this irreducible p, it has all its friends that differ by the units. So, this is some kind of class of irreducibles. And we just pick one of them.

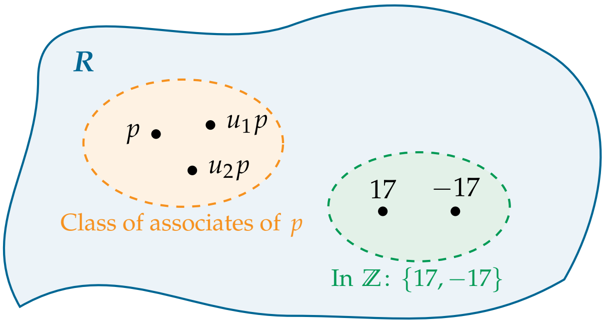
Because we don't need to worry, and we put all the units out here front. That's what we do with integers also; in the integers, this would be like 17 and -17. [draws small bubble with 17 and -17] That's irreducible, and that's the only other one, that differs by a sign.

So, if I look at this UFD, I can consider these classes of irreducibles that differ just by multiplication by units. Often they're called associates.

Equivalence Classes of Associates and Grouped Factorizations

Definition 6.34 (*Associate Elements*)**:** Let R be an integral domain. Two elements $p, \hat{p} \in R$ are called *"associates"* (denoted $p \sim \hat{p}$) if they differ only by multiplication by an invertible unit:

$$p \sim \hat{p} :\iff \exists u \in \mathcal{U}_R \quad \text{such that} \quad p = u \cdot \hat{p}. \tag{6.63}$$



Grouping Irreducibles into Canonical Factorizations: In any UFD, the relation of being associate is an equivalence relation on the set of irreducible elements. To represent factorizations systematically without duplicate associate factors:

- We choose ahead of time a fixed set of representatives $\mathcal{P} \subset R$, selecting exactly one irreducible element from each associate equivalence class (analogous to choosing positive primes in $\mathbb{Z}$).
- Any two non-zero elements $x, y \in R$ can then be simultaneously expressed over the same finite set of pairwise non-associate irreducibles $\{p_1, \ldots, p_k\} \subset \mathcal{P}$:
$$x = u \cdot p_1^{n_1} \cdot p_2^{n_2} \cdots p_k^{n_k}, \tag{6.64}$$
$$y = v \cdot p_1^{m_1} \cdot p_2^{m_2} \cdots p_k^{m_k}, \tag{6.65}$$
where $u, v \in \mathcal{U}_R$ are units and the exponents satisfy $n_i, m_i \geq 0$ (setting exponents to 0 if an irreducible divides one element but not the other).

Beyond the Lecture: Associates and Ideals

In an integral domain, $p \sim \hat{p} \iff (p) = (\hat{p})$.

Formulas for GCD and LCM in Terms of Factorizations

Spoken Transcript [01:21:04 - 01:23:12] Prof. Rahul Pandharipande

[writes: $p \sim \hat{p}$ are associates if $p = \text{unit} \times \hat{p}$]

When we write this unique factorization, the typical way to do it is we, ahead of time, pick one irreducible element for every class of associates, so, we don't confuse ourselves.

For the primes, this is like picking always the positive prime number.

And then we raise that to the number of times it occurs. So, you write x as some unit, and when we write this, silently we've decided how to pick one irreducible element for every class.

And then we do the same for y. So, we write them like this.

This was a kind of confusing thing. It's not deep, and almost no one ever says it, but in a UFD, if you have two elements, if you look at a book, they'll just say x is equal to unit times this, and y is equal to unit times that.

And what they mean here, it's important that you pick the same p's here, just so you don't confuse yourself, right?

Somehow I feel like this wasn't as clear as it could be. It's the same issue about having elements with too many names.

And the reason — the punchline, of course — the reason why you want to do that is then you can write down the formulas for the GCD and the LCM, right?

So, here's the... [wipes part of the board with a wet cloth]

Spoken Transcript [01:23:12 - 01:24:46]

Prof. Rahul Pandharipande

[writes: $\mathrm{GCD} = p_1^{\min(n_1,m_1)} \dots p_k^{\min(n_k,m_k)}$]

What should I put here? What's the exponent? Yeah?

Student Answer

Minimum.

Spoken Transcript [01:24:47 - 01:25:34]

Prof. Rahul Pandharipande

The minimum, yeah. This is why we like to write it like this: minimum of — what did I call them? — n_1 and m_1 [writes "$\min(n_1, m_1)$" and "$\dots p_k^{\min(n_k,m_k)}$"].

Okay. And then, so, that's the GCD, and then you can see that there's also this confusion that I could have put any unit here also: [writes: any unit $\times$] any unit. And it'll still satisfy that property.

So, that's the other confusion: the actual fact of the matter is that GCDs exist in UFDs, [writes: GCD exist in UFD] meaning that the GCD of x and y — and as I said, we never talk about x and y being zero or units, so, if you take an x and y that's not zero or a unit, it has a GCD, the GCD exists, it's given by this formula —, but the GCD itself is only unique up to multiplication by a unit. [writes: only unique up to mult. by a unit] Okay?

So, like 6 and 4, the GCD, from our point of view, of 6 and 4... well, $6 = 2 \cdot 3$, and $4 = 2 \cdot 2$ — I hope this is not a surprise to anyone, 2×2. [writes: $6 = 2 \cdot 3$, $4 = 2^2 \cdot 3^0$] And so, the GCD, I have to take... so, there's a 3^0, so, the zero gives me only 3^0 here, and this gives me the min as 2, so, the GCD is 2, but actually it's ± 2 because I could have a unit anywhere. So, that's the answer. [writes: $\mathrm{GCD}(6,4) = 2 \cdot 3^0 = \pm 2$]

So, yeah, anyway.

Formula for the Greatest Common Divisor in a UFD

Theorem 6.35 (*GCD Formula via Prime Exponents*)**:** Let R be a unique factorization domain, and let $x, y \in R \setminus \{0\}$ be factored simultaneously as:

$$x = u \cdot \prod_{i=1}^{k} p_i^{n_i} \quad \text{and} \quad y = v \cdot \prod_{i=1}^{k} p_i^{m_i} \quad (u, v \in \mathcal{U}_R,\ n_i, m_i \geq 0).$$

Then the greatest common divisor $\mathrm{GCD}(x,y)$ exists in R and is given by:

$$\mathrm{GCD}(x,y) \sim \prod_{i=1}^{k} p_i^{\min(n_i,m_i)}. \tag{6.66}$$

Any greatest common divisor is unique up to multiplication by an invertible unit:

$$z \sim \mathrm{GCD}(x,y) \iff z = w \cdot \prod_{i=1}^{k} p_i^{\min(n_i,m_i)} \quad \text{for some } w \in \mathcal{U}_R. \tag{6.67}$$

Why the Formula Holds (beyond the lecture)

The lecture states the formula without proof. Write $d := \prod_i p_i^{\min(n_i,m_i)}$; it divides x and y. The key fact is: **in a UFD, a non-zero w divides x if and only if every irreducible factor of w is associate to some p_i, with exponent at most n_i.** (If $x = wt$, factor w and t into irreducibles; their concatenation is a factorization of x, and uniqueness matches it with $u\, p_1^{n_1} \cdots p_k^{n_k}$.) So a common divisor w of x and y has the form $w = v \prod_i p_i^{e_i}$ with $v \in \mathcal{U}_R$ and $e_i \le \min(n_i, m_i)$, hence $w \mid d$. If z is another greatest common divisor, then $z \mid d$ and $d \mid z$, so $z = wd$ and $d = w'z$ give $d = w'wd$, and cancelling $d \neq 0$ shows that w is a unit. **Q.E.D.**

Example 6.36 (*GCD Computation for 6 and 4 in $\mathbb{Z}$*): Over the domain of integers $\mathbb{Z}$, factor 6 and 4 using the non-associate primes 2 and 3:

$$6 = 2^1 \cdot 3^1,$$
$$4 = 2^2 \cdot 3^0.$$

Applying formula (6.66):

$$\mathrm{GCD}(6,4) = 2^{\min(1,2)} \cdot 3^{\min(1,0)} = 2^1 \cdot 3^0 = 2.$$

Because $\mathrm{Units}(\mathbb{Z}) = \{1, -1\}$, both 2 and -2 are greatest common divisors of 6 and 4 in the ring-theoretic sense: $\mathrm{GCD}(6,4) = \pm 2$.

Least Common Multiples and the Product Formula

Spoken Transcript [01:25:34 - 01:27:19] Prof. Rahul Pandharipande

So, that's how the GCD works for any UFD.

And when I said this confusion, it's always this confusion about everybody has a lot of names, and that always causes confusion in mathematics.

Okay, and then the formula for the least common multiple is the same formula, except now we have maybe a different unit — it's only well-defined up to a unit — and then what I put here is the max. That's what happens for integers, right [writes: $\mathrm{LCM} = \hat{w} \cdot p_1^{\max(n_1,m_1)} \dots p_k^{\max(n_k,m_k)}$]?

Okay?

Definition and Formula for the Least Common Multiple

Definition 6.37 (*Least Common Multiple (LCM)*): Let R be a unique factorization domain, and let $x, y \in R \setminus \{0\}$. An element $\ell \in R$ is called a "*least common multiple*" of

x and y (written $\ell = \mathrm{LCM}(x, y)$) if:

- **(i) Common Multiple:** $x \mid \ell$ and $y \mid \ell$.
- **(ii) Least Multiple:** If $w \in R$ satisfies $x \mid w$ and $y \mid w$, then $\ell \mid w$.

Theorem 6.38 (*LCM Formula via Prime Exponents*): Under the simultaneous factorizations (6.64) and (6.65), the least common multiple $\mathrm{LCM}(x, y)$ exists in R, is unique up to multiplication by units, and satisfies:

$$\mathrm{LCM}(x, y) \sim \prod_{i=1}^{k} p_i^{\max(n_i, m_i)}. \tag{6.68}$$

Spoken Transcript [01:27:19 - 01:28:17] Prof. Rahul Pandharipande

All right, so, that's the confusing discussion of GCDs and LCMs.

So, the summary of this is: once you have a UFD, since we have this factorization, the matter is just the same as for integers, except the confusion about the units is greater. That's all.

If you can conquer that confusion in your head, everything is just the same.

And then we get the standard equation from school (Recall: Fact 0.20): $\mathrm{GCD}(x, y) \times \mathrm{LCM}(x, y)$... Does anyone remember what this is? I should put it here: $\mathrm{GCD}(x, y) \times \mathrm{LCM}(x, y)$, what should that be equal to?

Student Answer

$x \times y$.

Spoken Transcript [01:28:18 - 01:28:56] Prof. Rahul Pandharipande

Yeah, $x \times y$, yeah. That's an equation I never used in my life, but nevertheless, it's a standard one. [writes: $\mathrm{GCD}(x, y) \cdot \mathrm{LCM}(x, y) = xy$]

And this equation, by the way, if we're honest, is up to a unit, of course, [adds the unit $\hat{w}$ before xy] because the left-hand side is only defined up to units.

For integers, you can get rid of all this by making positive choices everywhere.

The GCD–LCM Product Identity in a UFD

Proposition 6.39 (*GCD–LCM Product Identity*): Let R be a unique factorization domain. For any two non-zero elements $x, y \in R \setminus \{0\}$, the product of their greatest common divisor and least common multiple equals the product $x \cdot y$ up to an invertible unit:

$$\mathrm{GCD}(x, y) \cdot \mathrm{LCM}(x, y) \sim x \cdot y. \tag{6.69}$$

That is, there exists a unit $u \in \mathcal{U}_R$ such that $\mathrm{GCD}(x, y) \cdot \mathrm{LCM}(x, y) = u \cdot x \cdot y$.

Why the Identity Holds (beyond the lecture)

The lecture states this identity without proof. Using the canonical factorizations (6.64) and (6.65):

$$\begin{aligned}\mathrm{GCD}(x,y)\cdot\mathrm{LCM}(x,y) &\sim \left(\prod_{i=1}^{k} p_i^{\min(n_i,m_i)}\right)\cdot\left(\prod_{i=1}^{k} p_i^{\max(n_i,m_i)}\right)\\ &= \prod_{i=1}^{k} p_i^{\min(n_i,m_i)+\max(n_i,m_i)}.\end{aligned}$$

For any two real numbers (or non-negative integers), $\min(n_i,m_i)+\max(n_i,m_i)=n_i+m_i$. Therefore:

$$\prod_{i=1}^{k} p_i^{\min(n_i,m_i)+\max(n_i,m_i)}=\prod_{i=1}^{k} p_i^{n_i+m_i}=\left(\prod_{i=1}^{k} p_i^{n_i}\right)\cdot\left(\prod_{i=1}^{k} p_i^{m_i}\right)\sim x\cdot y.$$

This establishes the identity up to multiplication by units. **Q.E.D.**

Beyond the Lecture: The GCD as a Linear Combination

In a PID, the ideal (x,y) is principal, say $(x,y)=(d)$. Then d is a greatest common divisor of x and y: it divides both, as $x,y\in(d)$; and every common divisor w of x and y divides every element $ax+by$ of (x,y), in particular d. Since $d\in(x,y)$, we get *“Bézout’s identity”* $d=ax+by$ for some $a,b\in R$, as in $\mathbb{Z}$ (Fact 0.16).

In a UFD that is not a PID this fails. In $k[x,y]$, the elements x and y are non-associate irreducibles, so $\mathrm{GCD}(x,y)\sim 1$, but $(x,y)\neq(1)$ (Answer 6.3): 1 is not of the form $ax+by$. Likewise $\mathrm{GCD}(2,x)\sim 1$ in $\mathbb{Z}[x]$, while $(2,x)$ is a proper ideal (Examples 5.61). In a general UFD, greatest common divisors exist (Theorem 6.35) but need not be linear combinations.

Formal Power Series as a PID

Spoken Transcript [01:28:56 - 01:30:05] Prof. Rahul Pandharipande

Okay, so, anyway, we’ve proven something today which was that PID implies UFD.

And there was another really good question from last time, which I will address for one minute if I have time for it, which is: what other PIDs do we know?

And we know another one. I don’t think I explicitly said it, but it’s a very fun thing to think about. Well, relatively fun.

If I take a field, and I take the power series (Recall: Definition 3.41), this turns out to be a PID! [writes: $k[[x]]$ is a PID!]

Is that obvious to you? You should think about this. Maybe I leave this to you: prove this. [writes: Prove this]

And then in particular, this means that the power series ring is a UFD, because PID implies UFD. [writes: $k[[x]]$ is a UFD!]

And the hint for the proof is that... okay, no hint for the proof. You get to do it yourself. [laughter] You can see if you can figure it out. It's a good thing to think about.

All right. [unclips the microphone and packs up; students applaud]

The Formal Power Series Ring $k[[x]]$

Exercise 6.40 (*$k[[x]]$ is a Principal Ideal Domain*): Let k be a field, and let $k[[x]]$ denote the ring of formal power series in one variable over k.

(1) Prove that $k[[x]]$ is a principal ideal domain (PID).

(2) Deduce from Theorem A (Theorem 6.15) that $k[[x]]$ is a unique factorization domain (UFD).

The lecturer sketches the proof in Week 7 (Example 7.4, Exercise 7.5).

Structure of Ideals in $k[[x]]$: Going beyond the lecture: The ring $k[[x]]$ is an integral domain (Theorem 4.6), and a power series is a unit exactly when its constant term is non-zero (Theorem 3.43, as k is a field). Hence every non-zero formal power series $f \in k[[x]]$ can be uniquely written, by factoring out the lowest power of x that occurs, as:

$$f(x) = x^d \cdot u(x), \qquad d = \operatorname{ord}(f) \in \mathbb{N}_0, \quad u(x) \in \mathcal{U}_{k[[x]]}\ (u(0) \neq 0),$$

where $\operatorname{ord}(f)$, the *"order"* of f, is the smallest exponent with a non-zero coefficient. Consequently, if $I \subseteq k[[x]]$ is a non-zero ideal, let $d := \min\{\operatorname{ord}(f) \mid f \in I,\ f \neq 0\}$. Then $I = (x^d)$: an $f \in I$ of order d is $x^d u$ with u a unit, so $x^d = u^{-1} f \in I$; and every $g \in I \setminus \{0\}$ is $x^e v = x^d \cdot x^{e-d} v$ with $e \geq d$. Every non-zero ideal is generated by a single power of x. Thus, the only ideals of $k[[x]]$ are the powers $(x^0) = (1) \supsetneq (x) \supsetneq (x^2) \supsetneq (x^3) \supsetneq \cdots \supsetneq (0)$. This proves that $k[[x]]$ is a PID (and in fact a discrete valuation ring), which immediately implies that $k[[x]]$ is a UFD.

Summary of Key Concepts

This lecture established the core structural theorems linking principal ideal domains and unique factorization domains:

- **Ideals and Polynomials:** While univariate polynomial rings $k[x]$ over a field are PIDs, multivariate polynomial rings $k[x_1, \ldots, x_n]$ are never PIDs for $n \geq 2$ (e.g., (x, y) is non-principal, and (x^2, xy, y^2) requires three generators). By the Hilbert Basis Theorem, they are Noetherian rings (every ideal is finitely generated).
- **Factorization Terminology:** In an integral domain R, an irreducible element cannot be factored into two non-units ($x = yz \implies y \in \mathcal{U}_R \vee z \in \mathcal{U}_R$). A prime element generates a principal prime ideal ($p \mid ab \implies p \mid a \vee p \mid b$). In a PID, every irreducible element generates a prime ideal (Step 2).
- **Unique Factorization Domains:** A domain is a UFD if every non-zero, non-unit factors into irreducibles (Existence) uniquely up to permutation and unit associates (Uniqueness). The subring $R = \mathbb{Q}[x^2, x^3] \subset \mathbb{Q}[x]$ (whose picture is a cusp) fails to be a UFD because $x^6 = (x^2)^3 = (x^3)^2$.
- **Theorem A (PID $\implies$ UFD):**

Step 1: Uniqueness Criterion: Irreducible implies prime yields uniqueness of factorizations.

Step 2: Irreducible Implies Prime: In a PID, every irreducible element generates a prime ideal, establishing uniqueness.

Step 3: No Infinite Chains: In a PID, a strictly ascending chain $(x_0) \subsetneq (x_1) \subsetneq \dots$ cannot go on forever, because the union ideal $J = \bigcup_{n=0}^{\infty}(x_n) = (w)$ is principal, forcing $(x_N) = J$. This rules out infinite factorization branches and establishes existence.

- **GCD, LCM, and Power Series:** GCDs and LCMs exist in any UFD via minimum and maximum exponents of prime factorizations (unique up to units), satisfying $\mathrm{GCD}(x, y) \cdot \mathrm{LCM}(x, y) \sim x \cdot y$. Finally, formal power series rings $k[[x]]$ are PIDs, hence UFDs.

Wednesday, 21st October 2026

Gauss's Theorem on Polynomial Rings over UFDs

Lecture Context & Content Overview

This lecture proves Gauss's Theorem on polynomial rings (Theorem B of the previous lecture, Theorem 6.16): if R is a unique factorization domain (UFD), then the polynomial ring $R[x]$ is also a unique factorization domain. The last case of the proof is cut off by the end of the hour and finished in the next lecture; its completion is recorded at the end of this chapter.

The lecture is organized into two primary components:

- **Foundational Preliminaries:**
  - Review of the equivalence between irreducible elements and prime elements in any UFD.
  - Definition and computation of the *"content"* $C(f)$ of a non-zero polynomial, uniquely determined up to associates in R.
  - Canonical decomposition $f = C(f) \cdot f^*$ into content and a *"primitive"* polynomial f^*.
  - Proof of **Gauss's Lemma**: the product of primitive polynomials is primitive ($C(f) \sim 1 \wedge C(g) \sim 1 \implies C(fg) \sim 1$).
  - Divisibility property of primitive polynomials: for primitive $g \in R[x]$, $g \mid bf \implies g \mid f$ for any non-zero scalar $b \in R$.
- **Proof of Gauss's Theorem (Theorem B):**

  Step 1: Existence: Existence of irreducible factorizations in $R[x]$ via induction on $\deg(f)$, using primitivity to rule out non-unit degree-zero factors.

  Step 2: Uniqueness: Uniqueness via proving that every irreducible element in $R[x]$ is prime:

  - **Case 1 ($\deg(p) = 0$):** Constant irreducibles in R remain prime in $R[x]$ via coefficient divisibility and Gauss's Lemma.
  - **Case 2 ($\deg(p) > 0$):** Positive-degree irreducibles analyzed via minimal-degree elements in the ideal (p, f), division with remainder in $Q[x]$ over the field of fractions $Q = \mathrm{Frac}(R)$, and denominator clearing.

Gauss's Theorem and Preliminaries on Content

Spoken Transcript [00:00:00 - 00:00:41] Prof. Rahul Pandharipande

Okay, so, we start, and the goal today is this Theorem B that I said, which is that R is a UFD implies the polynomial ring is a UFD. [writes: Thm B: R is a UFD $\implies$ $R[x]$ is a UFD] And this theorem is, generally, well, it's a theorem of Carl Friedrich Gauss [writes: (C.F. Gauss)], whose ghost hovers over mathematics.

And it has also an argument which has some intricacy, like the last argument we had.

Doesn't it seem like there's an echo? Or not? No? Okay, good.

Recap: Theorem B was stated in the previous lecture (Theorem 6.16), together with its consequence that $k[x_1, \ldots, x_n]$ is a UFD (Example 6.20). The lecturer restates it.

Theorem 6.41 (*Gauss's Theorem on Polynomial Rings (Theorem B)*)**:** Let R be a unique factorization domain (UFD). Then the polynomial ring in one variable $R[x]$ is also a unique factorization domain.

Overview and Significance of Gauss's Theorem: Going beyond the lecture: Theorem B generalizes the classical result that if k is a field, then $k[x]$ is a UFD (in fact, a Principal Ideal Domain, Proposition 5.58). However, when R is merely a UFD (such as $R = \mathbb{Z}$), the polynomial ring $R[x]$ is a PID only if R is a field (for instance, the ideal $(2, x) \subset \mathbb{Z}[x]$ cannot be generated by a single element, Exercise 5.62; for a non-unit $a \neq 0$ of R, the ideal (a, x) plays the same role). Consequently, proving that $R[x]$ is nonetheless a UFD requires a delicate analysis of polynomial factorization using the concepts of **content** and **primitive polynomials**.

Irreducible Elements and Prime Ideals in a UFD

Spoken Transcript [00:00:41 - 00:01:51] Prof. Rahul Pandharipande

So, the first thing that I will say is an exercise which I really encourage you to do. It says that if R is a UFD, then we have an equivalence — because it's part of this language that the equivalence is that p is irreducible (that is the notion we had defined) if and only if the ideal is prime. [writes: [Ex. If R is a UFD, then p is irred. $\iff$ (p) is prime]]

So, this is an exercise; I don't do it here. I very much encourage you to do it because it has the ideas that we used a lot in the last lecture. You have to use, in one way, the property of being a UFD, and in the other way, the definition of a prime ideal.

And this exercise allows us to conflate, if we're in a UFD, then these two notions of being irreducible and p prime are the same notion. It's like the integers; the integers have this property. So, I encourage you to do that. Actually, I strongly encourage you to do this exercise. It's part of understanding the definitions, etc., in the previous lecture.

Exercise 6.42 (*Equivalence of Irreducible Elements and Prime Elements in a UFD*): Let R be a unique factorization domain, and let $p \in R \setminus \{0\}$. Prove that p is an irreducible element if and only if the principal ideal $(p) = pR$ is a prime ideal (or equivalently, p is a prime element).

Proof Outline for the Exercise (beyond the lecture)

The lecture leaves this as an exercise (*"I don't do it here"*). In any integral domain R, a prime element is always irreducible:

- **Prime implies Irreducible:** If (p) is prime and $p = ab$, then $ab \in (p)$, so $p \mid a$ or $p \mid b$. If $p \mid a$, write $a = pr$. Then $p = prb \implies 1 = rb$, so b is a unit in R. Thus, p is irreducible.
- **Irreducible implies Prime (requires UFD):** Suppose p is irreducible and $p \mid ab$. Then $ab = pc$ for some $c \in R$. If $a = 0$ or $b = 0$, there is nothing to prove; if a is a unit, then $p \mid a^{-1}ab = b$, and similarly if b is a unit. Otherwise, factoring a, b (and c, unless it is a unit) into products of irreducible elements, the irreducible element p appearing on the right-hand side must, by uniqueness of factorization up to associates, be associate to one of the irreducible factors of a or of b. Hence, $p \mid a$ or $p \mid b$, which implies (p) is a prime ideal.

The second half is the generalization of Euclid's lemma (Fact 0.17) to UFDs; for PIDs it was proved directly as Step 2 of Theorem A (Lemma 6.26). The first half, together with Step 1 of Theorem A (Claim 6.25), shows: a domain in which every non-zero non-unit has a factorization into irreducibles is a UFD if and only if every irreducible element is prime. **Q.E.D.**

III The Content of a Polynomial

Spoken Transcript [00:01:51 - 00:02:55] Prof. Rahul Pandharipande

Okay, so, there's a new notion here which is content, which is an extremely useful notion. [writes: Content] And the idea here is if I have a polynomial here — and for this discussion of content, I don't consider the zero polynomial [writes: $f \in R[x], f \neq 0$] —, I write f as a_0, a constant term, a linear term, and so on, and at some point it stops, because a polynomial always stops. [writes: $f = a_0 + a_1 x + \cdots + a_n x^n$]

Then, this is the definition: the content — so, the word for that is the content — is equal to the GCD of these elements here: $a_0, a_1, \ldots, a_n$. [writes: Def. $C(f) = \mathrm{GCD}(a_0, a_1, \ldots, a_n)$] Okay?

So, why are we allowed to consider GCD? It's because this R is a UFD. I should say this content, and for the whole lecture today, R is a UFD. [writes "R is a UFD" above] Okay?

Definition 6.43 (*Content of a Polynomial*): Let R be a unique factorization domain, and let $f \in R[x] \setminus \{0\}$ be a non-zero polynomial written as:

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \in R[x].$$

The *"content"* of f, denoted $C(f)$, is defined as the greatest common divisor of all the non-zero coefficients of f:

$$C(f) := \mathrm{GCD}(a_0, a_1, \dots, a_n). \tag{6.70}$$

Spoken Transcript [00:02:55 - 00:04:53] Prof. Rahul Pandharipande

And then, this is how it's typically written, and what do I mean by this? Well, we already discussed what GCD is. You're not allowed to have any zeros here. So, if any of these coefficients are zero, you just strike them out, because we don't do GCD of zero.

So, this is how it's typically written, but what it means is you take f, you put here the non-zero coefficients [writes "non-zero" under $a_0, a_1, \dots, a_n$] — non-zero coefficients —, and then take their GCDs, as we discussed yesterday (Recall: Definition 6.33, Theorem 6.35).

The way you take GCDs is for every one of these, like this one, if it's non-zero... There has to be some of these that are non-zero because f is not zero. So, there's at least one that's not zero. You put the list of all the ones that are non-zero, and for each one you take the prime factorization, which is, you know, some unit and then some product of p_i to the k_i — let's say k_{i0} for the zero one [writes: $a_0 = u_0 \prod p_i^{k_{i0}}$], and then a_1 would be some unit one, p_i to the k_{i1} [writes: $a_1 = u_1 \prod p_i^{k_{i1}}$].

And then to get the GCD — it's the greatest common divisor —, you take the minimal power in each prime. We discussed it for two, but you can also discuss it for several. It's just like for the integers.

Okay? So, that's the definition of the content: you just take your polynomial, put the coefficients which are not zero, take the GCD of them, which is well-defined since it's a UFD.

And there's another unpleasant ambiguity here: these GCDs are always defined only up to unit. So, this thing is defined up to unit. [boxes: up to unit] That's just how it is, but people will just talk about the content, even though implicitly it's defined only up to unit.

Explicit Computation of Content in a UFD

Let R be a UFD. Every non-zero coefficient $a_j \neq 0$ admits a unique prime factorization:

$$a_j = u_j \prod_{i=1}^{m} p_i^{k_{ij}},$$

where $u_j \in \mathcal{U}_R$ is a unit, $p_1, \dots, p_m$ are pairwise non-associate irreducible (prime) elements, and $k_{ij} \geq 0$. The greatest common divisor of several elements is defined as for two (Definition 6.33: a common divisor that every common divisor divides), and the argument of Theorem 6.35 gives it by minimal exponents. So the greatest common divisor of the non-zero coefficients is:

$$\mathrm{GCD}(a_0, a_1, \dots, a_n) = \prod_{i=1}^{m} p_i^{\min\{k_{ij} | a_j \neq 0\}}. \tag{6.71}$$

Remark 6.44 (*Ambiguity Up to Units*)**:** Because the greatest common divisor in an arbitrary UFD is uniquely determined only up to multiplication by a unit in $\mathcal{U}_R$, the content $C(f)$ is defined only up to a unit. People still just talk about the content.

Beyond the Lecture: The Content as an Associate Class

Formally, $C(f)$ is well defined as an associate class (Definition 6.34), an element of the quotient set of R by the equivalence relation $\sim$ (Definition 0.9), sometimes written $C(f) \in R/\mathcal{U}_R$. By convention, $C(f) = d$ means that $C(f)$ is associate to d, or one fixes a canonical choice of associates whenever one is naturally available (such as positive integers in $\mathbb{Z}$ or monic polynomials in $k[x]$).

III Canonical Decomposition of Polynomials

Spoken Transcript [00:04:53 - 00:06:10] Prof. Rahul Pandharipande

All right, some properties of the content. If I take this f — and whenever I consider contents, I always have $f \neq 0$ —, then I can write f as the content times f^* [writes: $f \neq 0, f = C(f) \cdot f^*$], because since it's a GCD, I can divide all the coefficients by the content, and I can write this equation. So, f is in polynomials in R, this content is a number, so it's in R, and this is also in polynomials in R [writes "$R[x]$" under f, "R" under $C(f)$, and "$R[x]$" under f^*]. Since the content is the GCD, I can just divide it out of all of them, and I get a polynomial here. Okay?

And if you think about this, what happens for this polynomial? What is the content of this polynomial? What do you think the answer to that is? It's just like I take the coefficients, they have a GCD, I divide out the GCD... yeah, one, yeah. [writes: What is $C(f^*) \overset{\text{unit}}{=} 1$] So, the answer is, this is one. And whenever I say something like this or whenever it's written, people will just write this, but what is meant is that this is up to multiplication by a unit, one, because that depends on how you did it. Because you've just taken out all of the common divisors, what's left is just units.

Proposition 6.45 (*Factorization into Content and Primitive Part*): Let R be a UFD and let $f \in R[x] \setminus \{0\}$. Then f can be factored as:

$$f = C(f) \cdot f^*, \tag{6.72}$$

where $C(f) \in R$ and $f^* \in R[x]$. Furthermore, the content of f^* satisfies:

$$C(f^*) \sim 1 \quad (\text{up to a unit in } R). \tag{6.73}$$

Algebraic Justification: Let $f(x) = \sum_{j=0}^{n} a_j x^j$ with $C(f) = d \in R$. By definition of the greatest common divisor, $d \mid a_j$ for all $j \in \{0, \dots, n\}$. Therefore, each quotient $a_j^* := a_j / d$ belongs to R, defining the polynomial:

$$f^*(x) := \sum_{j=0}^{n} a_j^* x^j \in R[x].$$

If an irreducible element $p \in R$ divided every non-zero coefficient a_j^* of f^*, then $p \cdot d$ would be a common divisor of all $a_j = d \cdot a_j^*$ in R, hence would divide d: $d = pdr$, and cancelling $d \neq 0$ gives $1 = pr$, so p would be a unit, which an irreducible is not. Thus no irreducible divides all a_j^*; since every non-unit has an irreducible factor, $\mathrm{GCD}(a_0^*, \dots, a_n^*) \sim 1$.

Spoken Transcript [00:06:10 - 00:07:13] Prof. Rahul Pandharipande

So, example. Let's say R is the integers, that's maybe our favorite ring that's a UFD, and I take some polynomial in the integers. Okay, let's say $17x^{17}+289$. [writes: Example: $R=\mathbb{Z}, f=17x^{17}+289$] Okay, how about this polynomial? What's the content of this polynomial?

Well, you can take the 17 out — this is maybe excessive 17s: $x^{17}+17$, because 289 is the square of 17, as you surely know. [writes: $=17(x^{17}+17)$] So, the content is 17. [writes: $C(f)=17$]

So, it's not a complicated notion, but it's going to be important. Whenever we have a polynomial with coefficients in R, we can somehow take out the GCD of all the coefficients, just like we did here — just take it out. That's all this content is.

Example 6.46 (*Computing Content over the Integers*)**:** Let $R=\mathbb{Z}$, which is a UFD. Consider the polynomial:

$$f(x)=17x^{17}+289\in\mathbb{Z}[x].$$

Since $289=17^2$, the coefficients are $a_{17}=17$, $a_0=289$, and all intermediate coefficients are 0. Factoring out the greatest common divisor of the non-zero coefficients yields:

$$f(x)=17\cdot(x^{17}+17).$$

Therefore, the content is $C(f)=17$ (up to the units ±1), and the remaining factor is $f^*(x)=x^{17}+17$.

Beyond the Lecture: Eisenstein's Criterion

The primitive part $x^{17}+17$ is even irreducible in $\mathbb{Q}[x]$, by **Eisenstein's criterion** at $p=17$: 17 divides every coefficient except the leading one, and $17^2\nmid 17$. The lecture does not cover this criterion.

Primitive Polynomials

Spoken Transcript [00:07:13 - 00:08:12] Prof. Rahul Pandharipande

So, there's a definition that's important. Definition: $g\in R[x]$, $g\neq 0$, is primitive — that's the word that's being defined — if the content of g is equal to one, and of course I always mean up to a unit, okay? [writes: Def. $g\in R[x]$, $g\neq0$, is primitive if $C(g)\overset{\text{unit}}{=}1$]

That's the notion, that's the definition of primitive. Primitive are those polynomials where, if I take their coefficients, there's no GCD. Like this one here. [points at $x^{17}+17$] 17 is GCD; this one has no GCD, because of a one and a 17.

So, in particular, if I take this $f\neq0$, and I take $f=C(f)\cdot f^*$, this one is primitive. [writes "primitive" under f^*] Okay?

Definition 6.47 (*Primitive Polynomial*)**:** Let R be a UFD. A non-zero polynomial $g\in$

$R[x] \setminus \{0\}$ is called *"primitive"* if its content is a unit in R:

$$C(g) \sim 1. \tag{6.74}$$

In other words, the non-zero coefficients of g share no common non-unit divisors in R.

Observation 6.48 (*Primitive Factor in Canonical Decomposition*)**:** For any non-zero polynomial $f \in R[x] \setminus \{0\}$, the polynomial factor f^* in the canonical decomposition:

$$f = C(f) \cdot f^*$$

is always a primitive polynomial in $R[x]$. For instance, in Example 6.46, the polynomial $x^{17} + 17 \in \mathbb{Z}[x]$ is primitive because $\mathrm{GCD}(1, 17) = 1$.

Spoken Transcript [00:08:12 - 00:10:10] Prof. Rahul Pandharipande

And the next simple observation is: suppose $f = \gamma \cdot g$, where γ is a scalar and g is a polynomial that is primitive. [writes: Suppose $f = \gamma \cdot g, \gamma \in R, g \in R[x]$ primitive] Okay, suppose I have any expression, where f is just some polynomial in $R[x]$, always not zero, and I write this polynomial as some scalar times a polynomial which is primitive. We already know one way to do that. Here, I've written it as some scalar times a polynomial that's primitive.

And the answer is that if I do this, it must be the case that this content is equal to γ up to a unit, and f^* is equal to g, again, up to a unit. [writes: $\implies C(f) \overset{\text{unit}}{=} \gamma, f^* \overset{\text{unit}}{=} g$] And the reason is because this is the GCD of all the coefficients. If I'm able to take this scalar out, then this scalar must divide the GCD, because it's a divisor and it should divide the greatest common divisor. And if there's anything left over, then this wouldn't be primitive.

So, you just have to think about it. There's actually nothing going on in these statements except understanding this move of taking out the GCD from all of the coefficients. So, this is going to be useful for us.

Those are the basic facts about content: this content exists, and when you take f, you can take out the content, and then you get a primitive. And anytime you see f as a scalar times a primitive, it's essentially the content and the primitive associated with the content. Okay. So, this is going to play a role in the proof.

Proposition 6.49 (*Uniqueness of Scalar and Primitive Factorization*)**:** Let R be a UFD and let $f \in R[x] \setminus \{0\}$. Suppose that f is written as:

$$f = \gamma \cdot g, \tag{6.75}$$

where $\gamma \in R \setminus \{0\}$ and $g \in R[x]$ is a primitive polynomial. Then:

$$\gamma \sim C(f) \quad \text{and} \quad g \sim f^* \quad (\text{up to units in } R). \tag{6.76}$$

Proof

Let $g(x) = \sum_{j=0}^{n} b_j x^j$. Then $f(x) = \sum_{j=0}^{n} (\gamma b_j) x^j$. Computing the content of f using the properties of the greatest common divisor yields:

$$C(f) = \mathrm{GCD}(\gamma b_0, \ldots, \gamma b_n) \sim \gamma \cdot \mathrm{GCD}(b_0, \ldots, b_n) = \gamma \cdot C(g).$$

Since g is primitive, $C(g) \sim 1$, which immediately implies:

$$C(f) \sim \gamma \cdot 1 = \gamma.$$

Writing $\gamma = u \cdot C(f)$ for some unit $u \in \mathcal{U}_R$, substituting back into (6.75) gives:

$$C(f) \cdot f^* = f = uC(f) \cdot g \implies C(f)(f^* - ug) = 0.$$

Since $R[x]$ is an integral domain and $C(f) \neq 0$, it follows that $f^* = ug$, so $g \sim f^*$.

Q.E.D.

Gauss's Lemma

Spoken Transcript [00:10:10 - 00:11:20] Prof. Rahul Pandharipande

So, then there's two observations. The first observation is by Gauss, and this is a rather famous observation. It says that suppose f and g are primitive, then $f \times g$ is primitive. [writes: $\implies$ Gauss: Suppose $f, g \in R[x]$ are primitive, then $fg \in R[x]$ is primitive]

Remember, primitive means that the coefficients of the polynomial have no non-trivial GCD; the GCD is just a unit. And it says if I take two of those and I multiply them, it stays primitive.

So, this is saying something extremely explicit. It's saying, for example, for integers, if you have a polynomial with integer coefficients — two of them — such that this one has no GCD — well, the GCD of its coefficients is one — and that one has a GCD of coefficients one, when you multiply them, the resulting polynomial has GCD of its coefficients one. That's something extremely explicit.

Okay, the proof of this by now for us is also very simple.

Lemma 6.50 (*Gauss's Lemma*): Let R be a unique factorization domain. If $f, g \in R[x]$ are primitive polynomials, then their product $fg \in R[x]$ is also a primitive polynomial:

$$C(f) \sim 1 \quad \text{and} \quad C(g) \sim 1 \implies C(fg) \sim 1. \tag{6.77}$$

Spoken Transcript [00:11:20 - 00:13:25] Prof. Rahul Pandharipande

So, we're going to proceed by contradiction. So, assume that the content of the product is not a unit. [writes: Proof: Contradiction. Assume $C(fg)$ is not a unit.] All right?

Then there exists some prime that divides this content by the unique factorization property. [writes: then $\exists p \mid C(fg)$ by UFD] Because if I have some element that's not a unit, I can factorize it. Primitive stuff is always non-zero, because to be primitive you're not zero. So, if it were not primitive, then this would not be a unit, because being primitive is if and only if this is a unit. So, then there exists some prime that divides it.

And then comes the idea in the proof: for this prime is irreducible, so that implies that the ideal is a prime ideal — that's that exercise (Recall: Exercise 6.42). [writes: p irr $\implies$ (p) is prime ideal] And then we do something interesting, which is we look at this map, a homomorphism from $R[x]$ to... So, there's a quotient map R to $R \pmod{p}$. [writes: $R \to R/(p)$] That's the quotient map from previous weeks (Recall: Definition 4.45). That's a homomorphism. And that also gives me a homomorphism from polynomials by just applying this quotient map to the coefficients of the polynomials. [writes: $R[x] \to$

$(R/(p))[x]$] And this is a ring homomorphism.

Now, this is an integral domain. [writes "int. domain" under $R/(p)$ and $(R/(p))[x]$] Why is it an integral domain? It's because this is a prime ideal. Prime ideal means this is an integral domain (Recall: Proposition 5.24). All right?

Spoken Transcript [00:13:25 - 00:15:16] Prof. Rahul Pandharipande

So, now I consider $\phi(f) \cdot \phi(g)$. That's equal to, because of the homomorphism property, $\phi(fg)$. [writes: $\phi(f) \cdot \phi(g) = \phi(fg)$] And by this hypothesis that p divides all the coefficients of fg, this is equal to zero. [writes: $= 0$ (since $p \mid$ all coeffs)] Since p divides all the coefficients of fg, it's equal to zero.

That means, since this is an integral domain... If I take a polynomial ring in one variable over an integral domain, the whole ring is an integral domain (Recall: Theorem 4.6). So, we have two things multiplying to zero in an integral domain; that means either $\phi(f) = 0$ or — let me erase that — $\phi(g) = 0$. [writes: $\implies \phi(f) = 0$ or $\phi(g) = 0$]

One of these two holds, but that contradicts the assumption. Because for either of these to hold, that means p divides all their coefficients. But the assumption is that these are primitive, so no irreducible divides all their coefficients.

So, that's the proof. It's a simple proof. Gauss must have been very happy when he thought of it. And it's something, yeah, you can think about.

To understand this proof, you have to remember all these constructions: that given this ring, I take this prime ideal, I get an integral domain; integral domains stay integral when I take a polynomial ring, and then I use that property here.

Proof of Lemma 6.50

We proceed by contradiction. Suppose that fg is not primitive. Then its content $C(fg)$ is not a unit in R. Since R is a UFD and $C(fg) \neq 0$ is a non-unit, there exists an irreducible element $p \in R$ such that:

$$p \mid C(fg).$$

By Exercise 6.42, because R is a UFD, the irreducible element p generates a prime ideal $(p) \subset R$. Consequently, the quotient ring:

$$R/(p)$$

is an integral domain (Proposition 5.24). The quotient map $R \to R/(p)$ (Definition 4.45) induces a ring homomorphism on polynomial rings (by the universal property, Theorem 3.38, sending $x \mapsto x$; it reduces every coefficient):

$$\phi\colon R[x] \to (R/(p))[x],$$

$$\sum_{j=0}^{m} c_j x^j \mapsto \sum_{j=0}^{m} \overline{c_j} x^j.$$

Since $R/(p)$ is an integral domain, the polynomial ring $(R/(p))[x]$ is likewise an integral domain (Theorem 4.6). Applying ϕ to the product fg and using the ring homomorphism property yields:

$$\phi(f) \cdot \phi(g) = \phi(fg).$$

Because $p \mid C(fg)$, the prime p divides every coefficient of fg. Therefore, every coefficient of fg maps to 0 in $R/(p)$, which implies:

$$\phi(fg) = 0 \in (R/(p))[x].$$

Thus, in the integral domain $(R/(p))[x]$, we have:

$$\phi(f) \cdot \phi(g) = 0 \implies \phi(f) = 0 \quad \text{or} \quad \phi(g) = 0.$$

If $\phi(f) = 0$, then every coefficient of f is divisible by p, meaning $p \mid C(f)$, which contradicts the assumption that f is primitive ($C(f) \sim 1$). Similarly, if $\phi(g) = 0$, then $p \mid C(g)$, contradicting the assumption that g is primitive. This contradiction proves that fg must be primitive. **Q.E.D.**

III A Divisibility Property of Primitive Polynomials

Spoken Transcript [00:15:16 - 00:16:14] Prof. Rahul Pandharipande

[wipes the right-hand boards clean with a sponge] Okay, and there's one more property of contents that we have to discuss before I go to the proof of Gauss's theorem, Theorem B. Oh, it's already gone.

So, the first property was Gauss; the second property that I need is the following claim, which is also a kind of pretty claim. [writes: Claim: g is primitive, $g \in R[x]$] Suppose that g is primitive, so g is in here, and g divides $b \times f$ [writes: and $g \mid bf$], where b is a scalar that's not equal to zero [writes: $b \in R, b \neq 0$], and f is some element here [writes: $f \in R[x]$]. So, these are the assumptions: it's primitive, and this assumption.

Then the conclusion is that g divides f. [writes: then $g \mid f$] Okay?

Claim 6.51 (*Divisibility Property of Primitive Polynomials*)**:** Let R be a UFD. Suppose $g \in R[x]$ is a primitive polynomial, and suppose that:

$$g \mid bf \quad \text{in } R[x], \tag{6.78}$$

for some non-zero scalar $b \in R \setminus \{0\}$ and some polynomial $f \in R[x]$. Then:

$$g \mid f \quad \text{in } R[x]. \tag{6.79}$$

Spoken Transcript [00:16:14 - 00:17:55] Prof. Rahul Pandharipande

The first time you see these things, you just have to follow, but then you can think about it and understand the arguments for this. So, let us try to prove this.

We know that g divides bf. That's the assumption. This is just language: g divides bf means that there exists some h such that $bf = gh$ — maybe hg, doesn't matter, but okay. [writes: Proof: $g \mid bf \implies \exists h \in R[x] : bf = hg$] That's what it means to say that g divides bf. All right.

And now, we take the content of both sides. Well, take the content of this side: it says this polynomial here is the content of bf times f^*, and I can take h and take the content of h times h^*g. [writes: $C(bf)f^* = C(h)h^*g$] Okay? So, this is just rewriting the left-hand side and the right-hand side in the contents.

Now, by the lemma of Gauss, we think about this object here: h^*g. h^* is primitive because that's what you do when you take h. h may not be primitive, but we took the content of h out, so h^* is primitive. g is primitive by assumption. So, then by Gauss upstairs (Recall: Lemma 6.50), $h^* \times g$ is primitive. [writes: h^*g : h^* primitive, g primitive $\implies$ h^*g is primitive by Gauss] Okay?

Spoken Transcript [00:17:55 - 00:19:24] Prof. Rahul Pandharipande

So, then we look at this equation: this is primitive, and this is primitive. And I said that whenever we have such an equation where we have the primitive part times the content, then the only possibility is that the primitives are equal up to a unit (Recall: Proposition 6.49). This is a content equation: this is a primitive times some content. That was one of the comments up there.

So, this means that f^*, up to units, is this. [writes: $f^* \overset{\text{unit}}{=} h^*g$] The consequence is that g divides f^*, because that's what this equation means. And this means g divides f, since $f = C(f)f^*$ (Recall: Proposition 6.45). [writes: so $g \mid f^* \implies g \mid f$ since $f = C(f)f^*$] If you divide f^*, you also divide f.

So, that's the proof of this too. Now with these preliminaries, we can finally prove Gauss's theorem.

Proof of Claim 6.51

By assumption, $g \mid bf$ in $R[x]$, which means there exists a polynomial $h \in R[x]$ such that:

$$bf = hg. \tag{6.80}$$

We express each side in terms of its canonical decomposition into content and primitive part:

$$\begin{aligned} \text{Left-hand side:} \quad & bf = C(bf)\cdot(bf)^* \sim b \cdot C(f) \cdot f^*, \\ \text{Right-hand side:} \quad & hg = (C(h)\cdot h^*)\cdot g = C(h)\cdot(h^*g). \end{aligned}$$

By Observation 6.48, h^* is primitive. Since g is primitive by assumption, Gauss's Lemma (Lemma 6.50) implies that the product h^*g is primitive:

$$C(h^*g) \sim 1.$$

Equating the two decompositions gives:

$$C(bf)\cdot f^* = C(h)\cdot(h^*g).$$

On both sides, we have a scalar in R multiplied by a primitive polynomial (f^* on the left, and h^*g on the right). By the uniqueness of the primitive decomposition (Proposition 6.49), the primitive parts must be associate in $R[x]$:

$$f^* \sim h^*g.$$

Hence, $g \mid f^*$ in $R[x]$. Since $f = C(f)\cdot f^*$, it follows immediately that:

$$g \mid f \quad \text{in } R[x].$$

Q.E.D.

Spoken Transcript [00:19:24 - 00:20:34] Prof. Rahul Pandharipande

It's a new idea, this content, and I've given you the definition and various properties of it. But it takes a little bit of time to understand how to use them and how to think about them, and you have to spend that time. But the idea of the content is in some sense a very simple idea: if you have this polynomial with coefficients in a UFD, you just take out the common divisor of all the coefficients, and you put it out.

And these properties I'm talking about are something about what happens to what's left. These primitive polynomials become a little bit special.

So, now we come to Theorem B, which is the second theorem for the week. And it also, as I said, has a little bit of an intricate logic. I did look at the book to try to follow more or less the notation. It's not going to be perfect, but I did try to at least match the notation. [finishes erasing the upper chalkboard]

Proof of Gauss's Theorem

Spoken Transcript [00:20:34 - 00:21:44] Prof. Rahul Pandharipande

Okay, so, we want to prove: R is a UFD implies $R[x]$ is a UFD. [writes: Thm. R UFD $\Longrightarrow$ $R[x]$ is a UFD] So, for $R[x]$, we must prove two things: we must prove the existence statement — existence of factorization into irreducibles in $R[x]$ [writes: for $R[x]$, we must prove: $\exists$ factorization into irreducibles in $R[x]$] —, and then we must prove the uniqueness statement: uniqueness [writes: ! uniqueness].

There's no shortcut to that: we have to prove existence and uniqueness to prove it's a UFD.

So, let us start this time with existence. Step 1 is the existence result. [writes: STEP I $\exists$]

Statement of Theorem B and Two-Part Strategy

The lecturer writes Theorem B on the board once more (Theorem 6.41):

$$R \text{ is a UFD } \Longrightarrow R[x] \text{ is a UFD.}$$

To establish that $R[x]$ is a UFD (Definition 6.13), we must prove two fundamental assertions for every non-zero, non-unit polynomial $f \in R[x]$:

(1) **Existence of Factorization:** Every non-zero, non-unit $f \in R[x]$ can be factored into a product of irreducible elements in $R[x]$.

(2) **Uniqueness of Factorization:** Such factorization into irreducibles is unique up to permutation and associate elements.

Proof of Gauss's Theorem (Theorem 6.41)

Step 1: Existence of Factorization

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For this there's a ground case and an induction.

The ground case is: suppose $f \in R[x]$... [writes: Ground case:] Here f is not zero, not a unit, and suppose the degree of f is zero. [writes: Suppose $f \in R[x], f \neq 0,$ unit and suppose $\deg(f) = 0$]

We're going to prove existence based on induction of the degree of the polynomial. All right? The polynomials have degree in any ring, so we're going to start with the case where the degree is zero. What does that mean, the degree is zero? If f is a degree-zero polynomial, then it's like a_0 plus nothing else. So, this means f is a scalar, actually. This means f is just a scalar and it lives in R. [writes: $f = a_0 \implies$ This means $f \in R$] A degree-zero polynomial is just a scalar.

So, you can imagine, in this ground case, we need to factor something in R. How are we going to do that? Well, we can't do that; we just assumed that from the beginning: R is a unique factorization domain. So, we can just factor it.

We can factor f, since it's a scalar, into some irreducible factors: [writes: We factor $f = q_1 \cdots q_n$, irreducible in R] irreducibles in R. And every irreducible in R — these are also irreducibles in $R[x]$, [writes: these are also irreducibles in $R[x]$] because irreducibility is a condition on what happens when you factor. And if you have a scalar, when you factor you don't get any x's, because the x's would never go.

All right? So, that's the ground case. In the ground case, we have degree zero, and then it's just a scalar. We don't really do anything; all we do is just take this assumption: it's a UFD, so we can factor the scalars. So, actually nothing happened in this ground case.

Existence: Ground Case ($\deg(f) = 0$)

Ground Case ($\deg(f) = 0$): Suppose $f \in R[x]$ is a non-zero, non-unit polynomial with $\deg(f) = 0$. Then $f = a_0 \in R$. Since f is not a unit in $R[x]$, a_0 is a non-zero, non-unit element of R.

Because R is a UFD by assumption, a_0 factors into a product of irreducible elements in R:

$$f = q_1 \cdot q_2 \cdots q_m, \quad \text{where each } q_i \in R \text{ is irreducible in } R.$$

Furthermore, every element $q \in R$ that is irreducible in R remains irreducible as a constant polynomial in $R[x]$.

Why Irreducibles in R Remain Irreducible in $R[x]$: Suppose an irreducible element $q \in R$ factors in $R[x]$ as $q = g(x)h(x)$ for some $g, h \in R[x]$. Since R is an integral domain, the degree of a product equals the sum of degrees (the product of the leading coefficients is non-zero):

$$0 = \deg(q) = \deg(g) + \deg(h) \implies \deg(g) = 0 \quad \text{and} \quad \deg(h) = 0.$$

Thus, $g, h \in R$. Because q is irreducible in R, either g or h must be a unit in R. Since the units of $R[x]$ are precisely the units of R (Theorem 3.39), it follows that either g or

h is a unit in $R[x]$. Hence, q remains irreducible as an element of $R[x]$.

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The induction case is slightly more interesting.

In the induction case, suppose we have this $f \in R[x]$, not zero, not equal to zero or a unit. [writes: Induction case: Suppose $f \in R[x] \neq 0, \text{unit}, \deg(f) > 0$] And now the degree of f is positive. We want to show that there exists a factorization into irreducibles in the polynomial ring. And now we actually have to do something. We didn't here, because that was just assumption.

Let us see.

As far as Theorem B is concerned, this is the first moment where an idea is needed. We need to prove existence. We're not necessarily a scalar, so we can't just use the assumption; we're some actual bona fide polynomial, so we have here f. And the idea here is we know one thing that I taught at the beginning of the class: we can write f as $C(f)$ times f^* (Recall: Proposition 6.45). [writes: $f = C(f) \cdot f^*$] That's the content, and this is primitive. [writes "primitive" above f^*] Okay?

So, we need to factor f. We need to show there exists a factorization of f into irreducibles for $R[x]$. We already have a kind of factorization here. And remember, this is a scalar [writes "R" under $C(f)$] and this is in $R[x]$ [writes "$R[x]$" under f^*]. And since the degree of f is positive, the degree of this is positive. So, I just write that explicitly: the degree of f^* is equal to the degree of f, which is positive [writes: $\deg(f^*) = \deg(f) > 0$], because this is just a scalar, okay?

Now, we have to think: what could happen here? So, what could happen? [writes: What could happen?]

Induction Step: Canonical Decomposition Setup

Induction Step ($\deg(f) > 0$): Suppose $f \in R[x]$ is a non-zero, non-unit polynomial with $\deg(f) > 0$. By Proposition 6.45, f can be decomposed into its content and primitive part:

$$f = C(f) \cdot f^*,$$

where $C(f) \in R \setminus \{0\}$ and $f^* \in R[x]$ is a primitive polynomial ($C(f^*) \sim 1$). Since $C(f)$ is a non-zero scalar, $\deg(C(f)) = 0$. By additivity of degrees in the integral domain $R[x]$:

$$\deg(f) = \deg(C(f)) + \deg(f^*) = 0 + \deg(f^*) \implies \deg(f^*) = \deg(f) > 0.$$

If $C(f) \in R$ is not a unit, the UFD property of R guarantees that $C(f)$ factors into a product of irreducible elements in R, each of which remains irreducible in $R[x]$. Therefore, to obtain a complete factorization of f into irreducibles in $R[x]$, it suffices to factor the primitive polynomial f^*.

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Well, one thing that could happen — the first case — is that f^* in $R[x]$ is irreducible. [writes: **(i)** $f^* \in R[x]$ is irred.] Okay?

If this happens, we're in really good shape. Because suppose this is irreducible, then we look at $f = C(f) \cdot f^*$. [writes: $f = C(f) \cdot f^*$] And in this branch, this is irreducible in $R[x]$. [writes "irred. in $R[x]$" under f^*] Now, this is a scalar, so we can apply the UFD property of R and factor that into irreducibles. [writes "UFD of R" under $C(f)$] Okay?

So, then we're done, because we have irreducibles and irreducibles, so we are done. We are done with this case. [writes: We are done with this case]

This is not really a surprise: at the very beginning, anything in R we can already factor into irreducibles, so we're interested in being able to factor that into irreducibles. If I just assume it's irreducible, then I'm done.

Case (i): Primitive Factor f^* is Irreducible

Case (i): f^* is irreducible in $R[x]$. If f^* is already irreducible in $R[x]$, then:

Unit Content: $C(f) \in \mathcal{U}_R$: If $C(f) \in \mathcal{U}_R$ is a unit, then $f = C(f)f^*$ is associate to f^* and is therefore itself irreducible in $R[x]$.

Non-Unit Content: $C(f) \notin \mathcal{U}_R$: If $C(f) \notin \mathcal{U}_R$, then $C(f)$ is a non-zero, non-unit scalar in the UFD R. Thus, $C(f)$ factors into a finite product of irreducible elements in R:

$$C(f) = q_1 \cdot q_2 \cdots q_k \quad (q_i \in R \text{ irreducible in } R).$$

Since each q_i remains irreducible in $R[x]$, the product:

$$f = q_1 \cdot q_2 \cdots q_k \cdot f^*$$

is a complete factorization of f into irreducible elements in $R[x]$.

In either sub-case, the existence of a factorization into irreducibles is established.

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The other case, of course, is that f^* in $R[x]$ is not irreducible. [writes: **(ii)** $f^* \in R[x]$ is not irred.] Okay, that's the case where I have to worry. And then again you can imagine what I'm going to do: if it's not irreducible, that means f^* factors into some A and B, [writes: $f^* = A \cdot B$] where A and B are in $R[x]$ [writes: $A, B \in R[x]$], and A is not a unit and B is not a unit. [writes: A is not a unit, B is not a unit] Because if it's not irreducible, then there's a legitimate factorization. Okay? That's the other world: either it's irreducible, or there's a legitimate factorization.

And we would like to now use induction on it. If A and B have degree less than f, we are done by induction. [writes: If A, B have degree $< \deg(f)$, we are done by induction] Because we're proceeding by induction on f, and if these are of lower degree, then we just use the induction statement to factor them. This is how you factor integers.

So, this is almost the whole proof, but now there's one case we have to worry about. Normally, if you factor a polynomial, these things will be of degree less, but there's one case they won't. The only remaining worry to consider, let's say it that way, [writes: Only remaining worry to consider is] is if one of these, say A, is a scalar — that means the degree of A is zero. [writes: $A \in R, \deg(A) = 0$] Because if the degree of A is zero, then the degree of B is the degree of f^*, and we can't use induction. Is that clear? That's actually the whole point of the argument, the only interesting point of the argument here. Up to now we've been just kind of shuffling the pawns around, but this is the actual point where something happens.

Case (ii): Reducible Primitive Factor and the Degree Obstacle

Case (ii): f^* is reducible in $R[x]$. If f^* is reducible in $R[x]$, then by definition there exist non-units $A, B \in R[x]$ such that:

$$f^* = A \cdot B. \tag{6.81}$$

Because R is an integral domain, the degree is additive on products of non-zero polynomials:

$$\deg(f^*) = \deg(A) + \deg(B).$$

- **The Desired Induction Step:** If both $\deg(A) < \deg(f^*) = \deg(f)$ and $\deg(B) < \deg(f^*) = \deg(f)$, then by the induction hypothesis, both A and B factor into products of irreducible elements in $R[x]$. Multiplying these factorizations together would immediately produce a factorization of f^*, and hence of $f = C(f)f^*$.
- **The Potential Degree Obstacle:** The induction hypothesis fails if one of the factors, say A, has degree zero:
  $$\deg(A) = 0 \implies A \in R.$$
  In that scenario, $\deg(B) = \deg(f^*) = \deg(f)$, meaning that the degree has not strictly decreased.

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So, this is the case to worry about. How can we exclude that? You tell me: how do we exclude this case?

Student Answer

Because f^* is primitive.

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Yes! f^* is primitive. There is no common factor for all of its coefficients. If this were not a unit, this would be a common factor for all the coefficients.

So, the only way this happens... Since f^* is primitive, this would imply A is a unit. [writes: Since f^* is primitive, A is a unit] But that's a contradiction, because A is not a unit; it's born not a unit. [writes: Contradiction]

The logic is: if this is reducible, I get A and B, neither of which are units. And they both have degree strictly less than f, unless one of them is a scalar, and that's excluded because this is primitive. This is why that definition was made in the first place for that step in the argument.

Okay, so, this finishes the existence!

Now there's uniqueness. Uniqueness is fun, maybe more fun.

Resolution via Primitivity and Conclusion of Step 1

Claim 6.52 (*Exclusion of Degree-Zero Factors*): In the factorization $f^* = AB$ of the primitive polynomial f^*, neither factor can have degree zero.

Proof of Claim 6.52

Suppose for contradiction that $\deg(A) = 0$, so $A = a \in R$. Since A is not a unit in $R[x]$ by assumption, a is not a unit in R. Because a divides $f^* = aB$ in $R[x]$, a must divide every coefficient of f^*. Consequently, a divides the content of f^*:

$$a \mid C(f^*) \quad \text{in } R.$$

However, f^* is primitive (Definition 6.47), which means $C(f^*) \sim 1$. Thus, $a \mid 1$ in R, implying that $a \in \mathcal{U}_R$ is a unit in R. This contradicts the fact that A is not a unit in $R[x]$. **Q.E.D.**

Conclusion of the Induction Step for Existence: Since degree-zero non-unit factors cannot occur, we have:

$$\deg(A) \geq 1 \quad \text{and} \quad \deg(B) \geq 1.$$

Because $\deg(A) + \deg(B) = \deg(f^*) = \deg(f)$, it follows that:

$$\deg(A) < \deg(f) \quad \text{and} \quad \deg(B) < \deg(f).$$

By the induction hypothesis on the degree, both A and B factor into products of irreducible elements in $R[x]$. Therefore, $f^* = AB$ factors into irreducibles, and combining this with the irreducible factorization of the scalar content $C(f) \in R$ yields a complete factorization of $f = C(f)f^*$ into irreducible elements in $R[x]$. This completes the proof of Step 1 (Existence of Factorization).

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So, you see how cunning this definition was. As I said, if you're working on this problem, the first thing you would not do is write down the definition. It comes only after you understand the argument. Then you can go back to the beginning and write down the definition for content. [wipes the lower-left chalkboard clean with a sponge]

Step 2: Uniqueness of Factorization

Spoken Transcript [00:31:28 - 00:33:17] Prof. Rahul Pandharipande

Okay, so, now we have to prove the uniqueness, and the path to proving uniqueness was already explained in the previous lecture.

Step 2: prove uniqueness. [writes: Step. Prove (!)] We already had shown a method to prove uniqueness: we already had shown that if p irreducible implies p prime, then uniqueness holds. [writes "We already have shown: If p irr $\implies$ $(p) \subset R[x]$ prime" and, on the next line, "the uniqueness holds"] Because that was one of our steps yesterday (Recall: Claim 6.25): if I have any ring where irreducible implies prime, then I get uniqueness.

So, I have to prove this now. That's the last thing I have to do.

You should look at the lecture from yesterday: we must prove that if $p \in R[x]$ is irreducible, [writes: We must prove: $p \in R[x]$ irred] then it's prime. The way you do that is you have to say that we have the assumption: p is irreducible, $fg \in (p)$, [writes: $fg \in (p)$] $f \notin (p)$, [writes: $f \notin (p)$] and we must prove under these hypotheses that $g \in (p)$. [writes: $\implies g \in (p)$] That's the last thing.

If we prove this — you assume it's irreducible, these things lie in (p), f is not in (p), and then g is in (p) —, then that finishes the proof of Gauss's theorem.

Strategy for Uniqueness: Irreducible Implies Prime in $R[x]$

Step 2 (Uniqueness): By Step 1 of the proof of Theorem A (Claim 6.25), in any integral domain where every irreducible element is a prime element, factorization into irreducibles is unique up to permutation and associate elements.

Therefore, to prove that $R[x]$ is a UFD, it suffices to prove that every irreducible element in $R[x]$ is prime:

$$p \in R[x] \text{ is irreducible} \implies p \text{ is a prime element in } R[x]. \tag{6.82}$$

In terms of principal ideals, this means that for any $f, g \in R[x]$:

$$fg \in (p) \quad \text{and} \quad f \notin (p) \implies g \in (p), \tag{6.83}$$

which is equivalent to:

$$p \mid fg \quad \text{and} \quad p \nmid f \implies p \mid g \quad \text{in } R[x].$$

We divide the verification of (6.83) into two distinct cases based on the degree of p:

Case 1: $\deg(p) = 0$: p is a constant scalar in R.

Case 2: $\deg(p) > 0$: p is a polynomial of positive degree.

Case 1: Degree Zero Irreducibles ($\deg(p) = 0$)

Spoken Transcript [00:33:17 - 00:34:58] Prof. Rahul Pandharipande

Okay, so, this is what we have to prove. And to start, there's a case that helps separately. Case: [writes: Case] just as before, $\deg(p) = 0$. [writes: $\deg(p) = 0$]

This case means that p is a scalar: $p \in R$. [writes: $p \in R$] All right.

What do we have here? We need to prove this implication. fg is equal to the content of f times f^* times the content of g times g^*. [writes: $fg = C(f)f^* \cdot C(g)g^*$] I just used the content equation for f, the content equation for g, and then I put them in order here: content of f, content of g, f^*, g^*. [writes: $= C(f)C(g)f^*g^*$] All right?

And the second thing says that p divides this: p divides the whole thing [writes: $p \mid C(f)C(g)f^*g^*$] — content of f, content of g, times these polynomials f^*, g^*.

Now, f^* is primitive because that's what happens when you extract the content. [writes "primitive" under f^*] This one is also primitive [writes "primitive" under g^*] — that's what happens when you extract the content. So, the whole thing is primitive. [writes "primitive" under f^*g^*] That's by the Gauss claim (Recall: Lemma 6.50): Gauss says that the whole thing is primitive. [writes: Gauss $\implies$ f^*g^* is primitive]

Case 1: Decomposition of the Product

Case 1 ($\deg(p) = 0$): Suppose $p \in R[x]$ is irreducible with $\deg(p) = 0$. Then $p = c \in R$. Since p is irreducible in $R[x]$, it is irreducible in R (a factorization in R is one in $R[x]$, and the units agree). Because R is a UFD, p is a prime element in R (Exercise 6.42).

Suppose $p \mid fg$ in $R[x]$ and $p \nmid f$. Decomposing f and g into their content and primitive parts:

$$f = C(f) \cdot f^* \quad \text{and} \quad g = C(g) \cdot g^*,$$

where $f^*, g^* \in R[x]$ are primitive, we express the product fg as:

$$fg = \big(C(f)f^*\big) \cdot \big(C(g)g^*\big) = \big(C(f)C(g)\big) \cdot (f^*g^*).$$

By Gauss's Lemma (Lemma 6.50), since f^* and g^* are primitive, their product:

$$f^*g^* \in R[x] \quad \text{is also primitive.}$$

Since $p \mid fg$, the scalar p divides the polynomial $\big(C(f)C(g)\big) \cdot (f^*g^*)$ in $R[x]$.

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But that's interesting. p divides... This is a scalar, that's a scalar, and it divides every coefficient of this polynomial, which is primitive.

There must be one coefficient of this polynomial which p does not divide. p cannot divide every coefficient of this polynomial, otherwise this polynomial would not be primitive.

So, there's some coefficient, let's say a_{17}. That's the coefficient [writes: a_{17} is the coeff.] of f^*g^* [writes: of f^*g^*] which p does not divide. [writes: which p does not divide] I could put a variable there too, but 17 in this class is kind of a variable anyway. p does not divide.

This means that p does not divide a_{17} [writes: $p \nmid a_{17}$], but p divides all of these things together: it divides $C(f)C(g)a_{17}$. [writes: $p \mid C(f)C(g)a_{17}$] Because it must divide all of this, but it doesn't divide a_{17}.

These are all scalars now, and we have unique factorization. So, p doesn't divide that.

p doesn't divide the content of f either, because if p divided the content of f, then f would be in this ideal [writes: $f \notin (p) \implies p \nmid C(f)$], because $f = C(f) \cdot f^*$. If p divided the content of f, f would be in that ideal, but it's not.

So, it doesn't divide that, doesn't divide that. By unique factorization, p must divide the content of g [writes: $p \mid C(g)$], and this implies p divides the polynomial g [writes: $\implies p \mid g$], which says that g is in the ideal. [writes: $\implies g \in (p)$]

So, that handles the degree-zero case, and I have five minutes, and I will do the last case, hopefully.

Case 1: Conclusion that $g \in (p)$

Completion of Case 1 ($\deg(p) = 0$)

Write the primitive polynomial $f^* g^*$ as:

$$f^* g^* = \sum_{j=0}^{N} d_j x^j \in R[x].$$

Since $f^* g^*$ is primitive, $\mathrm{GCD}(d_0, \ldots, d_N) \sim 1$. Therefore, the prime element $p \in R$ cannot divide all coefficients d_j. Thus, there exists an index k (the lecturer calls it 17, writing a_{17} for d_k) such that:

$$p \nmid d_k \quad \text{in } R.$$

Because $p \mid fg$ with $p \in R$ a constant, p must divide every coefficient of $fg = C(f)C(g)(f^* g^*)$ in R. In particular, the k-th coefficient gives:

$$p \mid C(f)C(g)d_k \quad \text{in } R. \tag{6.84}$$

Since p is a prime element in the UFD R, (6.84) implies:

$$p \mid C(f) \quad \text{or} \quad p \mid C(g) \quad \text{or} \quad p \mid d_k.$$

We already know $p \nmid d_k$. Moreover, if $p \mid C(f)$, then $C(f) = p \cdot r$ for some $r \in R$, which implies:

$$f = C(f) f^* = p \cdot (r f^*) \implies p \mid f \quad \text{in } R[x] \implies f \in (p),$$

directly contradicting the hypothesis that $f \notin (p)$. Consequently, we must have:

$$p \mid C(g) \quad \text{in } R.$$

Writing $C(g) = p \cdot s$ for some $s \in R$, we conclude:

$$g = C(g) g^* = p \cdot (s g^*) \implies p \mid g \quad \text{in } R[x] \implies g \in (p).$$

This proves that every irreducible constant element $p \in R$ generates a prime ideal in $R[x]$. **Q.E.D.**

Case 2: Positive Degree Irreducibles ($\deg(p) > 0$)

Spoken Transcript [00:36:57 - 00:39:16]

Prof. Rahul Pandharipande

The last case is the most interesting. Let's try to do this.

The last case is that we have those hypotheses, the same on the board, but Case: [writes: Case] now the degree of p is positive. [writes: $\deg(p) > 0$]

We make an ideal: I is the ideal generated by p and f. [writes: $I = (p, f) \subset R[x]$] This is an ideal in $R[x]$.

It's just some ideal. But we have the degree function here, and this is the first idea in the proof: choose an element here which is not zero — we're never interested in zero here — such that m has minimal degree. [writes: choose $m \in I, m \neq 0$ such that m has minimal degree]

We can do that, because you can always pick things of minimal degree; you can always pick the smallest number which occurs. It's important that m is not zero.

A crucial thing is we're going to show now: we will show the degree of m is zero.

[writes: We will show $\deg(m) = 0$] So, it's a scalar. And that has a very clever argument.

What we would like to do is use the division algorithm. But we cannot use the division algorithm. We would very much like to use the division algorithm for polynomial rings. [writes: We would like to use the div. alg.] We can't do it, because the division algorithm requires a field (Recall: Theorem 5.57). So, we give ourselves a field. What field? The field of fractions!

Let Q be the field of fractions. [writes: Let $Q = \text{Fraction }(R)$] R is a domain; we've seen that to every domain you associate a field of fractions (Recall: Definition 3.17). We let Q be this field of fractions: $Q = \text{Frac}(R)$.

So, then in $Q[x]$ we have the division algorithm. [writes: So in $Q[x]$ we have div. alg.]

Case 2: The Ideal (p, f) and the Field of Fractions Q

Case 2 ($\deg(p) > 0$): Suppose $p \in R[x]$ is irreducible with $\deg(p) > 0$. We assume $fg \in (p)$ and $f \notin (p)$, and we must show $g \in (p)$.
Consider the ideal in $R[x]$ generated by p and f:

$$I := (p, f) = \{Ap + Bf \mid A, B \in R[x]\} \subseteq R[x]. \tag{6.85}$$

Because $p \in I$ and $p \neq 0$, the set of degrees of non-zero polynomials in I:

$$\{\deg(h) \mid h \in I \setminus \{0\}\} \subset \mathbb{N}_0$$

is non-empty. By the Well-Ordering Principle of $\mathbb{N}_0$, there exists an element of minimal degree:

$$m \in I \setminus \{0\} \quad \text{such that } \deg(m) \leq \deg(h) \quad \text{for all } h \in I \setminus \{0\}. \tag{6.86}$$

Core Objective: We claim that $\deg(m) = 0$, meaning $m \in R \setminus \{0\}$ is a scalar.

Why We Introduce the Field of Fractions Q: In $k[x]$, where k is a field, division with remainder (Theorem 5.57) shows that an element of minimal degree in an ideal generates the entire ideal (Proposition 5.58). However, $R[x]$ is not a PID when R is not a field. To exploit division with remainder, we embed R into its field of fractions (Definition 3.17):

$$Q := \text{Frac}(R).$$

Then $R[x] \subset Q[x]$, and $Q[x]$ has division with remainder.

Spoken Transcript [00:39:16 - 00:41:26]

Prof. Rahul Pandharipande

What does that mean? We can write $f = \hat{q}m + \hat{r}$, [writes: $f = \hat{q}m + \hat{r}$] where these $\hat{q}, \hat{r}$ are in the field of fractions. [writes: $\hat{q}, \hat{r} \in Q[x]$] And the division algorithm has the usual properties that $\hat{r}$ is either zero or the degree of $\hat{r}$ is strictly less than the degree of m. [writes: $\hat{r} = 0$ or $\deg(\hat{r}) < \deg(m)$]

And we can also do it with p: $p = \hat{\hat{q}}m + \hat{\hat{r}}$. [writes: $p = \hat{\hat{q}}m + \hat{\hat{r}}$] Same thing: $\hat{\hat{q}}, \hat{\hat{r}} \in Q[x]$. [writes: $\hat{\hat{q}}, \hat{\hat{r}} \in Q[x]$] Maybe I just call it Q, I don't need to put that here. And again, now with $\hat{\hat{r}} = 0$ or the degree of $\hat{\hat{r}}$ is less than the degree of m. [writes: $\hat{\hat{r}} = 0$ or $\deg(\hat{\hat{r}}) < \deg(m)$]

So, that's one of the ideas: we use the division algorithm in the field of fractions.

But we want to be in R! We want to be in the ring R, and these things can have denominators. And it's not a problem: we just multiply by the denominators to clear the denominators. By clearing the denominators, we will multiply by some $b \in R$. [writes: $b \in R$]

So, if we just multiply by the denominators, I get an equation: $b'f = q'm + r'$, [writes: $b'f = q'm + r'$] and similarly $b''p = q''m + r''$. [writes: $b''p = q''m + r''$]

And now, if you look at this, everything is in R: all the terms here are in the ring R. [writes: Everything in R]

Division with Remainder in $Q[x]$ and Clearing Denominators

Applying division with remainder in $Q[x]$ (Theorem 5.57, with $k = Q$) to divide f and p by $m \neq 0$, there exist unique quotients and remainders $\hat{q}, \hat{r}, \hat{\hat{q}}, \hat{\hat{r}} \in Q[x]$ such that:

$$f = \hat{q}m + \hat{r}, \qquad \text{with } \hat{r} = 0 \text{ or } \deg(\hat{r}) < \deg(m), \tag{6.87}$$

$$p = \hat{\hat{q}}m + \hat{\hat{r}}, \qquad \text{with } \hat{\hat{r}} = 0 \text{ or } \deg(\hat{\hat{r}}) < \deg(m). \tag{6.88}$$

Since the coefficients of $\hat{q}, \hat{r}$ and $\hat{\hat{q}}, \hat{\hat{r}}$ are fractions in $Q = \operatorname{Frac}(R)$, we clear the denominators. By multiplying (6.87) by a common denominator $b' \in R \setminus \{0\}$ of all coefficients of $\hat{q}$ and $\hat{r}$ (for instance the product of their denominators) and multiplying (6.88) by $b'' \in R \setminus \{0\}$, we obtain equations with coefficients entirely in R:

$$b'f = q'm + r', \qquad \text{with } b' \in R \setminus \{0\} \text{ and } q', r' \in R[x], \tag{6.89}$$

$$b''p = q''m + r'', \qquad \text{with } b'' \in R \setminus \{0\} \text{ and } q'', r'' \in R[x], \tag{6.90}$$

where $q' := b'\hat{q}$, $r' := b'\hat{r}$, $q'' := b''\hat{\hat{q}}$, and $r'' := b''\hat{\hat{r}}$ all lie in $R[x]$ by the choice of b' and b''. Multiplying by non-zero scalars in R leaves polynomial degrees unchanged:

$$r' = 0 \quad \text{or} \quad \deg(r') < \deg(m), \qquad \text{and} \qquad r'' = 0 \quad \text{or} \quad \deg(r'') < \deg(m).$$

Spoken Transcript [00:41:26 - 00:42:45]

Prof. Rahul Pandharipande

When I clear the denominators, I don't change the degree here.

Now, this is in I, that is in I because m is in I, so r' is in I. But m is the least-degree element in I — you can't be lower than that.

So, these cases are not possible, and they must be zero. So, actually these terms go away. [erases $+r'$ and $+r''$]

So, we have $b'f = q'm$ [writes: $b'f = q'm$] and $b''p = q''m$. [writes: $b''p = q''m$]

And we can write this on the right-hand side as a content: this equals $q'C(m)m^*$ [writes: $= q'C(m)m^*$], and this is $q''C(m)m^*$. [writes: $= q''C(m)m^*$]

So, that means m^*, which is primitive, divides this product.

Ah, I failed! [the bell rings, marking the end of the lecture hour]

I'll finish it next time. We're basically almost... I'll do it next time, there's no reason to go over. I need another three minutes.

This lecture was maybe a little bit faster than usual.

But... [the lecture ends; students applaud and pack up]

Vanishing of Remainders and Divisibility by Primitive m^*

Claim 6.53 (*Vanishing of Remainders*): The cleared remainders in (6.89) and (6.90) must vanish identically:

$$r' = 0 \quad \text{and} \quad r'' = 0.$$

Proof of Claim 6.53

Rearranging (6.89) and (6.90) gives:

$$r' = b'f - q'm,$$
$$r'' = b''p - q''m.$$

Recall that $I = (p, f)$. Since $f \in I$ and $m \in I$, any $R[x]$-linear combination of f and m belongs to I, so $r' \in I$. Similarly, since $p \in I$ and $m \in I$, $r'' \in I$.

By definition (6.86), m was chosen to be a non-zero element of minimal degree in I. If $r' \neq 0$, then $r' \in I \setminus \{0\}$ with $\deg(r') < \deg(m)$, which contradicts the minimality of $\deg(m)$. Therefore, $r' = 0$. The identical argument shows that $r'' = 0$. **Q.E.D.**

Divisibility of $b'f$ and $b''p$ by the Primitive Factor m^*: Substituting $r' = 0$ and $r'' = 0$ into (6.89) and (6.90), we obtain exact factorization identities in $R[x]$:

$$b'f = q'm, \tag{6.91}$$
$$b''p = q''m. \tag{6.92}$$

Expressing m via its canonical content decomposition $m = C(m) \cdot m^*$, where $C(m) \in R$ and $m^* \in R[x]$ is primitive, equations (6.91) and (6.92) become:

$$b'f = q' \cdot C(m) \cdot m^*, \tag{6.93}$$
$$b''p = q'' \cdot C(m) \cdot m^*. \tag{6.94}$$

Thus, the primitive polynomial $m^* \in R[x]$ satisfies:

$$m^* \mid b'f \quad \text{and} \quad m^* \mid b''p \quad \text{in } R[x],$$

for non-zero scalars $b', b'' \in R \setminus \{0\}$. By the divisibility property of primitive polynomials (Claim 6.51), this immediately implies:

$$m^* \mid f \quad \text{and} \quad m^* \mid p \quad \text{in } R[x].$$

Here the lecture stops, at the bell. The proof that $\deg(m) = 0$, and from it $g \in (p)$, is continued in the next lecture (proof of Theorem 7.1); the remaining argument is given below.

Completion of the Argument for $\deg(p) > 0$: The bell ended the lecture a few minutes before the end of the proof (*"I'll do it next time"*). The remaining argument, which the next lecture gives in full (Theorem 7.1), runs as follows; it uses only the results of this lecture.

At the end of the lecture, $m^* \mid f$ and $m^* \mid p$ in $R[x]$, where m^* is the primitive part of the element $m \in I = (p, f)$ of minimal degree.

Step 1: m^* is a Unit: Write $p = m^*t$ with $t \in R[x]$. Since p is irreducible, m^* or t is a unit. If t were a unit, then $p = m^*t$ would divide f (as $m^* \mid f$), that is, $f \in (p)$, contradicting the hypothesis $f \notin (p)$. Hence m^* is a unit of $R[x]$, so $m^* \in \mathcal{U}_R$ has degree 0 (Theorem 3.39).

Step 2: m is a Non-Zero Constant: Since $m = C(m) \cdot m^*$ with both factors in R, $m =: c \in R \setminus \{0\}$; this is the claim $\deg(m) = 0$ written on the board.

Step 3: p is Primitive: $p = C(p) \cdot p^*$ with $\deg(p^*) = \deg(p) > 0$, so p^* is not a unit, and the irreducibility of p forces $C(p)$ to be a unit.

Step 4: Deducing $g \in (p)$: Because $c = m \in I = (p, f)$, there exist polynomials $A, B \in R[x]$ such that:

$$c = Ap + Bf.$$

Multiplying both sides by g yields:

$$cg = Apg + B(fg).$$

Since $p \mid pg$ and by hypothesis $p \mid fg$, it follows that $p \mid cg$ in $R[x]$. Applying the divisibility property of primitive polynomials (Claim 6.51) to the primitive polynomial p and $p \mid cg$ with $c \in R \setminus \{0\}$, we conclude:

$$p \mid g \quad \text{in } R[x] \implies g \in (p).$$

Together with Case 1, every irreducible element $p \in R[x]$ is prime (and $(p) \neq R[x]$, since p is not a unit). By Step 1 of Theorem A (Claim 6.25), factorizations in $R[x]$ are therefore unique; with the existence proved in Step 1 above, this concludes the proof of Gauss's Theorem (Theorem 6.41).

Q.E.D.

Summary of Key Concepts

- **Content of a Polynomial:** For a UFD R and non-zero polynomial $f(x) = \sum_{j=0}^{n} a_j x^j \in R[x] \setminus \{0\}$, the content is $C(f) := \mathrm{GCD}(a_0, \ldots, a_n)$, determined up to units in R.
- **Primitive Polynomials:** A polynomial $g \in R[x]$ is primitive if $C(g) \sim 1$. Every non-zero polynomial factors uniquely up to units as $f = C(f) \cdot f^*$, where f^* is primitive.
- **Gauss's Lemma:** If $f, g \in R[x]$ are primitive, then their product fg is primitive ($C(fg) \sim 1$). The proof utilizes reduction modulo a prime ideal $(p) \subset R$, exploiting the fact that $(R/(p))[x]$ is an integral domain.
- **Divisibility by Primitive Polynomials:** If $g \in R[x]$ is primitive and $g \mid bf$ for some non-zero scalar $b \in R$, then $g \mid f$ in $R[x]$.
- **Gauss's Theorem on Polynomial Rings:** If R is a UFD, then $R[x]$ is a UFD:

  Step 1: Existence of Factorization: Proven by induction on polynomial degree. Factoring $f = C(f)f^*$ reduces the problem to primitive polynomials; reducibility $f^* = AB$ cannot yield degree-zero non-unit factors because $C(f^*) \sim 1$, ensuring strictly decreasing degrees $\deg(A), \deg(B) < \deg(f)$.

  Step 2: Uniqueness of Factorization: Proven by showing that every irreducible element is prime. Constant irreducibles ($\deg(p) = 0$) are handled by Gauss's

Lemma and prime divisibility in R; positive-degree irreducibles ($\deg(p) > 0$) are analyzed using division with remainder in $Q[x] = (\mathrm{Frac}(R))[x]$ and clearing denominators. The bell ended the lecture just before the last step ($\deg(m) = 0$, hence $g \in (p)$), which the next lecture completes (Theorem 7.1); the aiNote at the end of the proof gives it.

Tuesday, 27th October 2026

Gauss's Theorem, Group Axioms, the Symmetric Group, and Element Orders

Course Content Overview

This lecture concludes the foundational development of ring theory and initiates the study of abstract group theory:

1. **Gauss's Theorem on Polynomial Rings:** Reviewing examples of UFDs and PIDs (multivariate polynomials, Gaussian integers $\mathbb{Z}[i]$, and formal power series $k[[x]]$), and completing the proof that R UFD $\implies$ $R[x]$ UFD by passing to the fraction field $Q(R)$ and clearing denominators.
2. **Pedagogical Reflections on Algebra:** Framing the study of mathematics around the triad of definitions, theorems, and examples, and discussing philosophical perspectives on independent learning (Schopenhauer, Confucius).
3. **Foundations of Group Theory:** Defining groups $(G, \cdot, e)$, establishing uniqueness of identity and inverses, deriving the left and right cancellation laws, and contrasting finite, non-commutative groups with classical rings.
4. **The Symmetric Group S_n and Cycle Notation:** Permutations, associativity and brackets, Catalan numbers C_n, right-to-left composition, non-commutativity, the multiplication table, and cycle decomposition notation.
5. **Element Orders and Cyclic Subgroups:** Applying the Pigeonhole Principle to establish that every element in a finite group has finite order, and introducing $\langle x \rangle \cong \mathbb{Z}/\operatorname{Ord}(x)\mathbb{Z}$.

Gauss's Theorem and Classical Examples of UFDs

Statement of Gauss's Theorem

Spoken Transcript [00:00:00 - 00:00:39] Prof. Rahul Pandharipande

Okay, so, at the last lecture, I was almost at the end of this theorem (Recall: Theorem 6.41), which I will state again. So, this is Carl Friedrich... and the theorem is: R is a UFD implies the polynomial ring is also a UFD [writes: C.F.G. R is a UFD $\implies$ $R[x]$ is a UFD].

And there were many steps in the logic here, and maybe I'll take a chance to review some of the things and also explain some of the ideas. So, this is an intricate proof, and as I said, at the end of the last lecture I came to within five minutes of completing it. But anyway, I'll take some time now.

Recap: Theorem B was stated in Week 6 (Theorem 6.16) and restated in the previous lecture (Theorem 6.41), where the bell cut its proof short. The lecturer restates it and completes the proof below.

Theorem 7.1 (*Gauss's Theorem on Polynomial Rings*): Let R be a unique factorization domain (UFD). Then the polynomial ring $R[x]$ is also a unique factorization domain:

$$R \text{ is a UFD} \implies R[x] \text{ is a UFD.} \tag{7.1}$$

Historical Context and Scope of the Theorem: Going beyond the lecture: Gauss proved a form of the key lemma on primitive polynomials (Lemma 6.50) for integer polynomials in his **Disquisitiones Arithmeticae** (1801, Art. 42); the statement for an arbitrary UFD R is the modern form of his argument. By induction on the number of variables n, the theorem extends to multivariate polynomial rings (as in Example 6.20):

$$R[x_1, \dots, x_n] \cong (R[x_1, \dots, x_{n-1}])[x_n].$$

Examples: Multivariable Polynomials, Gaussian Integers, and Power Series

Spoken Transcript [00:00:39 - 00:02:17] Prof. Rahul Pandharipande

So, the first thing to notice — so this is not about the proof, this was what was labeled before Theorem B last week [writes: Thm B] (Recall: Theorem 6.16) — the first thing to note, which is not about the proof, it's just about examples [writes: Examples:], is that if you know this result, then you can take any UFD you want, like $\mathbb{Z}$ is a UFD. So, this result implies that if I take the polynomial ring in one variable or two variables or any number of variables [writes: $\mathbb{Z} \to \mathbb{Z}[x], \mathbb{Z}[x,y], \dots$] — which you can just keep applying this, one after the other — that these are all UFDs.

And also, these are kind of the standard class of examples: if I take the finite field $\mathbb{Z}/p\mathbb{Z}$ [writes: $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$], every field is a UFD. So, that means if I take polynomials in finite fields [writes: $\implies \mathbb{Z}/p\mathbb{Z}[x,y,z,\dots]$], these, however many variables you want — as long as there are finitely many — these obey all the factorization rules that you are familiar with for the integers: the idea that if you have a polynomial, there is unique factorization into prime polynomials, and whatever you remember about the integers, not every single statement, but the parts about unique factorization, they apply here.

And the other classical examples: you take the fields $\mathbb{Q}$, $\mathbb{C}$, $\mathbb{R}$ [writes: $\mathbb{Q}, \mathbb{C}, \mathbb{R} \implies \mathbb{Q}[x,y,z]$]. These are like the classical examples. And it's one of the aspects of this course that if you see these rings, there's a huge study of these rings, and this course is only the very beginning of the study. But at least you'll know that you can factor things, and the rules of factorization work like in integers.

Example 7.2 (*Polynomial Rings as Unique Factorization Domains*): Applying Gauss's Theorem (Theorem 7.1) inductively yields that multivariate polynomial rings over familiar base rings are UFDs:

(1) Integers: Since $\mathbb{Z}$ is a UFD, repeated application implies:

$$\mathbb{Z} \text{ is a UFD} \implies \mathbb{Z}[x],\ \mathbb{Z}[x,y],\ \mathbb{Z}[x_1,\dots,x_n]\dots \quad \text{are all UFDs.}$$

(2) Finite Fields: For any prime p, the field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ is a field (hence trivially a UFD), so:

$$\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z} \text{ is a UFD} \implies \mathbb{F}_p[x],\ \mathbb{F}_p[x_1,\dots,x_n]\dots \quad \text{are all UFDs.}$$

(3) Classical Fields: For the fields $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$:

$$\mathbb{Q}, \mathbb{C}, \mathbb{R} \text{ are UFDs} \implies \mathbb{Q}[x],\ \mathbb{C}[x],\ \mathbb{R}[x],\dots \quad \text{are all UFDs.}$$

Contrast: UFDs versus PIDs in Multivariate Polynomial Rings: Going beyond the lecture: while the polynomial ring $k[x]$ in one variable over a field k is a principal ideal domain (PID) by division with remainder (Proposition 5.58), the rings $\mathbb{Z}[x]$ (coefficients not in a field) and $k[x,y]$ (two variables) are **not** PIDs. For example, the ideals

$$I = (2,x) \subset \mathbb{Z}[x] \quad \text{and} \quad J = (x,y) \subset k[x,y]$$

cannot be generated by any single polynomial (Exercise 5.62, Examples 5.61). Nevertheless, by Gauss's Theorem, both rings are unique factorization domains.

Spoken Transcript [00:02:17 - 00:03:57] Prof. Rahul Pandharipande

Okay, so, you could say: what about other rings? So, as it was kind of an example in this class, this ring of Gaussian integers [writes: $\mathbb{Z}[i]$] (Recall: Definition 1.1). What about this ring? That's not exactly in this context, and it's not... So, it turns out this ring of Gaussian integers is a PID [writes: $\mathbb{Z}[i]$ is a PID $\implies$ $\mathbb{Z}[i]$ is a UFD]. We haven't proven it. You can go look in the book. It was not in the topics in the course, but this is a PID, so it implies that this ring, the Gaussian integers, is also a UFD.

The argument that it's a PID is not hard, it's just a page in the book. There's a notion of Euclidean domains, so you have to look in the book for... there's a notion of a division algorithm here [writes: division alg.]. Then that shows it's a PID, and so, if you're interested, you can read it; you can find it's a page in the book basically. But that's also a UFD.

What about rings like, if I take a field, let's say k, k is a field [writes: k field]. What about the rings... so these are polynomial rings, what about power series [writes: $k[[x]]$?]? That's not covered here, and it's actually an interesting, complicated topic, but in this particular case it's not complicated. And the answer, and I hinted at this (Recall: Exercise 6.40), is that actually this power series in a field is a PID, so, it's also a UFD [writes: Ans: $k[[x]]$ is a PID $\implies$ $k[[x]]$ is a UFD].

And I said that I was not going to give a hint about the proof, but I will now basically give the proof, more or less.

Example 7.3 (*Gaussian Integers as a Euclidean Domain and UFD*): The ring of Gaussian integers $\mathbb{Z}[i] := \{a + bi \mid a, b \in \mathbb{Z}\}$ admits a division algorithm, so it is a Euclidean domain. Since every Euclidean domain is a PID, and every PID is a UFD:

$$\mathbb{Z}[i] \text{ is a Euclidean Domain } \implies \mathbb{Z}[i] \text{ is a PID } \implies \mathbb{Z}[i] \text{ is a UFD.}$$

Example 7.4 (*Formal Power Series Ring over a Field*): For any field k, the ring of formal power series $k[[x]]$ is also a principal ideal domain:

$$k \text{ is a field } \implies k[[x]] \text{ is a PID } \implies k[[x]] \text{ is a UFD.}$$

This was posed as Exercise 6.40; the lecturer now sketches the proof.

Spoken Transcript [00:03:57 - 00:05:31] Prof. Rahul Pandharipande

So, proof: let I... [writes: Proof: Let $I \subset k[[x]]$ ideal] So, I have to prove it's a PID, because we proved earlier that PIDs are UFDs (Recall: Theorem 6.15). So, let I be any ideal...

Then either I could be zero [writes: $I = (0)$], that's one possibility, or I is not the zero ideal [writes: $I \neq (0)$]. So, here, well, of course, there's nothing to prove here if it's the zero ideal. So, if it's not the zero ideal, choose a power series in here, so, choose f in here that has minimal lowest degree [writes: choose $f \in k[[x]]$ that has min. lowest degree].

Power series don't have highest degrees, you don't think about it, so you think about lowest degree. So, it's the minimal lowest degree. That means f starts somewhere, like $2x^{17} + x^{18} + 3x^{19} + \dots$ [writes: $f = 2x^{17} + x^{18} + 3x^{19} + \dots$]. So, f starts somewhere. And so here, or it could be, you know, there's some element in the field, anyway, something non-zero in the field, it starts somewhere... and you find the element here in I, so f is in I, of lowest degree. Okay? So, do you accept you can do that? Once it's non-zero, it has some elements, and then you can find the one of lowest degree. All right.

Proof Sketch that $k[[x]]$ is a PID

Let $I \subseteq k[[x]]$ be an ideal.

Case 1: $I = (0)$: If $I = (0)$, it is generated by the zero element, so I is principal.

Case 2: $I \neq (0)$: Since I contains non-zero elements, we can choose an element $f \in I \setminus \{0\}$ of minimal lowest degree m: the lowest degree of a non-zero power series $\sum_{n=0}^{\infty} a_n x^n$ is the smallest n with $a_n \neq 0$, and a non-empty set of natural numbers has a smallest element. Thus, f can be written in the form:

$$f(x) = a_m x^m + a_{m+1} x^{m+1} + \cdots = x^m \underbrace{(a_m + a_{m+1} x + \dots)}_{=:u(x)}, \quad a_m \in k \setminus \{0\}.$$

Since $a_m \neq 0$, the series $u(x)$ has an invertible constant term in k, which implies that $u(x)$ is a unit in $k[[x]]$ (i.e., $u(x) \in \mathcal{U}_{k[[x]]}$, Theorem 3.43).

The remaining step, $I = (x^m)$, is left to the students (Exercise 7.5). **Q.E.D.**

Spoken Transcript [00:05:31 - 00:05:45] Prof. Rahul Pandharipande

Then this is... now this is the part for you: exercise — I did most of it [writes: Ex: prove $I =$]. Prove I... What am I going to write here?

Student Answer

Generated by f.

Spoken Transcript [00:05:47 - 00:06:53] Prof. Rahul Pandharipande

Yeah, you could say that, but there's a more elegant way to say it. I went to all this trouble writing the minimal degree here. It's generated by the lowest term, in this case x^{17} [writes: (x^{17})]. It's generated by the lowest term.

So, in general, if you have f in I of minimal degree, then this is generated by the monomial x to the minimal, that's the one [writes: $f \in I$ min. degree $\implies I = (x^m)$]. The monomial where the first one that comes, okay?

And the reason is that... well, I mean, we discussed these things, I leave that as an exercise for you to think. So, this shows that not only is this power series ring a principal ideal domain, because every ideal is generated by one element, in this case 17, or some other power, but actually I just told you all the ideals in the ring. All the ideals are of the form x to some number, that's it. Those are the only ideals. So, there are very few... I mean, there's infinitely many, but in some conceptual sense very few ideals.

Okay, so... these rings are all PIDs, so, they're all UFDs.

Exercise 7.5 (*Classification of Ideals in $k[[x]]$*): Let k be a field. Prove that if $f \in I \setminus \{0\}$ has minimal lowest degree m, then:

$$f \in I \text{ of minimal degree } m \implies I = (x^m).$$

Consequently, the only non-zero ideals of $k[[x]]$ are the powers of the principal maximal ideal (x):

$$(0) \subset \cdots \subset (x^3) \subset (x^2) \subset (x) \subset k[[x]].$$

In particular, $k[[x]]$ is a PID and a UFD.

Review of Content and Gauss's Lemma

Content of a Polynomial and Primitive Polynomials

Spoken Transcript [00:06:53 - 00:08:12] Prof. Rahul Pandharipande

Okay, so, now let's go back to the proof. So, the ideas of this proof, there were several new ideas, which I'll just comment on, so, we're going to use them. If I have f as a polynomial [writes: $f \in R[x]$], and f is not zero [writes: $f \neq 0$] — we don't like factoring zeros — if f is not zero, then we defined the content [writes: $C(f) \overset{\text{unit}}{=} \text{GCD}(\text{nonzero coeffs of } f)$]. This is the content of f, that's the words for it [writes: content of f], and it equals the GCD of the non-zero coefficients of f. Okay? Since f is not zero, it has at least one coefficient that's non-zero, and that's the GCD, and it's only defined up to a unit.

And the ways we think about this is given any f — always non-zero — f non-zero, then we can write f is equal to the content, or we just pull out the GCD, and then we multiply by what's left, and this is what's called primitive [writes: $f \in R[x], f \neq 0 \implies f = C(f) \cdot f^*$ and labels f^* as primitive]. And the definition of the term primitive was the content is equal to 1, up to a unit [writes: $C(f^*) \overset{\text{unit}}{=} 1$]. Okay? So, this is review from last time (Recall: Definition 6.43, Definition 6.47).

Recap: Content and primitive polynomials were defined in the previous lecture (Definition 6.43, Definition 6.47), with the decomposition $f = C(f)f^*$ of Proposition 6.45.

Definition 7.6 (*Content and Primitive Polynomials*): Let R be a UFD and let $f \in R[x]$ be a non-zero polynomial, $f(x) = \sum_{i=0}^{n} a_i x^i$.

(1) The *"content"* of f, denoted $C(f)$, is defined as the greatest common divisor of its non-zero coefficients:

$$C(f) \overset{\text{unit}}{=} \mathrm{GCD}(a_0, a_1, \dots, a_n) \in R,$$

which is well-defined up to multiplication by a unit in $\mathcal{U}_R$.

(2) A polynomial $f^* \in R[x]$ is called *"primitive"* if its content is a unit:

$$C(f^*) \overset{\text{unit}}{=} 1.$$

(3) Every non-zero polynomial $f \in R[x]$ admits a canonical factorization:

$$f = C(f) \cdot f^*, \tag{7.2}$$

where $C(f) \in R$ and $f^* \in R[x]$ is a primitive polynomial.

Gauss's Lemma and Divisibility of Primitive Polynomials

Spoken Transcript [00:08:12 - 00:10:18] Prof. Rahul Pandharipande

And the two properties that we use about this, which are really important: the first one is another theorem by Carl Friedrich, that says that if f and g are primitive, then their product is primitive [writes: $*$ CFG: if f, g are primitive $\implies$ fg primitive]. Okay? So, we proved that (Recall: Lemma 6.50), and that was a very nice proof actually. It used the fact about polynomial rings over integral domains being integral domains (Recall: Theorem 4.6).

And what the content of this is, is if you take something like... let's take the integers: $(2x+5) \cdot (3x+17)$ [writes: $(2x+5)(3x+17)$]. So, like these two things, these two polynomials, this is f and this is g [labels f and g under the factors]. This one's primitive because 2 and 5 don't have any common factors. This one's also primitive because 3 and 17 don't have any common factors. And what our friend Carl Friedrich is asserting here is if I multiply these, it'll also have that property.

So, then to check it, we can multiply. So, we have to do some multiplication here: $6x^2 + \dots$, something, $+17 \times 5$, that's 85. And this is the one where you have to add, I think it's 49 [writes: $= 6x^2 + 49x + 85$]. Is that right? Anyway, you can do some multiplication. It's important to get this right. Looks reasonable.

Okay, so, then if you factor this, you get 2×3, 7^2, and this is 5×17 [writes "$2 \cdot 3$", "7^2", "$5 \cdot 17$" under the coefficients]. And you see they have no common factors. And that's the assertion he's saying: every single time you do this, that happens. And the thing that's tricky... I mean, we proved it, so I'm not repeating the proof, but the thing that's tricky here is that although these things are relatively prime, when you multiply there's all sorts of addition that happens. And he's saying that doesn't stop his claim from being true. Okay, that's one principle.

Recap: Gauss's Lemma was proved in the previous lecture (Lemma 6.50).

Lemma 7.7 (*Gauss's Lemma*): Let R be a UFD. If $f, g \in R[x]$ are primitive polynomials, then their product $f \cdot g$ is also primitive:

$$C(f) = 1 \text{ and } C(g) = 1 \implies C(f \cdot g) = 1. \tag{7.3}$$

Example 7.8 (*Explicit Verification in $\mathbb{Z}[x]$*): Consider $f(x) = 2x + 5$ and $g(x) = 3x + 17$ in $\mathbb{Z}[x]$:

- $C(f) = \text{GCD}(2, 5) = 1$, so f is primitive.
- $C(g) = \text{GCD}(3, 17) = 1$, so g is primitive.

Multiplying the two polynomials:

$$f(x) \cdot g(x) = (2x + 5)(3x + 17) = 6x^2 + 49x + 85.$$

Factoring the individual coefficients into prime factors:

$$6 = 2 \cdot 3, \qquad 49 = 7^2, \qquad 85 = 5 \cdot 17.$$

Since no prime divides 6, 49, and 85 simultaneously, we have $\text{GCD}(6, 49, 85) = 1$, confirming that $f \cdot g$ is primitive.

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The second one was that if g divides $b \cdot f$ [writes: $* \, g \mid bf$], and the assumptions are g is primitive [writes: g primitive], so that means g is some primitive polynomial [writes: $g \in R[x]$], b is a scalar not equal to 0 [writes: $b \in R, b \neq 0$], and f is some polynomial [writes: $f \in R[x]$], the fact that g is primitive implies g actually divides f [writes: $\implies g \mid f$].

So, those are the two things that we're going to use. I'm not going to prove them again; I proved them last time (Recall: Lemma 6.50, Claim 6.51). But if you've never seen the content, it's worth playing with. It's kind of a useful idea.

Recap: This is Claim 6.51 of the previous lecture.

Lemma 7.9 (*Primitive Divisor Lemma*): Let R be a UFD. Let $g \in R[x]$ be a primitive polynomial, $f \in R[x]$, and $b \in R \setminus \{0\}$. If g divides $b \cdot f$ in $R[x]$, then g divides f in $R[x]$:

$$g \mid b \cdot f \quad \text{with } C(g) = 1 \text{ and } b \in R \setminus \{0\} \implies g \mid f. \tag{7.4}$$

Beyond the Lecture: A Second Proof of the Primitive Divisor Lemma

The lecturer does not repeat his proof (Claim 6.51). With Lemma 7.7 it also follows directly. We may assume $f \neq 0$. Write $bf = gh$ with $h \in R[x]$. Taking contents, $C(bf) = b\,C(f)$, while $C(gh) = C(h)$ because g is primitive (Lemma 7.7, applied to g and the primitive part of h). Hence $b \mid C(h)$, so $h = bh'$ with $h' \in R[x]$, and cancelling $b \neq 0$ in the domain $R[x]$ gives $f = gh'$.

Proof of Gauss's Theorem: Uniqueness of Factorization

Reduction to Primeness of Irreducible Elements

Spoken Transcript [00:11:20 - 00:13:05] Prof. Rahul Pandharipande

Okay, so, now we come to the structure of the proof. It's intricate, and there are many steps we've already done, which I don't repeat, but I'll just show you the structure. So, we want to prove this: R UFD implies $R[x]$ UFD [writes: R UFD $\implies$ $R[x]$ UFD].

So, for $R[x]$ we have to prove the existence [writes: For $R[x]$, we have to prove: $\exists$ – factorization into irred. ✓], that's the existence for factorization into irreducibles, that's the existence part, and I did that all last time. That was some induction on degree. Then we also have to prove uniqueness. So, that's the uniqueness statement [writes: ! – uniqueness].

And for uniqueness, we use this principle that we proved last time, and this principle is that if we have p in $R[x]$ is irreducible, if this implies the ideal in $R[x]$ is prime [writes: [$p \in R[x]$ irred. $\implies$ $(p) \subset R[x]$ prime]] — so, if I can prove this statement, then that implies the uniqueness of factorization into irreducibles [writes: $\implies$! factorization into irred.].

So, this is a principle we proved last Tuesday (Recall: Claim 6.25): that if I have a ring such that the irreducibles are prime, then the uniqueness statement holds, if you remember that. So, we're going to use it now, so, we need to prove this.

Recap: The reduction in (2) is Claim 6.25 (Week 6 Tuesday); the existence part (1) was proved in the previous lecture.

Proposition 7.10 (*Reduction of Uniqueness to Prime Ideals*)**:** To prove that $R[x]$ is a UFD when R is a UFD, two components are required:

(1) **Existence of Factorization (✓):** Every non-zero, non-unit polynomial factors into a product of irreducible elements. This was established previously via induction on the degree of the polynomial.

(2) **Uniqueness of Factorization:** Uniqueness holds if every irreducible element generates a prime ideal:

$$p \in R[x] \text{ is irreducible} \implies (p) \subset R[x] \text{ is a prime ideal.} \tag{7.5}$$

Consequently, proving (7.5) completes the proof of Gauss's Theorem.

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So, I remind you... reminder: p not equal to 0 or a unit is irreducible if we have whenever $p = ab$, then at least one of these is a unit, then a is a unit or b is a unit [writes: Reminder: $p \neq 0$, unit is irred. if $p = ab$ then a is a unit or b is a unit].

So, that's just the definition, just to remind you. So, we have to prove that whenever I have an irreducible person here, the ideal generated is prime. That will prove the uniqueness, we already have existence, and then we will have proved Gauss's theorem.

Recap: Irreducible elements were defined in Week 6 (Definition 6.10).

Definition 7.11 (*Irreducible Element*): Let R be an integral domain. A non-zero, non-unit element $p \in R \setminus (\{0\} \cup \mathcal{U}_R)$ is called *"irreducible"* if:

$$p = a \cdot b \implies a \in \mathcal{U}_R \text{ or } b \in \mathcal{U}_R.$$

Proof of Theorem 7.1: Primeness of Irreducible Elements

Case 1: Degree Zero Irreducibles

Spoken Transcript [00:13:52 - 00:15:33] Prof. Rahul Pandharipande

All right, so, even this I started last time. So, our goal here is we must prove that implication: p in $R[x]$ irreducible implies the ideal generated by p is prime [writes: We must prove: $p \in R[x]$ irred. $\implies (p) \subset R[x]$ is prime]. This is the statement we must prove. And even for this, there were two cases.

Case 1 is that the degree of p is 0 [writes: CASE (i) $\deg(p) = 0$]. So, what does it mean if p is a polynomial of degree 0? This means really p is a scalar, it's in R [writes: $p \in R$]. And we discussed this case last time, but this case should certainly not surprise you, because once p is in R, we're assuming R is a UFD, so, we just factorize it in R. That's basically the summary of that proof. So, we discussed this last time, I'm not going to discuss it again. But this is essentially because R is a UFD, we're just assuming [writes: R is a UFD].

It's not a problem. So, the interesting case, and it's actually the most interesting case of the whole proof, which is why I try to do it carefully, is the degree positive one [writes: CASE (ii) $\deg(p) > 0$]. Okay, and that's the statement I was in the middle of, or actually almost at the end of last time. But it is an intricate... I mean, at least for the kind of arguments we've had in this class, it is a slightly intricate one. So, let's do it slowly, and it's pretty, actually. As I said, most of the arguments for these things are proofs that prove themselves. This is not one; you have to actually have some ideas.

Case 1: Constant Irreducible Polynomials

We must show that for any irreducible polynomial $p \in R[x]$, the principal ideal $(p) \subset R[x]$ is prime. We analyze two cases according to $\deg(p)$:

Case 1 ($\deg(p) = 0$): If $\deg(p) = 0$, then $p \in R$ is a non-zero, non-unit constant polynomial. Since p is irreducible in $R[x]$, it is also irreducible in R. Because R is a UFD, every irreducible element in R is a prime element (Exercise 6.42).

The lecturer does not repeat the argument here: this case was settled in the previous lecture (Week 6, Wednesday). Q.E.D.

Beyond the Lecture: A Second Proof for Constants

The lecture settles this case by looking at coefficients (Week 6, Wednesday). A shorter argument: the reduction homomorphism

$$\pi\colon R[x] \to (R/(p))[x], \quad \sum a_i x^i \mapsto \sum \bar{a}_i x^i$$

has kernel $(p) \subset R[x]$, so $R[x]/(p) \cong (R/(p))[x]$ by the First Isomorphism Theorem. Since p is prime in R, the quotient $R/(p)$ is an integral domain (Proposition 5.24), so $(R/(p))[x]$ is one too (Theorem 4.6), and therefore (p) is a prime ideal in $R[x]$.

Case 2: Positive Degree Irreducibles and the Field of Fractions

Spoken Transcript [00:15:33 - 00:16:44] Prof. Rahul Pandharipande

Okay, so, what do we have to do? Let's write the assumptions [writes: Assumptions:]. So, one is that p is irreducible and the degree of p is positive [writes: $p \in R[x]$ is irred, $\deg(p) > 0$]. That's where we are in the logic.

We want to prove that the ideal is prime, so, then we can assume fg is in the ideal [writes: $fg \in (p)$], and we assume f is not in the ideal [writes: $f \notin (p)$]. All right? And we need to show — this is our goal — to show that then g is in the ideal [writes: Goal: $g \in (p)$].

All right, so, if we can do this, that says this is our assumptions and this last part is it being prime: you take any two where the first one's not, then the second one has to be. So, that's what we have to do. Okay? That's the logical thing we have to do. Is that clear?

Case 2: Setup and Logical Assumptions

Case 2 ($\deg(p) > 0$): Let $p \in R[x]$ be an irreducible polynomial with $\deg(p) \geq 1$. To show that the principal ideal $(p) \subset R[x]$ is prime, we must establish:

$$f \cdot g \in (p) \quad \text{and} \quad f \notin (p) \implies g \in (p). \tag{7.6}$$

- **Assumptions:**

$$p \in R[x] \text{ is irreducible with } \deg(p) > 0, \tag{7.7}$$

$$f \cdot g \in (p) \iff p \mid f \cdot g, \tag{7.8}$$

$$f \notin (p) \iff p \nmid f. \tag{7.9}$$

- **Goal:** Prove that $g \in (p)$, i.e., $p \mid g$ in $R[x]$.

Spoken Transcript [00:16:44 - 00:18:50] Prof. Rahul Pandharipande

All right, so, the proof of this is: the first idea is we construct this ideal generated by f and p, and it's an ideal in this polynomial ring [writes: Proof: $I = (f, p) \subset R[x]$ ideal]. So, just to be clear here, everything here, p, f, g, are polynomials [writes: $p, f, g \in R[x]$]. All right? And so, I can take the ideal here.

And so, now I select this element, and this is an idea that we've used many times: if I have this ideal in a ring, like in this polynomial ring, it's a very good idea often to select the element of lowest degree. So, now we're here: select m in I so that m is of lowest degree among all non-zero elements of I [writes: Select $m \in I$ so that $m \neq 0$ is of lowest degree among all non-zero elements of I].

Okay? So, again, this ideal has some people in it. In particular, p is irreducible, so, it's not zero, so, this ideal is not zero. So, it has some non-zero elements; we choose the one of lowest degree, which we're allowed to. It's not unique, but we're allowed to choose an element of lowest degree. Are you happy with that?

And now the sub-goal here is that we will show that m is a constant, m is in R [writes: [We will show $m \in R$]]. In other words, m is of degree 0. We will show that. That's going to be the goal of the next part of the argument.

Auxiliary Ideal and Minimal Degree Element

To prove that $p \mid g$, we consider the ideal in $R[x]$ generated by f and p:

$$I := (f, p) = \{a \cdot f + b \cdot p \mid a, b \in R[x]\} \subset R[x]. \tag{7.10}$$

Since $p \in I$ and p is irreducible, $p \neq 0$, which ensures $I \neq (0)$.

Choice of Minimal Degree Element: Select an element $m \in I \setminus \{0\}$ of minimal degree among all non-zero polynomials in I:

$$m \in I \setminus \{0\} \quad \text{such that} \quad \deg(m) = \min_{h \in I \setminus \{0\}} \deg(h). \tag{7.11}$$

Such an element exists by the well-ordering of non-negative integer degrees.

Sub-goal: We will show that m is a non-zero constant in the base ring:

$$m \in R \quad (\text{i.e., } \deg(m) = 0). \tag{7.12}$$

Spoken Transcript [00:18:50 - 00:21:05] Prof. Rahul Pandharipande

Now, what we normally do in the case when we have a field is we use the division algorithm (Recall: Theorem 5.57). So, we'd very much like to use the division algorithm, and this is one of the ideas in this proof, and it's part of the reason why the notion of the fraction field was introduced. So, we would like to use the division algorithm [writes: We would like to use the division alg. in $R[x]$] ... the division algorithm...

But we can't because this is not a field. So, the solution here, and this is what I said, one of the ideas in the proof, is that... so, the problem is that R need not be a field [writes: But the problem is that R need not be a field.]. And for the division algorithm you need a field because you're dividing.

So, the idea here is to use the field of fractions. As I said, that's kind of why we introduced it. So, let $Q(R)$ be the field of fractions [writes: Idea: $Q(R)$ be the field of fractions of R] (Recall: Definition 3.17). So, remember, R sits inside... I'm just going to call it Q, sits inside Q [writes: $R \subset Q = \dots$], and this is the field of fractions. Every element of Q, an element of Q is some kind of a/b, where b is not zero [writes: $Q \ni \frac{r}{b}, b \neq 0$].

All right? And all of these things we can consider, we can view all of these people, p, f, g, to be inside this polynomial ring over this field [writes: We can view $p, f, g \in Q[x]$]. Because they live inside R, so, we just view them inside the field. Okay? And the advantage of this is we can run the division algorithm in the field. That's the whole purpose of this idea.

Passage to the Field of Fractions $Q(R)$

In the general polynomial ring $R[x]$, polynomial division with remainder cannot be performed because leading coefficients may not be invertible in R. To resolve this, we pass to the field of fractions of R:

$$Q := Q(R) = \left\{ \frac{r}{b} \;\middle|\; r, b \in R,\ b \neq 0 \right\}. \tag{7.13}$$

Since $R \subset Q$, we have a natural embedding of polynomial rings:

$$R[x] \subset Q[x].$$

Consequently, the polynomials $p, f, g, m \in R[x]$ can all be viewed as polynomials in $Q[x]$:

$$p, f, g, m \in Q[x]. \tag{7.14}$$

Since Q is a field, the division algorithm is fully applicable in $Q[x]$.

Spoken Transcript [00:21:05 - 00:23:05] Prof. Rahul Pandharipande

So, what does it mean we can run the division algorithm? This means that we can write... so, we can take f and m — I should put m also there in that list, sorry, just so I'm complete [adds m to the list $p, f, g, m \in Q[x]$]. Also we've already selected m too, so m is on this list.

So, we want to apply the division algorithm. We have f, I want to divide... division algorithm in here for division by m [writes: f in division alg. in $Q[x]$ division by m]. Okay? So, the consequence of that is I can write f is equal to the quotient times m plus some remainder [writes: $f = \hat{q}m + \hat{r}$].

And these hats signify that these elements may be in this ring, right? [writes: $\hat{q}, \hat{r} \in Q[x]$] Because f and m are also in $Q[x]$, but these things we know are actually in the smaller place, that they're here [writes: $\hat{q}, m \in Q[x]$ and indicates $f, m \in R[x]$]. Okay, but we can't apply the division algorithm in polynomials in R, but we can with polynomials with coefficients in Q. So, we apply the division algorithm, and we get this quotient and remainder, but they live here. Is that clear?

And the condition that comes out of the division algorithm is that either $\hat{r}$ is 0, or the degree of $\hat{r}$ is strictly less than the degree of m, okay? [writes: $\hat{r} = 0$ or $\deg \hat{r} < \deg m$]. Are you happy with that?

Division Algorithm in $Q[x]$

We apply the division algorithm in $Q[x]$ to divide f by m:

$$f = \hat{q} \cdot m + \hat{r}, \qquad \hat{q}, \hat{r} \in Q[x], \tag{7.15}$$

where $\hat{q}$ and $\hat{r}$ are polynomials whose coefficients a priori lie in the fraction field Q, satisfying:

$$\hat{r} = 0 \quad \text{or} \quad \deg(\hat{r}) < \deg(m). \tag{7.16}$$

Spoken Transcript [00:23:05 - 00:24:54]

Prof. Rahul Pandharipande

And now comes the next step, is this step which is called clear the denominators [writes: Clear the denoms:]. Which is... So, we don't want to work in polynomials with coefficients in this field. Somehow the theorem is about polynomials with coefficients in R, so, we'd like to get back to polynomials with coefficients in R.

And the only problem here... so, this is a polynomial with coefficients in $Q[x]$, so, this has some $c_6x^6 + c_5x^5 + \cdots + c_0$ [writes: $\hat{q} = c_6x^6 + c_5x^5 + \cdots + c_0$]. Now, all of these coefficients are in the field of fractions of R [writes: $c_i \in Q(R)$], so, I can write them instead as elements in R: a_6/b_6, a_5/b_5 — this is kind of explicit — a_0/b_0 [writes: $\hat{q} = \frac{a_6}{b_6}x^6 + \frac{a_5}{b_5}x^5 + \cdots + \frac{a_0}{b_0}$ and $a_i, b_i \in R, b_i \neq 0$], where all of these a_i's and b_i's are in R, and moreover the denominators are not zero, right? So, I can just write it like this.

So, that means that if I multiply by all these denominators, I can have $b \cdot \hat{q} = a_6'x^6 + a_5'x^5 + \cdots + a_0'$ [writes: $b\hat{q} = a_6'x^6 + a_5'x^5 + \cdots + a_0'$], where this b is the product of the b_i [writes: $b = \prod b_i$] — that's clearing the denominators — and then these are all in the ring also [writes: $a_i' \in R$].

Okay, so, that's clearing the denominators. I did this really explicitly, but you know how to clear denominators. We just clear the denominators.

Clearing Denominators of $\hat{q}$

To return from $Q[x]$ to the original polynomial ring $R[x]$, we clear the denominators of $\hat{q}(x) \in Q[x]$. Expanding $\hat{q}(x)$ explicitly:

$$\hat{q}(x) = c_kx^k + c_{k-1}x^{k-1} + \cdots + c_0, \qquad c_i \in Q(R).$$

Since each coefficient is a quotient of elements of R, we write $c_i = \frac{a_i}{b_i}$ with $a_i, b_i \in R$ and $b_i \neq 0$:

$$\hat{q}(x) = \frac{a_k}{b_k}x^k + \frac{a_{k-1}}{b_{k-1}}x^{k-1} + \cdots + \frac{a_0}{b_0}.$$

Defining the common denominator $b \in R \setminus \{0\}$ as the product of all denominators:

$$b := \prod_{i=0}^{k} b_i \in R \setminus \{0\}, \tag{7.17}$$

we clear denominators so that:

$$b \cdot \hat{q}(x) = a_k'x^k + a_{k-1}'x^{k-1} + \cdots + a_0' \in R[x], \qquad a_i' \in R. \tag{7.18}$$

Spoken Transcript [00:24:54 - 00:25:59] Prof. Rahul Pandharipande

And the price of clearing the denominators... so, what's good about clearing the denominators is you land in this ring; the price is you have to multiply by b.

So, I can just do that here. So, I take here, I have f... sorry... [moves to the middle lower blackboard] I have $f = \hat{q}m + \hat{r}$ [writes: $f = \hat{q}m + \hat{r}$].

So, there exists some b' in R, b' not equal to zero [writes: $\exists b' \in R, b' \neq 0$], which clears the denominators of this. That means I multiply by b': $b'f$, then I get a new q' and a new r' [writes: $b'f = q'm + r'$].

And now b' is in R, as I said, not equal to zero, and q' and r' are polynomials with coefficients here, okay? [writes: $b' \in R, b' \neq 0, q', r' \in R[x]$]

So, I'm saying it very explicitly, but all I do is I take this, there might be some denominators, I clear them. Which, you know, you've been clearing denominators for a long time by now, I assume.

Okay. So, are you happy with this?

Cleared Division Equation for f

Applying this clearing procedure simultaneously to both $\hat{q}$ and $\hat{r}$ in (7.15), there exists a non-zero scalar $b' \in R \setminus \{0\}$ such that multiplying through clears all denominators:

$$b' \cdot f = q' \cdot m + r', \tag{7.19}$$

where:

$$b' \in R \setminus \{0\}, \qquad q' := b'\hat{q} \in R[x], \qquad r' := b'\hat{r} \in R[x]. \tag{7.20}$$

Spoken Transcript [00:25:59 - 00:28:09] Prof. Rahul Pandharipande

And now I look at this r'.

So, r', when I clear denominators, I don't change the degree. So, we have the same thing: r' is equal to zero [writes: $r' = 0$], because when you clear denominators if it were zero, clearing the denominators will keep it zero; or... So, either r' is zero, or the degree of r' is less than the degree of m, okay? [writes: or $\deg(r') < \deg(m)$]. Because this multiplication by a scalar does not change the degree. We took $\hat{r}$ and we cleared the denominators, it does not change the degree.

All right, but when I look at this equation, I notice that m is in I, and f is in I [writes: $m \in I, f \in I$]. Because m was the lowest element in I, and f is born in I. So, this implies that r' is in I [writes: $\implies r' \in I$].

Right? But r' cannot be lower than m in I, because m is an element of the lowest degree in I. So, this possibility is impossible [crosses out $\deg(r') < \deg(m)$]. So, it must be that r' is zero [writes: $\implies r' = 0$].

Which is kind of what we were expecting, because that was the argument for a field if you use that. But this is a way to come to this conclusion even when R is not a field, by going to the field.

Okay, so, the conclusion here is that we have $b'f = q'm$ [writes: Conclude: $b'f = q'm$].

All right, and then if we write this in terms of contents, we have $b'f = q' \cdot C(m) \cdot m^*$

[writes: $b'f = q'C(m)m^*$]. So, the conclusion is that m^* divides f [writes "$m^* \mid f$" and draws a rectangular box around it]. This uses one of our principles, because m^* is primitive: the primitive is dividing this product, so, it has to divide one of them, all right? So, this is one of our conclusions. Okay.

Vanishing of the Remainder and Divisibility of f

Because $b' \in R \setminus \{0\}$ is a non-zero scalar, multiplication by b' does not alter the degree of polynomials:

$$\deg(r') = \deg(b'\hat{r}) = \deg(\hat{r}).$$

Therefore, condition (7.16) carries over directly to r':

$$r' = 0 \quad \text{or} \quad \deg(r') < \deg(m). \tag{7.21}$$

Ideal Containment and Contradiction: From (7.19), we isolate the remainder:

$$r' = b' \cdot f - q' \cdot m.$$

Since $f \in I$ and $m \in I$, and I is an ideal of $R[x]$, it follows that:

$$r' \in I.$$

By definition of m, it is a non-zero element of minimal degree in I:

$$\deg(m) = \min_{h \in I \setminus \{0\}} \deg(h).$$

Consequently, no non-zero element of I can have degree strictly less than $\deg(m)$. This rules out the second alternative in (7.21):

$$\deg(r') < \deg(m) \quad \text{is impossible for } r' \neq 0.$$

Thus, the remainder must vanish identically:

$$r' = 0. \tag{7.22}$$

Deducing Divisibility via Primitive Divisor Lemma: Substituting $r' = 0$ into (7.19) gives:

$$b' \cdot f = q' \cdot m. \tag{7.23}$$

Factoring m into its content and primitive part:

$$m = C(m) \cdot m^*, \qquad \text{where } C(m) \in R \setminus \{0\} \text{ and } m^* \in R[x] \text{ is primitive.}$$

Equation (7.23) becomes:

$$b' \cdot f = \big(q' \cdot C(m)\big) \cdot m^* \implies m^* \mid b' \cdot f.$$

Since m^* is primitive and $b' \in R \setminus \{0\}$, the Primitive Divisor Lemma (Lemma 7.9) implies:

$$\boxed{m^* \mid f} \quad \text{in } R[x]. \tag{7.24}$$

Spoken Transcript [00:28:09 - 00:29:04] Prof. Rahul Pandharipande

So, this proof is now going at about half the speed as I was going last time. Which I don't know is good or bad.

You can also just watch the video at half the speed, I guess, or twice the speed, or whatever you want.

What I'm telling you... more details of the argument [erases the lower left blackboard to prepare for repeating the argument with p].

Spoken Transcript [00:29:04 - 00:30:34] Prof. Rahul Pandharipande

All right, so, we did this whole board with the division algorithm, we divided f by m. So, do the whole board again, so, repeat for p divided by m [writes: Repeat for p divided by m:].

So, what happened there? So, we take now p here, and now we have some different quotients. They're double hats because we've already used one hat: $\hat{\hat{q}}m + \hat{\hat{r}}$ [writes: $p = \hat{\hat{q}}m + \hat{\hat{r}}$]. And again, these two are in the polynomial ring with coefficients in the field, just as they were there [writes: $\hat{\hat{q}}, \hat{\hat{r}} \in Q[x]$]. We just do the same division algorithm.

And now we clear, just as before: clear denominators [writes: Clear denoms:].

And the conclusion is we get $b''p = q''m + r''$ [writes: $b''p = q''m + r''$], where now, all the same conclusions: $0 \neq b'' \in R$, and q'', r'' are here, okay? [writes: $0 \neq b'' \in R$, $q'', r'' \in R[x]$]

Division of p by m and Clearing Denominators

We now replicate the identical division and clearing procedure for the irreducible polynomial p in place of f:

(1) Division in $Q[x]$: Dividing p by m in $Q[x]$ yields:

$$p = \hat{\hat{q}} \cdot m + \hat{\hat{r}}, \qquad \hat{\hat{q}}, \hat{\hat{r}} \in Q[x], \tag{7.25}$$

with:

$$\hat{\hat{r}} = 0 \quad \text{or} \quad \deg(\hat{\hat{r}}) < \deg(m). \tag{7.26}$$

(2) Clearing Denominators: Clearing denominators by multiplying through by a suitable non-zero scalar $b'' \in R \setminus \{0\}$:

$$b'' \cdot p = q'' \cdot m + r'', \tag{7.27}$$

where:

$$b'' \in R \setminus \{0\}, \qquad q'' := b''\hat{\hat{q}} \in R[x], \qquad r'' := b''\hat{\hat{r}} \in R[x]. \tag{7.28}$$

Spoken Transcript [00:30:34 - 00:31:52] Prof. Rahul Pandharipande

So, here I just do just the same thing I did upstairs, I just do it with p instead of f. And I reach the same conclusion, of course with different possible coefficients.

But then, again, the next step was we notice that m is in I, and p is in I [writes: $m \in I, p \in I$]. Because p is also born inside I — remind you I is (f, p) [writes "$I = (f, p)$" above].

And so, the same conclusion holds, that r'' is in I [writes: $\implies r'' \in I$], and as a consequence r'' must be zero by the same logic [writes: $\implies r'' = 0$].

So, we get the equation... conclude, just as above — in fact this whole board is just writing that whole argument again — is that $b''p = q''m$ [writes: Conclude: $b''p = q''m$].

Okay?

So, we conclude here that this m^* has to divide p, for the same reason that this m^*

divides f, because m... we can write it further, the next step if you want, but it's the same thing: $q''C(m)m^*$ [writes: $b''p = q''C(m)m^*$]; this is primitive [draws an arrow to m^* and writes: prim], it divides the product, so, it must divide, all right? [writes: $m^* \mid p$ and frames it in a box]

So, it's just the same. It's not surprising the conclusion is the same because the steps are the same.

Vanishing of r'' and Divisibility of p

Since $p \in I$ and $m \in I$, the remainder:

$$r'' = b'' \cdot p - q'' \cdot m$$

belongs to I. By the minimality of $\deg(m)$, we cannot have $\deg(r'') < \deg(m)$ for any non-zero element of I. Hence:

$$r'' = 0. \tag{7.29}$$

Substituting $r'' = 0$ into (7.27):

$$b'' \cdot p = q'' \cdot m = \left(q'' \cdot C(m)\right) \cdot m^*. \tag{7.30}$$

Because m^* is primitive and $b'' \in R \setminus \{0\}$, the Primitive Divisor Lemma (Lemma 7.9) implies:

$$\boxed{m^* \mid p} \quad \text{in } R[x]. \tag{7.31}$$

Spoken Transcript [00:31:52 - 00:33:44] Prof. Rahul Pandharipande

All right, so, what does this mean? This means p is equal to m^* times something [writes: $p = m^* \cdot B$]. That's what $m^* \mid p$ means: $p = m^*$ times something.

So, now we have to use the irreducibility. All right?

So, since p is irreducible [writes: p is irred.], this implies one of these: so, either m^* is a unit or B is a unit [writes: $\implies \begin{cases} m^* \text{ unit} \in R[x] \\ B \text{ unit} \in R[x] \end{cases}$]. Since it's irreducible, you have to use it's irreducible now. So, one of these is true.

And the claim is... this is impossible [crosses out the second branch B unit with an X]. Because if B were a unit, then p would be m^* up to a unit. But m^* divides f, so, if this were true, then p would divide f [writes: $p \mid f$]. That would mean f is in the ideal generated by p [writes: $f \in (p)$].

But our hypothesis is exactly f is not in the ideal generated by p. This contradicts that hypothesis [writes: CONTRA].

Okay?

So, here's where we start using the hypotheses. So, irreducible means we have this dichotomy, and then if we go on the second branch, B is a unit, we get a contradiction. It contradicts that... so, one of the hypotheses, if you remember, assumption on that board was that f is not in (p) [writes: (Ass $f \notin (p)$)]. That was one of our assumptions. But if we go on this branch, we get f is in (p).

So, it has to be that m^* is a unit, okay? [ticks the branch m^* unit $\in R[x]$]

Irreducibility of p and Unit Property of m^*

From (7.31), there exists a polynomial $B \in R[x]$ such that:

$$p = m^* \cdot B. \tag{7.32}$$

Since p is assumed to be irreducible in $R[x]$ (Definition 7.11), every factorization of p must contain a unit. This yields a dichotomy:

$$m^* \in \mathcal{U}_{R[x]} \quad \text{or} \quad B \in \mathcal{U}_{R[x]}.$$

Exclusion of $B \in \mathcal{U}_{R[x]}$: Suppose for contradiction that B is a unit in $R[x]$:

$$B \in \mathcal{U}_{R[x]} \implies m^* = B^{-1} \cdot p \implies p \sim m^*.$$

Since $m^* \mid f$ from (7.24), this would imply:

$$p \mid f \implies f \in (p).$$

However, this directly contradicts our foundational assumption (7.9) that $f \notin (p)$:

$$f \in (p) \quad \text{contradicts} \quad f \notin (p). \quad \lightning$$

Therefore, B cannot be a unit. Consequently, the first branch must hold:

$$m^* \in \mathcal{U}_{R[x]}. \tag{7.33}$$

Spoken Transcript [00:33:44 - 00:34:34] Prof. Rahul Pandharipande

How can m^* be a unit? m^* is a polynomial. The only way it can be a unit is if it's a constant.

So, this implies... that's the conclusion we were seeking.

Conclusion: m^* is in R [writes: Conclude: $m^* \in R \implies m \in R$]. The only way you can be a unit in a polynomial ring over an integral domain is if you're a constant, right? (Recall: Theorem 3.39)

And this implies that m was actually in R too, because they differ just by the content [writes: $m \in R$].

That's what I said with the goal here. So, after all this work, we selected this m to be an element in the ideal of lowest degree, and we concluded that it has to be in R.

So, we're almost done now.

Attainment of the Sub-Goal: $m \in R$

Since R is an integral domain, the units of the polynomial ring $R[x]$ are exactly the units of the base ring (Theorem 3.39):

$$\mathcal{U}_{R[x]} = \mathcal{U}_R.$$

Therefore, (7.33) implies:

$$m^* \in \mathcal{U}_R \subset R.$$

Since $m = C(m) \cdot m^*$ and $C(m) \in R$, we obtain:

$$\boxed{m \in R \setminus \{0\}} \quad (\text{i.e., } \deg(m) = 0). \tag{7.34}$$

This successfully proves our sub-goal (7.12).

Spoken Transcript [00:34:34 - 00:36:24] Prof. Rahul Pandharipande

[erases the middle lower board]

Okay, so, we have m is in the ideal [writes: $m \in I$]. This is the ideal generated by f and p [writes: $I = (f, p)$]. So, this means we have $m = Af + Bp$ [writes: $m = Af + Bp$], where everything here, m, A, f, B, p, are in the ring [writes: $m, A, f, B, p \in R[x]$].

Okay? So, this is just because it's an ideal generated by them.

But now we know something more: we've proved that... we also know that this is in R [underlines m and writes "$0 \neq m \in R$" on the left]. This is a scalar. This m is in R, it's a non-zero element of R, okay?

So, we have to get g in the picture. The whole goal is to prove g is in the ideal generated by p, and we haven't seen g at all. So, how are we going to get g in this picture? This is a trick we've used many, many times.

Student Answer

Multiply by g.

Spoken Transcript [00:36:24 - 00:37:08] Prof. Rahul Pandharipande

Yeah, multiply by g, exactly. We've used this many times. Multiply by g. [writes: $mg = Afg + Bgp$]

Okay? And this is our non... not equal to zero...

Okay. Now we know this is in the ideal generated by p [points at Afg and writes "(p)" underneath]. That was one of our assumptions: fg is in the ideal generated by p. So, this term is in the ideal generated by p.

This term is in the ideal generated by p [circles Bgp].

So, this means that p divides $m \cdot g$ [writes: $p \mid mg$], where m is a non-zero scalar [writes: $m \neq 0, m \in R$].

Okay?

Expressing m in the Ideal and Introducing g

Since $m \in I = (f, p)$, there exist polynomials $A, B \in R[x]$ such that:

$$m = A \cdot f + B \cdot p, \qquad A, B \in R[x]. \tag{7.35}$$

(This B is new, as on the board; the cofactor B of (7.32) is not used again.) By (7.34), m is a non-zero scalar: $m \in R \setminus \{0\}$.

Multiplying both sides of (7.35) by the target polynomial g:

$$m \cdot g = A \cdot (f \cdot g) + B \cdot g \cdot p. \tag{7.36}$$

We analyze both terms on the right-hand side:

- $A \cdot (f \cdot g) \in (p)$ because $f \cdot g \in (p)$ by assumption (7.8).
- $B \cdot g \cdot p \in (p)$ because it is an explicit multiple of p.

Since (p) is an ideal, the sum lies in (p):

$$m \cdot g \in (p) \iff p \mid m \cdot g \quad \text{in } R[x], \quad \text{with } m \in R \setminus \{0\}.$$

Spoken Transcript [00:37:08 - 00:39:19] Prof. Rahul Pandharipande

So, now the last thing is: what is the assumption on p? p is irreducible [writes: p is irred.].

If we knew p were primitive, we'd be done.

If p were primitive [writes: If p were primitive], then we could use the principle that primitive p divides $m \cdot g$, m not equal to zero, primitive... [writes "$p \mid mg, m \neq 0, p$ prim. $\implies p \mid g \implies g \in (p)$" and checks it ✓] we use this principle: it implies p divides g, and so g is in (p) and we're done.

So, all we need to show is: we must show that p irreducible, and now we use the assumption the degree of p is positive, implies p is primitive [writes: We must show: p irred., $\deg(p) > 0 \implies p$ is primitive].

Okay? That's the last step. The last step in this intricate argument: p irreducible, degree positive, it's primitive.

So, we factor p as the content of p times p^* [writes: $p = C(p) \cdot p^*$], all right? That's the content factorization of p.

The fact that p is irreducible, one of these two must be units: p irreducible implies either $C(p)$ or p^* is a unit [writes: p irred. $\implies$ either $C(p)$ or p^* is a unit]. That's what it means to be irreducible: one of these two is a unit.

This thing cannot be a unit [circles p^*]. Why is that? Because it has the same degree as p, which is positive, and positive degree things are not units [writes: $\deg(p^*) = \deg(p) > 0 \implies$ cannot be unit].

So, this thing has to be a unit [circles $C(p)$]. So, this cannot be a unit, because it's positive degree, so, that has to be a unit [writes: $C(p)$ unit]. And that means that p is primitive [writes: $\implies p$ is primitive].

So, we used that at the very last stage, we used the hypothesis the degree is greater than zero.

And that's the end of the proof.

Primitiveness of p and Conclusion of Gauss's Theorem

To deduce $p \mid g$ from $p \mid m \cdot g$ via the Primitive Divisor Lemma (Lemma 7.9), we must establish that p is primitive.

Claim: If $p \in R[x]$ is irreducible and $\deg(p) > 0$, then p is primitive.

Proof of Claim

Factor p into its content and primitive part:

$$p = C(p) \cdot p^*, \qquad C(p) \in R, \quad C(p^*) = 1.$$

Since p is irreducible in $R[x]$:

$$C(p) \in \mathcal{U}_{R[x]} \quad \text{or} \quad p^* \in \mathcal{U}_{R[x]}.$$

Since $C(p)$ is a constant scalar, $\deg(p^*) = \deg(p) > 0$. Any element of positive degree in $R[x]$ cannot be a unit (units in $R[x]$ must have degree 0). Thus:

$$p^* \notin \mathcal{U}_{R[x]}.$$

Therefore, $C(p)$ must be a unit in R:

$$C(p) \in \mathcal{U}_R.$$

By definition (Definition 7.6), this means that p is primitive:

$$C(p) = 1 \quad \text{(up to units)}.$$

Q.E.D.

Final Conclusion of the Proof: Since p is primitive, $m \in R \setminus \{0\}$, and $p \mid m \cdot g$, the Primitive Divisor Lemma (Lemma 7.9) implies:

$$p \mid g \implies g \in (p).$$

This completes Case 2, establishing that (p) is prime whenever p is irreducible. Consequently, by Proposition 7.10, uniqueness of factorization holds, and Gauss's Theorem (Theorem 7.1) is proven.

Q.E.D.

Spoken Transcript [00:39:19 - 00:40:39] Prof. Rahul Pandharipande

So, that's the whole argument.

And, as I said, it has a certain intricacy. It uses most of the things we've discussed about rings. Yeah.

And that, unfortunately, is the last result about rings in this class.

That's not really true, there's a little bit at the end, but as far as the development of the theory of rings, we are finished. So, you can bid a tearful goodbye to rings now. [washes the entire blackboard with a sponge]

Reflections on Ring Theory and Mathematical Learning

The Triad of Definitions, Theorems, and Examples

Spoken Transcript [00:40:39 - 00:41:34] Prof. Rahul Pandharipande

Yeah, so the next topic is groups, but I want to say a few things because people asked me some questions, and they're good questions.

Like the first question is: how are you supposed to study this material? That's a really good question.

Another question is: how did I learn the material? That's not such a good question, it's not so relevant, but...

Maybe a deeper question which goes to the heart of the matter is: you know, what are we doing here? That's a really good question.

And, you know, if you're studying you should take some time to think about these questions. There's different ways. So, maybe I try to answer these questions in different... I have a few minutes, but maybe it's worth giving a few...

So, the first question about how to study the material: you can say, what have I taught in this class about rings? And then you can say there's kind of three things:

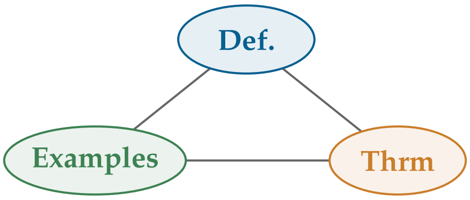
One is definitions, and as I've said many, many times, those are incredibly important. It's what makes all these proofs work perfectly. And they're not tautological. In the end we prove some actual non-trivial results, but the definitions are... there's a lot of mathematical insight and knowledge in the definition. So, I would say, study the definitions.

Then there's theorems, and then there's examples [writes "Def.", "Thrm", "Examples" in circles on the board and connects them].

So, these are like the three things you should really kind of try to understand. I don't know how to say more than that, but they are somehow equal weights in the class.

If you understand the definitions... these definitions and the objects are like chess pieces, and each of them have certain moves, and the theorems involve playing out those moves. So, if you understand this, you'll understand that also.

That's like at the lowest level.



The Pedagogical Triad of Algebra: The lecturer frames mathematical mastery as balancing three equally weighted pillars:

1. **Definitions (The Chess Pieces):** Definitions are not mere conventions or tautologies; they condense deep mathematical insight and define the *"rules of the game"* and allowed moves.
2. **Theorems (The Gameplay):** Theorems represent the execution of mathematical possibilities, playing out the logical consequences of the definitions.
3. **Examples (The Intuition):** Concrete instances ground abstract definitions and test the boundaries of theorems.

Philosophical Perspectives on Learning and Education

Spoken Transcript [00:41:34 - 00:43:02] Prof. Rahul Pandharipande

But this question about what are we doing here, that's an interesting question.

And many people have written about it. It's been, you know, 2000 years of intellectual history about that, and you can go look at it.

So, one thing that I recommend my students read is an essay by Schopenhauer, which is called *"On Thinking for Yourself"*, which I like a lot. He's a very dense, repetitive author.

So, you can read just half the essay, the short essay — the German is *"Selbstdenken"*. It's a nice essay. Most of the students don't like it after they read it. But anyway, it's a really good essay, it's kind of about this topic.

And, you know, all the greats have written about it. If you read the past... I mean, there's a lot of things to say, I could give a whole lecture, but...

If you read Plato, in his Republic he talks about what education is about. Socrates, he didn't write, but there's a lot of stuff there.

The one sentence which I like the best is Confucius. And you think: is he going to quote Confucius in this class? But I will quote, it's I think the statement that best captures what we do: Confucius wrote about teaching. He says that — so, roughly speaking is: The truth has four corners. As a teacher I can show you one; the other three are for the student to discover themselves.

That's what we're doing here. Okay. That's it.

***Insight:* Confucius and the Four Corners of Truth**

The lecturer quotes Confucius on teaching, as the statement that best captures what is done here:

> *"The truth has four corners. As a teacher I can show you one; the other three are for the student to discover themselves."*

Beyond the Lecture: The Source of the Quote

The saying goes back to the **Analects** of Confucius (VII.8).

Retrospective: The Role of the Hypothesis $\deg(p) > 0$

Spoken Transcript [00:43:02 - 00:44:09] Prof. Rahul Pandharipande

Okay, so, there were some questions about the proof. The question roughly speaking was: in that whole hour I was assuming the degree of p is positive [writes "$\deg(p) > 0$" on the left board], where do I use it?

And it was just... it's like a player who sits on the bench for the whole game. But at the end, we use this: p irreducible, degree of p positive implies p primitive [writes: p irred., $\deg(p) > 0 \implies p$ primitive].

That was all we needed at the last moment. So, this player just sits on the bench, and at the last moment scores the winning goal. That's the only time it's used, at the end.

You can look at the argument, but it's the only time it's used.

Yeah, so, if anyone reads Schopenhauer, you should tell me.

Finally, the question about: how did I learn it? The answer is: at a certain point in my life, I studied very hard. That's how you learn it. That's the only way.

So, that's not very helpful, I think.

Okay, so, now we switch to groups.

Remark 7.12 (*The "Player on the Bench" Hypothesis*): Throughout the extensive division and clearing arguments of Case 2, the assumption $\deg(p) > 0$ was never invoked until the very final step:

$$p = C(p) \cdot p^* \quad \text{with } p \text{ irreducible.}$$

Because $\deg(p) > 0$, the primitive part p^* has $\deg(p^*) = \deg(p) > 0$, which prevents p^* from being an invertible constant unit in $\mathcal{U}_{R[x]} = \mathcal{U}_R$. This forced $C(p) \in \mathcal{U}_R$, proving that p is primitive. As the lecturer notes, this hypothesis acts like a player who sits on the bench for the whole game and at the last moment scores the winning goal.

Introduction to Group Theory

Definition and Axioms of a Group

Spoken Transcript [00:44:09 - 00:45:55] Prof. Rahul Pandharipande

And we've already seen the definition (Recall: Definition 2.1), so, I don't really say so much about it. A group is... the definition: so, first groups, that's the topic now [writes "Groups"]. So, G with a binary operation and a neutral element is a group [writes: $(G, \cdot, e)$ is a group].

So, here we have associativity, the unit property, and the existence of inverse [writes: Associativity, Unit property, Existence of inv.].

So, we've seen all of this; I'm not going to say so much about it except mostly about notation.

Associativity... they're all interesting and they're worth thinking about again now, because I defined all of this maybe at the beginning of the semester, and you've probably seen the definition of a group even before.

So, associativity is something that says that if I do the binary operation in two different ways, it's the same [writes: $(x \cdot y) \cdot z = x \cdot (y \cdot z)$].

We haven't said anything about associativity also for rings, and maybe I'll say in a moment, I'll say more.

The unit property says that there exists a unit such that $e \cdot x = x \cdot e = x$ for all x in the group [writes: $\exists e$ s.t. $e \cdot x = x \cdot e = x \quad \forall x \in G$].

And existence of inverse says that for all x in the group, there exists y such that $xy = yx = e$, the unit [writes: $\forall x \in G, \exists y$ s.t. $xy = yx = e$].

And there are many things to say and I leave this to you. You can say that the unit is unique [writes: [unit is unique]].

Spoken Transcript [00:45:55 - 00:46:04] Prof. Rahul Pandharipande

I will say one thing here because... [writes: [inverse is unique]] unique...

Recap: Groups were defined in Week 2 (Definition 2.1), together with the uniqueness of the neutral element and of inverses (Proposition 2.4, Proposition 2.7). The lecturer reviews them.

Definition 7.13 (*Group*): A *"group"* is a triple $(G, \cdot, e)$, where G is a set, $\cdot \colon G \times G \to G$ is a binary operation (often written $x \cdot y$ or xy), and $e \in G$ is a distinguished neutral element, satisfying the following axioms:

(G1) Associativity: For all $x, y, z \in G$:

$$(x \cdot y) \cdot z = x \cdot (y \cdot z). \tag{7.37}$$

(G2) Identity / Unit Property: The element $e \in G$ satisfies:

$$e \cdot x = x \cdot e = x \quad \forall x \in G. \tag{7.38}$$

(G3) Existence of Inverses: For every $x \in G$, there exists an element $y \in G$ such that:

$$x \cdot y = y \cdot x = e. \tag{7.39}$$

The axioms only demand existence; the neutral element and the inverse of each x are unique (Proposition 7.14, Proposition 7.16), and the inverse is denoted x^{-1}.

Uniqueness of Identity, Inverses, and Cancellation Laws

Spoken Transcript [00:46:04 - 00:48:17] Prof. Rahul Pandharipande

These proofs are all very simple.

So, the idea is that if we have both... these are both units [writes "($e, \hat{e}$ units)" under "unit is unique"], this is a simple proof: then just I take $e \cdot \hat{e}$ [writes: $e \cdot \hat{e}$]. Using the unit property of e, I get this is equal to $\hat{e}$ [writes: $= \hat{e}$]. And using the unit property of $\hat{e}$, I get this is equal to e [writes: $= e$]. So, this shows the unit is unique.

Something that we use a lot in groups is left and right cancellation [writes: Left/right cancellation:]. This says that if I have an equation $xa = xb$ [writes: $xa = xb$], then I can cancel x. This implies $a = b$ [writes: $\implies a = b$]. In a group you can always cancel.

And the reason is because I take this equation $xa = xb$ and I multiply by the inverse of x. I can then use $y = x^{-1}$ [writes: $y = x^{-1}$], and I act on this. So, I get $yxa = yxb$ [writes: $yxa = yxb$]. Then these are units [underlines yx and writes "e" underneath: $ea = eb$], it's the neutral element, and this implies $a = b$ [writes: $\implies a = b$]. So, we'll use this all the time, also on the other side. That's the left and right cancellation property.

The final thing is that... I'll say it here, I've said all these things before, but maybe just something... The inverse is unique [writes above: "inv. of x is unique:"]. That means if I have $xy = e$, and this also equals $x\hat{y}$ [writes: $xy = e = x\hat{y}$] — so, y and $\hat{y}$ are now two putative inverses — then I get $xy = x\hat{y}$ [writes: $xy = x\hat{y}$], and then I use the cancellation property [writes: $\implies y = \hat{y}$ by cancellation].

Okay, so, those are the really basic aspects of it.

Proposition 7.14 (*Uniqueness of the Identity Element*)**:** In any group $(G, \cdot, e)$, the neutral element e is uniquely determined.

Proof

Let e and $\hat{e}$ be two neutral elements in G. Consider the product $e \cdot \hat{e}$:

- Applying the identity property of e yields:
$$e \cdot \hat{e} = \hat{e}.$$
- Applying the identity property of $\hat{e}$ yields:
$$e \cdot \hat{e} = e.$$

Combining both equalities gives $e = \hat{e}$. **Q.E.D.**

Proposition 7.15 (*Left and Right Cancellation Laws*)**:** Let $(G, \cdot, e)$ be a group. For all elements $x, a, b \in G$:

$$xa = xb \implies a = b \qquad \text{(Left Cancellation)}, \tag{7.40}$$
$$ax = bx \implies a = b \qquad \text{(Right Cancellation)}. \tag{7.41}$$

Proof

Suppose $xa = xb$. By axiom **(G3)**, there exists an inverse $y := x^{-1} \in G$ satisfying $yx = e$. Multiplying on the left by y:

$$\begin{aligned} y(xa) &= y(xb) \\ (yx)a &= (yx)b && \text{(by Associativity (G1))} \\ ea &= eb && \text{(since } yx = e) \\ a &= b && \text{(by Identity (G2)).} \end{aligned}$$

Right cancellation is verified analogously by multiplying on the right by x^{-1}.

Q.E.D.

Proposition 7.16 (*Uniqueness of Inverses*)**:** For each element x in a group $(G, \cdot, e)$, the inverse element is uniquely determined.

Proof

Let y and $\hat{y}$ be two inverses of x, so that $xy = e$ and $x\hat{y} = e$. Then:

$$xy = x\hat{y}.$$

Applying left cancellation (Proposition 7.15) by x immediately implies:

$$y = \hat{y}.$$

Q.E.D.

Beyond the Lecture: A Second Proof of Uniqueness of Inverses

The lecturer proves this here with cancellation. Without cancellation as a separate step, uniqueness of the inverse follows directly from associativity, which is his argument of

Week 2 (Proposition 2.7):

$$y = y \cdot e = y \cdot (x \cdot \hat{y}) = (y \cdot x) \cdot \hat{y} = e \cdot \hat{y} = \hat{y}.$$

Contrast: Finite and Non-Commutative Groups

Spoken Transcript [00:48:17 - 00:49:46] Prof. Rahul Pandharipande

So, about groups: in some sense, it's algebra, just like about rings. Rings have more structure.

We were very much interested, because of the classical examples, primarily in rings that have infinite numbers of elements. Not exclusively, but primarily our classical examples.

Our focus on groups will be primarily on groups with a finite number of elements. So, groups can have, of course, infinite numbers of elements.

So, we will primarily be interested in groups with finite numbers of elements [writes on the lower left board: → groups with finite numbers of elements]. So, that's a different perspective than our study of rings, where I would say we were primarily interested in rings that were infinite rings.

And we will primarily be interested in groups that are not commutative [writes: → groups which are not commutative]. That's another difference.

The Two Methodological Shifts in Group Theory

The transition from ring theory to group theory involves two fundamental shifts in mathematical focus:

(1) **Finiteness versus Infinitude:** Ring theory in this course was driven by infinite classical structures ($\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, polynomials $R[x]$, and formal power series $k[[x]]$). In contrast, the study of group theory will focus primarily on *"finite groups"*.

(2) **Non-Commutativity (Non-Abelian Structures):** The rings of this course were commutative (Remark 1.17). In group theory, the focus shifts decisively to *"non-commutative"* (non-abelian) groups, where the order of factors in a product cannot be altered.

Notational Conventions and the Additive Group of a Ring

Spoken Transcript [00:49:46 - 00:51:49] Prof. Rahul Pandharipande

So, we have now for this period about groups to be extremely careful never to switch order. I was cavalier about that; I never cared about the order because our rings were commutative, but now when we apply the group operation, you have to be very careful about the order. You're not allowed to switch the order unless you have a special reason for it.

Okay. And the examples, of course, the examples we've seen before: if you take R is any ring [writes: $(R, +, \cdot, 0, 1)$], that's a ring, then you can make a group [writes: $\implies$ $(R, +, 0)$

is a group]. That was our definition of a ring (Recall: Definition 1.8): the additive structure is always a group. These are abelian groups, they're commutative groups, they won't be primarily what we're interested in, but that's a good solid example of a group.

And you see here the notation for the group, it's a good point here: I said G and dot is the standard for the operation in the group, and e is standard for the neutral element [writes: $(G, \cdot, e)$]. The neutral element is also sometimes denoted by 1 [writes "1" under e], because the dot kind of suggests multiplication, and 1 psychologically is helpful, I don't know. Sometimes people will write 1_G [writes: 1_G]. These are just accepted terms when you use dot.

But you can also name this anything you want. You could also name it plus [writes "$(G, +, 0)$" underneath]. If you name it plus, usually the neutral element is called 0 [writes "0" under e]. But this is just a question about names. You can name this binary operation anything you want, and the neutral element anything you want. If you do too crazy choices, you'll confuse everybody, but these are kind of the standard. So, that's why when I write plus, it's 0 as the neutral element.

So, there's lots of examples here.

Example 7.17 (*Additive Group of a Ring*): Let $(R, +, \cdot, 0, 1)$ be any ring. By the ring axioms (Definition 1.8), the additive structure $(R, +, 0)$ forms an abelian (commutative) group.

Notation 7.18 (*Multiplicative versus Additive Group Notation*): Depending on the context, group operations adopt multiplicative or additive conventions:

Convention	Operation	Identity Element	Inverse of x
Multiplicative (Default)	$x \cdot y$ or xy	e, 1, or 1_G	x^{-1}
Additive	$x + y$	0	$-x$

The names are a choice. In a possibly non-commutative group the order of the factors must be kept: $xy \neq yx$ in general.

Beyond the Lecture: Additive Notation

By convention, additive notation is used only for commutative groups, and a possibly non-commutative group is written multiplicatively. The neutral element is then also written 0_G.

The Symmetric Group and Associativity

Associativity and Brackets

Spoken Transcript [00:51:49 - 00:53:39] Prof. Rahul Pandharipande

The example that's more relevant for the way we think about groups, the interesting things we're going to prove about groups, is the symmetric group S_n, that's the permutation group [writes: S_n = permutation group on n elements]. I'm going to review this; I think I mentioned this in passing. Permutation group on n elements. And in some sense this is like the most interesting finite group; I will make that claim.

And you should think about these... you know, I already presented the definitions earlier, but you should think about them. There's a lot of interesting things here which we don't think... like I didn't talk at all about this associativity property. And in some sense in your education, it's kind of late to talk about associativity, because you've been using operations which were associative since childhood, and not thinking about them.

So, maybe we take a moment for like mathematical culture. If I have this product of four elements in the group [writes on the middle board: $x_1 \cdot x_2 \cdot x_3 \cdot x_4$], we just write it like this, if you're multiplying numbers, etc. But what do you mean by this? Because you're only allowed to do two elements at a time.

So, one thing you could mean is this: you do these two [writes: $(x_1 \cdot x_2)$], get an answer, and do these two [writes: $(x_3 \cdot x_4)$], get an answer, and that is equal to your answer [writes: $(x_1 \cdot x_2) \cdot (x_3 \cdot x_4)$]. That makes sense because I do these two at a time and those two at a time, right?

Or I could do these two [writes: $(x_1 \cdot x_2)$], and then I could do that one [writes: $((x_1 \cdot x_2) \cdot x_3)$], and then I could send it to that one at the end [writes: $(((x_1 \cdot x_2) \cdot x_3) \cdot x_4)$]. That's a different way of doing it.

The Problem of Multi-Element Products and Parenthesizations

A binary operation $\cdot : G \times G \to G$ is defined strictly for pairs of elements. When writing a product of four elements:

$$x_1 \cdot x_2 \cdot x_3 \cdot x_4,$$

the expression is formally ambiguous unless brackets specify the sequential order of pairwise evaluations. For example, two valid bracketings are:

$$\text{Tree evaluation by pairs:} \quad (x_1 \cdot x_2) \cdot (x_3 \cdot x_4), \tag{7.42}$$

$$\text{Iterative left-to-right evaluation:} \quad \big((x_1 \cdot x_2) \cdot x_3\big) \cdot x_4. \tag{7.43}$$

Catalan Numbers and the Combinatorics of Bracketing

Spoken Transcript [00:53:39 - 00:55:39] Prof. Rahul Pandharipande

And the associativity axiom says that these two are the same, which you prove by induction. Which probably you've never done, or I don't know whether you've done it or not. Unless you're really deeply paranoid, you probably have never done that. But it might be fun to do it.

And if you do it, you discover some beautiful things.

So, this is now nothing to do with the class, but mathematical culture. You can ask: how many different ways can I put these brackets? [writes on the right board: "How many ways can I put these brackets?"] Meaning if I have $x_1 \ldots x_n$ [writes: $x_1 \ldots x_n$], I can group them, I can do this operation in many different ways. How many ways?

And that has an extremely famous answer, which is part of math culture, which I am going to write. The hint to this is... I'm kind of really bad at languages, but anyway, it's... [writes on the board in Catalan words: "u, u, dos, cinc, catorze, ..."] maybe like that.

Does anyone recognize these numbers? No one? Aren't there any Catalan speakers in this room? Does anyone speak Catalan? You speak Catalan? So, you'll not recognize if I got it right. Spanish, Spanish is close enough, but... You seriously don't recognize these numbers?

Well, this was supposed to be a huge hint: $1, 1, 2, 5, 14$ [writes digits: $1, 1, 2, 5, 14$ underneath].

So, these are the Catalan numbers. That's actually Catalan, that's my attempt to write them in Catalan. They don't come from the language; there's a French mathematician named Catalan.

Spoken Transcript [00:55:39 - 00:57:49] Prof. Rahul Pandharipande

The formula for this is $C_n = \frac{1}{n+1}\binom{2n}{n}$ [writes: $C_n = \frac{1}{n+1}\binom{2n}{n}$], and these are some of the most famous numbers in mathematics. Which doesn't mean... they're some of the most famous numbers in mathematics, but they probably don't occur in any course, I don't know.

So, that's the n-th Catalan number. It's a totally beautiful object. And if you solve this problem correctly, the number of ways to bracket n things, this is equal to C_{n-1} [writes "C_{n-1}" under $x_1 \ldots x_n$]. It's the $(n-1)$-st Catalan number.

Anyway. So, that's already just trying to unpack associativity, what it means. If you do it, and you follow your nose, eventually you get to the Catalan numbers.

But have you seen the definition of Catalan numbers before? Yeah. That's a tragedy.

Well, I don't think they occur in any class, I don't know. Maybe they do. Hard to say. You can't actually learn mathematics by going to class, that's the other problem.

So, the number of ways you can do three things is 2, and the number of ways you can do four things is 5 [points at the equations for 4 elements]. That's what they are... [takes the sponge and cleans the middle and right boards completely]

Okay, so, then there's other things that also follow from what I already explained. If I have a group... but let me just... so, if I have a group, another way to think about the operations: there's this binary operation, and we have the neutral element, and it's nice to think also about inverse being a map, inverse from the group to the group [writes on the lower left board: G, $\cdot$ binary, $e \in G$, $\text{inv}\colon G \to G, x \mapsto x^{-1}$]. Meaning if I have x, I have a unique inverse, x inverse. So, that's a function from the group to a group: it takes an element to its inverse, right?

Theorem 7.19 (*Associativity for n Elements*): Let $(G,\cdot,e)$ be a group. Any product $x_1 \cdot x_2 \ldots x_n$ of $n \geq 3$ elements in a fixed order evaluates to the exact same element in G, regardless of the placement of parentheses.

Combinatorics of Parenthesizations via Catalan Numbers: Although the final product is independent of parentheses by Theorem 7.19, the number of distinct formal ways to fully parenthesize a product of n factors is given by the $(n-1)$-th *"Catalan number"*:

$$\#(\text{Parenthesizations of } x_1 \ldots x_n) = C_{n-1}.$$

The Catalan numbers are defined by:

$$C_n := \frac{1}{n+1}\binom{2n}{n} = \frac{(2n)!}{(n+1)!\,n!}. \tag{7.44}$$

The first values are:

n	0	1	2	3	4	5
C_n	1	1	2	5	14	42

- For $n = 3$ elements $(x_1x_2x_3)$, there are $C_2 = 2$ bracketings:
$$(x_1x_2)x_3 \quad \text{and} \quad x_1(x_2x_3).$$
- For $n = 4$ elements $(x_1x_2x_3x_4)$, there are $C_3 = 5$ bracketings:
$$((x_1x_2)x_3)x_4, \quad (x_1(x_2x_3))x_4, \quad (x_1x_2)(x_3x_4), \quad x_1((x_2x_3)x_4), \quad x_1(x_2(x_3x_4)).$$

Definition 7.20 (*The Inverse Map*): In addition to the binary operation $\cdot\colon G \times G \to G$ and the neutral element $e \in G$, a group comes with the *"inverse map"*:

$$\operatorname{inv}\colon G \to G, \quad x \mapsto x^{-1}. \tag{7.45}$$

Beyond the Lecture: Generalized Associativity

Textbooks call Theorem 7.19 the generalized associative law.

Beyond the Lecture: Catalan Numbers and Eugène Charles Catalan

The Catalan numbers are named after the Belgian/French mathematician Eugène Charles Catalan (1814–1894). In combinatorics, C_n also counts the number of Dyck paths of length $2n$, rooted binary trees with n internal vertices, and triangulations of a convex polygon with $n+2$ vertices.

Subgroups and Group Homomorphisms

Definition of a Subgroup

Spoken Transcript [00:57:49 - 00:59:44] Prof. Rahul Pandharipande

And so, there's a notion of a subgroup [writes: Subgroup: $S \subset G$]. So, it's a subset, and it should be closed under the operation, meaning if I take two elements of the subset and I use the group operation, I stay in it [writes: $\cdot\colon S \times S \to S$]. It should contain the neutral element, so, this should be $S \times S \to S$, and the neutral element should be in S [writes: $e \in S$]. And one more thing is S should have inverses: that inverse maps S to S [writes: $\mathrm{inv}\colon S \to S$]. So, every element in S has its inverse in S. That's what a subgroup is. Okay?

And we have a notion of group homomorphisms [writes on the middle board: Homs:]. If $(G_1, \cdot, e_1)$ and $(G_2, \cdot, e_2)$ are groups [writes: $(G_1, \cdot, e_1), (G_2, \cdot, e_2)$ are groups], then f from G_1 to G_2 [writes: $f\colon G_1 \to G_2$] is — this is a definition — is a group homomorphism [writes: is a group hom.] (when I underline things I always mean definition), is a group homomorphism if $f(xy) = f(x)f(y)$ [writes: if $f(xy) = f(x)f(y)$]. So, here I use the group operation on G_1, and here I use the group operation on G_2 [underlines the operations in G_1 and G_2].

So, that's the whole definition, just this condition. So, you could wonder: what happens to the neutral element?

Recap: Subgroups and group homomorphisms were defined in Week 2 (Definition 2.19, Definition 2.10), where $f(e_1) = e_2$ was also shown (Proposition 2.11).

Definition 7.21 (*Subgroup*)**:** Let $(G, \cdot, e)$ be a group. A subset $S \subseteq G$ is called a *"subgroup"* of G (denoted $S \leq G$) if it satisfies the following three conditions:

(S1) Closure under Multiplication: For all $x, y \in S$, the product lies in S:

$$\cdot\colon S \times S \to S \quad (\text{i.e., } x, y \in S \implies x \cdot y \in S).$$

(S2) Identity Containment: The neutral element belongs to S:

$$e \in S.$$

(S3) Closure under Inverses: For each $x \in S$, its inverse lies in S:

$$\mathrm{inv}\colon S \to S \quad (\text{i.e., } x \in S \implies x^{-1} \in S).$$

Definition 7.22 (*Group Homomorphism*)**:** Let $(G_1, \cdot_{G_1}, e_1)$ and $(G_2, \cdot_{G_2}, e_2)$ be groups. A map $f\colon G_1 \to G_2$ is called a *"group homomorphism"* if it respects the group operations:

$$f(x \cdot_{G_1} y) = f(x) \cdot_{G_2} f(y) \quad \forall x, y \in G_1. \tag{7.46}$$

Preservation of Identities and Inverses

Spoken Transcript [00:59:44 - 01:01:49] Prof. Rahul Pandharipande

So, what happens to e_1? [writes: What happens to e_1?]

So, I just plug it in: $f(e_1e_1)$ has to be $f(e_1)f(e_1)$, right? [writes: $f(e_1e_1) = f(e_1)f(e_1)$] But this thing, by using the unit property, is equal to $f(e_1)$, because $e_1e_1 = e_1$, right? [writes: $f(e_1) = f(e_1)f(e_1)$] And then I can use cancellation.

So, I look at this equation, I just write it: $f(e_1) = f(e_1)f(e_1)$ [writes: $f(e_1) = f(e_1)f(e_1)$]. So, this is a group equation in G_2, right? Everything's in G_2. But now I can cancel. If I want, with cancellation I could put an e_2 here [writes: $f(e_1)e_2 = f(e_1)f(e_1)$], because that uses the identity in e_2, and then I can cancel [cancels $f(e_1)$ on both sides].

So, that's the answer: what happens to the neutral element? It has to go to the neutral element in the second one [writes "$\implies e_2 = f(e_1)$" and draws a frame around it]. Okay? It's implied by this condition. I don't have to impose it as an additional condition. I could, it would be harmless to do it, but I don't have to do it.

Then I leave for you to check yourself: what happens to f of an inverse? [writes on the right board: What happens to $f(x^{-1})$?] And the answer, it's just the same argument. Answer is that f of an inverse is equal to the inverse here: $f(x^{-1}) = f(x)^{-1}$ [writes: Answer: $f(x^{-1}) = f(x)^{-1}$ and boxes it]. I leave this for you to check, okay? It's the same argument. It respects the inverse operation also.

So, we're going to study the kernels and images just as before for groups, and that'll lead to a theorem which is parallel to what we called the First Isomorphism Theorem (Recall: Theorem 4.27). But we're not doing that today.

Proposition 7.23 (*Preservation Properties of Homomorphisms*): Let $f\colon G_1 \to G_2$ be a group homomorphism. Then f automatically preserves identity elements and inverses:

$$f(e_1) = e_2, \tag{7.47}$$

$$f(x^{-1}) = \big(f(x)\big)^{-1} \quad \forall x \in G_1. \tag{7.48}$$

Proof

Since $e_1 \cdot e_1 = e_1$ in G_1, applying the homomorphism property (7.46) gives:

$$f(e_1) = f(e_1 \cdot e_1) = f(e_1) \cdot f(e_1).$$

Rewriting the left-hand side as $e_2 \cdot f(e_1)$ in G_2:

$$e_2 \cdot f(e_1) = f(e_1) \cdot f(e_1).$$

By right cancellation in G_2 (Proposition 7.15):

$$e_2 = f(e_1).$$

This proves (7.47). The lecturer leaves (7.48) to the students: it is the same argument.

Q.E.D.

Preservation of Inverses (beyond the lecture)

The lecture leaves this to the students. For any $x \in G_1$, by definition $x \cdot x^{-1} = e_1$. Applying f:

$$f(x) \cdot f(x^{-1}) = f(x \cdot x^{-1}) = f(e_1) = e_2.$$

Similarly, $f(x^{-1}) \cdot f(x) = f(x^{-1} \cdot x) = f(e_1) = e_2$. By the uniqueness of the inverse of $f(x)$ in G_2 (Proposition 7.16), we conclude:

$$f(x^{-1}) = \big(f(x)\big)^{-1}.$$

Q.E.D.

The Symmetric Group S_n

Definition and Concrete Examples of S_1, S_2, S_3

Spoken Transcript [01:01:49 - 01:03:59] Prof. Rahul Pandharipande

[washes the upper middle board with a sponge] I thought I would do some example, think about the symmetric group, because it's totally crucial to understand that for your future mathematical life. But the symmetric group must have come up in many classes of yours, I would guess. But I won't assume you know anything about it. [washes the remaining lower boards with a sponge]

So, if we start formally with the discussion of the symmetric group... And this is in the trichotomy I said about the subject: this is not definition, this is not theorem, it's example. But it's an incredibly important example: the symmetric group [writes on the lower left board: "Symmetric groups:"].

So, the first symbol which is kind of standard is this square brackets n [writes: $[n] = \{1, \ldots, n\}$], to be the set of numbers from 1 to n. It's the n-element set, so it's exactly the integers from 1 to n.

And the symmetric group S_n equals the set of functions f from $[n]$ to $[n]$ such that f is bijective [writes: $S_n = \{f \colon [n] \to [n] \mid f \text{ is bijection}\}$]. That's as a set. So, it's just the set of bijections. I hope that's clear.

I need to tell you the operation, and the operation is composition of functions [writes: $(S_n, \circ, \mathrm{id})$ def]. And I have to give you the neutral element, and it's the identity. That's the identity: there's a unique identity which takes 1 to 1, 2 to 2, and so on. And that is the definition of the symmetric group. That's the full definition, and you have to think about it, make sure you understand.

Notation 7.24 (*Standard n-Element Set*)**:** For any integer $n \geq 1$, we denote by $[n]$ the canonical n-element set:

$$[n] := \{1, 2, \ldots, n\}. \tag{7.49}$$

Definition 7.25 (*Symmetric Group S_n*)**:** The *"symmetric group on n elements"*, denoted S_n, is the group whose underlying set is the collection of all bijections (permutations) of

$[n]$ onto itself:

$$S_n := \{f\colon [n] \to [n] \mid f \text{ is bijective}\}, \tag{7.50}$$

equipped with the binary operation of function composition $\circ$ and identity element $\mathrm{Id}_{[n]}$:

$$(S_n, \circ, \mathrm{Id}_{[n]}).$$

Verification of Group Axioms for S_n (beyond the lecture)

The lecture gives the definition and leaves the group axioms to the students. The triple $(S_n, \circ, \mathrm{Id}_{[n]})$ satisfies the group axioms:

- **Closure:** The composition of two bijections is a bijection: $f, g \in S_n \implies f \circ g \in S_n$.
- **Associativity:** Function composition is universally associative: $(f \circ g) \circ h = f \circ (g \circ h)$.
- **Identity:** The identity function $\mathrm{Id}_{[n]}(i) = i$ satisfies $\mathrm{Id}_{[n]} \circ f = f \circ \mathrm{Id}_{[n]} = f$ for all $f \in S_n$.
- **Inverses:** Every bijection f admits a unique two-sided inverse function $f^{-1} \in S_n$.

Q.E.D.

III Composition Order and Diagrams of Bijections

Spoken Transcript [01:03:59 - 01:05:31] Prof. Rahul Pandharipande

One of the important things to think about is: what does composition mean? If I take f composed with g [writes: $f \circ g$], where g is from $[n]$ to $[n]$ and f is from $[n]$ to $[n]$ [writes: $g\colon [n] \to [n]$, $f\colon [n] \to [n]$]. This means... so, this acts on some number [writes: $f \circ g(i)$]. And the answer is you do g first, and then you do f [writes: $= f(g(i))$]. That's the definition of composition, right?

Even though, you know, it's kind of... you can confuse yourself by this, because you see f first. But nevertheless, the mathematical convention for this symbol is you do g first, then you do f. Okay?

So, that's the full definition of the group. And you could say, have I checked the axioms? Well, you're supposed to understand that composition of bijective functions gives you a bijective function, you have to also say that bijective functions have an inverse, and you have to show that composition is associative. I leave all of this to you. None of those are surprises, I think.

So, that's the definition of S_n. But then there's a question about how to think about it. And that's an interesting question, that's a question about names: how are we going to talk about elements of S_n?

So, the primitive way is say like if you're taking S_1 [writes: S_1]: there's nothing to do, S_1 has one element [writes: $S_1 = \{1 \to 1\}$], because it's the number of bijections from 1 to 1. So, you could say, okay, I could say: here's 1 and here's the map from 1 to 1 [draws bipartite dots 1 and 1 and horizontal arrow $1 \to 1$], it's a bijection, it has one element. That's not so exciting.

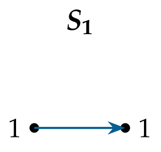
In S_2 [writes: S_2], we have bijections from the set of two elements. And one bijection is the identity. Do you accept this drawing as a function? 1 goes to 1, 2 goes to 2 [draws two columns of dots $1, 2$ and parallel horizontal lines $1 \to 1, 2 \to 2$]. That's one element. Another element is 1 goes to 2 and 2 goes to 1 [draws two columns $1, 2$ and crossing lines $1 \to 2, 2 \to 1$]. And that's pretty much the only two bijections, right?

But this is kind of unfortunate: if we're going to have to draw a picture for every element, that's kind of unfortunate. Like S_3 [writes: S_3], things get more complicated. You could have $1, 2, 3$ [draws two columns $1, 2, 3$ and $1, 2, 3$]. You could have like 1 goes to 2, 2 goes to 3, 3 goes to 1 [draws lines $1 \to 2, 2 \to 3, 3 \to 1$]. That's a nice element. But you can imagine this is not a very good way of naming the function by just giving its full... somehow it takes too much chalk, let's say. So, this is one element.

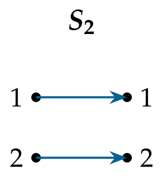
Remark 7.26 (*The Right-to-Left Convention of Composition*): When evaluating a composition $f \circ g$ on an input $x \in [n]$, the inner function g is evaluated first:

$$(f \circ g)(x) := f\big(g(x)\big). \tag{7.51}$$

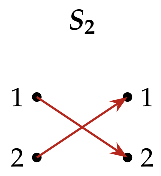
Although f appears on the left in $f \circ g$, the mapping sequence proceeds right-to-left: $[n] \xrightarrow{g} [n] \xrightarrow{f} [n]$.



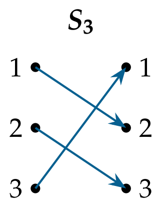
$|S_1| = 1$



identity



swap $1 \leftrightarrow 2$



$1 \mapsto 2,\ 2 \mapsto 3,\ 3 \mapsto 1$

Limitations of Bipartite String Diagrams: Going beyond the lecture: representing permutations as bipartite matching graphs provides immediate visual clarity for small n (such as S_1 and S_2). However, for larger n, this diagrammatic representation quickly becomes unmanageable: it consumes excessive blackboard space and obscures algebraic properties such as element orders, parity, and products. This motivates the introduction of a compact algebraic notation: *"cycle notation"*.

Order of S_n and Prelude to Cycle Decomposition

Spoken Transcript [01:05:31 - 01:06:07] Prof. Rahul Pandharipande

I should say that it should be clear that S_n is a finite group [writes: S_n is finite group], and its order, which is given by... the standard notation for order is putting these straight lines, is n factorial [writes: $|S_n| = n!$]. That's how many bijections there are, and that's a standard formula because you can take the first one to any n, and then that position is occupied, the second one has to go to $n - 1$, and so on. That's kind of a proof.

Proposition 7.27 (*Order of S_n*): For every integer $n \geq 1$, the symmetric group S_n is a finite group of order:

$$|S_n| = n! = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1. \tag{7.52}$$

Proof

An element $f \in S_n$ is uniquely determined by its values on $[n] = \{1, 2, \ldots, n\}$:

- There are n possible choices for $f(1) \in [n]$.
- Since f must be injective, $f(2)$ can be any element of $[n] \setminus \{f(1)\}$, yielding $n-1$ choices.
- Inductively, for each $k \in \{1, \ldots, n\}$, there remain $n-(k-1)$ choices for $f(k)$.

By the fundamental counting principle, the total number of bijections is:

$$|S_n| = n \cdot (n-1) \cdot (n-2) \cdots 1 = n!$$

Q.E.D.

Motivation for Cycle Decomposition: Going beyond the lecture: since $|S_n| = n!$ grows super-exponentially ($|S_3| = 6$, $|S_4| = 24$, $|S_5| = 120$, $|S_{10}| = 3{,}628{,}800$), working with two-line Cauchy arrays or bipartite graphs is practically infeasible. The canonical algebraic language for studying permutations is the *"cycle decomposition"*, which factors each permutation into disjoint cycles.

Spoken Transcript [01:06:07 - 01:07:01] Prof. Rahul Pandharipande

Okay, so, the question is: how are we going to talk about these elements of the symmetric group? And there's a standard way, which is about the cycle decomposition. So, I'll try to write it, and it's really useful notation. In some sense, it's again about names. We need to find some nice way to name these elements that will allow us to manipulate them, both in our head and on the board and on paper.

Logically, one doesn't have to do this. Logically, you can view those diagrams as the data type. [washes the middle board with a sponge]

But there is a standard way to do it, which is the cycle decomposition, and once you understand it — and maybe you already understand it — you'll find this a very nice way to do it.

Cycle Decomposition Notation

Spoken Transcript [01:07:01 - 01:08:35] Prof. Rahul Pandharipande

So... how to name elements of S_n [writes: How to name elements of S_n].

And the standard answer is: use the cycle decomposition [writes: Cycle decomp. notation].

And how this works, so, let me do an example. I'll do it in a couple of examples and then you should be able to see.

So, if I take the three case, so, let's do this one: 1 goes to 2, 2 goes to 1, 3 goes to 3. So, this is one of the bijections. I flip 1 and 2, and 3 stays the same, okay? [draws bipartite matching for $(1\,2)(3)$ on the board]

And the way to represent it in the cycle decomposition is I start by... I open parentheses [writes: (], and I put 1 here [writes: 1]. And then next to 1, I write where 1 goes. So, where does 1 go? Goes to 2 [writes: 2]. Okay.

And then I look, and where does 2 go? Well, 2 goes back to 1, so, then that closes the cycle [writes:)].

And then now I take the first thing that's not here, which is 3. So, I open the parentheses, and I see where 3 goes, but 3 goes back to itself. So, the cycle decomposition of this would be that: $(1\,2)(3)$ [writes: (3)]. All right?

Definition 7.28 (*Cycle*): A permutation $\sigma \in S_n$ is called a *"k-cycle"* (or cycle of length k), denoted $(i_1\,i_2\,\ldots\,i_k)$, if:

$$\sigma(i_1) = i_2, \quad \sigma(i_2) = i_3, \quad \ldots, \quad \sigma(i_{k-1}) = i_k, \quad \sigma(i_k) = i_1,$$

while fixing all other elements: $\sigma(j) = j$ for all $j \in [n] \setminus \{i_1, \ldots, i_k\}$.

Example 7.29 (*A 2-Cycle in S_3*): Consider the permutation $\sigma \in S_3$ swapping 1 and 2 while fixing 3:

$$\sigma(1) = 2, \qquad \sigma(2) = 1, \qquad \sigma(3) = 3.$$

Traced into cycle notation:

- Start with 1: $1 \mapsto 2$, and $2 \mapsto 1$, closing the cycle: $(1\,2)$.
- Move to the remaining element 3: $3 \mapsto 3$, closing the 1-cycle: (3).

Thus, the complete disjoint cycle decomposition is $(1\,2)(3)$.

Spoken Transcript [01:08:35 - 01:10:01] Prof. Rahul Pandharipande

Another example: the one I had there, this one [draws bipartite matching for $1 \to 2, 2 \to 3, 3 \to 1$].

So, again, I start, I open the parentheses, and there's 1 [writes: (1]. 1 goes to 2 again [writes: 2]. Now 2 goes to 3 [writes: 3]. And then 3 goes back to 1, so, I close the cycle [writes:)].

So, if I write down now what is S_3, I can write down all the cycle decompositions. The first one is the trivial one, that's where no one goes anywhere [writes: $S_3 = \{(1)(2)(3),$].

Then there's these ones: $(1\,2)(3)$, $(1\,3)(2)$, $(2\,3)(1)$ [writes: $(1\,2)(3), (1\,3)(2), (1)(2\,3)$]. And that's already four, I need two more. And there's $(1\,2\,3)$ and $(1\,3\,2)$ [writes: $(1\,2\,3), (1\,3\,2)\}$]. So, here I've listed all six elements.

So, do you understand this notation? Like if I pick this, these cycles just tell you where everything goes. Like if I take this one, and I want to know where 2 goes, I find 2, and then it goes to the next one. And if I want to know 3, whenever the cycle ends, it means you go to the first one. And if you're by yourself, it means you don't go anywhere, you're a fixed point.

Example 7.30 (*Complete List of Elements in S_3*)**:** Since $|S_3| = 3! = 6$, the symmetric group S_3 consists of exactly six permutations, partitioned by cycle structure:

(1) Identity Permutation (Three 1-cycles):

$$\mathrm{Id} = (1)(2)(3).$$

(2) One 2-cycle and One 1-cycle:

$$(1\,2)(3), \qquad (1\,3)(2), \qquad (2\,3)(1).$$

(3) 3-Cycles (One 3-cycle):

$$(1\,2\,3), \qquad (1\,3\,2).$$

Thus, explicitly:

$$S_3 = \{(1)(2)(3),\ (1\,2)(3),\ (1\,3)(2),\ (2\,3)(1),\ (1\,2\,3),\ (1\,3\,2)\}. \tag{7.53}$$

Spoken Transcript [01:10:01 - 01:11:13] Prof. Rahul Pandharipande

So, are you satisfied with this? You have to play with it. But it's much more efficient, because I've already written for you the six elements in S_3.

Traditionally, I mean, the custom is you don't write these fixed points, because they're kind of a waste of chalk also. So, the standard is that if they don't appear, they're fixed.

So, the standard way I would write this [writes: Standard:], would be that I would just call it e, the neutral element [writes: e]. And this I'd say $(1\,2)$, $(1\,3)$, $(2\,3)$, $(1\,2\,3)$, $(1\,3\,2)$ [writes: $(1\,2), (1\,3), (2\,3), (1\,2\,3), (1\,3\,2)$]. That would be the standard. If they're not there, it means that nothing's happening to them. That's kind of the standard way to represent it.

And, you know, this way I picked some way to break symmetry, but you could also write this $(3\,2)$ as $(2\,3)$, because 2 goes to 3 and 3 goes to 2, and here 3 goes to 2, 2 goes to 3. You can also write it like that if you want [writes: $(2\,3) = (3\,2)$]. But this is a canonical way of doing it. All right.

Notation 7.31 (*Suppression of 1-Cycles*)**:** In standard mathematical practice, 1-cycles (fixed points) are omitted from the cycle decomposition:

- The identity permutation is denoted e or Id.
- A 2-cycle with a fixed point, $(i\,j)(k)$, is written compactly as $(i\,j)$, with the tacit understanding that k is fixed.

Under this standard convention, the elements of S_3 are recorded as:

$$S_3 = \{e,\ (1\,2),\ (1\,3),\ (2\,3),\ (1\,2\,3),\ (1\,3\,2)\}. \tag{7.54}$$

Furthermore, cycles are invariant under cyclic shifts of their entries:

$$(2\,3) = (3\,2), \qquad (1\,2\,3) = (2\,3\,1) = (3\,1\,2).$$

Notation: Transpositions

Many books call a 2-cycle a **transposition**. The lecturer says two-cycle.

Multiplication and Inverses in Cycle Notation

Spoken Transcript [01:11:13 - 01:11:31] Prof. Rahul Pandharipande

So, that works, of course, for any n, and the rule is if you have this kind of function, it tells you the cycle decomposition; if you have the cycle decomposition, it tells you the function. But it's much more efficient. And so, let us try to...

In this notation, what is the inverse map? [writes: Inv:]

So, of course, the inverse of e is, of course, e, right? [writes: $e \mapsto e$] Neutral element. What's the inverse of this element: $(1\,2)$? [points at $(1\,2)$]

Student Answer

Itself.

Spoken Transcript [01:11:32 - 01:12:15] Prof. Rahul Pandharipande

Yeah, itself [writes: $(1\,2) \mapsto (1\,2)$]. If I apply it twice... so, now I have to tell you how to multiply in the cycle notation. So, here I take $(1\,2)$, that's one time... I can just put the cycles together, that's it, I just put them together [writes: $(1\,2)(1\,2) =$]. And now I have to figure out what cycle to write here.

So, then I have to find out where 1... I open it up, I find out where 1 goes. 1 goes to 2, and then I go to the next one, now 2 goes back to 1, so 1 stays by itself [writes: (1)]. And I start here: 2. 2 goes to 1, 1 goes back to 2, so 2 goes to 2 [writes: (2)]. That's the identity [writes: $= (1)(2) = e$].

Also, you can just see it in functions. But there's just rules. So, of course, these are all the same, so, this is the inverse, that's the inverse [writes: $(1\,3) \mapsto (1\,3), (2\,3) \mapsto (2\,3)$].

What's the inverse of this person: $(1\,2\,3)$?

Student Answer

$(1\,3\,2)$.

Spoken Transcript [01:12:16 - 01:12:31] Prof. Rahul Pandharipande

Yeah, the other one. So, they flip around here: $(1\,3\,2)$ and $(1\,2\,3)$ [writes: $(1\,2\,3) \mapsto (1\,3\,2)$ and $(1\,3\,2) \mapsto (1\,2\,3)$]. So, that's the inverse map.

Example 7.32 (*The Inverse Map on S_3*)**:** The neutral element is its own inverse. For the other elements of S_3:

- The 2-cycles are their own inverses:

$$(1\,2)^{-1} = (1\,2), \qquad (1\,3)^{-1} = (1\,3), \qquad (2\,3)^{-1} = (2\,3).$$

- The two 3-cycles are inverse to each other:

$$(1\,2\,3)^{-1} = (1\,3\,2), \qquad (1\,3\,2)^{-1} = (1\,2\,3).$$

Beyond the Lecture: The Inverse of a Cycle

The inverse of any cycle $(i_1\, i_2\, \dots\, i_k)$ is obtained by reversing the order of its entries:

$$(i_1\, i_2\, \dots\, i_k)^{-1} = (i_k\, i_{k-1}\, \dots\, i_1).$$

For example, $(1\,2\,3)^{-1} = (3\,2\,1) = (1\,3\,2)$.

Spoken Transcript [01:12:31 - 01:14:49] Prof. Rahul Pandharipande

So, these are more complicated: $(1\,2\,3)$, let us say $(1\,3\,2)$... So, let's do this one slowly, just for fun [writes: $(1\,2\,3)(1\,3\,2) =$]. I don't know how fun it is, but anyway: to figure out where 1 goes, you start with 1 in the first cycle you see where 1 is, and this cycle tells you 1 goes to 3. And then you keep going left: oh, you still see a 3, so 3 is going somewhere, 3 goes back to 1, that's kind of nice. So, that's 1, and you'll see $(2)(3)$, so, it's equal to e [writes: $= (1)(2)(3) = e$].

Okay, let's try some... so, that's the inverse map. But we should be able to multiply anything, like what is $(1\,3)$ times $(2\,3)$? [writes: $(1\,3)(2\,3) =$]

So, to do this, as I said, you start out with 1. 1 doesn't do anything here, but it goes to 3 over there. So, now we do 3. 3 goes to 2, 2 doesn't go anywhere here, right? So, it stays at 2.

2 goes to 3, and 3 goes to 1, so, it's that cycle: $(1\,3\,2)$ [writes: $= (1\,3\,2)$]. That's the product.

And as I said, this is notation, you have to see how it works. You could, if you're suspicious about this notation, every time you see this, make one of those diagrams and put them together. But it's not a good idea. It's better to understand this notation. That's how people actually work with the cycles.

So, an exercise that you should do if you've never seen it... but have you seen this kind of cycle notation in other classes? Some people have, some people haven't. It's just notation, and there's not much more I can say. You have to become familiar with it yourself.

Example 7.33 (*Computing the Product of Permutations in Cycle Notation*): To compute the product of two permutations, we evaluate the cycles from right to left:

(1) Verification that $(1\,2\,3)(1\,3\,2) = e$:

- Under $(1\,3\,2)$, $1 \mapsto 3$. Then under $(1\,2\,3)$, $3 \mapsto 1$. Thus, $1 \mapsto 1$.
- Under $(1\,3\,2)$, $2 \mapsto 1$. Then under $(1\,2\,3)$, $1 \mapsto 2$. Thus, $2 \mapsto 2$.
- Under $(1\,3\,2)$, $3 \mapsto 2$. Then under $(1\,2\,3)$, $2 \mapsto 3$. Thus, $3 \mapsto 3$.

Hence, $(1\,2\,3)(1\,3\,2) = (1)(2)(3) = e$.

(2) Computing $(1\,3)(2\,3)$:

- Track 1: $(2\,3)$ fixes 1; then $(1\,3)$ sends $1 \mapsto 3$. Thus, $1 \mapsto 3$.
- Track 3: $(2\,3)$ sends $3 \mapsto 2$; then $(1\,3)$ fixes 2. Thus, $3 \mapsto 2$.
- Track 2: $(2\,3)$ sends $2 \mapsto 3$; then $(1\,3)$ sends $3 \mapsto 1$. Thus, $2 \mapsto 1$, closing the cycle.

Combining these transitions yields the 3-cycle:

$$(1\,3)(2\,3) = (1\,3\,2). \tag{7.55}$$

Spoken Transcript [01:14:49 - 01:15:08] Prof. Rahul Pandharipande

What if I do it in a different way: $(2\,3)(1\,3) =$ [writes: $(2\,3)(1\,3) =$]

So, now you tell me: 1, where does 1 go?

Student Answer

2.

Spoken Transcript [01:15:09 - 01:15:58] Prof. Rahul Pandharipande

Yeah, 1 goes to 2, because 1 goes to 3, then 3 gets bumped to 2.

And where does 2 go? Well, 2 doesn't see anything here, but then gets bumped to 3.

And then 3 gets bumped to 1 here, so, that's it: $(1\,2\,3)$ [writes: $= (1\,2\,3)$].

So, actually, this already shows that it's not commutative, because you got two different elements when you do them in different order [underlines $(1\,3\,2)$ and $(1\,2\,3)$].

So, what I advocate, what I was trying to say, is I advocate if you are learning this cycle notation is to go home and write the multiplication table of S_3.

Observation 7.34 (*Non-Commutativity in S_3*): Reversing the order of multiplication of the 2-cycles $(1\,3)$ and $(2\,3)$ yields:

$$(2\,3)(1\,3) = (1\,2\,3). \tag{7.56}$$

Comparing (7.55) and (7.56):

$$(1\,3)(2\,3) = (1\,3\,2) \neq (1\,2\,3) = (2\,3)(1\,3).$$

Consequently, S_3 is a non-abelian (non-commutative) group.

Beyond the Lecture: The Smallest Non-Abelian Group

S_3, of order 6, is the smallest non-abelian group.

Spoken Transcript [01:15:58 - 01:16:58] Prof. Rahul Pandharipande

So, for S_3, it's not that many. I mean, most of them we've already done. For S_3, the multiplication table, of course, is this table where I write the elements of S_3 here, and I write dot, and then I write them again here [draws the header of the multiplication table on the middle board].

So, there's 36 elements, and you just fill it out. That's a good exercise. Most of them we've already done actually. After you do that, you'll understand.

You could do it for S_4 too, but it has more elements. That has 576 elements; you might take the afternoon off and do that. [washes the middle board with a sponge]

Exercise 7.35 (*Multiplication Table of S_3*): Write down the full multiplication table of S_3, with its 36 entries.

Beyond the Lecture: The Multiplication Table of S_3

The solution of Exercise 7.35, also called the Cayley table of S_3, with the entry in row σ and column τ equal to $\sigma \circ \tau$ (first τ, then σ):

$\circ$	e	$(1\,2)$	$(1\,3)$	$(2\,3)$	$(1\,2\,3)$	$(1\,3\,2)$
e	e	$(1\,2)$	$(1\,3)$	$(2\,3)$	$(1\,2\,3)$	$(1\,3\,2)$
$(1\,2)$	$(1\,2)$	e	$(1\,3\,2)$	$(1\,2\,3)$	$(2\,3)$	$(1\,3)$
$(1\,3)$	$(1\,3)$	$(1\,2\,3)$	e	$(1\,3\,2)$	$(1\,2)$	$(2\,3)$
$(2\,3)$	$(2\,3)$	$(1\,3\,2)$	$(1\,2\,3)$	e	$(1\,3)$	$(1\,2)$
$(1\,2\,3)$	$(1\,2\,3)$	$(1\,3)$	$(2\,3)$	$(1\,2)$	$(1\,3\,2)$	e
$(1\,3\,2)$	$(1\,3\,2)$	$(2\,3)$	$(1\,2)$	$(1\,3)$	e	$(1\,2\,3)$

Spoken Transcript [01:16:58 - 01:18:11]

Prof. Rahul Pandharipande

But one way or the other, you should make sure you understand what this symmetric group is, how to think about its elements, how to compose. It's kind of an important part of this class.

And whatever direction your future mathematical life will go, it will have the symmetric group. It's like one of the really basic objects in mathematics. You cannot get away from the symmetric group.

Insight: The Universality of the Symmetric Group

The lecturer emphasizes that the symmetric group S_n is inescapable in mathematics: whatever direction one's mathematical life takes, it will have the symmetric group.

Beyond the Lecture: Cayley's Theorem

This is made precise by Cayley's Theorem: every finite group of order n is isomorphic to a subgroup of S_n. Thus, permutations capture the complete structural diversity of all finite group operations.

Orders of Elements in Finite Groups

Powers of Elements and the Pigeonhole Principle

Spoken Transcript [01:18:11 - 01:19:33] Prof. Rahul Pandharipande

Okay, so, let G be a finite group [writes: Let G be a finite group].

So, this will be the assumption for many statements now for the next couple weeks.

And always by this, you know, I talk about this $(G, \cdot, e)$ [writes: $(G, \cdot, e)$], and I do not assume it's commutative [writes: but I do not assume commutative!]. I never assume it — I mean, sorry, I never silently assume it. Sometimes I'll say *"let G be a commutative group"*, but I'll never silently assume it. If I just say a group, it means that I am considering in my head a group possibly non-commutative.

I should say, and I made this definition before, that a commutative group, when $x \cdot y = y \cdot x$ for all x, y, then we say G is abelian [writes: When $x \cdot y = y \cdot x \quad \forall x, y$ then G is Abelian]. That's the standard: a commutative group, the definition, it's abelian. So, as I said, if I say G is a group, you should not assume it's abelian.

Remark 7.36 (*Standing Convention: Groups are not Silently Commutative*): Throughout the subsequent study of group theory:

- A generic group $(G, \cdot, e)$ is **never** assumed to be commutative unless explicitly designated as such.
- A group is called *"abelian"* (or commutative) if:

$$x \cdot y = y \cdot x \quad \forall x, y \in G. \tag{7.57}$$

Spoken Transcript [01:19:33 - 01:20:53] Prof. Rahul Pandharipande

So, we take G is a finite group, and let x be an element of G [writes: and let $x \in G$].

All right, so, the first thing I can do is consider this: x^1, which is just x, x^2, x^3, and so on [writes: (i) $x, x^2, x^3, \dots$].

And what I mean by this, which is what everybody means: x^2 is just $x \cdot x$, and x^3 is $x \cdot x \cdot x$, and so on, okay? [writes: $x^2 = x \cdot x, x^3 = x \cdot x \cdot x$] That's what this notation means: we just apply the group operation to itself, all right?

And I just consider all of these: x^4, x^5, and eventually there will be some x^{17}, and then eventually x^{691}, and so on [writes: $x^4, x^5, \dots, x^{17}, \dots, x^{691}, \dots$].

And you will see that there are infinitely many elements in this list. But G is a finite group. So, some of these must be coincident; they cannot all be distinct elements of G [writes: cannot all be distinct elements of G!].

Spoken Transcript [01:20:53 - 01:21:43] Prof. Rahul Pandharipande

They can't all be distinct, because G is finite and this list is infinite. I hope you view that as a solid argument. Sometimes this is called something like the pigeonhole principle or something like that. It seems like a fancy word for it, but...

So, if they're not distinct, that means there exists some $k < \ell$ such that $x^k = x^\ell$ [writes on the right board: $\exists k < \ell$ such that $x^k = x^\ell$]. That's what it means that they're not all distinct: there must be at least two that are the same, and then I just name them k and ℓ with $k < \ell$, okay?

And now, by cancellation, I can cancel all the powers of k. Since $k < \ell$, by cancellation, this means the neutral element is equal to $x^{\ell-k}$ [writes: by cancell. $\implies e = x^{\ell-k}$]. I just keep canceling powers of x until I have none here. Okay?

Spoken Transcript [01:21:43 - 01:22:38] Prof. Rahul Pandharipande

So, this is what we have proven: we have proven that for $x \in G$, where G is finite, there exists a positive number such that x^m is the neutral element [writes in a box: We have proven that for $x \in G$, G finite, $\exists m > 0$, such that $x^m = e$].

So, for x in G, for any element of a finite group, there's some positive number that when I raise the element to that power, I get the neutral element. We've proven it. Okay?

This is not true for infinite groups.

Proposition 7.37 (*Existence of a Power Equal to the Identity*): Let G be a finite group and let $x \in G$. Then there exists an integer $m > 0$ such that:

$$x^m = e. \tag{7.58}$$

Proof

Consider the infinite sequence of positive integer powers of x:

$$x^1,\ x^2,\ x^3,\ x^4, \dots,\ x^k, \dots \in G. \tag{7.59}$$

Since the index set $\mathbb{Z}_{>0}$ is infinite while G is finite, by the Pigeonhole Principle the elements of this sequence cannot all be distinct. Thus, there exist integers $k, \ell \in \mathbb{Z}_{>0}$ with $k < \ell$ such that:

$$x^k = x^\ell.$$

Expanding x^ℓ as $x^k \cdot x^{\ell-k}$:

$$x^k \cdot e = x^k \cdot x^{\ell-k}.$$

Applying left cancellation (Proposition 7.15) k times to cancel x^k:

$$e = x^{\ell-k}.$$

Setting $m := \ell - k$, since $\ell > k$, we have $m \in \mathbb{Z}_{>0}$ with $x^m = e$. **Q.E.D.**

Spoken Transcript [01:22:38 - 01:23:33] Prof. Rahul Pandharipande

Because you could take the infinite group to be the integers, and you could take our element x to be 1.

And then a power for the integers is adding it more and more times to itself, and you can add 1 to itself as many times as you want, you're never going to get 0.

And the reason the proof fails, of course, is that for the integers these are all distinct.

And you have to be clear when I'm talking about the integers with addition. So, it's easy to confuse yourself.

All right, so, this is an extremely basic property of groups, and it's a very important one: if I have a finite group, and I have an element, if I apply powers of that element, eventually I get back to the unit, the neutral element.

Example 7.38 (*The Integers under Addition*)**:** Consider the infinite additive group $(\mathbb{Z}, +, 0)$ with element $x = 1$. The corresponding sequence of *"powers"* under additive notation is:

$$n \cdot 1 = \underbrace{1 + 1 + \cdots + 1}_{n \text{ times}} = n.$$

For any $m \in \mathbb{Z}_{>0}$:

$$m \cdot 1 = m \neq 0.$$

Thus, no positive multiple of 1 equals the neutral element 0. The pigeonhole argument does not apply: the *"powers"* $1, 2, 3, \ldots$ are pairwise distinct.

Definition of the Order of an Element

Spoken Transcript [01:23:33 - 01:24:33] Prof. Rahul Pandharipande

And so, we come to a definition. So, again here: Def: let $x \in G$, G finite [writes: Def: Let $x \in G$, G finite].

Then the order of x is the minimal positive m such that x^m is the neutral element [writes: then $\operatorname{Ord}(x)$ is the minimal positive m such that $x^m = e$]. All right?

So, once we have some, we can always take the minimal. And so, that's a canonical number associated... anytime you have a group that's finite and you have an element, that element has a number which is its order, and it's an extremely important aspect of that element.

It's how many times you have to apply the group operation to itself to get back the neutral element.

Definition 7.39 (*Order of a Group Element*)**:** Let G be a finite group and let $x \in G$. The *"order"* of x, denoted $\operatorname{Ord}(x)$, is the smallest positive integer $m \in \mathbb{Z}_{>0}$ such that:

$$\operatorname{Ord}(x) := \min\{m \in \mathbb{Z}_{>0} \mid x^m = e\}. \tag{7.60}$$

By Proposition 7.37, the set $\{m \in \mathbb{Z}_{>0} \mid x^m = e\}$ is non-empty, so $\operatorname{Ord}(x)$ is well-defined by the well-ordering of $\mathbb{N}$.

Spoken Transcript [01:24:33 - 01:24:51] Prof. Rahul Pandharipande

So, let us try to think about S_3. What is the order of e? [writes: $e \to \operatorname{Ord}(e) = 1$]

Well, the order of e is 1, because $e^1 = e$ — it's kind of silly, all right, anyway.

So, what's the order of this element: $(1\,2)$? [writes: $(1\,2) \rightarrow$]

Student Answer

2.

Spoken Transcript [01:24:52 - 01:25:17] Prof. Rahul Pandharipande

Yeah, 2, because if I take this one, it's not e, so the order is not 1 [writes: $(1\,2) \neq e \quad \text{ord} \neq 1$]. And if I apply it to itself, I have to figure out what the square is, and the square is that, and we calculated that's e, so the order is 2 [writes: $(1\,2)(1\,2) = e \quad \text{ord} : 2$].

And, of course, for all the two-cycles — this thing is called a two-cycle — so, these are all order 2 [writes "$(1\,3) \rightarrow \text{ord} = 2$" with "$(2\,3)$" below it]. These are all order 2, for all of them.

And then the other one is this one: $(1\,2\,3)$. What's the order of this? [writes: $(1\,2\,3) \rightarrow$]

Student Answer

3.

Spoken Transcript [01:25:18 - 01:25:43] Prof. Rahul Pandharipande

Three, yeah, three. Good guess [writes: $\text{ord} = 3$].

Because you have to apply this thing three times to get back to 1: you take 1, when you apply it once it goes to 2, if you apply it twice you get 3, it's not that anyway, but if you do it three times it gets back to 1. So, that's that.

Example 7.40 (*Computing Orders of Elements in S_3*): Evaluating the powers of elements in $S_3 = \{e, (1\,2), (1\,3), (2\,3), (1\,2\,3), (1\,3\,2)\}$:

- **Identity element:**
$$e^1 = e \implies \text{Ord}(e) = 1.$$
- **2-cycles:** For $\sigma = (1\,2)$: $\sigma^1 = (1\,2) \neq e$, and $\sigma^2 = (1\,2)(1\,2) = e$. Thus:
$$\text{Ord}((1\,2)) = \text{Ord}((1\,3)) = \text{Ord}((2\,3)) = 2.$$
- **3-cycles:** For $\tau = (1\,2\,3)$: $\tau^1 = (1\,2\,3) \neq e$, $\tau^2 = (1\,3\,2) \neq e$, and $\tau^3 = e$. Thus:
$$\text{Ord}((1\,2\,3)) = \text{Ord}((1\,3\,2)) = 3.$$

The Powers of an Element

Spoken Transcript [01:25:43 - 01:27:08] Prof. Rahul Pandharipande

Okay, and you will notice here that part of the definition... so, let $x \in G$ be an element where G is finite [writes on the lower right board: $x \in G$, G finite].

Then I can consider this set, which is usually called this [writes: $\langle x\rangle \overset{\text{def}}{=}$]. And it's the set that contains the neutral element, x, and all the way up to x to the order of x minus 1, okay? [writes: $\{e, x, \ldots, x^{\text{Ord}(x)-1}\}$].

I don't put order because order is the neutral element. That's the definition of the order, so, I leave out the next one. All right?

And the claim is... so, proposition, if I'm with this definition, this is the definition, and now maybe lemma [writes: Lemma:]... the size of this set is exactly the order of x [writes: $|\langle x\rangle| = \text{Ord}(x)$].

So, this is some set, and the size of this is exactly order of x. And by that I mean certainly I've written that many symbols, order of x, because there's 1 to the $\text{Ord}(x) - 1$ plus the neutral element. So, there's certainly that many things I've written, but to say that this set has this order means that they're all distinct.

So, the lemma, what I have to prove, to prove this lemma I have to show all these people here are distinct elements [writes: I have to show all these are distinct elts.].

Spoken Transcript [01:27:08 - 01:27:29] Prof. Rahul Pandharipande

Because over here I wrote down a set of infinite elements, but they weren't distinct.

So, why is it that when I go from the neutral up to $\text{Ord}(x) - 1$ that they're distinct? Why is that?

Student Answer

Because the order was defined as the minimal positive exponent...

Spoken Transcript [01:27:30 - 01:28:08] Prof. Rahul Pandharipande

Yeah, so, look, I mean, it could be like what happens... this neutral element, it certainly can't be one of these, because if it was like three here, it can't be that, because then that would be lower than the order. So, nothing here could be the neutral element.

More interesting is: why can none of these two be equal? And again for the same reason: if these two were equal, then I could use my cancellation argument and I would get a power m lower that's the neutral. So, you should think about that; it's an elementary argument, but it's certainly crucial. So, these things are just totally crucial aspects about group theory, we'll discuss them.

So, let me say the final thing, then the bell's going to ring. Actually, this set is a subgroup [writes: is a subgroup]. This is a subgroup. And moreover, this subgroup is isomorphic to something. And what is it isomorphic to?

Student Answer

$\mathbb{Z}/\operatorname{Ord}(x)\mathbb{Z}$.

Spoken Transcript [01:28:08 - 01:28:18] Prof. Rahul Pandharipande

Yeah, $\mathbb{Z}_{\mathrm{Ord}(x)}$, $\mathbb{Z}/\operatorname{Ord}(x)\mathbb{Z}$ [writes: $\cong \mathbb{Z}/\operatorname{Ord}(x)\mathbb{Z}$].

So, it's a subgroup: it says that if I multiply anything here, or their inverse... I have to first show that it's closed under multiplication, you have to close it under inverse. I'll do that next time. It's kind of basic, I mean, it's kind of important, so, I'll do it next time.

But the first thing is you have to show when you multiply these together, you stay in this set, and you have to show the inverses are there, and finally, what the subgroup is, is just this with addition and zero [writes "$(+,0)$" on the right].

So, I'll explain that next time. [students applaud and pack up; the lecturer puts away chalk and sponge as the lecture concludes]

Definition 7.41 (*The Set $\langle x\rangle$*): Let G be a finite group and let $x \in G$ with $m := \operatorname{Ord}(x)$. The set $\langle x\rangle$ is the set of powers:

$$\langle x\rangle \overset{\text{def}}{=} \{e, x, x^2, \dots, x^{m-1}\}. \tag{7.61}$$

Lemma 7.42 (*Size of $\langle x\rangle$*): The m powers in (7.61) are pairwise distinct. Consequently:

$$|\langle x\rangle| = \operatorname{Ord}(x). \tag{7.62}$$

Proof

Suppose for contradiction that $x^i = x^j$ for some $0 \le i < j \le m-1$. Applying left cancellation (Proposition 7.15) i times:

$$x^i \cdot e = x^i \cdot x^{j-i} \implies e = x^{j-i}.$$

Setting $d := j - i$, we have:

$$0 < d \le j \le m-1 < m = \operatorname{Ord}(x), \quad \text{with } x^d = e.$$

This contradicts the definition of $m = \operatorname{Ord}(x)$ as the **minimal** positive integer satisfying $x^m = e$. Hence, all elements $e, x, x^2, \dots, x^{m-1}$ are distinct, proving that $|\langle x\rangle| = m = \operatorname{Ord}(x)$. **Q.E.D.**

Proposition 7.43 (*Isomorphism with the Additive Cyclic Group*): For any element $x \in G$ of order $m = \operatorname{Ord}(x)$:

(1) $\langle x\rangle$ is a subgroup of G: $\langle x\rangle \le G$.

(2) $\langle x\rangle$ is isomorphic to the additive cyclic group of integers modulo m:

$$\langle x\rangle \cong (\mathbb{Z}/m\mathbb{Z}, +, 0). \tag{7.63}$$

Subgroup Properties and Isomorphism Map: The lecturer proves this in the next lecture (Proposition 7.49, Corollary 7.50). In outline:

- **Closure under Multiplication:** For any $x^i, x^j \in \langle x\rangle$, by division with remainder $i + j = qm + r$ with $0 \le r < m$. Then $x^i \cdot x^j = x^{i+j} = (x^m)^q \cdot x^r = e^q \cdot x^r = x^r \in \langle x\rangle$.
- **Inverses:** The inverse of x^i is $x^{m-i} \in \langle x\rangle$ because $x^i \cdot x^{m-i} = x^m = e$.
- **Isomorphism:** The map $\phi\colon \mathbb{Z}/m\mathbb{Z} \to \langle x\rangle$ defined by $\bar{k} \mapsto x^k$ is well-defined because $x^m = e$, surjective by the definition of $\langle x\rangle$, hence bijective because both sides have m elements (Lemma 7.42), and a homomorphism:

$$\phi(\bar{a} + \bar{b}) = x^{a+b} = x^a \cdot x^b = \phi(\bar{a}) \cdot \phi(\bar{b}).$$

Beyond the Lecture: Cyclic Subgroups

The subgroup $\langle x \rangle$ is also called the **cyclic subgroup** generated by x, and it can be written as $\{x^k \mid k \in \mathbb{Z}\}$.

Summary of Key Concepts

This lecture bridged the culmination of ring theory with the foundational principles of abstract group theory:

(1) Gauss's Theorem on Polynomial Rings:

- Establishes that R UFD $\implies R[x]$ UFD, extending inductively to multivariate polynomial rings $R[x_1, \ldots, x_n]$.
- The proof reduces uniqueness of factorization to showing that every irreducible polynomial $p \in R[x]$ generates a prime ideal (p).
- For $\deg(p) > 0$, the proof constructs the ideal $I = (f, p)$, selects a nonzero element $m \in I$ of minimal degree, passes to the fraction field $Q(R)[x]$ to execute the division algorithm, clears denominators to show $m^* \mid f$ and $m^* \mid p$, and uses the irreducibility and $\deg(p) > 0$ hypothesis to deduce that $m \in R \setminus \{0\}$, forcing $p \mid g$ via the Primitive Divisor Lemma.

(2) Pedagogical Triad of Algebra:

- Conceptual mastery balances definitions (the rules and pieces), theorems (the gameplay), and examples (the grounding intuition).

(3) Group Axioms and Mechanics:

- A group $(G, \cdot, e)$ satisfies associativity **(G1)**, the identity property **(G2)**, and existence of two-sided inverses **(G3)**; identity and inverses are then unique.
- Left and right cancellation laws ($xa = xb \implies a = b$) hold universally.
- The pedagogical focus shifts to finite, non-commutative (non-abelian) groups.

(4) The Symmetric Group S_n and Cycle Notation:

- S_n has order $|S_n| = n!$.
- Function composition is applied right-to-left: $(f \circ g)(x) = f(g(x))$.
- Products of permutations are efficiently manipulated using cycle notation; S_3 is not commutative.

(5) Element Orders and Cyclic Subgroups:

- In a finite group, the Pigeonhole Principle ensures that powers of an element satisfy $x^m = e$ for some $m > 0$.
- The order $\operatorname{Ord}(x)$ is the minimal positive exponent achieving the identity.
- The set of distinct powers $\langle x \rangle = \{e, x, \ldots, x^{\operatorname{Ord}(x)-1}\}$ forms a subgroup of size $\operatorname{Ord}(x)$, isomorphic to $(\mathbb{Z}/\operatorname{Ord}(x)\mathbb{Z}, +, 0)$.

Week 7

Wednesday, 28th October 2026

Lagrange's Theorem, Symmetric Groups, and Geometric Symmetries

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Course Content Overview

This lecture explores the foundational structure of finite groups, culminating in Lagrange's Theorem and its geometric applications:

- **Perspectives on Group Theory:** Focusing on finite, non-Abelian groups alongside prototype extreme families: symmetric groups S_n and cyclic groups ${}^{\mathbb{Z}}\!/\!_{n\mathbb{Z}}$.
- **Element Orders and Cyclic Subgroups:** Well-definedness of $\operatorname{Ord}(x)$ and verification that $\langle x\rangle \leq G$ is isomorphic to $(\mathbb{Z}_{\operatorname{Ord}(x)}, +, 0)$.
- **Lagrange's Theorem:** Formulation and proof that $|H|$ divides $|G|$ for any subgroup $H \leq G$, establishing that element orders divide group orders and that prime-order groups have no non-trivial proper subgroups.
- **The Equivalence Relation:** Partitioning G into equivalence classes under $x \sim y \iff \exists h \in H\colon xh = y$ and establishing the bijection $\phi\colon H \to E$, $\phi(h) = xh$.
- **Cycle Decompositions and Partitions:** Dynamic visualization of cycles in S_n and the deep connection between cycle decompositions and integer partitions.
- **Rotational Symmetry Groups of Geometric Solids:** Counting symmetries of the Platonic solids by counting faces, establishing the geometric isomorphisms $\operatorname{Rot}(\text{Cube}) \cong S_4$ via body diagonals and $\operatorname{Rot}(\text{Tetrahedron}) \cong A_4$ via vertex permutations.

Perspectives on Group Theory

Finite and Non-Abelian Groups

Spoken Transcript [00:00:00 - 00:01:25] Prof. Rahul Pandharipande

Okay, so, we start. I hope you have now concluded your study of Confucius, Plato, and Schopenhauer, and we can come back to the discussion of groups.

So, I defined last time group — G is a group [writes: G group $(G, \cdot, e)$] — which I don't repeat now (Recall: Definition 7.13). And I remind you, because it's kind of important for the perspective of the discussion about group theory: we will normally... our interest will be here mainly, but not exclusively, focused on finite groups [writes: $\implies$ mostly focused on finite groups], but not always.

And also our main interest, but not always, is on non-commutative groups, so non-Abelian [writes: $\implies$ on non-Abelian]. But not always. So, what you should have in your mind when I say group: it's a finite group and not necessarily Abelian.

The two families of examples on extremes are the family of the symmetric groups S_n — we'll say more; I said a little bit about this before, but I will say more — symmetric group [writes: S_n Symmetric group $|S_n| = n!$]. And this is, of course, a finite group of size n factorial. And that's somehow the richest family that we will speak about, and in some sense... in a sense I'll make more precise later, really a very rich family of finite groups, maybe the richest family of finite groups.

Standing Focus in Group Theory

Let $(G, \cdot, e)$ be a group. Throughout this part of the course, our primary analytical focus rests upon:

- **Finite Groups:** $|G| < \infty$ (though infinite groups will occasionally appear).
- **Non-Abelian Groups:** Groups where the operation is generally non-commutative, i.e., $\exists x, y \in G$ with $x \cdot y \neq y \cdot x$.

Example 7.44 (*The Symmetric Group*): For $n \geq 1$, the *"symmetric group"* S_n consists of all bijections (permutations) of the set $\{1, 2, \ldots, n\}$ under function composition (Definition 7.25). Its order is given by (Proposition 7.27):

$$|S_n| = n!$$

Why Symmetric Groups Are Considered the Richest Family: The lecturer highlights S_n as *"the richest family"* of finite groups. Going beyond the lecture: this perspective is formalized by Cayley's Theorem, which establishes that every finite group of order n is isomorphic to a subgroup of the symmetric group S_n. Thus, understanding subgroups of symmetric groups in principle encompasses all finite group theory. For $n \geq 3$, S_n is non-Abelian: it contains $(1\,3)$ and $(2\,3)$, which do not commute (Observation 7.34).

Cyclic Abelian Groups

Spoken Transcript [00:01:25 - 00:02:04] Prof. Rahul Pandharipande

And then there's the simplest family, $\mathbb{Z}/n\mathbb{Z}$ [writes: $({}^{\mathbb{Z}}/_{n\mathbb{Z}}, +, 0)$]. That's with plus and zero. These are the Abelian groups we've seen before, and these are called cyclic [writes: cyclic Abelian]. So, this is a word we haven't really used before: cyclic Abelian.

If you meet someone and they say they're considering a cyclic Abelian group of order n, they mean this group, exactly this group. It's cyclic — well, it's because the one element generates it all in a big cycle.

The Simplest Family: Cyclic Groups

At the opposite structural extreme from S_n lies the family of finite cyclic groups:

Example 7.45 (*Additive Integers Modulo n*): The set of residue classes modulo n,

$$\mathbb{Z}/n\mathbb{Z} = \{\bar{0}, \bar{1}, \dots, \overline{n-1}\},$$

equipped with addition modulo n and neutral element $0 = \bar{0}$, forms a finite Abelian group of order n.

Definition 7.46 (*Cyclic Group*): A group G is called *"cyclic"* if there exists an element $g \in G$ such that every element of G can be written as a power (or multiple in additive notation) of g. In particular, $(\mathbb{Z}/n\mathbb{Z}, +, 0)$ is generated by $\bar{1}$.

Review of Element Orders

Spoken Transcript [00:02:04 - 00:03:20] Prof. Rahul Pandharipande

So, we were discussing in the second period last time... the setup of the discussion was G is a group [writes: G is group, finite], a finite one, and x is an element [writes: $x \in G$]. And we had proven some things already, so we had defined — it's kind of a theorem-definition — we had seen that the order of x is well-defined [writes: (1) we had seen that order of x is well-defined].

And I'll just remind you what the order is. This order of x [writes: $\mathrm{Ord}(x) =$] is equal... is the minimal m positive integer such that [writes: $\min_{m>0}$ s.t. $x^m = e$] x^m is the neutral element in G. So, to say it's well-defined, you have to first show that there exists some, and that it's a non-empty set and you can pick the minimal one. So, we proved all of this last time (Recall: Proposition 7.37, Definition 7.39).

Recap: The order of an element was defined in the previous lecture (Definition 7.39), which showed that it is well-defined (Proposition 7.37).

Order of a Group Element

Let G be a finite group and let $x \in G$.

Definition 7.47 (*Order of an Element*): The *"order"* of $x \in G$, denoted by $\mathrm{Ord}(x)$, is the smallest positive integer $m \geq 1$ such that $x^m = e$:

$$\mathrm{Ord}(x) := \min\{m \in \mathbb{N}_{>0} \mid x^m = e\}.$$

Why the Order is Well-Defined in Finite Groups: Since G is finite, the infinite sequence of powers

$$x^1, x^2, x^3, x^4, \dots$$

cannot consist of distinct elements. By the Pigeonhole Principle, there must exist integers $j > i \geq 1$ such that $x^j = x^i$. By cancellation (Proposition 7.15), $x^{j-i} = e$. Since $j - i > 0$, the set $\{m \in \mathbb{N}_{>0} \mid x^m = e\}$ is non-empty. By the Well-Ordering Principle of the natural numbers, a unique minimal positive integer m exists.

The Subgroup Generated by an Element

Spoken Transcript [00:03:20 - 00:04:06] Prof. Rahul Pandharipande

And I think that the point where I stopped was this discussion. Another definition is I use these triangular parentheses, the angle brackets. This is by definition [writes: (ii) $\langle x\rangle \overset{\text{def}}{=}$] the set of elements that includes the neutral element, x, x^2, all the way to x to the order of x minus one [writes "$\langle e, x, x^2, \ldots, x^{\mathrm{Ord}(x)-1}\rangle$", a slip for $\{e, x, x^2, \ldots\}$].

Okay? This is a subset of G [writes: $\subset G$]. That's the definition. Sometimes in words this is the... well, let me say the words later. So, that's where we were last time (Recall: Definition 7.41). Are there questions about this? That's in my memory where we were last time.

Recap: This is Definition 7.41 of the previous lecture, where Lemma 7.42 showed that the listed powers are distinct. That $\langle x\rangle$ is a subgroup isomorphic to $\mathbb{Z}/\operatorname{Ord}(x)\mathbb{Z}$ was stated there (Proposition 7.43) and is proved now.

The Set of Powers of an Element

Definition 7.48 (*The Set* $\langle x\rangle$): For any element $x \in G$ in a finite group, we define the set of its distinct powers using angle brackets:

$$\langle x\rangle \overset{\text{def}}{=} \{e, x, x^2, \ldots, x^{\mathrm{Ord}(x)-1}\} \subset G.$$

Subgroups Generated by Elements

Verification of the Subgroup Criteria

Spoken Transcript [00:04:06 - 00:05:35] Prof. Rahul Pandharipande

So, the statement now, proposition, is that this subset in G is a subgroup [writes: Prop. $\langle x\rangle \subset G$ is a subgroup].

All right, and what's the proof? Well, to check it's a subgroup [writes: Proof. $\langle x\rangle$ is closed under $\cdot$], I have to check several things. The first is that this subset is closed under the group product, okay? So, I have to check that.

Is this true? That means I take some person here, x^k, and take another one, x^ℓ [writes: $x^k \cdot x^l$], and I apply the group product. And this is just... this is x k times, this is x ℓ times [writes: $x \cdots x \quad x \cdots x$], so, when I put them together, I get x $k+\ell$ times [writes: $= x^{k+l}$].

And then I get slightly worried because $k+\ell$ might be bigger than this, but of course we know that x to the order [writes: $x^{\mathrm{Ord}(x)} = e$] is e, is the neutral element. So, I can keep removing this, so, all that matters is that's a residue modulo this number. Is that clear?

So, I can just, yeah. I hope that's clear it's closed under product, because if I come up over here, I can always keep taking... I can take out the $\mathrm{Ord}(x)$ part, because it's just the neutral element.

Spoken Transcript [00:05:35 - 00:06:33] Prof. Rahul Pandharipande

Okay, so, then we have to show that — well, I should have said this first: e is in here, but that was [writes: $e \in \langle x \rangle \checkmark$] just on the list.

And the last one we have to show is that if someone is here, then the inverse is also here [writes: if $y \in \langle x \rangle$, then $y^{-1} \in \langle x \rangle$]. So, this should also be clear.

If I pick one of these elements, x^k, where k is in this list, I can always multiply it by $x^{\mathrm{Ord}(x)-k}$ [writes: $x^k \cdot x^{\mathrm{Ord}(x)-k}$], and it will also be in this list. Unless k is zero — there's a boundary case you can check, but it's fine also. But this times this is equal to $x^{\mathrm{Ord}(x)}$, which is e [writes: $= x^{\mathrm{Ord}(x)} = e$].

So, that means the inverse is also in this list, okay? So, all these three things are satisfied...

Spoken Transcript [00:06:33 - 00:07:49] Prof. Rahul Pandharipande

...and it is a fact that this construction, if I take x and I take the neutral element and I go all the way up to the order minus one, this is a nice subset and it is a subgroup.

And moreover, if you understood this product rule where you take this and this — and maybe I just write it: is equal to x^r where r is $k+\ell$ mod the order [writes: $= x^r$ where $r \equiv k+l \pmod{\mathrm{Ord}(x)}$]. If you understand this product rule, as a... maybe I write it as a corollary, corollary to the proof [writes: Cor $\langle x \rangle \cong (\mathbb{Z}/_{\mathrm{Ord}(x)\mathbb{Z}}, +, 0)$]: actually, this group is isomorphic to $\mathbb{Z}/\mathrm{Ord}(x)\mathbb{Z}$ with addition and zero, okay?

And the words for this — now I tell you the words because I didn't want to bias the issue — is that this symbol, the words for it is the subgroup generated by the element x [writes: $\langle x \rangle \subset G$ subgroup generated by x]. Okay? I didn't want to say at the beginning because I didn't want to bias the issue, but now we know it's a subgroup, I can say.

Proposition 7.49 (*Subgroup Generated by an Element*)**:** Let G be a finite group and $x \in G$. Then the set $\langle x \rangle$ is a subgroup of G, termed the *"subgroup generated by x"*:

$$\langle x \rangle \leq G.$$

Proof

Let $m := \mathrm{Ord}(x)$. We verify the three standard subgroup criteria:

(a) Identity Element: By construction, $e = x^0 \in \langle x \rangle$.

(b) Closure under Multiplication: Let $x^k, x^\ell \in \langle x \rangle$, where $0 \leq k, \ell < m$. The product in G satisfies:

$$x^k \cdot x^\ell = \underbrace{(x \cdot \cdots \cdot x)}_{k \text{ times}} \cdot \underbrace{(x \cdot \cdots \cdot x)}_{\ell \text{ times}} = x^{k+\ell}.$$

By division with remainder, we can write $k+\ell = q \cdot m + r$ with $0 \leq r < m$, so $r \equiv k+\ell \pmod{m}$. Since $x^m = e$, we compute:

$$x^{k+\ell} = (x^m)^q \cdot x^r = e^q \cdot x^r = x^r.$$

Since $0 \leq r < m$, we have $x^r \in \langle x \rangle$.

(c) Closure under Inverses: Let $y \in \langle x \rangle$, so $y = x^k$ for some $0 \leq k < m$.

- **Case $k = 0$:** then $y = e$ and $y^{-1} = e \in \langle x \rangle$.

- **Case $1 \leq k < m$:** consider $y' = x^{m-k}$. Since $0 < m-k < m$, we have $y' \in \langle x \rangle$. Multiplying yields:
  $$y \cdot y' = x^k \cdot x^{m-k} = x^m = e.$$
  Thus, $y^{-1} = x^{m-k} \in \langle x \rangle$.

Hence, $\langle x \rangle$ is a subgroup of G. **Q.E.D.**

Corollary 7.50 (*Isomorphism with Additive Cyclic Groups*): Let G be a finite group and $x \in G$ with $m = \mathrm{Ord}(x)$. Then the subgroup $\langle x \rangle$ is isomorphic to the additive cyclic group $(\mathbb{Z}/m\mathbb{Z}, +, 0)$:

$$\langle x \rangle \cong (\mathbb{Z}/m\mathbb{Z}, +, 0).$$

In particular, $|\langle x \rangle| = \mathrm{Ord}(x)$.

Why the Map Is an Isomorphism: The lecture reads this off the product rule. The assignment $\psi\colon \mathbb{Z}/m\mathbb{Z} \to \langle x \rangle$ defined by $\psi(\bar{k}) := x^k$ is well-defined because $x^m = e$. It is surjective by definition of $\langle x \rangle$, and since both sets have cardinality m (Lemma 7.42), it is bijective. Furthermore,

$$\psi(\bar{k} + \bar{\ell}) = \psi(\overline{k+\ell}) = x^{k+\ell} = x^k \cdot x^\ell = \psi(\bar{k}) \cdot \psi(\bar{\ell}),$$

which proves that ψ is a group isomorphism.

Lagrange's Theorem

Divisibility of Orders in Symmetric Groups

Spoken Transcript [00:07:49 - 00:09:18] Prof. Rahul Pandharipande

All right, so, like in our example for S_3 (Recall: Example 7.40), we had some elements like these two-cycles, this element [writes: S_3, $(1\,2)(3) \rightsquigarrow \mathrm{ord} = 2$], and I said sometimes I write this, sometimes I don't, it's just, you know, however I feel. But often people don't write the singletons.

If I take this element, we saw it has order 2. So, this has order 2, and this [writes: $(1\,2\,3) \rightsquigarrow \mathrm{ord} = 3$] has order 3.

And one of the things you notice here is: what is the order of this group? It's 3 factorial, which is 6 [writes: $|S_3| = 3! = 6$]. So, these orders, we find here the orders of the elements, at least in this example, orders of elements seem to divide the order of the group [writes: Orders of elem. $\mid |S_3|$].

At least in this case. So, you can wonder whether this is always the case. So, this is kind of an interesting question. As I said, like this kind of question is a question about finite groups. Because if G has infinite order, we can't really talk about... well, if G was a group with infinite order, the elements may not have finite order, and in any case we can't talk about this division property. That's why I say we're focusing on finite groups.

Motivating Example: Element Orders in S_3

Consider the symmetric group on three letters, S_3, of order:

$$|S_3| = 3! = 6.$$

Evaluating the orders of representative elements:

$$\begin{aligned} \sigma = (1\,2)(3) &\rightsquigarrow \mathrm{Ord}(\sigma) = 2, && \text{and } 2 \mid 6, \\ \tau = (1\,2\,3) &\rightsquigarrow \mathrm{Ord}(\tau) = 3, && \text{and } 3 \mid 6, \\ \mathrm{Id} &\rightsquigarrow \mathrm{Ord}(\mathrm{Id}) = 1, && \text{and } 1 \mid 6. \end{aligned}$$

In every case, the order of the element divides the order of the group:

$$\mathrm{Ord}(x) \mid |S_3| \quad \forall x \in S_3.$$

Statement of Lagrange's Theorem

Spoken Transcript [00:09:18 - 00:10:50] Prof. Rahul Pandharipande

So, anyway, in this case we find that the order of the elements divide the order of the group, so, one can wonder: is this always the case? [writes: Is this always true?]

Specifically, if x is an element of G, a finite group, does the order of x divide the order of G? [writes: (If $x \in G$ finite group, does $\mathrm{Ord}(x) \mid |G|$?)] That's a very specific question, and the answer is going to be yes [writes: YES].

And actually the proof is that a much bigger property, a much more general property holds, and that's the main topic of today's lecture, which is really in some sense usually the first theorem in finite group theory: it's a theorem of Joseph-Louis Lagrange [writes: J. L. Lagrange].

And it's that if H is a subgroup of G — H is a subgroup and G is finite [writes: Thm: If $H \subset G$ is a subgroup and G is finite] — so that's the hypothesis: we have a finite group and a subgroup. I don't have to say the subgroup is finite, because once the group is finite, the subgroup is also finite. So, then, this is what Joseph-Louis says: the order of H always divides the order of G [writes: then $|H| \mid |G|$]. Okay?

Question 7.51 (*Element Order Divisibility*)**:** If G is a finite group and $x \in G$, does $\mathrm{Ord}(x)$ always divide $|G|$?

Answer 7.52 (*to Question 7.51*)**:** **Yes**. In fact, this follows as an immediate corollary of a far more general theorem governing all subgroups of a finite group.

Theorem 7.53 (*Lagrange's Theorem*)**:** Let G be a finite group and let $H \leq G$ be a subgroup of G. Then the order of H divides the order of G:

$$|H| \mid |G|.$$

Connection to Element Orders: By Corollary 7.50, for any $x \in G$, the generated subgroup $H = \langle x \rangle$ has order $|H| = \mathrm{Ord}(x)$. Therefore, Theorem 7.53 immediately implies:

$$\mathrm{Ord}(x) \mid |G| \quad \forall x \in G.$$

Historical Context of the Theorem

Spoken Transcript [00:10:50 - 00:12:05] Prof. Rahul Pandharipande

This is usually the very first actual theorem about finite groups. And I will prove it; the proof is simple, it's conceptual.

One interesting thing is that he lived in the 18th century, and the definition of a group came like half a century after he passed away, so you can wonder: how did he prove this theorem? And, you know, of course he didn't prove this theorem. He proved something about symmetric groups that people have interpreted to be this theorem.

But this is the same Lagrange that's responsible for Lagrangian mechanics and the calculus of variations, which is a fantastic topic. Whether that's covered in the math curriculum, I don't know. Part of my life was as physicists and I like that part. That's a great topic. So, he's done a lot of amazing stuff.

But you've taken this stuff like Lagrangian mechanics at some point? Calculus of variations? Do you take that? Yeah. Those are really great subjects.

Beyond the Lecture: Historical Context of Lagrange's Theorem

Joseph-Louis Lagrange (1736–1813) published his foundational work on the solvability of algebraic equations, **Réflexions sur la résolution algébrique des équations**, in 1770. In that treatise, Lagrange analyzed permutations of roots of polynomial equations and showed that the number of different values that a rational function of n variables can take under all $n!$ permutations must divide $n!$. The abstract definition of a group was not formulated until the work of Arthur Cayley in 1854, more than four decades after Lagrange's death. Modern algebraists later abstracted Lagrange's permutation calculation into the universal statement $|H| \mid |G|$ for any subgroup of a finite group.

Proof of Lagrange's Theorem

The Subgroup Equivalence Relation

Spoken Transcript [00:12:05 - 00:13:17] Prof. Rahul Pandharipande

Okay, so, we're going to prove this. Proof [writes: Proof.]. Is there any question about this theorem?

And you see this theorem certainly implies this statement, because if I have an element, the order of this element is the size of that subgroup. So, in particular, if every subgroup divides the order of G, the order always divides the order of G, okay? But then this is a much more general result.

So, how are we going to prove it? It's an extremely nice proof, and yeah. So, the proof is, we start by defining an equivalence relation on the set G [writes: We start by defining an equivalence relation on set G.]. And then we will count... we will count G by counting up the size of the equivalence classes.

And this equivalence relation is just incredibly important for group theory. So, we will define it.

Spoken Transcript [00:13:17 - 00:14:52] Prof. Rahul Pandharipande

The notion of a group, while I erase I say, is that, you know, it has many fewer axioms than a ring, and in some sense it's a more primitive notion in mathematics, which means that it's more widely applicable and in some sense more fundamental than the notion of a ring. The ring is motivated by number systems, so somehow it is a natural choice for the first topic in the course, since most people have familiarity with number systems. But the group is a more ubiquitous notion, because of course it has fewer axioms, so you can find it more generally. And it's motivated — I mean, it has many motivations, but a very strong motivation about the group is about symmetries of geometric objects. It's like a totally different motivation than a ring, which is about number systems that you encounter in school.

So, while they're in the same class, and in some sense one thinks about them algebraically in the same space, the motivations that led to them are pretty much different, that I would say. But they're both really amazingly interesting, intricate topics. And a lot of the algebraic moves are similar. Maybe that justifies having them in the same course. Sometimes the course is taught with groups first and then rings. I've taught it with rings first, then groups. It doesn't really matter, of course.

Spoken Transcript [00:14:52 - 00:16:00] Prof. Rahul Pandharipande

All right, so, for the proof of this theorem, I'm going to define an equivalence relation. So, that means if you have x and y in G [writes: $x, y \in G$], I have to tell you how to define this relation. The definition [writes: $x \sim y \overset{\text{def}}{\Longleftrightarrow}$] is that we say x is related to y if and only if — the definition — if there exists an h in the subgroup [writes: $\exists h \in H$] such that $xh = y$ [writes: s.t. $xh = y$]. All right?

Now, you could say: why do I put h on the right side? Could I put it on the left side? And you could do that. That'd be your own equivalence relation, and it's totally equivalent for this proof, and the proof will be fine. But since the groups are not commutative, not necessarily commutative, I have to decide, and I've decided to put it on the right, okay? That's the first comment.

The Equivalence Relation Induced by H

Let G be a group and $H \leq G$ a subgroup. For elements $x, y \in G$, we define the relation $\sim$ by:

$$x \sim y \overset{\text{def}}{\Longleftrightarrow} \exists h \in H \text{ such that } xh = y. \tag{7.64}$$

Left vs. Right Coset Conventions: Going beyond the lecture: because G is not assumed to be Abelian, the equation $xh = y$ places h on the right, defining what is conventionally known as the *"left coset"* relation (since $y \in xH$). Placing h on the left ($hx = y$) yields an entirely symmetric theory of right cosets (Hx). The proof of Lagrange's Theorem works identically under either convention.

Verifying the Equivalence Relation Axioms

Spoken Transcript [00:16:00 - 00:17:28] Prof. Rahul Pandharipande

So, then, I have to check for an equivalence relation, I have to check the properties. And normally I leave this to you, but here this is so fundamental I'm going to do it. It's probably a mistake, but nevertheless.

The first property is that x is related to itself [writes: • $x \sim x$]. And that's if and only if I can find an h in this group that satisfies that [writes: $\iff$], and which one is that? Yeah, neutral element, because e has to be here, because it's a subgroup, and then of course I have this equation [writes: $e \in H \quad xe = x \checkmark$], so that's fine.

Now, that's the reflexive property. Then the symmetric property is if x is related to y, then y is related to x [writes: • $x \sim y$ then $y \sim x$]. So, we have to prove this. x is related to y means that there exists an h such that $xh = y$ [writes: $\iff \quad xh = y$]. And then what do I write on the then-side? If I have this, I can just multiply by h inverse, right? So, then, I get... sorry... so, I can multiply by h inverse, which will make this h go away and put h inverse there [writes: then $x = yh^{-1}$]. And then this means that y is related to x. I can write this again if you really want to fit the definition [writes: $yh^{-1} = x$]: this says that y is related to x because if you have an element of H, the inverse is also in H, that's a property of being in a subgroup, right?

Spoken Transcript [00:17:28 - 00:19:02] Prof. Rahul Pandharipande

And the third one is transitivity, which says that if I have x related to y and y related to z, then I have to show that x is related to z [writes: • If $x \sim y$ and $y \sim z \implies x \sim z$]. And if I do this using the definition, this means that there exists some h_1 that satisfies this equation [writes: $xh_1 = y$], and this one means there's some h_2 that satisfies this equation [writes: $yh_2 = z$]. And then I have to find this relation equation, and you can imagine what I do: I just take here $xh_1h_2 = z$ [writes: $xh_1h_2 = z$], all right?

And why does this work? It's because it's a subgroup, so, H is closed under product [writes: $h_1h_2 \in H$ as H is closed under product $\checkmark$]. So, that's the proof that it's an equivalence relation.

And you see the proof is perfect in a way that I need to check three properties to be an equivalence relation, and those were exactly the same three properties that H is a subgroup. I've used everything exactly: the existence of the neutral element is there, it's closed under inverse, and it's closed under product. So, in some sense, the statement that this definition is... on one hand, on this side it's an equivalence relation, and on that side H is a subgroup, they're just perfectly aligned. All right, so, we have proven that.

Proposition 7.54 (*Equivalence Relation Induced by a Subgroup*): The relation $\sim$ defined in (7.64) is an equivalence relation on G.

Proof

We verify the three defining axioms of an equivalence relation:

- **Reflexivity ($x \sim x$):** We require an element $h \in H$ such that $xh = x$. Since H is a subgroup, the identity element satisfies $e \in H$, and $xe = x$. Thus, $x \sim x$.

- **Symmetry ($x \sim y \implies y \sim x$):** Suppose $x \sim y$. By definition, there exists $h \in H$ such that $xh = y$. Multiplying on the right by $h^{-1} \in G$ yields:
  $$x = yh^{-1} \iff yh^{-1} = x.$$
  Since H is a subgroup, $h \in H \implies h^{-1} \in H$. Therefore, setting $h' := h^{-1} \in H$ gives $yh' = x$, which proves $y \sim x$.
- **Transitivity ($x \sim y$ and $y \sim z \implies x \sim z$):** Suppose $x \sim y$ and $y \sim z$. Then there exist elements $h_1, h_2 \in H$ such that:
  $$xh_1 = y \quad \text{and} \quad yh_2 = z.$$
  Substituting the expression for y into the second equation yields:
  $$(xh_1)h_2 = z \implies x(h_1h_2) = z.$$
  Since H is closed under multiplication, $h_3 := h_1h_2 \in H$. Thus, $x \sim z$.

Hence, $\sim$ is an equivalence relation on G. **Q.E.D.**

Correspondence between Subgroup Axioms and Equivalence Axioms: Notice the exact structural alignment highlighted by the lecturer:

$$\begin{aligned} e \in H &\iff \text{Reflexivity of } \sim \\ (h \in H \implies h^{-1} \in H) &\iff \text{Symmetry of } \sim \\ (h_1, h_2 \in H \implies h_1h_2 \in H) &\iff \text{Transitivity of } \sim \end{aligned}$$

The definition of the relation translates the three subgroup axioms directly into the three equivalence relation axioms.

Partitioning into Equivalence Classes and the Counting Formula

Spoken Transcript [00:19:02 - 00:20:33] Prof. Rahul Pandharipande

Okay, so, now we have a picture. Here is the group G, the finite group, so there's some people here [draws a contour for G with dots inside]. And I have an equivalence relation. Whenever you have an equivalence relation, it divides the set into equivalence classes.

So, our equivalence relation divides it into equivalence classes. Maybe here's one equivalence class, E_1 [draws boundary for E_1], and then there's some equivalence class here, E_2 [draws boundary for E_2], and then there's another equivalence class, E_3 [draws boundary for E_3], I don't know. There's finitely many — it's a finite set, so it's finitely many equivalence classes.

So, we can write an equation, and this equation you can always write for equivalence classes on finite sets, and it's an equation that is so naive you think it couldn't possibly be useful: the equation is that the size of G is equal to the sum from $i = 1$ to k [writes: $|G| = \sum_{i=1}^{k} |E_i|$], where k is the number of equivalence classes [writes: $k = \#$ of equiv classes] — there are some finite number of equivalence classes, it's a finite group — you sum over the number of equivalence classes and it's the size of the equivalence class.

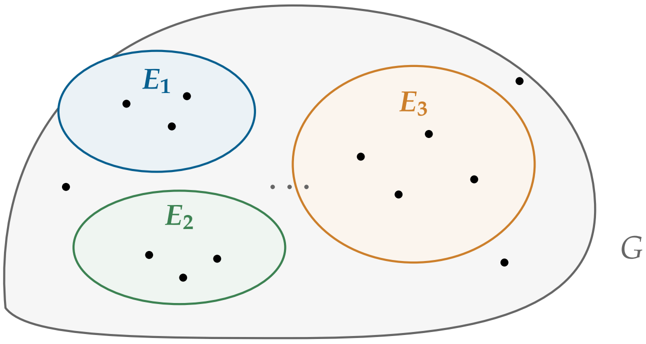
So, that's the equation. It's a pretty naive equation, but as I said, it looks so silly it seems hard to believe that's useful, but it's just saying you take this G, equivalence relation makes some equivalence classes, and you account for everybody in G, and that equation just says you've accounted for everybody exactly once.

Partition of G into Equivalence Classes

Since $\sim$ is an equivalence relation, G is divided into disjoint, non-empty equivalence classes:

$$G = E_1 \,\dot\cup\, E_2 \,\dot\cup\, \ldots \,\dot\cup\, E_k,$$

where k is the total number of distinct equivalence classes.



Since the equivalence classes are pairwise disjoint and their union exhausts G, the sum of their individual sizes accounts for every element in G exactly once:

$$|G| = \sum_{i=1}^{k} |E_i|, \tag{7.65}$$

where k denotes the total number of equivalence classes.

Deducing Lagrange's Theorem from the Cardinality Claim

Spoken Transcript [00:20:33 - 00:21:22] Prof. Rahul Pandharipande

This fellow Joseph-Louis, he was actually born Italian, and his name at birth was Giuseppe Luigi, as far as I can tell. But he became French somehow in his life. His name was always Lagrange as far... but he was born in Turin.

The standard place in the undergraduate curriculum where you learn calculus of variations would be a mathematical class on classical mechanics, which is often taught probably in the physics department, I don't know. But that's probably his greatest work, Lagrange's greatest work. This is not his greatest work. But it's very well known.

Spoken Transcript [00:21:22 - 00:22:28] Prof. Rahul Pandharipande

All right, so, what are we going to do here? We have to prove this, and so, the observation, which is an incredibly simple observation... So, the claim, which will prove everything, is that for all equivalence classes, the size of this equivalence class is actually the size of H [writes: $\boxed{\text{Claim: } \forall i,\ |E_i| = |H|}$]. All right?

So, that sum is actually just adding up the order of H. And so, of course, that proves it. Claim: if I prove this — so, I have to prove this claim — then Lagrange's Theorem follows [writes: Then Lagrange's Theorem follows], since we rewrite that equation as: the size of G is equal to the sum $i = 1$ to k of $|E_i|$ [writes: $|G| = \sum_{i=1}^{k} |E_i|$], but all of these are just the size of H, so this is actually $k \times |H|$ [writes: $= k \cdot |H|$].

So, that shows that, as a consequence, the order of H divides the order of G [writes: $\implies |H| \mid |G| \ \square$]. So, that's the end of the theorem, and I just have to prove this. Okay?

Claim 7.55 (*Cardinality of Equivalence Classes*): Every equivalence class E_i has the same size as the subgroup H:

$$\boxed{\forall i \in \{1, \dots, k\}\colon\ |E_i| = |H|.}$$

Proof of Theorem 7.53 assuming Claim 7.55

Assuming that Claim 7.55 holds, we substitute $|E_i| = |H|$ into the counting equation (7.65):

$$\begin{aligned} |G| &= \sum_{i=1}^{k} |E_i| \\ &= \sum_{i=1}^{k} |H| \\ &= k \cdot |H|. \end{aligned} \tag{7.66}$$

Since k is an integer (the number of equivalence classes), equation (7.66) directly implies:

$$|H| \mid |G|.$$

This completes the proof of Lagrange's Theorem, modulo proving Claim 7.55.

Q.E.D.

The Concept of Group Index: Going beyond the lecture: the integer k, representing the number of distinct equivalence classes, is called the *"index"* of H in G, conventionally written as:

$$[G : H] := k = \frac{|G|}{|H|}.$$

Proof of the Cardinality Claim

Spoken Transcript [00:22:28 - 00:23:53] Prof. Rahul Pandharipande

So, let us see: how are we going to prove that? Let $E = E_i$ be an equivalence class in G [writes: Let $E = E_i$ be an equiv. class in G]. So, we have some equivalence class in G. Here's the picture of G. Here is E [draws set G and subset E], and I need to show that the cardinality of E is exactly $|H|$.

The first thing I do is I reach into E and I pick a person in it, which of course I call x, right? [draws point x in E and writes: $x \in E$] So, I have x in E. I can do that, because it's an equivalence class, it's not empty; we don't consider the empty equivalence class.

Okay, so, then I would like to make a map from H to E [writes: $\phi\colon H \to E$], and I define this: $\phi(h) = \ldots$ [writes: $\phi(h) =$] And what do you think I'm going to write here?

I have to write something that's definitely in E, and all I know about E, it has x in it. So, you have to remember what this equivalence relation... So, yeah, what would you propose I write? I can't... yeah.

Student Answer

xh

Spoken Transcript [00:23:54 - 00:25:12] Prof. Rahul Pandharipande

xh [writes: xh]. I have to write something.

And whatever I write has to be in E. And the things that are in E, remember: x is equivalent to y if and only if we have this equation, $xh = y$ for some h in H [writes: $x \sim y \iff \exists h \in H$ s.t. $xh = y$]. So, these things for sure are equivalent to x.

Is that clear? This is more or less the only thing you could possibly write in this situation.

So, that gives a map, so that at least ϕ maps H to E [writes: At least $\phi \colon H \to E$]. That means if you put in any h in here, you get something in the equivalence class.

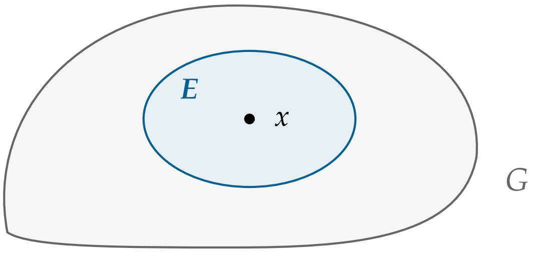
So, now we want to prove — the last step is we should prove, we will prove, let's be confident here — prove that ϕ is a bijection [writes: We will prove that ϕ is a bijection].

Construction of the Bijection between H and E

Let $E = E_i$ be an arbitrary equivalence class of $\sim$. Since equivalence classes are non-empty, we choose an arbitrary element $x \in E$.

We define the mapping:

$$\phi \colon H \to E, \quad \phi(h) := xh. \tag{7.67}$$



Well-Definedness of ϕ: For every $h \in H$, setting $y := xh$ gives $xh = y$ with $h \in H$, which satisfies the definition of the equivalence relation $x \sim y$. Since $x \in E$, it follows that $xh \in E$. Thus, $\phi(H) \subseteq E$, confirming that $\phi \colon H \to E$ is well-defined.

Spoken Transcript [00:25:12 - 00:25:57] Prof. Rahul Pandharipande

So, if we do that, then we're happy, because we need to show that the cardinality of

E is the same as the cardinality of H, and if I show this map is a bijection, then I will have shown that.

All right, so, for this we need to start... we need to prove that ϕ is surjective [writes: • ϕ is surjective]. And this is true because anybody in E is someone that's related to x, and all people related to x are of this form. So, it's surjective by definition. And this is true by definition of this equivalence relation [writes: (True by def of $\sim$)], meaning the only way to be in this equivalence class of x is to satisfy such a relation for some h.

Spoken Transcript [00:25:57 - 00:27:37] Prof. Rahul Pandharipande

Okay, then we have to prove ϕ is injective [writes: • ϕ is injective], and then we will be done.

Okay, so, we have to prove this. Why? Why would this be true? Injectivity means you have to study the equation $\phi(h_1) = \phi(h_2)$ [writes: $\phi(h_1) = \phi(h_2)$]. We have to study that equation, right? So, let's see: suppose $\phi(h_1) = \phi(h_2)$, where h_1 and h_2 are in H [writes: $h_1, h_2 \in H$].

We suppose this is true. Well, what do we do next? We write down the definitions. This thing is xh_1, and that should equal xh_2 [writes: $G \ni xh_1 = xh_2$]. All right. But this is an equation that holds in the group.

So, we now use the cancellation property (Recall: Proposition 7.15). By the cancellation property, we can get rid of this [cancels x from both sides]. This implies $h_1 = h_2$ [writes: $\implies h_1 = h_2$], and that's injectivity.

So, we finished the proof. Yeah, that's the whole proof. It's an extremely simple proof, and if you understand it, you'll never forget it.

When you first see this, it's a very strange property that if you have these finite groups and you assemble these subgroups, you can try as hard as you want, but you'll never be able to make a subgroup that doesn't have order which divides the group.

In particular, if your group has prime order, it's extremely hard to make subgroups, because there's very few numbers that divide prime numbers. So, this is going to be used over and over again in the study of finite groups.

Verification of Bijectivity and Completion of the Proof

Proof of Claim 7.55

To prove $|E| = |H|$, we show that the map $\phi\colon H \to E$, $\phi(h) := xh$, is a bijection:

(a) **Surjectivity:** Let $y \in E$. Since E is the equivalence class of x, we have $x \sim y$. By definition of $\sim$, there exists an element $h \in H$ such that:

$$xh = y \iff \phi(h) = y.$$

Hence, every element $y \in E$ has a pre-image in H, proving that ϕ is surjective.

(b) **Injectivity:** Let $h_1, h_2 \in H$ satisfy $\phi(h_1) = \phi(h_2)$. By definition of ϕ:

$$xh_1 = xh_2.$$

Multiplying on the left by $x^{-1} \in G$ (left cancellation in the group G, Proposition 7.15) gives:

$$x^{-1}(xh_1) = x^{-1}(xh_2) \implies (x^{-1}x)h_1 = (x^{-1}x)h_2 \implies h_1 = h_2.$$

Thus, ϕ is injective.

Since ϕ is both injective and surjective, it is a bijection. Therefore, $|E| = |H|$. This completes the proof of Claim 7.55, and consequently proves Lagrange's Theorem (Theorem 7.53).

Q.E.D.

Corollary 7.56 (*Subgroups of Groups of Prime Order*): Let G be a finite group whose order $|G| = p$ is a prime number. Then the only subgroups of G are the trivial subgroup $\{e\}$ and G itself.

Classification of Groups of Prime Order: Going beyond the lecture: By Lagrange's Theorem, if $H \leq G$, then $|H| \mid p$. Since p is prime, the only positive divisors of p are 1 and p. Thus:

- **Case $|H| = 1$:** then $H = \{e\}$.
- **Case $|H| = p$:** then $H = G$.

Furthermore, choosing any non-identity element $x \in G \setminus \{e\}$, its generated subgroup $\langle x \rangle$ must have order > 1, forcing $|\langle x \rangle| = p$, and thus $\langle x \rangle = G$. Consequently, every group of prime order is cyclic: $G \cong \mathbb{Z}/p\mathbb{Z}$.

Symmetric Groups and Geometric Symmetries

Cycle Decomposition in S_n and Integer Partitions

Spoken Transcript [00:28:02 - 00:29:27] Prof. Rahul Pandharipande

Okay, I wanted to do some examples of groups, because that's the fun part of — at least for me, the fun part of group theory is this rich source of examples.

Okay, so, this S_n ... I mean, I can't somehow say enough how much you should make sure you understand S_n. Understand the cycle notation, etc. It's extremely important. If you learn anything in this class that you'll learn later in your life, you'll regret not thinking about S_n.

So, let me say some more comments about it. We said that... so, maybe I say it again, just because I don't know whether this is all things you've studied before or not, but...

So, S_n I write now [writes: S_n] — this is not S_n, this is the set of n points. So, I should maybe write it differently [erases doodle]. I'll try to say one more thing about the cycle notation, because, as I said, it's extremely important, and it's not like I gave you a totally formal presentation of cycle notation (Recall: Definition 7.28). I gave you a little bit, and it's kind of your responsibility to work out the cycle notation from that. But I'll give you one more hint.

Spoken Transcript [00:29:27 - 00:30:52] Prof. Rahul Pandharipande

So, if I take here the set $[n]$, which is the set $\{1,2,3,\ldots,n\}$ [writes: $[n] = \{1,2,3,\ldots,n\}$], and now I write them as dots... Okay, let's make it 7. Is that 7? 4, 5, 6... Ah, very good [draws 7 points].

Okay, so, if this is my set, if I take an element in here, in S_7 [writes: $S_7 \ni \sigma$], then it maps this set [writes: $\sigma\colon [n] \to [n]$]; it's a bijection, that's by definition, it's a bijection from this set to itself.

So, how do I see the cycles if I want to see it as a picture? I pick 1, and σ takes it to somewhere. So, maybe σ takes 1 to 4 [draws an arrow from 1 to 4 with σ]. It has to go somewhere. And then maybe it goes to 6 [draws an arrow from 4 to 6]. It keeps going. But eventually this path has to repeat, right? Because there's only finitely many elements. It can't keep going to different points; eventually it has to repeat.

Spoken Transcript [00:30:52 - 00:32:32] Prof. Rahul Pandharipande

Could it repeat like this? [gestures to a branching/intersecting path] Could it be like going here, here, here, here? The answer is no! Because it's a bijection. We already know who comes to that; there can't be something else that comes to that. So, it cannot repeat like that. It has to repeat, and the only way it can repeat is to go back to the beginning. That's the only way it can repeat.

So, another way to think about these cycles is just you look at it dynamically: there's the seven points, or however many points, you pick one, it goes, and eventually it comes back to 1. And that's the cycle for 1. In this case, it would be $(1\,4\,6)$ [writes: $(1\,4\,6)$].

And then we take this stuff out, and we go with the points that are remaining. Again, now we start with 2. 2 has to go, and has to be a cycle, and has to come back to 2. And it can never interfere with the cycle here because of the bijection property. So, you just take out another cycle. You do that, and that's how you get the cycle decomposition.

Associated to the cycle decomposition, let's pretend here 2 goes to 3 and comes back, then this would be $(2\;3)$ [writes: $(2\,3)$]. And what's left? 7 by himself, all right? That would be in this case. But that's how you visually see it.

You notice — and this will be for σ — out of this cycle decomposition, you get a certain set of numbers: the lengths of the cycles. And you get $7 = \ldots$ I don't know, it's not working, we missed one! [chuckles] 5, sorry, here [draws cycle connecting 5 and 7]. There! You get $7 = 3+2+2$ [writes: $7 = 3+2+2$].

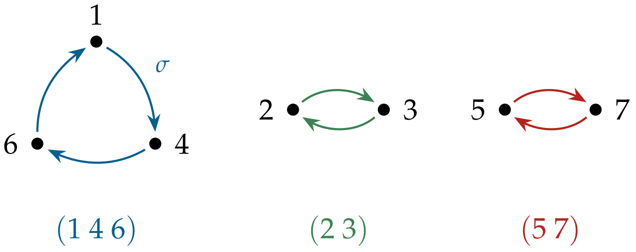
Whenever you have... because at the end you have to account for all the points. Every point has to be in one of the cycles, so you get all the points, and that's the lengths of the cycles.

So, this gives you a connection to partitions, and that's an extremely deep connection. So, this is a partition. Do you know that? That's like an object: how to take a number and break it up into other numbers. There's a very deep connection between the symmetric group and the study of partitions [writes: $S_n \longleftrightarrow \mathrm{Part}(n)$].

Maybe I'll say a little bit about this later. But have you ever thought about the study of partitions? Maybe I'll say a little bit about it. But not today. That's a great subject. Okay, but not today.

Cycle Decomposition in S_7 and Integer Partitions

Let $[n] := \{1,2,\dots,n\}$. Any permutation $\sigma \in S_n$ is a bijection $\sigma\colon [n] \to [n]$.



In cycle notation, this permutation is written as a product of disjoint cycles:

$$\sigma = (1\,4\,6)(2\,3)(5\,7) \in S_7.$$

The lengths of the disjoint cycles partition the total number of elements:

$$7 = 3 + 2 + 2.$$

Connection between Symmetric Groups and Integer Partitions: Every permutation $\sigma \in S_n$ decomposes uniquely (up to the order of the cycles) into a product of disjoint cycles. The multiset of cycle lengths forms an *"integer partition"* of n, denoted:

$$n = \lambda_1 + \lambda_2 + \cdots + \lambda_r, \quad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_r \geq 1.$$

Going beyond the lecture: Two permutations in S_n are conjugate if and only if they possess the identical cycle type. Therefore, the conjugacy classes of the symmetric group S_n are in bijective correspondence with the integer partitions of n:

$$S_n/\text{conjugacy} \longleftrightarrow \text{Partitions}(n).$$

III Rotational Symmetries of the Platonic Solids

Spoken Transcript [00:32:32 - 00:34:15] Prof. Rahul Pandharipande

So, I want to do some examples. And the examples should be symmetry groups of geometric objects. And the most famous geometric objects are the Platonic solids, in some sense the most famous.

So, you could take the tetrahedron... [draws a tetrahedron] the cube... [draws a cube] and then there's the octahedron... [draws an octahedron] I'm not gifted, but that's not terrible. And then there's the dodecahedron, which is much harder to draw. You know, I don't know. Well, you know what this thing is, right? But that's not bad. So, this is the one with 12 sides, and then of course there's the icosahedron [draws an icosahedron], which is even more complicated to draw.

Those are the five Platonic solids. And one thing in this class, at some point, you should be familiar with what the groups of rotational symmetries of these objects are.

And if you remember a long time ago — well, couldn't be that long, since the beginning of the class — we discussed this. For every one of these, there's a group of rotational symmetries, which means you imagine this object, and you look at the rotations that move the faces to the faces.

And we discussed this once before. So, this one, let's just review this: how many rotational symmetries does this have? [points to the cube, writes "how many rotat. symm." under it]

Spoken Transcript [00:34:15 - 00:35:37] Prof. Rahul Pandharipande

You just imagine you have a cube, and it's really you could just pick up a cube and do this: how many ways can you rotate it? And the group is rotation and composition. The neutral element is do nothing, and the inverse element is undo what you did. That's the proof it's a group.

So, how many rotational symmetries does this have? You can just rotate the cube in your head.

The standard way to count this — there's many different ways to count it, but one way to count it is you take the top side, and suppose just take the top of the box. You can ask: how many ways can I rotate it, keeping that the top of the box? And the answer is clearly four. If you want to keep it to be the top, there's four things you can do. Do you agree? So, that's four [writes "4" under the cube].

And what else can you do? I can move the top of the box to any other side. And how many sides are there? Six. And it turns out that's all you can do, because you can just think about it. You should think about this yourself. That's the only thing you can do: you can either rotate the box or move it to the other side [writes "$4 \cdot 6 = 24$" under the cube].

But these things, as I said, this is the geometric part of the class, and geometry is all about the camera in your head, and at some point, if you want to understand geometry, you have to develop the camera in your head. There's no other way in some sense. Okay, and I can't really help you do that.

So, what about this one? How many rotational symmetries does this have? [points to the tetrahedron] This time you should use the same logic.

Student Answer

12

Spoken Transcript [00:35:38 - 00:35:49] Prof. Rahul Pandharipande

Yeah, 12. 12 is good. Why do you say it's 12?

Student Answer, continued

If you fix the bottom side, you can rotate it in three ways, and then there are four faces.

Spoken Transcript [00:35:50 - 00:36:52] Prof. Rahul Pandharipande

Yeah, three ways, exactly. And then four faces, yeah, in this way. So, there's lots of different ways, there's no unique way to do it. This one is: $3 \times 4 = 12$ [writes "$3 \cdot 4 = 12$" under the tetrahedron].

And what about this one? [points to the octahedron] So, let's just do this together. We could pick one of the triangles... You know what these things are, right? You've seen them, it's not...

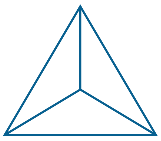
So, if you take one of the triangle sides, there's three motions around, and eight ways it can go, so, it's 24 also.

We leave these... these are more fun, we'll discuss them later [points to the dodecahedron and icosahedron]. These are easy to think about, relatively. This one is a super fun thing to think about.

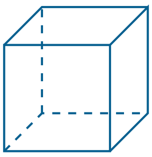
So, that's a group of order 24.

The Five Platonic Solids and Their Rotational Symmetry Groups

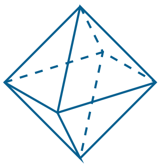
The five *"Platonic solids"* are the tetrahedron, the cube, the octahedron, the dodecahedron and the icosahedron.



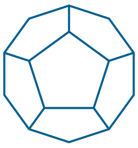
Tetrahedron
$3 \cdot 4 = 12$



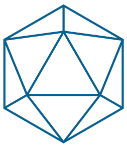
Cube
$4 \cdot 6 = 24$



Octahedron
$3 \cdot 8 = 24$



Dodecahedron
$5 \cdot 12 = 60$



Icosahedron
$3 \cdot 20 = 60$

Counting the Rotations Face by Face: Going beyond the lecture: for each Platonic solid P, the rotational symmetry group $\mathrm{Rot}(P)$ can move any face to any other face (it acts transitively on the faces), and the rotations that keep a chosen face in place are the rotations about the axis through its centre. By the orbit-stabilizer theorem, the order of the group therefore satisfies:

$|\mathrm{Rot}(P)| = (\text{rotational symmetries preserving a chosen face}) \times (\text{total number of faces})$.

- **Tetrahedron:** 4 triangular faces. Fixing one face allows 3 in-plane rotations (120°, 240°, 360°):
$$|\mathrm{Rot}(\text{Tetrahedron})| = 3 \times 4 = 12.$$
- **Cube:** 6 square faces. Fixing one face allows 4 in-plane rotations (90°, 180°, 270°, 360°):
$$|\mathrm{Rot}(\text{Cube})| = 4 \times 6 = 24.$$
- **Octahedron:** 8 triangular faces (dual to the cube). Fixing one face allows 3 rotations:
$$|\mathrm{Rot}(\text{Octahedron})| = 3 \times 8 = 24.$$
- **Dodecahedron:** 12 pentagonal faces. Fixing one face allows 5 rotations:
$$|\mathrm{Rot}(\text{Dodecahedron})| = 5 \times 12 = 60.$$
- **Icosahedron:** 20 triangular faces (dual to the dodecahedron). Fixing one face allows 3 rotations:
$$|\mathrm{Rot}(\text{Icosahedron})| = 3 \times 20 = 60.$$

> The lecture counts the first three and leaves the dodecahedron and the icosahedron for later; their counts are added here for completeness.

Beyond the Lecture: What a Platonic Solid Is

A Platonic solid is a regular, convex polyhedron in three-dimensional space $\mathbb{R}^3$ whose faces are congruent regular polygons and whose vertices have identical valency.

The Rotational Symmetry Group of the Cube and S_4

Spoken Transcript [00:36:52 - 00:38:42] Prof. Rahul Pandharipande

So, these rotations of this cube is a group of order 24 [writes: Rot(□) is a group of order 24].

What group is it? Do you know any groups of order 24? Well, 24 happens to be a factorial: 24 happens to be 4 factorial. So, that's a group of order 24, 4 factorial. We know another group of order 24, is S_4. S_4 is also a group of order 24.

So, this is now a beautiful geometric question: is this rotational group, these rotations here of this cube — this is bad terminology, but anyway, it's not great notation, but it's very convincing if I do it: is rotations of the... is it isomorphic to S_4? [writes: Is Rot(□) $\cong S_4$?] That's a beautiful question!

And I don't know whether I should solve it for you or not. Maybe I solve it for you; I do this one for you.

There's, again, many ways you can do this. One way to do it is you write down the multiplication table for this rotation group, you name all 24 elements — you have to make up your own, now you can be like Fred, Bob, Jane, whatever you want, you name 24 elements here, you name the 24 elements for that S_4, and those are already named by the cycle decompositions — and then you find some bijection that respects the product.

That is a hard way to do it. But you could do it that way. It's a hard way to do it because you have two sets of 24 elements, and very few of them are the correct bijection. So, that's not a great way of doing it.

There is a beautiful way of doing it, which I will show you in a moment. And it's good to show this because it shows, you know, the power of geometry here. [erases the middle blackboard]

Spoken Transcript [00:38:42 - 00:40:22] Prof. Rahul Pandharipande

So, it helps here to have a professor who's artistically gifted, and you do not have this luck, unfortunately. [laughter] This is not a hard diagram, I should be able to handle it.

Here is the... there's the cube, all right? [draws a 3D wireframe cube] And it turns out there is a set of four things here. There's a beautiful geometric set of four things, and those are the four long diagonals.

So, if I take this point of the cube, there's an opposite point, and there's a long diagonal [draws first long diagonal through the center]. Okay? If you imagine a box, for every point in the cube, there's an opposite point. Meaning if that point has coordinates $(+1,+1,+1)$ in the cube, it's $(-1,-1,-1)$ if the cube has side length 2. So, it has an opposite, and there's one of them.

And there's eight vertices, so there's four long diagonals. So, I'll try to draw them, I don't know if it's so helpful. It's not so good, they all pass through the origin. I think I missed one, too. One, two... where is it? One, two, three... this one... No, this one's hidden, yeah, it's there [draws all four diagonals intersecting at origin]. Okay, I told you.

So, there's $1, 2, 3, 4$. But still, as I said, it's much better to have the camera in your head. There's the box, and there's four long diagonals; they all go through the middle, this origin. Is that clear?

Spoken Transcript [00:40:22 - 00:41:47] Prof. Rahul Pandharipande

And now, you label these as, say, a, b, c, d. Where is d? d [labels the four diagonals a, b, c, d]. So, there's four of them; you give them different names.

And you can see now, when you rotate this cube, you permute these four long diagonals. That's just a fact. I mean, you move them, the long diagonals go to other long diagonals.

So, that's kind of an amazing thing. That shows that a rotation of the cube gives you a permutation of the long diagonals. And the proof is just that: the proof is that this rotation of this cube... this is a group, and there's a homomorphism to the permutation group of the long diagonals, which is S_4 [writes: $\mathrm{Rot}(\square) \xrightarrow{\phi} S_4$].

And the claim is that the rotations go to permutations of the long diagonals [writes: $\phi\colon$ rotations $\to$ permutations of the long diagonals]. I say long diagonals all the time here, but I mean, you know, if you pick two points on the same face, then there's kind of a short diagonal that's on the side. But every time you rotate, you permute the long diagonals, and the claim is this is an isomorphism [writes: This is an isom.].

Spoken Transcript [00:41:47 - 00:42:34] Prof. Rahul Pandharipande

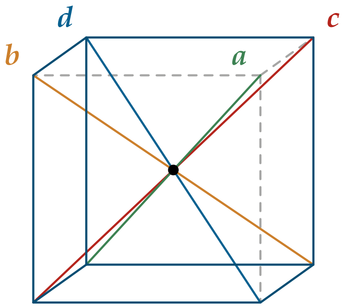
So, how are you going to prove this is an isomorphism? It's not going to be too hard. You just have to check every time you rotate, you do move some long diagonal; it's not trivial. Because both groups are already of order 24.

So, if you prove that it's injective, then it has to be surjective. Or you could also prove it's surjective. But you have to think about how when you move this cube, it cannot be the case that if you do a non-trivial rotation, all the long diagonals stay fixed.

So, I leave that to you to think about. But that's a totally geometric proof, a beautiful geometric proof, which is way better than trying to write down this table, which, you know, probably you will not succeed in doing.

Theorem 7.57 (*Rotational Symmetry Group of the Cube*)**:** The group of rotational symmetries of a cube is isomorphic to the symmetric group on four letters:

$$\mathrm{Rot}(\mathrm{Cube}) \cong S_4.$$



Proof of Theorem 7.57

A cube has 8 vertices, which form 4 antipodal pairs. The line segments connecting each antipodal pair through the origin are the 4 *"long diagonals"* (space diagonals), denoted by the set:

$$\Delta := \{a, b, c, d\}.$$

Every rotation $g \in \mathrm{Rot}(\mathrm{Cube})$ maps vertices to vertices and preserves antipodal pairs, thereby permuting the set Δ. This defines a group homomorphism:

$$\phi\colon \mathrm{Rot}(\mathrm{Cube}) \to S_4, \quad g \mapsto \sigma_g.$$

We claim that ϕ is an isomorphism. Both groups have order 24, so if ϕ is injective, then it is also surjective. For this, a non-trivial rotation must not fix all the long diagonals, which is left to the students. **Q.E.D.**

Injectivity of ϕ (beyond the lecture)

The lecture leaves this to the students. Suppose $g \in \ker(\phi)$, so g maps each long diagonal to itself, possibly reversing it. For the cube $[-1,1]^3$ the diagonals have directions $v_1 = (1,1,1)$, $v_2 = (1,1,-1)$, $v_3 = (1,-1,1)$, $v_4 = (-1,1,1)$, so $gv_i = \varepsilon_i v_i$ with $\varepsilon_i = \pm 1$. Since v_1, v_2, v_3 form a basis and $v_4 = v_1 - v_2 - v_3$, comparing $gv_4 = \varepsilon_1 v_1 - \varepsilon_2 v_2 - \varepsilon_3 v_3$ with $\varepsilon_4 v_4$ forces $\varepsilon_1 = \varepsilon_2 = \varepsilon_3 = \varepsilon_4$, so $g = \pm \mathrm{Id}$. As $-\mathrm{Id}$ has determinant -1, it is not a rotation; hence $g = \mathrm{Id}$, $\ker(\phi) = \{\mathrm{Id}\}$, and ϕ is injective. **Q.E.D.**

III The Rotational Symmetry Group of the Tetrahedron and A_4

Spoken Transcript [00:42:34 - 00:43:53] Prof. Rahul Pandharipande

Okay, so, we have one more moment here, so, what about this one? [draws a tetrahedron]

This has order 12, right? We said that the rotations of this has order 12 [writes: $\mathrm{Rot}(\triangle)$ has order 12]. This sits inside... there's a homomorphism here to S_4 again! [writes: $\mathrm{Rot}(\triangle) \xrightarrow{\phi} S_4$] Because any rotation permutes the four vertices. It's a different geometric object. This is a much more sophisticated thing, the long diagonals. This one's kind of obvious: every time I rotate, I permute the four vertices.

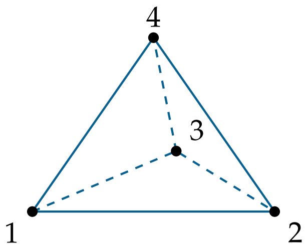
And this is also injective. But this is not an isomorphism: this is 24, this is order 12, it goes to a subgroup. And that subgroup is a very well-known important subgroup, it's called A_4. So, the answer — I'm not defining this now, but it's just the name — this thing here is actually isomorphic to what's called A_4 [writes: $A_4 \cong \mathrm{Rot}(\triangle)$]. We'll talk about it more.

It's not the whole thing. You could say: why didn't we do that here? The rotation group also has a homomorphism to the permutation group of the eight vertices! It does. But that's clearly not an isomorphism, because 8 factorial is a lot bigger than 24.

Okay, so, I stop. [applause from students; video ends]

Theorem 7.58 (*Rotational Symmetry Group of the Tetrahedron*): The group of rotational symmetries of a regular tetrahedron is isomorphic to a subgroup A_4 of S_4:

$$\mathrm{Rot}(\text{Tetrahedron}) \cong A_4.$$



Comparison: Cube vs. Tetrahedron Permutations:

- **Tetrahedron Action on 4 Vertices:** Any rotation of the tetrahedron permutes its 4 vertices $\{1,2,3,4\}$, yielding an injective homomorphism:
  $$\tilde{\phi}\colon \mathrm{Rot}(\text{Tetrahedron}) \hookrightarrow S_4.$$
  Since $|\mathrm{Rot}(\text{Tetrahedron})| = 12$ and $|S_4| = 24$, this is not an isomorphism: the image is a subgroup, which the lecture only names A_4, deferring the definition to the next week (Definition 8.15, Theorem 8.19). Going beyond the lecture: $\tilde{\phi}$ is injective because a rotation fixing all four vertices fixes three linearly independent vectors and is the identity. The image has index 2, and A_4 is the *"alternating group"*, the group of even permutations. Concretely, the image consists of the identity, the eight 3-cycles (rotations by $\pm 120^\circ$ about the axis through a vertex and the centre of the opposite face) and the three products of two disjoint 2-cycles such as $(1\,2)(3\,4)$ (rotations by 180° about the axis through the midpoints of two opposite edges): $1+8+3=12$ even permutations.
- **Why Not Act on the 8 Vertices of the Cube?** Permuting the 8 vertices of a cube gives an injective homomorphism $\mathrm{Rot}(\text{Cube}) \hookrightarrow S_8$. However, $|S_8| = 8! = 40\,320 \gg 24$, so the image is a tiny subgroup of S_8. The 4 long diagonals give a set whose permutation group has exactly the right size, $|S_4| = 24$.

Summary of Key Concepts

- **Standing Focus in Finite Group Theory:** Emphasis on finite, non-Abelian groups alongside prototype extreme families: symmetric groups S_n (order $n!$) and cyclic groups ${}^{\mathbb{Z}}\!/_{n\mathbb{Z}}$ (order n, Abelian).

- **Element Orders and Cyclic Subgroups:** For any element $x \in G$ in a finite group, its order $\operatorname{Ord}(x) := \min\{m \geq 1 \mid x^m = e\}$ is well-defined, and the generated subgroup $\langle x \rangle = \{e, x, \ldots, x^{\operatorname{Ord}(x)-1}\}$ satisfies $|\langle x \rangle| = \operatorname{Ord}(x)$ with $\langle x \rangle \cong (\mathbb{Z}_{\operatorname{Ord}(x)}, +, 0)$.
- **Lagrange's Theorem:** For any finite group G and subgroup $H \leq G$, the order of H divides the order of G, i.e., $|H| \mid |G|$. Consequently, $\operatorname{Ord}(x) \mid |G|$ for all $x \in G$, and a group of prime order has no subgroups other than $\{e\}$ and itself.
- **The Equivalence Relation and the Proof:** The relation $x \sim y \iff \exists h \in H \colon xh = y$ defines an equivalence relation whose equivalence classes partition G. For any class E and representative $x \in E$, the map $\phi \colon H \to E$, $\phi(h) = xh$, is a bijection (injective by group cancellation, surjective by definition of $\sim$). Thus, $|E| = |H|$, yielding $|G| = k \cdot |H|$, where k is the number of equivalence classes.
- **Cycle Decompositions and Partitions:** Every permutation in S_n decomposes into disjoint cycles. The cycle lengths form a partition of n, and there is a very deep connection between S_n and the study of partitions.
- **Rotational Symmetries of Platonic Solids:**
  - **Cube:** Order $24 = 4 \times 6$. The rotations permute the 4 long diagonals, establishing the isomorphism $\operatorname{Rot}(\text{Cube}) \cong S_4$.
  - **Tetrahedron:** Order $12 = 3 \times 4$. Rotations permute the 4 vertices, establishing the isomorphism $\operatorname{Rot}(\text{Tetrahedron}) \cong A_4 \leq S_4$.
  - **Octahedron:** Order $24 = 3 \times 8$.

Tuesday, 3rd November 2026

Homomorphisms of Groups, Quotient Groups, and Conjugacy Classes

Course Content Overview

This lecture develops the foundational theory of group homomorphisms, quotient groups, and conjugacy classes, drawing systematic parallels with linear algebra and ring theory:

- **Group Homomorphisms:** Definition of group homomorphisms, the subgroup properties of the image $\mathrm{Im}(f) \leq G_2$ and kernel $\ker(f) \leq G_1$, and injectivity/surjectivity criteria.
- **The Matrix Homomorphism & the Sign:** Construction of the matrix homomorphism $\Phi_{\mathrm{mat}}\colon S_n \to \mathrm{GL}_n(\mathbb{Q})$ and its composition with the determinant to yield the sign homomorphism $\mathrm{Sign}\colon S_n \to \{+1, -1\}$.
- **The Alternating Group A_n:** Characterization of even and odd permutations, cycle decomposition into transpositions, the structure of A_4 ($|A_4| = 12$), and its geometric isomorphism with the rotational symmetry group of the regular tetrahedron ($\mathrm{Rot}(\text{Tetrahedron}) \cong A_4$).
- **Normal Subgroups & Quotient Groups:** The definition of normal subgroups, the non-normal subgroup $K = \{e, (1\,2)\} \leq S_3$, the construction of the quotient group G/N from the equivalence classes of N, and the proof that normality makes its product well defined.
- **The First Isomorphism Theorem:** Statement of $G_1/\ker(f) \cong \mathrm{Im}(f)$ (the proof is left for the next lecture), presented alongside its counterparts in linear algebra (vector spaces) and ring theory (ideals).
- **Conjugacy Classes & Integer Partitions:** The conjugacy equivalence relation, Jordan normal form in $\mathrm{GL}_n(\mathbb{C})$, the bijection between conjugacy classes of S_n and integer partitions of n, Euler's generating function product formula, and Ramanujan's partition congruences.

Homomorphisms of Groups

Definition and Basic Properties

Spoken Transcript [00:00:00 - 00:00:37] Prof. Rahul Pandharipande

Okay, so, we start, and the topic this week is about the study of homomorphisms [writes: Homs of Groups].

Homs of groups. So, this means we think about the image, the kernel, the first isomorphism theorem. And in some sense, this is the third time for you: you did this study for vector spaces last year, and then a few weeks ago we did the same study for rings — ring homomorphisms. And you'll see the pattern here is very similar, but there are little differences, so you should be sensitive to those differences. And of course, also the examples are different. So...

Recap: Group homomorphisms were defined in Week 2 (Definition 2.10) and again in the previous week (Definition 7.22), which showed $f(e_1) = e_2$ and left $f(x^{-1}) = f(x)^{-1}$ to the students (Proposition 7.23).

Definition 8.1 (*Group Homomorphism*): Let $(G_1, \cdot_1, e_1)$ and $(G_2, \cdot_2, e_2)$ be groups. A mapping $f\colon G_1 \to G_2$ is called a *"group homomorphism"* (or simply *"homomorphism"*) if for all $x, y \in G_1$:

$$f(x \cdot_1 y) = f(x) \cdot_2 f(y). \tag{8.1}$$

Convention on Products: When the operations are understood from context, we drop the subscripts and write $f(xy) = f(x)f(y)$. The operation on the left-hand side takes place within G_1, whereas the operation on the right-hand side takes place within G_2.

Spoken Transcript [00:00:37 - 00:01:16] Prof. Rahul Pandharipande

So, we start with a homomorphism from one group to another group [writes: $f\colon G_1 \to G_2$]. So, this group here is some G_1 with some product and some neutral element [writes: $(G_1, \cdot_1, e_1)$], and this is G_2 with some product and some neutral element [writes: $(G_2, \cdot_2, e_2)$].

And I remind you that the homomorphism condition is that $f(x \cdot y)$ is equal to $f(x) \cdot f(y)$ [writes: $f(xy) = f(x) \cdot f(y)$] for all x, y in G_1 [writes: $\forall x, y \in G_1$]. And this product, of course, then is in G_1, and this product is in G_2. So, that's the homomorphism definition. Okay.

The Image of a Homomorphism

Recap: Week 2 stated this and left the proof to the students (Proposition 2.21).

Proposition 8.2 (*The Image is a Subgroup*): Let $f\colon G_1 \to G_2$ be a group homomorphism. Then the image of f, defined as

$$\operatorname{Im}(f) := f(G_1) = \{f(x) \in G_2 \mid x \in G_1\},$$

is a subgroup of G_2, denoted $\operatorname{Im}(f) \leq G_2$.

Proof of Proposition 8.2

Spoken Transcript [00:01:16 - 00:02:45]

Prof. Rahul Pandharipande

So, the first proposition is not a surprise, and Proposition 1 is that the image of f — this is a subset of G_2 — is a subgroup [writes: Prop 1 $\operatorname{Im}(f) \subset G_2$ is subgp].

I remind you: for a linear map, the image is a sub-vector space, and for a ring homomorphism, the image was a subring. So, this is exactly in that line.

And so, the proof of this: well, what do we have to prove? We have to check the properties of a subgroup [writes: proof: Check properties of subgp].

The first one is that $f(e_1)$ is equal to e_2 (i.e., $f(e_1) = e_2$) [writes: $f(e_1) = e_2$]. That's something we had already derived about homomorphisms (Recall: Proposition 7.23). So, this implies that e_2 is in the image of f [writes: $\implies e_2 \in \operatorname{Im}(f)$]. That's one of the properties: in order for a subset to be a subgroup, the neutral element must be there, and the neutral element is the image of the neutral element.

The second is that if I take... let's say, I take a and b in the image [writes: $a, b \in \operatorname{Im}(f)$], we must check that $a \cdot b$ is in the image [writes: we must check that $a \cdot b \in \operatorname{Im}(f)$]. And the proof of this is that: let x and y in G_1 be such that [writes: let $x, y \in G_1$ s.t.]...

Subgroup Criterion: Identity and Setup for Closure

To prove that $\operatorname{Im}(f) \leq G_2$, we verify the three standard subgroup axioms:

(1) **Neutral Element:** The neutral element $e_2 \in G_2$ belongs to $\operatorname{Im}(f)$:

$$f(e_1) = e_2 \implies e_2 \in \operatorname{Im}(f). \tag{8.2}$$

(2) **Closure under Multiplication:** For all $a, b \in \operatorname{Im}(f)$, we must show that $a \cdot b \in \operatorname{Im}(f)$. Since $a, b \in \operatorname{Im}(f)$, there exist $x, y \in G_1$ such that:

$$a = f(x) \quad \text{and} \quad b = f(y). \tag{8.3}$$

Spoken Transcript [00:02:45 - 00:03:26]

Prof. Rahul Pandharipande

...let's go over here... $f(x) = a$ and $f(y) = b$ [writes: $f(x) = a, f(y) = b$]. This is possible because we're saying a and b are in the image.

Okay, then, we'll just look at $f(x \cdot y)$. This is equal to $f(x) \cdot f(y)$ — that's the homomorphism property — and this is equal to $a \cdot b$, and so, this says that $a \cdot b$ is in the image [writes: $f(x \cdot y) = f(x) \cdot f(y) = a \cdot b \implies a \cdot b \in \operatorname{Im}(f)$]. Okay? So, that's that.

And the last one, I won't do all these arguments so carefully.

Subgroup Criterion: Closure under Products

Using the homomorphism property (8.1) and the chosen pre-images from (8.3):

$$a \cdot b = f(x) \cdot f(y) = f(x \cdot y). \tag{8.4}$$

Since $x, y \in G_1$, the product $x \cdot y$ lies in G_1, which implies:

$$a \cdot b \in \mathrm{Im}(f).$$

Spoken Transcript [00:03:26 - 00:04:36]

Prof. Rahul Pandharipande

The last one is... the third property for the subgroup we have to check is: suppose a is in the image of f, we have to check that a^{-1} is in the image of f [writes: Suppose $a \in \mathrm{Im}(f)$, we must check $a^{-1} \in \mathrm{Im}(f)$].

And so, how to do that? It's just like all of these other ones: we first select, choose some x in G_1 such that $f(x) = a$ [writes: Choose $x \in G_1$ such that $f(x) = a$].

And then we have to find something in G_1 that goes to the inverse, and the natural guess is: consider $f(x^{-1})$ [writes: Consider $f(x^{-1})$]. This is a very reasonable thing to consider for the guess.

So, consider $f(x^{-1})$, and what equation do we have? The homomorphism property equation is $f(x \cdot x^{-1})$ is equal to $f(x) \cdot f(x^{-1})$ [writes: $f(x \cdot x^{-1}) = f(x) \cdot f(x^{-1})$], that's the hom property [writes: hom prop]. Okay? And then we just analyze this.

Spoken Transcript [00:04:36 - 00:05:48]

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So, as I said, in these equations, you have to understand where the products are occurring here. That's just something, when you read this, you have to see that this product means it's before f, so, it's in G_1; this product is after f, so, it's in G_2. So, when you read it, you should just be able to read that.

You know, as I said, formally one could make a different symbol for the product in G_1 and G_2, but nobody does that. It's just too much work. And generally speaking, in all of these equations, if you just read them, so to speak, whether the thing is happening before or after f, you can tell whether it's G_1 or G_2. Is that clear?

Okay, so, then this is happening before, and that's x and x inverse, that's $f(e_1)$, right? [writes: $f(e_1)$ under $f(x \cdot x^{-1})$] And this is $f(x) \cdot f(x^{-1})$. And $f(e_1)$ is e_2 [writes: $e_2 =$]. So, then, we conclude that $f(x^{-1})$ is equal to $f(x)^{-1}$ [writes: $f(x^{-1}) = f(x)^{-1}$]. And this is the inverse in G_1, and that's the inverse in G_2, because I have that times that is equal to e_2.

So, that concludes the three properties.

Subgroup Criterion: Closure under Inverses

(3) Closure under Inverses: Suppose $a \in \mathrm{Im}(f)$. We must verify that $a^{-1} \in \mathrm{Im}(f)$.

Choose $x \in G_1$ such that $f(x) = a$. We evaluate $f(x^{-1})$:

$$\begin{aligned} e_2 &= f(e_1) && \text{(by (8.2))} \\ &= f(x \cdot x^{-1}) && \text{(since } x \cdot x^{-1} = e_1 \text{ in } G_1\text{)} \\ &= f(x) \cdot f(x^{-1}) && \text{(by homomorphism property)} \\ &= a \cdot f(x^{-1}). && (8.5) \end{aligned}$$

Similarly, $f(x^{-1}) \cdot a = e_2$. By uniqueness of inverses in G_2, this implies:

$$f(x^{-1}) = \big(f(x)\big)^{-1} = a^{-1}. \tag{8.6}$$

Since $x^{-1} \in G_1$, its image $f(x^{-1})$ belongs to $\mathrm{Im}(f)$, establishing $a^{-1} \in \mathrm{Im}(f)$.

Therefore, $\mathrm{Im}(f)$ satisfies all three axioms and is a subgroup of G_2.

Q.E.D.

Spoken Transcript [00:05:48 - 00:06:22] Prof. Rahul Pandharipande

So, we have to check these kinds of things a couple of times today, and I did the first one carefully; the later ones I will not do so carefully.

This is not a surprise: it says when you have this homomorphism, the image is a subgroup. So, more interesting than that claim — which is, as I said, there's nothing to this proof, every line just writes itself, you just have to make sure you don't fall off the path...

The Matrix Homomorphism of the Symmetric Group

The Matrix Homomorphism from S_n

Spoken Transcript [00:06:22 - 00:08:04] Prof. Rahul Pandharipande

So, much more interesting are some examples. As I said, the guiding, most interesting example is the symmetric group.

And there is an extremely important homomorphism in mathematics: it's from the symmetric group to GL_n — and let me just put $\mathbb{Q}$ here; you have a lot of choices for what this is, but let's just put $\mathbb{Q}$ because you studied linear algebra over fields [writes: $\Phi_{\mathrm{mat}} : S_n \to \mathrm{GL}_n(\mathbb{Q})$].

So, first of all, do we remember what this is? I defined this before: this is the group of $n \times n$ invertible matrices with rational coefficients [writes: group of $n \times n$ invertible mat. with $\mathbb{Q}$-coeff.]. And it has matrix multiplication and it has the identity matrix [writes: $(\mathrm{GL}_n(\mathbb{Q}), \cdot, I_n)$]. So, are you happy with this group? We discussed it as an example in the beginning of class. But this is a group, in some sense, you should be happy with from linear algebra: it's just that the matrices are invertible and you multiply them.

And S_n, this is this group that, as I said, you should really spend some time making yourself comfortable with if you're not comfortable with it. It's one of the most important objects in mathematics, the symmetric group. And this is the group of bijections from the

n-element set to... so, this is bijections [writes: $[n] \to [n]$ bijections]. I won't define it again, we defined it several times.

So, elements here are bijections from the n-element set to the n-element set, and this is the GL group. So, what is this homomorphism? In words... Okay, let's not do words. In symbols: to write it down, this has a very simple formula, but you have to think about it.

The Groups S_n and $\mathrm{GL}_n(\mathbb{Q})$

Recap: The symmetric group was defined in the previous week (Definition 7.25), with $[n] = \{1, \ldots, n\}$ (Notation 7.24) and products read from right to left (Remark 7.26).

We consider the mapping

$$\Phi_{\text{mat}} \colon S_n \to \mathrm{GL}_n(\mathbb{Q}),$$

where the domain and codomain groups are:

- **Symmetric Group S_n:** The group of all bijections from the set $[n] = \{1, \ldots, n\}$ to itself under composition:
  $$S_n = \{\sigma \colon [n] \to [n] \mid \sigma \text{ is bijective}\}.$$
- **General Linear Group $\mathrm{GL}_n(\mathbb{Q})$:** The group of all invertible $n \times n$ matrices with coefficients in the field of rational numbers $\mathbb{Q}$, equipped with matrix multiplication and identity element I_n:
  $$\mathrm{GL}_n(\mathbb{Q}) := \{M \in \mathrm{Mat}_{n\times n}(\mathbb{Q}) \mid \det(M) \neq 0\}.$$

Spoken Transcript [00:08:04 - 00:09:44] Prof. Rahul Pandharipande

So, the first thing to write this down is I'm going to pick some basis vectors for this vector space. So, let v_1 be $(1, 0, \ldots, 0)$ [writes: $v_1 = (1, 0, \ldots, 0)$], v_2 is equal to $(0, 1, 0, \ldots, 0)$ and so on [writes: $v_2 = (0, 1, 0, \ldots, 0)$]. So, these are the standard basis vectors, and v_n is the last one, the coordinate basis vectors [writes: $v_n = (0, \ldots, 0, 1)$]. So, this is a basis of this vector space $\mathbb{Q}^n$ [writes: basis $\mathbb{Q}^n$], okay?

And in order to give you a matrix, I just have to tell you what it does on the basis vectors. That's linear algebra. So, what do I have to do? I have to define this object, this homomorphism. It eats an element — this is an element of the symmetric group [writes: $\Phi_{\text{mat}}(\sigma)$ with $\sigma \in S_n$]. So, you give it this homomorphism, it eats that and it gives you a matrix, okay? So, I have to explain what this matrix is.

So, how do I explain what the matrix is? Of course, I could write the matrix, and I'm going to do that in an example, but writing the matrix is the data of n^2 things, and I have to write it out, and most of them are zero. So, a better way to do it is just to explain what this does to each basis vector, okay?

So, again, when you read this, you have to understand how to read this: this is the homomorphism, it eats an element of the symmetric group, this whole thing is therefore a matrix, and so, I can act on this basis vector, okay? And in order to tell what this matrix is, I only have to tell you what the actions are on the basis vectors — that's some linear algebra.

Okay, what do you think I'm going to write here? Well, the thing about an element of S_n is it eats indices. These indices go from 1 to n, so you can just feed it that [writes: $= v_{\sigma(i)}$]. So, that's the answer. It's almost a solipsistic answer: it's totally inward-looking...

Spoken Transcript [00:09:44 - 00:10:45] Prof. Rahul Pandharipande

So, do you understand this? This is like the standard kind of thing a pure mathematician writes without thinking. And to understand it, you have to just understand how to take apart these things.

The notation is always a tiny bit sloppy, it's not perfect notation, because to write perfect notation is too much work. So, maybe a more perfect notation would be you take here... so, it eats this symmetric group, this gives a matrix, and maybe you should put parentheses here: that whole matrix acts on v_i [writes "$(\Phi_{\text{mat}}(\sigma))(v_i)$" and crosses it out], that's less ambiguous. But no one writes this, because it's just too much... it's too much people write that.

Definition of Φ_{mat} on Basis Vectors

Definition 8.3 (*The Matrix Homomorphism Φ_{mat}*): Let $\{v_1, \ldots, v_n\}$ denote the standard basis of the vector space $\mathbb{Q}^n$:

$$\begin{aligned} v_1 &:= (1,0,\ldots,0)^T, \\ v_2 &:= (0,1,0,\ldots,0)^T, \\ &\vdots \\ v_n &:= (0,\ldots,0,1)^T. \end{aligned}$$

For each permutation $\sigma \in S_n$, the linear map $\Phi_{\text{mat}}(\sigma) \in \mathrm{GL}_n(\mathbb{Q})$ is uniquely determined by its action on the basis vectors:

$$\Phi_{\text{mat}}(\sigma)(v_i) := v_{\sigma(i)} \quad \text{for all } i \in \{1,\ldots,n\}. \tag{8.7}$$

Disambiguating Notation: $\Phi_{\text{mat}}(\sigma)(v_i)$: The expression $\Phi_{\text{mat}}(\sigma)(v_i)$ represents a two-stage evaluation:

- **Stage 1 (Permutation to Matrix):** First, Φ_{mat} evaluates the permutation $\sigma \in S_n$ to output an $n \times n$ matrix $\Phi_{\text{mat}}(\sigma) \in \mathrm{GL}_n(\mathbb{Q})$.
- **Stage 2 (Matrix-Vector Action):** Second, this matrix acts on the i-th basis vector $v_i \in \mathbb{Q}^n$ via standard matrix-vector multiplication, written unambiguously with parentheses as $(\Phi_{\text{mat}}(\sigma))(v_i)$.
- **Resulting Column Vector:** The resulting vector is $v_{\sigma(i)}$, meaning the i-th column of the matrix $\Phi_{\text{mat}}(\sigma)$ is precisely the basis vector $v_{\sigma(i)}$.

Homomorphism Property of Φ_{mat}

Spoken Transcript [00:10:45 - 00:12:42] Prof. Rahul Pandharipande

Okay, so, this is just the definition, so, we have to check that this map is a homomorphism [writes: Check Φ_{mat} is hom]. So, we have to check that.

So, what's the homomorphism property? On the one hand, I have to see what happens when I send the composition in S_n. So, these are two elements of S_n, and I compose them in S_n, and then I send that to a vector [writes: $\Phi_{\text{mat}}(\hat{\sigma}\sigma)(v_i)$].

And the other thing I can do — this is the homomorphism property I have to check — is whether that's the same thing as this matrix... so, this is a matrix, now this is matrix multiplication, and then in the end it can also eat v_i [writes: $\overset{?}{=} \Phi_{\text{mat}}(\hat{\sigma}) \cdot \Phi_{\text{mat}}(\sigma)(v_i)$]. So, I have to check whether these two things are equal, all right?

Well, what is this by definition? This is v, and then this composition is still an element of S_n, so, this composition can eat an index [writes: $\overset{\text{def}}{=} v_{\hat{\sigma}\sigma(i)}$].

And what is this one? Well, first I do this [points at inner matrix multiplication], so, again by definition... so, we know what the answer for this is, this is $v_{\sigma(i)}$ [writes: $= \Phi_{\text{mat}}(\hat{\sigma})v_{\sigma(i)}$]. And then I apply the definition again: it's this one acting on that, it's v of $\hat{\sigma}$ eating $\sigma(i)$ [writes: $= v_{\hat{\sigma}(\sigma(i))}$].

So, simplifying the right-hand side gets $v_{\hat{\sigma}(\sigma(i))}$, and this is $\hat{\sigma} \circ \sigma$ eating i. But these things are equal because that's the definition of what composition is in the symmetric group. So, this is equal by definition of composition in S_n [writes "def $\circ$ in S_n" over a double arrow between the two results].

Spoken Transcript [00:12:42 - 00:13:38] Prof. Rahul Pandharipande

All right? So, the whole thing is kind of tautological. But, yeah, it's good to learn how to think about it this way. Once you understand the notation, it's very clear: there's only one thing you can do all the time. And the notation is essentially perfect, except for this kind of certain sloppiness about parentheses.

But it takes a while... as I said, it takes a while to understand how to write that.

Verification of the Homomorphism Property

Claim 8.4: The map $\Phi_{\text{mat}} \colon S_n \to \mathrm{GL}_n(\mathbb{Q})$ is a group homomorphism:

$$\Phi_{\text{mat}}(\hat{\sigma} \circ \sigma) = \Phi_{\text{mat}}(\hat{\sigma}) \cdot \Phi_{\text{mat}}(\sigma) \quad \text{for all } \hat{\sigma}, \sigma \in S_n.$$

Proof

Since a linear map is completely determined by its evaluation on a basis, it suffices to check that both matrices agree on every basis vector v_i ($i \in \{1, \dots, n\}$).

Left-Hand Side (Evaluating the Composite Permutation): Applying the definition (8.7) directly to the permutation $\hat{\sigma} \circ \sigma \in S_n$:

$$\Phi_{\text{mat}}(\hat{\sigma} \circ \sigma)(v_i) = v_{(\hat{\sigma} \circ \sigma)(i)} = v_{\hat{\sigma}(\sigma(i))}. \tag{8.8}$$

Right-Hand Side (Successive Matrix Multiplication): Evaluating the product of matrices on v_i:

$$\begin{aligned} \big(\Phi_{\text{mat}}(\hat{\sigma}) \cdot \Phi_{\text{mat}}(\sigma)\big)(v_i) &= \Phi_{\text{mat}}(\hat{\sigma})\Big(\Phi_{\text{mat}}(\sigma)(v_i)\Big) \\ &= \Phi_{\text{mat}}(\hat{\sigma})\big(v_{\sigma(i)}\big) \\ &= v_{\hat{\sigma}(\sigma(i))}. \end{aligned} \tag{8.9}$$

Comparing (8.8) and (8.9), we find:

$$\Phi_{\text{mat}}(\hat{\sigma}\circ\sigma)(v_i) = \big(\Phi_{\text{mat}}(\hat{\sigma})\cdot\Phi_{\text{mat}}(\sigma)\big)(v_i) \quad \text{for all } i\in\{1,\dots,n\}.$$

By linearity, $\Phi_{\text{mat}}(\hat{\sigma}\circ\sigma) = \Phi_{\text{mat}}(\hat{\sigma})\cdot\Phi_{\text{mat}}(\sigma)$. **Q.E.D.**

Explicit Matrix Calculations in S_3

Spoken Transcript [00:13:38 - 00:15:35] Prof. Rahul Pandharipande

So, that's the fancier way to think about it. What it is in words is much simpler. There's a question whether I should give the fancier way first. In words, it's much simpler: this S_n permutes the n indices, so, it just permutes the basis vectors. So, if you view it as a map that takes one basis vector to a permuted basis vector, that's the matrix, and that's what's going on here. That's what it is in words.

And then the last thing is to give an example. Let's do S_3.

So, S_3 has these elements like $(1\,2),(1\,3),(2\,3),(1\,2\,3),(1\,3\,2)$, and of course the neutral element [writes elements of S_3]. And this goes to $\mathrm{GL}_3(\mathbb{Q})$, so, they're 3×3 matrices, it's not so bad [writes: $\Phi_{\text{mat}}\colon S_3\to\mathrm{GL}_3(\mathbb{Q})$].

So, where does e go? Well, it had better go to the identity matrix, because it's a homomorphism. But you can also just see that: e is the σ that takes every index to itself, so, it just takes every vector to itself, so, it's the identity matrix. Is this clear? So, this is this matrix: $\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}$ [writes: $\Phi_{\text{mat}}(e)=\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}$].

All right, so, what about this one? [writes: $\Phi_{\text{mat}}((1\,2))=$] Okay, so, what do you think this is? Here, let's try the first column. What should the first column be? Yeah, someone say it.

Student Answer

$0,1,0$

Spoken Transcript [00:15:38 - 00:17:05] Prof. Rahul Pandharipande

Yeah, $0,1,0$, because it takes the first basis vector to the second one, and that's the matrix that does it. And then this one is that, and then it's that, all right? [writes: $\begin{pmatrix}0&1&0\\1&0&0\\0&0&1\end{pmatrix}$]

And all these other ones are the same idea: this takes the first one to the third, so, this is $0,0,1$; the second one stays the same, and the third one goes to the first, so, it's that [writes: $\Phi_{\text{mat}}((1\,3))=\begin{pmatrix}0&0&1\\0&1&0\\1&0&0\end{pmatrix}$].

Okay, and there's one more. Let's just do them all for completeness, why not? This one: the first one stays the same, the second one goes to the third, so, it looks like this [writes: $\Phi_{\mathrm{mat}}((2\,3)) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$], all right?

And now, there's only two more. So, what does this one look like? [writes: $\Phi_{\mathrm{mat}}((1\,2\,3)) =$] Well, 1 goes to 2, so, this one goes here, 2 goes to 3, 3 goes to 1, so, it's this matrix [writes: $\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$].

And the last one, $(1\,3\,2)$ [writes: $\Phi_{\mathrm{mat}}((1\,3\,2)) =$]... so, 1 goes to 3, 3 goes to 2, and 2 goes to 1 [writes: $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$].

So, that's totally explicit now. This is just doing that, writing it out. And as I said, you can think about it however you want, but that's the answer, and it's for all.

The Six Matrices of S_3

Example 8.5 (*The Matrices $\Phi_{\mathrm{mat}}(\sigma)$ for S_3*): The symmetric group S_3 consists of $3! = 6$ elements, written in cycle notation as:

$$S_3 = \{e, (1\,2), (1\,3), (2\,3), (1\,2\,3), (1\,3\,2)\}.$$

Under $\Phi_{\mathrm{mat}} \colon S_3 \to \mathrm{GL}_3(\mathbb{Q})$, each element maps to the 3×3 matrix whose j-th column is the standard basis vector $v_{\sigma(j)}$:

$$\Phi_{\mathrm{mat}}(e) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \Phi_{\mathrm{mat}}((1\,2)) = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \tag{8.10}$$

$$\Phi_{\mathrm{mat}}((1\,3)) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad \Phi_{\mathrm{mat}}((2\,3)) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \tag{8.11}$$

$$\Phi_{\mathrm{mat}}((1\,2\,3)) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad \Phi_{\mathrm{mat}}((1\,3\,2)) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}. \tag{8.12}$$

How to Read the Matrices by Columns: For any $\sigma \in S_3$, the j-th column of $\Phi_{\mathrm{mat}}(\sigma)$ is the image of the j-th basis vector v_j, which is $v_{\sigma(j)}$. For $\sigma = (1\,2\,3)$:

- **Column 1 ($1 \mapsto 2$):** $v_2 = (0, 1, 0)^T$.
- **Column 2 ($2 \mapsto 3$):** $v_3 = (0, 0, 1)^T$.
- **Column 3 ($3 \mapsto 1$):** $v_1 = (1, 0, 0)^T$.

This yields the matrix of $(1\,2\,3)$ in (8.12).

Beyond the Lecture: Textbook Names

Textbooks call Φ_{mat} the **permutation representation** of S_n, and its matrices **permutation matrices**. The lecturer says *"matrix homomorphism"* and *"the matrix for this $(1\,2)$"*.

The Determinant and Sign Homomorphisms

The Determinant Homomorphism

Spoken Transcript [00:17:05 - 00:18:32] Prof. Rahul Pandharipande

All right, so, that's really a basic thing, but its interest increases when we do one more thing. There's another homomorphism which you did learn also — maybe not in this language — it's the determinant.

If I take $\mathrm{GL}_n(\mathbb{Q})$ — and we could have picked other fields, $\mathbb{Q}$ wasn't so important here, we could have put $\mathbb{R}$ or $\mathbb{C}$, it doesn't matter, but we'll just do $\mathbb{Q}$ — there's the determinant, going to this group: $(\mathbb{Q}^*, \cdot, 1)$ [writes: $\det\colon \mathrm{GL}_n(\mathbb{Q}) \to (\mathbb{Q}^*, \cdot, 1)$]. That's the determinant.

Well, it's certainly true: if I have an invertible matrix, I take the determinant, I get a number. And if the matrix has rational coefficients, the number is a rational number. And it's not zero — that's one of the theorems in linear algebra: if the matrix is invertible, you can get lots of determinants, but you can never get zero, all right? So, that means it maps here.

And what is the homomorphism property? The homomorphism property is like the most important property of the determinant: it's that $\det(A \cdot B)$, if I have two matrices, is equal to that [writes: $\det(A \cdot B) = \det(A) \cdot \det(B)$]. So, this multiplicative property of the determinant exactly says this map is a homomorphism. It literally says that, says nothing more, nothing less.

So, this is the right way to understand the determinant: it's a homomorphism from this GL_n to $\mathbb{Q}^*$. I remind you of this notation: $\mathbb{Q}^*$ is everybody in $\mathbb{Q}$ except for 0 [writes: $\mathbb{Q}^* = \mathbb{Q} \setminus \{0\}$].

The Determinant as a Group Homomorphism

Proposition 8.6 (*Determinant Homomorphism*)**:** The determinant map defines a group homomorphism:

$$\det\colon \mathrm{GL}_n(\mathbb{Q}) \to (\mathbb{Q}^*, \cdot, 1), \tag{8.13}$$

where $\mathbb{Q}^* := \mathbb{Q} \setminus \{0\}$ is the multiplicative group of non-zero rationals.

Proof

If $A \in \mathrm{GL}_n(\mathbb{Q})$, then $\det(A) \in \mathbb{Q}$ and $\det(A) \neq 0$, so $\det(A) \in \mathbb{Q}^*$. Furthermore, for any $A, B \in \mathrm{GL}_n(\mathbb{Q})$, the multiplicative property of determinants states:

$$\det(A \cdot B) = \det(A) \cdot \det(B). \tag{8.14}$$

This is exactly the homomorphism property (8.1), so the determinant is a group homomorphism.

Q.E.D.

The Sign Homomorphism

Spoken Transcript [00:18:32 - 00:20:13] Prof. Rahul Pandharipande

So, if you put them together, you get something kind of interesting. I take here S_n, and I go to $\mathrm{GL}_n(\mathbb{Q})$, and then I go to $\mathbb{Q}^*, 1$ [writes: $S_n \xrightarrow{\Phi_{\mathrm{mat}}} \mathrm{GL}_n(\mathbb{Q}) \xrightarrow{\det} (\mathbb{Q}^*, \cdot, 1)$]. So, this is my matrix homomorphism, this is $\det$.

And if I put them together all the way, the direct flight here, this is $\hat{\Phi}_{\mathrm{mat}}$: first... so, det is second and this one first [writes: $\hat{\Phi}_{\mathrm{mat}} = \det \circ \Phi_{\mathrm{mat}}$], okay?

And it's a fact that if I take... so, this is a homomorphism and that's a homomorphism. If you compose homomorphisms, you also get a homomorphism. So, this guy is a homomorphism [writes: $\hat{\Phi}_{\mathrm{mat}} \colon S_n \to \ldots$ is a hom.]. This is a homomorphism, okay?

And now the question is: what could I get here? This is a finite group, right? We know it has $n!$. So, whatever the image is, it's a finite subgroup of $\mathbb{Q}^*$.

Composition and the Composite Homomorphism $\hat{\Phi}_{\mathrm{mat}}$

Definition 8.7 (*Composite Homomorphism $\hat{\Phi}_{\mathrm{mat}}$*)**:** The composition of Φ_{mat} with the determinant yields a group homomorphism:

$$\hat{\Phi}_{\mathrm{mat}} := \det \circ \Phi_{\mathrm{mat}} \colon S_n \to \mathbb{Q}^*. \tag{8.15}$$

$$\begin{tikzcd}
S_n \arrow[r, "\Phi_{\mathrm{mat}}"] \arrow[rr, bend right, "\hat{\Phi}_{\mathrm{mat}} = \det \circ \Phi_{\mathrm{mat}}"'] & \mathrm{GL}_n(\mathbb{Q}) \arrow[r, "\det"] & (\mathbb{Q}^*, \cdot, 1)
\end{tikzcd}$$

Fact 8.8 (*Composition of Homomorphisms*)**:** If $f \colon G_1 \to G_2$ and $g \colon G_2 \to G_3$ are group homomorphisms, then their composition $g \circ f \colon G_1 \to G_3$ is also a group homomorphism. In particular, $\hat{\Phi}_{\mathrm{mat}}$ is a group homomorphism.

Spoken Transcript [00:20:13 - 00:21:32] Prof. Rahul Pandharipande

So, I'm erasing it, but you see why this notation is much better than keeping data of the square matrix. That's clearer, but it's harder to manipulate in algebraic arguments. So, it's good to make sure you understand both of them [erases the lower-middle board].

Okay, so, I have constructed this homomorphism from the group S_n to $\mathbb{Q}^*$, and now I want to argue that the image is very simple.

So, the image of this $\hat{\Phi}_{\mathrm{mat}}$, it's a subgroup of $\mathbb{Q}^*$ [writes: $\mathrm{Im}(\hat{\Phi}_{\mathrm{mat}}) \subset \mathbb{Q}^*$], subgroup, because we proved that images are always subgroups, all right?

And it's not only a subgroup — let me write it out: this is a subgroup, but it's not only a subgroup, it's a finite subgroup [writes: finite subgroup].

So, what are the finite subgroups of $\mathbb{Q}^*$?

Student Answer

Just one.

Spoken Transcript [00:21:36 - 00:21:50] Prof. Rahul Pandharipande

So, one proposal: you mean just the element containing 1 [writes: $\{1\} \subset \mathbb{Q}^*$]? That is indeed a finite subgroup. That's a correct answer, but it's not the only one. Yeah?

Spoken Transcript [00:21:52 - 00:22:16] Prof. Rahul Pandharipande

Plus and minus 1 [writes: $\{1, -1\} \subset \mathbb{Q}^*$]. This is the full characterization. This is also a finite subgroup.

There are no other finite subgroups, because any other element, when you start taking powers of it, either goes to infinity or goes to zero. So, these are the only two.

Okay, so, this means that the image has to be one of these two groups. And which one do you think it is?

Student Answer

I think it is minus one, one.

Spoken Transcript [00:22:19 - 00:23:42] Prof. Rahul Pandharipande

Yeah, it has to be this one [circles $\{1, -1\}$].

To show it's this one, you just have to find one element that has determinant -1. And the claim is, for any S_n, if I take just the transposition, if I take just the element $(1\,2)$ [writes: $\hat{\Phi}_{\text{mat}}((1\,2)) = -1$], this thing is -1, because the matrix for this $(1\,2)$ is that these guys get switched [writes: $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$] and everybody else stays the same [writes diagonal 1's], and the determinant of this block is -1, all right?

So, that proves that the image of this... so, that's the proposition here: the image of this is equal to the subgroup $\{+1, -1\}$ [writes: Prop. $\operatorname{Im}(\hat{\Phi}_{\text{mat}}) = \{\pm 1\}$].

Okay, and so, then we change the notation for everything. This is the standard mathematical notation: it's called, of course, the sign. So, the standard notation is that there is a homomorphism from S_n to the group $\{+1, -1\}$ — this is the group with multiplication — and this homomorphism is called the sign, which really makes sense, because you're assigning every permutation a sign [writes: Standard notation: $\text{Sign}\colon S_n \to \{\pm 1\}$]. And in the language of this lecture, the sign is by definition this homomorphism $\hat{\Phi}_{\text{mat}}$ [writes: $\text{Sign} \overset{\text{def}}{=} \hat{\Phi}_{\text{mat}}$], okay?

The Image of $\hat{\Phi}_{\text{mat}}$ and the Sign Homomorphism

Proposition 8.9 (*Image of the Composite Homomorphism*)**:** For $n \geq 2$, the image of the composite homomorphism $\hat{\Phi}_{\text{mat}} = \det \circ \Phi_{\text{mat}}\colon S_n \to \mathbb{Q}^*$ is the two-element subgroup:

$$\operatorname{Im}(\hat{\Phi}_{\text{mat}}) = \{1, -1\} \leq \mathbb{Q}^*. \tag{8.16}$$

Proof

Since S_n is a finite group of order $n!$, its image $\operatorname{Im}(\hat{\Phi}_{\text{mat}})$ must be a finite subgroup of $(\mathbb{Q}^*, \cdot, 1)$. The only non-trivial finite subgroup of $\mathbb{Q}^*$ is $\{1, -1\}$, since any rational number $q \in \mathbb{Q}^* \setminus \{1, -1\}$ has infinite order (powers $|q|^k \to \infty$ or $|q|^k \to 0$ as $k \to \infty$).

To show that $\operatorname{Im}(\hat{\Phi}_{\text{mat}}) = \{1, -1\}$, it suffices to exhibit a single permutation $\sigma \in S_n$ with $\hat{\Phi}_{\text{mat}}(\sigma) = -1$. Consider the transposition $\tau = (1\,2) \in S_n$. Its matrix $\Phi_{\text{mat}}((1\,2))$ swaps the first two standard basis vectors v_1, v_2 and fixes $v_3, \ldots, v_n$:

$$\Phi_{\text{mat}}((1\,2)) = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix} = \begin{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & 0 \\ 0 & I_{n-2} \end{pmatrix}. \tag{8.17}$$

Using the determinant formula for block diagonal matrices:

$$\hat{\Phi}_{\text{mat}}((1\,2)) = \det(\Phi_{\text{mat}}((1\,2))) = \det \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \cdot \det(I_{n-2}) = (-1) \cdot 1 = -1.$$

Since $\hat{\Phi}_{\text{mat}}(e) = 1 \in \operatorname{Im}(\hat{\Phi}_{\text{mat}})$ and $\hat{\Phi}_{\text{mat}}((1\,2)) = -1 \in \operatorname{Im}(\hat{\Phi}_{\text{mat}})$, we conclude:

$$\operatorname{Im}(\hat{\Phi}_{\text{mat}}) = \{1, -1\}.$$

Q.E.D.

Definition 8.10 (*Sign Homomorphism*): The *"sign"* $\text{Sign}\colon S_n \to \{+1, -1\}$ is the homomorphism

$$\text{Sign} \overset{\text{def}}{=} \hat{\Phi}_{\text{mat}} = \det \circ \Phi_{\text{mat}}. \tag{8.18}$$

Notation: The Sign

The board writes Sign, for the map and for its values ($\text{Sign}(\sigma) = +1$), and these notes follow it. Textbooks write $\operatorname{sgn}(\sigma)$, and some call it the **signature** of σ.

Even and Odd Permutations

Spoken Transcript [00:23:44 - 00:24:35] Prof. Rahul Pandharipande

And so, that's kind of an interesting thing: for every permutation you get a sign, and the language is that permutations are even or odd.

If you have σ in S_n [writes: $\sigma \in S_n$], then σ is called even or odd; it's even if and only if the sign is $+1$ [writes: even $\iff$ $\text{Sign}(\sigma) = +1$], and it's odd if and only if the sign is -1 [writes: odd $\iff$ $\text{Sign}(\sigma) = -1$].

And what we've seen here is the two-cycle, where we just move two elements, that is always odd. And moreover, this is a homomorphism, so, that means that if you take a product of permutations, the sign is multiplicative. So, we'll get back to that.

Parity of Permutations

Definition 8.11 (*Even and Odd Permutations*): Let $\sigma \in S_n$ be a permutation. Then σ is called:

- **Even:** *"even"* if $\operatorname{Sign}(\sigma) = +1$,
- **Odd:** *"odd"* if $\operatorname{Sign}(\sigma) = -1$.

Observation 8.12 (*Sign of Transpositions and Multiplicativity*): Any transposition $(i\,j) \in S_n$ (with $i \neq j$) has sign -1 and is therefore an odd permutation. Furthermore, because Sign is a group homomorphism, the sign of a composition of permutations is multiplicative:

$$\operatorname{Sign}(\sigma \circ \tau) = \operatorname{Sign}(\sigma) \cdot \operatorname{Sign}(\tau) \quad \text{for all } \sigma, \tau \in S_n. \tag{8.19}$$

Consequently:

- **Even ∘ Even:** The composition of two even permutations is even: $(+1) \cdot (+1) = +1$.
- **Odd ∘ Odd:** The composition of two odd permutations is even: $(-1) \cdot (-1) = +1$.
- **Even ∘ Odd:** The composition of an even and an odd permutation is odd: $(+1) \cdot (-1) = -1$.

Spoken Transcript [00:24:35 - 00:25:42] Prof. Rahul Pandharipande

But, in some sense, if you see a permutation, there's now two things you can ask — and we'll get back to this theme. The first thing, if you see a permutation, you can ask: what is the cycle decomposition? That gives you a partition.

Another thing you can ask is: what is that sign? And these questions are related; I will say a little bit about it [erases the right section of the lower board].

Well, that's the absolute standard mathematical notation: the sign of a permutation, and the sign is a homomorphism to $\{\pm 1\}$.

The Alternating Group

The Kernel of the Sign Homomorphism and Definition of A_n

Spoken Transcript [00:25:42 - 00:27:00] Prof. Rahul Pandharipande

Okay, so, then we come back to the analysis of the homomorphisms.

So, we have here f from G_1 to G_2 [writes: $f\colon G_1 \to G_2$], and the first proposition was that the image is a subgroup [writes: Prop: $\operatorname{Im}(f) \subset G_2$ is a subgroup], and we checked this. That's already done. And I should have said that we never used anything about finite groups here, right? That was not used.

So, the next proposition you can guess is that the kernel — which I'll define in a moment — is a subgroup [writes: Prop: $\ker(f) \subset G_1$ is a subgroup]. And here, the definition — we've defined it before — the kernel of f is equal to all those people in G_1 such that when you apply f to them, you get e_2, right? [writes: Def: $\ker(f) = \{x \in G_1 \mid f(x) = e_2\}$]

So, that's the definition, and this kernel is a subgroup. And this is proven in the same way; I won't prove this one at all, I leave this for you, so you can check [writes: You can check]. There's nothing there to really do.

You have to check that e_1 is there, the product is there, so, I leave this to you. I did the first one very carefully.

Recap: The kernel was defined in Week 2 (Definition 2.17), which also stated that it is a subgroup (Proposition 2.20).

Definition 8.13 (*Kernel of a Group Homomorphism*): Let $f\colon G_1 \to G_2$ be a group homomorphism. The *"kernel"* of f, denoted $\ker(f)$, is the pre-image of the neutral element $e_2 \in G_2$:

$$\ker(f) := \{x \in G_1 \mid f(x) = e_2\}. \tag{8.20}$$

Proposition 8.14 (*The Kernel is a Subgroup*): Let $f\colon G_1 \to G_2$ be a group homomorphism. Then $\ker(f)$ is a subgroup of G_1, denoted $\ker(f) \leq G_1$.

Proof (beyond the lecture)

The lecture leaves this to the students. We check the three subgroup axioms.

(1) Identity Element: Since $f(e_1) = e_2$, we have $e_1 \in \ker(f)$.

(2) Closure under Products: Let $x, y \in \ker(f)$, so $f(x) = e_2$ and $f(y) = e_2$. By the homomorphism property:

$$f(x \cdot y) = f(x) \cdot f(y) = e_2 \cdot e_2 = e_2 \implies x \cdot y \in \ker(f).$$

(3) Closure under Inverses: Let $x \in \ker(f)$. Then $f(x) = e_2$. By (8.6):

$$f(x^{-1}) = \left(f(x)\right)^{-1} = e_2^{-1} = e_2 \implies x^{-1} \in \ker(f).$$

Therefore, $\ker(f)$ is a subgroup of G_1. **Q.E.D.**

Spoken Transcript [00:27:00 - 00:28:00] Prof. Rahul Pandharipande

Okay. So, if we go back to the example then: the sign from S_n to this $\{\pm 1\}$ [writes: Sign: $S_n \to \{\pm 1\}$]. I write it as $\{\pm 1\}$ to indicate it's a group with multiplication.

This $\{\pm 1\}$, this group is isomorphic to $\mathbb{Z}/2\mathbb{Z}$ with addition and 0 [writes: $\cong (\mathbb{Z}/2\mathbb{Z}, +, \bar{0})$], right? Where $+1$ goes to $\bar{0}$ and -1 goes to $\bar{1}$ [writes: $+1 \mapsto \bar{0}, -1 \mapsto \bar{1}$]. It's another way to think about it. Is that clear? But it's more natural to write it as $\{\pm 1\}$ here because we call it the sign.

Okay, so, what is the kernel? The kernel of the sign, the kernel of the sign homomorphism [writes: What is the ker of Sign hom.?]?

Well, we don't know what it is, but we know at least it's a subgroup, right? By that second proposition.

Spoken Transcript [00:28:00 - 00:29:07] Prof. Rahul Pandharipande

So, there's a definition: and this is a very important group, A_n [writes: Def: $A_n \subset S_n$].

You know, there's a little bit of some kind of botany in this group theory where all these groups have names — it's like collecting plants. And it goes a long way, but there's no way to get around it: there's some kind of collection of groups, mathematicians have studied them for a long time, and we've agreed upon names. And part of the subject, if you want to understand it, is you have to learn some names.

So, this is called A_n, and A_n is for alternating group, and this A_n is by definition the kernel of the sign homomorphism [writes: $A_n \overset{\text{def}}{=} \ker(\text{Sign})$].

In other words, equivalently, A_n is equal to the subgroup of even permutations [writes: A_n = the subgroup of even permutations]. Because the even ones are 1, that's in the kernel, so, that's another way to think about it.

So, let us see what they are here.

Definition 8.15 (*Alternating Group*): Let $n \geq 2$. The *"alternating group"* on n elements, denoted A_n, is the kernel of the sign homomorphism $\text{Sign}\colon S_n \to \{+1, -1\}$:

$$A_n \overset{\text{def}}{=} \ker(\text{Sign}) = \{\sigma \in S_n \mid \text{Sign}(\sigma) = +1\}. \tag{8.21}$$

Equivalently, A_n is the subgroup of all even permutations in S_n.

Remark 8.16 (*Multiplicative $\{\pm 1\}$ vs. Additive $\mathbb{Z}/2\mathbb{Z}$*): The multiplicative group $(\{+1, -1\}, \cdot, 1)$ is isomorphic to the group $(\mathbb{Z}/2\mathbb{Z}, +, \bar{0})$ via the isomorphism:

$$\psi\colon \{+1, -1\} \to \mathbb{Z}/2\mathbb{Z}, \qquad \psi(+1) = \bar{0}, \quad \psi(-1) = \bar{1}.$$

Under this identification, multiplication of signs corresponds to addition modulo 2:

$$(-1) \cdot (-1) = +1 \iff \bar{1} + \bar{1} = \bar{0}.$$

The Alternating Group A_4 and Permutations of S_4

Spoken Transcript [00:29:07 - 00:29:55] Prof. Rahul Pandharipande

[pushes the lower blackboard up and erases the upper-left chalkboard] So, the A is called alternating.

Okay, so, let's try to do an example, because it turns out, in some nice cases, these groups are ones you've seen before.

Cycle Partitions and Sign Determination

Spoken Transcript [00:29:55 - 00:31:25] Prof. Rahul Pandharipande

All right, so, let's try for S_4. S_4 has... [writes: $|S_4| = 4! = 24$] S_4 has 24 elements. All right, so, let's try to write them.

So, first of all, there's the neutral element [writes: e]. And we write the sign of it: the sign of this is $+1$ [writes: $+1$], because it does nothing, and it has determinant 1, is that okay?

So, then, there's the two-cycles: there's six of them: $(1\,2), (1\,3), (1\,4), (2\,3), (2\,4), (3\,4)$ [writes the six transpositions in a column].

So, that's 4 choose 2, there's six two-cycles. When I write this, I don't write the ones that don't do anything (Recall: Notation 7.31). I could write them, and maybe I will, sometimes. But here I've chosen not to write them, just to save chalk.

All right. And so, what we decided also: the signs of all of these, these all have sign -1 [writes: -1]. Maybe I'll try a new pedagogical strategy: I'll write the sign in red [rewrites $+1$ and -1 in red chalk].

So, all these have sign -1. Do you agree? Because we just said that they correspond to switching just two, and that's a theorem also in linear algebra: if you switch, you get a minus sign for the determinant, which I hope you remember. Okay.

Spoken Transcript [00:31:25 - 00:33:05] Prof. Rahul Pandharipande

And this corresponds to the partition, by the way... oh, I can use red again. We'll see how long this continues [writes: $1+1+1+1$ in red chalk].

This corresponds to the partition $1+1+1+1$, because if I wrote them all out, there'd just be a bunch of ones.

This corresponds to the partition $2+1+1$ [writes: $2+1+1$ in red chalk], because if I wrote them out, there'd be two singletons. Let me write this one out so I don't get... you know [writes: $(1\,2)(3)(4)$]. So, if I wrote this one fully out, it would be like that, and so, it has a partition $2+1+1$, okay?

So, what other partitions of 4 are there? Well, let's do the $2+2$ partition [writes: $2+2$ in red chalk].

All right, so, what are these? Then I have to pick a two-cycle and then another one, and they're disjoint, right? [writes: $(1\,2)(3\,4)$]

How many ways can I do this, by the way? Do you know?

Well, you have to pick any two-cycle here, so, I could pick another one. What's another one I could... well, they're here, like $(1\,3)$, but then this one determines the opposite one, $(2\,4)$ [writes: $(1\,3)(2\,4)$]. And there's, well, one more: $(1\,4)$ and $(2\,3)$ [writes: $(1\,4)(2\,3)$]. And that's all I can do.

This is some kind of partition combinatorics, because I've just figured out every way I can write the cycle decomposition in $2, 2$. If I put them in different order, it doesn't matter, because those cycles commute. So, these are the only ones. Are you happy with that?

And what's the sign of this?

Student Answer

Plus one.

Spoken Transcript [00:33:06 - 00:33:30] Prof. Rahul Pandharipande

Plus 1, because each one of these has sign -1, and sign is a homomorphism, so, there's two of them, so, it becomes $+1$, okay? [writes: $+1$ in red chalk]

Spoken Transcript [00:33:30 - 00:34:23] Prof. Rahul Pandharipande

All right, so, what else can I do? $3+1$ [writes: $3+1$ in red chalk].

So, I could do something like: I keep 4 fixed, and I do $(1\,2\,3)$ [writes: $(4)(1\,2\,3)$]. I keep 4 fixed, I do $(1\,3\,2)$ [writes: $(4)(1\,3\,2)$]. All right.

And there's more of them. The total list here is 8, because I could decide which one to keep fixed, and once I decide to fix k, there's the three-cycle and there's two of the three-cycles. So, it's going to be 2×4, which is 8.

So, this would be $(3)(1\,2\,4), (3)(1\,4\,2)$... (2) and then I skip 2 here, that's $(2)(1\,3\,4), (2)(1\,4\,3)$, and what am I missing? $(1)(2\,3\,4), (1)(2\,4\,3)$ [writes all eight 3-cycles in two columns].

Okay, that's eight of them. And what's their sign? This is a more complicated question, and you have to think.

Spoken Transcript [00:34:23 - 00:34:27] Prof. Rahul Pandharipande

Okay, if you don't think, you can guess. All right. What? Yeah?

Student Answer

Minus one, because we are swapping three times, so we have to do three multiplications to get this.

Spoken Transcript [00:34:32 - 00:34:38] Prof. Rahul Pandharipande

It's a convincing argument, but it's not correct.

Student Answer

Plus one.

Spoken Transcript [00:34:40 - 00:34:48] Prof. Rahul Pandharipande

Yeah, it's very good! If that one's not correct, that's a plus one! [laughter] This was a good application of logic.

Decomposition of Cycles into Transpositions

Spoken Transcript [00:34:48 - 00:36:22] Prof. Rahul Pandharipande

So, the reason here is that if you have an $(a\,b\,c)$ [writes: $(a\,b\,c)$], I have to figure out how I can write this as a product of two-cycles. So, we could try something. I mean, you have to play around. Let's see: what if I write that: $(a\,b)(b\,c)$ [writes: $(a\,b)(b\,c)$]?

So, if I look at what happens to a here: a goes to... wait, a goes to b. What happens to a here? There's no a, so, a goes to b. What happens to b here? b is supposed to go to c. So, b goes to c, and it's fine. And what happens to c? c goes to b, but then here b goes to a. So, this actually works.

And it works in general. There's a very nice formula for how to write a long cycle in terms of two-cycles, and you have to concentrate — or I have to concentrate — and you just do it in the simplest way possible. You write: $(a_1\,a_2\ldots a_\ell) = (a_1\,a_2)(a_2\,a_3)\ldots(a_{\ell-1}\,a_\ell)$ [writes formula on board]. And this is a very good formula to understand. It's a way of writing this long chain in terms of two-cycles. And to prove it, you just check it works, that's all.

And what was wrong with the first argument was that to swap all three, you actually only need two swaps to do it.

So, what's the answer then finally? Plus 1. These are all plus 1 [writes: $+1$ in red chalk].

Cycle Decomposition into Transpositions and Sign of an ℓ-Cycle

Lemma 8.17 (*Factorization of Cycles into Transpositions*): Any cycle of length $\ell \geq 2$ can be factored into a product of $\ell - 1$ transpositions:

$$(a_1\,a_2\,\ldots\,a_\ell) = (a_1\,a_2)(a_2\,a_3)\cdots(a_{\ell-1}\,a_\ell). \tag{8.22}$$

Consequently, the sign of an ℓ-cycle is given by:

$$\operatorname{Sign}\big((a_1\,a_2\,\ldots\,a_\ell)\big) = (-1)^{\ell-1}. \tag{8.23}$$

Proof (beyond the lecture)

The lecture checks $(a\,b\,c) = (a\,b)(b\,c)$ and leaves the general case to the students. We trace the product from right to left (Remark 7.26) on each entry a_j; elements outside the cycle are fixed by every factor.

- **Element a_1:** The first $\ell - 2$ transpositions $(a_{\ell-1}\,a_\ell), \ldots, (a_2\,a_3)$ fix a_1. Then the final transposition $(a_1\,a_2)$ sends $a_1 \mapsto a_2$.
- **Element a_j $(1 < j < \ell)$:** The transpositions $(a_{\ell-1}\,a_\ell), \ldots, (a_j\,a_{j+1})$ fix a_j until $(a_j\,a_{j+1})$ is reached, which sends $a_j \mapsto a_{j+1}$. The subsequent transpositions $(a_{j-1}\,a_j), \ldots, (a_1\,a_2)$ do not contain a_{j+1} and thus fix it. Therefore $a_j \mapsto a_{j+1}$.
- **Element a_ℓ:** The first transposition $(a_{\ell-1}\,a_\ell)$ maps $a_\ell \mapsto a_{\ell-1}$. Then each successive transposition $(a_{k-1}\,a_k)$ maps $a_k \mapsto a_{k-1}$, cascading all the way down to $(a_1\,a_2)$, which maps $a_2 \mapsto a_1$. Hence $a_\ell \mapsto a_1$.

This exactly matches the permutation $(a_1\,a_2\,\ldots\,a_\ell)$. Since Sign is a group homomorphism and each transposition has sign -1:

$$\operatorname{Sign}\big((a_1\,a_2\,\ldots\,a_\ell)\big) = \prod_{k=1}^{\ell-1} \operatorname{Sign}\big((a_k\,a_{k+1})\big) = (-1)^{\ell-1}.$$

Q.E.D.

Parity versus Cycle Length: Notice the essential parity flip:

- **Odd-Length Cycles (ℓ odd):** An ℓ-cycle has an **odd** number of elements when ℓ is odd, but factors into an **even** number $(\ell - 1)$ of transpositions. Thus, odd-length cycles are even permutations (e.g., 3-cycles have sign $(-1)^2 = +1$).
- **Even-Length Cycles (ℓ even):** An ℓ-cycle has an **even** number of elements when ℓ is even, but factors into an **odd** number $(\ell - 1)$ of transpositions. Thus, even-length

cycles are odd permutations (e.g., 2-cycles have sign $(-1)^1 = -1$, and 4-cycles have sign $(-1)^3 = -1$).

The Knights of the Round Table Combinatorics

Spoken Transcript [00:36:22 - 00:38:01] Prof. Rahul Pandharipande

And I'm missing one partition of 4, and which one is it? It's the one that you normally forget about. Yeah?

Student Answer

4.

Spoken Transcript [00:36:34 - 00:38:01] Prof. Rahul Pandharipande

Four itself. It's the dumb partition [erases scratch calculations for 3-cycles].

This is all somehow stuff that you have to learn, you know, you have to make your own peace with. If you look at a book, maybe it has this formula, maybe not. But in order to learn the subject, you have to become comfortable with these things. It's just like learning addition and subtraction, except it's now multiplication of elements in the symmetric group, and it has these twists where it's not commutative... All right, so, we have to do 4 [writes: 4 in red chalk].

So, first of all, and these are all the partitions. The numbers we have are 1 [points to e], and then there is 6 [points to transpositions], and then there's 3 here [points to $2+2$ column], and here is 8 [points to $3+1$ column].

And then we have to find the last partitions... that's the only possible cycle left, and this equals 24 [writes: $1+6+3+8+\square = 24$]. So, you can figure out how many things I have to write here, right? And the answer is 6. So, I have to write 6.

And of course, one of them is $(1\,2\,3\,4)$ [writes: $(1\,2\,3\,4)$]. That's one of them.

Spoken Transcript [00:38:01 - 00:39:40] Prof. Rahul Pandharipande

And the reason this is 6 — so, I'm not going to write all six, I could, but... there's a way to think about this problem which is kind of basic.

It's the question of: if you have a long cycle like this, how many different ones can you write with the same indices? And this problem, if you study combinatorics, is the problem of knights on the round table. Have you seen this problem?

It's a kind of standard high school problem where you have a table, and you have some knights with different names: A, B, C... I don't know [draws a circle representing a round table with seated knights A, B, C, D, E]. You have some knights on the round table. And you want to know, if you're arranging a dinner party, how many different ways can you seat them?

And the idea is that the rotational symmetry doesn't matter, because they're still seated in the same way. And the answer is: if you have k knights, you can put them anywhere you want, that's $k!$, but if you rotate it, it's equivalent. So, a bunch of these are the same, and that's divided by k. So, the answer to this knight problem is $(k-1)!$ [writes: $\frac{k!}{k} = (k-1)!$].

And that's the exact same problem here: we have these $1, 2, 3, 4$, and we want to put them in a long cycle, all together. The number of ways you can do it is $4!/4$, which is exactly 6. That's the way you can do it. So, that's all of them, and, by the way, what's the sign of that?

Student Answer

Minus one.

Spoken Transcript [00:39:35 - 00:39:40] Prof. Rahul Pandharipande

Minus 1, because we just said that this is a product of three two-cycles. So, these signs are all minus 1 [writes: -1 in red chalk].

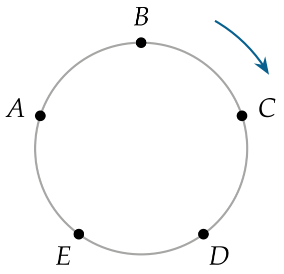
Counting Cycles and the Knights of the Round Table

Proposition 8.18 (*Number of Circular Arrangements*): The number of distinct ℓ-cycles that can be constructed using all elements of an ℓ-element set is:

$$\frac{\ell!}{\ell} = (\ell - 1)!. \tag{8.24}$$

In particular, for $\ell = 4$, the number of full 4-cycles in S_4 is:

$$(4-1)! = 3! = 6.$$



Combinatorics of Circular Equivalences: In linear permutations, there are $k!$ ways to order k distinct elements. However, cycle notation is invariant under cyclic shifts:

$$(a_1\, a_2\, \ldots\, a_k) = (a_2\, a_3\, \ldots\, a_k\, a_1) = \cdots = (a_k\, a_1\, \ldots\, a_{k-1}).$$

Because each cycle can be written in k equivalent ways by choosing any of its k elements as the starting entry, the total number of distinct k-cycles is $\frac{k!}{k} = (k-1)!$.

Spoken Transcript [00:39:40 - 00:40:27] Prof. Rahul Pandharipande

All right. So, now we're in a position to say what the alternating group here is: it's the ones with $+1$.

So, it's this one [circles e in red chalk] — in the alternating group, I circle in red — this is in the alternating group, this is in the alternating group [circles the three $(2+2)$ permutations in red chalk], and these guys are in the alternating group [circles the eight 3-cycles in red chalk].

So, that's totally explicit. This A_4 inside S_4 has order 12 [writes: $A_4 \subset S_4$, order 12], right? Because $8+3+1=12$.

And by the theorem we proved last week (Recall: Theorem 7.53), whatever its order is, it has to divide the order of S_4, and the order of S_4 is 24, so, it's at least reasonable, because 12 does divide 24.

Classification of the Elements of S_4 and Structure of A_4

We summarize the 24 elements of S_4 classified by their cycle partitions:

Partition	Permutations	Count	Sign	In A_4?
$1+1+1+1$	$e=(1)(2)(3)(4)$	1	$+1$	**Yes**
$2+1+1$	$(1\,2), (1\,3), (1\,4), (2\,3), (2\,4), (3\,4)$	6	-1	No
$2+2$	$(1\,2)(3\,4), (1\,3)(2\,4), (1\,4)(2\,3)$	3	$+1$	**Yes**
$3+1$	$(4)(1\,2\,3), (4)(1\,3\,2), (3)(1\,2\,4), (3)(1\,4\,2), (2)(1\,3\,4), (2)(1\,4\,3), (1)(2\,3\,4), (1)(2\,4\,3)$	8	$+1$	**Yes**
4	$(1\,2\,3\,4), (1\,2\,4\,3), (1\,3\,2\,4), (1\,3\,4\,2), (1\,4\,2\,3), (1\,4\,3\,2)$	6	-1	No
Total	$1+6+3+8+6=24$	24		$\lvert A_4\rvert = \mathbf{12}$

The alternating group $A_4 \leq S_4$ is precisely the union of all even permutations:

$$A_4 = \{e\} \cup \{(1\,2)(3\,4), (1\,3)(2\,4), (1\,4)(2\,3)\} \cup \{\text{all eight 3-cycles}\}. \tag{8.25}$$

The order of A_4 is:

$$|A_4| = 1+3+8 = 12.$$

In accordance with Lagrange's theorem (Theorem 7.53), $|A_4| = 12$ divides $|S_4| = 24$: exactly half of the permutations in S_4 are even.

Geometric Realization: Rotational Symmetries of the Tetrahedron

Spoken Transcript [00:40:27 - 00:42:00] Prof. Rahul Pandharipande

And you've seen this group! If you take the rotational symmetries of the tetrahedron... I'll try to draw it slightly better this time [draws a regular tetrahedron].

That's the tetrahedron. So, we discussed this group of rotational symmetries of the tetrahedron [writes: $\mathrm{Rot}(\triangle)$], and we said that the order of this group is 12 [writes: $|\mathrm{Rot}(\triangle)| = 12$]. And we also know that this group sits inside the symmetric group S_4 [writes: $\mathrm{Rot}(\triangle) \subset S_4$], because every rotation moves the four vertices. Is that clear? It's a subgroup of S_4, and has order 12.

So, it seems pretty likely it's going to be this group, but all you have to do to check it's this group is just to show — maybe *"see"* is a better word — to see every rotation gives an even permutation of the four vertices [writes: To check $\mathrm{Rot}(\triangle) = A_4$, all you have to do is see every rotation gives an even permut. of 4 verts].

Spoken Transcript [00:42:00 - 00:44:00] Prof. Rahul Pandharipande

And so, let's just see that. Is that clear? Because then it sits inside it, and it's order 12, and it has to be just the same. So, this is one of the basic ways this A_4 shows up in nature: it's the actual rotational symmetries of that tetrahedron. So, why is it even?

Well, there's kind of two types of symmetries. If I hold the top of this tetrahedron fixed, then I can rotate it by 120 degrees. And if you see what is that, that's the three-cycle: if this is A, B, C, D, if I rotate this, that's the B goes to C goes to D three-cycle, and A is fixed. So, that's this type [points to the $3+1$ column on the board]. And those are all $+1$, right?

But that's only eight. Of course, there's the one when you do nothing — that's also the rotational symmetry where you were just about to decide to rotate it and decided to do nothing; that's a legitimate symmetry, and it's this one [points to e].

So, what is this [points to the $2+2$ column]?

That's the one where you take the midpoint here... that's a cool one. You take the midpoint here, you take the midpoint here [marks midpoints of opposite edges on the drawing], and you draw a line through them, and you rotate it by 180 degrees — it's like a skewer. If you have like a tetrahedral kebab, and you rotate it... then these guys get flipped and those guys get flipped. Is that clear?

Anyway, you should think about it. I said part of geometry is about a camera in your head. And so, that rotation flips these two and those two simultaneously, so, it's of this form [points to $(12)(34)$ column].

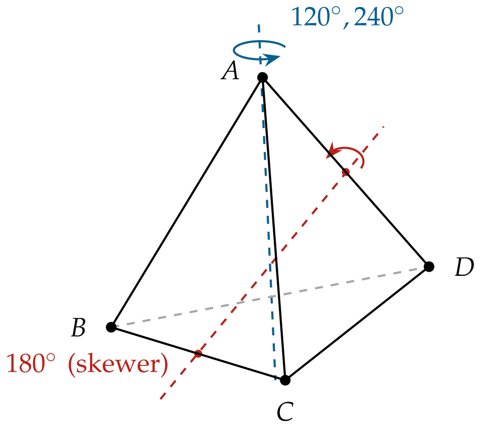
And you can get three of them, because there's three different ways to do it. So, from the point of view of mathematics, I've given you a totally complete proof that the rotational symmetries is A_4, but there's a lot of little things you should check for yourself to make sure you believe every step of it.

Geometric Isomorphism: $\mathrm{Rot}(\text{Tetrahedron}) \cong A_4$

Recap: The previous week found that the rotations of the tetrahedron form a group of order 12 inside S_4, acting on the four vertices, and named it A_4 before A_4 was defined (Theorem 7.58).

Theorem 8.19 (*Rotational Symmetries of the Regular Tetrahedron*): The group of rotational symmetries of a regular tetrahedron, $\mathrm{Rot}(\text{Tetrahedron})$, is isomorphic to the alternating group A_4:

$$\mathrm{Rot}(\text{Tetrahedron}) \cong A_4. \tag{8.26}$$



Proof that $\mathrm{Rot}(\text{Tetrahedron}) \cong A_4$

Every rotational symmetry of the regular tetrahedron induces a permutation of its four vertices $\{A, B, C, D\}$; as in the previous week (Theorem 7.58), this gives an injective group homomorphism:

$$\rho\colon \mathrm{Rot}(\text{Tetrahedron}) \hookrightarrow S_4.$$

We classify the 12 rotational symmetries by their geometric axes:

(1) **Identity:** The trivial rotation fixes all 4 vertices, corresponding to $e \in A_4$ (1 element).

(2) **Face-Vertex Rotations (3-Cycles):** There are 4 vertices, each paired with an opposite triangular face. Rotation by 120° or 240° around the axis connecting a vertex to the centroid of the opposite face fixes that vertex and cyclically permutes the remaining 3 vertices. This yields $4 \times 2 = 8$ distinct 3-cycles. By Lemma 8.17, every 3-cycle has sign $(-1)^{3-1} = +1$, so all 8 lie in A_4.

(3) **Skewer Rotations (Type $2+2$):** There are 6 edges of the tetrahedron, forming 3 pairs of mutually opposite, disjoint edges (e.g., AD and BC). Rotating by 180° around the axis passing through the midpoints of two opposite edges simultaneously swaps the endpoints of both edges:

$$(A\,D)(B\,C).$$

This yields 3 rotations of type $2+2$. The sign is $(-1)\cdot(-1) = +1$, so all 3 lie in A_4.

Because every rotational symmetry induces an even permutation, the image of ρ lies entirely inside the alternating group:

$$\mathrm{Im}(\rho) \subseteq A_4.$$

Since $|\mathrm{Rot}(\text{Tetrahedron})| = 1 + 8 + 3 = 12$ and $|A_4| = 12$, the injective map ρ is surjective, establishing the isomorphism $\mathrm{Rot}(\text{Tetrahedron}) \cong A_4$. **Q.E.D.**

Spoken Transcript [00:44:00 - 00:44:10] Prof. Rahul Pandharipande

Okay, maybe we take the break now. I have to do something else.

Mid-Lecture Break

The lecturer takes a brief mid-lecture pause. While stepping away, he pushes the lower blackboard upward, leaving a clean chalkboard visible for the next section.

Normal Subgroups and General Properties of Homomorphisms

Kernel, Image, and Injectivity Criteria

Spoken Transcript [00:44:18 - 00:45:55] Prof. Rahul Pandharipande

Okay, so, we go back to the general situation. We have G_1 is a group, G_2 is a group, and we have f from G_1 to G_2 [writes: $f\colon G_1 \to G_2$]. And in these general statements that I wrote, there was no condition that G_1 and G_2 are finite; they're just groups.

So, the first statement was: the image of f in G_2 is a subgroup [writes: $\mathrm{Im}(f) \subset G_2$ is subgp].

The second one was: the kernel of f in G_1 is a subgroup [writes: $\ker(f) \subset G_1$ is subgp].

And I just remind you: the kernel is those elements that go to the neutral element here. There was a question about that.

So, there are some lemmas here that I leave to you to prove, and they're the same in the other things we've studied: the lemma is that f is surjective if and only if the image of f is G_2 [writes: Lemma: f is surjective $\iff$ $\mathrm{Im}(f) = G_2$]. It's hardly even reasonable to call that a lemma, it's almost just the definition.

The second one is: f is injective [writes: f is injective $\iff$]... So, what am I going to write here?

Student Answer

The kernel is trivial.

Spoken Transcript [00:45:34 - 00:45:37] Prof. Rahul Pandharipande

The kernel is... and what does trivial mean?

Student Answer

Just the neutral element.

Spoken Transcript [00:45:38 - 00:45:55] Prof. Rahul Pandharipande

Yeah, it consists of just the neutral element [writes: $\ker(f) = \{e_1\}$].

Okay. So, this one is just by definition of surjective. This one uses a little argument which I leave to you. So, these are the most basic facts of the images and kernels here, except for one.

The most interesting fact comes now, and that's a new definition — that's the most interesting definition of the week: that...

Injectivity and Surjectivity of Group Homomorphisms

Lemma 8.20 (*Injectivity and Surjectivity via Kernel and Image*): Let $f\colon G_1 \to G_2$ be a group homomorphism.

(1) f is surjective if and only if $\operatorname{Im}(f) = G_2$.
(2) f is injective if and only if $\ker(f) = \{e_1\}$.

Proof of (1)

By definition, f is surjective if every $y \in G_2$ has a pre-image $x \in G_1$ with $f(x) = y$, which is precisely the statement $\operatorname{Im}(f) = G_2$. **Q.E.D.**

Proof of (2) (beyond the lecture)

The lecture leaves this part to the students.

($\Longrightarrow$): Assume f is injective. If $x \in \ker(f)$, then $f(x) = e_2 = f(e_1)$. Since f is injective, this forces $x = e_1$. Hence $\ker(f) = \{e_1\}$.

($\Longleftarrow$): Assume $\ker(f) = \{e_1\}$. Let $x, y \in G_1$ such that $f(x) = f(y)$. Multiplying by $f(y)^{-1} = f(y^{-1})$ in G_2:

$$f(x) \cdot f(y)^{-1} = e_2 \implies f(x) \cdot f(y^{-1}) = e_2 \implies f(x \cdot y^{-1}) = e_2.$$

Thus $x \cdot y^{-1} \in \ker(f)$. Since $\ker(f) = \{e_1\}$, we obtain:

$$x \cdot y^{-1} = e_1 \implies x = y.$$

Therefore, f is injective. **Q.E.D.**

Definition of Normal Subgroups

Spoken Transcript [00:45:55 - 00:47:48] Prof. Rahul Pandharipande

... N ... let's maybe put N [writes: $N \subset G$ is a normal subgroup].

N inside G is a normal subgroup. So, I'm defining normal subgroup. This is the definition [writes: $\iff$ Def]:

The first point is, of course, N is a subgroup [writes: • $N \subset G$ is a subgroup]. That's hardly a surprise.

And the second is the actual interesting part of the definition, and it says: if I take any person in G and anyone in N [writes: $\forall x \in G, \forall n \in N$], then if I do a conjugation —

conjugation is this: take $x \cdot n \cdot x^{-1}$ — this stays in N [writes: $xnx^{-1} \in N$]. So, that's the conclusion, all right?

So, this is the part that has some teeth.

Spoken Transcript [00:47:48 - 00:49:00] Prof. Rahul Pandharipande

Okay? If you're a normal subgroup, you have to be a subgroup, of course. Then you have to have this peculiar property that every time I pick any element in the subgroup and I conjugate it by anybody in the full group, I have to stay in the subgroup, okay?

So, are there any questions about this definition?

It's a definition, but it's super important for group theory, in particular for today. And the reason is...

Proposition: the kernel of f inside G_1 is a normal subgroup [writes: Prop: $\ker(f) \subset G_1$ is a normal subgroup].

All right, so, I will say a little bit about... when we study these homomorphisms for vector spaces, then the kernel and image are both vector spaces, and there's nothing more than that. When we study this for rings, the kernel is not a ring, it's an ideal. There was already a twist there. And for groups, the kernel is a subgroup, but it's not any subgroup: it has to be a normal subgroup. It's a very special kind of subgroup.

Definition 8.21 (*Normal Subgroup*): Let G be a group. A subgroup $N \leq G$ is called a *"normal subgroup"* of G if it is stable under conjugation by all elements of G:

$$\forall x \in G,\ \forall n \in N: \quad x \cdot n \cdot x^{-1} \in N. \tag{8.27}$$

Conjugation and Normal Subgroups: Conjugating an element $n \in N$ by $x \in G$ yields xnx^{-1}. In a commutative group conjugation does nothing ($xnx^{-1} = nxx^{-1} = n$), but in general xnx^{-1} differs from n. The normality condition requires that although xnx^{-1} may differ from n, it must still remain inside the subset N.

Notation: Normal Subgroups

The lecturer writes *"$N \subset G$ is a normal subgroup"* in words, and these notes keep to words. Textbooks write $N \trianglelefteq G$ or $N \triangleleft G$, the sign many of them also use for ideals.

Beyond the Lecture: Other Forms of Normality

Textbooks often state the condition for all of N at once: $xNx^{-1} \subseteq N$ for every $x \in G$, where $xNx^{-1} = \{xnx^{-1} \mid n \in N\}$. Applied to x^{-1} as well, it gives $xNx^{-1} = N$, which is the same as $xN = Nx$. The class xN of x (Definition 8.27) is called a **left coset** of N, and Nx a **right coset**; so a subgroup is normal exactly when its left and right cosets agree. Lemma 8.31 uses this equality one element at a time: $n_1 y \in Ny = yN$.

The Kernel as a Normal Subgroup

Spoken Transcript [00:49:00 - 00:50:36] Prof. Rahul Pandharipande

So, that's one of the first things about homomorphisms of groups: the kernels are subgroups, but they're not any subgroups, they are these normal subgroups, very special subgroups.

So, what's the proof of it? There's nothing to this proof: we have to check the kernel has this property.

So, the proof is: let x be in G and let n be in the kernel [writes: proof: Let $x \in G$ and $n \in \ker(f)$].

All right, so, then we have to consider this conjugation: consider xnx^{-1} [writes: Consider $xnx^{-1} \in G$]. That's certainly in G, and we have to decide whether it's in the kernel. How do we test whether it's in the kernel? We just apply f to it and see where it goes.

All right? So, we apply f to xnx^{-1} [writes: $f(xnx^{-1}) =$], and here we use the homomorphism property — actually, I guess we use it twice — and it just gives $f(x)f(n)f(x^{-1})$ [writes: $\overset{\text{hom}}{=} f(x)f(n)f(x^{-1})$], okay?

And now what do we do? Well, we know that n is in the kernel, so, this is the neutral element for the second one, right? So, this is equal to $f(x)$, and now we put the neutral element here, e_2 [writes: $= f(x)e_2f(x^{-1})$], because n is in the kernel [writes: $n \in \ker(f)$], okay?

And the neutral element doesn't do anything, we can just erase it, right? So, it goes away — that's the property of the neutral element, because e_2 is neutral [writes: $e_2 \in G_2$].

And this thing goes away too, because we know that $f(x^{-1})$ is the inverse of $f(x)$ (Recall: Eq. (8.6)). We did that at the beginning of the class. Or if you really want to be pedantic, you can then use this property [writes: $= f(xx^{-1}) = f(e_1) = e_2$].

So, that's the complete proof that says that when I take any element of the kernel and I take any element of G, if I consider the conjugation, I stay in the kernel, because when I apply f to it, I still get e_2.

The Kernel is Always a Normal Subgroup

Proposition 8.22 (*Normality of the Kernel*): Let $f\colon G_1 \to G_2$ be a group homomorphism. Then $\ker(f)$ is a normal subgroup of G_1.

Proof

We already established in Proposition 8.14 that $\ker(f) \leq G_1$ is a subgroup. It remains to verify the normality condition (8.27).

Let $x \in G_1$ and $n \in \ker(f)$ be arbitrary. To determine whether the conjugated element $x \cdot n \cdot x^{-1}$ belongs to $\ker(f)$, we evaluate f on it:

$$\begin{aligned} f(x \cdot n \cdot x^{-1}) &= f(x) \cdot f(n) \cdot f(x^{-1}) && \text{(homomorphism property applied twice)} \\ &= f(x) \cdot e_2 \cdot f(x^{-1}) && \text{(since } n \in \ker(f) \implies f(n) = e_2\text{)} \\ &= f(x) \cdot f(x^{-1}) && \text{(since } e_2 \text{ is the neutral element in } G_2\text{)} \\ &= f(x \cdot x^{-1}) = f(e_1) && \text{(homomorphism property; or use (8.6))} \\ &= e_2. \end{aligned} \tag{8.28}$$

Since $f(x \cdot n \cdot x^{-1}) = e_2$, we conclude that $x \cdot n \cdot x^{-1} \in \ker(f)$. Therefore, $\ker(f)$ is a normal subgroup of G_1. **Q.E.D.**

***Insight:* Kernels Across Algebraic Structures**

Notice the escalating hierarchy of kernels across abstract algebra:

- **Linear Algebra:** A linear map between vector spaces $T\colon V \to W$ has a kernel $\ker(T)$ that is an arbitrary sub-vector space; every sub-vector space is the kernel of a projection map.
- **Ring Theory:** A ring homomorphism $\phi\colon R \to S$ has a kernel that is **not** merely a subring, but an **ideal**.
- **Group Theory:** A group homomorphism $f\colon G_1 \to G_2$ has a kernel that is **not** an arbitrary subgroup, but a **normal subgroup**. Normal subgroups play precisely the role in group theory that ideals play in ring theory.

A Non-Normal Subgroup in S_3

Spoken Transcript [00:50:36 - 00:51:40] Prof. Rahul Pandharipande

So, in some sense... yeah, I don't want to say things that are distracting, but anyway, it is the case that you get subgroups for kernels, but you get very special subgroups, these normal subgroups.

So, you might ask: is it the case that every subgroup is normal? And I'm kind of implying that that's not the case. So, what's an example?

Example: if I take S_3 and I take the subgroup which is the neutral element of S_3 and $(1\,2)$ [writes: Example: S_3, $K = \{e, (1\,2)\} \subset S_3$], this is a subgroup in S_3, because if I take the square of this, it's the neutral element.

Do you accept this is a subgroup of S_3?

And so, then the claim is: this subgroup K is not normal in S_3 [writes: Claim: K is not normal in S_3].

Spoken Transcript [00:51:40 - 00:53:13] Prof. Rahul Pandharipande

And to show it's not normal, I just have to take one element of it, $(1\,2)$ [writes: $(1\,2)$], and take any element of S_3 and show the conjugation is not here. So, which element of S_3 should I take?

Most elements will work. If you're unlucky, you'll pick the wrong one.

It would be bad to take $(1\,2)$, but if you take anything else, like $(1\,3)$ [writes: $(1\,3)$], the inverse of $(1\,3)$ is $(1\,3)$ — so, this is x and that's x^{-1} [writes: $(1\,3)(1\,2)(1\,3)$] — and if you multiply this out, you get 1 goes to 3, 3 goes to 1, so, 1 is fixed here, 2 goes to 1, 1 goes to 3, so, this is $(2\,3)$, okay? [writes: $= (2\,3)$]

So, this is an element of our subgroup K, and this is the conjugation, and it's not in K [writes: $\notin K$].

Are you happy with this calculation?

So, this shows that this subgroup is not a normal subgroup. It just fails the criterion of being a normal subgroup.

In particular, this shows that there is no homomorphism of S_3 to anything whose kernel is this group. It's not possible, because the kernels are always normal subgroups.

If you meet a subgroup of a group, roughly speaking, the very first question to ask is: is it normal or not? It just changes how you think about it.

Example of a Non-Normal Subgroup

Example 8.23 (*A Non-Normal Subgroup of S_3*): Consider the symmetric group S_3 and the two-element subset:

$$K := \{e, (1\,2)\} \leq S_3.$$

Since $(1\,2)^2 = e$, K is a valid subgroup of order 2.

Claim 8.24: K is not a normal subgroup of S_3.

Proof

To disprove normality, it suffices to find a single element $x \in S_3$ and $n \in K$ such that $xnx^{-1} \notin K$. Take $n = (1\,2) \in K$ and choose $x = (1\,3) \in S_3$. Since every transposition is its own inverse, $x^{-1} = (1\,3)$. We evaluate the conjugation:

$$\begin{aligned}(1\,3) \circ (1\,2) \circ (1\,3)^{-1} &= (1\,3) \circ (1\,2) \circ (1\,3) \\ &= (2\,3).\end{aligned}$$

Since $(2\,3) \notin \{e, (1\,2)\} = K$, the subgroup K is not stable under conjugation. Thus K is not normal in S_3. **Q.E.D.**

Observation 8.25 (*Impossibility of K as a Kernel*): By Proposition 8.22, the kernel of any group homomorphism must be a normal subgroup. Consequently, there cannot exist **any** group homomorphism $f\colon S_3 \to H$ into any group H such that $\ker(f) = \{e, (1\,2)\}$.

Quotient Groups and the Quotient Map

Construction and Statement of the Quotient Theorem

Spoken Transcript [00:53:13 - 00:55:00] Prof. Rahul Pandharipande

Okay, so, once one understands that the kernels are always normal, then one proceeds to prove this First Isomorphism Theorem.

Let's do this slowly. We start with an abstract statement:

Let N in G be a normal subgroup [writes: Let $N \subset G$ be a normal subgroup].

Then we will construct — and this will be very similar to the way we construct quotient rings and the quotient map (Recall: Definition 4.44, Definition 4.45) — a quotient group, which is called G/N [writes: Then we will construct $(G/N, \cdot, \bar{e})$], which has its own product and its own neutral element, and a quotient homomorphism [writes: and quotient hom $q\colon G \to G/N$] with the property that the kernel of the quotient homomorphism is exactly the normal subgroup we started with [writes: with the property that $\ker(q) = N$].

This says that if you're not normal, you cannot be a kernel of a homomorphism.

This construction will show you that if you are normal, then you do arise as the kernel of a group homomorphism. In fact, we will construct the quotient and this kernel.

And this is very, very similar to what happened with rings. There were ideals, and there are some slight differences.

So, as usual, you have to concentrate to pick up the slight differences.

The Quotient Group Theorem

Theorem 8.26 (*Existence of the Quotient Group and Quotient Map*): Let G be a group and $N \leq G$ a normal subgroup. Then:

(1) The set G/N of equivalence classes of $\sim_N$ (Definition 8.27) is a group $(G/N, \cdot, \bar{e})$ with the product

$$\bar{x} \cdot \bar{y} \overset{\text{def}}{=} \overline{x \cdot y} \quad \text{for all } x, y \in G. \tag{8.29}$$

(2) The quotient map

$$q\colon G \to G/N, \qquad q(x) := \bar{x} \tag{8.30}$$

is a surjective group homomorphism whose kernel is precisely N:

$$\ker(q) = N. \tag{8.31}$$

Equivalence of Normal Subgroups and Kernels: Combining Proposition 8.22 and Theorem 8.26 establishes a fundamental equivalence in group theory:

N is the kernel of a homomorphism from G $\iff$ N is a normal subgroup of G.

A subgroup can be quotiented out if and only if it is normal.

Step 1: The Underlying Set G/N via Equivalence Classes

Spoken Transcript [00:55:00 - 00:56:55] Prof. Rahul Pandharipande

All right, so, Step 1 is we have to construct the quotient group [writes: Step 1: Construct quotient group $(G/N, \cdot, \bar{e})$]. And we've done this actually already: this is always done by an equivalence relation, and we've seen the equivalence relation before.

Remember from last week [writes: Remember the $\sim_N$] the equivalence relation with respect to a subgroup N: that says that x is equivalent to y if and only if $y = xn$ for some n in the subgroup, okay? (Recall: Proposition 7.54) [writes: $x \sim y \iff y = xn$ for some $n \in N$]

That was the equivalence relation from last week. Do you remember it?

And the proof that this was an equivalence relation required N to be a subgroup. It did not require N to be a normal subgroup; it just required N to be a subgroup. The normality condition is crucial, and it's going to come later.

Anyway, just because it's a subgroup, we have this equivalence relation, and so, we define the set G/N to be the set of equivalence classes of this relation [writes: Def: $G/N =$

set of equiv. classes of $\sim_N$], okay? We used this construction to prove Lagrange's theorem last time.

But the discussion today is slightly different in two ways: one is that we're going to be using it for a normal subgroup, not just any subgroup. And the second is, for Lagrange's theorem, the whole point of it was proving some divisibility, which had to do with finite groups, and there's nothing here about finite groups — I haven't made any assumptions about finite groups here.

The Equivalence Relation of a Subgroup

Recap: This is the equivalence relation of the previous lecture (Eq. (7.64), Proposition 7.54), where it served the proof of Lagrange's theorem (Theorem 7.53).

Definition 8.27 (*The Set G/N*): Let G be a group and $N \leq G$ a subgroup. We define the equivalence relation $\sim_N$ on G by:

$$x \sim_N y \iff x^{-1}y \in N \iff y = xn \quad \text{for some } n \in N. \tag{8.32}$$

The underlying set of the quotient G/N is defined as the set of all equivalence classes under $\sim_N$:

$$G/N := \{[x] \mid x \in G\} = \{xN \mid x \in G\}. \tag{8.33}$$

Lagrange versus Quotient Groups: In the proof of Lagrange's theorem (Theorem 7.53), all classes of this relation have $|H|$ elements, which gives $|H| \mid |G|$ for a finite group G. Here nothing is assumed finite, and N will be normal: that is what makes the product on G/N, defined next, well defined.

Blackboard Transition

The lecturer pauses speaking from 00:56:55 to 00:57:35 to completely erase the right half of the lower chalkboard with a sponge, preparing a clean writing surface to define the product on G/N and verify its well-definedness.

Step 2: Defining the Multiplication on G/N

Spoken Transcript [00:57:35 - 00:58:55] Prof. Rahul Pandharipande

Okay, so, if you accept this definition of the set for G/N, then I have to give you the group structure, which means I have to tell you how to do the group multiplication.

Let's try to do that.

There's the usual discussion of names, how to name things in the equivalence classes.

Here I have G/N [writes: $\bar{x} \in G/N$], and we name things by saying $\bar{x}$ here, where x is in G and $\bar{x}$ is the equivalence class of x under the equivalence relation [writes: $x \in G, \bar{x}$ is the equiv. class of x under $\sim_N$].

That's one convention. Another convention is you just write xN is an element of G/N [writes: $xN \in G/N$], because the actual equivalence class, as we said before, is just all the people who are x times some element of N. So, it's just another way of writing it.

I'll tend to use this notation because it's slightly more efficient, but if you look, sometimes people will write this.

Okay, so, the main issue [writes: Main issue:] is to define the group product. We take some element here: we want to define $\bar{x} \cdot \bar{y}$ [writes: $\bar{x} \cdot \bar{y} \overset{\text{def}}{=}$].

All right, so, what do you think we're going to define this as? This is not a hard question. Yeah?

Student Answer

$\overline{xy}$

Spoken Transcript [00:58:55 - 01:01:00] Prof. Rahul Pandharipande

Yeah, what else could you do? Nothing else! You just have to write this: $\overline{xy}$ [writes: $\overline{xy}$].

Okay. So, that's how you define it. But the issue is whether this is well-defined.

So, the heart of the matter is not this, but: is this rule well-defined? [writes: Is this rule well-defined?]

And this has to do with the problem about names, which I've discussed several times, because every equivalence class has many names when I use the bar notation. For this to be a definition, when I change the names here, I have to end up in the same equivalence class. So, that's really the heart of the matter, and this is where all the action is in this definition.

Let us check what happens when we change the names [writes: Let us check when we change the names:].

This means that we have $\bar{x} = \overline{xn_1}$ [writes: $\bar{x} = \overline{xn_1}, n_1 \in N$], where n_1 is in N, because that's the definition of the equivalence class. Instead of taking x, I can take any element of N, n_1, and just call it that, right?

And $\bar{y} = \overline{yn_2}$ [writes: $\bar{y} = \overline{yn_2}, n_2 \in N$] for some n_2 in N.

So, in order for this definition to make sense, what do I have to check? I must check that [writes: I must check that] if I take this, then this should be in the same... I wonder: is this in the same equivalence class as this? [writes: $\overline{xy} \overset{?}{=} \overline{xn_1yn_2}$ are these the same equiv. classes] That's what I wonder.

Because I'm trying to define a multiplication on the level of equivalence classes, and the rule is: I select one representative of the first equivalence class, another representative of the second one, and I just say it's the equivalence class of the product. But I could have picked different representatives, and the only way this is well-defined is if the equivalence class of the product is the same. Is that clear?

So, now we have to check. It looks extremely unlikely. Are these the same equivalence classes?

Names for the Classes and the Proposed Multiplication

Notation 8.28 (*Names for the Classes in G/N*): Let $N \leq G$ and $x \in G$. Two notations name the equivalence class of x under $\sim_N$:

(1) Bar Notation: $\bar{x} \in G/N$.

(2) Product Notation: $xN = \{xn \mid n \in N\} \in G/N$, the elements x times an element of N.

In this lecture, we mostly use the bar notation $\bar{x}$.

Definition 8.29 (*Multiplication on G/N*): For any $\bar{x}, \bar{y} \in G/N$, we define their group product by:

$$\bar{x} \cdot \bar{y} \overset{\text{def}}{=} \overline{x \cdot y}. \tag{8.34}$$

Question 8.30 (*The Well-Definedness Problem*): Since a class has many names ($\bar{x} = \overline{xn_1}$ and $\bar{y} = \overline{yn_2}$ for any $n_1, n_2 \in N$), does the product depend on the choice of representatives? That is, does the equality

$$\overline{x \cdot y} \overset{?}{=} \overline{xn_1 \cdot yn_2} \tag{8.35}$$

hold for all $x, y \in G$ and all $n_1, n_2 \in N$?

Using Normality to Establish Well-Definedness

Spoken Transcript [01:01:00 - 01:03:20] Prof. Rahul Pandharipande

It looks extremely unlikely, because we cannot switch these orders using the group axioms.

If we could switch these orders, we would be happy, because I could switch this over here and then this would be $xy \cdot n_1n_2$, and that would be in the same equivalence class. Is that clear? But I'm not allowed to do that.

So, I have one thing in my pocket, which is the fact that it's a normal subgroup. I have to use it.

So, we're going to use the fact that N is normal [writes: Use N is normal]. And we want to switch this order.

The idea is you look at n_1, and y is an element of the group: this is a conjugation [writes: $y^{-1}n_1y$]. Do you agree this is a conjugation? And because it's normal, this thing is in N [writes: $\in N$]. That uses the normality: that this exact conjugation is equal to something in the subgroup.

Is that clear? This is actually the crucial step.

And this is where the normality condition comes in: we have to confront this naming problem and the well-definedness of the product. And there's just no way to do it besides switching this order. You have to somehow be able to switch this order. And this normality gives you a sneaky way to switch the order.

Because the normality says that this element, whatever it is, is some other element $\hat{n}_1$ in N. So, that means I get this equation: $y^{-1}n_1y = \hat{n}_1$ [writes: $y^{-1}n_1y = \hat{n}_1$].

I could have called it n_3 or anything, I'm just calling it $\hat{n}_1$.

Then I can multiply by y. If I multiply by y, it's yy^{-1}, so, that goes away, but this is

$y\hat{n}_1$ [writes: $n_1 y = y\hat{n}_1$].

And this is extremely pleasant for me, because we are now allowed to switch the order. Well, it's not switching the order exactly, it's giving us a rule for switching the order: if we want to move the y to this side, we can get something else which is still in N, that's okay.

So, let's do this carefully, since this is kind of the crucial part.

The Normality Commutation Trick

Lemma 8.31 (*Commutation via Normality*): Let N be a normal subgroup of G. For any $y \in G$ and any $n_1 \in N$, there exists an element $\hat{n}_1 \in N$ such that:

$$n_1 \cdot y = y \cdot \hat{n}_1. \tag{8.36}$$

Proof

Since N is normal, it is stable under conjugation by all elements of G; we conjugate $n_1 \in N$ by $y^{-1} \in G$:

$$\hat{n}_1 := y^{-1} \cdot n_1 \cdot y \in N. \tag{8.37}$$

Multiplying both sides on the left by y:

$$y \cdot \hat{n}_1 = y \cdot (y^{-1} \cdot n_1 \cdot y) = (y \cdot y^{-1}) \cdot n_1 \cdot y = e \cdot n_1 \cdot y = n_1 \cdot y.$$

This gives the commutation relation $n_1 \cdot y = y \cdot \hat{n}_1$ with $\hat{n}_1 \in N$. **Q.E.D.**

Why Normality is Indispensable for Multiplication: In general, elements $n_1 \in N$ and $y \in G$ do not commute ($n_1 y \neq y n_1$). We cannot simply swap n_1 and y, but normality lets us slide y to the left at the cost of replacing n_1 with a conjugated partner $\hat{n}_1 \in N$. This modified swap is sufficient to resolve the well-definedness problem.

Spoken Transcript [01:03:20 - 01:05:28] Prof. Rahul Pandharipande

[erases the middle chalkboard area to make room for the derivation] And it's the standard thing with this kind of math: this problem was anticipated before the whole argument started, and the definition was arranged precisely correctly to solve this problem when you found it.

Okay, so, the last step here, which maybe is already implicit: we have to think about whether this equals $\overline{xn_1yn_2}$ [writes: $\overline{xy} \overset{?}{=} \overline{xn_1yn_2}$].

So, we write here: xn_1yn_2 equals — now using that rule I'm allowed to switch this — $xy\hat{n}_1n_2$ [writes: $xn_1yn_2 = xy\hat{n}_1n_2$]. That was using the normality for some $\hat{n}_1$.

So, this means that $\overline{xn_1yn_2} = \overline{xy\hat{n}_1n_2}$ [writes: $\overline{xn_1yn_2} = \overline{xy\hat{n}_1n_2}$].

And that is equal to $\overline{xy}$, as we want [writes: $= \overline{xy}$].

So, we're very happy with that, because these two things are in the same equivalence class, because they differ by an N multiplication on the right. These are two elements in N, but when you multiply two elements in N, you stay in N because it's a group. Okay?

So, that's the proof that if I take this set of equivalence classes, at least we've defined this: this is well-defined [writes: $(G/N, \cdot)$ well def.].

Resolution of the Well-Definedness Question

Answer 8.32 (*to Question 8.30*): The multiplication defined in (8.34) is well-defined.

Proof

Let $\bar{x} = \overline{xn_1}$ and $\bar{y} = \overline{yn_2}$ with $n_1, n_2 \in N$. We compute the product of these representatives in G:

$$(xn_1) \cdot (yn_2) = x \cdot (n_1 y) \cdot n_2.$$

By Lemma 8.31, there exists $\hat{n}_1 \in N$ such that $n_1 y = y\hat{n}_1$. Substituting this identity:

$$\begin{aligned}(xn_1) \cdot (yn_2) &= x \cdot (y\hat{n}_1) \cdot n_2 \\ &= (xy) \cdot (\hat{n}_1 n_2).\end{aligned}$$

Since $N \leq G$ is closed under multiplication, $n_3 := \hat{n}_1 \cdot n_2 \in N$. Therefore:

$$(xn_1) \cdot (yn_2) = (xy) \cdot n_3 \quad \text{with } n_3 \in N.$$

By the definition of $\sim_N$ in (8.32), this implies:

$$xn_1yn_2 \sim_N xy \implies \overline{xn_1yn_2} = \overline{xy}.$$

Thus, the product $\overline{xn_1} \cdot \overline{yn_2} = \overline{xy}$ is completely independent of the chosen representatives.

Q.E.D.

III The Quotient Homomorphism

Spoken Transcript [01:05:28 - 01:08:00] Prof. Rahul Pandharipande

And of course, what is going to be the neutral element? It's the equivalence class of the neutral element [writes: $(G/N, \cdot, \bar{e})$].

And the rest is just checking a bunch of compatibilities for the construction, which I leave to you. This was the exciting part.

So, then, I leave the other checks to you [writes: I leave the other checks to you]. There's not much to do here. You have to check that $(\bar{x})^{-1} = \overline{x^{-1}}$ [writes: $(\bar{x})^{-1} = \overline{x^{-1}}$], and some other things like that to check the group axioms. But all of it is straightforward. The interesting step is that it's well-defined, and that's where we use the normality.

Okay, so, we have this G/N, and then there's the map $G \to G/N$ [writes: $G \xrightarrow{q} G/N$]. This is the definition of the quotient map [writes: def of q:]: I have to define it for every element of G, and how are you going to define this?

$q(x) = \bar{x}$ [writes: $q(x) = \bar{x}$].

It's again some kind of tautological definition, okay? That's the definition. It's exactly what we did for rings, and also what happened for vector spaces.

And then there's the claim that q is a group homomorphism [writes: claim: q is a group hom], q is surjective [writes: q is surjective], and the kernel of q is exactly N [writes: $\ker(q) = N$], all right?

So, those are the three claims.

And they're proven just the same way we proved it for rings, and you've proved it before for vector spaces.

So, again, maybe I leave it to you this time.

The Quotient Group and the Quotient Map

Proposition 8.33 (*Group Axioms for G/N*): Let N be a normal subgroup of G. The pair $(G/N,\cdot)$ with the multiplication $\bar{x}\cdot\bar{y}=\overline{xy}$ is a group with neutral element $\bar{e}=N$ and inverses given by:

$$(\bar{x})^{-1}=\overline{x^{-1}}. \tag{8.38}$$

Proof (beyond the lecture)

The lecture leaves these checks to the students.

1. **Associativity:** For all $\bar{x},\bar{y},\bar{z}\in G/N$:
$$(\bar{x}\cdot\bar{y})\cdot\bar{z}=\overline{xy}\cdot\bar{z}=\overline{(xy)z}=\overline{x(yz)}=\bar{x}\cdot\overline{yz}=\bar{x}\cdot(\bar{y}\cdot\bar{z}).$$
2. **Neutral Element:** For any $\bar{x}\in G/N$:
$$\bar{x}\cdot\bar{e}=\overline{x\cdot e}=\bar{x}\quad\text{and}\quad\bar{e}\cdot\bar{x}=\overline{e\cdot x}=\bar{x}.$$
3. **Inverses:** For any $\bar{x}\in G/N$:
$$\bar{x}\cdot\overline{x^{-1}}=\overline{x\cdot x^{-1}}=\bar{e}\quad\text{and}\quad\overline{x^{-1}}\cdot\bar{x}=\overline{x^{-1}\cdot x}=\bar{e}.$$

Hence $(\bar{x})^{-1}=\overline{x^{-1}}$. **Q.E.D.**

Proposition 8.34 (*Properties of the Quotient Map*): The quotient map

$$q\colon G\to G/N,\qquad q(x):=\bar{x}$$

satisfies:

- **(1)** q is a group homomorphism.
- **(2)** q is surjective.
- **(3)** $\ker(q)=N$.

Proof (beyond the lecture)

The lecture leaves this to the students.

1. **Homomorphism:** For all $x,y\in G$, by definition of the multiplication on G/N:
$$q(x\cdot y)=\overline{x\cdot y}=\bar{x}\cdot\bar{y}=q(x)\cdot q(y).$$
2. **Surjectivity:** Any element of G/N is of the form $\bar{x}$ for some $x\in G$. Then $q(x)=\bar{x}$, proving surjectivity.
3. **Kernel:** By definition of kernel:
$$\begin{aligned}\ker(q)&=\{x\in G\mid q(x)=\bar{e}\}\\&=\{x\in G\mid \bar{x}=\bar{e}\}\\&=\{x\in G\mid x\sim_N e\}\\&=\{x\in G\mid x=e\cdot n\text{ for some }n\in N\}\\&=N.\end{aligned}$$

Q.E.D.

Spoken Transcript [01:08:00 - 01:08:45] Prof. Rahul Pandharipande

[pauses to erase the right section of the lower chalkboard with a sponge]

Okay, so, that's the construction of the quotient. I skipped some steps, but I think I did the crucial steps, and to understand that you have to go through all the steps of the construction of the quotient. As I said, we'd already gone through this once with the quotient ring, and it's very, very similar.

The main issue is this normality, and with groups you should always remember: you can't switch things around without explaining.

All right, so, then we come to this First Isomorphism Theorem, which is exactly parallel.

The First Isomorphism Theorem for Groups

Spoken Transcript [01:08:45 - 01:10:50] Prof. Rahul Pandharipande

So, let $f\colon G_1 \to G_2$ be a group homomorphism [writes: First Isom Thm for Groups] [writes: let $f\colon G_1 \to G_2$ be a group hom.].

Then the First Isomorphism Theorem is that $G_1/\ker(f)$ is isomorphic to the image of f [writes "then $G/\ker(f) \cong \mathrm{Im}(f)$", a slip for $G_1/\ker(f)$]. Okay?

So, that was what we had for rings, and also for vector spaces. In fact, the diagram here... and the proof is just the same in all of these things. But maybe I just review the diagram: this is a review of three times first isomorphism [writes: Review $3 \times 1^{\text{st}}$ isom].

The first time you see it is if $f\colon V_1 \to V_2$ is a linear transformation [writes: $f\colon V_1 \to V_2$ lin. trans. of vector spaces/F] of vector spaces over some field F.

Then the First Isomorphism Theorem says that we have here V_1, V_2 [writes: $V_1 \longrightarrow V_2$], and then inside here is the kernel of f — that is a subspace [writes: $\ker(f) \subset V_1$] — then we have the quotient map to V_1 modulo the kernel of f [writes: $V_1 \to V_1/\ker(f)$], and this is the image of f, that's inside V_2 [writes: $\mathrm{Im}(f) \subset V_2$], and this is an isomorphism [writes: $\cong$ arrow from $V_1/\ker(f)$ to $\mathrm{Im}(f)$].

Okay? So, that's the First Isomorphism Theorem in linear algebra. Was it called that? Or I don't know, maybe not?

Theorem 8.35 (*First Isomorphism Theorem for Groups*)**:** Let $f\colon G_1 \to G_2$ be a group homomorphism. Then the quotient group $G_1/\ker(f)$ is isomorphic to the image subgroup $\mathrm{Im}(f)$:

$$G_1/\ker(f) \cong \mathrm{Im}(f). \tag{8.39}$$

Proof (beyond the lecture)

The lecture says that the proof is the same as for rings and gives it in the next lecture (Theorem 8.50). Set $K := \ker(f)$, a normal subgroup of G_1 by Proposition 8.22, and define $\bar{f}\colon G_1/K \to \operatorname{Im}(f)$ by

$$\bar{f}(\bar{x}) := f(x) \quad \text{for all } x \in G_1. \tag{8.40}$$

1. Well-Definedness and Injectivity: Let $\bar{x}, \bar{y} \in G_1/K$. Then:

$$\begin{aligned} \bar{x} = \bar{y} &\iff x^{-1}y \in K \\ &\iff f(x^{-1}y) = e_2 \\ &\iff f(x)^{-1}f(y) = e_2 \\ &\iff f(x) = f(y) \\ &\iff \bar{f}(\bar{x}) = \bar{f}(\bar{y}). \end{aligned}$$

Reading from left to right establishes that $\bar{f}$ is well-defined; reading from right to left establishes that $\bar{f}$ is injective.

2. Homomorphism Property: Using the definition of multiplication in G_1/K and the homomorphism property of f:

$$\bar{f}(\bar{x}\cdot\bar{y}) = \bar{f}(\overline{x\cdot y}) = f(x\cdot y) = f(x)\cdot f(y) = \bar{f}(\bar{x})\cdot\bar{f}(\bar{y}).$$

3. Surjectivity: By definition, any element in $\operatorname{Im}(f)$ has the form $f(x)$ for some $x \in G_1$. Then $\bar{f}(\bar{x}) = f(x)$, so $\bar{f}$ is surjective onto $\operatorname{Im}(f)$. Thus, $\bar{f}$ is a bijective group homomorphism, proving $G_1/\ker(f) \cong \operatorname{Im}(f)$. With $q\colon G_1 \to G_1/K$ the quotient map and $\iota\colon \operatorname{Im}(f) \hookrightarrow G_2$ the inclusion, $f = \iota \circ \bar{f} \circ q$: this is the triangle the lecturer draws for all three cases below. **Q.E.D.**

The Three-Fold Parallel: Vector Spaces, Rings, and Groups

Spoken Transcript [01:10:50 - 01:12:40] Prof. Rahul Pandharipande

That's for linear algebra.

For ring theory, we had the same thing: here R_1 is a ring [writes: R_1], there's a ring homomorphism, that's the image of f [writes: $R_1 \longrightarrow \operatorname{Im}(f) \subset R_2$] — that's in R_2 — and there's the kernel; the kernel is now an ideal [writes: $\ker(f) \subset R_1$, Ideal].

So, then, there's the quotient map from R_1 to R_1 modulo the kernel of f [writes: $R_1 \to R_1/\ker(f)$], and that's isomorphic to the image [writes: $\cong$ arrow to $\operatorname{Im}(f)$]. That was the First Isomorphism Theorem for rings (Recall: Theorem 4.27).

And for groups, that's the one we're in the middle of proving. The issue here was: the kernel is always just a vector space, it's exactly the same flavor as all the objects. Here the kernel is different, it's an ideal — that was the twist here.

For groups, again, it's the same picture: we have G_1, it maps to the image of f inside G_2 [writes: $G_1 \longrightarrow \operatorname{Im}(f) \subset G_2$], and G_1 maps to G_1 mod $\ker(f)$ [writes: $G_1 \to G_1/\ker(f)$]. The kernel of f is a subgroup, but what's special here is it's a normal subgroup [writes: $\ker(f) \subset G_1$, Normal], and this is isomorphic [writes: $\cong$ arrow].

So, that's three times the First Isomorphism Theorem, and they're formally the same. One thinks of them as basically the same structure, and the part you have to be careful about is that the nature of the kernel is a little bit different in the three cases.

Comparative Overview: The First Isomorphism Theorem Across Algebraic Categories

Vector Spaces / F:
$\ker(f) \subset V_1$ (subspace)

$$\begin{tikzcd}
V_1 \arrow[rr, "f"] \arrow[dr, "q"'] & & \mathrm{Im}(f) \subset V_2 \\
& V_1/\ker(f) \arrow[ur, "\cong"'] &
\end{tikzcd}$$

Ring Theory:
$\ker(f) \subset R_1$ (**ideal**)

$$\begin{tikzcd}
R_1 \arrow[rr, "f"] \arrow[dr, "q"'] & & \mathrm{Im}(f) \subset R_2 \\
& R_1/\ker(f) \arrow[ur, "\cong"'] &
\end{tikzcd}$$

Group Theory:
$\ker(f) \subset G_1$ (**normal**)

$$\begin{tikzcd}
G_1 \arrow[rr, "f"] \arrow[dr, "q"'] & & \mathrm{Im}(f) \subset G_2 \\
& G_1/\ker(f) \arrow[ur, "\cong"'] &
\end{tikzcd}$$

The Essential Difference: The Nature of the Kernel: In all three settings the diagram is the same triangle: the quotient map q goes down to the quotient, and the isomorphism goes from the quotient to the image of f. The difference lies in the nature of $\ker(f)$:

- **Vector Spaces:** The kernel is a sub-vector space, the same kind of object as V_1.
- **Rings:** The kernel is not a subring but an **ideal**: it absorbs multiplication by all of R_1.
- **Groups:** The kernel is a subgroup, and more: a **normal subgroup**, stable under conjugation by all of G_1.

Spoken Transcript [01:12:40 - 01:13:38] Prof. Rahul Pandharipande

Okay, so, maybe I'll leave it at this. I have one more lecture on this topic tomorrow, so I'll come back to say more about the proof here. The proof is all tautological. But I want to use the last few minutes to do some example stuff.

And somehow, since we already went through this path once, it seems unnecessary to spend a lot of time going through the path again. But I will do a little bit tomorrow.

[pushes the lower board up and erases the right lower board]

Conjugacy and Conjugacy Classes

The Conjugacy Relation

Spoken Transcript [01:13:38 - 01:15:15] Prof. Rahul Pandharipande

Okay, so, let's go back to S_n.

The notion here that comes up is conjugation. It already came up in the definition of a normal subgroup.

So, if we have x and y in a group G are conjugate [writes: x and y in G are conjugate] — this is the definition of this term [writes: Def:] — if and only if there exists some other element in the group [writes: $\exists z \in G$] such that z conjugates x to y [writes: $zxz^{-1} = y$]. Okay? So, that's not a surprising definition.

And one can say that a normal subgroup is a group that, if it contains an element, contains all of the element's conjugates. That's another way of just saying what a normal subgroup is. So, I can maybe say that in words:

N in G is a normal subgroup [writes: $N \subset G$ is a normal subgroup] if and only if, of course, N is a subgroup [writes: N subgroup], and if n is in N, then all the conjugates of n are also in N [writes: if $n \in N$ then all conjugates of n are also in N].

That's just a restatement of the definition of normal subgroup.

Conjugacy in Groups and Normal Subgroups

Definition 8.36 (*Conjugate Elements*): Let G be a group. Two elements $x, y \in G$ are called *"conjugate"* in G, denoted

$$x \sim_{\text{conj}} y,$$

if there exists an element $z \in G$ such that:

$$z \cdot x \cdot z^{-1} = y. \tag{8.41}$$

Proposition 8.37 (*Normality via Conjugate Closure*): A subgroup $N \leq G$ is normal if and only if N contains all conjugates of its elements:

$$n \in N \implies z \cdot n \cdot z^{-1} \in N \quad \text{for all } z \in G. \tag{8.42}$$

Conjugacy Classes as an Equivalence Relation

Spoken Transcript [01:15:15 - 01:17:00] Prof. Rahul Pandharipande

In every group there's a notion of conjugacy, and this is an equivalence relation: $x \sim_{\text{conj}} y$ if and only if x and y are conjugate in G [writes: $x \sim_{\text{conj}} y \iff x$ and y are conjugate in G].

Conjugacy defines an equivalence relation on G [writes: Conjugacy defines an equivalence relation on G].

This is a different kind of equivalence relation: we already said that if you have a subgroup, that defines an equivalence relation; if you have a normal subgroup, that defines an equivalence relation. But on the group G, there's a notion of equivalence that has nothing to do with any choice of subgroups: it's just when two elements are conjugate.

So, here's the group G [draws a large potato/bean shape representing G partitioned into conjugacy classes], in some kind of equivalence classes. These are called the conjugacy classes [writes: Conjugacy classes].

You can decompose G into the conjugacy classes, which are the equivalence classes of this conjugacy relation.

And you'll maybe be surprised at this moment to learn...

Well, one thing you're not surprised by: the neutral element is by itself. The neutral element's conjugacy class is very small. The only way to be conjugate to the neutral element is if you are the neutral element, because if you try to conjugate the neutral element, you don't go anywhere. So, that's a very small conjugacy class [draws a small singleton cell containing e inside the diagram of G]. Other conjugacy classes could be large.

Partition of a Group into Conjugacy Classes

Proposition 8.38 (*Conjugacy is an Equivalence Relation*): The relation $\sim_{\text{conj}}$ is an equivalence relation on G.

Proof (beyond the lecture)

The lecture states this without proof.

(1) Reflexivity: For any $x \in G$, $e \cdot x \cdot e^{-1} = x$, so $x \sim_{\text{conj}} x$.

(2) Symmetry: If $y = zxz^{-1}$, then $x = z^{-1}yz = z^{-1}y(z^{-1})^{-1}$, so $y \sim_{\text{conj}} x$.

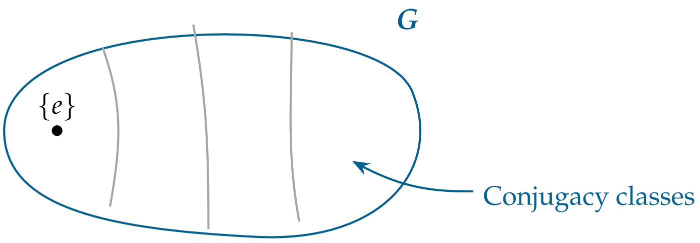
(3) Transitivity: If $y = z_1xz_1^{-1}$ and $w = z_2yz_2^{-1}$, then $w = (z_2z_1)x(z_2z_1)^{-1}$, so $x \sim_{\text{conj}} w$.

Q.E.D.

Definition 8.39 (*Conjugacy Class*): The equivalence class of an element $x \in G$ under $\sim_{\text{conj}}$ is called the *"conjugacy class"* of x:

$$\{z \cdot x \cdot z^{-1} \mid z \in G\}. \tag{8.43}$$

As for every equivalence relation, G is the disjoint union of its conjugacy classes.



Observation 8.40 (*The Class of the Neutral Element*): For the neutral element $e \in G$, conjugation by any $z \in G$ yields $zez^{-1} = zz^{-1} = e$. Thus the conjugacy class of e is $\{e\}$.

III Conjugacy in Matrix Groups and Jordan Normal Form

Spoken Transcript [01:17:00 - 01:19:02] Prof. Rahul Pandharipande

The fact of the matter is that most of linear algebra is the study of conjugacy classes. That's just the fact of the matter, probably at least 50 percent of the course. It's usually not said that way, but that's what it is.

There's a theorem in linear algebra: what are the conjugacy classes of $\mathrm{GL}_n(\mathbb{C})$? That's basically the best theorem in the linear algebra class. It's a matter of taste, I guess. But you know what theorem I'm talking about?

The problem in general can be hard, but the one place where there's a nice classical solution is matrices with complex coefficients. And you can ask: when are two matrices in the same conjugacy class? Or how are the conjugacy classes indexed?

And what is the answer? How are conjugacy classes indexed in $\mathrm{GL}_n(\mathbb{C})$? Yeah?

[inaudible student response]

Exactly: Jordan normal form! That's the answer. That's in fact the whole point of that!

So, if you look at $\mathrm{GL}_n(\mathbb{C})$ [writes: $(\mathrm{GL}_n(\mathbb{C}), \cdot, \mathrm{Id})$], this is a group with matrix multiplication and identity, and the statement is that the conjugacy classes [writes: Conjugacy classes] of this group are in bijective correspondence with the Jordan normal form [writes: $\overset{\text{bij}}{\longleftrightarrow}$ JNF]. That's the whole purpose of the Jordan normal form: how to index the conjugacy classes.

So, that's a really cool answer, and that's the group-theoretic meaning of it. You're just trying to figure out what the conjugacy classes are. Because in linear algebra this conjugacy stuff is about change of basis, or whatever it's called there. But it's just conjugacy in the group.

Conjugacy in $\mathrm{GL}_n(\mathbb{C})$ and Jordan Normal Form

Observation 8.41 (*Conjugacy as Change of Basis in Linear Algebra*): In the general linear group $\mathrm{GL}_n(\mathbb{C})$, two invertible matrices $A, B \in \mathrm{GL}_n(\mathbb{C})$ are conjugate if and only if there exists an invertible matrix $P \in \mathrm{GL}_n(\mathbb{C})$ such that:

$$P \cdot A \cdot P^{-1} = B.$$

In linear algebra, this is a change of basis: A and B describe the same linear map in two bases.

Fact 8.42 (*Classification of Conjugacy Classes via Jordan Normal Form*): The conjugacy classes of the group $\mathrm{GL}_n(\mathbb{C})$ are in bijective correspondence with Jordan normal forms (up to permutation of Jordan blocks):

$$\text{Conjugacy classes of } \mathrm{GL}_n(\mathbb{C}) \xleftrightarrow{1-1} \text{Jordan Normal Forms (JNF).} \tag{8.44}$$

Every matrix $A \in \mathrm{GL}_n(\mathbb{C})$ is conjugate to a Jordan matrix J, unique up to the order of its blocks, so J can stand for the conjugacy class of A.

Beyond the Lecture: More on Conjugacy Classes

Textbooks write the conjugacy class of x as $\mathrm{Cl}(x)$ or x^G. An element x is alone in its class exactly when $zxz^{-1} = x$ for all $z \in G$, that is, when x commutes with every element of G; these elements form the **center** $Z(G)$, which the next lecture meets as the kernel of conjugation (Definition 8.59). The restatement in Proposition 8.37 says that a subgroup is normal exactly when it is a union of conjugacy classes. And conjugate matrices in $\mathrm{GL}_n(\mathbb{C})$ are called **similar** in linear algebra.

Conjugacy Classes in the Symmetric Group and Partitions

Spoken Transcript [01:19:02 - 01:19:30] Prof. Rahul Pandharipande

So, you could ask now for S_n the parallel result: $(S_n, \cdot, e)$ [writes: $(S_n, \cdot, e)$]. What are the conjugacy classes here?

Do you have a guess for that? How do you index the conjugacy classes of this? This question is: which elements are conjugate? How about guessing? Yeah?

Student Guess

My guess would be that it's one of every type that was listed up earlier, so we have...

Spoken Transcript [01:19:35 - 01:19:58] Prof. Rahul Pandharipande

Yeah, and that type is called what?

[student speaks: "The size of the..."]

When we had the board, which red parts are you saying?

[student murmurs]

I don't know, we need some more specific guessing. Yeah?

Student Guess

Are the even and the odd ones each in a class?

Spoken Transcript [01:20:03 - 01:20:07] Prof. Rahul Pandharipande

You would say there's only two classes?

Student Answer

Yeah.

Spoken Transcript [01:20:09 - 01:20:26] Prof. Rahul Pandharipande

No, there's many more than that. I did one thing: if you take this two-cycle and you conjugate it, you get another two-cycle. I did this calculation. This means that this cycle and that cycle are in the same conjugacy class.

Yeah?

Student Answer

Partitions?

Spoken Transcript [01:20:28 - 01:22:35] Prof. Rahul Pandharipande

Partitions! This is the correct answer. This is one of the reasons why the partitions are so intimately related here.

So, the answer to this is that the conjugacy classes of S_n are in bijective correspondence with the partitions of n [writes: Conjugacy classes of $S_n \longleftrightarrow$ Partitions of n].

And this correspondence is by the cycle decomposition: if you have an element, you get a cycle decomposition, which is a partition. And so, you have to show that if you have an element with a cycle decomposition and you conjugate it, it stays the same cycle decomposition. That's what you have to show: that conjugacy does not change the cycle decomposition.

And in a sense, this is a confusing thing. Conjugacy, it turns out, is confusing: humans aren't good at conjugacy, because it has to do with some aspect of something non-commutative.

But the fact that if you take a permutation and you conjugate it, the cycle decomposition stays the same is... there's almost nothing to this fact, except keeping yourself not confused about what's being said.

So, I'll try to explain a little bit about it.

Suppose we have some set from 1 to n, and let's just take the cycle $(1\,2\,3)$, all right? [writes: $(1\,2\,3)$] So, there's a cycle here $(1\,2\,3)$, and then let's say our conjugacy element is some z that maps this $1, 2$ to some other numbers. So, it maps things like, I don't know, maybe 1 goes to 6, 2 goes to 8, 3 goes to 17 [writes $1 \mapsto 6$, $2 \mapsto 8$, $3 \mapsto 17$ with arrows labeled z].

And it does everything, so, this is maybe for S_{17}, it doesn't really matter. I'm interested in these three, and I want to calculate this conjugacy: $z(1\,2\,3)z^{-1}$ [writes: $z(1\,2\,3)z^{-1}$].

So, this is a three-cycle here, and then z is some rule that gets me from these indices to those indices, however many there are. So, let's just say there's at least 17, and then I fill it out somewhere irrelevant, but you fill it out how you want.

I want to calculate this three-cycle. What happens to 6 here?

z^{-1} goes to 1. So, 6 under z^{-1} goes to 1, 1 goes to 2, then 2 goes to 8.

Conjugation as Relabeling of Cycle Entries

$$z \circ (1\,2\,3) \circ z^{-1} = \big(z(1)\; z(2)\; z(3)\big) = (6\;8\;17)$$

$$\begin{tikzcd}
1 \arrow[r, "z"] \arrow[d] & 6 = z(1) \arrow[d] \\
2 \arrow[r, "z"] \arrow[d] & 8 = z(2) \arrow[d] \\
3 \arrow[r, "z"] \arrow[uu, bend left] & 17 = z(3) \arrow[uu, bend right]
\end{tikzcd}$$

Lemma 8.43 (*Conjugation Formula for Permutations*): Let $\sigma \in S_n$ be a permutation and $z \in S_n$. If σ has the disjoint cycle decomposition

$$\sigma = (a_1\, a_2\, \ldots\, a_\ell)(b_1\, b_2\, \ldots\, b_m)\cdots,$$

then its conjugate $z\sigma z^{-1}$ is obtained simply by applying the permutation z to every entry inside each cycle:

$$z\sigma z^{-1} = \big(z(a_1)\, z(a_2)\, \ldots\, z(a_\ell)\big)\big(z(b_1)\, z(b_2)\, \ldots\, z(b_m)\big)\cdots. \tag{8.45}$$

Proof (beyond the lecture)

The lecture works out one example, $z(1\,2\,3)z^{-1} = (6\,8\,17)$, and leaves the general case to the students until the next lecture (Theorem 8.65). Let a_k be an entry of one of the cycles of σ, with a_{k+1} the next entry of that cycle (a_1 after the last one), and let $j = z(a_k)$. Then

$$(z\sigma z^{-1})(j) = z\big(\sigma(z^{-1}(j))\big) = z\big(\sigma(a_k)\big) = z(a_{k+1}).$$

If j is not of this form, then $z^{-1}(j)$ is fixed by σ, so $(z\sigma z^{-1})(j) = z(z^{-1}(j)) = j$. Hence $z\sigma z^{-1}$ is the product of the cycles $\big(z(a_1)\, \ldots\, z(a_\ell)\big)\big(z(b_1)\, \ldots\, z(b_m)\big)\cdots$. **Q.E.D.**

Spoken Transcript [01:22:35 - 01:23:40] Prof. Rahul Pandharipande

So, if you just calculate this, you'll find this goes to $(6\,8\,17)$ [writes: $= (6\,8\,17)$].

If I conjugate a three-cycle by some map, all I do is relabel the numbers there. That's all I'm doing.

And in general, if I take any cycle decomposition and I take the conjugate, all that happens in this conjugacy is relabeling the numbers, so, the cycle decomposition stays the same.

And moreover, if I take any two with the same cycle decomposition, there is a relabeling that will take one to the other. You have to think about that. This is part of understanding how the symmetric group works.

So, if you think about this example carefully, you'll see that there's nothing happening with conjugation, it's just a relabeling. That proves this statement: the conjugacy classes of S_n are in bijective correspondence with partitions.

Classification of Conjugacy Classes in S_n via Partitions

Theorem 8.44 (*Conjugacy Classes in the Symmetric Group*): Let $n \geq 1$. Two permutations $\sigma, \tau \in S_n$ are conjugate in S_n if and only if their disjoint cycle decompositions have the same cycle lengths, that is, give the same partition of n. Consequently, there is a natural bijective correspondence:

$$\text{Conjugacy classes of } S_n \xleftrightarrow{1-1} \text{Partitions of } n. \tag{8.46}$$

Why the Correspondence is Bijective:

- **Well-Definedness:** By Lemma 8.43, conjugating σ by z preserves the lengths of all cycles in its disjoint cycle decomposition. Thus, all elements in a given conjugacy class share the same cycle partition of n.
- **Surjectivity:** Given any partition $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_r)$ of n, one can immediately write down a permutation having disjoint cycles of lengths $\lambda_1, \ldots, \lambda_r$.
- **Injectivity:** If two permutations σ, τ have the same cycle lengths, $\sigma = (a_1 \ldots a_k) \ldots$ and $\tau = (b_1 \ldots b_k) \ldots$ (with the cycles of length 1 written out too), we explicitly construct the conjugating element $z \in S_n$ by declaring $z(a_i) := b_i$. Then by (8.45), $z\sigma z^{-1} = \tau$.

III The Partition Function, Euler's Product Formula, and Ramanujan Congruences

Spoken Transcript [01:23:40 - 01:25:05] Prof. Rahul Pandharipande

I'll say more about it. I don't expect you to fully understand rigorously the whole argument from what I said, although there is enough to understand it; but it takes some time to think about these things.

I promised to say a few words about partitions, and partitions are one of the coolest objects in mathematics.

If we look at this partition function: $p(n)$ equals the number of partitions of n [writes: $p(n) = \#\,\text{partitions of } n$], where n should be positive:

$$\begin{aligned} p(1) &= 1 \text{[writes: } p(1) = 1\text{]} \\ p(2) &= 2 \quad (2, 1+1) \text{[writes: } p(2) = 2 \quad 2, 1+1\text{]} \\ p(3) &= 3 \text{[writes: } p(3) = 3\text{]} \\ p(4) &= 5 \text{[writes: } p(4) = 5\text{]} \\ p(5) &= 7 \text{[writes: } p(5) = 7\text{]} \\ p(6) &= 11 \text{[writes: } p(6) = 11\text{]} \end{aligned}$$

I don't have them all memorized, but...

These are interesting numbers. Even if you're interested in group theory, they give you the number of conjugacy classes.

And you can say: well, how do you calculate this partition number? There's been a lot of work on those things, but the most famous formula is due to none other than Euler.

That's a great formula, which I'll write down in a moment.

The Integer Partition Function $p(n)$

Definition 8.45 (*Partition Function $p(n)$*): For an integer $n \geq 1$, the *"partition function"* $p(n)$ counts the number of distinct ways to write n as a sum of positive integers without regard to order:

$$p(n) := \#\left\{(\lambda_1,\dots,\lambda_k)\in\mathbb{N}^k \,\middle|\, \lambda_1\geq\lambda_2\geq\cdots\geq\lambda_k\geq 1,\ \sum_{i=1}^{k}\lambda_i = n\right\}. \tag{8.47}$$

💡 **Example 8.46 (*Values of $p(n)$ for Small n*):** The first few values of $p(n)$ are:

n	$p(n)$	**Partitions**
1	1	1
2	2	$2,\ 1+1$
3	3	$3,\ 2+1,\ 1+1+1$
4	5	$4,\ 3+1,\ 2+2,\ 2+1+1,\ 1+1+1+1$
5	7	$5,\ 4+1,\ 3+2,\ 3+1+1,\ 2+2+1,\ 2+1+1+1,\ 1+1+1+1+1$
6	11	$6,\ 5+1,\ 4+2,\ 4+1+1,\ 3+3,\ 3+2+1,\ 3+1+1+1,\ 2+2+2,\ 2+2+1+1,\ 2+1+1+1+1,\ 1+1+1+1+1+1$

Spoken Transcript [01:25:05 - 01:27:25] Prof. Rahul Pandharipande

By what I was saying, this is telling me how many conjugacy classes the symmetric group has. For S_4, we did it by hand: if you look, if you remember, look at your notes, or look in your brain, there were five of them.

[erases the chalkboard to the left of the partition table with a sponge]

Now, the Euler product formula [writes: Euler:]: he says that you should consider the function

$$1+\sum_{n\geq 1} p(n)q^n \text{[writes: } 1+\sum_{n\geq 1} p(n)q^n\text{]}.$$

This is what's called a generating series: he's just writing down a series whose Taylor expansion gives the partitions.

And his formula gives a product — this is a summation, he gives a product formula. This Euler product formula says that instead of taking the sum — because we don't know what this is, right? On the right-hand side, for equality, it's most useful when there's one side that you don't know and the other side you know; that's the bread and butter of mathematics.

So, this side should be stuff we know, and he says that you should do this:

$$\prod_{n\geq 1}\frac{1}{1-q^n}\text{[writes: } = \prod_{n\geq 1}\frac{1}{1-q^n}\text{]}.$$

That's an infinite product. He says this infinite sum equals that infinite product.

If you've never thought about it, you should play with this. It's a really cool formula. Have you seen this before?

This is the standard product formula, and I just leave it to you to play with. You might think: how does this make sense? You're doing an infinite product, and that can be a problem, but the coefficients of q get higher and higher. So, if you cut off this product at any finite power of q, then there's only finitely many terms.

And then I leave it to you: you can walk in Euler's footsteps and prove this thing. It's not hard.

Euler's Generating Function for Partitions

Theorem 8.47 (*Euler's Product Formula for the Partition Function*)**:** The generating function of the partition function $p(n)$ is given by the formal power series identity:

$$1 + \sum_{n\geq 1} p(n)q^n = \prod_{n\geq 1} \frac{1}{1-q^n}. \tag{8.48}$$

Algebraic Derivation via Geometric Series (beyond the lecture)

The lecture leaves this to the students. Write k for the index of the product and expand each factor with the geometric series $\frac{1}{1-q^k} = \sum_{m_k=0}^{\infty} (q^k)^{m_k}$:

$$\begin{aligned}\prod_{k=1}^{\infty} \frac{1}{1-q^k} &= \prod_{k=1}^{\infty} \left(1 + q^k + q^{2k} + q^{3k} + \dots\right) \\ &= \sum_{m_1,m_2,m_3,\dots \geq 0} q^{1\cdot m_1 + 2\cdot m_2 + 3\cdot m_3 + \dots}.\end{aligned}$$

The coefficient of q^n in this product counts the number of non-negative integer sequences $(m_1, m_2, \dots)$ such that:

$$1 \cdot m_1 + 2 \cdot m_2 + 3 \cdot m_3 + \dots = n.$$

Interpreting m_k as the number of parts of size k, this is precisely the definition of a partition of n. Hence the coefficient of q^n is $p(n)$. **Q.E.D.**

Spoken Transcript [01:27:25 - 01:29:00] Prof. Rahul Pandharipande

Some other crazy things... That's a long time ago.

Ramanujan found lots of crazy things about partitions; I'll tell you a little bit [writes: Ramanujan:].

If you look at the 5, you can see he says that if you take $p(5k+4)$, this should be congruent to 0 mod 5 [writes: $p(5k+4) = 0 \bmod 5$]. That's his first congruence, this Ramanujan congruence. If you take $k = 0$, that's what's on the board: $p(4) = 5$.

And then he also proves that if you take $7k+6$, this is congruent to 0 mod 7 [writes: $p(7k+6) = 0 \bmod 7$]. That's a really strange thing: that's the first time it occurs ($p(5) = 7$), and he says it's always like that.

And also $p(11k+7)$ should be congruent to 0 mod 11 [writes: $p(11k+7) = 0 \bmod 11$].

Those are totally wild statements about the properties of partitions. There's a whole list of studies about properties of this partition function; those are just three.

And this one is really hard to prove: this one is related to modular forms and a totally different subject.

This one you can prove if you have played with power series at all. Have you played with power series? Didn't you learn about power series in some class?

Anyway, you can try to prove this.

And as I said, the whole subject of partitions is intimately connected to the symmetric group because of this.

Okay, good. I'll see you tomorrow. [students applaud]

Spoken Transcript [01:29:00 - 01:29:21] Prof. Rahul Pandharipande

This was wrong by the way [corrects $7k+6$ to $7k+5$ and $11k+7$ to $11k+6$]. Ha! 6, look. [video cuts off]

Ramanujan's Partition Congruences

Proposition 8.48 (*Ramanujan's Partition Congruences*): For all non-negative integers $k \geq 0$, the partition function satisfies the three famous Ramanujan congruences:

$$p(5k+4) \equiv 0 \pmod{5}, \tag{8.49}$$

$$p(7k+5) \equiv 0 \pmod{7}, \tag{8.50}$$

$$p(11k+6) \equiv 0 \pmod{11}. \tag{8.51}$$

Ramanujan Base Cases: For $k=0$: $p(4) = 5 \equiv 0 \pmod{5}$, $p(5) = 7 \equiv 0 \pmod{7}$, and $p(6) = 11 \equiv 0 \pmod{11}$.

Board Correction: Two Ramanujan Congruences

The lecturer first writes $p(7k+6) = 0 \bmod 7$ and $p(11k+7) = 0 \bmod 11$. After the students applaud at the end of class, he walks back to the board and corrects them to $p(7k+5)$ and $p(11k+6)$, which match the cases $k=0$ in the partition table: $p(5) = 7$ and $p(6) = 11$. On the board each congruence reads "$= 0 \bmod m$"; the box writes $\equiv 0 \pmod{m}$.

Summary of Key Concepts

- **Group Homomorphisms:** A map $f\colon G_1 \to G_2$ with $f(xy) = f(x)f(y)$ sends e_1 to e_2 and inverses to inverses, $f(x^{-1}) = f(x)^{-1}$. Its image $\mathrm{Im}(f)$ is a subgroup of G_2; its kernel $\ker(f)$ is a subgroup of G_1, and even a normal subgroup. Moreover, f is surjective $\iff \mathrm{Im}(f) = G_2$, and f is injective $\iff \ker(f) = \{e_1\}$.
- **The Matrix Homomorphism and the Sign:** The map $\Phi_{\mathrm{mat}}\colon S_n \to \mathrm{GL}_n(\mathbb{Q})$, $\Phi_{\mathrm{mat}}(\sigma)(v_i) = v_{\sigma(i)}$, is a homomorphism. Composed with the determinant, it gives $\mathrm{Sign} = \det \circ \Phi_{\mathrm{mat}}\colon S_n \to \{\pm 1\}$, whose image is $\{\pm 1\}$. Two-cycles have sign -1, and an ℓ-cycle is a product of $\ell - 1$ two-cycles, so its sign is $(-1)^{\ell-1}$.
- **The Alternating Group:** $A_n = \ker(\mathrm{Sign})$ is the subgroup of even permutations. In S_4 it has $1+3+8 = 12$ elements, and it is the group of rotations of the tetrahedron: $\mathrm{Rot}(\mathrm{Tetrahedron}) \cong A_4$.
- **Normal Subgroups and Quotient Groups:** A subgroup $N \leq G$ is normal if $xnx^{-1} \in N$ for all $x \in G$ and $n \in N$. Kernels are normal, so $\{e, (1\,2)\} \leq S_3$, which

is not normal, is not a kernel. For N normal, the product $\bar{x} \cdot \bar{y} = \overline{xy}$ on G/N is well defined, and the quotient map $q\colon G \to G/N$ is a surjective homomorphism with $\ker(q) = N$.

- **The First Isomorphism Theorem:** For a group homomorphism $f\colon G_1 \to G_2$, $G_1/\ker(f) \cong \operatorname{Im}(f)$: the same triangle as for vector spaces and rings, where only the kind of kernel differs. The proof is left for the next lecture.
- **Conjugacy Classes and Partitions:** Conjugacy ($x \sim_{\text{conj}} y \iff \exists z \in G\colon zxz^{-1} = y$) is an equivalence relation, and its classes are the conjugacy classes. In $\mathrm{GL}_n(\mathbb{C})$ they correspond to the Jordan normal forms; in S_n to the partitions of n, because conjugation only relabels the entries of the cycles. The partition function satisfies Euler's formula $1 + \sum_{n\geq 1} p(n)q^n = \prod_{n\geq 1} \frac{1}{1-q^n}$ and Ramanujan's congruences modulo 5, 7 and 11.

Wednesday, 4th November 2026

The First Isomorphism Theorem, Alternating Groups, Automorphisms, and Conjugation in S_n

Lecture Context & Overview

This lecture returns to the kernel, the quotient group and the First Isomorphism Theorem of the previous lecture, proves the theorem, and looks at conjugation more closely:

- **Recap on Normal Subgroups and Quotients:** The kernel of a homomorphism is a normal subgroup, and a normal subgroup N gives the quotient group G/N with its quotient map $q\colon G \to G/N$.
- **The First Isomorphism Theorem:** For a group homomorphism $f\colon G_1 \to G_2$, the map $g(\bar{x}) = f(x)$ is a group isomorphism $G_1/\ker(f) \xrightarrow{\cong} \mathrm{Im}(f)$, checked in four questions: well defined, homomorphism, surjective, injective.
- **Order of the Alternating Group:** The First Isomorphism Theorem for the sign $\mathrm{Sign}\colon S_n \to \{\pm 1\}$, together with Lagrange's Theorem, gives $|A_n| = \frac{n!}{2}$.
- **Conjugation and Automorphisms:** Conjugation $c_z(x) = zxz^{-1}$ is the identity in abelian groups, and in every group it is a group isomorphism. The map $\mathrm{Conj}\colon G \to \mathrm{Aut}(G)$, $z \mapsto c_z$, is a homomorphism with kernel the center $Z(G)$; its image, the inner automorphisms, is a normal subgroup of $\mathrm{Aut}(G)$, and it need not be all of $\mathrm{Aut}(G)$.
- **Conjugation in the Symmetric Group S_n:** The formula for $z\sigma z^{-1}$ in cycle notation, the conjugacy classes of S_n as partitions of n, and the sign $\mathrm{Sign}(\sigma) = (-1)^{n-r}$ for σ with r cycles.

Recap: Normal Subgroups and Quotient Groups

Spoken Transcript [00:00:00 - 00:01:20] Prof. Rahul Pandharipande

Okay, so, we return to the topic we started yesterday, where we have a homomorphism from two groups [writes: $f\colon G_1 \to G_2$ hom], and the kernel of f is equal to those people in G_1 such that the image is the neutral element [writes: $\ker(f) = \{x \in G_1 \mid f(x) = e_2\}$].

We had seen yesterday that there were lots of ideas actually in the lecture yesterday. So, the first was that this kernel is not only a subgroup [writes "$\ker(f) \subset G$", a slip for G_1] — it's a subgroup, but also a normal subgroup [draws an arrow up to it and writes: "Subgroup, but

Normal Subgroup"].

And I just remind you that normal means that for all elements n in this group and for all x in G, if we conjugate — that was another idea, and we'll talk more about it — the conjugate lives inside the kernel [writes "$\forall n \in \ker(f), \forall x \in G \quad xnx^{-1} \in \ker(f)$", again G for G_1].

Kernels and Normal Subgroups

Recap: The previous lecture defined the kernel (Definition 8.13) and normal subgroups (Definition 8.21), and proved that kernels are normal (Proposition 8.22).

Let $f\colon G_1 \to G_2$ be a group homomorphism. The *"kernel"* of f is the preimage of the neutral element $e_2 \in G_2$:

$$\ker(f) := \{x \in G_1 \mid f(x) = e_2\} \subseteq G_1. \tag{8.52}$$

Proposition 8.49 (*Normality of the Kernel*): The kernel $\ker(f)$ is a normal subgroup of G_1, that is, it is closed under conjugation by arbitrary elements of G_1:

$$\forall n \in \ker(f),\ \forall x \in G_1\colon \quad xnx^{-1} \in \ker(f). \tag{8.53}$$

Verification of Normality: The previous lecture proved this (Proposition 8.22); in short, for $n \in \ker(f)$ and $x \in G_1$:

$$f(xnx^{-1}) = f(x)f(n)f(x^{-1}) = f(x)e_2f(x)^{-1} = e_2.$$

Spoken Transcript [00:01:20 - 00:02:40]

Prof. Rahul Pandharipande

So, that was the beginning, and then one of the main parts from last time was that whenever we have a normal subgroup, in particular the kernel, we constructed a quotient group.

We constructed this quotient group with its multiplication and its neutral element [writes: $(G_1/\ker(f), \cdot, \bar{e}_1)$]. And moreover, there is a quotient map that takes that group G_1 to this quotient group [writes: $q\colon G_1 \to G_1/\ker(f)$], and this quotient map takes x to the equivalence class of x, that is to say, $q(x)$ is equal to the equivalence class of x [writes: $q\colon x \longrightarrow \bar{x}$ and below it $q(x) = \bar{x}$].

And just to remember the terminology from last time: as a set [writes: "as a set"], G_1 over... this is equal to the set of equivalence classes under this equivalence relation that's defined by the subgroup [writes: $G_1/\ker(f) =$ set equiv classes, with $\sim_{\ker(f)}$ below].

This equivalence relation defined by a subgroup has come up twice: once in the proof of Lagrange's theorem, and in that case the subgroup was any subgroup; we did not require it to be nor— yeah. [inaudible question from a student]

The Quotient Group and the Quotient Map

Recap: The previous lecture built the quotient group of a normal subgroup (Definition 8.27, Theorem 8.26, Proposition 8.33) and its quotient map (Proposition 8.34).

Let N be a normal subgroup of G_1, such as $N = \ker(f)$. Its equivalence relation is

$$x \sim_N y \iff x^{-1}y \in N \iff y \in xN. \tag{8.54}$$

As a set, the quotient is the collection of all such equivalence classes:

$$G_1/N := \{\bar{x} \mid x \in G_1\} = \{xN \mid x \in G_1\}. \tag{8.55}$$

Because N is normal, the product $\bar{x} \cdot \bar{y} := \overline{xy}$ is well defined, and the quotient is a group $(G_1/N, \cdot, \bar{e}_1)$.

The map sending every element to its equivalence class, $q(x) := \bar{x}$, is the *"quotient map"*:

$$\begin{aligned} q\colon G_1 &\to G_1/\ker(f) \\ x &\mapsto \bar{x}. \end{aligned} \tag{8.56}$$

Spoken Transcript [00:02:40 - 00:03:50] Prof. Rahul Pandharipande

Okay, so, anyway, the point I was trying to make is: we've used this equivalence class twice. If you have a subgroup of a group, we've used this idea of this equivalence relation and the set of equivalence classes twice.

Once in the proof of Lagrange's theorem. And in that proof, we did not care whether the subgroup was normal. That was not an issue. Although in Lagrange's theorem, since that was a matter of division of integers, what was important was that the objects were finite groups.

Now we're using this equivalence relation in the context where we're not assuming these groups are finite groups; they have not been assumed to be finite groups. But this subgroup is normal. That was very important to define the multiplication on the quotient.

So, this relation was used twice, slightly different perspectives both times, and it's good to keep that clear in your head. And it was crucial — it was a point from yesterday — it was crucial that if you want to define the product here, and to have that well-defined, that this subgroup has to be a normal subgroup.

***Insight:* Two Roles of the Same Equivalence Relation**

The equivalence relation $\sim_H$ of a subgroup H of G serves two different purposes:

(1) **Counting in finite groups (Lagrange's Theorem):** For any subgroup H of a finite group G, the equivalence classes xH split G into disjoint subsets with $|H|$ elements each. This needs **no normality** and gives $|G| = (\text{number of classes}) \cdot |H|$.

(2) **Building a new group (quotient groups):** To multiply classes by $(xH)(yH) = (xy)H$, the product must not depend on the chosen representatives, and this is where H must be a **normal** subgroup. Here finiteness plays no role.

The First Isomorphism Theorem for Groups

Spoken Transcript [00:03:50 - 00:05:13] Prof. Rahul Pandharipande

All right. So, the situation for the First Isomorphism Theorem is: G_1 maps by q to G_1 mod the kernel of f [draws arrow q from G_1 to $G_1/\ker(f)$], and here f maps to the image of f, which is always a subgroup inside G_2 [draws arrow f from G_1 to $\operatorname{Im}(f) \subset G_2$].

And, of course, inside here there's the kernel of f [writes: $\ker(f) \subset G_1$]. So, this is kind of the picture for the First Isomorphism Theorem, and the First Isomorphism Theorem says that there exists a group isomorphism g [draws a downward arrow from $G_1/\ker(f)$ to $\operatorname{Im}(f)$, marked $\wr$ and labeled g], which now I will just write. And this is the same that we did with the rings.

So, the First Isomorphism Theorem says that we have this isomorphism g from the source group modulo the kernel to the image [writes: "The 1st ISOM Thrm says that $\exists g\colon G_1/\ker(f) \cong \operatorname{Im}(f)$"]. I just wanted to define it and talk about some of its properties. Sorry.

Recap: The previous lecture stated this theorem (Theorem 8.35) and left the proof for this lecture, as the same proof as for rings (Theorem 4.27).

$$\begin{tikzcd}
& & G_1/\ker(f) \arrow[d, "g", "\wr"'] \\
\ker(f) \subset & G_1 \arrow[ur, "q"] \arrow[r, "f"'] & \operatorname{Im}(f) \subset G_2
\end{tikzcd}$$

Theorem 8.50 (*First Isomorphism Theorem for Groups*): Let $f\colon G_1 \to G_2$ be a group homomorphism, and $q\colon G_1 \to G_1/\ker(f)$ the quotient map. Then there exists a group isomorphism

$$g\colon G_1/\ker(f) \xrightarrow{\cong} \operatorname{Im}(f) \tag{8.57}$$

with $g(q(x)) = f(x)$ for all $x \in G_1$. In particular:

$$G_1/\ker(f) \cong \operatorname{Im}(f). \tag{8.58}$$

Proof of the First Isomorphism Theorem

Definition and Well-Definedness of the Map

Spoken Transcript [00:05:13 - 00:05:41] Prof. Rahul Pandharipande

The first step is to define it. How are we going to define g? So, we have to define [writes: "Define g:"] g.

We have to say: what is g of somebody in the quotient? A person in the quotient, we say, is the equivalence class of x [writes: $g(\bar{x}) =$], and we need to define g of this. So, how do you think we're going to do this? Yeah?

Student Proposal

Takes x to the equivalence class...

Spoken Transcript [00:05:45 - 00:05:46] Prof. Rahul Pandharipande

Equivalence class of what?

Student Clarification

Wait...

Spoken Transcript [00:05:47 - 00:05:54] Prof. Rahul Pandharipande

Look, so what is the goal? g better live in the image of f, because that's what we're trying to do.

Student Proposal, continued

Yeah, sorry, I meant, uh, map it to something in the image of f, so...

Spoken Transcript [00:05:59 - 00:06:01] Prof. Rahul Pandharipande

Yeah, what?

Student Answer

$f(x)$.

Spoken Transcript [00:06:02 - 00:06:21] Prof. Rahul Pandharipande

$f(x)$, yes! What else could we do here? [writes: $g(\bar{x}) = f(x)$]

Do you agree? It's more or less the only way. You have to get to the image of f. If you have to start with somebody in x and get to the image of f, there's only one way: it's that.

Definition of the candidate map: To construct the map $g\colon G_1/\ker(f) \to \operatorname{Im}(f)$, we define the assignment on any equivalence class $\bar{x} \in G_1/\ker(f)$ by:

$$g(\bar{x}) := f(x). \tag{8.59}$$

Spoken Transcript [00:06:21 - 00:07:09] Prof. Rahul Pandharipande

All right, so then we come to the normal problems: Is this well-defined? It's the usual questions: Is this well-defined? [writes: "is this well-def?"]

And the problem is that we could have a different name for this equivalence class. It could be xn [writes: $g(\overline{xn})$], where n is in the kernel [writes: $n \in \ker(f)$], because that's how the equivalence class is. Another name for the same equivalence class is that.

So, we have to check this gives the same thing. Let's see what this would give: by that definition, this would give $f(xn)$ [writes: $= f(xn)$], but since f is a homomorphism, this is $f(x)f(n)$ [writes: $= f(x)f(n)$], and since n is in the kernel, this is $f(x)e_2$ [writes: $= f(x)e_2$], which is $f(x)$ [writes: $= f(x)$].

So, we're extremely happy: it gives the same, so at least it's well-defined. Okay?

Well-definedness: Since the element $\bar{x} \in G_1 / \ker(f)$ is an equivalence class, it possesses multiple representative elements. Any other representative of the same class is of the form xn for some $n \in \ker(f)$. Evaluating g on the representative xn yields:

$$\begin{aligned} g(\overline{xn}) &= f(xn) \\ &= f(x)f(n) && \text{(since } f \text{ is a homomorphism)} \\ &= f(x)e_2 && \text{(since } n \in \ker(f)\text{)} \\ &= f(x) = g(\bar{x}). \end{aligned} \tag{8.60}$$

Thus, the value $g(\bar{x})$ is independent of the choice of representative, proving that g is well-defined.

Homomorphism Property, Surjectivity, and Injectivity

Spoken Transcript [00:07:09 - 00:08:15] Prof. Rahul Pandharipande

So, then you have to check... okay, that's the first check: Is it well-defined? Is this a homomorphism? [writes: "is this a hom?"]

Then we have to check, what is $g(\bar{x} \cdot \bar{y})$ [writes: $g(\bar{x} \cdot \bar{y})$]? And this is, by definition of the product in the quotient, $g(\overline{xy})$ [writes: $= g(\overline{xy})$], and then by the rule, this is $f(xy)$ [writes: $= f(xy)$].

And by the fact that f is a homomorphism, this is $f(x)f(y)$ [writes: $= f(x)f(y)$], and then we undo the process of the definitions: this is $g(\bar{x})g(\bar{y})$ [writes: $= g(\bar{x})g(\bar{y})$].

So, yes, the answer is yes, it's a homomorphism. Mostly these kinds of checks, they don't mean anything unless you do them yourselves, you know, because I'm just going through the motions and telling you the argument at each step, but you should be able to do this yourself. So, first, it's well-defined, second it's a homomorphism...

Homomorphism property: For any classes $\bar{x}, \bar{y} \in G_1 / \ker(f)$, applying the definition of the quotient group operation followed by the definition of g and the homomorphism property of f gives:

$$\begin{aligned} g(\bar{x} \cdot \bar{y}) &= g(\overline{xy}) && \text{(definition of multiplication in } G_1/\ker(f)\text{)} \\ &= f(xy) && \text{(definition of } g\text{)} \\ &= f(x)f(y) && \text{(since } f \text{ is a homomorphism)} \\ &= g(\bar{x}) \cdot g(\bar{y}) && \text{(definition of } g\text{).} \end{aligned} \tag{8.61}$$

Therefore, g is a group homomorphism.

Spoken Transcript [00:08:15 - 00:08:52] Prof. Rahul Pandharipande

But we need to prove it's an isomorphism. So, is it surjective? [writes: "is g surjective?"]

That's the third question. That means, can we get everything in the image of f? And the answer here is very simply yes, because everything in the image of f is f of something. And if it's f of something, then we can take g of that something's equivalence class here.

So, this is essentially tautological. So, yes, and I might just say, essentially tautological [writes: "Yes (essent. tautological)"].

Surjectivity: Let $y \in \operatorname{Im}(f)$. By definition of the image, there exists some $x \in G_1$ such that $y = f(x)$. Considering the class $\bar{x} = q(x) \in G_1/\ker(f)$, we have:

$$g(\bar{x}) = f(x) = y. \tag{8.62}$$

Hence, every element of $\operatorname{Im}(f)$ has a preimage under g, establishing that g is surjective.

Spoken Transcript [00:08:52 - 00:10:56] Prof. Rahul Pandharipande

Okay, so, what's the last question? Is it surjective? Is it injective? [writes: "is it injective?"]

If it's injective... there's the principle that a group homomorphism is injective if and only if the kernel is trivial, if the kernel consists only of the neutral element. I stated that, you can check it.

So, here, we have to check what's in the kernel [writes: "we have check $\ker(g)$"]. Check the kernel of g. Okay, so who's in the kernel? The kernel would be some element $g(\bar{x})$ which goes to the neutral element here, right? e_2 [writes: $g(\bar{x}) = e_2$]. That's how we get in the kernel.

So, what does this mean actually? This means that we have $f(x) = e_2$ [writes: $f(x) = e_2$]. This means x is in the kernel of f [writes: $x \in \ker(f)$]. And the kernel of f is the neutral element in the quotient.

So, this says that this equivalence class is the equivalence class of e_1, right? So, here, you have to understand what the... maybe I should have emphasized this: the neutral element in this $G_1/\ker(f)$ is the equivalence class which corresponds to the entire kernel.

And to show that the kernel is trivial, we have to show this x lies in the kernel, and that's what we have done here. So, you have to go through these steps somehow carefully in your head to believe all of them.

Injectivity: Recall that a group homomorphism is injective if and only if its kernel is trivial (Lemma 8.20). Let $\bar{x} \in \ker(g)$. Then:

$$g(\bar{x}) = e_2 \implies f(x) = e_2 \implies x \in \ker(f). \tag{8.63}$$

In the quotient group $G_1/\ker(f)$, the neutral element is the class of the identity, $\bar{e}_1 = \ker(f)$. Since $x \in \ker(f)$, its class satisfies:

$$\bar{x} = x\ker(f) = \ker(f) = \bar{e}_1. \tag{8.64}$$

Thus, $\ker(g) = \{\bar{e}_1\}$, which proves that g is injective.

Summary of the Four Proof Steps: We have proven:

(1) **Well-definedness:** $g(\bar{x})$ does not depend on the choice of representative x.
(2) **Homomorphism:** $g(\bar{x} \cdot \bar{y}) = g(\bar{x}) \cdot g(\bar{y})$.
(3) **Surjectivity:** Every element of $\operatorname{Im}(f)$ is attained by g.
(4) **Injectivity:** $\ker(g) = \{\bar{e}_1\}$, so distinct classes map to distinct elements.

Consequently, g is a bijective homomorphism, completing the proof that $G_1/\ker(f) \cong \operatorname{Im}(f)$.

Q.E.D.

Application: The Order of the Alternating Group

Spoken Transcript [00:10:56 - 00:11:35] Prof. Rahul Pandharipande

So, the outcome of that is: we have this First Isomorphism Theorem. And then I can tell you what some people were asking me after the class yesterday about the number of elements of the alternating group in general.

Let's just do that quickly. There's other ways to do it, but we can use this First Isomorphism Theorem; we should use it for something.

Did I turn that off? Okay. [pushes the lower board up and prepares the newly revealed board]

Spoken Transcript [00:11:35 - 00:13:16] Prof. Rahul Pandharipande

Okay, so we defined — and this was one of the other ideas from yesterday, which is a very important idea — the sign homomorphism.

The sign homomorphism goes from the symmetric group to the group plus minus one [writes: $\text{Sign}\colon S_n \to \{\pm 1\}$], and this group is the group where one is the neutral element and minus one is the only other element.

And the definition here was the alternating group. And this map is surjective, we also know that. This is surjective [writes: "surjective, hom."]. If it weren't surjective, it would be a really dumb map: it would just take every element to one, and we wouldn't really be talking about it. But it is surjective.

The definition of this is equal to the kernel of the sign map [writes: "Def:" $A_n = \ker(\text{Sign})$], the sign homomorphism. It's surjective and it's a homomorphism, both things. So, the kernel of this sign homomorphism... that says that A_n is a group [writes: A_n "is group, normal subgr."], and not only that, it's a normal subgroup. Because kernels are always normal subgroups. It's a special kind of group.

And so, we did this calculation by listing — or more honestly, I did the calculation by listing all of the elements of A_4 — and that's of course not a very good way to do the calculation. But now we can just do this in general.

So, the proposition is that the order of A_n is equal to the order of S_n over two, which is n factorial over two [writes: Prop : $|A_n| = \frac{|S_n|}{2} = \frac{n!}{2}$]. And what is the proof of that? The proof is that...

Recap: The sign was defined in the previous lecture (Definition 8.10), and A_n as its kernel (Definition 8.15).

Definition 8.51 (*Sign Homomorphism and Alternating Group*): For $n \geq 2$, the *"sign homomorphism"* is the surjective group homomorphism

$$\text{Sign}\colon S_n \to \{\pm 1\}, \tag{8.65}$$

where $\{\pm 1\} = \{1, -1\}$ is the group with neutral element 1. The *"alternating group"* A_n is the kernel of the sign homomorphism:

$$A_n := \ker(\text{Sign}). \tag{8.66}$$

Since kernels are always normal subgroups (Proposition 8.49), A_n is a normal subgroup of the symmetric group S_n.

Proposition 8.52 (*Order of the Alternating Group*): For any integer $n \geq 2$, the order of the alternating group A_n is half the order of the symmetric group S_n:

$$|A_n| = \frac{|S_n|}{2} = \frac{n!}{2}. \tag{8.67}$$

Spoken Transcript [00:13:16 - 00:14:26] Prof. Rahul Pandharipande

...we saw from Lagrange's theorem that this order of A_n has to divide the order of S_n, right? [writes: $|A_n| \mid |S_n|$] And moreover, that this division is equal to the number of equivalence classes of this $\sim_{A_n}$ [writes: $\frac{|S_n|}{|A_n|} = \#$ equiv. classes of $\sim_{A_n}$].

That's exactly how we proved Lagrange's theorem: if you have any subgroup of a group, there's a division, and moreover, the quotient is the number of equivalence classes.

So, all we have to do is figure out the number of equivalence classes. But the number of equivalence classes here is this quotient, S_n/A_n, and by the First Isomorphism Theorem, that's isomorphic to plus minus one [writes: $S_n/A_n \cong \{\pm 1\}$], so there's two equivalence classes [writes and circles: 2].

Okay? So, that's the proof.

Proof using the First Isomorphism Theorem and Lagrange's Theorem

By Lagrange's Theorem (Theorem 7.53), applied to the subgroup $A_n \leq S_n$, the order $|A_n|$ divides $|S_n|$, and the quotient is the number of equivalence classes of $\sim_{A_n}$:

$$\frac{|S_n|}{|A_n|} = \text{number of equivalence classes of } \sim_{A_n}. \tag{8.68}$$

These classes are the elements of S_n/A_n. By the First Isomorphism Theorem (Theorem 8.50), since $\text{Sign}\colon S_n \to \{\pm 1\}$ is surjective with $\ker(\text{Sign}) = A_n$, we have:

$$S_n/A_n \cong \text{Im}(\text{Sign}) = \{\pm 1\}. \tag{8.69}$$

Therefore, the number of classes is:

$$|S_n/A_n| = |\{\pm 1\}| = 2. \tag{8.70}$$

Substituting (8.70) into (8.68) yields:

$$\frac{|S_n|}{|A_n|} = 2 \implies |A_n| = \frac{|S_n|}{2} = \frac{n!}{2}.$$

This completes the proof. **Q.E.D.**

Elegance of the Homomorphism Approach: Previously, calculating $|A_4| = \frac{4!}{2} = 12$ required manually enumerating all even permutations (the identity, eight 3-cycles, and three products of disjoint transpositions). Using the First Isomorphism Theorem, the result holds instantly for all $n \geq 2$ without any combinatorial casework, purely from the fact that the sign map has a two-element image.

Spoken Transcript [00:14:26 - 00:15:48]

Prof. Rahul Pandharipande

Okay, so that was some discussion about the First Isomorphism Theorem. And as I said, the proofs on this level, I sometimes do them, sometimes I don't. Maybe the principle is that I don't spend the whole class doing them, because it'd be kind of boring. But if you want to understand the subject, you really have to understand every step of this proof.

So, by the way, Lycka sent me some link: someone of you put some 500-page transcript of my lectures, and there's a link for that. I don't know if it's helpful, I don't know if it's correct. But it was AI-generated, of course.

And I think anyway, the 500 pages is trying to check all of this; I don't know if that's really useful. But if you like it, you can look at it. In my research group, we have some kind of AI maximalists and AI minimalists. I am an AI pragmatist. But at least the check in the group of the 500 pages seemed to say that it was kind of correct, I don't know.

But if you use it, maybe you can tell me whether it is useful. But I think it just expands all of these things. [erases the lower chalkboard completely]

Classroom Aside: AI Lecture Transcripts and Proof Pedagogics

The lecturer comments on a 500-page AI-generated transcript of his lectures, which the head TA, Lycka Drakengren, sent him. He notes with amusement that while he personally does not present every single routine verification on the board to avoid slowing down the lecture, students must master every step of the algebraic derivations. He characterizes himself as an *"AI pragmatist"* amidst AI maximalists and minimalists in his research group, noting that the AI transcript expands all intermediate calculations.

Conjugation and Automorphisms

The Conjugation Map

Spoken Transcript [00:15:48 - 00:17:10] Prof. Rahul Pandharipande

Okay, so, one of the ideas from yesterday was this idea of a normal subgroup, the quotient, and the First Isomorphism Theorem, and I'm finished talking about that, more or less.

The other idea, which is a very interesting idea, what comes up here, is the idea of conjugation [writes: "Conjugation:"]. So, I'll say a little bit about that.

This came up in the definition of a normal subgroup: a normal subgroup is a subgroup such that all the conjugates of its elements lie in the subgroup. But the notion of conjugation is deeper, and I'll try to say a few words about this.

Suppose z is in the group. Here G is a group [writes: $(G, \cdot, e)$ "group"], and suppose I pick an element z in G [writes: $z \in G$]. Then I can conjugate everybody by z.

So, that gives me a map c_z, which is conjugation by z [writes: $c_z\colon G \to G$, "conj by z"], and it's a map that takes a group element and gives you back an element of the group.

And the definition is: c_z of any element x is equal to zxz^{-1} [writes: "Def" $c_z(x) = zxz^{-1}$, $x \in G$], okay? If you fix your z, you can just conjugate everybody.

Definition of Conjugation by an Element

Definition 8.53 (*Conjugation Map*): Let $(G, \cdot, e)$ be a group and let $z \in G$ be a fixed element. The *"conjugation map by z"* is the mapping:

$$\begin{aligned} c_z\colon G &\to G \\ x &\mapsto c_z(x) := zxz^{-1}. \end{aligned} \tag{8.71}$$

Connection to Normal Subgroups: In terms of these maps, a subgroup $N \le G$ is normal (Definition 8.21) exactly when every c_z maps N into itself:

$$c_z(N) \subseteq N \quad \text{for all } z \in G.$$

Conjugation in Abelian Groups

Spoken Transcript [00:17:10 - 00:18:55] Prof. Rahul Pandharipande

It's pointless to think about this conjugation if the group is abelian. So, the first observation is: if G is abelian, c_z is the identity map from G to G [writes: "If G is abelian,

$c_z = \mathrm{Id}\colon G \to G$"], because if it's abelian, then I can commute this z^{-1} by z and then it meets its inverse and it goes away, so it's just the identity map. Is that clear?

So, it is pointless to talk about this conjugation in the case of abelian groups. Pointless may not be the right word, but you're not going to profit from it: it's just always trivial.

And in particular, this shows, as a consequence of this line of thought: every subgroup of an abelian group is normal [writes: "(Every subgroup of abelian group is Normal)"]. That's a consequence of the fact that conjugation never does anything for abelian groups, so in particular the only conjugates of any element are itself. There's no other elements that are conjugate to it.

So, every subgroup of an abelian group is normal, which is, in a way, not bad news. It means if you have an abelian group, it's easy to find groups where there's quotients, because you can take any subgroup and you can make a quotient. If you have a non-abelian group, you're only allowed to use this quotient construction for subgroups that are normal, so it's a higher bar. But if you have an abelian group, you can just quotient by any subgroup.

That's the good side of it. The bad side of it is this interesting idea of conjugation doesn't really tell you anything in the abelian case.

Observation 8.54 (*Triviality of Conjugation in Abelian Groups*)**:** If G is an abelian group, then for every $z \in G$ and every $x \in G$:

$$c_z(x) = zxz^{-1} = xzz^{-1} = xe = x. \tag{8.72}$$

Hence, for all $z \in G$:

$$c_z = \mathrm{Id}\colon G \to G. \tag{8.73}$$

Corollary 8.55 (*Subgroups of Abelian Groups are Normal*)**:** Every subgroup H of an abelian group G is a normal subgroup (Definition 8.21).

The Abelian and the Non-Abelian Case: In abelian groups, every element is conjugate only to itself, and every subgroup is normal, so every subgroup gives a quotient group G/H. For non-abelian groups, conjugation c_z can move elements, and only normal subgroups give quotients; this is where conjugacy classes and normal subgroups become interesting.

III Conjugation as a Group Isomorphism

Spoken Transcript [00:18:55 - 00:20:35] Prof. Rahul Pandharipande

Okay, but this is just to warn you that when we talk about these ideas, they're rich for non-abelian groups.

All right, so the next lemma for conjugation, which is an interesting one — and I don't know if it's completely obvious to you right here — does anyone know what I'm going to write here?

Well, we have a map from a group to a group, and if you have such a map, you might think whether it's a homomorphism. Not only is this a homomorphism, it's an isomorphism; it's a group isomorphism [writes: "Lemma: c_z is group isom."].

So, that's kind of interesting: if you have a group, you can conjugate by anybody and get some group isomorphism. The proof of this... there is nothing to the proof of this. You just have to think: is this an isomorphism or not?

So, how do we want to prove it? Well, you have to prove like $c_z(xy) = c_z(x)c_z(y)$ [writes: $c_z(xy) = c_z(x)c_z(y)$]. That would be the homomorphism property. So, is it a homomorphism? [writes: "is it hom?"]

This is like the Socratic method of proofs: you ask questions, right? Is it a homomorphism? So, you have to check this, right? We just expand the definition. The definition of this is $zxyz^{-1}$ [writes: $zxyz^{-1}$], and the definition of this side is zxz^{-1}, and this one is zyz^{-1} [writes: $zxz^{-1}zyz^{-1}$]. And you see that these nicely cancel [circles $z^{-1}z$], and you get the fact that it is a homomorphism, okay?

You must have done this in your life before, but maybe you didn't realize you were doing this calculation.

Lemma 8.56 (*Conjugation Automorphism*): Let $(G, \cdot, e)$ be a group and $z \in G$. The conjugation map $c_z \colon G \to G$ is a group isomorphism (an "*automorphism*" of G).

Verification of Homomorphism Property

Question 1: Is c_z a homomorphism?

For any $x, y \in G$, expanding the definitions yields:

$$\begin{aligned} c_z(xy) &= z(xy)z^{-1} \\ &= zxeyz^{-1} \\ &= zx(z^{-1}z)yz^{-1} \\ &= (zxz^{-1})(zyz^{-1}) \\ &= c_z(x) \cdot c_z(y). \end{aligned} \tag{8.74}$$

Hence, c_z preserves the group multiplication, so it is a group homomorphism.

Q.E.D.

Spoken Transcript [00:20:35 - 00:21:28]

Prof. Rahul Pandharipande

So, it is a homomorphism; then you have to find out whether it's an isomorphism. Is it surjective? [writes: "is it surj?"] Is it a homomorphism? Yes. Is it surjective?

That means that we start... I can raise the board. [raises the board slightly] We start: suppose we have y in G, can we find some $\hat{y}$ such that $c_z(\hat{y}) = y$? [writes: "Suppose $y \in G$, can we find $\hat{y}$ $c_z(\hat{y}) = y$?"]

That's the assertion of surjectivity. And the answer is yes. And what am I going to write for $\hat{y}$? The answer is yes, and we've got $\hat{y}$ equals what?

Student Answer

$z^{-1}yz$.

Spoken Transcript [00:21:30 - 00:22:36] Prof. Rahul Pandharipande

Yeah, you flip them around: $z^{-1}yz$ [writes: "YES $\hat{y} = z^{-1}yz$"]. And then when you stick it into c, the z's will arrange so they will disappear on both sides.

Finally, you have to check whether it's injective, and for that, you have to see... Okay, let's use here: is c_z injective? [writes in a box: "is c_z injective?"]

I don't propose you really follow this method of writing a proof, but it's the way I think about it. I am proving it, but, you know. Is it injective? Now you ask this question.

So, then you have to check: who's in the kernel? We have to check what happens if I have $c(x) = e$, right? [writes: $c_z(x) = e$] Then this implies $zxz^{-1} = e$ [writes: $\implies zxz^{-1} = e$], and now I bring all the z's to the other side, where they go away. Is that clear?

So, this implies $x = e$ [writes $x = e$ below it, with a check mark], so the answer is yes. Actually, answers to all the questions are yes, so it's a group isomorphism.

Verification of Bijectivity

Question 2: Is c_z surjective?
Given any $y \in G$, we seek an element $\hat{y} \in G$ such that $c_z(\hat{y}) = y$. Setting:

$$\hat{y} := z^{-1}yz \in G, \tag{8.75}$$

we compute:

$$c_z(\hat{y}) = z(z^{-1}yz)z^{-1} = (zz^{-1})y(zz^{-1}) = eye = y.$$

Thus, every element $y \in G$ is in the image of c_z, so c_z is surjective.

Question 3: Is c_z injective?
We verify that the kernel of c_z is trivial:

$$\begin{aligned} c_z(x) = e &\implies zxz^{-1} = e \\ &\implies zx = z \\ &\implies x = z^{-1}z = e. \end{aligned} \tag{8.76}$$

Since $\ker(c_z) = \{e\}$, the homomorphism c_z is injective.

Conclusion: Being a bijective group homomorphism from G to itself, c_z is an isomorphism (specifically, an automorphism of G).

Q.E.D.

Beyond the Lecture: The Inverse of c_z

Bijectivity also follows in one step: $c_{z^{-1}}$ is a two-sided inverse of c_z, since $(c_z \circ c_{z^{-1}})(x) = z(z^{-1}xz)z^{-1} = x$ and likewise $(c_{z^{-1}} \circ c_z)(x) = x$ for all $x \in G$. The lecture's $\hat{y} = z^{-1}yz$ is $c_{z^{-1}}(y)$.

Spoken Transcript [00:22:36 - 00:23:32] Prof. Rahul Pandharipande

So, that's kind of an interesting thing. As I said, one of the origins of groups is symmetries of other objects. But you can also do this kind of introspective thing: if you have a group, think about its own symmetries, symmetries as a group.

And if the group is an interesting group... Like for example, if it's an abelian group, this whole line of thought is totally pointless, because it's true this is a group isomorphism, but it's the identity isomorphism, so it's not that interesting. But if you have an interesting group, it will give you interesting isomorphisms.

Before we do an example, maybe I state some things that are fun to think about [erases the upper chalkboard completely].

As I said, I don't view my purpose in this class as teaching you how to write proofs, so you don't have to follow my example.

Board Transition

The lecturer completely erases the upper chalkboard (which previously displayed the results on the alternating group A_n) to begin a broader, structural discussion of the automorphism group and the general mapping properties of conjugation.

The Automorphism Group and the Conjugation Homomorphism

Spoken Transcript [00:23:32 - 00:25:28] Prof. Rahul Pandharipande

Okay, let's do some fun stuff before we do the example. If you have a group... as I said, you can think about the symmetries of that group. So, you can associate to this group the automorphisms [writes: $G \rightsquigarrow \mathrm{Aut}(G) = \{f : G \to G \mid f \text{ is a group isom}\}$].

So, this is $\mathrm{Aut}(G)$, and this is the set of f from G to G where f is a group isomorphism. So, that's the set of all group isomorphisms.

Does that make sense?

If you have a group, you can think about maps from G to G, and they may or may not be isomorphisms, and then we just consider the set of all isomorphisms. So, there's no difficulty.

And now you will see — and this always happens with automorphisms and symmetries, so this should not be a surprise to you — $\mathrm{Aut}(G)$ is a group [writes: $\mathrm{Aut}(G)$ is group].

Do you believe that? It's just because you can compose: there's the identity here, you can compose them, there's inverses. So, the set of group isomorphisms of a group is itself a group. So, it's a group.

And conjugation gives a map from G to $\mathrm{Aut}(G)$, where an element here goes to c_z [writes: $\mathrm{Conj} : G \longrightarrow \mathrm{Aut}(G)$, $z \longrightarrow c_z$], right?

Does that make sense? And the claim is this is a group homomorphism [writes: "grp hom. (you can check)"].

I leave that for you to check. There is nothing to this check. You just have to write it. So, given any group, there's a natural map to its automorphisms by conjugation. That's again even further understanding of conjugation. It's a deep idea, conjugation. And it makes this a homomorphism.

Definition 8.57 (*Automorphism Group*): Let $(G, \cdot, e)$ be a group. An *"automorphism"* of G is a group isomorphism from G to itself. The collection of all automorphisms of G is denoted:

$$\mathrm{Aut}(G) := \{f \colon G \to G \mid f \text{ is a group isomorphism}\}. \tag{8.77}$$

The pair $(\mathrm{Aut}(G), \circ)$ forms a group under function composition, with neutral element Id_G.

Proposition 8.58 (*The Conjugation Homomorphism*): For any group G, the map that assigns each element $z \in G$ to its conjugation automorphism c_z:

$$\begin{aligned} \mathrm{Conj} \colon G &\to \mathrm{Aut}(G) \\ z &\mapsto c_z \end{aligned} \tag{8.78}$$

is a group homomorphism. That is, for all $z_1, z_2 \in G$:

$$c_{z_1 z_2} = c_{z_1} \circ c_{z_2}. \tag{8.79}$$

Verification of the Homomorphism Property (beyond the lecture)

The lecture leaves this check to the students (*"you can check"*). For all $x \in G$, expanding the definition of conjugation gives:

$$\begin{aligned} c_{z_1 z_2}(x) &= (z_1 z_2) x (z_1 z_2)^{-1} \\ &= z_1 z_2 x z_2^{-1} z_1^{-1} \\ &= c_{z_1}(z_2 x z_2^{-1}) \\ &= c_{z_1}(c_{z_2}(x)) = (c_{z_1} \circ c_{z_2})(x). \end{aligned} \tag{8.80}$$

Thus, $c_{z_1 z_2} = c_{z_1} \circ c_{z_2}$. **Q.E.D.**

Verification of the Group Axioms for $\mathrm{Aut}(G)$:

(1) **Closure under composition:** If $f, g \in \mathrm{Aut}(G)$, then both are bijective homomorphisms, so their composition $f \circ g \colon G \to G$ is also a bijective homomorphism, meaning $f \circ g \in \mathrm{Aut}(G)$.

(2) **Associativity:** Composition of mappings is associative: $(f \circ g) \circ h = f \circ (g \circ h)$.

(3) **Neutral element:** The identity map $\mathrm{Id}_G \colon G \to G$, defined by $\mathrm{Id}_G(x) = x$, is an isomorphism and satisfies $f \circ \mathrm{Id}_G = \mathrm{Id}_G \circ f = f$ for all $f \in \mathrm{Aut}(G)$.

(4) **Inverses:** Every bijective homomorphism $f \in \mathrm{Aut}(G)$ possesses an inverse mapping $f^{-1} \colon G \to G$, which is also a bijective group homomorphism, so $f^{-1} \in \mathrm{Aut}(G)$ and $f \circ f^{-1} = f^{-1} \circ f = \mathrm{Id}_G$.

Spoken Transcript [00:25:28 - 00:26:10]

Prof. Rahul Pandharipande

And if you have a homomorphism, you should immediately think about who is in the kernel, and who's not, and think about the First Isomorphism Theorem.

So, the next step in this line is: well, what is the kernel of this map? [writes: "Next step:" $\ker(\mathrm{Conj}) = \{z \mid c_z = \mathrm{Id}\}$] It's some normal subgroup of G.

So, how can you be in the kernel? The kernel consists of some elements z such that c_z is Id, right? Yeah.

Student Question

This happens if z commutes, right?

Spoken Transcript [00:26:12 - 00:27:25] Prof. Rahul Pandharipande

Exactly. If c_z is Id, this means $c_z(x) = zxz^{-1} = x$ for all x [writes: $zxz^{-1} = x\ \forall x$], and that means that $zx = xz$ [writes: $zx = xz$]. It's perfect!

So, you're in the kernel if and only if z commutes with everybody.

And the kernel, in general, is the set of elements of G that commute with every other person, and that has a name.

So, that's just the definition of the kernel, and this is called the center of G, usually it's called $Z(G)$, the center [writes below $\ker(\mathrm{Conj})$, joined by a vertical $=$: "Center(G)", circled, and $Z(G)$ under it]. That's the name of this thing.

So, it's the elements that commute with everybody.

The Kernel of Conjugation: The Center of a Group

Definition 8.59 (*Center of a Group*): The *"center"* of a group G, denoted $Z(G)$ (or $\mathrm{Center}(G)$), is the set of all elements that commute with every element in G:

$$Z(G) := \{z \in G \mid \forall x \in G,\ zx = xz\}. \tag{8.81}$$

Proposition 8.60 (*The Center as Kernel of Conjugation*): The kernel of the conjugation homomorphism $\mathrm{Conj}\colon G \to \mathrm{Aut}(G)$ is precisely the center $Z(G)$:

$$\ker(\mathrm{Conj}) = Z(G). \tag{8.82}$$

Consequently, because the kernel of any group homomorphism is a normal subgroup (Proposition 8.49), the center $Z(G)$ is always a normal subgroup of G.

Algebraic Derivation of the Center Condition: For an element $z \in G$, membership in the kernel is determined by:

$$\begin{aligned} z \in \ker(\mathrm{Conj}) &\iff \mathrm{Conj}(z) = \mathrm{Id}_G \\ &\iff c_z = \mathrm{Id}_G \\ &\iff \forall x \in G\colon c_z(x) = x \\ &\iff \forall x \in G\colon zxz^{-1} = x \\ &\iff \forall x \in G\colon zx = xz \\ &\iff z \in Z(G). \end{aligned}$$

This confirms that z acts trivially by conjugation on G if and only if z commutes with every element of G.

Spoken Transcript [00:27:25 - 00:29:09] Prof. Rahul Pandharipande

And now you could think, well, what about the other side? There's also the image [writes: $\mathrm{Im}(\mathrm{Conj}) \subset \mathrm{Aut}(G)$].

This is a subgroup of the automorphism group, and this has a name too: this is called the subgroup of what are called inner automorphisms [writes and circles: "Subgroup Inner Auts"].

Inner automorphisms — that's the name of this group. The image of this is called the inner automorphisms.

And you could ask: well, is it surjective? Or is this map... And those are all interesting questions.

So, the answer is, of course it can't be surjective, because you could take an abelian group. For an abelian group, this conjugation map is just trivial, so you're not getting anything interesting here. So, if you find one abelian group with an interesting automorphism, that'll show that this map is not surjective.

And of course there are: like you take $\mathbb{Z}/n\mathbb{Z}$, and the automorphism is like x goes to $-x$ [writes: $\mathbb{Z}/n\mathbb{Z}$, $x \mapsto -x$]. That's an automorphism of that group, and anyway, I leave this to you. So, it's not always surjective. There can be automorphisms that are not inner automorphisms.

The Image of Conjugation: Inner Automorphisms

Definition 8.61 (*Inner and Outer Automorphisms*)**:** The image of the conjugation homomorphism $\mathrm{Conj}\colon G \to \mathrm{Aut}(G)$ is a subgroup of $\mathrm{Aut}(G)$ (Proposition 8.2), the subgroup of *"inner automorphisms"*:

$$\mathrm{Im}(\mathrm{Conj}) = \{c_z \mid z \in G\} \leq \mathrm{Aut}(G). \tag{8.83}$$

An automorphism $\varphi \in \mathrm{Aut}(G) \setminus \mathrm{Im}(\mathrm{Conj})$, one that is not conjugation by any element of G, is called an *"outer automorphism"*.

Example 8.62 (*An Automorphism of $\mathbb{Z}/n\mathbb{Z}$ That Is Not Inner*)**:** Let $G = \mathbb{Z}/n\mathbb{Z}$ with addition, $n \geq 3$. Since G is abelian, every c_z is the identity (Observation 8.54), so $\mathrm{Im}(\mathrm{Conj}) = \{\mathrm{Id}\}$. The map

$$\psi\colon \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}, \quad x \mapsto -x$$

is an automorphism other than Id, so it is an outer automorphism, and Conj is not surjective in general.

Proof that ψ is an automorphism (beyond the lecture)

The lecture leaves this to the students (*"I leave this to you"*). Since addition is commutative, $\psi(x+y) = -(x+y) = (-x)+(-y) = \psi(x)+\psi(y)$, so ψ is a homomorphism, and $\psi \circ \psi = \mathrm{Id}$ makes it bijective. For $n \geq 3$, $\psi(\bar{1}) = \overline{n-1} \neq \bar{1}$, so $\psi \neq \mathrm{Id}$. **Q.E.D.**

Spoken Transcript [00:29:09 - 00:30:38] Prof. Rahul Pandharipande

And okay, I'll just say some things now. You can check them if you want. They're not, I think, really part of this class, but they're fun to know [erases: $\mathbb{Z}/n\mathbb{Z}$, $x \mapsto -x$].

So, this is not always surjective. The automorphisms here that are not inner automorphisms, you can imagine they're called outer automorphisms. So, there's the outer automorphisms.

You can always ask if a subgroup is a normal subgroup or not. That's a reasonable question.

And the answer is that... Question: Is the image of conjugation inside all automorphisms a normal subgroup? [writes: "Question: Is $\operatorname{Im}(\mathrm{Conj}) \subset \operatorname{Aut}(G)$ a normal subgr?" and a circled "YES"]

That's a good question. And the answer is yes. And this is a kind of fun thing to prove. It requires a little idea. I leave this to you to prove.

And then if we say one more comment about this, which is fun: what about for the symmetric group? Because the symmetric group is a very non-commutative group. So, it's interesting to think about conjugation for the symmetric group. We're going to do this explicitly here. I did it also yesterday, but it was kind of fast [sprays water and wipes the lower chalkboard].

Proposition 8.63 (*Normality of the Inner Automorphisms*)**:** The subgroup $\operatorname{Im}(\mathrm{Conj}) \leq \operatorname{Aut}(G)$ of inner automorphisms is a normal subgroup of $\operatorname{Aut}(G)$.

Proof (beyond the lecture)

The lecture leaves this to the students (*"I leave this to you to prove"*). Let $\varphi \in \operatorname{Aut}(G)$, and let c_z be an inner automorphism, $z \in G$. For any $x \in G$:

$$\begin{aligned}(\varphi \circ c_z \circ \varphi^{-1})(x) &= \varphi(c_z(\varphi^{-1}(x))) \\ &= \varphi(z \cdot \varphi^{-1}(x) \cdot z^{-1}) \\ &= \varphi(z) \cdot \varphi(\varphi^{-1}(x)) \cdot \varphi(z^{-1}) \qquad \text{(since } \varphi \text{ is a homomorphism)} \\ &= \varphi(z) \cdot x \cdot \varphi(z)^{-1} \\ &= c_{\varphi(z)}(x).\end{aligned}$$

Thus $\varphi \circ c_z \circ \varphi^{-1} = c_{\varphi(z)}$ is again an inner automorphism, so $\operatorname{Im}(\mathrm{Conj})$ is closed under conjugation by every element of $\operatorname{Aut}(G)$, and it is a normal subgroup. **Q.E.D.**

Beyond the Lecture: $G/Z(G)$, $\operatorname{Inn}(G)$ and $\operatorname{Out}(G)$

The lecturer suggests thinking about the First Isomorphism Theorem for every homomorphism. For Conj, with kernel $Z(G)$ (Proposition 8.60), Theorem 8.50 gives

$$G/Z(G) \cong \operatorname{Im}(\mathrm{Conj}).$$

Textbooks write $\operatorname{Inn}(G)$ for the subgroup $\operatorname{Im}(\mathrm{Conj})$ of inner automorphisms. Since it is a normal subgroup of $\operatorname{Aut}(G)$ (Proposition 8.63), the quotient $\operatorname{Out}(G) := \operatorname{Aut}(G)/\operatorname{Inn}(G)$ is a group, the **outer automorphism group**.

Spoken Transcript [00:30:38 - 00:31:45] Prof. Rahul Pandharipande

So, I'll do it explicitly for the symmetric group, but I'll say that for the symmetric group, this question about whether the inner automorphisms are the full automorphism group of the symmetric group has a fantastically interesting answer.

And the answer is yes in all cases, except for S_6. S_6 has some exotic automorphism.

S_6 has a strange automorphism that's not related to conjugation. And you know, this is one of the fun things about group theory: in some sense there's principles, patterns that hold for many groups, but then there's some exceptions, and the exceptions are always very interesting cases.

Okay, so that's the end of this idea of conjugation in general, and as I said, if you're interested in that, you can go look.

So, let's look at conjugation again, now more carefully for the symmetric group, because that really is part of this class — at least I think it is, I don't know.

If you are interested in mathematics, you will not waste any time thinking about the symmetric group. Any amount of time thinking about it will be useful, because it comes up everywhere. And it's a fantastically interesting group [finishes washing the lower chalkboard completely].

***Insight:* The Exceptional Outer Automorphism of S_6**

For almost all symmetric groups, the inner automorphisms exhaust the full automorphism group:

$$\operatorname{Aut}(S_n) = \operatorname{Inn}(S_n) \cong S_n \quad \text{for all } n \geq 3 \text{ with } n \neq 6.$$

The symmetric group on 6 elements, S_6, is the unique exception in the entire family: it possesses an *"exotic"* outer automorphism that does not arise from conjugation by any permutation. Consequently:

$$\operatorname{Out}(S_6) = \operatorname{Aut}(S_6) / \operatorname{Inn}(S_6) \cong \mathbb{Z}/2\mathbb{Z}.$$

Conjugation in the Symmetric Group S_n

Cycle Decomposition and the Total Length Formula

Spoken Transcript [00:31:45 - 00:33:34] Prof. Rahul Pandharipande

Okay, so, let's talk about conjugation again, and now more carefully than last time. So, I'm in S_n, and I pick an element in S_n [writes: $\sigma \in S_n$], and I've said many times: to this element I can associate a cycle decomposition.

This cycle decomposition has a certain number of cycles. The first one, now I try to... I didn't want to put the notation last time, but $a_{11}, a_{12}, \ldots, a_{1k_1}$ [writes: $\sigma \mapsto (a_{11}a_{12}\ldots a_{1k_1})$]. That's the first cycle.

I can't tell you what the elements are, because I don't know what the thing is. But if I take a particular element of S_n, I can write it in cycle decomposition, and this will be the first cycle, then there'll be the next cycle [writes: $(a_{21}a_{22}\ldots a_{2k_2})$]... k_2, and then there'll be some more cycles, and then there'll be the last cycle: $a_{r1}\ldots a_{rk_r}$ [writes: $\ldots(a_{r1}\ldots a_{rk_r})$].

Okay, are you happy with this notation? So, it means that when I write this out, there are r cycles, and the lengths of the cycles are k_1, k_2, all the way to k_r, okay? [writes: r-cycles $k_1, k_2, \ldots, k_r$]

And when I do it now, I'm going to write everything, meaning I include all the 1-cycles too. In this thing, I include... because I said sometimes, depending on mood, you can include the 1-cycles or not, depending on how you want to write. But here I include all 1-cycles [writes: "include all 1-cycles"].

Okay? So, that's the cycle decomposition, and we've discussed that several times. You should make sure you're able to do it, and you understand it.

So, first of all, then the question is: what is this number, by the way, if I take here the sum of k_i from i equals 1 to r? What is this number? [writes: $\sum_{i=1}^{r} k_i =$]

This is if you understand the notation. Yeah?

n. Because this is saying that every element appears somewhere in one of the cycles, exactly once [writes: n]. Okay?

Cycle Decomposition in the Symmetric Group

Let $\sigma \in S_n$ be a permutation of the set $\{1, 2, \ldots, n\}$. We represent σ as a product of r disjoint cycles, explicitly including all 1-cycles (fixed points):

$$\sigma = (a_{11}\, a_{12} \ldots a_{1k_1})(a_{21}\, a_{22} \ldots a_{2k_2}) \cdots (a_{r1}\, a_{r2} \ldots a_{rk_r}), \tag{8.84}$$

where:

- **Number of cycles:** $r \geq 1$ denotes the total number of disjoint cycles.
- **Cycle lengths:** $k_1, k_2, \ldots, k_r \geq 1$ denote the lengths of the respective cycles.
- **Disjoint partition:** The elements appearing in all cycles collectively partition $\{1, \ldots, n\}$:

$$\bigcup_{i=1}^{r} \{a_{i1}, a_{i2}, \ldots, a_{ik_i}\} = \{1, 2, \ldots, n\}. \tag{8.85}$$

Proposition 8.64 (*Sum of Cycle Lengths*): Because every element of $[n]$ appears in exactly one of the cycles, the sum of all cycle lengths equals n:

$$\sum_{i=1}^{r} k_i = n. \tag{8.86}$$

Why 1-Cycles Must Be Included: The previous week allowed leaving out the 1-cycles (x), the fixed points $\sigma(x) = x$ (Notation 7.31). However, to maintain the exact relation $\sum_{i=1}^{r} k_i = n$ and to establish a precise connection with integer partitions of n, all 1-cycles must explicitly be retained.

Action of Conjugation on Disjoint Cycles

Spoken Transcript [00:33:34 - 00:34:55] Prof. Rahul Pandharipande

All right, so, now let us take z as a map from n to n, right? So, z is some element of S_n [writes: $z\colon [n] \to [n]$, $z \in S_n$]. It's a map from n to n, so we have $1, 2, 3, \ldots, n$, and then here's $1, 2, 3, \ldots, n$, and z is this rule that takes a bijection from n to n [draws columns $1, 2, 3, \ldots, n$ with arrow z]. And the goal, to study the conjugation, is we want to understand the conjugate [writes: "We want to understand"], and I will write it as $z\sigma z^{-1}$, right? [writes: $z\sigma z^{-1}$]

So, this is explicit: we have this explicit cycle, we have some explicit bijection, which is also an element of S_n, and I want to write this.

And so this is what I was hinting at last time. I kind of said a little bit and waved my hands, but here I'm going to actually write the formula, in a moment of honesty.

So, yeah, what am I going to write?

All the data is there: z is this data, how to move from this side to this side, and I have that cycle. I hinted at it last time.

The Conjugate Permutation Setup

Let $z \in S_n$ be a permutation, viewed as a bijection of $[n] := \{1, 2, \ldots, n\}$ to itself:

$$\begin{array}{ccc} 1 & & 1 \\ 2 & \xrightarrow{z} & 2 \\ 3 & & 3 \\ \vdots & & \vdots \\ n & & n \end{array}$$

Our goal is to explicitly determine the cycle decomposition of the conjugate permutation:

$$z\sigma z^{-1} \in S_n. \tag{8.87}$$

Spoken Transcript [00:34:55 - 00:36:28]

Prof. Rahul Pandharipande

So, let us see why. Suppose I take here... suppose I want to find out — and this is already a hint — what happens to $z(a_{11})$?

So, I start at my first cycle. These a_{ij}'s give me the set 1 to n [writes: $\{a_{ij}\} = \{1, \ldots, n\}$]. So, z eats them. Yeah, so z of this.

Where is this going to go? [writes: $(z(a_{11})$]

If I want to find out where this index goes, I first apply z^{-1}: it goes to a_{11}, right?

Then I apply σ: a_{11} goes to a_{12}.

And then I apply z again, so this goes to $z(a_{12})$ [writes: $z(a_{12})$]. Okay? Is that clear when I start doing this cycle?

I'm trying to find the cycle notation of this, so my first idea is to see: where does this $z(a_{11})$ go? And I go through this argument and it goes to $z(a_{12})$, which is kind of nice.

And so, you think: where does $z(a_{12})$ go? And it's not going to be a surprise: it goes to $z(a_{13})$, and so on, until you get to $z(a_{1k_1})$, and the cycle closes [writes: $\ldots z(a_{1k_1}))$].

That's what happens if you just apply this to $z(a_{11})$: you get this cycle, right?

Tracing the Action on the First Cycle

To determine the cycle representation of $z\sigma z^{-1}$, we evaluate its action on the element

$z(a_{11})$:

$$\begin{aligned}(z\sigma z^{-1})(z(a_{11})) &= z\Big(\sigma\big(z^{-1}(z(a_{11}))\big)\Big)\\ &= z(\sigma(a_{11}))\\ &= z(a_{12}) \qquad\qquad (\text{since } \sigma(a_{11}) = a_{12}).\end{aligned} \tag{8.88}$$

Continuing this evaluation inductively along the elements of the first cycle:

$$\begin{aligned}(z\sigma z^{-1})(z(a_{12})) &= z(a_{13}),\\ &\vdots\\ (z\sigma z^{-1})(z(a_{1,k_1-1})) &= z(a_{1k_1}),\\ (z\sigma z^{-1})(z(a_{1k_1})) &= z(a_{11}).\end{aligned}$$

Therefore, the original cycle $(a_{11}\, a_{12} \ldots a_{1k_1})$ in σ maps under conjugation by z directly to the cycle:

$$(z(a_{11})\, z(a_{12}) \ldots z(a_{1k_1})).$$

Spoken Transcript [00:36:28 - 00:37:56] Prof. Rahul Pandharipande

As I said, you pretty much have to do this yourself and convince yourself of this. That's the first cycle.

What about the second cycle? Here I wrote them very nicely all in a row; here I kind of ran out of room, so it's going to be on different lines, but you think about it as one element of the group.

So, what's going to happen to the next one? Well, this is $z(a_{21})$, and it's just going to be z's applied to everybody: $z(a_{2k_2})$ [writes: $(z(a_{21}) \ldots z(a_{2k_2}))$]. That's the second cycle, and then so on, and then we get to the last cycle.

I'll just talk a little bit about it, but this is $r1$, z... [writes: $\ldots(z(a_{r1}) \ldots z(a_{rk_r}))$]

I have never in my whole mathematical life liked indices... That's why I mostly try to avoid writing formulas: it's hard to write them correctly.

So, you should look at it, see if you like it, see if you believe it's correct. All the indices are correct.

Okay, so, this is a better presentation than last time, where I just said you should do it yourself. Now I've mostly done it for you, but still, for you to really understand it, you have to understand what I'm doing.

It says that if you take this explicit element in the cycle decomposition, you write some bijection, and you calculate the conjugation, all you do is take this cycle, and you apply z to the elements there, and you get a new cycle decomposition. Okay?

Recap: The previous lecture stated this formula with one example and left the general case to the students (Lemma 8.43); the derivation above is the general case.

Theorem 8.65 (*Conjugation Formula in the Symmetric Group*): Let $\sigma \in S_n$ have disjoint cycle decomposition:

$$\sigma = (a_{11} \ldots a_{1k_1})(a_{21} \ldots a_{2k_2}) \cdots (a_{r1} \ldots a_{rk_r}). \tag{8.89}$$

For any permutation $z \in S_n$, the conjugate permutation $z\sigma z^{-1}$ is obtained by replacing each element in the cycle decomposition of σ by its image under z:

$$z\sigma z^{-1} = \big(z(a_{11}) \dots z(a_{1k_1})\big)\big(z(a_{21}) \dots z(a_{2k_2})\big) \cdots \big(z(a_{r1}) \dots z(a_{rk_r})\big). \tag{8.90}$$

Conjugation as Relabeling: Conjugation by z corresponds to a purely bijective relabeling of the underlying elements:

- **Preservation of cycle lengths:** Each cycle $(a_{i1}\, a_{i2} \dots a_{ik_i})$ of length k_i in σ transforms into a cycle of the exact same length k_i in $z\sigma z^{-1}$.
- **Preservation of disjointness:** Because z is a bijection, disjoint cycles remain disjoint, and the total number of cycles r remains unchanged.
- **Same cycle lengths:** Consequently, σ and $z\sigma z^{-1}$ have the same cycle lengths $k_1, k_2, \dots, k_r$.

Spoken Transcript [00:37:56 - 00:38:22] Prof. Rahul Pandharipande

And there is actually nothing going on here except understanding the notation. Just some kind of... you have to concentrate for some amount of time and keep all the definitions in your head, maybe a few minutes, but that's all it is: it's understanding notation.

But the conclusion is extremely nice: it says that σ and the conjugate have the same cycle decom— yeah?

Student Question

Does that mean it gets rid of the z and z^{-1}?

Spoken Transcript [00:38:24 - 00:38:56] Prof. Rahul Pandharipande

We're not getting rid of it. We have to figure out what happens to this element. The first thing we do is apply z^{-1}; it gets rid of itself, yeah. It's a good question.

I mean, a lot of times in mathematics, the goal is just to not stay confused for a few minutes. That's the... If you can do that at certain junctures, you can get very far as a mathematician. Because it's very confusing.

Conjugacy Classes in S_n and Integer Partitions

Spoken Transcript [00:38:56 - 00:40:06] Prof. Rahul Pandharipande

Okay, so, this calculation shows several things. If you understand it correctly, it shows: the set of equivalence classes under conjugation of S_n [writes: "Set of equiv. classes under Conjugation of $S_n \longleftrightarrow \{\text{Partitions } n\}$"]...

that the set of equivalence classes is in bijective correspondence with the partitions of n. Because all that matters here is how the cycle lengths are.

It says that if I take any σ here and I conjugate it, I get somebody with the same cycle lengths. Conversely, if I have two people with the same cycle lengths, I can find a z.

So, it exactly shows that the conjugacy classes of S_n are the partitions. All right? And so that's another step that this one calculation shows.

Recap: The previous lecture stated this correspondence (Theorem 8.44); here it follows from Theorem 8.65.

Theorem 8.66 (*Conjugacy Classes of S_n and Partitions*): Two permutations $\sigma, \tau \in S_n$ are conjugate in S_n if and only if their cycle decompositions, with all 1-cycles included, have the same cycle lengths $k_1, \ldots, k_r$, up to their order. Since $\sum_{i=1}^{r} k_i = n$, these lengths form a partition of n, and there is a bijective correspondence:

$$\{\text{equivalence classes of } S_n \text{ under conjugation}\} \longleftrightarrow \{\text{partitions of } n\}. \tag{8.91}$$

Proof of the Bijective Correspondence:

(1) Forward direction ($\implies$): If $\tau = z\sigma z^{-1}$, Theorem 8.65 guarantees that the cycles of τ have lengths $k_1, \ldots, k_r$, identical to those of σ.

(2) Reverse direction ($\impliedby$): Suppose σ and τ share the same cycle lengths $k_1, \ldots, k_r$. Write out both cycle decompositions:

$$\sigma = (a_{11} \ldots a_{1k_1})(a_{21} \ldots a_{2k_2}) \cdots (a_{r1} \ldots a_{rk_r}),$$
$$\tau = (b_{11} \ldots b_{1k_1})(b_{21} \ldots b_{2k_2}) \cdots (b_{r1} \ldots b_{rk_r}).$$

Since every element in $\{1, \ldots, n\}$ appears exactly once in each decomposition, the assignment $z(a_{ij}) := b_{ij}$ defines a bijective map $z\colon \{1, \ldots, n\} \to \{1, \ldots, n\}$, i.e., $z \in S_n$. By (8.90), $z\sigma z^{-1} = \tau$, proving that σ and τ are conjugate.

Beyond the Lecture: Cycle Type

Textbooks call the list of cycle lengths of σ, with the 1-cycles included, its **cycle type**. Theorem 8.66 says that two permutations are conjugate exactly when they have the same cycle type.

The Sign of a Permutation from Cycle Decomposition

Spoken Transcript [00:40:06 - 00:40:45] Prof. Rahul Pandharipande

Okay, and I'll say some other things that are kind of useful here. If I have an element of S_n [writes: $\sigma \in S_n$], and I have the cycle decomposition — I won't write it again, but just all those cycle decompositions — you can say: what is the sign of σ? [writes: "What is sign of σ?"]

So, can you tell me what that is? Meaning, if I take σ and I write it in terms of this cycle data, what is the sign?

This requires a small amount of understanding of the notation again.

Student Answer

$n - r$.

Spoken Transcript [00:40:46 - 00:41:55] Prof. Rahul Pandharipande

$n-r$, very good! [writes "$\mathrm{Sign}(\sigma) = n - r$", a slip for $(-1)^{n-r}$ that stays on the board]

And why do you say that, with such confidence? Yeah.

So, the proof of this is: sign is multiplicative, so I only need to calculate the sign for each cycle.

The sign for each cycle is... this cycle is length k_1. The cycle's length k_1, so its sign, we said before, is $k_1 - 1$, the sign of this [writes above the first cycle: $-1^{(k_1-1)}$].

Do you remember that? A cycle of length k_1 is $k_1 - 1$, okay?

And then it's multiplicative, so I just have to take $k_1 - 1, k_2 - 1 \ldots$ [writes above the second and the last cycle: $-1^{(k_2-1)}$ and $(-1)^{k_r-1}$]

These are the exponents! The sign is minus one to the k_1 minus one! And this sign is minus one to the $k_2 - 1$, and this is minus one to the $k_r - 1$.

So, that's the sign of each one of these, and then to get the sign of the whole thing, I multiply these all together. And the k parts, they add up to n...

Spoken Transcript [00:41:55 - 00:42:20] Prof. Rahul Pandharipande

Is that my phone? Yeah, wow. That's too bad. Minus one is... they all add up to minus r.

Huh, I guess the class is over.

Okay, anyway, it's over. Let's see who that is [chime rings, lecturer checks phone at desk while students pack up].

Proposition 8.67 (*Sign from Cycle Decomposition*): Let $\sigma \in S_n$ be decomposed into r disjoint cycles of lengths $k_1, k_2, \ldots, k_r$ (including all 1-cycles). Then the sign of σ is given by:

$$\mathrm{Sign}(\sigma) = (-1)^{n-r}. \tag{8.92}$$

Proof

By the previous lecture (Lemma 8.17), a cycle of length $k \geq 2$ has sign

$$\mathrm{Sign}\big((x_1 \ldots x_k)\big) = (-1)^{k-1}, \tag{8.93}$$

and a 1-cycle is the identity, with sign $1 = (-1)^{1-1}$. Since $\mathrm{Sign}\colon S_n \to \{\pm 1\}$ is a group homomorphism, it is multiplicative across the cycles:

$$\begin{aligned}
\mathrm{Sign}(\sigma) &= \prod_{i=1}^{r} \mathrm{Sign}\big((a_{i1} \ldots a_{ik_i})\big) \\
&= \prod_{i=1}^{r} (-1)^{k_i - 1} \\
&= (-1)^{\sum_{i=1}^{r}(k_i-1)} \\
&= (-1)^{\sum_{i=1}^{r} k_i - \sum_{i=1}^{r} 1}.
\end{aligned} \tag{8.94}$$

By Proposition 8.64, the sum of the cycle lengths is $\sum_{i=1}^{r} k_i = n$. Furthermore, the sum of r ones is $\sum_{i=1}^{r} 1 = r$. Substituting these values into (8.94) yields:

$$\operatorname{Sign}(\sigma) = (-1)^{n-r}. \tag{8.95}$$

This completes the proof. **Q.E.D.**

Beyond the Lecture: Even and Odd from the Number of Cycles

With the even and odd permutations of the previous lecture (Definition 8.11), Proposition 8.67 says that σ is even, that is $\sigma \in A_n$, exactly when n and the number r of its cycles have the same parity, and odd when they differ.

Board Correction: The Sign Formula

The board line $\operatorname{Sign}(\sigma) = n - r$ is the student's answer as first written, and the lecture ends before it is changed: the phone rings while he multiplies the signs $(-1)^{k_i - 1}$ written above the cycles. Their product is $(-1)^{\sum_i (k_i - 1)} = (-1)^{n-r}$, so $n - r$ is the exponent, and Proposition 8.67 states the corrected formula.

Summary of Key Concepts

- **Normal Subgroups and Quotient Groups:** Kernels of homomorphisms are normal subgroups, and for a normal subgroup N of G the product $\bar{x} \cdot \bar{y} = \overline{xy}$ on the equivalence classes of $\sim_N$ is well defined, which makes G/N a group with quotient map $q(x) = \bar{x}$.
- **First Isomorphism Theorem for Groups:** For a group homomorphism $f \colon G_1 \to G_2$, the map $g \colon G_1 / \ker(f) \to \operatorname{Im}(f)$, $g(\bar{x}) = f(x)$, is well defined, a homomorphism, surjective and injective, so
$$G_1 / \ker(f) \cong \operatorname{Im}(f).$$
- **Order of the Alternating Group:** For $n \geq 2$, $A_n = \ker(\operatorname{Sign})$ and $S_n / A_n \cong \{\pm 1\}$, so by Lagrange's Theorem $|A_n| = \frac{|S_n|}{2} = \frac{n!}{2}$.
- **Conjugation and Automorphisms:** Each $c_z(x) = zxz^{-1}$ is a group isomorphism $G \to G$, and $c_z = \operatorname{Id}$ when G is abelian, so every subgroup of an abelian group is normal. The homomorphism $\operatorname{Conj} \colon G \to \operatorname{Aut}(G)$, $z \mapsto c_z$, has kernel the center $Z(G)$; its image, the inner automorphisms, is a normal subgroup of $\operatorname{Aut}(G)$, and $x \mapsto -x$ on $\mathbb{Z}/n\mathbb{Z}$, $n \geq 3$, shows that it can be smaller than $\operatorname{Aut}(G)$.
- **Conjugation in S_n:** $z\sigma z^{-1}$ is obtained by applying z to every entry of the cycles of σ, so conjugate permutations have the same cycle lengths, and the conjugacy classes of S_n correspond to the partitions of n.
- **Sign from the Cycles:** If $\sigma \in S_n$ has r cycles, the 1-cycles included, then $\operatorname{Sign}(\sigma) = (-1)^{n-r}$.

Tuesday, 10th November 2026

Cosets, Normal Subgroups, Group Actions, and Cayley's Theorem

Course Content Overview

This lecture explores foundational structural concepts in group theory, bridging quotient structures, symmetry groups, and group actions:

- **Cosets and Normality:** We review the equivalence relation from Lagrange's theorem, formalize left versus right cosets (xH and Hx), prove that $xH = Hx$ whenever H is normal in G, and prove that every subgroup of index 2 is automatically normal.
- **The Subgroup Correspondence Theorem:** In direct parallel to the ideal correspondence for quotient rings, we establish the lattice bijection between subgroups of G containing a normal subgroup N of G and all subgroups of ${}^{G}\!/_{N}$, noting that the correspondence preserves normality.
- **Groups of Prime Order and Classical Symmetries:** We demonstrate that every group of prime order is cyclic ($G \cong \mathbb{Z}/p\mathbb{Z}$), explore rotational symmetry groups of the Platonic solids (A_4, S_4, A_5) via geometric duality, and introduce planar dihedral groups ($D_6 \cong S_3, D_8$).
- **Testing Group Isomorphisms:** We analyze classical isomorphism questions distinguishing groups via structural invariants (divisibility, cyclic generation, torsion elements, vector space dimensions over $\mathbb{F}_2$ via modulo 2 reduction, and Hamel bases over $\mathbb{Q}$ yielding $\mathbb{R} \cong \mathbb{C}$ under the Axiom of Choice), concluding with the structure theorem $(\mathbb{Q}^*, \cdot, 1) \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}^\infty$.
- **Group Actions and Permutation Representations:** We define group actions $G \curvearrowright T$ via action maps $\phi \colon G \times T \to T$ and prove their equivalence to permutation representations $\sigma \colon G \to \mathrm{Sym}(T)$. We analyze canonical examples (S_n on $\{1, \dots, n\}$, rotational actions on simplicial features of the tetrahedron, $\mathrm{GL}_n(\mathbb{C})$ on $\mathbb{C}^n$, and $\mathrm{U}(1)$ on $\mathbb{C}$).
- **Actions on the Group Itself and Cayley's Theorem:** We study the left translation action ($g \cdot h = gh$) and prove Cayley's Theorem that every finite group of order n embeds into S_n. Finally, we investigate the conjugation action ($g \cdot h = ghg^{-1}$), establishing its triviality on abelian groups and previewing its role in the Sylow theorems.

Cosets and Quotient Sets

Equivalence Relations and Coset Partitions

Spoken Transcript [00:00:00 - 00:02:05] Prof. Rahul Pandharipande

Okay, so, for the past week or two, we've been thinking about groups. So, G is a group [writes: G is group] and the idea of a subgroup [writes: $H \subset G$ subgroup].

I wanted to say some things: some things we've said before and also some things maybe said now slightly differently. So, H is a subgroup. We proved this theorem of Lagrange that, in the case where everything is finite, the order of the subgroup divides the order of the group. And in the proof, there was a construction that's very important. So, in the proof of Lagrange [writes: In the proof of Lagrange], we defined an equivalence relation which had to do with H, and this was if there exists an h in H such that this holds: [writes: $x \sim_H y \overset{\text{def}}{\iff} xh = y, \exists h \in H$].

And if you remember, the outcome of this equivalence relation was that it divided the set G into equivalence classes [writes: G into equiv. classes] for this equivalence relation, and actually all of those equivalence classes had a size which was in bijective correspondence to H. So, the part I wanted to just say here is that here we have G... It's very common to use this symbol, which is a quotient symbol. That's a little unfortunate, but it's very common to use this symbol for the set of equivalence classes [writes: ${}^{G}\!/_{H} \longleftarrow$ set of equiv. classes of $x \sim_H y$] of our equivalence relation.

And there are two unfortunate things about this notation. One is because, later, we defined a group structure on the set of equivalence classes, but not in the context of a subgroup, but in the context of a normal subgroup.

Review: The Equivalence Relation from Lagrange's Theorem

Let G be a group and $H \subset G$ a subgroup.

Definition 9.1 (*Left Coset Equivalence Relation*): We define a binary relation $\sim_H$ on the elements of G by:

$$x \sim_H y \overset{\text{def}}{\iff} xh = y \quad \text{for some } h \in H. \tag{9.1}$$

Equivalently, $x \sim_H y \iff x^{-1}y \in H$.

Equivalence Classes and the Partition of G: The relation $\sim_H$ is an equivalence relation on G:

1. **Reflexivity:** For all $x \in G$, $x \cdot e = x$ with $e \in H$, so $x \sim_H x$.
2. **Symmetry:** If $x \sim_H y$, then $xh = y$ for some $h \in H$. Multiplying by $h^{-1} \in H$ on the right gives $yh^{-1} = x$, hence $y \sim_H x$.
3. **Transitivity:** If $x \sim_H y$ and $y \sim_H z$, then $xh_1 = y$ and $yh_2 = z$ for $h_1, h_2 \in H$. Substituting yields $x(h_1h_2) = z$. Since $h_1h_2 \in H$, it follows that $x \sim_H z$.

Consequently, $\sim_H$ partitions the set G into disjoint equivalence classes. The equivalence class containing $x \in G$ is the *"left coset"*:

$$[x]_{\sim_H} = \{y \in G \mid x \sim_H y\} = \{xh \mid h \in H\} =: xH.$$

Each equivalence class xH is in bijection with H via the mapping $h \mapsto xh$, so $|xH| = |H|$ for every $x \in G$.

Notation 9.2 (*The Quotient Set* ${}^{G}\!/_{H}$): We denote the set of all equivalence classes of $\sim_H$ by the quotient notation:

$${}^{G}\!/_{H} \longleftarrow \text{set of equivalence classes of } \sim_H .$$

Left Cosets vs. Right Cosets

Spoken Transcript [00:02:05 - 00:03:37] Prof. Rahul Pandharipande

So, as I said, there are two drawbacks for this notation. First is: if H is a normal subgroup of G [writes: • If H is normal subgp of G,], then we proved that this set of equivalence classes carries a product and a neutral element, and this is the quotient group, sorry [writes: then we prove $({}^{G}\!/_{N}, \cdot, \bar{e}) \longleftarrow$ quotient group, with N for H].

When it happens that H is normal, the set of equivalence classes has this additional structure of being a group. If H is not normal, we can still think about the set of equivalence classes, but it's not necessarily a group. So, this is a little confusing. Is this more or less okay?

And there's another level of confusion, which is going to come today, which is: why did we define the set of equivalence classes this way?

Here, the way we defined it is G mod H is the set of equivalence classes, and they're what are called the left cosets [writes: Left cosets]. They're the sets xH [writes: xH], because the way the equivalence relation works is y is equivalent to x if it differs by a multiplication by h on the right. So, this is called a left coset because x sits on the left, okay? Is that clear?

Quotient Set vs. Quotient Group

Remark 9.3 (*The Two Drawbacks of the Notation* ${}^{G}\!/_{H}$): The notation ${}^{G}\!/_{H}$ carries two potential sources of confusion:

(1) Group Structure vs. Plain Set: The set of equivalence classes ${}^{G}\!/_{H}$ is **always** well-defined as a set of subsets partitioning G. However, ${}^{G}\!/_{H}$ inherits a natural group operation:

$$(xH) \cdot (yH) := (xy)H$$

if and only if H is a *"normal subgroup"* of G. When H is not normal, this operation is ill-defined (depends on the choice of representatives), and ${}^{G}\!/_{H}$ remains merely a partitioned set without group structure.

(2) Left vs. Right Handedness: Defining equivalence by multiplying by $h \in H$ on the right produces sets of the form xH, in which the representative x stands on the left.

Definition 9.4 (*Left Coset*): For $x \in G$, the *"left coset"* of H containing x is:

$$xH := \{xh \mid h \in H\}.$$

The set of all left cosets is denoted by $({}^{G}\!/_{H})_L := \{xH \mid x \in G\}$.

Spoken Transcript [00:03:37 - 00:05:35]

Prof. Rahul Pandharipande

But we could have defined another equivalence relation [writes: We could have defined]. So, now we go back and say, be very clear, that this was about the left cosets. We could define another equivalence relation, and maybe I put a hat here or something like this to show it's a different equivalence relation [writes: $x \mathrel{\hat{\sim}}_H y$]. It's now just the same equation, but you put h on the other side, which in group theory is always a different thing in general. So, this is where h is on this side: [writes: $\overset{\text{def}}{\iff}$ $hx = y$].

We could have just started the whole thing, and if you did that, you would prove Lagrange's theorem, just as we did, because left or right doesn't matter that much: you stay all on the other side. You'd get a different proof of Lagrange's theorem, but you'd have a different notion of these cosets.

This is a definition, and if we use this, we get a different notion of cosets [writes: $\hat{\sim}$ "different notion of cosets"]. And you can imagine that these are called right cosets. So, this is like G mod H, and now maybe it's nice to somehow indicate that the theory we're doing is left cosets, and you can do now G mod H with right cosets [writes: $({}^{G}\!/_{H})_L = \{\text{Left cosets } xH\}$ and $({}^{G}\!/_{H})_R =$]. This is the set of right cosets [writes: $\{$Right cosets$\}$].

The equivalence classes here, the set of right cosets, look not like this, but where the element is on the right: [writes: $\{Hx\}$]. Okay? We could have just proven Lagrange's theorem using right cosets. There's nothing particularly important about left or right, okay?

So, this is essentially a confusing discussion that's not telling you any more math, it's just confusing you.

Left and Right Coset Formulations

Definition 9.5 (*Right Coset Equivalence Relation*): We can define an alternative equivalence relation $\sim_{\hat{H}}$ on G by multiplying by $h \in H$ on the left:

$$x \sim_{\hat{H}} y \overset{\text{def}}{\iff} hx = y \quad \text{for some } h \in H. \tag{9.2}$$

Equivalently, $x \sim_{\hat{H}} y \iff yx^{-1} \in H$.

Definition 9.6 (*Right Coset*): The equivalence class of $x \in G$ under $\sim_{\hat{H}}$ is the *"right coset"*:

$$Hx := \{hx \mid h \in H\}.$$

The set of all right cosets is denoted by $({}^{G}\!/_{H})_R := \{Hx \mid x \in G\}$.

Comparison of Coset Collections: On the blackboard, the two parallel notions of coset spaces are summarized as:

$$\begin{cases} ({}^{G}\!/_{H})_L = \{\text{left cosets}\} = \{xH \mid x \in G\}, \\ ({}^{G}\!/_{H})_R = \{\text{right cosets}\} = \{Hx \mid x \in G\}. \end{cases}$$

Both collections partition G into subsets of identical cardinality $|H|$. Lagrange's theorem could be proved identically using right cosets instead of left cosets, since the bijection $h \mapsto hx$ demonstrates $|Hx| = |H|$ for every $x \in G$.

Equality of Left and Right Cosets for Normal Subgroups

Spoken Transcript [00:05:35 - 00:06:38] Prof. Rahul Pandharipande

So, then we come to some kind of new perspectives on that. As I said, the way to think about it is if you have a subgroup of a group, you can consider this equivalence relation on the left, the equivalence relation on the right; you get left cosets, right cosets.

If it happens you're so lucky as your group is normal, then you get a group structure on the set of cosets, and you could say, *"Well, do you get a group structure on the right cosets or left cosets?"* That's a good question.

And what do you think the answer to that is?

It's a strange question.

So, the answer is a very good answer: if H is a normal subgroup, then the set of left cosets equals the set of right cosets. That's the right answer.

Is that what you were going to say?

Student Answer

You already gave the answer.

Spoken Transcript [00:06:39 - 00:06:40] Prof. Rahul Pandharipande

What?

Student Answer, continued

You already gave the answer.

Spoken Transcript [00:06:41 - 00:07:35] Prof. Rahul Pandharipande

I just gave it now! You're too late.

So, if H is a normal subgroup, then we're going to get the left cosets are the same as right cosets [writes: If H is a normal subgp of G, the lefts are same as right cosets]. This is very philosophically comforting, because it says that you could have taken either of these paths, and you're going to get the same group at the end. And what's the proof of that? So, we have to have a proof [writes: Proof:].

Suppose we have here H inside G is normal [writes: $H \subset G$ is normal], and I take an x in G [writes: $x \in G$], then I consider [writes: then I consid.] this left coset [writes: xH].

Spoken Transcript [00:07:35 - 00:09:50] Prof. Rahul Pandharipande

So, I consider this left coset and I want to show that it's the right coset. What's an element in here? An element in here is something xh, all right? [writes: $xH \ni xh$, circles xh]

But then I know, since H is normal [writes: H is normal], this implies that xhx^{-1} is in

H [writes: $\implies xhx^{-1} \in H$]. That's the definition of normal, right? So, this means that xhx^{-1} is equal to somebody else in H, $\widehat{h}$ [writes: $xhx^{-1} = \widehat{h}$]. And if I rewrite this, if I right-multiply by x, I get this: [writes: $xh = \widehat{h}x$].

So, this means that this element, which was in the left coset, is actually also in the right coset [writes: $xh \in Hx$], okay? And then you can do it the other way around, too, by symmetry.

There's something that's kind of satisfying about that: there are these two paths, you can go left or right, but if you're normal, the left cosets are just the same as the right cosets. And so there's not like two different ideas of quotient groups; it's only one idea of quotient group.

Okay. This was a kind of pedantic point, but it's important. And I would say that some other things about language is that if someone writes H is a subgroup of G [writes: $H \subset G$ subgroup], then you have to decide whether this person means a normal subgroup or any subgroup, and it's a very different way of thinking about groups.

The subgroups, whether it's normal or not, is extremely important. So, when you meet a subgroup, you should decide whether it's normal — or maybe it's a hard question, you don't know the answer —, but you should at least consider it, because it changes the way you think about it.

If someone writes N is a subgroup of G [writes: $N \subset G$], this is a strong indication that person means it's normal, but it doesn't always. I'd say that's kind of the convention. So, this usually means normal [writes: $\longleftarrow$ usually means normal], but you should ask!

And then often there's some notation with triangles and everything, but, you know, I think the best is when you meet a subgroup, then you should investigate yourself whether the person means it's normal or not.

Left and Right Coset Coincidence for Normal Subgroups

Proposition 9.7 (*Equality of Left and Right Cosets*)**:** Let G be a group and $H \subset G$ a subgroup. If H is normal in G, then for every $x \in G$:

$$xH = Hx. \tag{9.3}$$

Consequently, $({}^{G}\!/\!_{H})_L = ({}^{G}\!/\!_{H})_R$.

Proof

Let $x \in G$. We demonstrate mutual containment $xH \subseteq Hx$ and $Hx \subseteq xH$:

Inclusion $xH \subseteq Hx$: Let $y \in xH$. Then $y = xh$ for some $h \in H$. Since H is normal in G, the conjugate xhx^{-1} belongs to H:

$$xhx^{-1} \in H.$$

Hence, there exists an element $\widehat{h} \in H$ such that:

$$xhx^{-1} = \widehat{h}.$$

Multiplying both sides on the right by x, we obtain:

$$xh = \widehat{h}x.$$

Because $\widehat{h} \in H$, the element $\widehat{h}x$ lies in the right coset Hx. Thus, $y = xh \in Hx$, which proves $xH \subseteq Hx$.

Inclusion $Hx \subseteq xH$: By symmetry, let $y \in Hx$, so $y = hx$ for some $h \in H$. Since H is normal, $x^{-1}h(x^{-1})^{-1} = x^{-1}hx \in H$. Setting $h' := x^{-1}hx \in H$ and multiplying by x on the left yields $xh' = hx = y \in xH$.

Therefore, $xH = Hx$ for all $x \in G$. **Q.E.D.**

Notational Conventions for Subgroups and Normal Subgroups

Notation 9.8 (*Subgroup and Normal Subgroup Symbols*): Mathematical literature employs the following naming conventions:

- Writing $H \subset G$ or $H \leq G$ designates an arbitrary subgroup of G.
- The letter N (as in $N \subset G$) is conventionally reserved for a *"normal subgroup"*.
- Many texts write $N \triangleleft G$ or $N \trianglelefteq G$ to explicitly denote normality, whereas the lecture emphasizes verifying the definition $gNg^{-1} = N$ directly.

Subgroups of Index 2

Normality of Index-2 Subgroups

Spoken Transcript [00:09:50 - 00:11:22] Prof. Rahul Pandharipande

Okay, and now if you really understood this well, there's a little proposition which is totally standard in group theory [writes: Prop.].

It says: suppose I have a subgroup of index 2 [writes: Suppose $H \subset G$ of index 2].

As an aside, index 2 means that the equivalence class G mod H — remember, I'm not assuming it's normal, it's just the set of equivalence classes — is equal to two things: H and one more [writes: (index 2 means ${}^{G}\!/_{H} = \{H, xH\}$)].

There's just two equivalence classes. That's what index 2 means. The index is the number of equivalence classes. So, index need not be finite, but if it's finite, it just means there's two equivalence classes. Is that clear? That's what index 2... So, if you find a subgroup of index 2, then this implies H must be normal [writes: $\implies$ then H must be normal].

This is the cheapest way to show a subgroup is normal: if you ever have a subgroup that's half the size of the group, it has to be normal.

And the proof of that — there's actually nothing to this proof except understanding what I said. Let's try to do it, though [erases the right chalkboard to prepare for the proof].

Subgroups of Index 2

Definition 9.9 (*Index of a Subgroup*): Let H be a subgroup of G. The *"index"* of H in G, denoted $[G : H]$, is the cardinality of the quotient set of left cosets:

$$[G : H] := |({}^{G}\!/_{H})_L|.$$

In particular, $[G : H] = 2$ means that there are exactly two left cosets in G. Since $eH = H$ is always one coset, the partition consists of H and exactly one other disjoint coset:

$$G/H = \{H, G \setminus H\}.$$

Proposition 9.10 (*Subgroups of Index 2 Are Normal*): Suppose $H \subset G$ is a subgroup of index $[G : H] = 2$. Then H is a normal subgroup of G.

Spoken Transcript [00:11:22 - 00:13:25] Prof. Rahul Pandharipande

Okay, so, we have to prove this result. The proof [writes: Proof:] is: let x in G be any element [writes: let $x \in G$]... I already used x there, so maybe let $y \in G$ be any element [corrects x to y, writes: let $y \in G$], and we consider the one possibility. The first possibility is y is in N [writes: • $y \in N$]. Then, of course, $yNy^{-1} = N$ [writes: $\implies yNy^{-1} = N$], because when you multiply by y, it's already in N, so it doesn't get out of N, all right?

Then suppose the second possibility is y is not in N [writes: • $y \notin N$]. Then I consider the left coset yN [writes: then I consider yN], and I consider the right coset Ny [writes: and I consider Ny].

This coset is not equal to N [writes: $yN \neq N$] because y is not in N, and this one's not equal to N [writes: $Ny \neq N$].

Although it's index 2, so that means that both of these things are all the things that are not in N, so they have to be equal. The index 2 hypothesis implies that both the left coset is the other one from N and the right coset is the other one; but there's only two, so they're the same set of elements. This equals this set of elements: [writes: Index 2 $\implies yN = Ny$].

And this actually implies it's stable under conjugation: [writes: $\implies yNy^{-1} = N$].

So, you can check this. This proof is written with some notational convenience. It's correct, but you should go check through it carefully. If it's index 2, the point is there's only one other coset besides the coset of the neutral element. And if they're equal, that means it stabilizes.

Okay. So, that's one aspect I wanted to emphasize.

Proof of Proposition 9.10

Let $y \in G$. To prove that N is normal in G, we show that $yNy^{-1} = N$ (or equivalently $yN = Ny$). We consider the two disjoint cases for y:

Case 1 ($y \in N$): Since $y \in N$ and N is a subgroup, multiplying N on the left by y and on the right by y^{-1} produces elements entirely within N. Thus:

$$yNy^{-1} = N.$$

Case 2 ($y \notin N$): Consider the left coset yN and the right coset Ny. Since $y \notin N$, neither coset is equal to the subgroup N:

$$yN \neq N \quad \text{and} \quad Ny \neq N.$$

By the hypothesis $[G : N] = 2$, G is partitioned by left cosets into exactly two disjoint subsets: N and $G \setminus N$. Because yN is disjoint from N, it must be the complement:

$$yN = G \setminus N.$$

Similarly, the right cosets partition G into N and $G \setminus N$, so:

$$Ny = G \setminus N.$$

Comparing the two identities yields:

$$yN = Ny.$$

Multiplying by y^{-1} on the right, we conclude:

$$yNy^{-1} = N.$$

Thus, $yNy^{-1} = N$ for every $y \in G$, proving that N is normal in G. **Q.E.D.**

Notation: Variable Switch from H to N

While the lecturer initially stated Proposition 9.10 using the letter H on the upper board, during the proof on the lower board he switches to N to reinforce the convention introduced moments earlier that N denotes a normal subgroup.

The Correspondence Theorem for Groups

Review: The First Isomorphism Theorem

Spoken Transcript [00:13:25 - 00:15:04] Prof. Rahul Pandharipande

The other thing is, there's one more thing about the isomorphism theorems. The next thing we proved was that if we have a homomorphism from one group to another [writes: If $f_1 : G_1 \to G_2$], then we know that the kernel [writes: $G_1/\ker(f_1)$] — this is not only a subgroup, but a normal subgroup, so this quotient exists as a group; this is a group quotient [writes: group quot. under $G_1/\ker(f_1)$].

And that first isomorphism theorem says that this group quotient is isomorphic to the image of f_1 [writes: $\cong \mathrm{Im}(f_1)$], which is a subgroup, all right? That's just repeating.

And in the context of rings, we proved this, but we proved another property about quotient rings.

So, it's a second property I want to emphasize also for groups. I'm not going to prove it, because the proof is just identical. This is one property we proved for both groups and rings. Then there was another property we proved for rings, which I remind you of here, which is: if R is a ring and I is an ideal [writes: $I \subset R$ id.], then we had this quotient map R goes to R mod I [writes: $R \xrightarrow{q} R/I$].

And for our groups, we have here N in G and N is a normal subgroup [writes: $N \subset G$, N norm. sub.], and we have a quotient map G goes to G mod N [writes: $G \xrightarrow{q} G/N$]. This is the quotient.

And the part I wanted to remind you of, which we didn't discuss — which we did discuss for rings, and it's entirely parallel here, and it's really important —, is this correspondence.

Theorem 9.11 (*First Isomorphism Theorem for Groups*): Let $f_1 \colon G_1 \to G_2$ be a group homomorphism. Then the kernel $\ker(f_1)$ is a normal subgroup of G_1, and the quotient group ${}^{G_1}\!/_{\ker(f_1)}$ is isomorphic to the image $\mathrm{Im}(f_1)$:

$$ {}^{G_1}\!/_{\ker(f_1)} \cong \mathrm{Im}(f_1). \tag{9.4}$$

The Lattice Correspondence for Rings and Groups

Spoken Transcript [00:15:04 - 00:17:00] Prof. Rahul Pandharipande

We put here the set of ideals of R which contain I [writes: $\{$set of ideals of R which contain $I\}$], and the result was that there's a bijection here [draws double-headed arrow labeled: biject.], and this is the set of ideals of R mod I [writes: $\{$set of ideals of ${}^{R}\!/_{I}\}$]. Okay, so, do you remember this?

And the idea here is if we start on this side, we have an ideal J which contains I in R [writes "$I \supset J \supset R$", a slip for $I \subset J \subset R$], the motion here is you just take $q(J)$ [writes: $\longmapsto q(J)$]. On the other hand, if you have some $\widehat{J}$ which is an ideal of R mod I, the motion back is you take $q^{-1}(\widehat{J})$ [writes: $q^{-1}(\widehat{J}) \longleftarrow \widehat{J} \subset {}^{R}\!/_{I}$]. And this was a bijection. We proved this — actually, we proved this kind of carefully.

So, I hope you remember that. And it is the absolute natural way to think about and to use backwards and forwards both ways. If you have a quotient object and you want to think about the ideals, more or less, one way or the other, you have to use this correspondence. It'll come up naturally.

And the exact parallel result holds for groups with subgroups. It's just literally the same, and the proof is the same, so I'm not going to give the proof here, but I will at least state it.

Okay, so you can imagine what I'm going to say here. On the left side — so here's the G side, and that's the quotient goes to G mod N [writes: $G \xrightarrow{q} {}^{G}\!/_{N}$] —, on this side is the set of subgroups of G which contain N [writes: $\{$set of subgroups of G which contain $N\}$], and this is going to be in bijective correspondence [draws double-headed arrow labeled: biject.] with here the set of all subgroups of G mod N [writes: $\{$set of all subgroups of ${}^{G}\!/_{N}\}$], and the...

Spoken Transcript [00:17:00 - 00:18:50] Prof. Rahul Pandharipande

...movements are exactly the same. So, if I start with a subgroup here that contains N [writes: $N \subset S \subset G$] — so this is the subgroup —, I get to this side by just taking q of the subgroup: [writes: $\longmapsto q(S)$]. And if I start here with any subgroup [writes: $\widehat{S} \subset {}^{G}\!/_{N}$], I get back by taking q^{-1} [writes: $q^{-1}(\widehat{S}) \longleftarrow$]. And the proof that these are bijections is just exactly the same as we did before, and I encourage you to think about that.

And there's even a stronger statement here, because this is just about subgroups, so you could imagine: what happens to the normal subgroups? That's also reasonable. This is just saying this set is bijective to that set. But inside all subgroups of G which contain N are those which are normal. And inside here is also distinguished those which are normal, and this bijection preserves that normality. So, I leave these two things for you to check, all right?

> Maybe I just write it: bijection here preserves normality [writes: Bijection here preserves normality].
>
> If you really check all the details here, it's some work, but it's worth it in the sense of understanding the subject. But at this level, the results are kind of parallel to the isomorphism theorems for rings.

The Correspondence Theorem for Rings and Groups

Theorem 9.12 (*Ideal Correspondence Theorem for Rings*): Let R be a ring and $I \subset R$ an ideal, with canonical projection $q\colon R \to {}^{R}\!/_{I}$. There is an inclusion-preserving bijection:

$$\{\text{ideals } J \subset R \text{ with } I \subseteq J\} \overset{\text{bijection}}{\longleftrightarrow} \left\{\text{ideals of } {}^{R}\!/_{I}\right\}$$
$$J \longmapsto q(J)$$
$$q^{-1}(\widehat{J}) \longleftarrow \widehat{J}$$

Theorem 9.13 (*Subgroup Correspondence Theorem for Groups*): Let G be a group, N a normal subgroup of G, and $q\colon G \to {}^{G}\!/_{N}$ the canonical quotient homomorphism ($g \mapsto gN$). There is an inclusion-preserving bijection:

$$\{\text{subgroups } S \le G \text{ with } N \subseteq S\} \overset{\text{bijection}}{\longleftrightarrow} \left\{\text{subgroups of } {}^{G}\!/_{N}\right\}$$
$$S \longmapsto q(S) = {}^{S}\!/_{N} \tag{9.5}$$
$$q^{-1}(\widehat{S}) \longleftarrow \widehat{S}$$

Furthermore, this bijection preserves normality:

$$S \text{ is normal in } G \iff q(S) \text{ is normal in } {}^{G}\!/_{N}. \tag{9.6}$$

Explicit Actions of the Correspondence: The explicit assignments between the subgroup lattices are:

- **Forward Direction ($S \mapsto q(S)$):** For a subgroup $S \le G$ containing N, its image is $q(S) = \{sN \mid s \in S\} = {}^{S}\!/_{N} \le {}^{G}\!/_{N}$.
- **Inverse Direction ($\widehat{S} \mapsto q^{-1}(\widehat{S})$):** For any subgroup $\widehat{S} \le {}^{G}\!/_{N}$, its preimage $q^{-1}(\widehat{S}) = \{g \in G \mid gN \in \widehat{S}\}$ is a subgroup of G containing $N = q^{-1}(\{eN\})$.
- **Normality Preservation:** Going beyond the lecture: if S is normal in G, then $q(S) = {}^{S}\!/_{N}$ is normal in ${}^{G}\!/_{N}$ by the Third Isomorphism Theorem (quotients of quotients), which establishes ${}^{({}^{G}\!/_{N})}\!/_{({}^{S}\!/_{N})} \cong {}^{G}\!/_{S}$. Conversely, preimages of normal subgroups under homomorphisms are always normal.

Groups of Prime Order and Platonic Solids

Groups of Prime Order Are Cyclic

Spoken Transcript [00:18:50 - 00:20:09] Prof. Rahul Pandharipande

Okay, then finally a comment which probably should have been made last week: suppose G is a finite group and has order p, where p is a prime [writes: Suppose G has order p, p prime].

So, it's a group of order 17, then the claim is: then G must be the cyclic group [writes: Then G must be isomorphic to $(\mathbb{Z}/p\mathbb{Z}, +, 0)$].

There's very few groups of prime order. And the proof of this is: pick any non-neutral element of G. Pick any one you want. It has to have one, because prime means two, three... you know, at least two. So, proof [writes: Proof:]: p is at least 2 [writes: $p \geq 2$], so, pick any element $x \in G$ where x is not the neutral element [writes: so pick any element $x \in G, x \neq e$].

Spoken Transcript [00:20:09 - 00:21:45] Prof. Rahul Pandharipande

You can, because it's at least 2. Then the subgroup generated by x [writes: then $\langle x \rangle \subset G$] — this subgroup has to have an order dividing the order of G. There's very few things that divide prime numbers, so it has to actually be the whole thing. It could be 1 or the prime; it can't be 1 because it has some element that's not 1 (i.e., not the neutral element e). So, this then has to have the order divide p, which means that the order of this could be 1 or it could be p [writes: has to $\langle x \rangle \mid p \implies |\langle x \rangle| = 1$].

This one's ruled out, because we have at least two elements, the one that's not the neutral element and x, so it must be this: [crosses out $|\langle x \rangle| = 1$, writes and circles $|\langle x \rangle| = p$].

So, this means that this must be the whole group [writes: $\langle x \rangle = G$], and then that's the cyclic group, okay?

Those are two pieces of luck you can encounter with groups: if you find a group that's of prime order, then you know what group it is. If you find a subgroup that is half the order of the group, then you know it's normal.

There's a whole subject about classifying groups, and the order of the group is extremely important, of course: the prime divisors of the order. And this is the first step in that direction.

It's not true, by the way, if you have a group of order $p \times q$, where p and q are prime, that it has to be cyclic, because, for example, S_3 has order 6, which is 2×3, and it's not even commutative.

Proposition 9.14 (*Groups of Prime Order*): Let G be a finite group of order p, where p is a prime number. Then G is cyclic and isomorphic to the additive group of integers modulo p:

$$G \cong (\mathbb{Z}/p\mathbb{Z}, +, 0). \tag{9.7}$$

Proof

Since p is prime, $p \geq 2$, so there exists at least one element $x \in G$ distinct from the identity e ($x \neq e$). Consider the cyclic subgroup generated by x:

$$\langle x \rangle = \{x^k \mid k \in \mathbb{Z}\} \subseteq G.$$

By Lagrange's theorem, the order of the subgroup $|\langle x \rangle|$ divides the order of the group $|G| = p$:

$$|\langle x \rangle| \mid p.$$

Since p is a prime number, its only positive divisors are 1 and p:

$$|\langle x \rangle| \in \{1, p\}.$$

Because $x \neq e$, the subgroup contains at least two distinct elements (e and x), so $|\langle x \rangle| \neq 1$. Therefore:

$$|\langle x \rangle| = p.$$

Since $\langle x \rangle$ is a subgroup of G with the exact same finite cardinality as G, it must be the entire group:

$$\langle x \rangle = G.$$

Thus, G is generated by x, so G is cyclic of order p, which implies $G \cong \mathbb{Z}/p\mathbb{Z}$.

Q.E.D.

Example 9.15 (*Non-Cyclic Group of Order pq*): The property fails for composite orders pq with distinct primes p and q. For example, the symmetric group S_3 has order $|S_3| = 6 = 2 \cdot 3$. Since S_3 is non-abelian, it is not cyclic ($S_3 \ncong \mathbb{Z}/6\mathbb{Z}$).

Rotational Symmetries of Platonic Solids and Duality

Spoken Transcript [00:21:45 - 00:24:02] Prof. Rahul Pandharipande

Okay, all right. So, then, I'm not going to say anything more about the Platonic solids, but I'll say some things, because it seems like people have never thought about these Platonic solids before. So, I'll just say a few words. We talked about this one [draws a tetrahedron on the board] and we had the rotation group. We've proved that this is A_4 [writes: A_4].

We proved that this is S_4 [draws a cube, writes: S_4], if you believe that proof, which was geometric. Then the other one is the eight-sided one [draws an octahedron], which is hard to draw... That's not actually terrible, all right? And the claim is this is also S_4 [writes: S_4]. This is obvious because the Platonic solids — it's one of their most basic properties, probably due already to the Greeks, certainly due to the Greeks — have a duality: if you have a Platonic solid, you can take the middle of every face and make that the vertices and connect them, and you get another Platonic solid.

And if you do that with the cube, if you take the midpoint of every face — there's six faces, right? — take the midpoint of every face, then you get this octahedron. Can you see that? And if you take this one and you take the midpoint of every triangle — there are eight of them —, you get back the cube. That's the duality.

So, you should think about it; it's fun to think about it if you've never seen it. You could say, *"Why doesn't this one have a friend?"* And the answer is: if you take the midpoint of all of them, you just get back the tetrahedron, so it's self-dual.

This shows that the automorphism groups, if you think about it, have to be the same, because they just tell you how to get from one automorphism to the other.

Then the other ones that are harder to draw are the 12-sided one [writes: 12-sided], the dodecahedron, and the 20-sided one [writes: 20-sided], the icosahedron. This one has a face that looks like this [draws a pentagon], and this has a triangular face [draws a triangle]. Then they have lots more faces.

These things are also dual. That's kind of fun to think about, all right? I recommend thinking about that. They're also dual, and so their rotational groups are the same. And what is the rotational group? Well, it was an exercise, or maybe it was on the quiz, that this rotational group has order 60 [writes: Rot group $\implies$ order 60], and the only interesting group you know of order 60...

Spoken Transcript [00:24:02 - 00:25:17] Prof. Rahul Pandharipande

...is the alternating group inside S_5 [writes: $A_5 \subset S_5$], because this has order 120 and that's exactly half, so it's 60 [writes: 60 under A_5 and 120 under S_5], and it's a very nice geometric fact that this is A_5 [writes: $\cong A_5$].

And the proof of this requires a little bit of work, so I recommend doing it. The best way to do this is go get yourself a model of one of these and think about it. And if you look hard enough — you know, the way this was proven was finding these four long diagonals and the rotations permuting them, and that giving an isomorphism. Here it's trickier: you have to find five objects in here, and it turns out that the dodecahedron has five inscribed cubes. When you rotate it, you swap around these five inscribed cubes, and you have to do an even permutation; that's an isomorphism. It's incredibly beautiful.

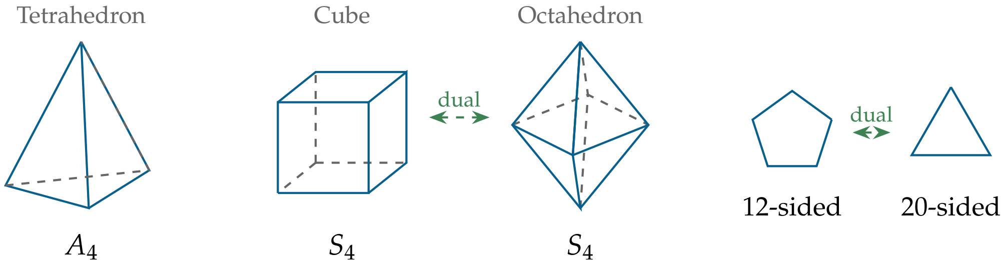
I recommend, if you're interested in it, just go look. You'll find either a reference which will draw you this dodecahedron and the five inscribed cubes, or you can find them yourself, but you have to play with the actual model.

Okay, that's more or less all I want to say about that. And basically, what I'm saying here is that we did all of this except for the actual isomorphism to A_5, which is an interesting thing, and I leave that to you to do.

Rotational Symmetries and Duality of Platonic Solids

Theorem 9.16 (*Rotational Symmetries of the Five Platonic Solids*)**:** The orientation-preserving rotational symmetry groups $\mathrm{Rot}(P) \leq \mathrm{SO}(3)$ of the five Platonic solids are given by:

Solid	Faces	Vertices	Rotation group	Order
Tetrahedron (self-dual)	4 triangles	4	A_4	12
Cube	6 squares	8	S_4	24
Octahedron (dual: cube)	8 triangles	6	S_4	24
Dodecahedron	12 pentagons	20	A_5	60
Icosahedron (dual: dodec.)	20 triangles	12	A_5	60



$$\text{Rotational group has order } 60: \quad A_5 \leq S_5, \quad |A_5| = \frac{5!}{2} = 60 \tag{9.8}$$

$$\text{Rot}(\text{Dodecahedron}) \cong \text{Rot}(\text{Icosahedron}) \cong A_5 \tag{9.9}$$

Geometric Duality and Rotational Symmetries: Geometric duality connects Platonic solids by mapping face midpoints of one solid to the vertices of its dual:

- **Cube and Octahedron:** A cube has 6 faces and 8 vertices; connecting the centers of its 6 faces forms an octahedron. Conversely, the 8 triangular faces of an octahedron yield the 8 vertices of a cube. Any rotational symmetry preserving the cube must preserve its face midpoints, and hence acts as an orientation-preserving symmetry of the dual octahedron. Therefore:

  $$\text{Rot}(\text{Cube}) \cong \text{Rot}(\text{Octahedron}) \cong S_4.$$

  The group S_4 acts faithfully by permuting the 4 principal body diagonals connecting opposite pairs of vertices in the cube (or opposite pairs of triangular faces in the octahedron).
- **Tetrahedron (Self-Dual):** Connecting the centers of the 4 triangular faces of a regular tetrahedron yields another regular tetrahedron. The rotational group is the alternating group A_4 of order $\frac{4!}{2} = 12$, acting on the 4 vertices.
- **Dodecahedron and Icosahedron:** Connecting the midpoints of the 12 pentagonal faces of a regular dodecahedron produces an icosahedron with 12 vertices and 20 triangular faces. Their common rotational symmetry group has order 60 and is isomorphic to the alternating group A_5:

  $$\text{Rot}(\text{Dodecahedron}) \cong \text{Rot}(\text{Icosahedron}) \cong A_5.$$

  This group acts by permuting the 5 regular cubes that can be inscribed inside a dodecahedron.

Geometric Realization of the Isomorphism Rot(Dodecahedron) $\cong A_5$: To construct the isomorphism with the alternating group A_5:

(1) **Inscribed Cubes:** Exactly 5 regular cubes can be inscribed within a regular dodecahedron, using subsets of its 20 vertices (each cube uses 8 vertices, with each dodecahedral vertex shared by two cubes).

(2) **Permutation Action:** Every rotational symmetry of the dodecahedron maps the set of 20 vertices to itself, permuting the 5 inscribed cubes. This defines a homomorphism $\phi\colon \text{Rot}(\text{Dodecahedron}) \to S_5$.

(3) **Faithful and Even:** No non-trivial rotation preserves all 5 cubes simultaneously ($\ker(\phi) = \{e\}$), so the action is faithful. Furthermore, rotations correspond to even permutations, establishing an injective homomorphism into A_5.

(4) Order Comparison: Since $|\mathrm{Rot}(\text{Dodecahedron})| = 60$ and $|A_5| = 60$, the map ϕ is an isomorphism.

Symmetries of Polygons and Dihedral Groups

Rotations and the Dihedral Group D_{2n}

Spoken Transcript [00:25:17 - 00:27:10] Prof. Rahul Pandharipande

Some more classical symmetry groups that come from objects are if I take an n-gon in the plane... [draws a polygon] I don't know, that's supposed to be an n-gon, right? [writes: n-gon] Like, for example, the triangle: I can draw a solid triangle, right? [draws an equilateral triangle]

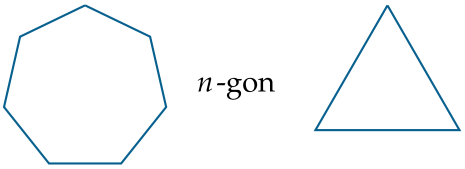
I can rotate it, just rotate it in the plane, just its rotational symmetries. These beautiful Platonic solids are three-dimensional objects, but now I'm talking about things in the plane.

So, I can just rotate it. If I just look at the rotational symmetries of this n-gon [writes: Rot(n-gon)], that's not that interesting, because that's just a cyclic group. This is isomorphic to $\mathbb{Z}/n\mathbb{Z}$ [writes: $\cong \mathbb{Z}/n\mathbb{Z}$], because I just rotate by that angle $2\pi/n$. Is that clear? Like here, you just have three rotations; that's a cyclic group.

That's not interesting. Much more interesting is what's called the dihedral group [writes: Dihedral]. And the usual indexing of this is dihedral D_{2n} [writes: D_{2n}], and this is now just a name: the dihedral group [writes: Dihedral group] of order $2n$ [writes: of order $2n$].

And what's the definition? The definition is: this is the symmetries of the n-gon [writes: Def:] [writes: Symmetries of n-gon] — the same one there — obtained by both rotations [writes: obtained by both rotations] and flips [writes: and flips].

Definition of the Dihedral Group



Definition 9.17 (*Rotational Symmetries of a Regular n-gon*): Let $P_n \subset \mathbb{R}^2$ be a regular n-gon centered at the origin. The group of purely rotational symmetries in the plane is cyclic of order n:

$$\mathrm{Rot}(P_n) = \langle \rho_{2\pi/n} \rangle \cong (\mathbb{Z}/n\mathbb{Z}, +, 0), \tag{9.10}$$

generated by the counter-clockwise rotation by angle $\frac{2\pi}{n}$.

Definition 9.18 (*Dihedral Group D_{2n}*): The *"dihedral group"* D_{2n} is the full symmetry group of a regular n-gon in $\mathbb{R}^2$, generated by both planar rotations and reflectional flips across lines of symmetry:

$$D_{2n} := \mathrm{Sym}(P_n) = \langle r, s \mid r^n = e,\ s^2 = e,\ srs = r^{-1} \rangle. \tag{9.11}$$

> The group D_{2n} has order $2n$, comprising n rotations and n reflections.

Notation: D_{2n} vs. D_n

Conventions in the literature differ regarding the subscript of dihedral groups:

- Many modern authors write D_n for the dihedral group of order $2n$, indexing by the number of vertices n.
- The convention used on the blackboard in this lecture explicitly indices by the group order: D_{2n} denotes the dihedral group of order $2n$.

We strictly follow the lecture's notation D_{2n} throughout these notes.

The Dihedral Group D_6 of the Equilateral Triangle

Spoken Transcript [00:27:10 - 00:28:40] Prof. Rahul Pandharipande

Okay, so, we can just think about this for a moment, but I invite you to think about... this is more interesting. In the language of the class, it's a new group, which is to say we didn't have a name for it before.

The flips are: you draw a line in the plane — I'll just show it on the board — and you flip it along that line. And if you do that, you get twice as many elements in the group, because it has order $2n$.

For example, if you have the triangle, [draws triangle] there are three rotations: the rotation where you don't do anything, the rotation by 120°, and one by 240°. But there are also three flips: you can flip it here [draws vertical axis].

If you flip it, you're really changing it from the view of the plane, meaning that what was on the upside of the plane is now on the downside. Is that clear? So, then I can also flip it here [draws second axis], and I can flip it here [draws third axis].

Well, this should have been like...

Spoken Transcript [00:28:40 - 00:29:45] Prof. Rahul Pandharipande

So, there are six rotations and flips. I can rotate it and I can flip it: I can rotate it a bit, then flip it, then rotate it back, etc.

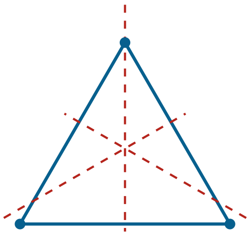
If it's not obvious to you, you should first think: is it really a group? Meaning that if I do some flips and rotations and flips and rotations, I don't get any more than the six elements. That's true. Is that obvious to you? It's about how does a flip commute with a rotation, so you should think about that.

And so, if you believe that, the order of this group — well, also by definition — this is D_6 [writes: D_6] and has order 6 [writes: $|D_6| = 6$].

And which group do you think this is?

Well, it's going to be S_3. There's not so many groups of order 6, and this D_6, it happens, is isomorphic to the symmetric group S_3 [writes: $D_6 \cong S_3$] by looking at where the vertices go. When you rotate, you could only get a 3-cycle. But now when you flip, you could also get the 2-cycles. So, you get the whole group.

Symmetries of the Equilateral Triangle and $D_6 \cong S_3$



Proposition 9.19 (*Isomorphism $D_6 \cong S_3$*): The dihedral group D_6 of the equilateral triangle has order 6 and is isomorphic to the symmetric group on three letters:

$$D_6 \cong S_3. \tag{9.12}$$

Proof

Label the three vertices of the equilateral triangle as $\{1, 2, 3\}$. Any symmetry $g \in D_6$ is an isometry of the plane mapping the set of vertices into itself, yielding an injective homomorphism $\Psi\colon D_6 \to S_3$:

- The 3 planar rotations by $0°, 120°, 240°$ correspond to the identity and the two 3-cycles $(1\,2\,3)$ and $(1\,3\,2)$ in $A_3 \leq S_3$.
- The 3 axial reflections (flips) each fix one vertex and swap the other two, corresponding to the three 2-cycles (transpositions) $(2\,3)$, $(1\,3)$, and $(1\,2)$.

Since $|D_6| = 6 = |S_3|$ and Ψ is injective, it is an isomorphism. **Q.E.D.**

The Dihedral Group D_8 of the Square

Spoken Transcript [00:29:45 - 00:30:40] Prof. Rahul Pandharipande

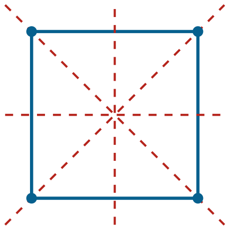
But that's not true for the next case, meaning the next case is the square [draws a square]. This is now D_8 [writes: D_8], and it has of course four rotations, and it has 1, 2, 3, 4 flips [draws the four reflection axes: horizontal, vertical, and two diagonals]. You can flip in any one of these four axes.

Anyway, that's the dihedral group. And this one now is not S_4, because S_4 has order 24 and this only has order 8.

But that's a group. You can play with it; it's another standard classical group. So, I leave that for you to think about.

That's kind of fun if you've never thought about it before. In all of these things, if you're interested in the symmetries of this, you should go make a model and look at it.

Symmetries of the Square and D_8



D_8

Observation 9.20 (*D_8 Is a Proper Subgroup of S_4*): For $n = 4$, the dihedral group D_8 has order:

$$|D_8| = 2 \cdot 4 = 8,$$

consisting of 4 rotations ($0^\circ, 90^\circ, 180^\circ, 270^\circ$) and 4 reflections (2 edge bisectors and 2 vertex diagonals). While D_8 embeds faithfully into the symmetric group on the 4 vertices S_4, it is a proper subgroup since $|S_4| = 4! = 24$:

$$D_8 \lneq S_4 \quad (\text{index } [S_4 : D_8] = 3).$$

The Rubik's Cube as a Group Theory Exercise

Spoken Transcript [00:30:40 - 00:32:40] Prof. Rahul Pandharipande

Actually, you know, a really good exercise — I remember, I forgot about this, I used to say this — a really good exercise in group theory is: go get the Rubik's Cube and solve it. That's just like straight group theory. You know, if you've never done that, just go get the Rubik's Cube and solve it.

The way you solve it — I mean, you can solve it with education or without education, but now it's too late: it's only for you with education. The whole thing is working in a certain kind of group, and the main issue with the group — do you know what I'm talking about, the Rubik's Cube?

The main issue with the Rubik's Cube is understanding the commutators. But that's actually a really good exercise: just go buy a Rubik's Cube and solve it. If you can do that, it means you understood a little bit of group theory.

***Insight:* The Rubik's Cube as a Group Theory Exercise**

The Rubik's Cube group $\mathcal{R}$ is generated by the six basic 90° face rotations $\{F, B, U, D, L, R\}$. A fundamental problem in solving the cube is moving a small set of pieces (e.g., swapping two edge pieces or twisting two corner pieces) without scrambling the rest of the puzzle. Mathematically, this is achieved via *"commutators"*:

$$[g, h] := ghg^{-1}h^{-1}.$$

If the support of an operation g and the support of h overlap in only a small subset of the cube, the commutator $[g, h]$ acts non-trivially **only** on that small intersection and acts as the identity everywhere else.

Testing Group Isomorphisms: Classical Questions

Is $(\mathbb{Q},+,0)$ Isomorphic to $(\mathbb{Z},+,0)$?

Spoken Transcript [00:31:00 - 00:32:00] Prof. Rahul Pandharipande

So, now some questions that are kind of embarrassing questions to you. We can ask some questions. Question 1 [writes: Questions: / Q1] — and it's important kind of math culture to know the answers to these questions:

Is $\mathbb{Q}$ as an additive group — so everything here is an additive group, is $\mathbb{Q}$... let me just write it: [writes: is $(\mathbb{Q},+,0) \overset{?}{\cong} (\mathbb{Z},+,0)$] — is $\mathbb{Q}$ isomorphic to $\mathbb{Z}$?

That's kind of an embarrassing question. What's the answer to this?

Well, they're both additive groups, they're both commutative — if one were not commutative, then they couldn't be isomorphic —, but they're both commutative, they're both countable.

So, what's the answer? What do you think?

Student Answer

I would say yes, because you could just...

Spoken Transcript [00:32:00 - 00:33:00] Prof. Rahul Pandharipande

Aha, yeah, so this is why it's embarrassing, because that answer is, while slightly convincing, completely wrong.

The simplest way to see it's wrong is: this group has a very special property. It's called a divisible group [writes: Divisible group, with an arrow to $\mathbb{Q}$].

A divisible group has the property that if I take any element of it, and I take any number — for example, 17 [writes: $x \in \mathbb{Q}$, $17 \in \mathrm{Num}$] is a number —, then I can solve this equation: $17 \cdot y = x$ [writes: Sol. $17 \cdot y = x$, checks y]. I can solve it.

And this makes sense in a group, because this 17, remember, is just $y + y + \cdots + y$ 17 times. So, this is an equation in the group, and I can always solve it for any number in $\mathbb{Q}$. You agree with that?

I cannot do that here [points to $\mathbb{Z}$].

So, this actually proves they're not isomorphic.

Question 1: Divisibility of the Additive Group of Rationals

Question 9.21 (*Additive Isomorphism between $\mathbb{Q}$ and $\mathbb{Z}$*)**:** Is the additive group of rational numbers $(\mathbb{Q},+,0)$ isomorphic to the additive group of integers $(\mathbb{Z},+,0)$?

$$(\mathbb{Q},+,0) \overset{?}{\cong} (\mathbb{Z},+,0) \tag{9.13}$$

Definition 9.22 (*Divisible Group*)**:** An abelian group $(G,+,0)$ is called *"divisible"* if for every element $x \in G$ and every non-zero integer $n \in \mathbb{Z}_{\neq 0}$, the equation:

$$n \cdot y = \underbrace{y + y + \cdots + y}_{n \text{ times}} = x \tag{9.14}$$

has at least one solution $y \in G$.

Answer 9.23 (*to Question 9.21*): No, $(\mathbb{Q}, +, 0) \ncong (\mathbb{Z}, +, 0)$.

Proof of Non-Isomorphism via Divisibility

The property of being a divisible group is an isomorphism invariant.

- $(\mathbb{Q}, +, 0)$ is divisible: Given any $x \in \mathbb{Q}$ and $n \in \mathbb{Z}_{\neq 0}$, the element $y := \frac{x}{n} \in \mathbb{Q}$ solves $n \cdot y = x$.
- $(\mathbb{Z}, +, 0)$ is not divisible: For $x = 1 \in \mathbb{Z}$ and $n = 2$, there is no integer $y \in \mathbb{Z}$ satisfying $2y = 1$.

Since divisibility is preserved under group isomorphisms, $(\mathbb{Q}, +, 0) \ncong (\mathbb{Z}, +, 0)$.

Q.E.D.

Is $(\mathbb{Q}^*, \cdot, 1)$ Isomorphic to $(\mathbb{Z}, +, 0)$?

Spoken Transcript [00:33:00 - 00:33:35] Prof. Rahul Pandharipande

So, another embarrassing question is: if I take the rational numbers and I take out zero, and I look at this group [writes: $(\mathbb{Q}^*, \cdot, 1) \overset{?}{\cong} (\mathbb{Z}, +, 0)$], is this now isomorphic to $(\mathbb{Z}, +, 0)$? What about this one? This is a question [adds question mark]. So, this is the question, and the answer is no. What happened to my red? Ah, here it is [crosses out the $\cong$ of Q1 in red].

The answer is no.

So, what about this one? This is now — I take here multiplication. This is a very interesting thing to think about. Do you have an opinion about this?

Student Answer

Yeah, I think that element, like to some power, or...

Spoken Transcript [00:33:35 - 00:34:45] Prof. Rahul Pandharipande

Yeah, this is generated by one element. You could have used this argument here also. The argument here is that $\mathbb{Z}$ is actually generated by one element, this element: [writes: $\langle 1 \rangle$]. So, it's a subgroup generated by a single element.

And here, there's no element that generates everything, because you can have bigger and bigger prime divisors. You could have used this argument here also, by the way. So, it's a different argument.

The argument here is that $\mathbb{Z}$ is generated by a single element [writes: $\mathbb{Z}$ is generated by a single element, $\mathbb{Z} = \langle 1 \rangle$] — for example, $\mathbb{Z}$ is equal to the subgroup generated by 1 —, and $\mathbb{Q}^*$ is not generated by a single element [writes: $\mathbb{Q}^*$ is not generated by a single element].

And so, they can't be isomorphic. And why is $\mathbb{Q}^*$ not generated by a single element? Because if you pick your favorite single element, it'll be some rational number n/m [writes: n/m], and since it's multiplication, the generation is by powers. So, in the bottom you will never see primes that weren't already there in m. But $\mathbb{Q}$ has in principle any prime you want in the bottom.

So, that's an argument.

Spoken Transcript [00:34:45 - 00:35:58] Prof. Rahul Pandharipande

A different argument, by the way — I mean, there's many ways you can prove it; these are not hard questions, but it forces you to think about some things. Another way to prove this is that there's a neutral element here. That's the neutral element. An isomorphism would have to take that neutral element to 0. That's what isomorphisms do, right? It's confusing to think about this because this is multiplication on this side, and addition on that side. Formally there's no difficulty, but in practice it's confusing.

So, $\mathbb{Q}^*$ has a special element: it has the element -1 [writes: $(-1)^2 = 1$]. -1 has the property that if I apply the group law to itself, I get 1, right? And -1 is not 1.

So, that would mean if there were an isomorphism, it would mean that there should be here some element that if I apply the group law to itself, I get 0, which is not 0. That's not true, right? So, this is another proof.

This is a way you can prove things are different: by looking for special elements in the structure. I don't know if you understood this proof, but it just said the existence of -1. There's no element here that can play that role.

Question 2: Multiplicative Group of Rationals vs. Integers

Question 9.24 (*Multiplicative Isomorphism between $\mathbb{Q}^*$ and $\mathbb{Z}$*)**:** Is the multiplicative group of non-zero rational numbers $(\mathbb{Q}^*, \cdot, 1)$ isomorphic to the additive group of integers $(\mathbb{Z}, +, 0)$?

$$(\mathbb{Q}^*, \cdot, 1) \overset{?}{\cong} (\mathbb{Z}, +, 0) \tag{9.15}$$

Answer 9.25 (*to Question 9.24*)**:** **No**, $(\mathbb{Q}^*, \cdot, 1) \ncong (\mathbb{Z}, +, 0)$.

First Proof: Invariance of Cyclic / Singly Generated Property

The additive group $(\mathbb{Z}, +, 0)$ is cyclic, generated by a single element:

$$\mathbb{Z} = \langle 1 \rangle.$$

If $\phi\colon \mathbb{Z} \to \mathbb{Q}^*$ were an isomorphism, $\mathbb{Q}^* = \langle \phi(1) \rangle$ would also be cyclic. Let $g = \frac{n}{m} \in \mathbb{Q}^*$ with $\gcd(n, m) = 1$ be any candidate generator. Then:

$$\langle g \rangle = \left\{ \left(\frac{n}{m}\right)^k \,\middle|\, k \in \mathbb{Z} \right\} = \left\{ \frac{n^k}{m^k} \,\middle|\, k \in \mathbb{Z} \right\}.$$

Any prime p appearing in the prime factorization of an element in $\langle g \rangle$ must divide either n or m. Since there are infinitely many primes, choose a prime $p \nmid nm$. Then $p \in \mathbb{Q}^*$ cannot be written as g^k for any $k \in \mathbb{Z}$. Thus, $\mathbb{Q}^*$ is not cyclic, so $(\mathbb{Q}^*, \cdot, 1) \ncong (\mathbb{Z}, +, 0)$. **Q.E.D.**

Second Proof: Invariance of Torsion Elements

In $(\mathbb{Q}^*, \cdot, 1)$, the element -1 is a non-trivial element of finite order 2:

$$(-1) \neq 1 \quad \text{and} \quad (-1)^2 = (-1) \cdot (-1) = 1.$$

If an isomorphism $\phi\colon \mathbb{Q}^* \to \mathbb{Z}$ existed, the element $a := \phi(-1) \in \mathbb{Z}$ would satisfy:

$$2a = a + a = \phi((-1)^2) = \phi(1) = 0, \quad \text{with } a \neq 0.$$

However, the only integer satisfying $2a = 0$ is $a = 0$. This contradiction demonstrates that $(\mathbb{Q}^*, \cdot, 1) \not\cong (\mathbb{Z}, +, 0)$. **Q.E.D.**

Direct Sums of Integers and Invariance of Dimension

Spoken Transcript [00:35:58 - 00:36:40] Prof. Rahul Pandharipande

Okay, what about this one? Is $\mathbb{Z}$ isomorphic to $\mathbb{Z} \oplus \mathbb{Z}$? [writes: $\mathbb{Z} \overset{?}{\cong} \mathbb{Z} \oplus \mathbb{Z}$]

What's your view on that?

Well, we had a bunch of ways to check, and you can use some of them now.

Yeah?

Student Answer

$\mathbb{Z}$ is still generated by one element, but...

Spoken Transcript [00:36:40 - 00:37:15] Prof. Rahul Pandharipande

Yeah, this one is generated by 1 [writes: $\langle 1 \rangle$ under $\mathbb{Z}$], and this one is not singly generated [writes: not singly gen. under $\mathbb{Z} \oplus \mathbb{Z}$]. Because you have to pick two numbers and they have some ratio, and, you know, not everybody has that ratio, right? Is that clear? So, this is also not [crosses through $\cong$].

So, a trickier question...

The answer to this is no. Sadly, all the answers to all these questions are no, it seems.

How about if I take $\mathbb{Z}$ to some powers, like 17 — direct sum of 17 [writes: $\mathbb{Z}^{17}$] —: is this isomorphic to $\mathbb{Z}$ to something else, like 691? [writes: $\mathbb{Z}^{17} \overset{?}{\cong} \mathbb{Z}^{691}$] How about that? That's kind of unlikely. Is that true or not? I mean, this argument is not going to work, because neither of them are singly generated.

So, you need a new argument. Yeah?

Student Answer

The first one is generated by 17 elements, and the other one by 691.

Spoken Transcript [00:37:15 - 00:38:35] Prof. Rahul Pandharipande

You're right. So, if it's true, the first one is generated by 17 elements. But how do you know this one's not generated by 17 elements?

There are many ways to see this, and the argument is correct. Let me try to give the simplest way.

The argument is that, roughly speaking, there's a 17 on this side, which is roughly speaking how many generators you need, and this is 691; you can't somehow match those two. And the difficulty... you know, if these were vector spaces, there's an invariant of a vector space, which is the dimension. You proved — at least I hope you proved — if you have two vector spaces over a field, the only way they can be isomorphic as vector spaces is if they have the same dimension. And it's kind of the argument you want to use, but you can't, because these are not vector spaces. So, the trick is: you can make them vector spaces.

Spoken Transcript [00:38:35 - 00:39:55] Prof. Rahul Pandharipande

We can look: if you have an abelian group A [writes: A], we can consider $2A$ [writes: $2A \subset A$]. That's the subgroup of all things you get by taking twice things in A, all right? So, the argument is: if we had an isomorphism [writes: If $\mathbb{Z}^{17} \cong \mathbb{Z}^{691}$], then this would imply that there would be an isomorphism between 2 × 17 and 2 × 691 [writes: $\implies \mathbb{Z}^{17}/2\mathbb{Z}^{17} \cong \mathbb{Z}^{691}/2\mathbb{Z}^{691}$]. And now, what are these? This one is actually $(\mathbb{Z}/2\mathbb{Z})^{17}$ [writes: $(\mathbb{Z}/2\mathbb{Z})^{17}$], and this on the right side is $(\mathbb{Z}/2\mathbb{Z})^{691}$ [writes: $(\mathbb{Z}/2\mathbb{Z})^{691}$]. Those are what the quotient groups are. And now these are vector spaces! And then your argument can apply. So, then these would have to be isomorphic like this, and then you have to make an argument that this would imply these vector spaces are isomorphic. $\mathbb{Z}/2\mathbb{Z}$ is a field, that's the point: actual vector spaces [writes: actual vector spaces / dim as v.s. over $\mathbb{Z}/2\mathbb{Z}$]. And then you could say that's impossible because of the theory of dimension in linear algebra [writes: impossible because of dim in lin. alg.].

So, this is the way to make your argument complete, because then you really have a field, and then you can really talk about dimension. So, the consequence of this is they're never isomorphic unless these numbers are the same.

Questions 3 and 4: Products of Free Abelian Groups

Question 9.26 (*Isomorphism between $\mathbb{Z}$ and $\mathbb{Z} \oplus \mathbb{Z}$*): Is the additive group of integers $(\mathbb{Z}, +, 0)$ isomorphic to the direct sum $(\mathbb{Z} \oplus \mathbb{Z}, +, (0,0))$?

$$\mathbb{Z} \overset{?}{\cong} \mathbb{Z} \oplus \mathbb{Z} \tag{9.16}$$

Answer 9.27 (*to Question 9.26*): No, $\mathbb{Z} \ncong \mathbb{Z} \oplus \mathbb{Z}$.

Proof

$\mathbb{Z}$ is cyclic (generated by 1), whereas $\mathbb{Z} \oplus \mathbb{Z}$ requires at least two generators. Any single element $(a, b) \in \mathbb{Z} \oplus \mathbb{Z}$ generates a cyclic subgroup $\langle (a,b) \rangle = \{(ka, kb) \mid k \in \mathbb{Z}\}$, which lies on a single line in $\mathbb{Z}^2$ and cannot span the full lattice. **Q.E.D.**

Question 9.28 (*Isomorphism between $\mathbb{Z}^{17}$ and $\mathbb{Z}^{691}$*): Let $k, \ell \in \mathbb{N}_{>0}$ with $k \neq \ell$ (for instance $k = 17$ and $\ell = 691$). Is:

$$\mathbb{Z}^{17} \overset{?}{\cong} \mathbb{Z}^{691} \tag{9.17}$$

Answer 9.29 (*to Question 9.28*): No, $\mathbb{Z}^{17} \ncong \mathbb{Z}^{691}$.

Proof via Reduction Modulo 2 and Vector Space Dimension

Let A and B be abelian groups. Any group isomorphism $\phi\colon A \xrightarrow{\cong} B$ maps $2A$ onto $2B$, because $\phi(2a) = \phi(a+a) = \phi(a) + \phi(a) = 2\phi(a)$. Consequently, ϕ descends to a group isomorphism of quotients:

$$A/_{2A} \cong B/_{2B}. \tag{9.18}$$

For $A = \mathbb{Z}^{17}$ and $B = \mathbb{Z}^{691}$, we evaluate the quotients:

$$\mathbb{Z}^{17}/_{2\mathbb{Z}^{17}} \cong (\mathbb{Z}/2\mathbb{Z})^{17} \quad \text{and} \quad \mathbb{Z}^{691}/_{2\mathbb{Z}^{691}} \cong (\mathbb{Z}/2\mathbb{Z})^{691}.$$

The field $\mathbb{F}_2 := \mathbb{Z}/2\mathbb{Z}$ has characteristic 2, making $(\mathbb{Z}/2\mathbb{Z})^{17}$ and $(\mathbb{Z}/2\mathbb{Z})^{691}$ into vector spaces over $\mathbb{F}_2$. Any group isomorphism between them is automatically $\mathbb{F}_2$-linear:

$$\phi(0 \cdot v) = 0, \quad \phi(1 \cdot v) = \phi(v) = 1 \cdot \phi(v).$$

By the fundamental theorem of linear algebra, two finite-dimensional vector spaces over the same field are isomorphic if and only if they possess identical dimensions. However:

$$\dim_{\mathbb{F}_2}\left((\mathbb{Z}/2\mathbb{Z})^{17}\right) = 17 \neq 691 = \dim_{\mathbb{F}_2}\left((\mathbb{Z}/2\mathbb{Z})^{691}\right).$$

Thus, $(\mathbb{Z}/2\mathbb{Z})^{17} \ncong (\mathbb{Z}/2\mathbb{Z})^{691}$, which implies $\mathbb{Z}^{17} \ncong \mathbb{Z}^{691}$. In general, $\mathbb{Z}^k \cong \mathbb{Z}^\ell \iff k = \ell$. **Q.E.D.**

Is $(\mathbb{R}, +, 0)$ Isomorphic to $(\mathbb{C}, +, 0)$?

Spoken Transcript [00:40:00 - 00:40:12] Prof. Rahul Pandharipande

Okay, so, let's see. How about this question: are $\mathbb{R}$ and the complex numbers isomorphic? [writes: is $(\mathbb{R}, +, 0) \overset{?}{\cong} (\mathbb{C}, +, 0)$]

What do you think about this one?

Student Answer

They are not isomorphic as real vector spaces because of dimension.

Spoken Transcript [00:40:15 - 00:40:43] Prof. Rahul Pandharipande

It's a very convincing argument, but it shows that they're not isomorphic as real vector spaces — but that's not the question!

So, it's true that these two things are not isomorphic as real vector spaces, which you gave an argument for using dimension theory. But it's not the question. I'm asking whether they're isomorphic as abelian groups!

This is a tricky question. Actually, secretly I knew all of you would be tricked by these questions, because I've done this before in my life.

So, can someone give me the right answer then?

Student Answer

No?

Spoken Transcript [00:40:46 - 00:42:18] Prof. Rahul Pandharipande

Well, no is the wrong answer.

Yes! It's right! This one happens to be yes: [writes and underlines "Yes" under $(\mathbb{R},+,0) \overset{?}{\cong} (\mathbb{C},+,0)$]."

So, why is that?

Do you understand why your answer is incorrect? Because you gave a correct answer that they're not isomorphic as real vector spaces, but real vector spaces is more structure than I'm asking for. I'm asking for just an abelian group.

The proof of this is a weird proof. The proof is: let's consider them as $\mathbb{Q}$-vector spaces [writes: Let's consider them as $\mathbb{Q}$-vector spaces].

I have the right to do that! They're both $\mathbb{Q}$-vector spaces.

Every $\mathbb{Q}$-vector space has a basis, and they both have an uncountable basis. It's actually the first uncountable ordinal... and so, they're both $\mathbb{Q}$-vector spaces with bases of the same cardinality! So, they have to be isomorphic. If they're isomorphic as $\mathbb{Q}$-vector spaces, that implies they're also isomorphic [writes: $\implies$ addit. gp. $\cong$] as additive groups.

That's a weird proof. Non-constructive, but it uses the fact that every vector space has a basis. If you don't believe that, then there's no proof of this; but if you believe that every vector space has a basis, then I view this as a $\mathbb{Q}$-vector space: it has a basis which is in bijection with the reals, the first uncountable ordinal; this one also has a basis which is in bijective correspondence with the reals. And if you have two vector spaces with bases which are in bijective correspondence, then they're isomorphic.

It's a really weird proof and it doesn't respect the $\mathbb{R}$-structure. Okay, good, that's it.

Question 5: Additive Isomorphism of Reals and Complex Numbers

Question 9.30 (*Additive Isomorphism between $\mathbb{R}$ and $\mathbb{C}$*)**:** Is the additive group of real numbers $(\mathbb{R},+,0)$ isomorphic to the additive group of complex numbers $(\mathbb{C},+,0)$?

$$(\mathbb{R},+,0) \overset{?}{\cong} (\mathbb{C},+,0) \tag{9.19}$$

Answer 9.31 (*to Question 9.30*)**: Yes**, $(\mathbb{R},+,0) \cong (\mathbb{C},+,0)$.

Proof via $\mathbb{Q}$-Vector Spaces and Hamel Bases

Both $(\mathbb{R},+,0)$ and $(\mathbb{C},+,0)$ are torsion-free, divisible abelian groups. By defining scalar multiplication for $\frac{a}{b} \in \mathbb{Q}$ and $v \in \mathbb{R}$ (respectively $\mathbb{C}$) as the unique solution y to $b \cdot y = a \cdot v$, both $\mathbb{R}$ and $\mathbb{C}$ acquire the structure of vector spaces over the field of rational numbers $\mathbb{Q}$.

By the Axiom of Choice (Zorn's Lemma), every vector space over a field possesses a basis (a Hamel basis):

- Let $\mathcal{B}_{\mathbb{R}}$ be a $\mathbb{Q}$-basis of $\mathbb{R}$.
- Let $\mathcal{B}_{\mathbb{C}}$ be a $\mathbb{Q}$-basis of $\mathbb{C}$.

We determine the cardinalities of these bases. Since $\mathbb{Q}$ is countable ($|\mathbb{Q}| = \aleph_0$) and every element of a vector space is a finite $\mathbb{Q}$-linear combination of basis vectors:

$$|\mathbb{R}| = \max\{|\mathbb{Q}|, |\mathcal{B}_{\mathbb{R}}|\} = |\mathcal{B}_{\mathbb{R}}|.$$

Because $|\mathbb{R}| = 2^{\aleph_0} = \mathfrak{c}$ (the cardinality of the continuum), it follows that:

$$|\mathcal{B}_{\mathbb{R}}| = \mathfrak{c}.$$

Similarly, as a set, $|\mathbb{C}| = |\mathbb{R}^2| = \mathfrak{c} \cdot \mathfrak{c} = \mathfrak{c}$, so:

$$|\mathcal{B}_{\mathbb{C}}| = \mathfrak{c}.$$

Since $|\mathcal{B}_{\mathbb{R}}| = |\mathcal{B}_{\mathbb{C}}| = \mathfrak{c}$, there exists a bijection between the two bases:

$$\psi \colon \mathcal{B}_{\mathbb{R}} \xrightarrow{\cong} \mathcal{B}_{\mathbb{C}}.$$

Extending ψ by $\mathbb{Q}$-linearity yields an isomorphism of $\mathbb{Q}$-vector spaces:

$$\Psi \colon \mathbb{R} \xrightarrow{\cong} \mathbb{C}.$$

In particular, Ψ preserves addition:

$$\Psi(x+y) = \Psi(x) + \Psi(y) \quad \forall x, y \in \mathbb{R},$$

so Ψ is an isomorphism of the underlying additive abelian groups:

$$(\mathbb{R}, +, 0) \cong (\mathbb{C}, +, 0).$$

Q.E.D.

Beyond the Lecture: Non-Constructivity and the Axiom of Choice

The existence of an additive group isomorphism $\mathbb{R} \cong \mathbb{C}$ depends strictly on the Axiom of Choice (AC). Without AC, it is consistent with ZF set theory that every $\mathbb{Q}$-linear map $\mathbb{R} \to \mathbb{R}$ is Lebesgue measurable (which forces all homomorphisms to be of the form $x \mapsto cx$), in which case $\mathbb{R} \ncong \mathbb{C}$ as additive groups.

The Structure of the Multiplicative Group $(\mathbb{Q}^*, \cdot, 1)$

Spoken Transcript [00:42:21 - 00:43:32] Prof. Rahul Pandharipande

[raises the middle chalkboard, revealing a fresh clean board] Okay, so, the last thing I wanted to point out was — this is something I leave for an exercise — is that there's actually a nice understanding of what this is: that's the rational numbers without zero, and multiplication, this group: [writes: $(\mathbb{Q}^*, \cdot, 1)$]. And at the end of the last session, we proved that this was not the integers, but this is what it is. So, this is actually something true, as opposed to all the false things I was mostly writing there: this is $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}^\infty$ [writes: $\cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}^\infty$].

I haven't even defined these terms, but I leave this as an exercise for you [draws a box around $(\mathbb{Q}^*, \cdot, 1) \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}^\infty$], in the theme of thinking for yourself.

So, the exercise [writes: Ex: Understand what is meant by $\mathbb{Z}^\infty$ and prove this.]: understand what is meant by $\mathbb{Z}^\infty$, and prove this. And the hint is: it only uses high school mathematics. All right? But that's an actual answer.

The Structure of the Multiplicative Group of Rationals

Chalkboard Box: Structure of $\mathbb{Q}^*$

$$(\mathbb{Q}^*, \cdot, 1) \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}^\infty \tag{9.20}$$

Exercise 9.32 (*Structure Theorem for $\mathbb{Q}^*$*): Understand what is meant by the notation $\mathbb{Z}^\infty$ in (9.20) and prove that:

$$(\mathbb{Q}^*, \cdot, 1) \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}^\infty.$$

Hint: The argument only uses high school mathematics (fundamental theorem of arithmetic).

Proof of Exercise 9.32 (beyond the lecture)

To interpret and prove (9.20):

Meaning of $\mathbb{Z}^\infty$: The symbol $\mathbb{Z}^\infty$ denotes the direct sum of countably infinitely many copies of $\mathbb{Z}$ (one for each prime $p \in \mathbb{P} = \{2, 3, 5, 7, 11, \dots\}$):

$$\mathbb{Z}^\infty := \bigoplus_{p \in \mathbb{P}} \mathbb{Z} = \left\{ (v_p)_{p \in \mathbb{P}} \in \mathbb{Z}^{\mathbb{P}} \;\middle|\; v_p = 0 \text{ for all but finitely many } p \right\}.$$

Unique Prime Factorization: By the Fundamental Theorem of Arithmetic, every non-zero rational number $q \in \mathbb{Q}^*$ can be factored uniquely as:

$$q = (-1)^s \prod_{p \in \mathbb{P}} p^{v_p(q)},$$

where $s \in \{0, 1\} \cong \mathbb{Z}/2\mathbb{Z}$ determines the sign ($s = 0$ if $q > 0$ and $s = 1$ if $q < 0$), and each exponent $v_p(q) \in \mathbb{Z}$ is the p-adic valuation of q.

Group Isomorphism: Multiplying two rational numbers corresponds to adding the signs modulo 2 and adding the prime exponents component-wise:

$$\phi \colon \mathbb{Q}^* \longrightarrow \mathbb{Z}/2\mathbb{Z} \oplus \bigoplus_{p \in \mathbb{P}} \mathbb{Z}, \quad q \longmapsto \left(s \bmod 2,\ (v_p(q))_{p \in \mathbb{P}} \right).$$

Because prime factorization is unique, ϕ is an isomorphism of groups. **Q.E.D.**

Introduction to Group Actions

Motivation: Symmetries and Groups *"in the Wild"*

Spoken Transcript [00:43:32 - 00:44:24] Prof. Rahul Pandharipande

Okay, so, the next topic in the class is like one of the coolest topics in the class, and that's group actions [writes: Group actions].

As I said, groups come from kind of a different space in mathematics than rings, and a lot of how we treat them so far has been similar to the way we think about rings. But they actually come from a different place: they come from symmetries.

And the way this is normally found: if you meet a group that's, so to speak, not in captivity — a wild group —, it will almost always occur as a symmetry object. And this is made explicit in this notion of group action, which is another way to say: when you meet a group typically in mathematics, it will be acting.

Spoken Transcript [00:44:24 - 00:45:35] Prof. Rahul Pandharipande

So, for a group action, you need a group [writes: G group], and this could be finite or infinite. A special part of this discussion will be for finite groups, but group actions are for both finite and infinite groups, and they could be abelian or non-abelian; it's just a group.

And then you need a set, T — this is a set [writes: T set] —, this is the set on which G acts [writes: on which G acts].

And the symbol for a group acting on a set is a great symbol: it's G, T, and it's like this: [writes: $G \curvearrowright T$]. That's kind of fun to draw: that's G acting on T.

All right, so, I have to tell you what a group action is.

Formal Definition and Axioms of a Group Action

Spoken Transcript [00:45:35 - 00:47:24] Prof. Rahul Pandharipande

So, a group action of G on T is [writes: A group action of G on T is] nothing more or less than a map from $G \times T$ to T [writes: $\phi\colon G \times T \to T$]. That's it, okay?

And it satisfies some conditions. The first condition [writes: (i)] is that ϕ of the neutral element times any element t is just t: [writes: $\phi(e, t) = t$].

This is where $e \in G$ is the neutral element [writes: $e \in G$ neutral] — I don't have to write this, because that's what my convention has —, and t is anything [writes: $\forall t \in T$].

This means that when the group acts, the action by the neutral element doesn't do anything.

And the second one [writes: (ii)] is this associativity property. For the second one, this is now some notation: we write [writes: We write] $\phi(g, t)$ as just $g \cdot t$ [writes: $\phi(g, t) = g \cdot t$].

In algebra, typically every operation is just labeled by dot. So, you have to concentrate when you see like $g \cdot t$: this means t is an element of T [writes: $t \in T$ under t] and g

is an element of G [writes: $g \in G$ under g], and the dot is that, okay?

To write the second property, it's the associativity property [writes: (ii) Associative]. And it's much easier to write in this notation: that if I take $g_1 \cdot t$ and I have g_2 [writes: $g_2 \cdot (g_1 \cdot t)$], this is equal to $(g_2 g_1) \cdot t$ [writes: $= (g_2 g_1) \cdot t$].

And to read this, we can put everything explicit: this is the action [draws an arrow from ϕ to the dot in $g_1 \cdot t$], this is group multiplication [draws an arrow from "Group" to the dot in $g_2 \cdot g_1$]. Here, this is the action [draws an arrow from ϕ to the last dot], and this is group multiplication [labels the dot after g_2 "group"].

Okay? Does that make sense? When you have, yeah.

Student Question

How is that group multiplication? $g_1 \cdot t$ is an element of T.

Spoken Transcript [00:47:28 - 00:47:30] Prof. Rahul Pandharipande

Which one?

Student Question, continued

g_2... on the left.

Spoken Transcript [00:47:33 - 00:47:40] Prof. Rahul Pandharipande

Yeah, sorry. I confused myself, yes. [erases "group", draws a second arrow from ϕ to the dot after g_2] They're both actions.

Spoken Transcript [00:47:40 - 00:48:58] Prof. Rahul Pandharipande

So, you can write this out in terms of that ϕ, but it's a little unpleasant: this would be [writes: $\phi(g_2, \phi(g_1, t)) = \phi(g_2 g_1, t)$].

That's the associativity here, written out explicitly in terms of the action. This is just a translation of that.

In principle, there's never any reason to be confused, because when you have a dot, when you consider group actions, there'll be two things and there could be dots between them. And in principle, there's no reason to be confused, because if both things are group elements, then the dot is group multiplication. If one thing is a group element and the other one is an element of the set T, then it's the action.

You're not allowed to put dots between two elements of T, because we do not have a map that takes two things in T to something else.

Okay? That's it. A group action is this map that takes an element of the group, an element of T to T; for the neutral element, it does nothing, and it has this associativity. The consequences of associativity are great.

Definition 9.33 (*Group Action*): Let $(G, \cdot, e)$ be a group and T a non-empty set. A *"group action"* of G on T (denoted $G \curvearrowright T$) is a mapping:

$$\phi \colon G \times T \longrightarrow T, \quad (g, t) \longmapsto \phi(g, t) =: g \cdot t \tag{9.21}$$

satisfying the following two axioms:

(i) Identity Action: For the neutral element $e \in G$ and all $t \in T$:

$$\phi(e, t) = e \cdot t = t. \tag{9.22}$$

(ii) Compatibility / Associativity: For all $g_1, g_2 \in G$ and all $t \in T$:

$$\phi(g_2, \phi(g_1, t)) = \phi(g_2 g_1, t), \tag{9.23}$$

or equivalently in dot notation:

$$g_2 \cdot (g_1 \cdot t) = (g_2 g_1) \cdot t. \tag{9.24}$$

Decoding the Dot Notation in Equation (9.24): In equation (9.24), the symbols represent distinct operations:

- **Left-Hand Side $g_2 \cdot (g_1 \cdot t)$:**
  - The inner operation $g_1 \cdot t$ is the *"group action"* evaluating to an element of T.
  - The outer operation $g_2 \cdot (g_1 \cdot t)$ is again the *"group action"* of $g_2 \in G$ on the element $(g_1 \cdot t) \in T$.
- **Right-Hand Side $(g_2 g_1) \cdot t$:**
  - The product $g_2 g_1$ is *"group multiplication"* inside G.
  - The outer dot is the *"group action"* of the resulting group element $(g_2 g_1) \in G$ on $t \in T$.

No operation is defined between elements of T: an expression of the form $t_1 \cdot t_2$ for $t_1, t_2 \in T$ is strictly meaningless.

Permutations Induced by a Group Action

Spoken Transcript [00:49:15 - 00:50:23] Prof. Rahul Pandharipande

[erases the left chalkboard] Let me say that first. There are very few hypotheses here, so the consequences of associativity are pretty big.

Let's say $\phi \colon G \times T \to T$ is a group action [writes: $\phi \colon G \times T \to T$ is a group action.], and then let g be an element of G [writes: and then let $g \in G$,]. Then we can define, say, σ_g to be a map from T to T [writes: then we can define $\sigma_g \colon T \to T$], where $\sigma_g(t)$ is equal to $g \cdot t$ [writes: where $\sigma_g(t) = g \cdot t$].

Okay? So, given any element in the group, it kind of spins elements of T, all right?

Spoken Transcript [00:50:23 - 00:52:17] Prof. Rahul Pandharipande

And the claim is, because of what we know, that if I take the neutral element, this is the identity map from T to T: [writes: $\sigma_e = \mathrm{Id} \colon T \to T$]. That's from **(1)** [writes: (1)]. When I take the neutral element, it does nothing to T.

And then the other thing is if I take $\sigma_g \circ \sigma_{g^{-1}}$ [writes: $\sigma_g \cdot \sigma_{g^{-1}}$] — this is a composition

—, this is equal to $\sigma_{gg^{-1}}$ [writes: $= \sigma_{gg^{-1}}$]. So, this is equal to σ_e [writes: $= \sigma_e$], and this is equal to the identity: [writes: $= \mathrm{Id}$].

So, if you expand this: if I do $\sigma_g \circ \sigma_{g^{-1}}(t)$ [writes: $\sigma_g \cdot \sigma_{g^{-1}}(t)$], this is equal to first doing that [writes: $= \sigma_g(\sigma_{g^{-1}}(t))$], and then doing the next one [writes: $= g \cdot (g^{-1} \cdot t)$], and then by associativity, this is $(gg^{-1}) \cdot t$ [writes: $= (gg^{-1}) \cdot t$], and then this is $e \cdot t$ [writes: $= e \cdot t$], which is t [writes: $= t$].

So, that means for every element, I have some map from T to T, and its inverse map is given by doing the group action for the inverse element, right?

In particular, these elements are all bijections. So, the corollary here is that σ_g from T to T is a bijection [writes: Cor. $\sigma_g : T \to T$ is bijection.].

To show that it's a bijection has these two parts: you need to know that the neutral element does nothing — it's an identity —, and then use associativity.

Spoken Transcript [00:52:17 - 00:53:27] Prof. Rahul Pandharipande

So, you can say that for every element of G, I get some bijection, which is kind of a symmetry. And that's why this group action can be viewed as some kind of symmetry.

This is all kind of formal. Let's do some examples. But are there questions about this?

Once you get used to this notation, you just use these dots liberally and just don't worry about them. If you're doing things correctly, the dots will always line up in the correct way.

And the main point is: you're not allowed to put any dots between two elements of T, because there's no rule, given two elements of T, to get a new element. If you ever study group actions and you're writing dots between two elements of T, you've done something terribly wrong.

Proposition 9.34 (*Group Elements Act as Bijections*): Let $\phi\colon G \times T \to T$ be a group action. For each $g \in G$, define the map:

$$\sigma_g : T \longrightarrow T, \quad t \longmapsto \sigma_g(t) := g \cdot t. \tag{9.25}$$

Then:

(1) $\sigma_e = \mathrm{id}_T$.

(2) For all $g, h \in G$, $\sigma_g \circ \sigma_h = \sigma_{gh}$.

(3) For every $g \in G$, σ_g is a bijection of T, with two-sided inverse given by:

$$(\sigma_g)^{-1} = \sigma_{g^{-1}}.$$

Proof

We verify the three claims directly from the group action axioms:

(1) By axiom **(i)**, $\sigma_e(t) = e \cdot t = t = \mathrm{id}_T(t)$ for all $t \in T$.

(2) For any $t \in T$, using axiom **(ii)**:

$$(\sigma_g \circ \sigma_h)(t) = \sigma_g(\sigma_h(t)) = g \cdot (h \cdot t) = (gh) \cdot t = \sigma_{gh}(t).$$

Thus, $\sigma_g \circ \sigma_h = \sigma_{gh}$.

(3) Setting $h = g^{-1}$ in property (2) and invoking property (1):

$$\begin{aligned}(\sigma_g \circ \sigma_{g^{-1}})(t) &= \sigma_g(\sigma_{g^{-1}}(t)) \\ &= g \cdot (g^{-1} \cdot t) \\ &= (gg^{-1}) \cdot t \\ &= e \cdot t = t = \mathrm{id}_T(t).\end{aligned}$$

Hence, $\sigma_g \circ \sigma_{g^{-1}} = \mathrm{id}_T$. Replacing g by g^{-1} shows symmetrically that $\sigma_{g^{-1}} \circ \sigma_g = \mathrm{id}_T$.

Since σ_g possesses a two-sided inverse map $\sigma_{g^{-1}} \colon T \to T$, it is a bijection. **Q.E.D.**

Corollary 9.35 (*Permutation Representation*): Every group action $G \curvearrowright T$ is equivalent to a group homomorphism into the symmetric group of all permutations on T:

$$\Psi \colon G \longrightarrow \mathrm{Sym}(T), \quad g \longmapsto \sigma_g. \tag{9.26}$$

Conceptual Meaning: Symmetries of a Set: This theorem reveals the philosophical essence of group actions:

- Each individual group element $g \in G$ acts as a permutation (*"symmetry"*) of the set T, transforming each state t into $g \cdot t$.
- Group multiplication in G corresponds precisely to functional composition of bijections: $\sigma_{g_1 g_2} = \sigma_{g_1} \circ \sigma_{g_2}$.
- The group G is realized concretely as a subgroup of symmetries $\mathrm{Im}(\Psi) \leq \mathrm{Sym}(T)$, modulo the kernel of elements that act trivially on every element of T.

Examples of Group Actions

Canonical Example: The Symmetric Group Action

Spoken Transcript [00:53:40 - 00:55:38] Prof. Rahul Pandharipande

[erases the middle chalkboard] So, we can do some examples, and usually the examples clear up some of the issues in understanding. The canonical example, the most basic example, is one that you already know.

The most fundamental example — I don't know, maybe that's an exaggeration, but anyway for this class — [writes: The most fundamental] is: you take S_n, that's our group [writes: $\phi\colon S_n$], and it acts on the set T, and the set T is the set of numbers 1, 2 to n [writes: $\times\{1,2,\ldots,n\}$]. This is T [writes: $= T$ under $\{1,2,\ldots,n\}$].

There's a canonical action of the symmetric group on the set 1 to n, and so, what do we have to do? We have to say what happens when I have an element of the symmetric group, γ, and an element of the set from 1 to n: [writes: $\phi(\gamma, t) = \gamma(t)$]. We just feed this element to the symmetric group, because the symmetric group is the set of bijections from the n-element set to itself. So, this defines a group action.

Okay? And then you have to check the two properties. Well, if this were the neutral element of the symmetric group, that neutral element doesn't move anybody, so that's fine. And then you have to check the associativity of the group action, and that is because that's exactly how composition is defined in the symmetric group: the group law in the symmetric group is composition, and that shows the associativity property of this group action.

So, here there are two things to check: check **(1)** and **(2)** [writes: Check (i) + (ii)]. And when you do it, they both come because of the definitions of the neutral element and composition in the symmetric group [writes: both are because of defs. of $e, \circ$ in S_n].

It's good to make sure you understand that. It's kind of the motivation example. You can see to go from this example to the whole definition is not a big leap, actually. So, it's very good to understand this example.

Example 9.36 (*Canonical Action of S_n on $\{1,\ldots,n\}$*): Let $G = S_n$ be the symmetric group of degree n, and let $T = \{1,2,\ldots,n\}$. The *"canonical action"* of S_n on T is defined by evaluating the permutation on the index:

$$\phi\colon S_n \times \{1,\ldots,n\} \longrightarrow \{1,\ldots,n\}, \quad \phi(\gamma,t) := \gamma(t). \tag{9.27}$$

Verification of Axioms:

(1) Neutral Element: The neutral element of S_n is the identity permutation $e = \mathrm{id}$. Thus:

$$\phi(e,t) = e(t) = t \quad \forall t \in \{1,\ldots,n\}.$$

(2) Associativity: The group operation in S_n is functional composition $\gamma_2 \circ \gamma_1$. Thus, for all $\gamma_1, \gamma_2 \in S_n$ and $t \in \{1,\ldots,n\}$:

$$\phi(\gamma_2, \phi(\gamma_1,t)) = \phi(\gamma_2, \gamma_1(t)) = \gamma_2(\gamma_1(t)) = (\gamma_2 \circ \gamma_1)(t) = \phi(\gamma_2 \circ \gamma_1, t).$$

Both conditions hold tautologically by the very definition of the symmetric group.

Geometric Actions: Symmetries of the Tetrahedron

Spoken Transcript [00:55:38 - 00:57:12] Prof. Rahul Pandharipande

Another example that we've talked about is if I take, let's say, the tetrahedron [draws a tetrahedron, labels vertices A, B, C, D]. I can take G to be the rotation group of the tetrahedron [writes: $G = \mathrm{Rot}(\triangle)$, with a small tetrahedron inside], which we know is A_4, but I don't say what it is, it's not necessary; and T to be the set of vertices, let's say. Let's name these vertices: A, B, C, D. So, T to be the set $\{A, B, C, D\}$ [writes: $T = \{A, B, C, D\}$].

And then this action, $G \times T \to T$ [writes: $\phi\colon G \times T \to T$], means if I have a rotation and I have a vertex, then what do I get here? Well, I can just rotate that vertex and get another vertex: [writes: $r \times v \to r(v)$].

Is that clear? This is how it's acting as a symmetry of the object. Again, to do this, you have to go check carefully all those things are true, and when you do that, you'll find that it's all somehow tautologically true: the neutral element here was the rotation that doesn't do anything. If it doesn't do anything, it just takes the vertex to itself.

And then you have to check the associativity, and the associativity in these groups of symmetries is always a consequence because the group structure is defined by composition, and that's what's needed there. So, you have to check that for yourself carefully that you believe it.

Spoken Transcript [00:57:12 - 00:58:28] Prof. Rahul Pandharipande

Okay, but why should I take the vertices? Why not take some other set? $\widehat{T}$ is the set of edges [writes: $\widehat{T} = \{\text{set of edges}\}$, with 6 under it]. I can do that also: the set of edges. Can you see all of the edges here? There's one missing: this one [draws missing internal edge in tetrahedron].

So, the set of edges — I don't label them all, but there's the six edges here: set of edges, six edges.

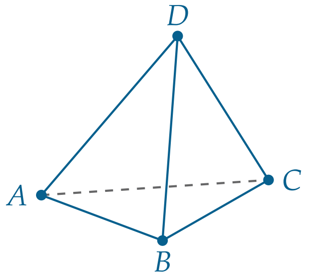
And then there's another action, which is the rotation group [writes: $G \times \widehat{T} \to \widehat{T}$]. So, here it's spinning around the edges. Just as before, if you give a rotation and an edge, then you can rotate that edge and get another edge: [writes: $r\,e \to r(e)$].

That's a different group action with the same geometry.

And there's no reason to stop here: you can also take the set of faces [writes: $\hat{\hat{T}}$ = Set of Faces]. There's four faces, and you can also take $G \times \hat{\hat{T}} \to \hat{\hat{T}}$ [writes: $G \times \hat{\hat{T}} \to \hat{\hat{T}}$], where again, I can rotate a face, and I get the rotated face: [writes: $r \times f \to r(f)$].

Those are all group actions from this...

Symmetries of the Tetrahedron Acting on Vertices, Edges, and Faces



Example 9.37 (*Actions of the Tetrahedron Rotation Group*)**:** Let $G = \mathrm{Rot}(\text{Tetrahedron}) \cong A_4$ be the rotation group of the regular tetrahedron ($|G| = 12$). The single symmetry group G acts naturally on three distinct geometric feature sets:

(1) Action on Vertices ($|T| = 4$): Let $T = \{A, B, C, D\}$. Any rotation $r \in G$ maps vertices to vertices:

$$\phi_V : G \times T \longrightarrow T, \quad (r, v) \longmapsto r \cdot v := r(v).$$

This action defines an isomorphism $\Psi_V : G \xrightarrow{\cong} A_4 \le S_4$.

(2) Action on Edges ($|\widehat{T}| = 6$): Let $\widehat{T}$ be the set of the 6 edges $\{AB, AC, AD, BC, BD, CD\}$. Each rotation maps edges to edges:

$$\phi_E : G \times \widehat{T} \longrightarrow \widehat{T}, \quad (r, e) \longmapsto r \cdot e := r(e).$$

This action defines an injective homomorphism $\Psi_E : G \hookrightarrow S_6$.

(3) Action on Faces ($|\hat{\hat{T}}| = 4$): Let $\hat{\hat{T}}$ be the set of the 4 triangular faces $\{ABC, ABD, ACD, BCD\}$. Each rotation maps faces to faces:

$$\phi_F : G \times \hat{\hat{T}} \longrightarrow \hat{\hat{T}}, \quad (r, f) \longmapsto r \cdot f := r(f).$$

> This action induces a permutation representation into S_4.

Linear Algebra: Matrix Action on Vector Spaces

Spoken Transcript [00:58:53 - 01:00:03] Prof. Rahul Pandharipande

[erases the left chalkboard] Another place symmetries come up is in linear algebra. In a sense, a lot of that course is about symmetry; it's just that it's not said that way for some historical reasons. But a lot of the mathematical content about linear algebra is about symmetries of the vector space.

So, the standard action from a linear algebra class is the following group action: you take GL_n [writes: $\mathrm{GL}_n(\mathbb{C})$] — whatever field, it doesn't really matter, but let's just pick one: $\mathrm{GL}_n(\mathbb{C})$.

Remember, this is the group of invertible $n \times n$ matrices with complex coefficients [writes: group of inv. $n \times n$ matrices with $\mathbb{C}$-coeff.], and it has the neutral element usually called the identity matrix: it's this one [writes: $\mathrm{Id} = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$]. We've discussed this group before, right?

Okay. It acts on the vector space $\mathbb{C}^n$ [writes: $\mathrm{GL}_n(\mathbb{C}) \times \mathbb{C}^n$]. And what is this action? It's not going to surprise you what this action is.

Given an element of this group, which is a matrix, and an element here, which is a vector: the matrix acts on the vector [writes: $\phi\colon \mathrm{GL}_n(\mathbb{C}) \times \mathbb{C}^n \to \mathbb{C}^n$, $\phi(g,v) = gv$]. That's also dot.

So, this is the matrix acting on the vector [draws a square matrix and a column vector]. That's a group action.

Action of the General Linear Group on Vector Spaces

Example 9.38 (*Canonical Action of $\mathrm{GL}_n(\mathbb{C})$ on $\mathbb{C}^n$*)**:** Let $G = \mathrm{GL}_n(\mathbb{C})$ be the general linear group of invertible $n \times n$ matrices with complex entries:

$$\mathrm{GL}_n(\mathbb{C}) = \{M \in \mathrm{Mat}_{n\times n}(\mathbb{C}) \mid \det(M) \neq 0\},$$

with neutral element $\mathrm{Id}_n = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$. The group G acts on the vector space $T = \mathbb{C}^n$ by matrix-vector multiplication:

$$\phi\colon \mathrm{GL}_n(\mathbb{C}) \times \mathbb{C}^n \longrightarrow \mathbb{C}^n, \quad (g,v) \longmapsto g \cdot v. \tag{9.28}$$

Visually, the action corresponds to multiplying an $n \times n$ matrix by a column vector:

$$\begin{pmatrix} g_{11} & \cdots & g_{1n} \\ \vdots & \ddots & \vdots \\ g_{n1} & \cdots & g_{nn} \end{pmatrix} \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} \sum_{j=1}^n g_{1j} v_j \\ \vdots \\ \sum_{j=1}^n g_{nj} v_j \end{pmatrix}.$$

Verification of Axioms:

(1) $\mathrm{Id}_n \cdot v = v$ for all $v \in \mathbb{C}^n$.

(2) For all $g_1, g_2 \in \mathrm{GL}_n(\mathbb{C})$ and $v \in \mathbb{C}^n$, associative matrix multiplication yields:

$$g_2 \cdot (g_1 \cdot v) = (g_2 g_1) \cdot v.$$

Complex Analysis: The Circle Group $\mathrm{U}(1)$ Acting on $\mathbb{C}$

Spoken Transcript [01:00:03 - 01:01:10] Prof. Rahul Pandharipande

A group action which is kind of natural from complex analysis is: you can take G to be... this is just a list of examples, we're going to stop soon.

If you take G to be the unit complex numbers [writes: G = unit complex numbers] — have you thought about this group? It's usually called $\mathrm{U}(1)$ [writes: $\mathrm{U}(1)$], this is the group of complex numbers such that the norm is equal to 1 [writes: $\mathrm{U}(1) = \{z \mid |z| = 1\}$].

It has a neutral element, because 1 is in it [writes: $1 \in \mathrm{U}(1)$], and its multiplication is multiplication of complex numbers. Are you happy with this group?

If you draw a picture of this group, it's complex numbers of norm 1, so it looks like this: [draws unit circle in the complex plane].

An element here is equal to $e^{i\theta}$ [writes: $z = e^{i\theta}$], where θ is a real number.

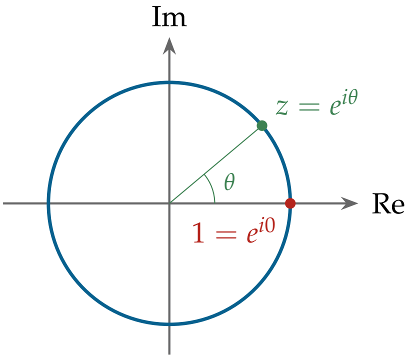
Spoken Transcript [01:01:10 - 01:02:25] Prof. Rahul Pandharipande

So, it's a nice group. In fact, it's a circle. We haven't discussed this group, but it's a fun group to think about: it gives a group structure on the circle.

The neutral element is here [marks $z = e^{i0} = 1$ on real axis], and so on.

And this thing naturally acts on the complex numbers [writes: $G \times \mathbb{C} \to \mathbb{C}$], again, just by multiplication [writes: $g \cdot z \to gz$]. It's just multiplication: it just rotates all the complex numbers.

The Unit Circle Group $\mathrm{U}(1)$ Acting on $\mathbb{C}$



Example 9.39 (*Action of the Circle Group $\mathrm{U}(1)$ on $\mathbb{C}$*): The *"circle group"* $\mathrm{U}(1)$ consists of all unit complex numbers:

$$\mathrm{U}(1) := \{z \in \mathbb{C} \mid |z| = 1\} = \{e^{i\theta} \mid \theta \in \mathbb{R}\}, \tag{9.29}$$

equipped with standard complex multiplication and neutral element $1 = e^{i0}$. The group $\mathrm{U}(1)$ acts on $T = \mathbb{C}$ by standard complex multiplication:

$$\phi\colon\ \mathrm{U}(1) \times \mathbb{C} \longrightarrow \mathbb{C}, \quad (z, w) \longmapsto z \cdot w := zw. \tag{9.30}$$

Geometrically, for $z = e^{i\theta}$ and $w = re^{i\varphi}$, the action yields:

$$z \cdot w = e^{i\theta} \cdot re^{i\varphi} = re^{i(\theta+\varphi)},$$

which rotates every point in the complex plane counter-clockwise by angle θ around the origin.

Groups Acting on Themselves

Left Translation Action

Spoken Transcript [01:02:25 - 01:03:35] Prof. Rahul Pandharipande

There's an endless list of group actions, but as I said, when you encounter groups in mathematics, typically it's because they're acting as symmetries.

Here, the group is acting as the symmetry of the vector space: GL is the symmetry of the vector space. Here, the unit complex numbers are rotational symmetries of $\mathbb{C}$. S_n is just defined to be the symmetries of that set 1 to n, so it's basically defined by that group action.

And then there's a different source of group actions, which are very important for us, which are the ways the group can act on itself.

These are slightly solipsistic group actions, but they're very important.

In order to understand this definition, which is kind of simple, the simplest way is to go look at these examples and make sure you believe that the two properties that are required for a group action are true in all of these cases.

So, another source of examples is that of the group acting on itself [writes: Another source of examples is that of the G-acting on itself].

Spoken Transcript [01:03:35 - 01:04:13] Prof. Rahul Pandharipande

As I said, this is kind of the group looking inward, because the group itself has a lot of symmetries, so the group can act on itself.

The most basic one here is Example 1 [writes: Example 1]: [writes: $\phi\colon G \times G \to G$]...

Now one has to concentrate a little bit: this is the group, and it's acting on itself. So, T is also the group [writes: G with arrow to T].

How do you think I'm going to be able to define this? [writes: $\phi(g, h) =$]

How am I going to define this? Yeah.

Student Answer

Group multiplication.

Spoken Transcript [01:04:22 - 01:05:40] Prof. Rahul Pandharipande

Yeah, what else could I do? All I know about the group is group multiplication; this is group multiplication: [writes: $= gh$].

So, now it's even more confusing.

That's the group acting on itself, just by group multiplication. Sometimes this is called the left action of the group on itself, because G is acting on the left.

Does this make sense? Formally it makes sense. To give an action, I have to have the group, I have to have who it's acting on, and I have to give a map. And in this case, everybody's G. That's logically all right, although confusing. And so, I have to give a rule when I take a group element acting on another element of G, and it's just multiplication. And this rule better satisfy **(1)**, which is that if I put the neutral element and it acts on somebody, then that's the somebody. But that's true, because that's how the neutral element acts in G [writes: (1) $e \cdot h = h$].

And the other one is associativity [writes: (ii)].

If I have this action of g_1 on h, and I put g_2 here, this better be this: [writes: $g_2 \cdot (g_1 \cdot h) = (g_2 g_1) \cdot h$]. And that's true, because it's associativity in the group. And now all the dots are just group multiplication. Again, there's nothing to check; it's just rewriting the definitions, okay?

Left Translation Action of a Group on Itself

Example 9.40 (*Example 1: Left Translation Action*): Let $(G, \cdot, e)$ be a group. We set the target set to be the underlying set of the group itself, $T := G$. The *"left translation action"* (or left regular action) of G on itself is defined by group multiplication:

$$\phi\colon G \times G \longrightarrow G, \quad \phi(g, h) := gh. \tag{9.31}$$

Verification of Action Axioms:

(1) Identity Axiom: For the neutral element $e \in G$ and any $h \in G$:

$$\phi(e, h) = e \cdot h = h.$$

(2) Associativity Axiom: For all $g_1, g_2 \in G$ and any $h \in G$:

$$\phi(g_2, \phi(g_1, h)) = g_2 \cdot (g_1 \cdot h) = g_2(g_1 h) = (g_2 g_1)h = (g_2 g_1) \cdot h = \phi(g_2 g_1, h),$$

which holds directly from the associativity of the binary operation in G.

Spoken Transcript [01:05:40 - 01:05:43] Prof. Rahul Pandharipande

This is actually, as silly as it looks, kind of a fundamental action.

Are there any questions about this action: the G-action on itself by left multiplication?

Student Question

What is the purpose of this example?

Spoken Transcript [01:05:43 - 01:07:15] Prof. Rahul Pandharipande

Okay, it's a good question. I'm going to give you the answer. That's not a question about understanding this example; you're asking why do we think about this example. All right, I will answer that directly.

If you accept this as an example, that everything fits, then I explained also... [moves to the right board] that if I take a group action — so let G be a group, and T be a set [writes: Let G be a group and T be a set], and if I have this group action [writes: $G \curvearrowright T$ / $\phi\colon G \times T \to T$] —, then I explained that from this group action you get a certain object.

Given this group action — this just means I have this multiplication map, this action map —, I explained that to every element g in G [writes: $\forall g \in G \longrightarrow \sigma_g\colon T \to T$], I can associate a bijection, σ_g, from T to T. And the rule was that $\sigma_g(t)$ is equal to $g \cdot t$ [writes: $\sigma_g(t) = g \cdot t$].

So, that's a way of associating to every element a bijection, okay?

The set of bijections is a group.

This is an aside [writes: Let $\mathrm{Sym}(T)$ be the set of bijections]: let $\mathrm{Sym}(T)$ be the set of bijections [writes: $\{f\colon T \to T \text{ bijection}\}$].

If T is a finite set, then this is kind of like the things we've thought about; but anyway, for any set, this is a group: [writes: $(\mathrm{Sym}(T), \circ, \mathrm{Id})$ group].

Review: The Symmetric Group $\mathrm{Sym}(T)$

Definition 9.41 (*Symmetric Group of a Set*)**:** For any non-empty set T, the set of all bijections from T to itself is denoted:

$$\mathrm{Sym}(T) := \{f\colon T \to T \mid f \text{ is bijective}\}.$$

Equipped with functional composition $\circ$ and identity element id_T, $(\mathrm{Sym}(T), \circ, \mathrm{id}_T)$ forms a group. When $|T| = n$ is finite, $\mathrm{Sym}(T) \cong S_n$.

Two Equivalent Perspectives on Group Actions

Spoken Transcript [01:07:15 - 01:08:59] Prof. Rahul Pandharipande

There's a different way to think about group actions — I'm getting to your question, this just takes a little while. [erases the board to the left where Example 1 had been written]

There's a different way to think about group actions which is extremely useful. If you ask how do I think about group actions: half the time it's the way that I already told you, and half the time it's this other way. They're of course logically equivalent, but they have some kind of different connotation.

Way 1 [writes: Way 1] is that a group action is given by $\phi\colon G \times T \to T$ [writes: $\phi\colon G \times T \to T$] that satisfies **(1)** and **(2)** [writes: that satisfies (1) + (2)].

Way 2 [writes: Way 2] is: a group action is a homomorphism — let's call it σ [writes: $\sigma\colon G \to \mathrm{Sym}(T)$] — from G to $\mathrm{Sym}(T)$, where now we just ask this to be a homomorphism [writes: σ is a homomorphism].

It's a different way of thinking about it: just a homomorphism.

And I claim these are equivalent. Let's try to prove they're equivalent.

Spoken Transcript [01:08:59 - 01:10:30] Prof. Rahul Pandharipande

If we're given Action 1 [writes: If given Action 1] — that's the top one —, then we have this ϕ from $G \times T$ to T [writes: $\phi\colon G \times T \to T$], and then I define [writes: then I define] σ by: $\sigma_g\colon T \to T$ [writes: σ by] [writes: $\sigma_g\colon T \to T$] the way I said, $\sigma_g(t) = g \cdot t$ [writes: $\sigma_g(t) = g \cdot t$], okay? That gives me a bijection, and so I have this data: I get a bijection for every element of G, exactly here, which I've been saying. And the condition of **(1)** and **(2)** translates — I leave this to you to check — to this being a homomorphism, all right?

The other way [writes: If given Action 2] is: if I have given Action in form 2, then I start with this homomorphism $\sigma\colon G \to \mathrm{Sym}(T)$ [writes: $\sigma\colon G \to \mathrm{Sym}(T)$], and then I define [writes: then I define] the action $\phi(g,t)$ [writes: $\phi(g,t) =$]. Okay, what am I going to write here?

Spoken Transcript [01:10:30 - 01:11:48] Prof. Rahul Pandharipande

What I have is this homomorphism. So, what am I going to write here? Well, I need to find a way to have g act on t. What I know is how to make g in this $\mathrm{Sym}(T)$, but $\mathrm{Sym}(T)$ is a set of bijections from T to T, so it can eat t. So, it's that: [writes: $\phi(g,t) = \sigma_g(t)$].

Okay? This is just a tautological recasting of the data.

Here is the idea of the group as an actor that's spinning T, which is kind of a dynamic view of the group. And then you have these two conditions.

The other way is: all you do is take a homomorphism to this set $\mathrm{Sym}(T)$. The set $\mathrm{Sym}(T)$ are the spins of T, they're the bijections from T to T, so you just take that as a homomorphism. Then the homomorphism has its own conditions, and they turn out to be equivalent to this. And this is the way to go back and forth, okay?

That's a useful thing to think about: slightly philosophical perspective. And as I said, personally I spend about half my time thinking about these two. It depends on what you want to emphasize. But they are just equivalent.

The Two Equivalent Formulations of Group Actions

Theorem 9.42 (*Equivalence of Action Maps and Permutation Representations*)**:** Let G be a group and T a non-empty set. There is a canonical bijection between:

Way I: Action Maps: Mappings $\phi\colon G \times T \to T$ satisfying:

$$\phi(e,t) = t \quad \forall t \in T, \tag{9.32}$$

$$\phi(g_2, \phi(g_1,t)) = \phi(g_2 g_1, t) \quad \forall g_1, g_2 \in G,\ t \in T. \tag{9.33}$$

Way II: Permutation Representations: Group homomorphisms:

$$\sigma\colon G \longrightarrow \mathrm{Sym}(T). \tag{9.34}$$

Proof

We construct explicit maps in both directions and verify they are mutually inverse:

Direction Way I $\implies$ Way II: Given an action map ϕ, define for each $g \in G$ the map $\sigma_g \colon T \to T$ by:

$$\sigma_g(t) := \phi(g, t) = g \cdot t.$$

By Proposition 9.34, each σ_g is a bijection on T (with inverse $\sigma_{g^{-1}}$), so $\sigma_g \in \mathrm{Sym}(T)$. Furthermore, condition (9.33) implies:

$$\sigma(g_2 g_1)(t) = (g_2 g_1) \cdot t = g_2 \cdot (g_1 \cdot t) = \sigma_{g_2}(\sigma_{g_1}(t)) = (\sigma_{g_2} \circ \sigma_{g_1})(t).$$

Hence, $\sigma(g_2 g_1) = \sigma(g_2) \circ \sigma(g_1)$, proving that $\sigma \colon G \to \mathrm{Sym}(T)$ is a group homomorphism.

Direction Way II $\implies$ Way I: Given a homomorphism $\sigma \colon G \to \mathrm{Sym}(T)$, define the action map ϕ by:

$$\phi(g, t) := \sigma(g)(t) = \sigma_g(t).$$

We verify the two action axioms:

- Since σ is a homomorphism, $\sigma(e) = \mathrm{id}_T$, so $\phi(e, t) = \mathrm{id}_T(t) = t$.
- For $g_1, g_2 \in G$ and $t \in T$:
$$\phi(g_2 g_1, t) = \sigma(g_2 g_1)(t) = (\sigma(g_2) \circ \sigma(g_1))(t) = \sigma(g_2)(\sigma(g_1)(t)) = \phi(g_2, \phi(g_1, t)).$$

These two constructions are mutually inverse by inspection. **Q.E.D.**

☰ **Dynamic vs. Static Perspectives:** The two formulations provide complementary intuitions:

- **Way I (Dynamic/Geometric):** Views G as an *"actor"* actively transforming and permuting the elements of the set T.
- **Way II (Structural/Algebraic):** Views the action as a global homomorphism realizing G as a concrete subgroup of permutations inside $\mathrm{Sym}(T)$.

Cayley's Theorem

Every Finite Group Embeds into a Symmetric Group

Spoken Transcript [01:11:48 - 01:13:45] Prof. Rahul Pandharipande

So, we're coming to your question. You asked it 10 minutes too early in the lecture. [erases the upper-left board where S_n and the tetrahedron were drawn]

We come to Example 1, which is the group acting on itself, and let us restrict ourselves to finite groups here. Because we can prove something which, I think, the first time one sees it, is slightly surprising — especially since we haven't really been doing anything here; all I've been doing is talking. It's been all definitions, but really important definitions.

Let G be a finite group of order n, okay? [writes: Let G be a finite group of order n.] We have a finite group of some order.

Then Example 1 was this group action of G on itself, right? [writes: Ex 1: $\phi \colon G \times G \to G$, $\phi(g, h) = gh$] This was by $\phi(g, h) = gh$.

By the equivalence, I can consider this rather as a homomorphism: I translate this to a homomorphism from G to $\mathrm{Sym}(G)$, okay? [writes: By equivalence, $\sigma\colon G \to \mathrm{Sym}(G)$]

Because that's what that equivalence was: given this actual group action, I can instead view it as a homomorphism from G to $\mathrm{Sym}(G)$.

Now, what is $\mathrm{Sym}(G)$? $\mathrm{Sym}(G)$ is just $\mathrm{Sym}(n)$, because if it has n elements, the group of bijections from G to G is S_n [writes: $\mathrm{Sym}(G) \cong S_n$]. Do you agree?

Spoken Transcript [01:13:45 - 01:15:28] Prof. Rahul Pandharipande

So, it seems like almost from nowhere, I have a homomorphism σ from G to S_n [writes: I have a $\sigma\colon G \to S_n$]. Like: where did that come from?

But I have it. The data of the group knows some homomorphism from G to S_n, okay? And as I said, it's just thinking about the data in a different way. Actually, you can look: I haven't really done anything; it's just about viewing the data.

Now, the thing that's kind of amazing is that this is an injection: [writes: Claim: This is an injection $\Updownarrow$ $\mathrm{Ker}(\sigma) = e$].

That's a claim: an injection, which is to say the kernel of this is just the neutral element. Nobody is going to the kernel here. And why is that?

Let $x \in G$ [writes: Let $x \in G$]. We have to answer ourselves: can this bijection from G to G be the identity? [writes: Can $\sigma_x\colon G \to G$ be the identity?]

What is the identity? The identity is someone that moves nobody, right?

We can just calculate this bijection on e, the neutral element. And what is that? [writes: $\sigma_x(e) = x \cdot e = x$]

Well, by definition, it's this. And that's x!

And if x were not the neutral element, then σ_x is not the identity, okay? [writes: if $x \neq e \implies \sigma_x \neq \mathrm{Id}$]

Because, well, just look: it moves the neutral element to x!

Spoken Transcript [01:15:28 - 01:17:10] Prof. Rahul Pandharipande

Given just the data of the group, not only do we have this homomorphism of the group to S_n in the finite case, this is an injection.

If it's an injection, as a consequence [writes: $G \cong {}^{G}\!/_{\mathrm{Ker}\,\sigma} \cong \mathrm{Im} \subset S_n$], G is isomorphic to G mod ker, because the kernel is just the neutral element, and that's isomorphic to the image inside S_n.

So, we have proven, doing almost nothing really, the following theorem, which is a basic theorem: every finite group [writes in a chalk box: Thm: Every finite group G is a subgroup of S_n, $n = |G|$.] is a subgroup of some symmetric group, of S_n, where n is equal to the order of G.

This also justifies why the symmetric groups — I said they're somehow the richest groups: any group you find, you can always put it in S_n.

And you can put it in by doing almost nothing, just understanding what it means to be a group action: it just sits there as a subgroup.

In other words, there's a sense in which if you're studying group theory — this is not a very useful sense, but there's a sense in which when you're studying group theory —, you never have to look at any groups besides the symmetric group and their subgroups. That covers every single finite group in the world.

So, this is one of the answers to your question, right?

Claim 9.43 (*Injectivity of the Left Regular Representation*): Let G be a group acting on itself by left translation, $\phi(g,h) = gh$. The associated homomorphism:

$$\sigma\colon G \longrightarrow \mathrm{Sym}(G), \quad g \longmapsto \sigma_g$$

is injective, i.e., $\ker(\sigma) = \{e\}$.

Proof

Let $x \in \ker(\sigma)$. By definition of the kernel, $\sigma_x = \mathrm{id}_G$. Evaluating the bijection σ_x on the neutral element $e \in G$:

$$\sigma_x(e) = xe = x. \tag{9.35}$$

On the other hand, since $\sigma_x = \mathrm{id}_G$, we must have:

$$\sigma_x(e) = \mathrm{id}_G(e) = e.$$

Comparing the two expressions gives $x = e$. Conversely, if $x \neq e$, then $\sigma_x(e) = x \neq e = \mathrm{id}_G(e)$, which proves that $\sigma_x \neq \mathrm{id}_G$. Thus, $\ker(\sigma) = \{e\}$, establishing that σ is injective. **Q.E.D.**

Theorem 9.44 (*Cayley's Theorem*): Every finite group G of order $n = |G|$ is isomorphic to a subgroup of the symmetric group S_n:

$$G \cong \mathrm{Im}(\sigma) \leq S_n. \tag{9.36}$$

Proof

Let G be a finite group with $|G| = n$. By Theorem 9.42, the left translation action of G on itself corresponds to a group homomorphism:

$$\sigma\colon G \longrightarrow \mathrm{Sym}(G).$$

Since $|G| = n$, the symmetric group $\mathrm{Sym}(G)$ is isomorphic to S_n. By Claim 9.43, $\ker(\sigma) = \{e\}$, so σ is an injective homomorphism. Applying the First Isomorphism Theorem for groups (Theorem 9.11):

$$G \cong {}^{G}\!/_{\ker(\sigma)} = {}^{G}\!/_{\{e\}} \cong \mathrm{Im}(\sigma) \leq S_n.$$

Therefore, G is isomorphic to the subgroup $\mathrm{Im}(\sigma)$ of S_n. **Q.E.D.**

III Significance and Limitations of Cayley's Theorem

Spoken Transcript [01:17:10 - 01:18:50] Prof. Rahul Pandharipande

You should look at this proof. It's a very strange thing, because when you first see it, you think: how could that be? Because almost nothing has happened. All I've done is somehow taken the definitions and rewritten them in a way that there's been almost no complicated logic at all.

Which is true.

Okay, so that's one of the ways the group acts on itself. [erases the lower-left board]

By the way, let me make a comment: why this result was kind of important for the view of groups. If you have a finite group, it's not like some unknowable thing: it has to sit inside the symmetric group. It's not like a total mystery what groups are: they're subgroups of the symmetric groups.

But the problem with this, as you see, is that this group is sitting inside a symmetric group on the size of the group. If you take A_4 — we know A_4 already sits in S_4 —, suppose you take a group of order 12, then this theorem shows that it's a subgroup of the symmetric group S_{12}, and that's a pretty big group.

So, there's a kind of price you pay for this theorem, which is: the symmetric group is very big. And that price generally means that, you know, you could say that instead of studying groups of order 12, I could study subgroups of S_{12}. It's better to study groups of order 12, because S_{12} is too big.

So, it's not so useful in practice. But as a kind of understanding of where groups sit, it's kind of useful.

Spoken Transcript [01:18:50 - 01:19:30] Prof. Rahul Pandharipande

This is considered a kind of basic result in group theory: that if you have a finite group, you can find it as a subgroup of the symmetric group. In particular, you could write some program that enumerates all groups by just taking one symmetric group at a time and enumerating all the subgroups. This program won't get very far, but nevertheless.

***Insight:* The Philosophical Power and Practical Limit of Cayley's Theorem**

Cayley's theorem provides a universal classification: **no exotic finite groups exist outside of permutation groups**. Every abstract finite group can be realized concretely as permutations of its own elements.

However, the cost of this universality is astronomical growth in group size:

- A group of order 12 (such as A_4) embeds via Cayley's theorem into S_{12}, which has order:
$$12! = 479{,}001{,}600.$$
Yet we already know from geometry that A_4 embeds into S_4, which has order only $4! = 24$.
- Consequently, Cayley's regular embedding $\sigma\colon G \hookrightarrow S_{|G|}$ is rarely the most efficient representation. Rather than studying groups of order 12 by searching for subgroups inside S_{12}, it is far more effective to analyze groups intrinsically or seek representations in much smaller symmetric groups S_k with $k \ll |G|$.

The Conjugation Action

Definition and Verification of the Action by Conjugation

Spoken Transcript [01:19:30 - 01:20:15] Prof. Rahul Pandharipande

Okay, the next way the group acts on itself, we've also seen. Example 2 of the group acting on itself is something you've also seen: [writes: Example 2]

Example 2: G acts on itself by conjugation [writes: G acts on itself by conjugation].

That one's called the left action of G on itself, because G is acting on the left. This one is: G acts on itself by conjugation. We've also studied this.

And the formula for the action then is $\phi\colon G \times G \to G$ [writes: $\phi\colon G \times G \to G$], and it's $\phi(g,h) = ghg^{-1}$ [writes: $\phi(g,h) = ghg^{-1}$], okay?

Spoken Transcript [01:20:15 - 01:21:55] Prof. Rahul Pandharipande

So, that's the formula. And then you have to go check whether this formula works. You have to check that **(1)** is that $\phi(e,h) = ehe^{-1} = h$ [writes: (1) $\phi(e,h) = ehe^{-1} = h$]. We have to go check that: well, that's ehe^{-1}, that's h, so that's the identity, that's good.

And then you have to check if I do this g_1 on h, and then g_2: [writes: (ii) $g_2(g_1 \cdot h) = (g_2 g_1) \cdot h$].

By definition, this is $g_1 h g_1^{-1}$, and then I do that one: that's $g_2(g_1 h g_1^{-1})g_2^{-1}$. That's this side, and you have to check that that equals $g_2 g_1$ acting on h: [writes: $= (g_2 g_1)h(g_2 g_1)^{-1}$].

And then you need to know that this is $(g_2 g_1)h$, and what's the question is: what's the inverse of $g_2 g_1$? And that's that: [writes: $= g_2 g_1 h g_1^{-1} g_2^{-1}$]. Okay? And that's that.

The Conjugation Action of a Group on Itself

Example 9.45 (*Example 2: Group Acting on Itself by Conjugation*)**:** Let $(G,\cdot,e)$ be a group, and set $T := G$. The *"conjugation action"* of G on itself is defined by:

$$\phi\colon G \times G \longrightarrow G, \quad \phi(g,h) := ghg^{-1}. \tag{9.37}$$

Verification of Action Axioms:

(1) Identity Axiom: For the neutral element $e \in G$ and any $h \in G$:

$$\phi(e,h) = ehe^{-1} = ehe = h.$$

(2) Associativity Axiom: Let $g_1, g_2 \in G$ and $h \in G$. Evaluating the repeated action using the shoe-sock rule $(g_2 g_1)^{-1} = g_1^{-1} g_2^{-1}$:

$$\begin{aligned} g_2 \cdot (g_1 \cdot h) &= g_2 \cdot (g_1 h g_1^{-1}) \\ &= g_2 (g_1 h g_1^{-1}) g_2^{-1} \\ &= (g_2 g_1) h (g_1^{-1} g_2^{-1}) \\ &= (g_2 g_1) h (g_2 g_1)^{-1} \\ &= (g_2 g_1) \cdot h. \end{aligned}$$

Hence, $\phi(g_2, \phi(g_1, h)) = \phi(g_2 g_1, h)$, satisfying the associativity axiom.

Contrast Between Left Translation and Conjugation

Spoken Transcript [01:21:55 - 01:23:48] Prof. Rahul Pandharipande

So, that's a whole different action. It's a very different action than Example 1.

And maybe, since I only have a few minutes, instead of writing some theorems, I'll say a little bit about that. They have totally different flavors.

This left action of G on itself is always kind of an interesting action, because every element of G acts non-trivially on itself: it pushes G around. If you have an element of G, it can't act trivially on G by left multiplication; it would have to be the identity. So, it's something non-trivial: it's always spinning G around.

Conjugation is different. The amount that an element spins G by conjugation depends on how commutative it is. For example, if G is a commutative group, then nothing is happening here: nothing at all! If G is a commutative group, then... yeah: [writes on the board to the right: If G is abelian,]

If G is abelian, then this conjugation action is silly. We would say it is trivial [writes: Conjugation action is trivial].

Which means that the action takes (g, h) just to h: [writes: $\phi(g, h) = h$]. It just doesn't care what g is. Another way of saying it's trivial is: the corresponding homomorphism from G to $\mathrm{Sym}(G)$ [writes: $\sigma\colon G \to \mathrm{Sym}(G)$] is the trivial homomorphism. It takes σ_g equal to the neutral element everywhere: [writes: $\sigma_g = e$]. Nothing at all is happening with it.

Conjugation is a different kind of action: it is interesting only when the group is non-abelian, in which case it can be very interesting, because it tells you in some sense how the non-abelianness of the group is working.

So, they have completely different flavors, even though they just come from the definition of the group.

In particular, you can't use the conjugation action to prove the group is a subgroup of the symmetric group in the same way, because it could be that you're an abelian group, in which case the statement that fails here is this: can you have a bijection which is the identity? For the left group action, you cannot, unless it's the neutral element. But here you could, so you couldn't use this to prove that.

Comparison: Left Regular Action vs. Conjugation Action

Proposition 9.46 (*Triviality of Conjugation on Abelian Groups*)**:** If G is an abelian group, then the conjugation action of G on itself is trivial:

$$\phi(g, h) = ghg^{-1} = hgg^{-1} = h \quad \forall g, h \in G. \tag{9.38}$$

Equivalently, the induced homomorphism into the symmetric group:

$$\sigma\colon G \longrightarrow \mathrm{Sym}(G), \quad g \longmapsto \sigma_g = \mathrm{id}_G \tag{9.39}$$

is the trivial homomorphism mapping every element of G to the identity permutation.

Fundamental Differences Between the Two Actions: Although both actions use $T = G$, they possess fundamentally distinct algebraic properties:

- **Left Translation Action ($\phi(g, h) = gh$):**
  - **Always Faithful:** The only element acting as the identity is $g = e$.

  - **Transitive:** For any $h_1, h_2 \in G$, the element $g = h_2 h_1^{-1}$ maps h_1 to h_2.
  - **Embeds G into S_n:** Generates Cayley's theorem for any group, abelian or non-abelian.
- **Conjugation Action ($\phi(g,h) = ghg^{-1}$):**
  - **Measures Non-Commutativity:** The kernel of the action is the *"center"* of the group:
$$\ker(\sigma) = \{g \in G \mid ghg^{-1} = h \; \forall h \in G\} = Z(G).$$
  - **Trivial for Abelian Groups:** If G is abelian, $Z(G) = G$, so the kernel is the entire group and the action carries zero information.
  - **Orbits are Conjugacy Classes:** For non-abelian groups, the partition into orbits corresponds to conjugacy classes, playing a central role in character theory and the Sylow theorems.

Course Outlook: The Sylow Theorems

Spoken Transcript [01:23:48 - 01:24:20] Prof. Rahul Pandharipande

For this course, we have something like a week and a half more about groups, and it will all be about actions. And the fanciest theorems there traditionally are the Sylow theorems,which have to do with p-subgroups. They're proved by studying group actions, and they're all proved in kind of magical ways, in the sense that the funny thing about this group action idea is that it's such a simple idea, but it's incredibly powerful.

Okay, good. So, we stop here. [puts down chalk] [lecture concludes]

Summary of Key Concepts

This lecture completed foundational structural topics in group theory before pivoting to group actions:

- **Cosets and Index 2:** A subgroup $H \leq G$ partitions G into left cosets xH or right cosets Hx. If H is normal in G, left and right cosets coincide ($xH = Hx$). Furthermore, any subgroup of index 2 is automatically normal ($[G : H] = 2 \implies H \triangleleft G$).
- **Subgroup Correspondence Theorem:** The canonical projection $q\colon G \to {}^{G}\!/_{N}$ induces an inclusion-preserving bijection between subgroups of G containing N and subgroups of ${}^{G}\!/_{N}$, which preserves normality in both directions.
- **Classification of Prime Order Groups and Polyhedral Symmetries:** Every group of prime order p is cyclic ($G \cong \mathbb{Z}/p\mathbb{Z}$). The rotational symmetry groups of the Platonic solids are A_4 (tetrahedron), S_4 (cube and octahedron via geometric duality), and A_5 (dodecahedron and icosahedron via duality and inscribed cubes). Planar regular n-gons have rotation group $\mathbb{Z}/n\mathbb{Z}$ and full symmetry dihedral group D_{2n}, with $D_6 \cong S_3$ and $D_8 \lneq S_4$.
- **Isomorphism Invariants:** Groups are distinguished by divisibility ($(\mathbb{Q},+,0) \ncong (\mathbb{Z},+,0)$), singly generated / cyclic properties ($(\mathbb{Q}^*,\cdot,1) \ncong (\mathbb{Z},+,0)$), torsion elements (order 2 element $-1 \in \mathbb{Q}^*$), and vector space dimension via mod-2 re-

duction ($\mathbb{Z}^k \ncong \mathbb{Z}^\ell$ for $k \neq \ell$). Under the Axiom of Choice, $(\mathbb{R}, +, 0) \cong (\mathbb{C}, +, 0)$ as additive groups via Hamel bases over $\mathbb{Q}$, while $(\mathbb{Q}^*, \cdot, 1) \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}^\infty$ by unique prime factorization.

- **Group Actions and Permutation Representations:** An action $G \curvearrowright T$ via $\phi\colon G \times T \to T$ is equivalent to a homomorphism $\sigma\colon G \to \mathrm{Sym}(T)$ into the symmetric group of bijections on T.
- **Actions on the Group Itself and Cayley's Theorem:** The left regular action $\phi(g,h) = gh$ is faithful, proving Cayley's Theorem: every finite group of order n embeds into S_n ($G \hookrightarrow S_n$). In contrast, the conjugation action $\phi(g,h) = ghg^{-1}$ is trivial on abelian groups and measures non-commutativity in general groups.